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ILI9806G

a-Si TFT LCD Single-Chip Driver
480(RGB) x 864 Resolution
16.7M-color with Internal GRAM

Specification
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Table of Contents

SECTION **PAGE**

1. Introduction 19
2. Features 20
3. Device Overview 22
3.1. Block Diagram 22
3.2. Block Function Description 23
3.2.1. System Interface 23
3.2.2. Parallel RGB Interface 23
3.2.3. Address Counter (AC) 24
3.2.4. Graphic RAM (GRAM) 24
3.2.5. Grayscale Voltage Generating Circuit 24
3.2.6. Timing Generating 24
3.2.7. Oscillator 24
3.2.8. Source Driver Circuit 25
3.2.9. Panel Control Circuit 25
3.2.10. Power Supply Circuit 25
3.2.11. MIPI DSI Controller Circuit 25
3.2.12. MDDI 25
3.2.13. CABC (Content Adaptive Brightness Control) 25
3.3. Pin Descriptions 26
3.4. Pin Assignment 33
3.5. Bump Arrangement 36
3.6. Pad Coordination 37
System Interface 46
3.7. DBI Type B Parallel Interface 46
3.7.1. Write Cycle Sequence 47
3.7.2. Read Cycle Sequence 48
3.7.3. DBI Type B Interface Set Table 49
3.8. DBI Type C Serial Interface 52
3.8.1. Write Cycle Sequence 52
3.8.2. Read Cycle Sequence 54
3.9. Data Transfer Break and Recovery 55
3.10. Data Transfer Pause 57
3.10.1. Serial Interface Pause 58
3.10.2. Parallel Interface Pause 59
3.10.3. Data Transfer Mode 60

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3.10.4. Method 1 60
3.10.5. Method 2 60
3.11. DPI (RGB) Interface 61
3.11.1. DPI Interface Selection 61
3.11.2. DPI Interface Timing 63
3.12. DSI System Interface 64
3.12.1. General Description 64
3.12.2. Interface Level Communication 64
3.12.3. General 64
3.12.4. DSI-CLK Lanes 65
3.12.5. Low Power Mode (LPM) 66
3.12.6. Ultra Low Power Mode (ULPM) 68
3.12.7. High-Speed Clock Mode (HSCM) 69
3.12.8. DSI-D1 and DSI-D0 Data Lanes 71
3.12.9. General 71
3.12.10. Escape Modes 71
3.12.11. Low-Power Data Transmission (LPDT) 73
3.12.12. Ultra-Low Power State (ULPS) 74
3.12.13. Remote Application Reset (RAR) 75
3.12.14. Tearing Effect (TEE) 76
3.12.15. Acknowledge (ACK) 77
3.12.16. High-Speed Data Transmission (HSDT) 78
3.12.17. Entering High-Speed Data Transmission (TSOT of HSDT) 78
3.12.18. Leaving High-Speed Data Transmission (TEOT of HSDT) 79
3.12.19. Burst of the High-Speed Data Transmission (HSDT) 80
3.12.20. Bus Turnaround (BTA) 83
3.12.21. Packet Level Communication 84
3.12.22. Short Packet (SPa) and Long Packet (LPa) Structures 84
3.12.23. Bit Transmission Order of Bytes in Packets 85
3.12.24. Byte Transmission Order of Multiple-Byte Information in Packets 85
3.12.25. Packet Header (PH) 86
3.12.26. Data Identification (DI) 87
3.12.27. Virtual Channel (VC) 88
3.12.28. Data Type (DT) 89
3.12.29. Packet Data (PD) of a Short Packet (SPa) 91
3.12.30. Word Count (WC) of a Long Packet (LPa) 92
3.12.31. Error Correction Code (ECC) 93
3.12.32. Packet Data (PD) of a Long Packet (LPa) 97

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3.12.33. Packet Footer (PF) of a Long Packet (LPa) 98
3.12.34. Packet Transmissions 100
3.12.35. Packet from the MPU to the Display Module 100
3.12.36. Display Command Set (DCS) 100
3.12.37. Display Command Set (DCS) Write, No Parameter (DCSWN-S) 101
3.12.38. Display Command Set (DCS) Write, 1 Parameter (DCSW1-S) 102
3.12.39. Display Command Set (DCS) Write Long (DCSW-L) 103
3.12.40. Display Command Set (DCS) Read, No Parameter (DCSRN-S) 107
3.12.41. Null Packet, No Data (NP-L) 109
3.12.42. End of Transmission Packet (EoTP) 111
3.12.43. Packet from the Display Module to the MPU 113
3.12.44. Used Packet Types 113
3.12.45. Acknowledge with Error Report (AwER) 115
3.12.46. DCS Read Long Response (DCSRR-L) 118
3.12.47. DCS Read Short Response, 1 Byte Returned (DCSRR1-S) 120
3.12.48. DCS Read Short Response, 2 Bytes Returned (DCSRR2-S) 121
3.12.49. Communication Sequences 122
3.12.50. General 122
3.12.51. Sequences 123
3.12.52. DCS Write, 1 Parameter Sequence 123
3.12.53. DCS Write, No Parameter Sequence 124
3.12.54. DCS Write Long Sequence 125
3.12.55. DCS Read, No Parameter Sequence 127
3.12.56. Null Packet, No Data Sequence 129
3.12.57. End of Transmission Packet 129
3.12.58. 16 Bit/Pixel Writing 130
3.12.59. 24 Bit/Pixel Writing 131
3.12.60. 24 bit/pixel Reading 133
3.13. MDDI (Mobile Display Digital Interface) 137
3.14. Display Data Format 147
3.14.1. 3-line Serial Interface 147
3.14.2. SPI Data for 3-bit/pixel (RGB 1-1-1 bits input), 8-color 147
3.14.3. SPI Data for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 148
3.14.4. SPI Data for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 148
3.14.5. SPI Data for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 149
3.14.6. 8-bit Parallel MPU Interface 150
3.14.7. 8-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 151
3.14.8. 8-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 152

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3.14.9. 8-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 153
3.14.10. 9-bit Parallel MPU Interface 154
3.14.11. 9-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 155
3.14.12. 9-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 156
3.14.13. 9-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 157
3.14.14. 16-bit Parallel MPU Interface 158
3.14.15. 16-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 159
3.14.16. 16-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 160
3.14.17. 16-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 161
3.14.18. 18-bit Parallel MPU Interface 162
3.14.19. 18-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 163
3.14.20. 18-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 164
3.14.21. 18-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 165
3.14.22. 24-bit Parallel MPU Interface 166
3.14.23. 24-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 167
3.14.24. DPI (RGB) Interface 168
3.14.25. 16-bit/pixel 65K colors order on the DPI Interface 168
3.14.26. 18-bit/pixel 262K Colors Order on the DPI Interface 169
3.14.27. 24-bit/pixel 16.7M colors order on the DPI Interface 170
3.14.28. 16/18-bit Color Data Mapping to 24-bit Pixel Data Operation 171
3.14.29. DSI Transmission Data Format 173
3.14.30. 16-bit per Pixel, Long Packet, Data Type 00 1110 (0Eh) 173
3.14.31. 18-bit per Pixel, Long Packet, Data Type = 01 1110 (1Eh) 174
3.14.32. 18-bit per Pixel, Long Packet, Data Type = 10 1110 (2Eh) 175
3.14.33. 24-bit per Pixel, Long Packet, Data Type = 11 1110 (3Eh) 176
4. Command 177
4.1. Command List 177
4.1.1. Standard Command List 177
4.1.2. Extended Command List 184
4.2. Command Description 190
4.2.1. NOP (00h) 190
4.2.2. Software Reset (01h) 191
4.2.3. Read display identification information (04h) 192
4.2.4. Read Number of the Errors on DSI (05h) 193
4.2.5. Get Red Channel (06h) 194
4.2.6. Get Green Channel (07h) 195
4.2.7. Get Blue Channel (08h) 196
4.2.8. Read Display Status (09h) 197

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4.2.9. Read Display Power Mode (0Ah) 199
4.2.10. Read Display MADCTL (0Bh) 201
4.2.11. Read Display Pixel Format (0Ch) 203
4.2.12. Read Display Image Mode (0Dh) 204
4.2.13. Read Display Signal Mode (0Eh) 206
4.2.14. Read Display Self-Diagnostic Result (0Fh) 208
4.2.15. Sleep In (10h) 209
4.2.16. Sleep Out (11h) 210
4.2.17. Partial Mode On (12h) 211
4.2.18. Normal Display Mode On (13h) 212
4.2.19. Display Inversion Off (20h) 213
4.2.20. fDisplay Inversion On (21h) 214
4.2.21. All Pixel Off (22h) 215
4.2.22. All Pixel On (23h) 216
4.2.23. Gamma Set (26h) 217
4.2.24. Display Off (28h) 218
4.2.25. Display ON (29h) 219
4.2.26. Column Address Set (2Ah) 220
4.2.27. Page Address Set (2Bh) 222
4.2.28. Memory Write (2Ch) 224
4.2.29. Memory Read (2Eh) 225
4.2.30. Partial Area (30h) 226
4.2.31. Vertical Scrolling Definition (33h) 228
4.2.32. Tearing Effect Line OFF (34h) 232
4.2.33. Tearing Effect Line ON (35h) 233
4.2.34. Memory Access Control (36h) 235
4.2.35. Vertical Scrolling Start Address (37h) 238
4.2.36. Idle Mode Off (38h) 240
4.2.37. Idle Mode On (39h) 241
4.2.38. Interface Pixel Format (3Ah) 242
4.2.39. Memory Write Continue (3Ch) 243
4.2.40. Memory Read Continue (3Eh) 244
4.2.41. Write Tear Scan Line (44h) 245
4.2.42. Read Scan Line (45h) 246
4.2.43. Write Display Brightness Value (51h) 247
4.2.44. Read Display Brightness Value (52h) 248
4.2.45. Write CTRL Display Value (53h) 249
4.2.46. Read CTRL Display Value (54h) 251

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4.2.47. Write Content Adaptive Brightness Control Value (55h) 252
4.2.48. Read Content Adaptive Brightness Control Value (56h) 254
4.2.49. Write CABC Minimum Brightness (5Eh) 256
4.2.50. Read CABC Minimum Brightness (5Fh) 257
4.2.51. Read Automatic Brightness Control Self-diagnostic Result (68h) 258
4.2.52. Read Black/White Low Bits (70h) 259
4.2.53. Read Bkx (71h) 260
4.2.54. Read Bky (72h) 261
4.2.55. Read Wx (73h) 262
4.2.56. Read Wy (74h) 263
4.2.57. Read Red/Green Low Bits (75h) 264
4.2.58. Read Rx (76h) 265
4.2.59. Read Ry (77h) 266
4.2.60. Read Gx (78h) 267
4.2.61. Read Gy (79h) 268
4.2.62. Read Blue/A Color Low Bits (7Ah) 269
4.2.63. Read Bx (7Bh) 270
4.2.64. Read By (7Ch) 271
4.2.65. Read Ax (7Dh) 272
4.2.66. Read Ay (7Eh) 273
4.2.67. Read DDB Start (A1h) 274
4.2.68. Read DDB Continue (A8h) 276
4.2.69. Read First Checksum (AAh) 277
4.2.70. Read Continue Checksum (AFh) 278
4.2.71. Read Device Code (D3h) 279
4.2.72. Read ID1 (DAh) 280
4.2.73. Read ID2 (DBh) 281
4.2.74. Read ID3 (DCh) 282
4.3. Extend Command (EXTC) Description 283
4.3.1. Interface Mode Control (B0h) 283
4.3.2. Frame Rate Control 1 (In Normal Mode/Full Colors) (B1h) 284
4.3.3. Frame Rate Control 2 (In Idle Mode/8 colors) (B2h) 286
4.3.4. Frame Rate control 3 (In Partial Mode/Full Colors) (B3h) 288
4.3.5. Display Inversion Control (B4h) 290
4.3.6. Blanking Porch Control (B5h) 292
4.3.7. Display Function Control (B6h) 293
4.3.8. Entry Mode Set (B7h) 294
4.3.9. DBI Type B Interface Setting (B8h) 295

The information contained herein is the exclusive property of ILI Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of ILI Technology Corp.

4.3.10. Panel Control (B9h) 296
4.3.11. SPI Interface Setting (BAh) 298
4.3.12. Power Control 1 (C0h) 299
4.3.13. Power Control 2 (C1h) 301
4.3.14. Power Control 3 (C2h) 303
4.3.15. VCOM Control 1 (C7h) 305
4.3.16. Backlight Control 1 (C8h) 306
4.3.17. Backlight Control 2 (C9h) 307
4.3.18. Backlight Control 3 (CAh) 309
4.3.19. Backlight Control 4 (CBh) 310
4.3.20. Backlight Control 5 (CCh) 312
4.3.21. Backlight Control 6 (CDh) 313
4.3.22. Backlight Control 7(CEh) 314
4.3.23. Backlight Control 8 (CFh) 315
4.3.24. NV Memory Write (D0h) 316
4.3.25. NV Memory Protection Key (D1h) 317
4.3.26. NV Memory Status Read (D2h) 318
4.3.27. Engineering Setting (DFh) 320
4.3.28. Positive Gamma Control (E0h) 321
4.3.29. Negative Gamma Correction (E1h) 322
4.3.30. Digital Gamma Control 1 (E2h) 323
4.3.31. Digital Gamma Control 2 (E3h) 323
4.3.32. Digital 3 Gamma Enable (EAh) 324
4.3.33. VGMP / VGMN /VGSP / VGSN Voltage Measurement Set (EDh) 325
4.3.34. Panel Timing Control 1 (F1h) 326
4.3.35. Panel Timing Control 2 (F2h) 327
4.3.36. DVDD Voltage Setting (F3h) 329
4.3.37. Power Control 5 (F5h) 330
4.3.38. Panel Resolution Selection Set (F7h) 331
4.3.39. Read EXTC Command In SPI Mode (FBh) 332
4.3.40. LVGL Voltage Setting (FCh) 333
4.3.41. External Power Selection Set (FDh) 334
4.3.42. EXTC Command Set enable register (FFh) 335
5. Source, Gate and Memory Map 336
5.1. Memory Map of the 480 (RGB) x 864 Resolution 336
5.2. Memory Map of the 480 (RGB) x 854 Resolution 337
5.3. Memory Map of the 480 (RGB) x 800 Resolution 338
5.4. Memory Map of the 480 (RGB) x 640 Resolution 339

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6. Source Driver 340
7. Display Data RAM 341
7.1. Configuration 341
7.2. Memory to Display Address Mapping 342
7.2.1. Fully Display 342
7.2.2. Vertical Scrolling Display 344
7.2.3. Vertical Scroll 480(RGB) (H) X 864 (V) Example 344
7.2.4. Case1: TFA+VSA+BFA ≠ 864 344
7.2.5. Case2: TFA+VSA+BFA = 864 (Rolling Scrolling) 344
7.3. MPU to Memory Write/Read Direction 347
8. Tearing Effect Information 350
8.1. Tearing Effect Line 351
8.1.1. Tearing Effect Line Modes 351
8.1.2. Tearing Effect Line Timing 352
8.2. Tearing Effect Bus Trigger 353
8.2.1. Tearing Effect Bus Trigger Enable 353
8.2.2. Tearing Effect Bus Trigger Disable 354
8.2.3. Tearing Effect Bus Trigger Sequences 355
9. Content Adaptive Brightness Control (CABC) Function 363
10. Color Enhancement Function 364
11. Sleep Out Command and Self-Diagnostic Functions 365
11.1. Register Loading Detection 365
11.2. Functionality Detection 366
12. Power ON/OFF Sequence 367
12.1. Case 1 – RESX Line is Held High or Unstable by Host at Power ON 368
12.2. Case 2 – RESX Line is Held Low by Host at Power ON 369
12.3. Abnormal Power Off 370
13. Power Level Definition 371
13.1. Power Levels 371
13.2. Power Flow Chart 372
13.3. LCM Voltage Generation 373
13.4. Gamma Curves 374
13.4.1. Gamma Curve 1 (GC0), applies the function y = x^2.2 374
13.4.2. Gamma Curve 2 (GC1), applies the function y = x^1.8 374
13.4.3. Gamma Curve 3 (GC2), applies the function y = x^2.5 374
13.4.4. Gamma Curve 4 (GC3), applies the function y = x^1.0 375
14. Reset 376
14.1. Registers 376

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14.2. Driver IC Input and Output pins 377
14.2.1. Output Pins, I/O Pins 377
14.2.2. Reset Timing 378
15. NV Memory Programming 379
15.1. NV Memory Programming Flow 379
16. 3 Gamma Setting Flow 380
16.1. 3 Gamma Setting Flow 380
17. Gamma Correction 381
18. Deep Standby Mode Setting 399
19. Electrical Characteristics 400
19.1. Absolute Maximum Ratings 400
19.2. DC Characteristics for Panel Driving 401
19.3. DSI DC Characteristics 402
19.3.1. DC characteristics for Power Lines 402
19.3.2. DC Characteristics for DSI LP Mode 403
19.3.3. Spike/Glitch Rejection 403
19.3.4. DC Characteristics for DSI HS Mode 404
19.3.5. DC Characteristics for MDDI 407
AC Characteristics 408
19.3.6. Display Parallel 24/18/16/9/8-bit Interface Timing Characteristics (8080-series) 408
19.3.7. Display Serial Interface Timing Characteristics (3-line SPI system) 410
19.3.8. Parallel 24/18/16-bit RGB Interface Timing Characteristics 411
19.3.9. DSI Timing Characteristics 412
19.3.10. High Speed Mode – Clock Channel Timing 412
19.3.11. High Speed Mode – Data Clock Channel Timing 412
19.3.12. High Speed Mode – Rising and Falling Timings 413
19.3.13. Low Speed Mode – Bus Turn Around 414
19.3.14. Data Lanes from Low Power Mode to High Speed Mode 415
19.3.15. Data Lanes from High Speed Mode to Low Power Mode 416
19.3.16. DSI Clock Burst – High Speed Mode to/from Low Power Mode 417
19.3.17. MDDI Timing Characteristics 418
20. Application Circuit 419
20.1. Reference Circuit 419
20.2. External Component 420
21. Revision History 421

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List of Figures

Figure 1: Block Diagram 22
Figure 2: DBI Type B Write Cycle 47
Figure 3: DBI Type B Write Cycle Sequence 47
Figure 4: DBI Type B Read Cycle 48
Figure 5: DBI Type B Read Cycle Sequence 48
Figure 6: DBI Type B 24-bit Data Bus 49
Figure 7: DBI Type B 18-bit Data Bus 49
Figure 8: DBI Type B 16-bit Data Bus 50
Figure 9: DBI Type B 9-bit Data Bus 50
Figure 10: DBI Type B 8-bit Data Bus 51
Figure 11: DBI Type C Data Format 52
Figure 12: DBI Type C Protocol 53
Figure 13: DBI Type C Read Cycle Sequence 54
Figure 14: Data Transfer Break and Recovery 55
Figure 15: Data Transfer Break - Case 1 55
Figure 16: Data Transfer Break - Case 2 56
Figure 17: Data Transfer Pause 57
Figure 18: DBI Type C Data Transfer Pause 58
Figure 19: DBI Type B Data Transfer Pause 59
Figure 20: Data Transfer Mode - Method 1 60
Figure 21: Data Transfer Mode - Method 2 60
Figure 22: DPI Interface 16/18/24-bit Pixel Format Selection 61
Figure 23: General DPI Timing Diagram 62
Figure 24: DPI Interface Timing diagram 63
Figure 25: DSI System Interface Diagram 64
Figure 26: Power Modes of Clock Lanes 65
Figure 27: From ULPM to LPM 66
Figure 28: From High Speed Clock Mode (HSCM) to LPM 66
Figure 29: All the Changes of the Three Modes to the LPM 67
Figure 30: From LPM to ULPM 68
Figure 31: Mode Change from LPM to ULPM 68
Figure 32: From LPM to HSCM 69
Figure 33: Mode Change from LPM to HSCM 69
Figure 34: High Speed Clock Burst 70
Figure 35: General Escape Mode Sequence 72
Figure 36: Low-Power Data Transmission (LPDT) 73
Figure 37: Pause (Example) 73

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Figure 38: Ultra-Low Power State (ULPS) 74
Figure 39: Remote Application Reset (RAR) 75
Figure 40: Tearing Effect (TEE) 76
Figure 41: Acknowledge (ACK) 77
Figure 42: Entering High-Speed Data Transmission (TSOT of HSDT) 78
Figure 43: Leaving High-Speed Data Transmission (TEOT of HSDT) 79
Figure 44: Single Packet in High-Speed Data Transmissions 80
Figure 45: Multiple Packets in High-Speed Data Transmission – Examples 80
Figure 46: Single Packet in HSDT – Even Number of Bytes 81
Figure 47: Single Packet in HSDT – Odd Number of Bytes 81
Figure 48: Start of Transmission (SoT) in HSDT for Multiple Packets 81
Figure 49: Continuous Multiple Packets in HSDT when Number of Bytes is Equal on Both Data Lanes at the End of the Packet 81
Figure 50: Continuous Multiple Packets in HSDT when Number of Bytes is not Equal on Both Data Lanes at the End of the Packet 82
Figure 51: End of Transmission (EoT) in HSDT when Number of Bytes is Equal on Both Data Lanes at the End of the Packet 82
Figure 52: End of Transmission (EoT) in HSDT when Number of Bytes is not Equal on Both Data Lanes at the End of the Packet 82
Figure 53: Bus Turnaround Procedure 83
Figure 54: Short Packet (SPa) Structure 84
Figure 55: Long Packet (LPa) Structure 84
Figure 56: Bit Transmission Order of Bytes in Packets 85
Figure 57: Byte Transmission Order of the Multiple-Byte Information in Packets 85
Figure 58: Packet Header (PH) of a Short Packet (SPa) 86
Figure 59: Packet Header (PH) on Long Packet (LPa) 86
Figure 60: Data Identification (DI) Structure 87
Figure 61: Data Identification (DI) in the Packet Header (PH) 87
Figure 62: Virtual Channel (VC) on the Packet Header (PH) 88
Figure 63: Virtual Channel (VC) Configuration 88
Figure 64: Data Type (DT) in the Packet Header (PH) 89
Figure 65: Packet Data (PD) of a Short Packet (SPa), 2 Bytes Information 91
Figure 66: Packet Data (PD) of a Short Packet (SPa), 1 Byte Information 91
Figure 67: Word Count (WC) of a Long Packet (LPa) 92
Figure 68: Packet Data in Short and Long Packets 92
Figure 69: D [23…0] and P [7…0] of a Short Packet (SPa) 93
Figure 70: D [23…0] and P [7…0] of a Long Packet (LPa) 93
Figure 71: XOR Functionality in a Short Packet (SPa) 94

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Figure 72: XOR Functionality in a Long Packet (LPa) 95
Figure 73: Internal Error Correction Code (IECC) on the Display Module (the Receiver) 95
Figure 74: Internal XOR Calculation between ECC and IECC Values – No Error 96
Figure 75: Internal XOR Calculation between ECC and IECC Values - Error 96
Figure 76: 16-bit Cyclic Redundancy Check (CRC) Calculation 98
Figure 77: CRC Calculation – Packet Data (PD) is 01h 98
Figure 78: Packet Footer (PF) Example 99
Figure 79: Display Command Set (DCS) in Short Packet (SPa) and Long Packet (LPa) 100
Figure 80: Display Command Set (DCS) Write, No Parameter (DCSWN-S) - Example 101
Figure 81: Display Command Set (DCS) Write, 1 Parameter (DCSW1-S) – Example 102
Figure 82: Display Command Set (DCS) Write Long (DCSW-L) with DCS Only - Example 104
Figure 83: Display Command Set (DCS) Write Long with DCS and 1 Parameter - Example 105
Figure 84: Display Command Set (DCS) Write Long with DCS and 4 Parameters - Example 106
Figure 85 Set Maximum Return Packet Size (SMRPS-S) - Example 108
Figure 86 Display Command Set (DCS) Read, No Parameter (DCSRN-S) - Example 109
Figure 87: Null Packet, No Data (NP-L) - Example 110
Figure 88: End of Transmission Packet (EoTP) 112
Figure 89: End of Transmission Packet (EoTP) - Examples 112
Figure 90: Return Bytes in a Single Packet 113
Figure 91: Return Bytes in Several Packets – Not Allowed 113
Figure 92: Exception when Return Bytes in Several Packets 114
Figure 93: Acknowledge with Error Report (AwER) – Example 116
Figure 94: Errors Packets 117
Figure 95: Flow Chart for Errors on DSI 117
Figure 96: DCS Read Long Response (DCSRR-L) - Example 119
Figure 97: DCS Read Short Response, 1 Byte Returned (DCSRR1-S) - Example 120
Figure 98: DCS Read Short Response, 2 Bytes Returned (DCSRR2-S) - Example 121
Figure 99: One Pixel Bit and Write Color Orders 130
Figure 100: One Pixel Write (DCSW-L) – Example 1 130
Figure 101: Red/Green [0:2] Subpixel Write (DCSW1-S) – Example 2 131
Figure 102: Green [3:5]/Blue Subpixel Write (DCSW1-S) – Example 3 131
Figure 103: One Pixel Bit and Color Write Orders 131
Figure 104: One Pixel Write (DCSW-L) – Example 132
Figure 105: Blue Subpixel Write (DCSW1-S) – Example 2 132
Figure 106: Green Subpixel Write (DCSW1-S) – Example 3 133
Figure 107: Red Subpixel Write (DCSW1-S) – Example 4 133
Figure 108: One Pixel Bit and Color Read Order 133
Figure 109: One Pixel Read Response (DCSRR-L) – Example 1 134

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Figure 110: Red Subpixel Response (DCSRR1-S) – Example 2 134
Figure 111: Green Subpixel Response (DCSRR1-S) – Example 3 135
Figure 112: Blue Subpixel Response (DCSRR1-S) – Example 4 135
Figure 113: Red and Green Subpixels Response (DCSRR2-S) – Example 5 135
Figure 114: Green and Blue Subpixels Response (DCSRR2-S) – Example 6 136
Figure 115: Blue and Red Subpixels Response (DCSRR2-S) – Example 7 136
Figure 116: Forward Link Structure 139
Figure 117: Sub-Frame Header Packet 139
Figure 118: Example of Video Stream Packet Fields 140
Figure 119: Video Data Format 141
Figure 120: Register Access Packet 142
Figure 121: Example of Register Access Packet (write to the ILI9806G) 143
Figure 122: Link Shutdown Packet 144
Figure 123: Filler Packet 144
Figure 124: Hibernation Mode Sequence 145
Figure 125: Host Initiated Wake-up without Contention 146
Figure 126: 3-line Serial Interface 147
Figure 127: SPI Data for 3-bit/pixel (RGB 1-1-1 bits input), 8-color 147
Figure 128: SPI Data for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 148
Figure 129: SPI Data for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 148
Figure 130: SPI Data for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 149
Figure 131: 3-line SPI Mode Read Data 150
Figure 132: 8-bit Parallel MPU Interface 150
Figure 133: 8-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 151
Figure 134: 8-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 152
Figure 135: 8-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 153
Figure 136: 9-bit Parallel MPU Interface 154
Figure 137: 9-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 155
Figure 138: 9-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 156
Figure 139: 9-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 157
Figure 140: 16-bit Parallel MPU Interface 158
Figure 141: 16-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 159
Figure 142: 16-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 160
Figure 143: 16-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 161
Figure 144: 18-bit Parallel MPU Interface 162
Figure 145: 18-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color 163
Figure 146: 18-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color 164
Figure 147: 18-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 165

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Figure 148: 24-bit Parallel MPU Interface 166
Figure 149: 24-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color 167
Figure 150: 16-bit/pixel 65K Colors Order on the DPI Interface 168
Figure 151: 18-bit/pixel 262K colors order on the DPI Interface 169
Figure 152: 24-bit/pixel 16.7M Colors Order on the DPI Interface 170
Figure 153: EPF [1:0] = 10, 16-bit data mapping to 24-bit 172
Figure 154: EPF [1:0] = 10, 18-bit data mapping to 24-bit 172
Figure 155: 16-bit per Pixel, Data Type 00 1110 (0Eh) 173
Figure 156: 18-bit per Pixel, Data Type = 01 1110 (1Eh) 174
Figure 157: 18-bit per Pixel, Data Type = 10 1110 (2Eh) 175
Figure 158: 24-bit per Pixel, Data Type = 11 1110 (3Eh) 176
Figure 159: Memory Map of the 480 (RGB) x 864 Resolution, R36h MV bit D5 = 0 336
Figure 160: Memory Map of the 480 (RGB) x 854 Resolution, R36h MV bit D5 = 0 337
Figure 161: Memory Map of the 480 (RGB) x 800 resolution, R36h MV bit D5 = 0 338
Figure 162: Memory Map of the 480 (RGB) x 640 Resolution, R36h MV bit D5 = 0 339
Figure 163: Vertical Scroll Mode Function 344
Figure 164: MV, MX, MY setting (480 (RGB) x 864) 347
Figure 165: Tearing Effect Line Mode 1 351
Figure 166: Tearing Effect Line Mode 2 351
Figure 167: CABC Block Diagram 363
Figure 168: PWM OUT on/off period 363
Figure 169: Saturation Enhancement Image 364
Figure 170: Register Loading Detection 365
Figure 171: Functionality Detection 366
Figure 172: Case 1 – RESX Line is Held High or Unstable by Host at Power ON 368
Figure 173: Case 2 – RESX Line is Held Low by Host at Power ON 369
Figure 174: Power Mode Flow Chart 372
Figure 175: Power Stage Diagram 373
Figure 176: Gamma Curve 1 (GC0) 374
Figure 177: Gamma Curve 2 (GC1) 374
Figure 178: Gamma Curve 3 (GC2) 374
Figure 179: Gamma Curve 4 (GC3) 375
Figure 180: Reset Timing 378
Figure 181: Positive Noise Pulse during Reset Low 378
Figure 182: NV Memory Programming Flow 379
Figure 183: Positive Gamma Control (E0h) 381
Figure 184: Negative Gamma Control (E1h) 382
Figure 185: Deep Standby Mode Entry/Exit 399

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Figure 186: Noise on Power Supply Lines 402
Figure 187: Spike/Glitch Rejection 403
Figure 188: Differential Inputs Logical 0 and 1, Threshold High/Low, Differential Voltage Range 405
Figure 189: Common Mode Voltage on Clock and Data Channels 405
Figure 190: Differential Pair Termination Resistor on the Receiver Side 406
Figure 191: DSI Clock Channel Timing 412
Figure 192: DSI Data to Clock Channel Timings 412
Figure 193: Rising and Falling Timings on Clock and Data Channels 413
Figure 194: BTA from the MPU to the Display Module 414
Figure 195: BTA from the Display Module to the MPU 414
Figure 196: Data Lanes - Low Power Mode to High Speed Mode Timings 415
Figure 197: Data Lanes - High Speed Mode to Low Power Mode Timings 416
Figure 198: Clock Lanes - High Speed Mode to/from Low Power Mode Timings 417
Figure 199: Reference Circuit 419

The information contained herein is the exclusive property of ILI Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of ILI Technology Corp.

List of Tables

Table 1: System Operating Mode 23
Table 2: Bus Interface Pins 26
Table 3: Driver Output Pins 29
Table 4: Charge Pump Relative Pins 30
Table 5: DC/DC Converter Pins 30
Table 6: Power Pins 31
Table 7: Test Pins 32
Table 8: DBI Type B Parallel Interface 46
Table 9: DBI Type C Serial Interface 52
Table 10: DPI Interface Selection 61
Table 11: High Speed and Low-Power Lane Pair State Codes 65
Table 12: Entering and Leaving Sequences 71
Table 13: Escape Commands 72
Table 14: Abbreviations 80
Table 15: Data Types (DT) from the MPU to the Display Module (ILI9806G) 89
Table 16: Data Types (DT) from the Display Module (ILI9806G) to the MPU 90
Table 17: One Bit Error Value of the Error Correction Code (ECC) 97
Table 18: Display Command Set (DCS) Write, No Parameters (DCSWN-S) 101
Table 19: Display Command Set (DCS) Write, 1 Parameter (DCSW1-S) 102
Table 20: Display Command Set (DCS) Write Long (DCSW-L) 103
Table 21: Display Command Set (DCS) Read, No Parameter (DCSRN-S) 107
Table 22: Receiving and Transmitting EoTP during LPDT 111
Table 23: Acknowledge with Error Report (AwER) for Long Packet (LPa) Response 115
Table 24: Acknowledge with Error Report (AwER) for Short Packet (SPa) Response 115
Table 25: Interface Level Communication 122
Table 26: Packet Level Communication 122
Table 27: DCS Write, 1 Parameter Sequence – Example 1 123
Table 28: DCS Write, 1 Parameter Sequence – Example 2 123
Table 29: DCS Write, 1 Parameter Sequence – Example 3 123
Table 30: DCS Write, No Parameter Sequence – Example 1 124
Table 31: DCS Write, No Parameter Sequence – Example 2 124
Table 32: DCS Write, No Parameter Sequence – Example 3 124
Table 33: DCS Write Long Sequence – Example 1 125
Table 34: DCS Write Long Sequence – Example 2 125
Table 35: DCS Write Long Sequence – Example 3 125
Table 36: DCS Write Long Sequence – Example 4 127
Table 37: DCS Read, No Parameter Sequence – Example 1 127

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Table 38: DCS Read, No Parameter Sequence – Example 2 128
Table 39: Null Packet, No Data Sequence - Example 129
Table 40: End of Transmission Packet – Example 129
Table 41: List of Supported MDDI Packets 138
Table 42: Pixel Data Attributes 141
Table 43: 16/18-bit Color Data Mapping to 24-bit Pixel Data Operation 171
Table 44: Tearing Effect Line Timing 352
Table 45: Initial Values of Registers 376
Table 46: Output and I/O Pins 377
Table 47: Reset Timing 378
Table 48: Reset Description 378
Table 49: Absolute Maximum Ratings 400
Table 50: Spike/Glitch Rejection 403
Table 51: DSI Clock Channel Timing 412
Table 52: DSI Data to Clock Channel Timings 412
Table 53: Rising and Falling Timings on Clock and Data Channels 413
Table 54: Low Power State Period Timings – A 414
Table 55: Low Power State Period Timings – B 414
Table 56: Data Lanes - Low Power Mode to High Speed Mode Timings 415
Table 57: Data Lanes - High Speed Mode to Low Power Mode Timings 416
Table 58: Clock Lanes - High Speed Mode to/from Low Power Mode Timings 417
Table 59: External Component Table 420

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1. Introduction

The ILI9806G is a 16.7M single-chip SOC driver for a-Si TFT liquid crystal display panels with a resolution of up to 480 (RGB) x 864 memory size. The ILI9806G is comprised of a 1440-channel source driver, a gate-IC-less level shifter, a 1,244,160-byte GRAM for graphic data of 480 (RGB) x 864 dots, and a power supply circuit.

The ILI9806G supports parallel MPU 8-/9-/16-/18-/24-bit data bus interfaces and a 3-line serial peripheral interface to input commands. The ILI9806G supports a RGB (16-/18-/24-bit) data bus for video image display. For high-speed serial interface, the MIPI DSI and MDDI (Mobile Display Digital Interface) interface mode, the ILI9806G supports two data lanes and one clock lane for high-speed and low power transmission in both directions with low EMI noise.

The ILI9806G operates with a wide range of an analog power supply. The ILI9806G supports 8-color display, sleep mode and deep standby power management functions, ideal for medium or small size portable products where battery power conservation is desirable, such as digital cellular phones, smart phones, MP3 players, personal media players, and similar devices with color graphics display. Additionally, it has an internal DC/DC converter that generates the LCD driving voltage and the voltage follower circuit for LCD driver.

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2. Features

- Display resolution options:
  - 480 (RGB) (H) x 864 (V) with GRAM
  - 480 (RGB) (H) x 854 (V) with GRAM
  - 480 (RGB) (H) x 800 (V) with GRAM
  - 480 (RGB) (H) x 720 (V) with GRAM
  - 480 (RGB) (H) x 640 (V) with GRAM

- Display color modes
  - Full color mode:

    16.7M colors (24-bit data, R: 8-bit, G: 8-bit, B: 8-bit)
  - Reduced color modes:

    262K colors (18-bit data, R: 6-bit, G: 6-bit, B: 6-bit)

    65K colors (16-bit data, R: 5-bit, G: 6-bit, B: 5-bit)

    8 colors (Idle Mode On): 8 colors (3-bit data, R: 1-bit, G: 1-bit, B: 1-bit)

- Display module:
  - On-chip Frame Memory size 1,244,160 bytes, 480 (RGB) (H) x 864 (V) x 24 bits
  - Supports 1440 source channel outputs
  - Supports gate control signals to gate driver in the panel
  - Supports 1-dot/2-dot/3-dot/4-dot/column
  - Gamma correction (1 preset Gamma curve)
  - On module VCOM control (VCL+0.2V to 0V common electrode output voltage range)

- Display interface types:
  - MPU mode:

    MIPI-DBI Type B (Display Bus Interface, 80 System) interface, 8/9/16/18/24-bit bus

    MIPI-DBI Type C (Serial data transfer interface, 3-line SPI) interface
  - MIPI-DSI (Display Serial Interface) interface:

    Supports one data lane/maximum speed 850Mbps or

    Supports two data lanes/maximum speed 500Mbps

    Supports DSI version 1.01.00

    Supports D-PHY version 1.00.00

    Supports DCS version 1.02.00
  - MDDI (Mobile Display Serial Interface):

    Supports one data lane/maximum speed 800Mbps or

    Supports two data lanes/maximum speed 400Mbps

    Supports MDDI V1.2 1 strobe
  - MIPI-DPI (Display Pixel Interface) interface:

    16 bit/pixel (R: 5-bit, G: 6-bit, B: 5-bit)

    18 bit/pixel (R: 6-bit, G: 6-bit, B: 6-bit)

    24 bit/pixel (R: 8-bit, G: 8-bit, B: 8-bit)

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- Power saving modes:
  - Deep-standby mode
  - Sleep mode

- Other on-chip functions/Miscellaneous
  - Supports inversion mode
  - Software programmable color depth mode
  - Oscillator for display clock generation
  - Supports DC VCOM driving
  - DC VCOM voltage generator and adjustment
  - OTP (One-Time Programming) memory store initialization register settings
  - Provide 3 times to store DC VCOM value setting, ID setting
  - Supports CABC (Content Adaptive Brightness Control) function
  - Supports 3-Gamma DGC (Digital Gamma Correction) function
  - Color enhancement function
  - DC/DC converter
  - VGH/VGLO voltage generator for gate control signal in panel
  - Supports gate control signals to gate driver in panel (GIP)

- Input power:
  - I/O supply voltage range：VDDI - VSSI = 1.65V ~ 3.3V (VDDI)
  - Analog supply voltage range：VDDB - VSSB/VDDA - VSSA/VDDR - VSSR = 2.5V ~ 3.3V
  - MIPI/MDDI regulator range：VDDAM - VSSAM = 2.5V ~ 3.3V
  - OTP programming voltage, VPP = 7.5V

- Source/VCOM/Gate power supply voltage:
  - AVDD-AVSS = 4.75 to 6.5V (Step-up 1 output voltage range)
  - AVEE-AVSS = -6.5 to -4.75V (Step-up 2 output voltage range)
  - VCL-GND = -VCI to -1.8V (Step-up 3 output voltage range)
  - DC VCOM = VCL+0.2V to 0V, a step 12.5mV (Common electrode voltage range)
  - VGMP = 3.0V to 6.1875V (AVDD-0.3V) (Positive gamma high voltage range)
  - VGSP = 0.0V to 3.475V (Positive gamma low voltage range)
  - VGMN = -3.0V to -6.1875V (AVEE+0.3V) (Negative gamma high voltage range)
  - VGSN = 0.0V to -3.475V (Negative gamma low voltage range)
  - VGH-AVSS = 12V to 20V (Positive gate driver output voltage range)
  - VGL-AVSS = -7V to -15V (Negative gate driver output voltage range)
  - LVGL-AVSS = -7V to -15V (Negative gate driver output voltage range)
  - VRGH = 1.0V ~ 6.0V (AVDD-0.3V) (Panel voltage range)

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3. Device Overview

3.1. Block Diagram

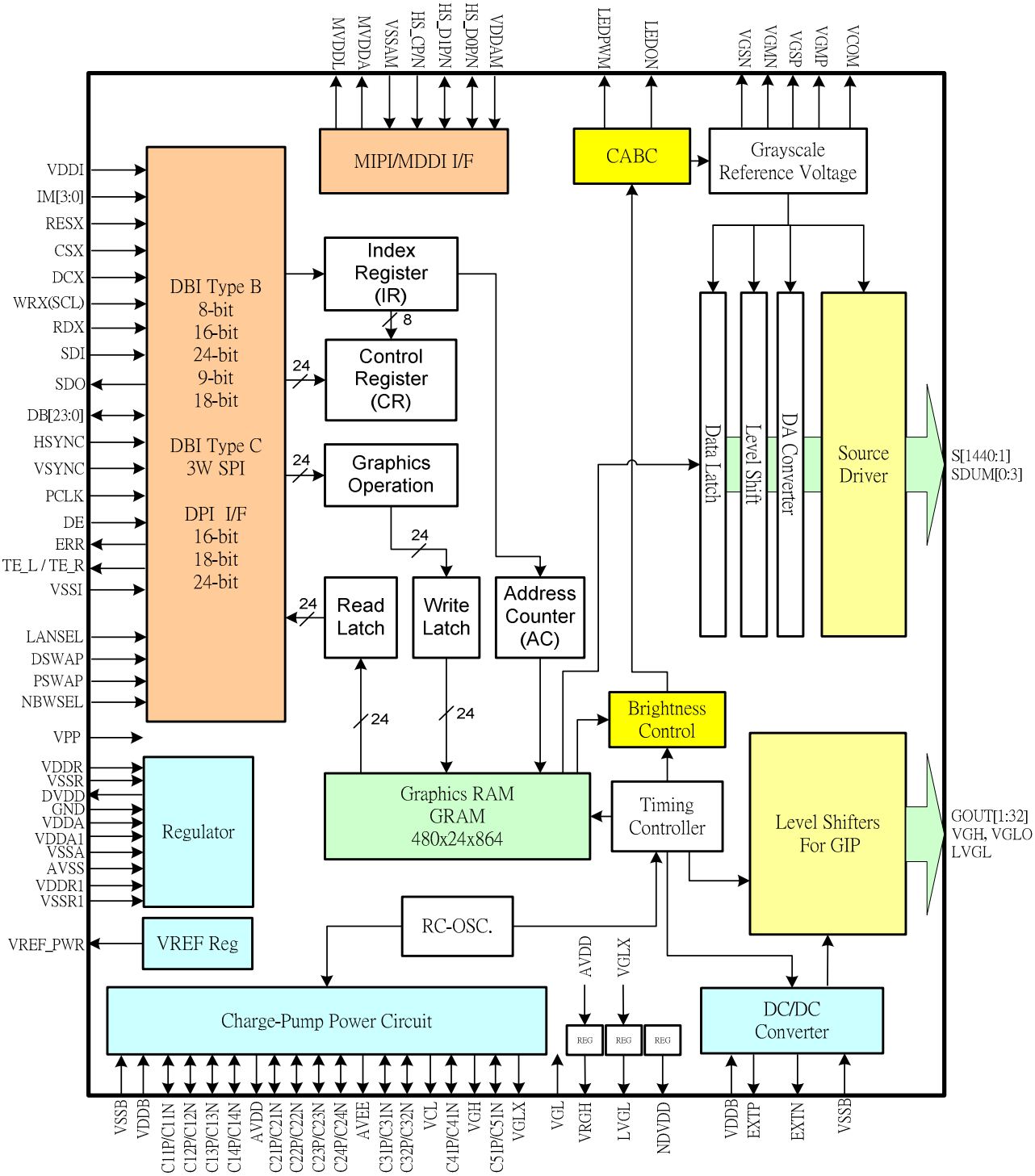


Figure 1: Block Diagram

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3.2. Block Function Description

3.2.1. System Interface

The interface operating mode (DBI, DPI or DSI) is selected by hardware pins IM [3:0], as shown in Table 1 below.

Table 1: System Operating Mode [Note 1]

IM3	IM2	IM1	IM0	Interface	IO Pin in Use
0	0	0	0	DBI Type B 8-bit bus interface	DB [7:0], WRX, RDX, CSX, DCX
0	0	0	1	DBI Type B 16-bit bus interface	DB [15:0], WRX, RDX, CSX, DCX
0	0	1	0	DBI Type B 24-bit bus interface	DB [23:0], WRX, RDX, CSX, DCX
1	1	0	0	DBI Type B 9-bit bus interface	DB [8:0], WRX, RDX, CSX, DCX
1	1	0	1	DBI Type B 18-bit bus interface	DB [17:0], WRX, RDX, CSX, DCX
X	0	1	1	DPI with DBI Type C 3-line 9-bit	DB [23:0], SDI, SDO, SCL, CSX
0	1	0	1	MIPI DSI Interface	HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_CP, HS_CN
0	1	1	0	MDDI Interface	HS_D0P, HS_D0N, HS_D1P, HS_D1N, HS_CP, HS_CN

3.2.2. Parallel RGB Interface

The RGB interface is used as the external interface for displaying moving pictures. When the RGB interface is selected, display operations are synchronized with the externally supplied signals, VSYNC, HSYNC, and PCLK. In the RGB interface mode, data (DB23-0) are written in synchronization with these signals according to the polarity of the Enable Signal (DE). This is done in order to prevent flicker on the display while updating display data.

The RGB interface allows transferring data only when updating the frames of a moving picture by writing all display data to the internal RAM. This method is a contributing factor for the low power requirement of moving picture display.

The ILI9806G includes an IR (index register) which stores the index data of internal control register and GRAM. When DCX = L, the command is written into the driver IC via the DBI interface. When DCX = H, GRAM data via the R2Ch register is written through the data bus. When the data is written into the GRAM from the MPU, it is first written into the write-data latch and then automatically written into the GRAM by internal operation. Data is read through the read-data latch when reading display data from the GRAM.

Note 1 Set Number of colors using set pixel format: 3Ah.

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3.2.3. Address Counter (AC)

The address counter (AC) assigns an address for writing pixel data to the GRAM, or for reading pixel data from the GRAM. Each time a pixel data is written into the GRAM, the X address or Y address of the AC is automatically increased by 1 (or decreased by 1), as determined by the register setting (MV, MX and MY bit). To simplify the address control of the GRAM access, the window address function allows writing data only to a window area of the GRAM specified by Column and Row address registers. After data is written to the GRAM, the AC will be increased or decreased within the setting window address range specified by the Column address register (start: SC, end: EC) and the Row address register (start: SP, end: EP). The window address function enables writing data only in the rectangular area set by users on the GRAM.

3.2.4. Graphic RAM (GRAM)

GRAM is the graphic RAM storing bit-pattern data of 1,244,160 bytes with 24 bits per pixel, enabling a maximum of 480(RGB) x 864 dots graphic display.

3.2.5. Grayscale Voltage Generating Circuit

The grayscale voltage generating circuit generates the LCD drive voltage which corresponds to 256 grayscale level set in the Gamma correction register. The ILI9806G can display up to 16.7M colors at maximum.

3.2.6. Timing Generating

The timing generator is used to generate timing signals for operating internal circuits. The timing generator generates timing signals for internal circuits, such as the internal GRAM. Timing for display operations (such as RAM read operation) and timing for internal operations (such as RAM access by the MPU) are outputted separately so that they do not interfere with each other.

3.2.7. Oscillator

The ILI9806G incorporates with an RC oscillator circuit. Command settings are used to change the frame frequency.

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3.2.8. Source Driver Circuit

The LCD display driver circuit consists of a 480-output source driver (S1~S1440). The display pattern data is latched when 480RGB pixels of data are input. The voltage is output from the source driver according to the latched data.

3.2.9. Panel Control Circuit

The panel control circuit outputs GOUT [1:32] signals at either the VGH or VGLO or AVDD level.

3.2.10. Power Supply Circuit

The LCD drive power supply circuit generates the voltage levels for driving a panel. Voltage levels are adjusted according to the register setting.

3.2.11. MIPI DSI Controller Circuit

The MIPI DSI controller circuit consists of the D-PHY controller, Protocol Control Unit (PCU), Packet Processing Unit (PPU), ECC generating circuit, internal data/command buffer and analog transceiver. The D-PHY controls communication with the analog block, and the ECC generating circuit generates the ECC to check the outgoing data stream for accuracy of the receiving data packet. The PCU controls outgoing and incoming data streams, and the PPU controls transmitting packet distribution and merging. The internal data and command buffer is used for temporary storage of incoming commands and display data.

3.2.12. MDDI

The ILI9806G supports MDDI as a differential small amplitude serial interface for high-speed data transfer.

3.2.13. CABC (Content Adaptive Brightness Control)

The CABC (Content Adaptive Brightness Control) dynamic backlight control function is used to reduce the power consumption of the luminance source.

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3.3. Pin Descriptions

Table 2: Bus Interface Pins

<table>
<tr><th colspan="3">Bus Interface Pins</th></tr>
<tr><th>Pin Name</th><th>I/O</th><th>Descriptions</th></tr>
<tr><td>IM [3:0]</td><td>I</td><td>- Select the interface mode
<table>
<tr><th>IM3</th><th>IM2</th><th>IM1</th><th>IM0</th><th>Interface</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>DBI Type B 8-bit</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>DBI Type B 16-bit</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>DBI Type B 24-bit</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>DBI Type B 9-bit</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>DBI Type B 18-bit</td></tr>
<tr><td>X</td><td>0</td><td>1</td><td>1</td><td>DPI with DBI Type C 3-line 9-bit</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>MIPI DSI</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>MDDI</td></tr>
</table></td></tr>
<tr><td>RESX</td><td>I</td><td>- The external reset input
- Initializes the chip with a low input. Be sure to execute a power-on reset after supplying power.</td></tr>
<tr><td>CSX</td><td>I</td><td>- A chip select signal.
Low: the chip is selected and accessible
High: the chip is not selected and not accessible
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>DCX</td><td>I</td><td>- The DBI Type B interface (DCX): The signal for command or parameter select.
Low: Command
High: Parameter
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>RDX</td><td>I</td><td>- Serves as a read signal and read data at the rising edge.
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>WRX
(SCL)</td><td>I</td><td>- The DBI Type B system (WRX): Serves as a write signal and writes data at the rising edge.
- Serial interface (SCL): Serial clock input
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>DB [23:0]</td><td>I/O</td><td>- A 24-bit parallel bi-directional data bus for MPU system
<table>
<tr><th>Interface Mode</th><th>Data Pin in Use</th></tr>
<tr><td>DBI Type B 8-bit</td><td>DB [7:0]</td></tr>
<tr><td>DBI Type B 16-bit</td><td>DB [15:0]</td></tr>
<tr><td>DBI Type B 24-bit</td><td>DB [23:0]</td></tr>
<tr><td>DBI Type B 9-bit</td><td>DB [8:0]</td></tr>
<tr><td>DBI Type B 18-bit</td><td>DB [17:0]</td></tr>
<tr><td>DPI with DBI Type C 3-line 9-bit</td><td>DB [23:0], SDI, SDO</td></tr>
</table>
Fix to VSSI level when not in use.</td></tr>
<tr><td>SDI
(SDA)</td><td>I/O</td><td>Serial data input pin used for the SPI Interface.
SDI: Serial data input pin</td></tr>
</table>

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<table>
<tr><td></td><td></td><td>SDA: Serial data input/output bidirectional pin
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>SDO</td><td>O</td><td>Serial data output pin used for the SPI Interface.
Leave the pin open when not in use.</td></tr>
<tr><td>TE_L / TE_R</td><td>O</td><td>-Tearing effect output
TE = TE_L= TE_R
Leave the pin open when not in use.</td></tr>
<tr><td>PCLK</td><td>I</td><td>- Dot clock signal for RGB interface operation
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>VSYNC</td><td>I</td><td>- Frame synchronizing signal for RGB interface operation
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>HSYNC</td><td>I</td><td>- Line synchronizing signal for RGB interface operation
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td>DE</td><td>I</td><td>- Data enable signal for RGB interface operation
Low: access enabled
High: access inhibited
Fix to VDDI or VSSI level when not in use.</td></tr>
<tr><td rowspan="2">HS_CP
HS_CN</td><td rowspan="2">I</td><td>MIPI DSI differential clock pair (DSI-CLK+/-)
If MIPI are not in use, they should be connected to VSSAM.</td></tr>
<tr><td>MDDI strobe signal lines (MDDI_Stb_P/M)
HS_CP (Stb+) and HS_CN (Stb-) are differential small swing signals. Make the wiring as short as possible so that the COG resistance becomes less than 10 ohm. The specifications of the interface must be compliant with the MDDI specification.
If MDDI are not in use, they should be connected to VSSAM.</td></tr>
<tr><td rowspan="2">HS_D0P
HS_D0N
HS_D1P
HS_D1N</td><td rowspan="2">I/O</td><td>MIPI DSI differential data pair (DSI-D0+/-、DSI-D1+/-)
If MIPI are not in use, they should be connected to VSSAM</td></tr>
<tr><td>MDDI data signal lines (MDDI_Data0_P/M、MDDI_Data1_P/M)
HS_D0P (Data+) and HS_D0N (Data-) are differential small swing signals. Make the wiring as short as possible so that the COG resistance becomes less than 10 ohm. The specifications of the interface must be compliant with the MDDI specification.
If MDDI are not in use, they should be connected to VSSAM.</td></tr>
<tr><td>ERR</td><td>O</td><td>- CRC and ECC error output pin for the MIPI interface, activated by S/W command. This pin is output low when it is not activated. When this pin is activated, it is output high if CRC/ECC error is found.
Leave the pin open when not in use.</td></tr>
<tr><td>LANSEL</td><td>I</td><td>- Input pin to select 1 data lane or 2 data lanes in MIPI/MDDI interface.
Low: 1 data lane
High: 2 data lanes
Fix to VSSI level when not in use.</td></tr>
<tr><td>DSWAP
PSWAP</td><td>I</td><td>- Differential clock polarity swap
For MIPI interface
<table>
<tr><th rowspan="2">DSWAP</th><th rowspan="2">PSWAP</th><th colspan="6">Pins</th></tr>
<tr><th>HS_CP</th><th>HS_CN</th><th>HS_D0P</th><th>HS_D0N</th><th>HS_D1P</th><th>HS_D1N</th></tr>
<tr><td rowspan="2">0</td><td>0</td><td>HS_CP</td><td>HS_CN</td><td>HS_D0P</td><td>HS_D0N</td><td>HS_D1P</td><td>HS_D1N</td></tr>
<tr><td>1</td><td>HS_CN</td><td>HS_CP</td><td>HS_D0N</td><td>HS_D0P</td><td>HS_D1N</td><td>HS_D1P</td></tr>
<tr><td rowspan="2">1</td><td>0</td><td>HS_CP</td><td>HS_CN</td><td>HS_D1P</td><td>HS_D1N</td><td>HS_D0P</td><td>HS_D0N</td></tr>
<tr><td>1</td><td>HS_CN</td><td>HS_CP</td><td>HS_D1N</td><td>HS_D1P</td><td>HS_D0N</td><td>HS_D0P</td></tr>
</table></td></tr>
</table>

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<table>
<tr><td></td><td></td><td>For MDDI interface
<table>
<tr><th rowspan="2">DSWAP</th><th rowspan="2">PSWAP</th><th colspan="6">Pins</th></tr>
<tr><th>HS_CP</th><th>HS_CN</th><th>HS_D0P</th><th>HS_D0N</th><th>HS_D1P</th><th>HS_D1N</th></tr>
<tr><td rowspan="2">0</td><td>0</td><td>HS_CP</td><td>HS_CN</td><td>HS_D0P</td><td>HS_D0N</td><td>HS_D1P</td><td>HS_D1N</td></tr>
<tr><td>1</td><td>HS_CN</td><td>HS_CP</td><td>HS_D0N</td><td>HS_D0P</td><td>HS_D1N</td><td>HS_D1P</td></tr>
</table>
Fix to VSSI level when not in use.</td></tr>
<tr><td>RGBBP</td><td>I</td><td>- Test Pin
Fix to VSSI level.</td></tr>
<tr><td>GPO [3:0]</td><td>O</td><td>- Test Pin
Leave the pin open.</td></tr>
<tr><td>EXB1T</td><td>I</td><td>- Input pin to select the external DC/DC circuit.
Low: Using internal DC/DC for AVDD / AVEE
High: Using external DC/DC for AVDD / AVEE
Fix to VSSI level when not in use.</td></tr>
<tr><td>NBWSEL</td><td>I</td><td>- Input pin to select the gamma voltage level sequence of V0~V255.
Low: V0 > V1 >…> V254 > V255, normally white
High: V255 > V254 >…> V1 > V0, normally black
Fix to VDDI level when not in use.</td></tr>
<tr><td>LEDON</td><td>O</td><td>- Used for turning On/Off external LED backlight control.
Leave the pin open when not in use.</td></tr>
<tr><td>LEDPWM</td><td>O</td><td>- The PWM frequency output for LED driver control.
Leave the pin open when not in use.</td></tr>
</table>

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Table 3: Driver Output Pins

Driver Output Pins			
Pin Name	I/O	Type	Descriptions
S [1:1440]	O	LCD	- Source output voltage signals applied to a LCD panel
GOUT [1:32]	O	LCD	- Gate control signals and the swing voltage level is VGH to VGLO
SDUM [0:3]	O	LCD	- Dummy Source ***Leave the pin open when not in use.***
VGH	O	LCD	- High voltage level for gate control signals and gate circuit of panel VGH is already connected to VGHO in IC.
VGLO (VGLO_L / VGLO_R)	O	LCD	- Low voltage level for gate control signals and gate circuit of panel
LVGL (LVGL_L / LVGL_R)	O	LCD	- Low voltage level for gate circuit of panel LVGL is already connected to VGL_REG in IC. (VGL_REG is the output voltage generated from VGL. LDO output used for panel voltage.) - Connect a capacitor for stabilization
VGMP (VGMP_PAD)	O	LCD	- Output voltage generated from AVDD. LDO output for positive gamma high voltage generator.
VGSP (VGSP_PAD)	O	LCD	- Output voltage generated from AVDD. LDO output for positive gamma low voltage generator.
VGMN (VGMN_PAD)	O	LCD	- Output voltage generated from AVEE. LDO output for negative gamma high voltage generator.
VGSN (VGSN_PAD)	O	LCD	- Output voltage generated from AVEE. LDO output for negative gamma low voltage generator.

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Table 4: Charge Pump Relative Pins

Charge Pump Relative Pins			
Pin Name	I/O	Type	Descriptions
AVDD	O	Stabilizing capacitor	- Output voltage from step-up circuit 1, generated from VDDB. - Connect to a stabilizing capacitor between AVDD and VSSB.
AVEE	O	Stabilizing capacitor	- Output voltage from step-up circuit 2, generated from VDDB. - Connect to a stabilizing capacitor between AVEE and VSSB.
VCL	O	Stabilizing capacitor	- Output voltage from step-up circuit 3, generated from VDDB. - Connect to a stabilizing capacitor between VCL and VSSB.
VGH	O	Stabilizing capacitor	- Output voltage from step-up circuit 4, generated from VDDB. - Connect to a stabilizing capacitor between VGH and VSSB.
VGL	O	Stabilizing capacitor	- Substrate voltage for driver IC - Output voltage from step-up circuit 5, generated from VDDB. - Connect to a stabilizing capacitor between VGL and VSSB.
C11P, C11N C12P, C12N C13P, C13N C14P, C14N	-	Flying capacitor	- Connect the charge-pumping capacitor for generating AVDD level. - Capacitor connection pins for the step-up circuit 1.
C21P, C21N C22P, C22N C23P, C23N C24P, C24N	-	Flying capacitor	- Connect the charge-pumping capacitor for generating AVEE level. - Capacitor connection pins for the step-up circuit 2.
C31P, C31N C32P, C32N	-	Flying capacitor	- Connect the charge-pumping capacitor for generating VCL level. - Capacitor connection pins for the step-up circuit 3.
C41P, C41N	-	Flying capacitor	- Connect the charge-pumping capacitor for generating VGH level. - Capacitor connection pins for the step-up circuit 4.
C51P, C51N	-	Flying capacitor	- Connect the charge-pumping capacitor for generating VGL level. - Capacitor connection pins for the step-up circuit 5.

Table 5: DC/DC Converter Pins

DC/DC Converter Pins			
Pin Name	I/O	Type	Descriptions
EXTN (CTRL_A)	O	Positive NMOS	-PWM control output for gate of NMOS in positive DC/DC converter (for AVDD) -CTRL_A : Control signal for an external charge pump IC (ex: ILI4002)
EXTP (CTRL_B)	O	Negative PMOS	-PWM control output for gate of PMOS in negative DC/DC converter (for AVEE) -CTRL_B : Control signal for an external charge pump IC (ex: ILI4002)

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Table 6: Power Pins

Power Pins			
Pin Name	I/O	Type	Descriptions
VRGH (VRGH_L / VRGH_R)	O	LDO Stabilizing Capacitor	- Output voltage generated from AVDD. - Connect a capacitor for stabilization. ***Leave the pin open when not in use.***
VREF_PWR	O	LDO Stabilizing Capacitor	- Reference voltage for regulator output. - Connect a capacitor for stabilization.
VDDA	P	Analog	- Power supply for analog system. - VDDA, VDDR, VDDA1, VDDB and VDDR1 should be the same input voltage level of 2.5 ~ 3.3V.
VDDA1	P	Analog	- Power supply for analog system. - VDDA, VDDR, VDDA1, VDDB and VDDR1 should be the same input voltage level of 2.5 ~ 3.3V.
VDDR	P	Regulator	- Power supply for regulator system. - VDDA, VDDR, VDDA1, VDDB and VDDR1 should be the same input voltage level of 2.5 ~ 3.3V.
VDDR1	P	Regulator	- Power supply for regulator system. - VDDA, VDDR, VDDA1, VDDB and VDDR1 should be the same input voltage level of 2.5 ~ 3.3V.
VDDB	P	DC/DC	- Power supply for DC/DC converter. - VDDA, VDDR, VDDA1, VDDB and VDDR1 should be the same input voltage level of 2.5 ~ 3.3V.
VDDI	P	I/O	- Power supply for I/O block.
DVDD	O	Stabilizing Capacitor	- Internal logic voltage output - Connect a capacitor for stabilization.
NDVDD	O	Stabilizing Capacitor	- Negative internal logic voltage output - Connect a capacitor for stabilization.
VDDAM	P	MIPI/MDDI	- Power supply for MIPI/MDDI analog system.
MVDDA	O	MIPI Stabilizing Capacitor	- Regulator output for internal MIPI / MDDI analog system (1.5V typical) - Connect a capacitor for stabilization.
MVDDL	O	MIPI Stabilizing Capacitor	- Regulator output for internal MIPI low power system (1.2V typical) - Connect a capacitor for stabilization.
VSSA	P	Analog	- System ground for analog circuit.
VSSR	P	Regulator	- System ground for regulator circuit.
VSSR1	P	Regulator	- System ground for regulator circuit.
VSSB	P	DC/DC	- System ground for DC/DC convertor.
VSSI	P	I/O	- System ground for interface system.
GND	P	Digital	- System ground for internal digital system.
AVSS	P	Source OP	- System ground for source OP system.
VSSAM	P	MIPI/MDDI	- System ground for MIPI/MDDI system.
VPP	I	OTP	- OTP programming power.
VCOM (VCOM_L /	O	LDO Stabilizing	- Output voltage generated from VCI / VCL - Connect a capacitor for stabilization.

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VCOM_R)		Capacitor	VCOM=VCOM_L=VCOM_R

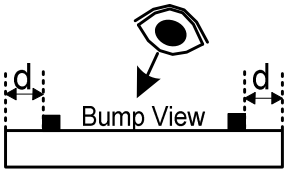
Table 7: Test Pins

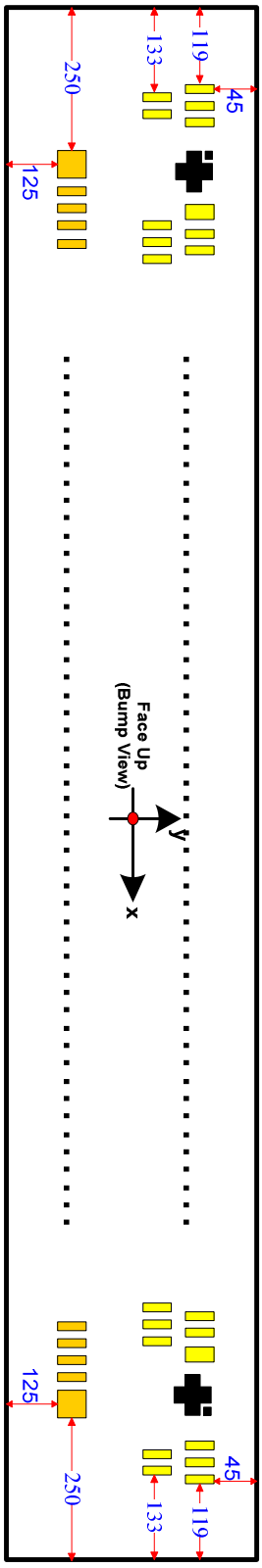
Test Pins			
Pin Name	I/O	Type	Descriptions
PADA [1:4] PADB [1:4]	I/O		- Dummy Pin.
CONTACT1A CONTACT1B	I/O		- Test pin, for test bonding quality, IC internal connect CONTACT1A with CONTACT1B.
KBBC	O		- Test pin, It's not accessible to user. MUST be left open.
VSEL	-		- Dummy pin.
TEST [0:3]	I		- Test pin, It's not accessible to user. MUST be left open. (Internal weakly pull low)
OSC_TEST	I		- Test pin, It's not accessible to user, MUST be left open (Internal weakly pull low)
VSSIDUM [0:106]	O		- These are dummy pins with VSSI potential (There are no function inside). - Signal traces cannot pass through on glass under these pads.
DUMMY	-		- Dummy pads. ***Leave the pin open.***

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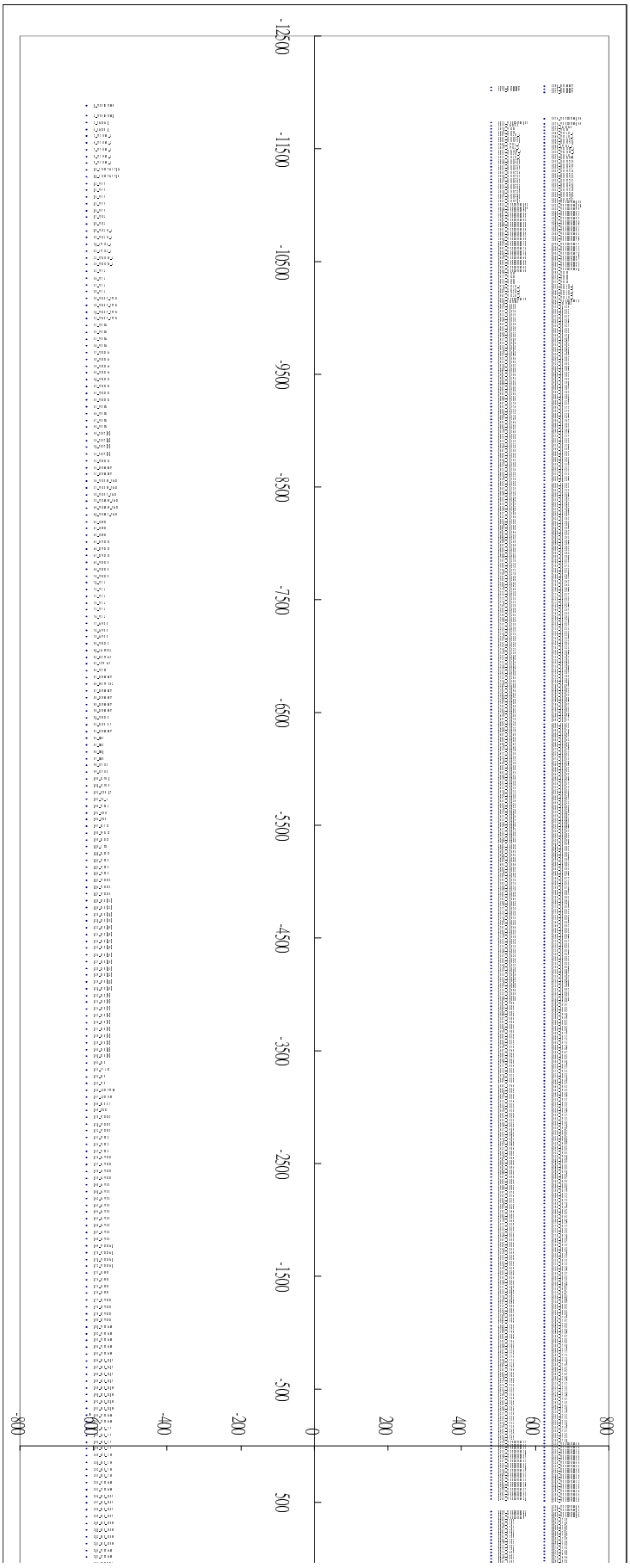
3.4. Pin Assignment

Chip Size：24360um X 1530um (include scribe line)
Chip Thickness：280um(typ.)
Pad Location：Pad Center
Coordinate Origin：Chip Center
Au Bump Size：
1. 14um X 115um
2. 40um X 120um

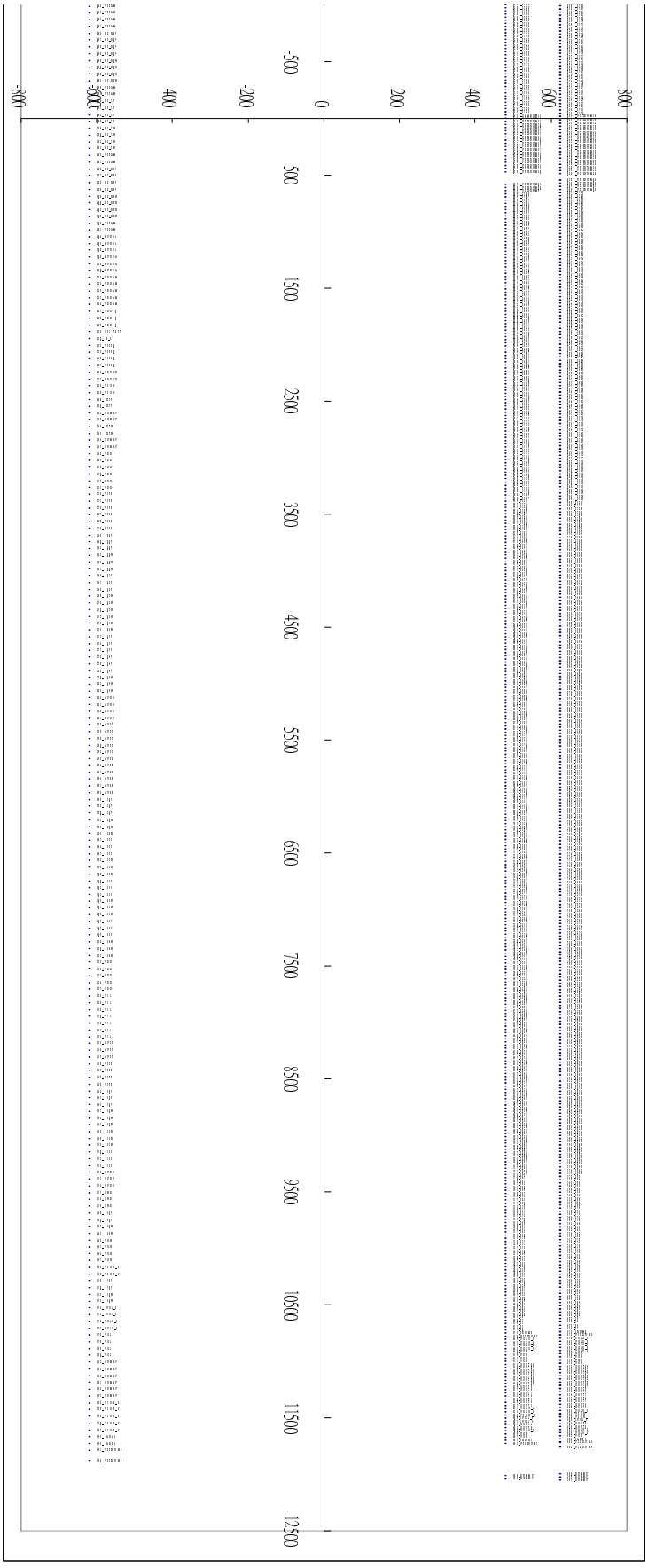




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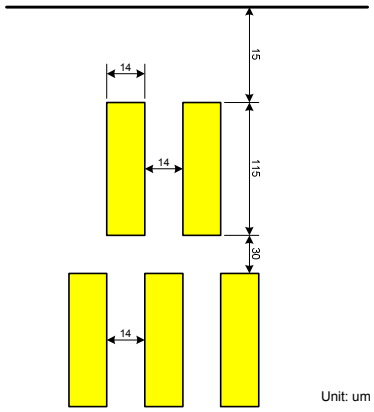
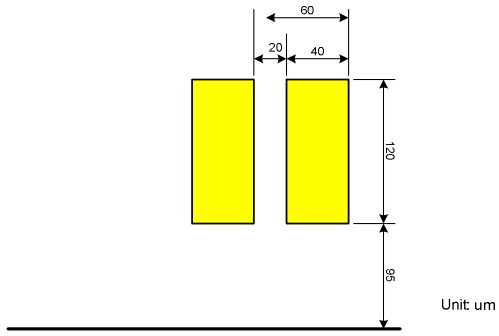
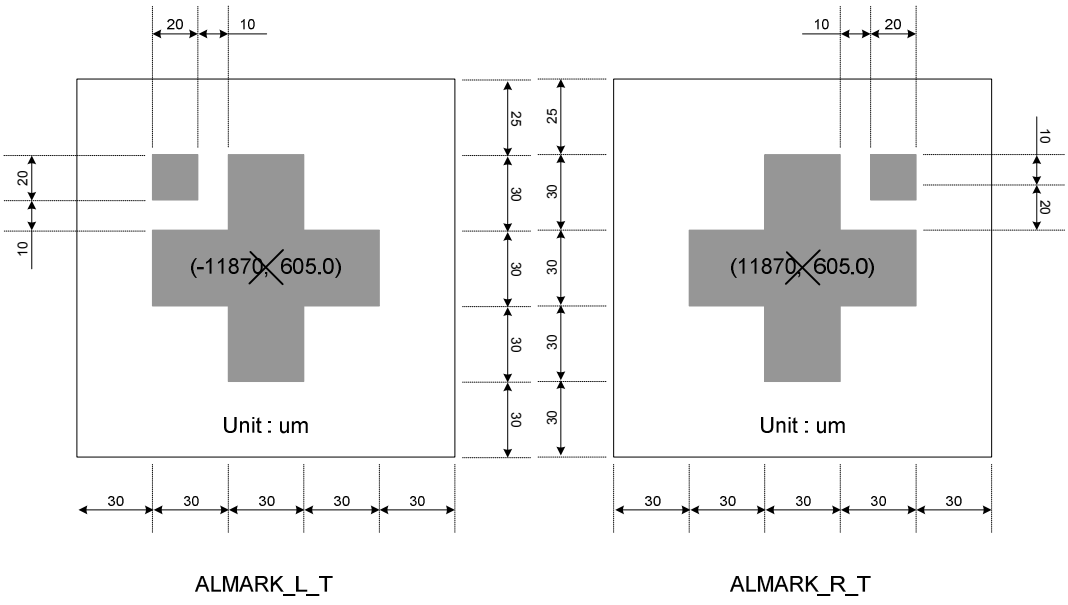


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3.5. Bump Arrangement

Output PAD	
Input PAD	
Alignment mark	

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3.6. Pad Coordination

No.	Text Name	X-axis	Y-axis
1	VSSIDUM0	-11880	-620
2	VSSIDUM1	-11790	-620
3	PADA1	-11730	-620
4	PADB1	-11670	-620
5	VCOM_L	-11610	-620
6	VCOM_L	-11550	-620
7	VCOM_L	-11490	-620
8	VCOM_L	-11430	-620
9	VCOM_L	-11370	-620
10	CONTACT1A	-11310	-620
11	CONTACT1B	-11250	-620
12	VPP	-11190	-620
13	VPP	-11130	-620
14	VPP	-11070	-620
15	VPP	-11010	-620
16	VPP	-10950	-620
17	VGL	-10890	-620
18	VGL	-10830	-620
19	VGLO_L	-10770	-620
20	VGLO_L	-10710	-620
21	LVGL_L	-10650	-620
22	LVGL_L	-10590	-620
23	VRGH_L	-10530	-620
24	VRGH_L	-10470	-620
25	VCL	-10410	-620
26	VCL	-10350	-620
27	VCL	-10290	-620
28	VCL	-10230	-620
29	VREF_PWR	-10170	-620
30	VREF_PWR	-10110	-620
31	VREF_PWR	-10050	-620
32	VREF_PWR	-9990	-620
33	VSSA	-9930	-620
34	VSSA	-9870	-620
35	VSSA	-9810	-620
36	VSSA	-9750	-620
37	VDDA	-9690	-620
38	VDDA	-9630	-620
39	VDDA	-9570	-620
40	VDDA	-9510	-620
41	VDDR	-9450	-620
42	VDDR	-9390	-620
43	VDDR	-9330	-620
44	VDDR	-9270	-620
45	VSSR	-9210	-620
46	VSSR	-9150	-620
47	VSSR	-9090	-620
48	VSSR	-9030	-620
49	TEST[0]	-8970	-620
50	TEST[1]	-8910	-620
51	TEST[2]	-8850	-620
52	TEST[3]	-8790	-620
53	VDDR	-8730	-620
54	DUMMY	-8670	-620
55	DUMMY	-8610	-620
56	VGSN_PAD	-8550	-620
57	VGSN_PAD	-8490	-620
58	VGSP_PAD	-8430	-620
59	VGMN_PAD	-8370	-620
60	VGMN_PAD	-8310	-620

No.	Text Name	X-axis	Y-axis
61	VGMP_PAD	-8250	-620
62	GND	-8190	-620
63	GND	-8130	-620
64	GND	-8070	-620
65	DVDD	-8010	-620
66	DVDD	-7950	-620
67	DVDD	-7890	-620
68	VDDB	-7830	-620
69	VDDB	-7770	-620
70	VDDB	-7710	-620
71	VCL	-7650	-620
72	VCL	-7590	-620
73	VCL	-7530	-620
74	VCL	-7470	-620
75	VCL	-7410	-620
76	VCL	-7350	-620
77	AVSS	-7290	-620
78	AVSS	-7230	-620
79	AVSS	-7170	-620
80	VDDI	-7110	-620
81	LANSEL	-7050	-620
82	DSWAP	-6990	-620
83	PSWAP	-6930	-620
84	VSSI	-6870	-620
85	DUMMY	-6810	-620
86	NBWSEL	-6750	-620
87	DUMMY	-6690	-620
88	DUMMY	-6630	-620
89	DUMMY	-6570	-620
90	DUMMY	-6510	-620
91	VDDI	-6450	-620
92	RGBBP	-6390	-620
93	DUMMY	-6330	-620
94	IM3	-6270	-620
95	IM2	-6210	-620
96	IM1	-6150	-620
97	IM0	-6090	-620
98	GPO3	-6030	-620
99	GPO2	-5970	-620
100	GPO1	-5910	-620
101	GPO0	-5850	-620
102	EXB1T	-5790	-620
103	TE_L	-5730	-620
104	VSEL	-5670	-620
105	SDO	-5610	-620
106	SDI	-5550	-620
107	DCX	-5490	-620
108	WRX	-5430	-620
109	RDX	-5370	-620
110	CSX	-5310	-620
111	RESX	-5250	-620
112	VSSI	-5190	-620
113	VSSI	-5130	-620
114	VSSI	-5070	-620
115	VDDI	-5010	-620
116	VDDI	-4950	-620
117	VDDI	-4890	-620
118	DB[23]	-4830	-620
119	DB[22]	-4770	-620
120	DB[21]	-4710	-620

No.	Text Name	X-axis	Y-axis
121	DB[20]	-4650	-620
122	DB[19]	-4590	-620
123	DB[18]	-4530	-620
124	DB[17]	-4470	-620
125	DB[16]	-4410	-620
126	DB[15]	-4350	-620
127	DB[14]	-4290	-620
128	DB[13]	-4230	-620
129	DB[12]	-4170	-620
130	DB[11]	-4110	-620
131	DB[10]	-4050	-620
132	DB[9]	-3990	-620
133	DB[8]	-3930	-620
134	DB[7]	-3870	-620
135	DB[6]	-3810	-620
136	DB[5]	-3750	-620
137	DB[4]	-3690	-620
138	DB[3]	-3630	-620
139	DB[2]	-3570	-620
140	DB[1]	-3510	-620
141	DB[0]	-3450	-620
142	DE	-3390	-620
143	PCLK	-3330	-620
144	HS	-3270	-620
145	VS	-3210	-620
146	LEDPWM	-3150	-620
147	LEDON	-3090	-620
148	KBBC	-3030	-620
149	ERR	-2970	-620
150	VDDI	-2910	-620
151	VDDI	-2850	-620
152	VDDI	-2790	-620
153	VSSI	-2730	-620
154	VSSI	-2670	-620
155	VSSI	-2610	-620
156	AVDD	-2550	-620
157	AVDD	-2490	-620
158	AVDD	-2430	-620
159	AVDD	-2370	-620
160	AVSS	-2310	-620
161	AVSS	-2250	-620
162	AVSS	-2190	-620
163	AVSS	-2130	-620
164	AVEE	-2070	-620
165	AVEE	-2010	-620
166	AVEE	-1950	-620
167	AVEE	-1890	-620
168	AVEE	-1830	-620
169	VDDA1	-1770	-620
170	VDDA1	-1710	-620
171	VDDA1	-1650	-620
172	VDDA1	-1590	-620
173	GND	-1530	-620
174	GND	-1470	-620
175	GND	-1410	-620
176	GND	-1350	-620
177	DVDD	-1290	-620
178	DVDD	-1230	-620
179	DVDD	-1170	-620
180	DVDD	-1110	-620

No.	Text Name	X-axis	Y-axis
181	VSSAM	-1050	-620
182	VSSAM	-990	-620
183	VSSAM	-930	-620
184	VSSAM	-870	-620
185	VSSAM	-810	-620
186	HS_D1P	-750	-620
187	HS_D1P	-690	-620
188	HS_D1P	-630	-620
189	HS_D1P	-570	-620
190	HS_D1N	-510	-620
191	HS_D1N	-450	-620
192	HS_D1N	-390	-620
193	HS_D1N	-330	-620
194	VSSAM	-270	-620
195	VSSAM	-210	-620
196	HS_CP	-150	-620
197	HS_CP	-90	-620
198	HS_CP	-30	-620
199	HS_CP	30	-620
200	HS_CN	90	-620
201	HS_CN	150	-620
202	HS_CN	210	-620
203	HS_CN	270	-620
204	VSSAM	330	-620
205	VSSAM	390	-620
206	HS_D0P	450	-620
207	HS_D0P	510	-620
208	HS_D0P	570	-620
209	HS_D0P	630	-620
210	HS_D0N	690	-620
211	HS_D0N	750	-620
212	HS_D0N	810	-620
213	HS_D0N	870	-620
214	VSSAM	930	-620
215	VSSAM	990	-620
216	MVDDL	1050	-620
217	MVDDL	1110	-620
218	MVDDL	1170	-620
219	MVDDA	1230	-620
220	MVDDA	1290	-620
221	MVDDA	1350	-620
222	VDDAM	1410	-620
223	VDDAM	1470	-620
224	VDDAM	1530	-620
225	VDDAM	1590	-620
226	VDDAM	1650	-620
227	VDDR1	1710	-620
228	VDDR1	1770	-620
229	VDDR1	1830	-620
230	OSC_TEST	1890	-620
231	TE_R	1950	-620
232	VSSR1	2010	-620
233	VSSR1	2070	-620
234	VSSR1	2130	-620
235	VSSR1	2190	-620
236	NDVDD	2250	-620
237	NDVDD	2310	-620
238	VRGH	2370	-620
239	VRGH	2430	-620
240	EXTP	2490	-620

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No.	Text Name	X-axis	Y-axis
241	EXTP	2550	-620
242	DUMMY	2610	-620
243	DUMMY	2670	-620
244	EXTN	2730	-620
245	EXTN	2790	-620
246	DUMMY	2850	-620
247	DUMMY	2910	-620
248	VDDB	2970	-620
249	VDDB	3030	-620
250	VDDB	3090	-620
251	VDDB	3150	-620
252	VDDB	3210	-620
253	VDDB	3270	-620
254	VSSB	3330	-620
255	VSSB	3390	-620
256	VSSB	3450	-620
257	VSSB	3510	-620
258	VSSB	3570	-620
259	VSSB	3630	-620
260	C11P	3690	-620
261	C11P	3750	-620
262	C11P	3810	-620
263	C11N	3870	-620
264	C11N	3930	-620
265	C11N	3990	-620
266	C12P	4050	-620
267	C12P	4110	-620
268	C12P	4170	-620
269	C12N	4230	-620
270	C12N	4290	-620
271	C12N	4350	-620
272	C13N	4410	-620
273	C13N	4470	-620
274	C13N	4530	-620
275	C13P	4590	-620
276	C13P	4650	-620
277	C13P	4710	-620
278	C14P	4770	-620
279	C14P	4830	-620
280	C14P	4890	-620
281	C14N	4950	-620
282	C14N	5010	-620
283	C14N	5070	-620
284	AVDD	5130	-620
285	AVDD	5190	-620
286	AVDD	5250	-620
287	AVDD	5310	-620
288	AVSS	5370	-620
289	AVSS	5430	-620
290	AVSS	5490	-620
291	AVSS	5550	-620
292	AVSS	5610	-620
293	AVEE	5670	-620
294	AVEE	5730	-620
295	AVEE	5790	-620
296	AVEE	5850	-620
297	AVEE	5910	-620
298	AVEE	5970	-620
299	C21P	6030	-620
300	C21P	6090	-620

No.	Text Name	X-axis	Y-axis
301	C21P	6150	-620
302	C21N	6210	-620
303	C21N	6270	-620
304	C21N	6330	-620
305	C22P	6390	-620
306	C22P	6450	-620
307	C22P	6510	-620
308	C22N	6570	-620
309	C22N	6630	-620
310	C22N	6690	-620
311	C23P	6750	-620
312	C23P	6810	-620
313	C23P	6870	-620
314	C23N	6930	-620
315	C23N	6990	-620
316	C23N	7050	-620
317	C24P	7110	-620
318	C24P	7170	-620
319	C24P	7230	-620
320	C24N	7290	-620
321	C24N	7350	-620
322	C24N	7410	-620
323	VDDB	7470	-620
324	VDDB	7530	-620
325	VDDB	7590	-620
326	VDDB	7650	-620
327	VDDB	7710	-620
328	VCL	7770	-620
329	VCL	7830	-620
330	VCL	7890	-620
331	VCL	7950	-620
332	VCL	8010	-620
333	VCL	8070	-620
334	VCL	8130	-620
335	AVSS	8190	-620
336	AVSS	8250	-620
337	AVSS	8310	-620
338	VSSB	8370	-620
339	VSSB	8430	-620
340	VSSB	8490	-620
341	VSSB	8550	-620
342	C31P	8610	-620
343	C31P	8670	-620
344	C31P	8730	-620
345	C31N	8790	-620
346	C31N	8850	-620
347	C31N	8910	-620
348	C32N	8970	-620
349	C32N	9030	-620
350	C32N	9090	-620
351	C32P	9150	-620
352	C32P	9210	-620
353	C32P	9270	-620
354	DVDD	9330	-620
355	DVDD	9390	-620
356	DVDD	9450	-620
357	GND	9510	-620
358	GND	9570	-620
359	GND	9630	-620
360	C41P	9690	-620

No.	Text Name	X-axis	Y-axis
361	C41P	9750	-620
362	C41N	9810	-620
363	C41N	9870	-620
364	VGH	9930	-620
365	VGH	9990	-620
366	VGH	10050	-620
367	VGH	10110	-620
368	VRGH_R	10170	-620
369	VRGH_R	10230	-620
370	C51P	10290	-620
371	C51P	10350	-620
372	C51N	10410	-620
373	C51N	10470	-620
374	LVGL_R	10530	-620
375	LVGL_R	10590	-620
376	VGLO_R	10650	-620
377	VGLO_R	10710	-620
378	VGL	10770	-620
379	VGL	10830	-620
380	VGL	10890	-620
381	VGL	10950	-620
382	DUMMY	11010	-620
383	DUMMY	11070	-620
384	DUMMY	11130	-620
385	DUMMY	11190	-620
386	DUMMY	11250	-620
387	DUMMY	11310	-620
388	VCOM_R	11370	-620
389	VCOM_R	11430	-620
390	VCOM_R	11490	-620
391	VCOM_R	11550	-620
392	VCOM_R	11610	-620
393	PADA2	11670	-620
394	PADB2	11730	-620
395	VSSIDUM2	11790	-620
396	VSSIDUM3	11880	-620
397	DUMMY	12054	622.5
398	DUMMY	12040	477.5
399	DUMMY	12026	622.5
400	DUMMY	12012	477.5
401	DUMMY	11998	622.5
402	VSSIDUM4	11760	622.5
403	VSSIDUM5	11732	477.5
404	VSSIDUM6	11718	622.5
405	PADA3	11704	477.5
406	PADB3	11690	622.5
407	VGH	11676	477.5
408	VGH	11662	622.5
409	VGH	11648	477.5
410	VGLO_R	11634	622.5
411	VGLO_R	11620	477.5
412	VGLO_R	11606	622.5
413	GOUT1	11592	477.5
414	GOUT1	11578	622.5
415	GOUT2	11564	477.5
416	GOUT2	11550	622.5
417	LVGL_R	11536	477.5
418	LVGL_R	11522	622.5
419	LVGL_R	11508	477.5
420	VRGH_R	11494	622.5

No.	Text Name	X-axis	Y-axis
421	VRGH_R	11480	477.5
422	VRGH_R	11466	622.5
423	VGLO_R	11452	477.5
424	VGLO_R	11438	622.5
425	VGLO_R	11424	477.5
426	GOUT3	11410	622.5
427	GOUT3	11396	477.5
428	GOUT4	11382	622.5
429	GOUT4	11368	477.5
430	GOUT5	11354	622.5
431	GOUT5	11340	477.5
432	GOUT6	11326	622.5
433	GOUT6	11312	477.5
434	GOUT7	11298	622.5
435	GOUT7	11284	477.5
436	GOUT8	11270	622.5
437	GOUT8	11256	477.5
438	GOUT9	11242	622.5
439	GOUT9	11228	477.5
440	GOUT10	11214	622.5
441	GOUT10	11200	477.5
442	GOUT11	11186	622.5
443	GOUT11	11172	477.5
444	GOUT12	11158	622.5
445	GOUT12	11144	477.5
446	GOUT13	11130	622.5
447	GOUT13	11116	477.5
448	GOUT14	11102	622.5
449	GOUT14	11088	477.5
450	GOUT15	11074	622.5
451	GOUT15	11060	477.5
452	GOUT16	11046	622.5
453	GOUT16	11032	477.5
454	VGH	11018	622.5
455	VGH	11004	477.5
456	VGH	10990	622.5
457	VGH	10976	477.5
458	VGH	10962	622.5
459	VGH	10948	477.5
460	VGH	10934	622.5
461	VGH	10920	477.5
462	VGLO_R	10906	622.5
463	VGLO_R	10892	477.5
464	VGLO_R	10878	622.5
465	VGLO_R	10864	477.5
466	VGLO_R	10850	622.5
467	VGLO_R	10836	477.5
468	VGLO_R	10822	622.5
469	VGLO_R	10808	477.5
470	VGLO_R	10794	622.5
471	VSSIDUM7	10780	477.5
472	VSSIDUM8	10766	622.5
473	SDUM0	10752	477.5
474	SDUM1	10738	622.5
475	S1	10724	477.5
476	S2	10710	622.5
477	S3	10696	477.5
478	S4	10682	622.5
479	S5	10668	477.5
480	S6	10654	622.5

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No.	Text Name	X-axis	Y-axis
481	S7	10640	477.5
482	S8	10626	622.5
483	S9	10612	477.5
484	S10	10598	622.5
485	S11	10584	477.5
486	S12	10570	622.5
487	S13	10556	477.5
488	S14	10542	622.5
489	S15	10528	477.5
490	S16	10514	622.5
491	S17	10500	477.5
492	S18	10486	622.5
493	S19	10472	477.5
494	S20	10458	622.5
495	S21	10444	477.5
496	S22	10430	622.5
497	S23	10416	477.5
498	S24	10402	622.5
499	S25	10388	477.5
500	S26	10374	622.5
501	S27	10360	477.5
502	S28	10346	622.5
503	S29	10332	477.5
504	S30	10318	622.5
505	S31	10304	477.5
506	S32	10290	622.5
507	S33	10276	477.5
508	S34	10262	622.5
509	S35	10248	477.5
510	S36	10234	622.5
511	S37	10220	477.5
512	S38	10206	622.5
513	S39	10192	477.5
514	S40	10178	622.5
515	S41	10164	477.5
516	S42	10150	622.5
517	S43	10136	477.5
518	S44	10122	622.5
519	S45	10108	477.5
520	S46	10094	622.5
521	S47	10080	477.5
522	S48	10066	622.5
523	S49	10052	477.5
524	S50	10038	622.5
525	S51	10024	477.5
526	S52	10010	622.5
527	S53	9996	477.5
528	S54	9982	622.5
529	S55	9968	477.5
530	S56	9954	622.5
531	S57	9940	477.5
532	S58	9926	622.5
533	S59	9912	477.5
534	S60	9898	622.5
535	S61	9884	477.5
536	S62	9870	622.5
537	S63	9856	477.5
538	S64	9842	622.5
539	S65	9828	477.5
540	S66	9814	622.5
541	S67	9800	477.5
542	S68	9786	622.5
543	S69	9772	477.5
544	S70	9758	622.5
545	S71	9744	477.5
546	S72	9730	622.5
547	S73	9716	477.5
548	S74	9702	622.5
549	S75	9688	477.5
550	S76	9674	622.5
551	S77	9660	477.5
552	S78	9646	622.5
553	S79	9632	477.5
554	S80	9618	622.5
555	S81	9604	477.5
556	S82	9590	622.5
557	S83	9576	477.5
558	S84	9562	622.5
559	S85	9548	477.5
560	S86	9534	622.5
561	S87	9520	477.5
562	S88	9506	622.5
563	S89	9492	477.5
564	S90	9478	622.5
565	S91	9464	477.5
566	S92	9450	622.5
567	S93	9436	477.5
568	S94	9422	622.5
569	S95	9408	477.5
570	S96	9394	622.5
571	S97	9380	477.5
572	S98	9366	622.5
573	S99	9352	477.5
574	S100	9338	622.5
575	S101	9324	477.5
576	S102	9310	622.5
577	S103	9296	477.5
578	S104	9282	622.5
579	S105	9268	477.5
580	S106	9254	622.5
581	S107	9240	477.5
582	S108	9226	622.5
583	S109	9212	477.5
584	S110	9198	622.5
585	S111	9184	477.5
586	S112	9170	622.5
587	S113	9156	477.5
588	S114	9142	622.5
589	S115	9128	477.5
590	S116	9114	622.5
591	S117	9100	477.5
592	S118	9086	622.5
593	S119	9072	477.5
594	S120	9058	622.5
595	S121	9044	477.5
596	S122	9030	622.5
597	S123	9016	477.5
598	S124	9002	622.5
599	S125	8988	477.5
600	S126	8974	622.5
601	S127	8960	477.5
602	S128	8946	622.5
603	S129	8932	477.5
604	S130	8918	622.5
605	S131	8904	477.5
606	S132	8890	622.5
607	S133	8876	477.5
608	S134	8862	622.5
609	S135	8848	477.5
610	S136	8834	622.5
611	S137	8820	477.5
612	S138	8806	622.5
613	S139	8792	477.5
614	S140	8778	622.5
615	S141	8764	477.5
616	S142	8750	622.5
617	S143	8736	477.5
618	S144	8722	622.5
619	S145	8708	477.5
620	S146	8694	622.5
621	S147	8680	477.5
622	S148	8666	622.5
623	S149	8652	477.5
624	S150	8638	622.5
625	S151	8624	477.5
626	S152	8610	622.5
627	S153	8596	477.5
628	S154	8582	622.5
629	S155	8568	477.5
630	S156	8554	622.5
631	S157	8540	477.5
632	S158	8526	622.5
633	S159	8512	477.5
634	S160	8498	622.5
635	S161	8484	477.5
636	S162	8470	622.5
637	S163	8456	477.5
638	S164	8442	622.5
639	S165	8428	477.5
640	S166	8414	622.5
641	S167	8400	477.5
642	S168	8386	622.5
643	S169	8372	477.5
644	S170	8358	622.5
645	S171	8344	477.5
646	S172	8330	622.5
647	S173	8316	477.5
648	S174	8302	622.5
649	S175	8288	477.5
650	S176	8274	622.5
651	S177	8260	477.5
652	S178	8246	622.5
653	S179	8232	477.5
654	S180	8218	622.5
655	S181	8204	477.5
656	S182	8190	622.5
657	S183	8176	477.5
658	S184	8162	622.5
659	S185	8148	477.5
660	S186	8134	622.5
661	S187	8120	477.5
662	S188	8106	622.5
663	S189	8092	477.5
664	S190	8078	622.5
665	S191	8064	477.5
666	S192	8050	622.5
667	S193	8036	477.5
668	S194	8022	622.5
669	S195	8008	477.5
670	S196	7994	622.5
671	S197	7980	477.5
672	S198	7966	622.5
673	S199	7952	477.5
674	S200	7938	622.5
675	S201	7924	477.5
676	S202	7910	622.5
677	S203	7896	477.5
678	S204	7882	622.5
679	S205	7868	477.5
680	S206	7854	622.5
681	S207	7840	477.5
682	S208	7826	622.5
683	S209	7812	477.5
684	S210	7798	622.5
685	S211	7784	477.5
686	S212	7770	622.5
687	S213	7756	477.5
688	S214	7742	622.5
689	S215	7728	477.5
690	S216	7714	622.5
691	S217	7700	477.5
692	S218	7686	622.5
693	S219	7672	477.5
694	S220	7658	622.5
695	S221	7644	477.5
696	S222	7630	622.5
697	S223	7616	477.5
698	S224	7602	622.5
699	S225	7588	477.5
700	S226	7574	622.5
701	S227	7560	477.5
702	S228	7546	622.5
703	S229	7532	477.5
704	S230	7518	622.5
705	S231	7504	477.5
706	S232	7490	622.5
707	S233	7476	477.5
708	S234	7462	622.5
709	S235	7448	477.5
710	S236	7434	622.5
711	S237	7420	477.5
712	S238	7406	622.5
713	S239	7392	477.5
714	S240	7378	622.5
715	S241	7364	477.5
716	S242	7350	622.5
717	S243	7336	477.5
718	S244	7322	622.5
719	S245	7308	477.5
720	S246	7294	622.5

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No.	Text Name	X-axis	Y-axis
721	S247	7280	477.5
722	S248	7266	622.5
723	S249	7252	477.5
724	S250	7238	622.5
725	S251	7224	477.5
726	S252	7210	622.5
727	S253	7196	477.5
728	S254	7182	622.5
729	S255	7168	477.5
730	S256	7154	622.5
731	S257	7140	477.5
732	S258	7126	622.5
733	S259	7112	477.5
734	S260	7098	622.5
735	S261	7084	477.5
736	S262	7070	622.5
737	S263	7056	477.5
738	S264	7042	622.5
739	S265	7028	477.5
740	S266	7014	622.5
741	S267	7000	477.5
742	S268	6986	622.5
743	S269	6972	477.5
744	S270	6958	622.5
745	S271	6944	477.5
746	S272	6930	622.5
747	S273	6916	477.5
748	S274	6902	622.5
749	S275	6888	477.5
750	S276	6874	622.5
751	S277	6860	477.5
752	S278	6846	622.5
753	S279	6832	477.5
754	S280	6818	622.5
755	S281	6804	477.5
756	S282	6790	622.5
757	S283	6776	477.5
758	S284	6762	622.5
759	S285	6748	477.5
760	S286	6734	622.5
761	S287	6720	477.5
762	S288	6706	622.5
763	S289	6692	477.5
764	S290	6678	622.5
765	S291	6664	477.5
766	S292	6650	622.5
767	S293	6636	477.5
768	S294	6622	622.5
769	S295	6608	477.5
770	S296	6594	622.5
771	S297	6580	477.5
772	S298	6566	622.5
773	S299	6552	477.5
774	S300	6538	622.5
775	S301	6524	477.5
776	S302	6510	622.5
777	S303	6496	477.5
778	S304	6482	622.5
779	S305	6468	477.5
780	S306	6454	622.5
781	S307	6440	477.5
782	S308	6426	622.5
783	S309	6412	477.5
784	S310	6398	622.5
785	S311	6384	477.5
786	S312	6370	622.5
787	S313	6356	477.5
788	S314	6342	622.5
789	S315	6328	477.5
790	S316	6314	622.5
791	S317	6300	477.5
792	S318	6286	622.5
793	S319	6272	477.5
794	S320	6258	622.5
795	S321	6244	477.5
796	S322	6230	622.5
797	S323	6216	477.5
798	S324	6202	622.5
799	S325	6188	477.5
800	S326	6174	622.5
801	S327	6160	477.5
802	S328	6146	622.5
803	S329	6132	477.5
804	S330	6118	622.5
805	S331	6104	477.5
806	S332	6090	622.5
807	S333	6076	477.5
808	S334	6062	622.5
809	S335	6048	477.5
810	S336	6034	622.5
811	S337	6020	477.5
812	S338	6006	622.5
813	S339	5992	477.5
814	S340	5978	622.5
815	S341	5964	477.5
816	S342	5950	622.5
817	S343	5936	477.5
818	S344	5922	622.5
819	S345	5908	477.5
820	S346	5894	622.5
821	S347	5880	477.5
822	S348	5866	622.5
823	S349	5852	477.5
824	S350	5838	622.5
825	S351	5824	477.5
826	S352	5810	622.5
827	S353	5796	477.5
828	S354	5782	622.5
829	S355	5768	477.5
830	S356	5754	622.5
831	S357	5740	477.5
832	S358	5726	622.5
833	S359	5712	477.5
834	S360	5698	622.5
835	S361	5684	477.5
836	S362	5670	622.5
837	S363	5656	477.5
838	S364	5642	622.5
839	S365	5628	477.5
840	S366	5614	622.5
841	S367	5600	477.5
842	S368	5586	622.5
843	S369	5572	477.5
844	S370	5558	622.5
845	S371	5544	477.5
846	S372	5530	622.5
847	S373	5516	477.5
848	S374	5502	622.5
849	S375	5488	477.5
850	S376	5474	622.5
851	S377	5460	477.5
852	S378	5446	622.5
853	S379	5432	477.5
854	S380	5418	622.5
855	S381	5404	477.5
856	S382	5390	622.5
857	S383	5376	477.5
858	S384	5362	622.5
859	S385	5348	477.5
860	S386	5334	622.5
861	S387	5320	477.5
862	S388	5306	622.5
863	S389	5292	477.5
864	S390	5278	622.5
865	S391	5264	477.5
866	S392	5250	622.5
867	S393	5236	477.5
868	S394	5222	622.5
869	S395	5208	477.5
870	S396	5194	622.5
871	S397	5180	477.5
872	S398	5166	622.5
873	S399	5152	477.5
874	S400	5138	622.5
875	S401	5124	477.5
876	S402	5110	622.5
877	S403	5096	477.5
878	S404	5082	622.5
879	S405	5068	477.5
880	S406	5054	622.5
881	S407	5040	477.5
882	S408	5026	622.5
883	S409	5012	477.5
884	S410	4998	622.5
885	S411	4984	477.5
886	S412	4970	622.5
887	S413	4956	477.5
888	S414	4942	622.5
889	S415	4928	477.5
890	S416	4914	622.5
891	S417	4900	477.5
892	S418	4886	622.5
893	S419	4872	477.5
894	S420	4858	622.5
895	S421	4844	477.5
896	S422	4830	622.5
897	S423	4816	477.5
898	S424	4802	622.5
899	S425	4788	477.5
900	S426	4774	622.5
901	S427	4760	477.5
902	S428	4746	622.5
903	S429	4732	477.5
904	S430	4718	622.5
905	S431	4704	477.5
906	S432	4690	622.5
907	S433	4676	477.5
908	S434	4662	622.5
909	S435	4648	477.5
910	S436	4634	622.5
911	S437	4620	477.5
912	S438	4606	622.5
913	S439	4592	477.5
914	S440	4578	622.5
915	S441	4564	477.5
916	S442	4550	622.5
917	S443	4536	477.5
918	S444	4522	622.5
919	S445	4508	477.5
920	S446	4494	622.5
921	S447	4480	477.5
922	S448	4466	622.5
923	S449	4452	477.5
924	S450	4438	622.5
925	S451	4424	477.5
926	S452	4410	622.5
927	S453	4396	477.5
928	S454	4382	622.5
929	S455	4368	477.5
930	S456	4354	622.5
931	S457	4340	477.5
932	S458	4326	622.5
933	S459	4312	477.5
934	S460	4298	622.5
935	S461	4284	477.5
936	S462	4270	622.5
937	S463	4256	477.5
938	S464	4242	622.5
939	S465	4228	477.5
940	S466	4214	622.5
941	S467	4200	477.5
942	S468	4186	622.5
943	S469	4172	477.5
944	S470	4158	622.5
945	S471	4144	477.5
946	S472	4130	622.5
947	S473	4116	477.5
948	S474	4102	622.5
949	S475	4088	477.5
950	S476	4074	622.5
951	S477	4060	477.5
952	S478	4046	622.5
953	S479	4032	477.5
954	S480	4018	622.5
955	S481	4004	477.5
956	S482	3990	622.5
957	S483	3976	477.5
958	S484	3962	622.5
959	S485	3948	477.5
960	S486	3934	622.5

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No.	Text Name	X-axis	Y-axis
961	S487	3920	477.5
962	S488	3906	622.5
963	S489	3892	477.5
964	S490	3878	622.5
965	S491	3864	477.5
966	S492	3850	622.5
967	S493	3836	477.5
968	S494	3822	622.5
969	S495	3808	477.5
970	S496	3794	622.5
971	S497	3780	477.5
972	S498	3766	622.5
973	S499	3752	477.5
974	S500	3738	622.5
975	S501	3724	477.5
976	S502	3710	622.5
977	S503	3696	477.5
978	S504	3682	622.5
979	S505	3668	477.5
980	S506	3654	622.5
981	S507	3640	477.5
982	S508	3626	622.5
983	S509	3612	477.5
984	S510	3598	622.5
985	S511	3584	477.5
986	S512	3570	622.5
987	S513	3556	477.5
988	S514	3542	622.5
989	S515	3528	477.5
990	S516	3514	622.5
991	S517	3500	477.5
992	S518	3486	622.5
993	S519	3472	477.5
994	S520	3458	622.5
995	S521	3444	477.5
996	S522	3430	622.5
997	S523	3416	477.5
998	S524	3402	622.5
999	S525	3388	477.5
1000	S526	3374	622.5
1001	S527	3360	477.5
1002	S528	3346	622.5
1003	S529	3332	477.5
1004	S530	3318	622.5
1005	S531	3304	477.5
1006	S532	3290	622.5
1007	S533	3276	477.5
1008	S534	3262	622.5
1009	S535	3248	477.5
1010	S536	3234	622.5
1011	S537	3220	477.5
1012	S538	3206	622.5
1013	S539	3192	477.5
1014	S540	3178	622.5
1015	S541	3164	477.5
1016	S542	3150	622.5
1017	S543	3136	477.5
1018	S544	3122	622.5
1019	S545	3108	477.5
1020	S546	3094	622.5

No.	Text Name	X-axis	Y-axis
1021	S547	3080	477.5
1022	S548	3066	622.5
1023	S549	3052	477.5
1024	S550	3038	622.5
1025	S551	3024	477.5
1026	S552	3010	622.5
1027	S553	2996	477.5
1028	S554	2982	622.5
1029	S555	2968	477.5
1030	S556	2954	622.5
1031	S557	2940	477.5
1032	S558	2926	622.5
1033	S559	2912	477.5
1034	S560	2898	622.5
1035	S561	2884	477.5
1036	S562	2870	622.5
1037	S563	2856	477.5
1038	S564	2842	622.5
1039	S565	2828	477.5
1040	S566	2814	622.5
1041	S567	2800	477.5
1042	S568	2786	622.5
1043	S569	2772	477.5
1044	S570	2758	622.5
1045	S571	2744	477.5
1046	S572	2730	622.5
1047	S573	2716	477.5
1048	S574	2702	622.5
1049	S575	2688	477.5
1050	S576	2674	622.5
1051	S577	2660	477.5
1052	S578	2646	622.5
1053	S579	2632	477.5
1054	S580	2618	622.5
1055	S581	2604	477.5
1056	S582	2590	622.5
1057	S583	2576	477.5
1058	S584	2562	622.5
1059	S585	2548	477.5
1060	S586	2534	622.5
1061	S587	2520	477.5
1062	S588	2506	622.5
1063	S589	2492	477.5
1064	S590	2478	622.5
1065	S591	2464	477.5
1066	S592	2450	622.5
1067	S593	2436	477.5
1068	S594	2422	622.5
1069	S595	2408	477.5
1070	S596	2394	622.5
1071	S597	2380	477.5
1072	S598	2366	622.5
1073	S599	2352	477.5
1074	S600	2338	622.5
1075	S601	2324	477.5
1076	S602	2310	622.5
1077	S603	2296	477.5
1078	S604	2282	622.5
1079	S605	2268	477.5
1080	S606	2254	622.5

No.	Text Name	X-axis	Y-axis
1081	S607	2240	477.5
1082	S608	2226	622.5
1083	S609	2212	477.5
1084	S610	2198	622.5
1085	S611	2184	477.5
1086	S612	2170	622.5
1087	S613	2156	477.5
1088	S614	2142	622.5
1089	S615	2128	477.5
1090	S616	2114	622.5
1091	S617	2100	477.5
1092	S618	2086	622.5
1093	S619	2072	477.5
1094	S620	2058	622.5
1095	S621	2044	477.5
1096	S622	2030	622.5
1097	S623	2016	477.5
1098	S624	2002	622.5
1099	S625	1988	477.5
1100	S626	1974	622.5
1101	S627	1960	477.5
1102	S628	1946	622.5
1103	S629	1932	477.5
1104	S630	1918	622.5
1105	S631	1904	477.5
1106	S632	1890	622.5
1107	S633	1876	477.5
1108	S634	1862	622.5
1109	S635	1848	477.5
1110	S636	1834	622.5
1111	S637	1820	477.5
1112	S638	1806	622.5
1113	S639	1792	477.5
1114	S640	1778	622.5
1115	S641	1764	477.5
1116	S642	1750	622.5
1117	S643	1736	477.5
1118	S644	1722	622.5
1119	S645	1708	477.5
1120	S646	1694	622.5
1121	S647	1680	477.5
1122	S648	1666	622.5
1123	S649	1652	477.5
1124	S650	1638	622.5
1125	S651	1624	477.5
1126	S652	1610	622.5
1127	S653	1596	477.5
1128	S654	1582	622.5
1129	S655	1568	477.5
1130	S656	1554	622.5
1131	S657	1540	477.5
1132	S658	1526	622.5
1133	S659	1512	477.5
1134	S660	1498	622.5
1135	S661	1484	477.5
1136	S662	1470	622.5
1137	S663	1456	477.5
1138	S664	1442	622.5
1139	S665	1428	477.5
1140	S666	1414	622.5

No.	Text Name	X-axis	Y-axis
1141	S667	1400	477.5
1142	S668	1386	622.5
1143	S669	1372	477.5
1144	S670	1358	622.5
1145	S671	1344	477.5
1146	S672	1330	622.5
1147	S673	1316	477.5
1148	S674	1302	622.5
1149	S675	1288	477.5
1150	S676	1274	622.5
1151	S677	1260	477.5
1152	S678	1246	622.5
1153	S679	1232	477.5
1154	S680	1218	622.5
1155	S681	1204	477.5
1156	S682	1190	622.5
1157	S683	1176	477.5
1158	S684	1162	622.5
1159	S685	1148	477.5
1160	S686	1134	622.5
1161	S687	1120	477.5
1162	S688	1106	622.5
1163	S689	1092	477.5
1164	S690	1078	622.5
1165	S691	1064	477.5
1166	S692	1050	622.5
1167	S693	1036	477.5
1168	S694	1022	622.5
1169	S695	1008	477.5
1170	S696	994	622.5
1171	S697	980	477.5
1172	S698	966	622.5
1173	S699	952	477.5
1174	S700	938	622.5
1175	S701	924	477.5
1176	S702	910	622.5
1177	S703	896	477.5
1178	S704	882	622.5
1179	S705	868	477.5
1180	S706	854	622.5
1181	S707	840	477.5
1182	S708	826	622.5
1183	S709	812	477.5
1184	S710	798	622.5
1185	S711	784	477.5
1186	S712	770	622.5
1187	S713	756	477.5
1188	S714	742	622.5
1189	S715	728	477.5
1190	S716	714	622.5
1191	S717	700	477.5
1192	S718	686	622.5
1193	S719	672	477.5
1194	S720	658	622.5
1195	VSSIDUM9	644	477.5
1196	VSSIDUM10	630	622.5
1197	VSSIDUM11	616	477.5
1198	VSSIDUM12	602	622.5
1199	VSSIDUM13	588	477.5
1200	VSSIDUM14	574	622.5

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No.	Text Name	X-axis	Y-axis
1201	VSSIDUM16	546	622.5
1202	VSSIDUM20	490	622.5
1203	VSSIDUM21	476	477.5
1204	VSSIDUM22	462	622.5
1205	VSSIDUM23	448	477.5
1206	VSSIDUM24	434	622.5
1207	VSSIDUM25	420	477.5
1208	VSSIDUM26	406	622.5
1209	VSSIDUM27	392	477.5
1210	VSSIDUM28	378	622.5
1211	VSSIDUM29	364	477.5
1212	VSSIDUM30	350	622.5
1213	VSSIDUM31	336	477.5
1214	VSSIDUM32	322	622.5
1215	VSSIDUM33	308	477.5
1216	VSSIDUM34	294	622.5
1217	VSSIDUM35	280	477.5
1218	VSSIDUM36	266	622.5
1219	VSSIDUM37	252	477.5
1220	VSSIDUM38	238	622.5
1221	VSSIDUM39	224	477.5
1222	VSSIDUM40	210	622.5
1223	VSSIDUM41	196	477.5
1224	VSSIDUM42	182	622.5
1225	VSSIDUM43	168	477.5
1226	VSSIDUM44	154	622.5
1227	VSSIDUM45	140	477.5
1228	VSSIDUM46	126	622.5
1229	VSSIDUM47	112	477.5
1230	VSSIDUM48	98	622.5
1231	VSSIDUM49	84	477.5
1232	VSSIDUM50	70	622.5
1233	VSSIDUM51	56	477.5
1234	VSSIDUM52	42	622.5
1235	VSSIDUM53	28	477.5
1236	VSSIDUM54	14	622.5
1237	VSSIDUM55	0	477.5
1238	VSSIDUM56	-14	622.5
1239	VSSIDUM57	-28	477.5
1240	S721	-42	622.5
1241	S722	-56	477.5
1242	S723	-70	622.5
1243	S724	-84	477.5
1244	S725	-98	622.5
1245	S726	-112	477.5
1246	S727	-126	622.5
1247	S728	-140	477.5
1248	S729	-154	622.5
1249	S730	-168	477.5
1250	S731	-182	622.5
1251	S732	-196	477.5
1252	S733	-210	622.5
1253	S734	-224	477.5
1254	S735	-238	622.5
1255	S736	-252	477.5
1256	S737	-266	622.5
1257	S738	-280	477.5
1258	S739	-294	622.5
1259	S740	-308	477.5
1260	S741	-322	622.5
1261	S742	-336	477.5
1262	S743	-350	622.5
1263	S744	-364	477.5
1264	S745	-378	622.5
1265	S746	-392	477.5
1266	S747	-406	622.5
1267	S748	-420	477.5
1268	S749	-434	622.5
1269	S750	-448	477.5
1270	S751	-462	622.5
1271	S752	-476	477.5
1272	S753	-490	622.5
1273	S754	-504	477.5
1274	S755	-518	622.5
1275	S756	-532	477.5
1276	S757	-546	622.5
1277	S758	-560	477.5
1278	S759	-574	622.5
1279	S760	-588	477.5
1280	S761	-602	622.5
1281	S762	-616	477.5
1282	S763	-630	622.5
1283	S764	-644	477.5
1284	S765	-658	622.5
1285	S766	-672	477.5
1286	S767	-686	622.5
1287	S768	-700	477.5
1288	S769	-714	622.5
1289	S770	-728	477.5
1290	S771	-742	622.5
1291	S772	-756	477.5
1292	S773	-770	622.5
1293	S774	-784	477.5
1294	S775	-798	622.5
1295	S776	-812	477.5
1296	S777	-826	622.5
1297	S778	-840	477.5
1298	S779	-854	622.5
1299	S780	-868	477.5
1300	S781	-882	622.5
1301	S782	-896	477.5
1302	S783	-910	622.5
1303	S784	-924	477.5
1304	S785	-938	622.5
1305	S786	-952	477.5
1306	S787	-966	622.5
1307	S788	-980	477.5
1308	S789	-994	622.5
1309	S790	-1008	477.5
1310	S791	-1022	622.5
1311	S792	-1036	477.5
1312	S793	-1050	622.5
1313	S794	-1064	477.5
1314	S795	-1078	622.5
1315	S796	-1092	477.5
1316	S797	-1106	622.5
1317	S798	-1120	477.5
1318	S799	-1134	622.5
1319	S800	-1148	477.5
1320	S801	-1162	622.5
1321	S802	-1176	477.5
1322	S803	-1190	622.5
1323	S804	-1204	477.5
1324	S805	-1218	622.5
1325	S806	-1232	477.5
1326	S807	-1246	622.5
1327	S808	-1260	477.5
1328	S809	-1274	622.5
1329	S810	-1288	477.5
1330	S811	-1302	622.5
1331	S812	-1316	477.5
1332	S813	-1330	622.5
1333	S814	-1344	477.5
1334	S815	-1358	622.5
1335	S816	-1372	477.5
1336	S817	-1386	622.5
1337	S818	-1400	477.5
1338	S819	-1414	622.5
1339	S820	-1428	477.5
1340	S821	-1442	622.5
1341	S822	-1456	477.5
1342	S823	-1470	622.5
1343	S824	-1484	477.5
1344	S825	-1498	622.5
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1346	S827	-1526	622.5
1347	S828	-1540	477.5
1348	S829	-1554	622.5
1349	S830	-1568	477.5
1350	S831	-1582	622.5
1351	S832	-1596	477.5
1352	S833	-1610	622.5
1353	S834	-1624	477.5
1354	S835	-1638	622.5
1355	S836	-1652	477.5
1356	S837	-1666	622.5
1357	S838	-1680	477.5
1358	S839	-1694	622.5
1359	S840	-1708	477.5
1360	S841	-1722	622.5
1361	S842	-1736	477.5
1362	S843	-1750	622.5
1363	S844	-1764	477.5
1364	S845	-1778	622.5
1365	S846	-1792	477.5
1366	S847	-1806	622.5
1367	S848	-1820	477.5
1368	S849	-1834	622.5
1369	S850	-1848	477.5
1370	S851	-1862	622.5
1371	S852	-1876	477.5
1372	S853	-1890	622.5
1373	S854	-1904	477.5
1374	S855	-1918	622.5
1375	S856	-1932	477.5
1376	S857	-1946	622.5
1377	S858	-1960	477.5
1378	S859	-1974	622.5
1379	S860	-1988	477.5
1380	S861	-2002	622.5
1381	S862	-2016	477.5
1382	S863	-2030	622.5
1383	S864	-2044	477.5
1384	S865	-2058	622.5
1385	S866	-2072	477.5
1386	S867	-2086	622.5
1387	S868	-2100	477.5
1388	S869	-2114	622.5
1389	S870	-2128	477.5
1390	S871	-2142	622.5
1391	S872	-2156	477.5
1392	S873	-2170	622.5
1393	S874	-2184	477.5
1394	S875	-2198	622.5
1395	S876	-2212	477.5
1396	S877	-2226	622.5
1397	S878	-2240	477.5
1398	S879	-2254	622.5
1399	S880	-2268	477.5
1400	S881	-2282	622.5
1401	S882	-2296	477.5
1402	S883	-2310	622.5
1403	S884	-2324	477.5
1404	S885	-2338	622.5
1405	S886	-2352	477.5
1406	S887	-2366	622.5
1407	S888	-2380	477.5
1408	S889	-2394	622.5
1409	S890	-2408	477.5
1410	S891	-2422	622.5
1411	S892	-2436	477.5
1412	S893	-2450	622.5
1413	S894	-2464	477.5
1414	S895	-2478	622.5
1415	S896	-2492	477.5
1416	S897	-2506	622.5
1417	S898	-2520	477.5
1418	S899	-2534	622.5
1419	S900	-2548	477.5
1420	S901	-2562	622.5
1421	S902	-2576	477.5
1422	S903	-2590	622.5
1423	S904	-2604	477.5
1424	S905	-2618	622.5
1425	S906	-2632	477.5
1426	S907	-2646	622.5
1427	S908	-2660	477.5
1428	S909	-2674	622.5
1429	S910	-2688	477.5
1430	S911	-2702	622.5
1431	S912	-2716	477.5
1432	S913	-2730	622.5
1433	S914	-2744	477.5
1434	S915	-2758	622.5
1435	S916	-2772	477.5
1436	S917	-2786	622.5
1437	S918	-2800	477.5
1438	S919	-2814	622.5
1439	S920	-2828	477.5
1440	S921	-2842	622.5

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No.	Text Name	X-axis	Y-axis
1441	S922	-2856	477.5
1442	S923	-2870	622.5
1443	S924	-2884	477.5
1444	S925	-2898	622.5
1445	S926	-2912	477.5
1446	S927	-2926	622.5
1447	S928	-2940	477.5
1448	S929	-2954	622.5
1449	S930	-2968	477.5
1450	S931	-2982	622.5
1451	S932	-2996	477.5
1452	S933	-3010	622.5
1453	S934	-3024	477.5
1454	S935	-3038	622.5
1455	S936	-3052	477.5
1456	S937	-3066	622.5
1457	S938	-3080	477.5
1458	S939	-3094	622.5
1459	S940	-3108	477.5
1460	S941	-3122	622.5
1461	S942	-3136	477.5
1462	S943	-3150	622.5
1463	S944	-3164	477.5
1464	S945	-3178	622.5
1465	S946	-3192	477.5
1466	S947	-3206	622.5
1467	S948	-3220	477.5
1468	S949	-3234	622.5
1469	S950	-3248	477.5
1470	S951	-3262	622.5
1471	S952	-3276	477.5
1472	S953	-3290	622.5
1473	S954	-3304	477.5
1474	S955	-3318	622.5
1475	S956	-3332	477.5
1476	S957	-3346	622.5
1477	S958	-3360	477.5
1478	S959	-3374	622.5
1479	S960	-3388	477.5
1480	S961	-3402	622.5
1481	S962	-3416	477.5
1482	S963	-3430	622.5
1483	S964	-3444	477.5
1484	S965	-3458	622.5
1485	S966	-3472	477.5
1486	S967	-3486	622.5
1487	S968	-3500	477.5
1488	S969	-3514	622.5
1489	S970	-3528	477.5
1490	S971	-3542	622.5
1491	S972	-3556	477.5
1492	S973	-3570	622.5
1493	S974	-3584	477.5
1494	S975	-3598	622.5
1495	S976	-3612	477.5
1496	S977	-3626	622.5
1497	S978	-3640	477.5
1498	S979	-3654	622.5
1499	S980	-3668	477.5
1500	S981	-3682	622.5

No.	Text Name	X-axis	Y-axis
1501	S982	-3696	477.5
1502	S983	-3710	622.5
1503	S984	-3724	477.5
1504	S985	-3738	622.5
1505	S986	-3752	477.5
1506	S987	-3766	622.5
1507	S988	-3780	477.5
1508	S989	-3794	622.5
1509	S990	-3808	477.5
1510	S991	-3822	622.5
1511	S992	-3836	477.5
1512	S993	-3850	622.5
1513	S994	-3864	477.5
1514	S995	-3878	622.5
1515	S996	-3892	477.5
1516	S997	-3906	622.5
1517	S998	-3920	477.5
1518	S999	-3934	622.5
1519	S1000	-3948	477.5
1520	S1001	-3962	622.5
1521	S1002	-3976	477.5
1522	S1003	-3990	622.5
1523	S1004	-4004	477.5
1524	S1005	-4018	622.5
1525	S1006	-4032	477.5
1526	S1007	-4046	622.5
1527	S1008	-4060	477.5
1528	S1009	-4074	622.5
1529	S1010	-4088	477.5
1530	S1011	-4102	622.5
1531	S1012	-4116	477.5
1532	S1013	-4130	622.5
1533	S1014	-4144	477.5
1534	S1015	-4158	622.5
1535	S1016	-4172	477.5
1536	S1017	-4186	622.5
1537	S1018	-4200	477.5
1538	S1019	-4214	622.5
1539	S1020	-4228	477.5
1540	S1021	-4242	622.5
1541	S1022	-4256	477.5
1542	S1023	-4270	622.5
1543	S1024	-4284	477.5
1544	S1025	-4298	622.5
1545	S1026	-4312	477.5
1546	S1027	-4326	622.5
1547	S1028	-4340	477.5
1548	S1029	-4354	622.5
1549	S1030	-4368	477.5
1550	S1031	-4382	622.5
1551	S1032	-4396	477.5
1552	S1033	-4410	622.5
1553	S1034	-4424	477.5
1554	S1035	-4438	622.5
1555	S1036	-4452	477.5
1556	S1037	-4466	622.5
1557	S1038	-4480	477.5
1558	S1039	-4494	622.5
1559	S1040	-4508	477.5
1560	S1041	-4522	622.5

No.	Text Name	X-axis	Y-axis
1561	S1042	-4536	477.5
1562	S1043	-4550	622.5
1563	S1044	-4564	477.5
1564	S1045	-4578	622.5
1565	S1046	-4592	477.5
1566	S1047	-4606	622.5
1567	S1048	-4620	477.5
1568	S1049	-4634	622.5
1569	S1050	-4648	477.5
1570	S1051	-4662	622.5
1571	S1052	-4676	477.5
1572	S1053	-4690	622.5
1573	S1054	-4704	477.5
1574	S1055	-4718	622.5
1575	S1056	-4732	477.5
1576	S1057	-4746	622.5
1577	S1058	-4760	477.5
1578	S1059	-4774	622.5
1579	S1060	-4788	477.5
1580	S1061	-4802	622.5
1581	S1062	-4816	477.5
1582	S1063	-4830	622.5
1583	S1064	-4844	477.5
1584	S1065	-4858	622.5
1585	S1066	-4872	477.5
1586	S1067	-4886	622.5
1587	S1068	-4900	477.5
1588	S1069	-4914	622.5
1589	S1070	-4928	477.5
1590	S1071	-4942	622.5
1591	S1072	-4956	477.5
1592	S1073	-4970	622.5
1593	S1074	-4984	477.5
1594	S1075	-4998	622.5
1595	S1076	-5012	477.5
1596	S1077	-5026	622.5
1597	S1078	-5040	477.5
1598	S1079	-5054	622.5
1599	S1080	-5068	477.5
1600	S1081	-5082	622.5
1601	S1082	-5096	477.5
1602	S1083	-5110	622.5
1603	S1084	-5124	477.5
1604	S1085	-5138	622.5
1605	S1086	-5152	477.5
1606	S1087	-5166	622.5
1607	S1088	-5180	477.5
1608	S1089	-5194	622.5
1609	S1090	-5208	477.5
1610	S1091	-5222	622.5
1611	S1092	-5236	477.5
1612	S1093	-5250	622.5
1613	S1094	-5264	477.5
1614	S1095	-5278	622.5
1615	S1096	-5292	477.5
1616	S1097	-5306	622.5
1617	S1098	-5320	477.5
1618	S1099	-5334	622.5
1619	S1100	-5348	477.5
1620	S1101	-5362	622.5

No.	Text Name	X-axis	Y-axis
1621	S1102	-5376	477.5
1622	S1103	-5390	622.5
1623	S1104	-5404	477.5
1624	S1105	-5418	622.5
1625	S1106	-5432	477.5
1626	S1107	-5446	622.5
1627	S1108	-5460	477.5
1628	S1109	-5474	622.5
1629	S1110	-5488	477.5
1630	S1111	-5502	622.5
1631	S1112	-5516	477.5
1632	S1113	-5530	622.5
1633	S1114	-5544	477.5
1634	S1115	-5558	622.5
1635	S1116	-5572	477.5
1636	S1117	-5586	622.5
1637	S1118	-5600	477.5
1638	S1119	-5614	622.5
1639	S1120	-5628	477.5
1640	S1121	-5642	622.5
1641	S1122	-5656	477.5
1642	S1123	-5670	622.5
1643	S1124	-5684	477.5
1644	S1125	-5698	622.5
1645	S1126	-5712	477.5
1646	S1127	-5726	622.5
1647	S1128	-5740	477.5
1648	S1129	-5754	622.5
1649	S1130	-5768	477.5
1650	S1131	-5782	622.5
1651	S1132	-5796	477.5
1652	S1133	-5810	622.5
1653	S1134	-5824	477.5
1654	S1135	-5838	622.5
1655	S1136	-5852	477.5
1656	S1137	-5866	622.5
1657	S1138	-5880	477.5
1658	S1139	-5894	622.5
1659	S1140	-5908	477.5
1660	S1141	-5922	622.5
1661	S1142	-5936	477.5
1662	S1143	-5950	622.5
1663	S1144	-5964	477.5
1664	S1145	-5978	622.5
1665	S1146	-5992	477.5
1666	S1147	-6006	622.5
1667	S1148	-6020	477.5
1668	S1149	-6034	622.5
1669	S1150	-6048	477.5
1670	S1151	-6062	622.5
1671	S1152	-6076	477.5
1672	S1153	-6090	622.5
1673	S1154	-6104	477.5
1674	S1155	-6118	622.5
1675	S1156	-6132	477.5
1676	S1157	-6146	622.5
1677	S1158	-6160	477.5
1678	S1159	-6174	622.5
1679	S1160	-6188	477.5
1680	S1161	-6202	622.5

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No.	Text Name	X-axis	Y-axis
1681	S1162	-6216	477.5
1682	S1163	-6230	622.5
1683	S1164	-6244	477.5
1684	S1165	-6258	622.5
1685	S1166	-6272	477.5
1686	S1167	-6286	622.5
1687	S1168	-6300	477.5
1688	S1169	-6314	622.5
1689	S1170	-6328	477.5
1690	S1171	-6342	622.5
1691	S1172	-6356	477.5
1692	S1173	-6370	622.5
1693	S1174	-6384	477.5
1694	S1175	-6398	622.5
1695	S1176	-6412	477.5
1696	S1177	-6426	622.5
1697	S1178	-6440	477.5
1698	S1179	-6454	622.5
1699	S1180	-6468	477.5
1700	S1181	-6482	622.5
1701	S1182	-6496	477.5
1702	S1183	-6510	622.5
1703	S1184	-6524	477.5
1704	S1185	-6538	622.5
1705	S1186	-6552	477.5
1706	S1187	-6566	622.5
1707	S1188	-6580	477.5
1708	S1189	-6594	622.5
1709	S1190	-6608	477.5
1710	S1191	-6622	622.5
1711	S1192	-6636	477.5
1712	S1193	-6650	622.5
1713	S1194	-6664	477.5
1714	S1195	-6678	622.5
1715	S1196	-6692	477.5
1716	S1197	-6706	622.5
1717	S1198	-6720	477.5
1718	S1199	-6734	622.5
1719	S1200	-6748	477.5
1720	S1201	-6762	622.5
1721	S1202	-6776	477.5
1722	S1203	-6790	622.5
1723	S1204	-6804	477.5
1724	S1205	-6818	622.5
1725	S1206	-6832	477.5
1726	S1207	-6846	622.5
1727	S1208	-6860	477.5
1728	S1209	-6874	622.5
1729	S1210	-6888	477.5
1730	S1211	-6902	622.5
1731	S1212	-6916	477.5
1732	S1213	-6930	622.5
1733	S1214	-6944	477.5
1734	S1215	-6958	622.5
1735	S1216	-6972	477.5
1736	S1217	-6986	622.5
1737	S1218	-7000	477.5
1738	S1219	-7014	622.5
1739	S1220	-7028	477.5
1740	S1221	-7042	622.5
1741	S1222	-7056	477.5
1742	S1223	-7070	622.5
1743	S1224	-7084	477.5
1744	S1225	-7098	622.5
1745	S1226	-7112	477.5
1746	S1227	-7126	622.5
1747	S1228	-7140	477.5
1748	S1229	-7154	622.5
1749	S1230	-7168	477.5
1750	S1231	-7182	622.5
1751	S1232	-7196	477.5
1752	S1233	-7210	622.5
1753	S1234	-7224	477.5
1754	S1235	-7238	622.5
1755	S1236	-7252	477.5
1756	S1237	-7266	622.5
1757	S1238	-7280	477.5
1758	S1239	-7294	622.5
1759	S1240	-7308	477.5
1760	S1241	-7322	622.5
1761	S1242	-7336	477.5
1762	S1243	-7350	622.5
1763	S1244	-7364	477.5
1764	S1245	-7378	622.5
1765	S1246	-7392	477.5
1766	S1247	-7406	622.5
1767	S1248	-7420	477.5
1768	S1249	-7434	622.5
1769	S1250	-7448	477.5
1770	S1251	-7462	622.5
1771	S1252	-7476	477.5
1772	S1253	-7490	622.5
1773	S1254	-7504	477.5
1774	S1255	-7518	622.5
1775	S1256	-7532	477.5
1776	S1257	-7546	622.5
1777	S1258	-7560	477.5
1778	S1259	-7574	622.5
1779	S1260	-7588	477.5
1780	S1261	-7602	622.5
1781	S1262	-7616	477.5
1782	S1263	-7630	622.5
1783	S1264	-7644	477.5
1784	S1265	-7658	622.5
1785	S1266	-7672	477.5
1786	S1267	-7686	622.5
1787	S1268	-7700	477.5
1788	S1269	-7714	622.5
1789	S1270	-7728	477.5
1790	S1271	-7742	622.5
1791	S1272	-7756	477.5
1792	S1273	-7770	622.5
1793	S1274	-7784	477.5
1794	S1275	-7798	622.5
1795	S1276	-7812	477.5
1796	S1277	-7826	622.5
1797	S1278	-7840	477.5
1798	S1279	-7854	622.5
1799	S1280	-7868	477.5
1800	S1281	-7882	622.5
1801	S1282	-7896	477.5
1802	S1283	-7910	622.5
1803	S1284	-7924	477.5
1804	S1285	-7938	622.5
1805	S1286	-7952	477.5
1806	S1287	-7966	622.5
1807	S1288	-7980	477.5
1808	S1289	-7994	622.5
1809	S1290	-8008	477.5
1810	S1291	-8022	622.5
1811	S1292	-8036	477.5
1812	S1293	-8050	622.5
1813	S1294	-8064	477.5
1814	S1295	-8078	622.5
1815	S1296	-8092	477.5
1816	S1297	-8106	622.5
1817	S1298	-8120	477.5
1818	S1299	-8134	622.5
1819	S1300	-8148	477.5
1820	S1301	-8162	622.5
1821	S1302	-8176	477.5
1822	S1303	-8190	622.5
1823	S1304	-8204	477.5
1824	S1305	-8218	622.5
1825	S1306	-8232	477.5
1826	S1307	-8246	622.5
1827	S1308	-8260	477.5
1828	S1309	-8274	622.5
1829	S1310	-8288	477.5
1830	S1311	-8302	622.5
1831	S1312	-8316	477.5
1832	S1313	-8330	622.5
1833	S1314	-8344	477.5
1834	S1315	-8358	622.5
1835	S1316	-8372	477.5
1836	S1317	-8386	622.5
1837	S1318	-8400	477.5
1838	S1319	-8414	622.5
1839	S1320	-8428	477.5
1840	S1321	-8442	622.5
1841	S1322	-8456	477.5
1842	S1323	-8470	622.5
1843	S1324	-8484	477.5
1844	S1325	-8498	622.5
1845	S1326	-8512	477.5
1846	S1327	-8526	622.5
1847	S1328	-8540	477.5
1848	S1329	-8554	622.5
1849	S1330	-8568	477.5
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1853	S1334	-8624	477.5
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1859	S1340	-8708	477.5
1860	S1341	-8722	622.5
1861	S1342	-8736	477.5
1862	S1343	-8750	622.5
1863	S1344	-8764	477.5
1864	S1345	-8778	622.5
1865	S1346	-8792	477.5
1866	S1347	-8806	622.5
1867	S1348	-8820	477.5
1868	S1349	-8834	622.5
1869	S1350	-8848	477.5
1870	S1351	-8862	622.5
1871	S1352	-8876	477.5
1872	S1353	-8890	622.5
1873	S1354	-8904	477.5
1874	S1355	-8918	622.5
1875	S1356	-8932	477.5
1876	S1357	-8946	622.5
1877	S1358	-8960	477.5
1878	S1359	-8974	622.5
1879	S1360	-8988	477.5
1880	S1361	-9002	622.5
1881	S1362	-9016	477.5
1882	S1363	-9030	622.5
1883	S1364	-9044	477.5
1884	S1365	-9058	622.5
1885	S1366	-9072	477.5
1886	S1367	-9086	622.5
1887	S1368	-9100	477.5
1888	S1369	-9114	622.5
1889	S1370	-9128	477.5
1890	S1371	-9142	622.5
1891	S1372	-9156	477.5
1892	S1373	-9170	622.5
1893	S1374	-9184	477.5
1894	S1375	-9198	622.5
1895	S1376	-9212	477.5
1896	S1377	-9226	622.5
1897	S1378	-9240	477.5
1898	S1379	-9254	622.5
1899	S1380	-9268	477.5
1900	S1381	-9282	622.5
1901	S1382	-9296	477.5
1902	S1383	-9310	622.5
1903	S1384	-9324	477.5
1904	S1385	-9338	622.5
1905	S1386	-9352	477.5
1906	S1387	-9366	622.5
1907	S1388	-9380	477.5
1908	S1389	-9394	622.5
1909	S1390	-9408	477.5
1910	S1391	-9422	622.5
1911	S1392	-9436	477.5
1912	S1393	-9450	622.5
1913	S1394	-9464	477.5
1914	S1395	-9478	622.5
1915	S1396	-9492	477.5
1916	S1397	-9506	622.5
1917	S1398	-9520	477.5
1918	S1399	-9534	622.5
1919	S1400	-9548	477.5
1920	S1401	-9562	622.5

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No.	Text Name	X-axis	Y-axis
1921	S1402	-9576	477.5
1922	S1403	-9590	622.5
1923	S1404	-9604	477.5
1924	S1405	-9618	622.5
1925	S1406	-9632	477.5
1926	S1407	-9646	622.5
1927	S1408	-9660	477.5
1928	S1409	-9674	622.5
1929	S1410	-9688	477.5
1930	S1411	-9702	622.5
1931	S1412	-9716	477.5
1932	S1413	-9730	622.5
1933	S1414	-9744	477.5
1934	S1415	-9758	622.5
1935	S1416	-9772	477.5
1936	S1417	-9786	622.5
1937	S1418	-9800	477.5
1938	S1419	-9814	622.5
1939	S1420	-9828	477.5
1940	S1421	-9842	622.5
1941	S1422	-9856	477.5
1942	S1423	-9870	622.5
1943	S1424	-9884	477.5
1944	S1425	-9898	622.5
1945	S1426	-9912	477.5
1946	S1427	-9926	622.5
1947	S1428	-9940	477.5
1948	S1429	-9954	622.5
1949	S1430	-9968	477.5
1950	S1431	-9982	622.5
1951	S1432	-9996	477.5
1952	S1433	-10010	622.5
1953	S1434	-10024	477.5
1954	S1435	-10038	622.5
1955	S1436	-10052	477.5
1956	S1437	-10066	622.5
1957	S1438	-10080	477.5
1958	S1439	-10094	622.5
1959	S1440	-10108	477.5
1960	SDUM2	-10122	622.5
1961	SDUM3	-10136	477.5
1962	VSSIDUM58	-10150	622.5
1963	VSSIDUM59	-10164	477.5
1964	VGLO_L	-10178	622.5
1965	VGLO_L	-10192	477.5
1966	VGLO_L	-10206	622.5
1967	VGLO_L	-10220	477.5
1968	VGLO_L	-10234	622.5
1969	VGLO_L	-10248	477.5
1970	VGLO_L	-10262	622.5
1971	VGLO_L	-10276	477.5
1972	VGLO_L	-10290	622.5
1973	VGH	-10304	477.5
1974	VGH	-10318	622.5
1975	VGH	-10332	477.5
1976	VGH	-10346	622.5
1977	VGH	-10360	477.5
1978	VGH	-10374	622.5
1979	VGH	-10388	477.5
1980	VGH	-10402	622.5

No.	Text Name	X-axis	Y-axis
1981	VSSIDUM60	-10416	477.5
1982	VSSIDUM61	-10430	622.5
1983	VSSIDUM62	-10444	477.5
1984	VSSIDUM63	-10458	622.5
1985	VSSIDUM64	-10472	477.5
1986	VSSIDUM65	-10486	622.5
1987	VSSIDUM66	-10500	477.5
1988	VSSIDUM67	-10514	622.5
1989	VSSIDUM68	-10528	477.5
1990	VSSIDUM69	-10542	622.5
1991	VSSIDUM70	-10556	477.5
1992	VSSIDUM71	-10570	622.5
1993	VSSIDUM72	-10584	477.5
1994	VSSIDUM73	-10598	622.5
1995	VSSIDUM74	-10612	477.5
1996	VSSIDUM75	-10626	622.5
1997	VSSIDUM76	-10640	477.5
1998	VSSIDUM77	-10654	622.5
1999	VSSIDUM78	-10668	477.5
2000	VSSIDUM79	-10682	622.5
2001	VSSIDUM80	-10696	477.5
2002	VSSIDUM81	-10710	622.5
2003	VSSIDUM82	-10724	477.5
2004	VSSIDUM83	-10738	622.5
2005	VSSIDUM84	-10752	477.5
2006	VSSIDUM85	-10766	622.5
2007	VSSIDUM86	-10780	477.5
2008	VSSIDUM87	-10794	622.5
2009	VSSIDUM88	-10808	477.5
2010	VSSIDUM89	-10822	622.5
2011	VSSIDUM90	-10836	477.5
2012	VSSIDUM91	-10850	622.5
2013	VSSIDUM92	-10864	477.5
2014	VSSIDUM93	-10878	622.5
2015	VSSIDUM94	-10892	477.5
2016	VSSIDUM95	-10906	622.5
2017	VSSIDUM96	-10920	477.5
2018	VSSIDUM97	-10934	622.5
2019	VSSIDUM98	-10948	477.5
2020	VSSIDUM99	-10962	622.5
2021	VSSIDUM100	-10976	477.5
2022	VSSIDUM101	-10990	622.5
2023	VSSIDUM102	-11004	477.5
2024	VSSIDUM103	-11018	622.5
2025	GOUT17	-11032	477.5
2026	GOUT17	-11046	622.5
2027	GOUT18	-11060	477.5
2028	GOUT18	-11074	622.5
2029	GOUT19	-11088	477.5
2030	GOUT19	-11102	622.5
2031	GOUT20	-11116	477.5
2032	GOUT20	-11130	622.5
2033	GOUT21	-11144	477.5
2034	GOUT21	-11158	622.5
2035	GOUT22	-11172	477.5
2036	GOUT22	-11186	622.5
2037	GOUT23	-11200	477.5
2038	GOUT23	-11214	622.5
2039	GOUT24	-11228	477.5
2040	GOUT24	-11242	622.5

No.	Text Name	X-axis	Y-axis
2041	GOUT25	-11256	477.5
2042	GOUT25	-11270	622.5
2043	GOUT26	-11284	477.5
2044	GOUT26	-11298	622.5
2045	GOUT27	-11312	477.5
2046	GOUT27	-11326	622.5
2047	GOUT28	-11340	477.5
2048	GOUT28	-11354	622.5
2049	GOUT29	-11368	477.5
2050	GOUT29	-11382	622.5
2051	GOUT30	-11396	477.5
2052	GOUT30	-11410	622.5
2053	VGLO_L	-11424	477.5
2054	VGLO_L	-11438	622.5
2055	VGLO_L	-11452	477.5
2056	VRGH_L	-11466	622.5
2057	VRGH_L	-11480	477.5
2058	VRGH_L	-11494	622.5
2059	LVGL_L	-11508	477.5
2060	LVGL_L	-11522	622.5
2061	LVGL_L	-11536	477.5
2062	GOUT31	-11550	622.5
2063	GOUT31	-11564	477.5
2064	GOUT32	-11578	622.5
2065	GOUT32	-11592	477.5
2066	VGLO_L	-11606	622.5
2067	VGLO_L	-11620	477.5
2068	VGLO_L	-11634	622.5
2069	VGH	-11648	477.5
2070	VGH	-11662	622.5
2071	VGH	-11676	477.5
2072	PADA4	-11690	622.5
2073	PADB4	-11704	477.5
2074	VSSIDUM104	-11718	622.5
2075	VSSIDUM105	-11732	477.5
2076	VSSIDUM106	-11760	622.5
2077	DUMMY	-11998	622.5
2078	DUMMY	-12012	477.5
2079	DUMMY	-12026	622.5
2080	DUMMY	-12040	477.5
2081	DUMMY	-12054	622.5

Alignment Mark	X-axis	Y-axis
ALMART_R_T	11870	605
ALMART_L_T	-11870	605

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System Interface

3.7. DBI Type B Parallel Interface

The ILI9806G supports an 8-/9-/16-/18-/24-bit MPU DBI Type B parallel interface. The chip-select CSX (active low) is used to enable or disable the ILI9806G chip. The RESX (active low) is an external reset signal, the WRX is parallel data write strobe, the RDX is parallel data read strobe, and DB [23:0] is parallel data bus.

The ILI9806G latches the input data at the rising edge of the WRX signal. The DCX is the signal for data/command selection. When DCX = 1, DB [23:0] bits are RAM data or command parameters. When DCX = 0, DB [23:0] bits are commands. The DBI Type B bi-directional interface is used for communication between the MPU controller and LCD driver chip. The selection of the parallel interface is shown in Table 8.

Table 8: DBI Type B Parallel Interface

IM3	IM2	IM1	IM0	MPU-Interface Mode	WRX	RDX	DCX	Function
0	0	0	0	DBI Type B 8-bit bus interface	↑	"H"	"L"	Write command code.
					"H"	↑	"H"	Read internal status.
					↑	"H"	"H"	Write parameter or display data.
					"H"	↑	"H"	Reads parameter or display data.
0	0	0	1	DBI Type B 16-bit bus interface	↑	"H"	"L"	Write command code.
					"H"	↑	"H"	Read internal status.
					↑	"H"	"H"	Write parameter or display data.
					"H"	↑	"H"	Reads parameter or display data.
0	0	1	0	DBI Type B 24-bit bus interface	↑	"H"	"L"	Write command code.
					"H"	↑	"H"	Read internal status.
					↑	"H"	"H"	Write parameter or display data.
					"H"	↑	"H"	Reads parameter or display data.
1	1	0	0	DBI Type B 9-bit bus interface	↑	"H"	"L"	Write command code.
					"H"	↑	"H"	Read internal status.
					↑	"H"	"H"	Write parameter or display data.
					"H"	↑	"H"	Reads parameter or display data.
1	1	0	1	DBI Type B 18-bit bus interface	↑	"H"	"L"	Write command code.
					"H"	↑	"H"	Read internal status.
					↑	"H"	"H"	Write parameter or display data.
					"H"	↑	"H"	Reads parameter or display data.

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3.7.1. Write Cycle Sequence

The WRX signal is driven from high to low then pulled back to high during the write cycle. The host processor provides information while the display module captures the information from the host processor on the rising edge of the WRX. When the DCX signal is driven to low level, the input data on the interface is interpreted as command information. The DCX signal can also be pulled to high level when the data is RAM data or command parameter.

Figure 2 shows the write cycle of the DBI Type B interface.

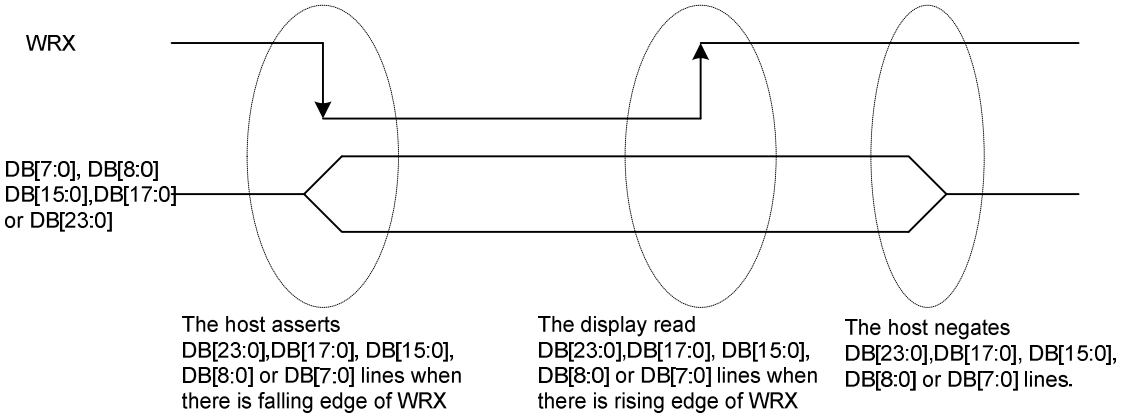


Figure 2: DBI Type B Write Cycle[Note]

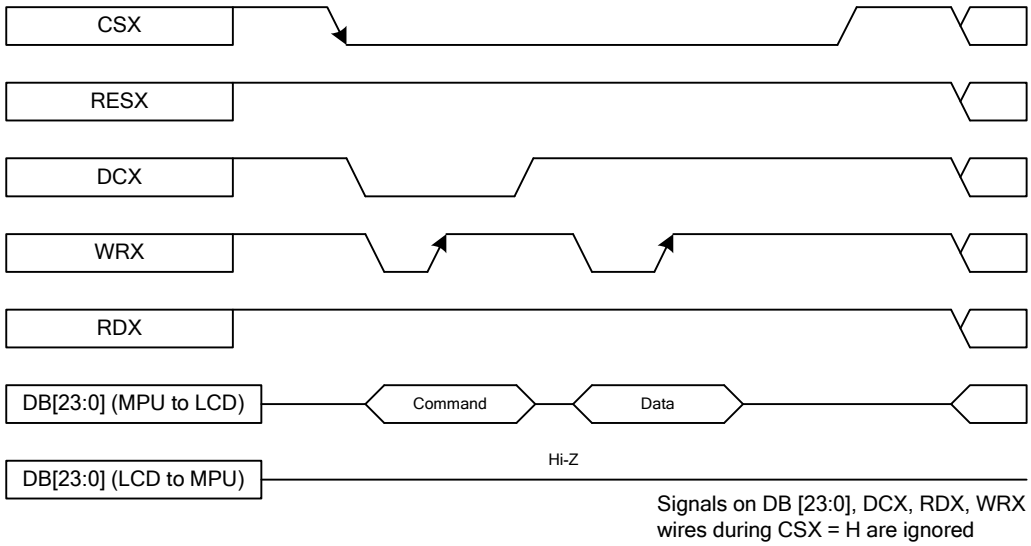


Figure 3: DBI Type B Write Cycle Sequence

Note WRX is an unsynchronized signal, which can be terminated when not in use.

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3.7.2. Read Cycle Sequence

The RDX signal is driven from high to low and then pulled back to high during the read cycle. The display module provides information to the host processor while the host processor reads the display module information on the rising edge of the RDX signal. When the DCX signal is driven to the low level, the input data on the interface is interpreted as internal status or parameter data. The DCX signal also can be pulled to high level when the data on the interface is RAM data or a command parameter data.

The following figure shows a read cycle of the DBI Type B interface.

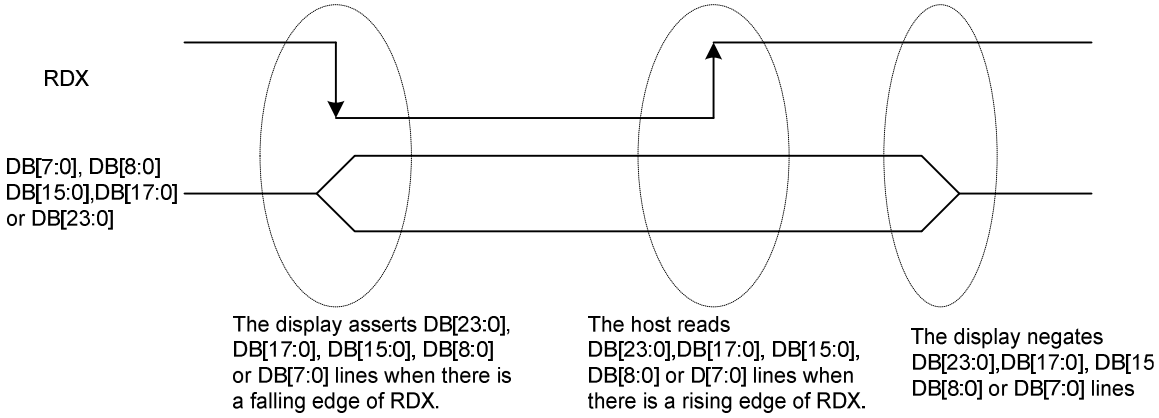


Figure 4: DBI Type B Read Cycle[Note 1]

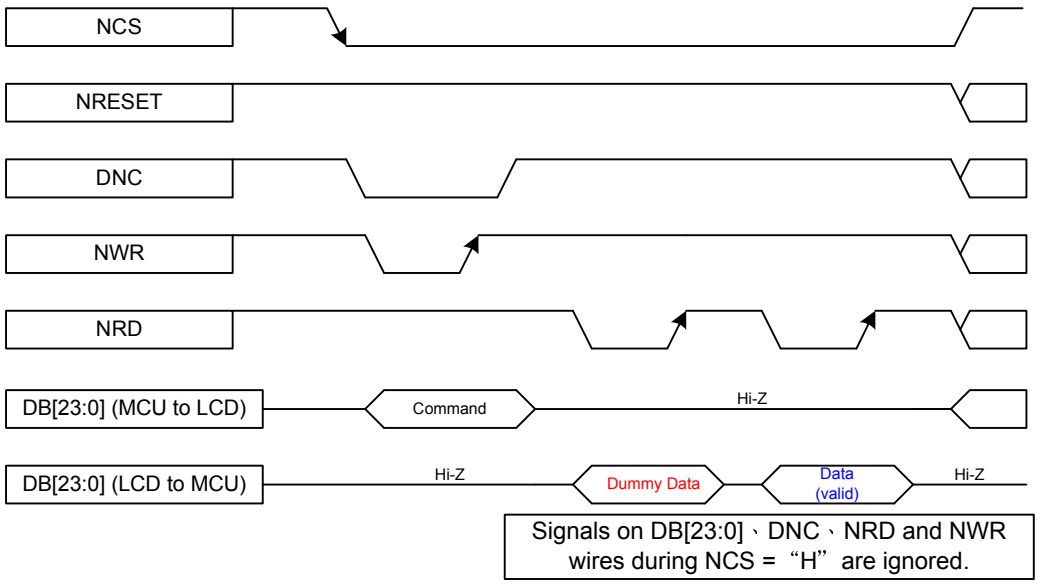


Figure 5: DBI Type B Read Cycle Sequence[Note 2]

Note 1 *RDX is an unsynchronized signal, which can be terminated when not in use.*

Note 2 *Read Data is only valid when the DCX input is pulled high. If the DCX signal is driven low during the read cycle then the display information outputs will be High-Z.*

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3.7.3. DBI Type B Interface Set Table

24-bit data bus DB [23:0] interface, IM [3:0] = 0010

	Register	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Command/Parameter Write	2Ch / 3Ch																	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
Command/Parameter Read	2Eh / 3Eh																	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]

	Set_pixel_format	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
24bpp Frame Memory Write	3'h7	R[7]	R[6]	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[7]	G[6]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[7]	B[6]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]
Frame Memory Read	*	r[7]	r[6]	r[5]	r[4]	r[3]	r[2]	r[1]	r[0]	g[7]	g[6]	g[5]	g[4]	g[3]	g[2]	g[1]	g[0]	b[7]	b[6]	b[5]	b[4]	b[3]	b[2]	b[1]	b[0]

Figure 6: DBI Type B 24-bit Data Bus

18-bit data bus DB [17:0] interface, IM [3:0] = 1101

	Register	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Command/Parameter Write	2Ch / 3Ch											D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
Command/Parameter Read	2Eh / 3Eh											D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]

	Set_pixel_format	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
18bpp Frame Memory Write	3'h5			R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[4]	B[3]	B[2]	B[1]	B[0]

	Set_pixel_format	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
18bpp Frame Memory Write	3'h6	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

		First Transfer			Second Transfer			Third Transfer		
18bpp Frame Memory Write	Set_pixel_format	DB[17:10]	DB[9:8]	DB[7:0]	DB[17:10]	DB[9:8]	DB[7:0]	DB[17:10]	DB[9:8]	DB[7:0]
	3'h7	R1[7:0]		G1[7:0]	B1[7:0]		R2[7:0]	G2[7:0]		B2[7:0]
Frame Memory Read	*	r1[7:0]		g1[7:0]	b1[7:0]		r2[7:0]	g2[7:0]		b2[7:0]

Figure 7: DBI Type B 18-bit Data Bus

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16-bit data bus DB [15:0] interface, IM [3:0] = 0001

	Set_pixel_format	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Command/Parameter Write	2Ch / 3Ch									D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
Command/Parameter Read	2Eh / 3Eh									D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]

	Set_pixel_format	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
16bpp Frame Memory Write	3'h5	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[4]	B[3]	B[2]	B[1]	B[0]

		First Transfer				Second Transfer				Third Transfer			
16bpp Frame Memory Write	Set_pixel_format	DB[15:10]	DB[9:8]	DB[7:2]	DB[1:0]	DB[15:10]	DB[9:8]	DB[7:2]	DB[1:0]	DB[15:10]	DB[9:8]	DB[7:2]	DB[1:0]
	3'h6	R1[5:0]		G1[5:0]		B1[5:0]		R2[5:0]		G2[5:0]		B2[5:0]	

		First Transfer		Second Transfer		Third Transfer	
16bpp Frame Memory Write	Set_pixel_format	DB[15:8]	DB[7:0]	DB[15:8]	DB[7:0]	DB[15:8]	DB[7:0]
	3'h7	R1[7:0]	G1[7:0]	B1[7:0]	R2[7:0]	G2[7:0]	B2[7:0]
Frame Memory Read	*	r1[7:0]	g1[7:0]	b1[7:0]	r2[7:0]	g2[7:0]	b2[7:0]

Figure 8: DBI Type B 16-bit Data Bus

9-bit data bus DB [8:0] interface, IM [3:0] = 1100

	Register	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Command/Parameter Write	2Ch / 3Ch		D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
Command/Parameter Read	2Eh / 3Eh		D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]

		First Transfer									Second Transfer								
9bpp Frame Memory Write	Set_pixel_format	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h5	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]		G[2]	G[1]	G[0]	B[4]	B[3]	B[2]	B[1]	B[0]	

		First Transfer									Second Transfer								
9bpp Frame Memory Write	Set_pixel_format	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h6	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

		First Transfer									Second Transfer									Third Transfer								
9bpp Frame Memory Write	Set_pixel_format	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h7	R1[7:0]									G1[7:0]									B1[7:0]								
Frame Memory Read	*	r1[7:0]									g1[7:0]									b1[7:0]								

Figure 9: DBI Type B 9-bit Data Bus

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8-bit data bus DB [7:0] interface, IM [3:0] = 0000

	Register	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Command/Parameter Write	2Ch / 3Ch	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
Command/Parameter Read	2Eh / 3Eh	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]

		First Transfer								Second Transfer							
8bpp Frame Memory Write	Set_pixel_format	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h5	R[4]	R[3]	R[2]	R[1]	R[0]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[4]	B[3]	B[2]	B[1]	B[0]

		First Transfer								Second Transfer								Third Transfer							
8bpp Frame Memory Write	Set_pixel_format	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h6	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]			G[5]	G[4]	G[3]	G[2]	G[1]	G[0]			B[5]	B[4]	B[3]	B[2]	B[1]	B[0]		

		First Transfer								Second Transfer								Third Transfer							
8bpp Frame Memory Write	Set_pixel_format	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	3'h7	R1[7:0]								G1[7:0]								B1[7:0]							
Frame Memory Read	*	r1[7:0]								g1[7:0]								b1[7:0]							

Figure 10: DBI Type B 8-bit Data Bus

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3.8. DBI Type C Serial Interface

The selection of this interface is done by the IM [3:0] pins. See Table 9.

Table 9: DBI Type C Serial Interface

IM3	IM2	IM1	IM0	DBI Type C Mode	CSX	SDA	SCL	Function
X	0	1	1	3-line serial interface	"L"	-	↑	Read/Write command, parameter or display data.

The ILI9806G uses a 3-line 9-bit serial interface for communication between the host and the ILI9806G. The 3-line serial interface consists of the chip enable input (CSX), the serial clock input (SCL), and serial data Input/Output (SDA). If the data bus (DB [23:0]) is not used for the DPI interface data transfer, the unused pins are unaffected. The Serial clock (SCL) is used only for the interface with the MPU, so it can be stopped when no communication is necessary.

3.8.1. Write Cycle Sequence

In the Write Mode of the interface, the host writes commands and data to the ILI9806G. The 3-line serial data packet contains a D/C (data/command) select bit and a transmission byte. If the D/C bit is "low", the transmission byte is interpreted as a command byte. If the D/C bit is "high", the transmission byte is stored in the GRAM as display data, or stored in the command register as a parameter data.

Any instruction can be sent in any order to the ILI9806G and the MSB is transmitted first. The serial interface is initialized when the CSX status is high. In this state, SCL clock pulse and SDI data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission. See below for the detail of data format for 3-line serial interface.

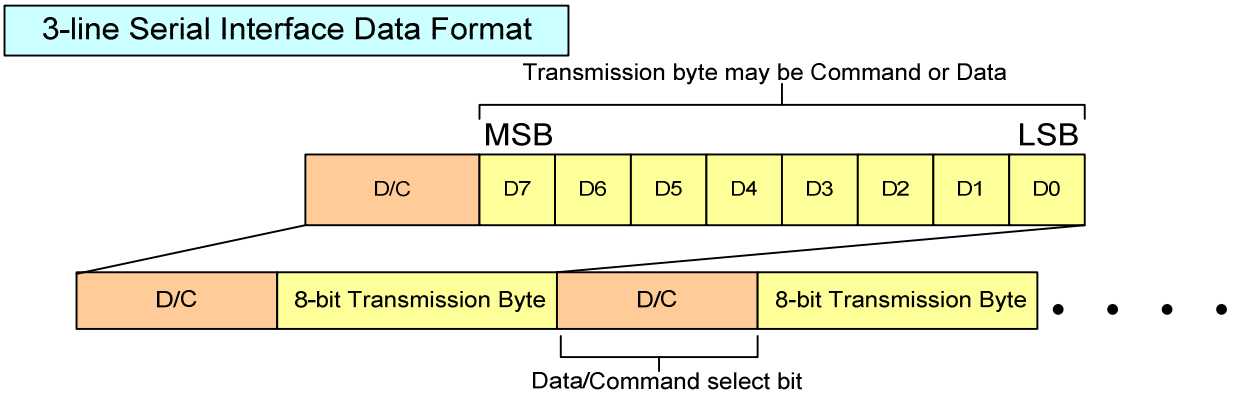


Figure 11: DBI Type C Data Format

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The host drives the CSX pin to low and setting the D/C bit on the SDI pin. The bit is read by the ILI9806G on the first rising edge of the SCL signal. On the next falling edge of the SCL, the MSB data bit (D7) is set on the SDI pin by the host. On the next falling edge of the SCL, the next bit (D6) is set on the SDI pin. If the optional D/C signal is used, a byte is eight read cycles long. The 3-line serial interface writes sequences as described in the Figure 12 below.

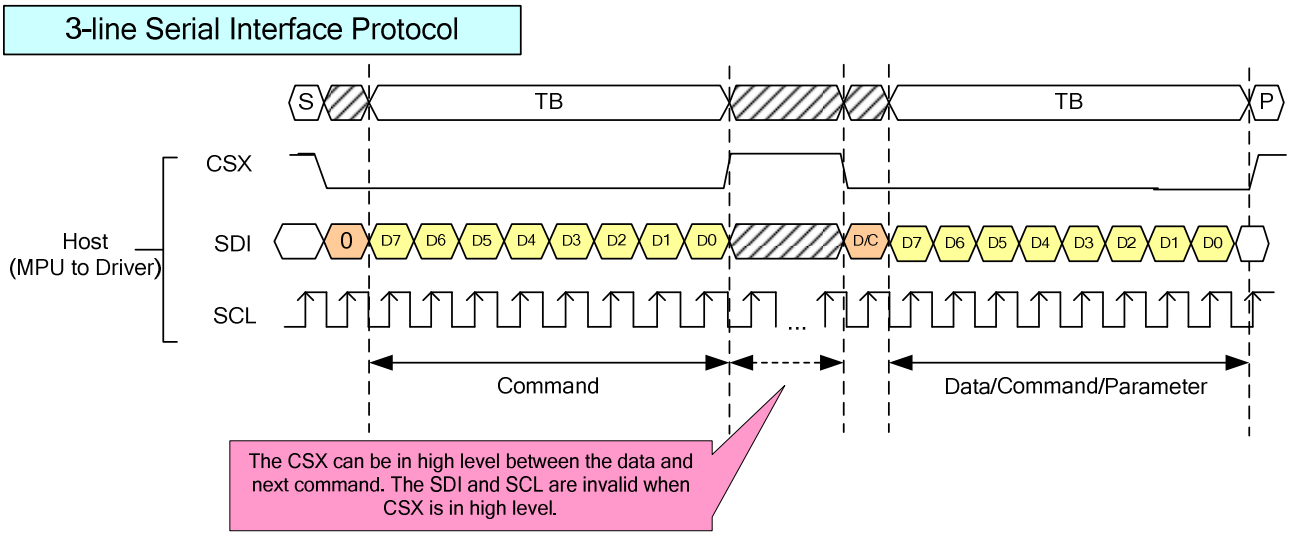


Figure 12: DBI Type C Protocol

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3.8.2. Read Cycle Sequence

In the Read Mode of the interface, the host reads the register value from the ILI9806G. The host sends a command (Read ID or register command), then a byte is (bytes are) transmitted in the opposite direction. The ILI9806G samples the SDI (input data) at the rising edge of the SCL (serial clock), and shifts SDO (output data) at the falling edge of the SCL (serial clock). The read mode has three types of transmitted command data (8-/24-/32-bit) according to the command code.

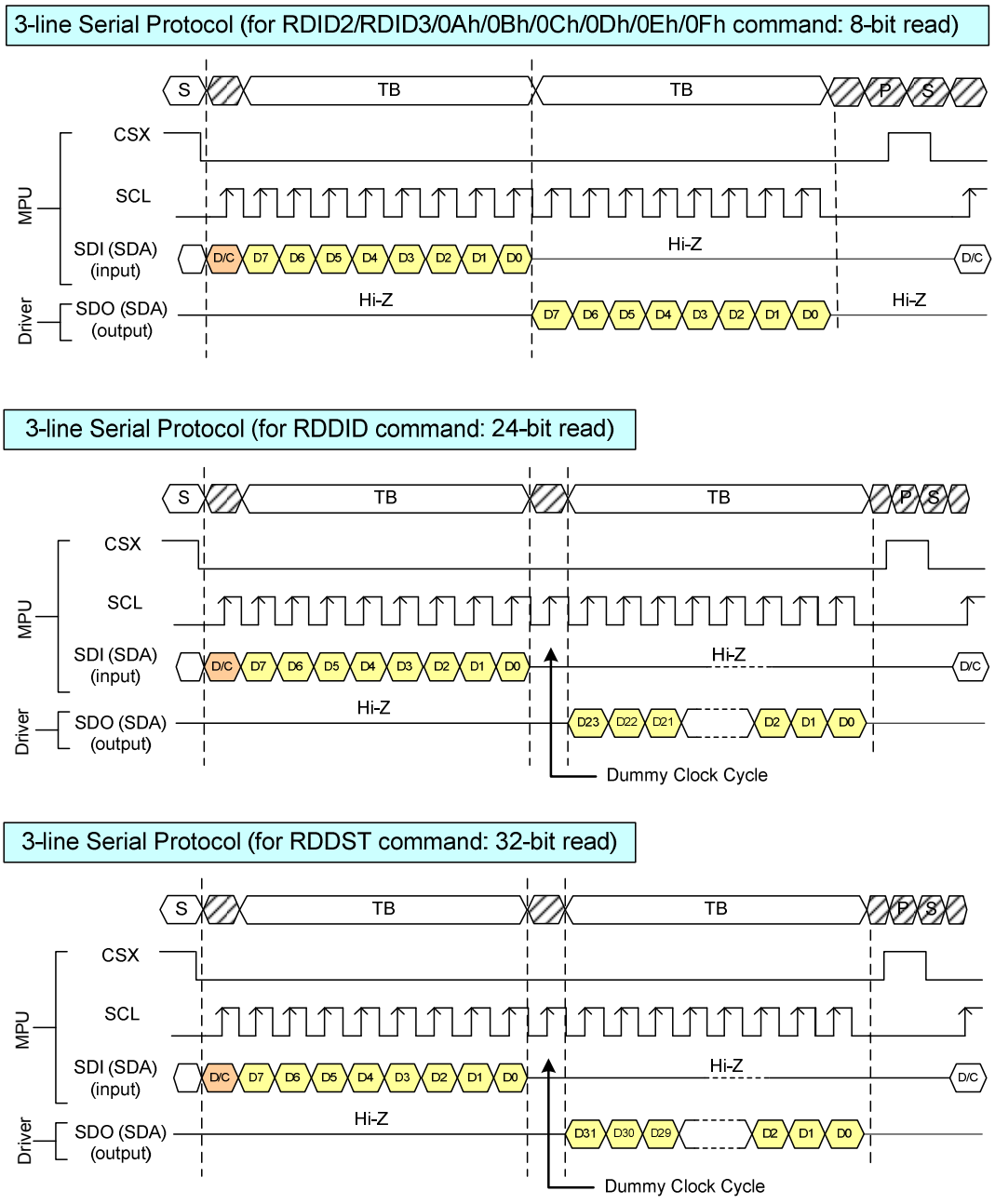


Figure 13: DBI Type C Read Cycle Sequence

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3.9. Data Transfer Break and Recovery

If data transmission is interrupted by the CSX pulse while transferring a Command, Frame Memory data or multiple parameter command before Bit D0 of the byte has been completed, then the driver will reject the previous bits and reset the interface so it will be ready to receive the same byte re-transmitted when the chip select pin (CSX) is activated again.

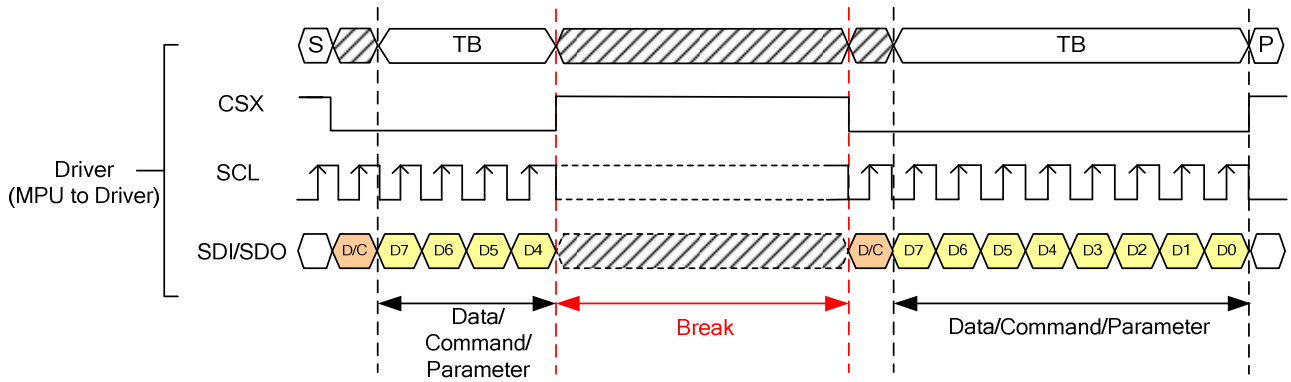


Figure 14: Data Transfer Break and Recovery

If there is a break when transmitting a command with multiple parameters and the host initiates transfer of a new command, the parameters that were successfully transferred are stored and the incomplete parameter data where the break occurred is dropped. The interface is ready to receive the next byte as shown in the figure below.

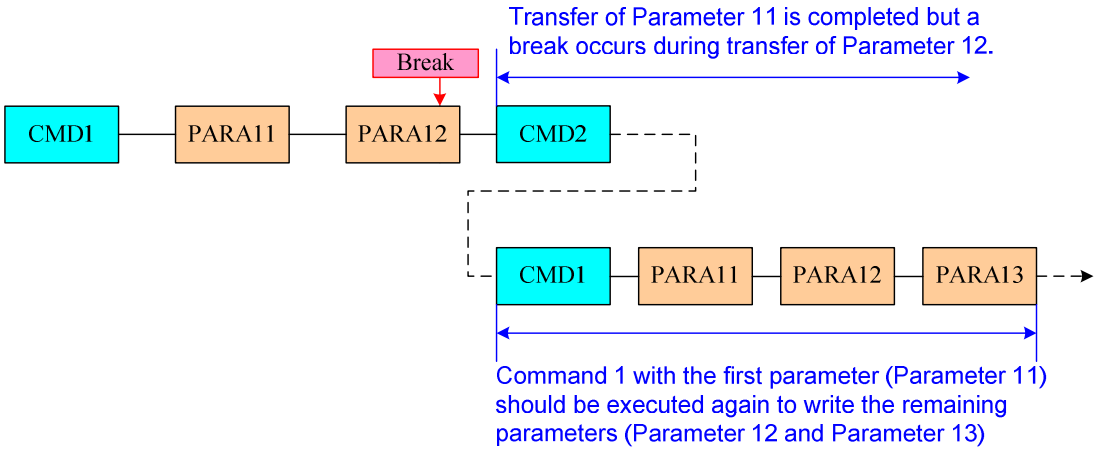


Figure 15: Data Transfer Break - Case 1

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If a command with multiple parameters is sent and a break occurs when a new command is sent before all the parameters are transferred, then the parameters that were successfully sent are stored and the remaining parameters of that command remain at the previous value.

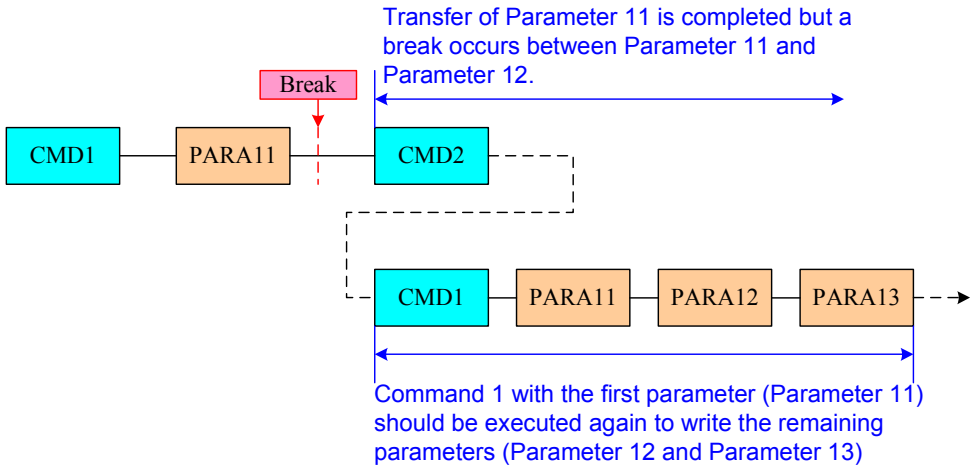


Figure 16: Data Transfer Break - Case 2

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3.10. Data Transfer Pause

Transferring a Command, Frame Memory Data, or Multiple Parameter Data might invoke a pause in the data transmission. If the Chip Select pin (CSX) is released after a whole byte of a Frame Memory Data or Multiple Parameter Data has been completed, then the ILI9806G will wait and continue the Frame Memory Data or Parameter Data Transmission from the point where it was paused. If the Chip Select pin is released after a whole byte of a command has been completely transmitted, then the display module will receive either the command's parameters or a new command when the Chip Select Line is enabled again, as shown below.

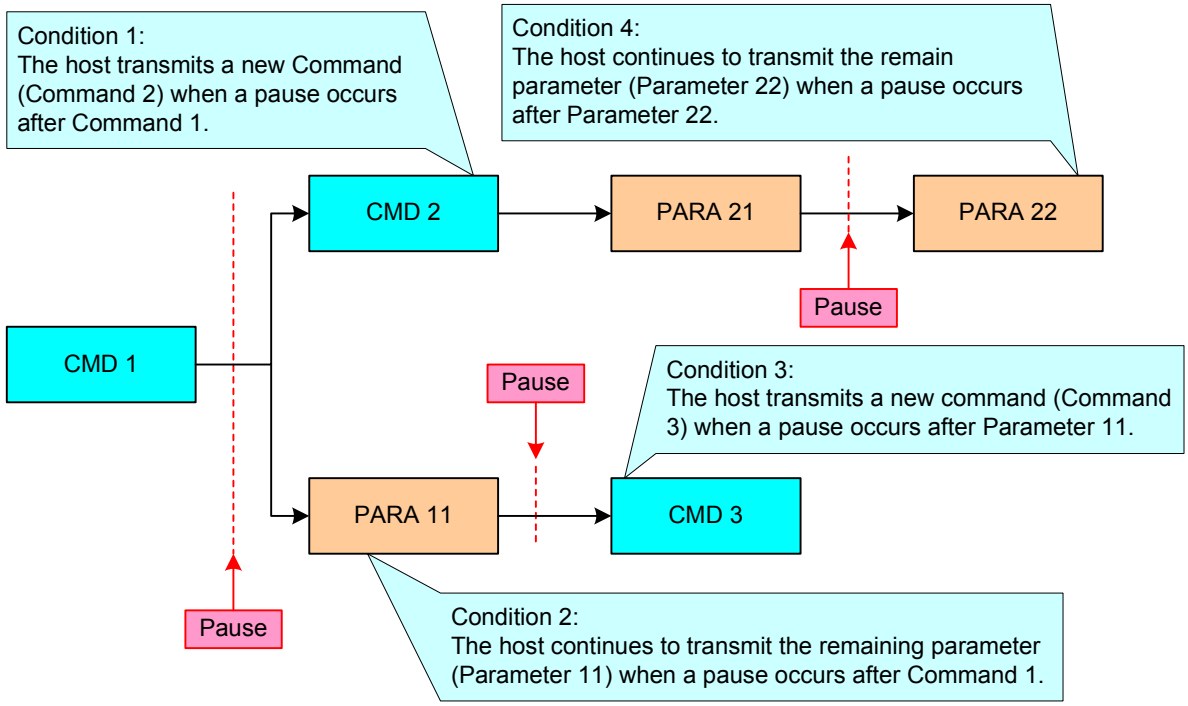


Figure 17: Data Transfer Pause

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3.10.1. Serial Interface Pause

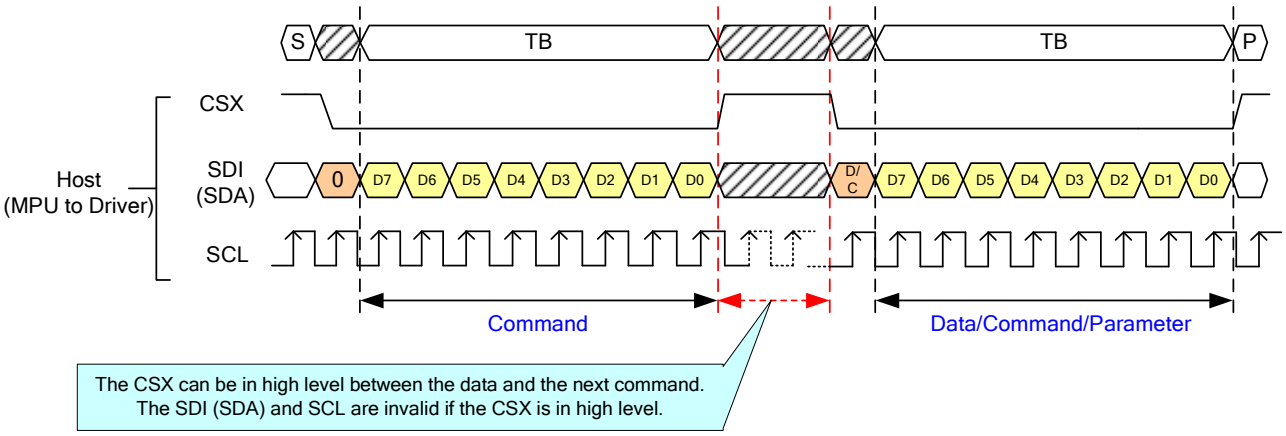


Figure 18: DBI Type C Data Transfer Pause

This applies to the following 4 conditions:

1) Command-Pause-Command
2) Command-Pause-Parameter
3) Parameter-Pause-Command
4) Parameter-Pause-Parameter

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3.10.2. Parallel Interface Pause

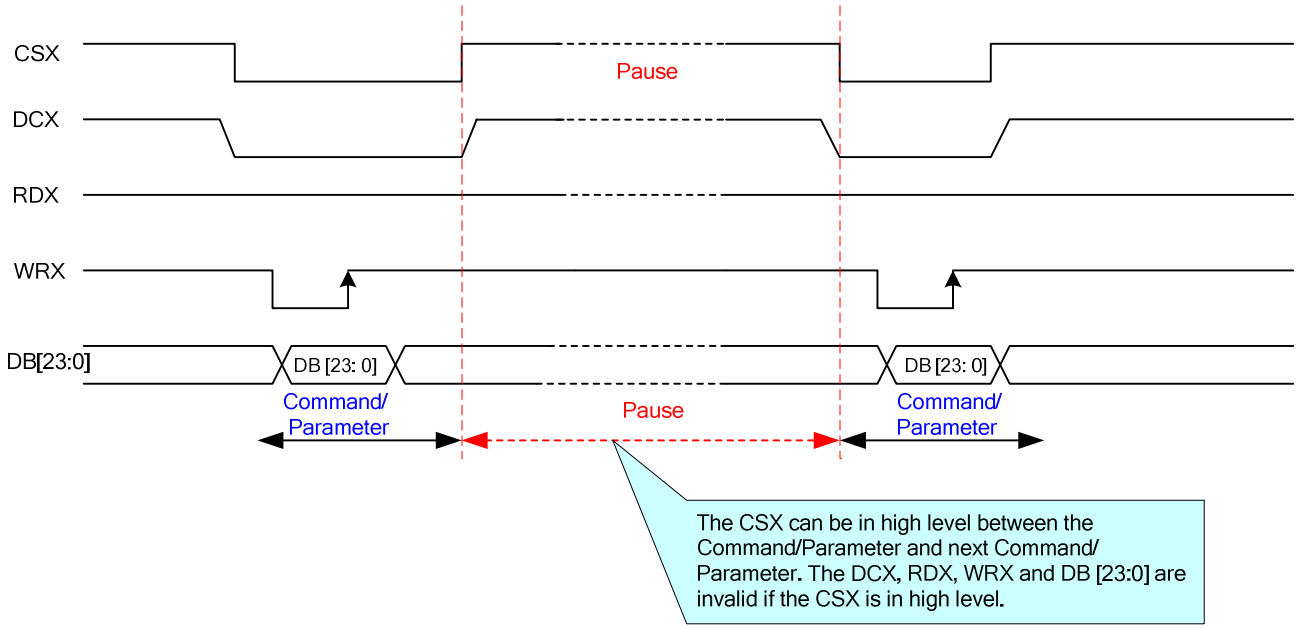


Figure 19: DBI Type B Data Transfer Pause

This applies to the following 4 conditions:

1) Command-Pause-Command
2) Command-Pause-Parameter
3) Parameter-Pause-Command
4) Parameter-Pause-Parameter

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3.10.3. Data Transfer Mode

The ILI9806G provides five different types of color depth: 8-bit/per pixel, 9-bit/per pixel, 16-bit/per pixel, 18-bit/per pixel, and 24-bit/pixel. The data format is described for each interface. Data can be downloaded to the Frame Memory by 2 methods[Note].

3.10.4. Method 1

The Image data is sent to the Frame Memory in the successive Frame writing, each time the Frame Memory is filled by image data, the Frame Memory pointer is reset to the start point and the next Frame is written.

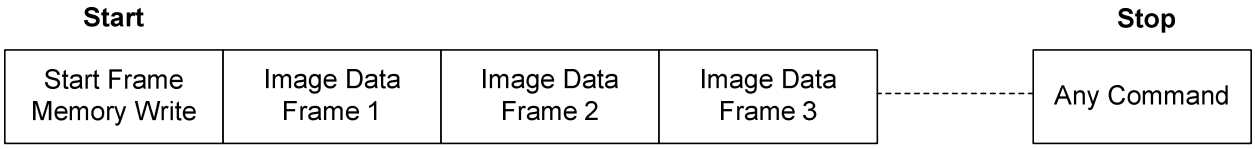


Figure 20: Data Transfer Mode - Method 1

3.10.5. Method 2

Image Data is sent and at the end of each Frame Memory download, a command is sent to stop the Frame Memory Writing. Then Start Memory Write command is sent, and a new Frame is downloaded.

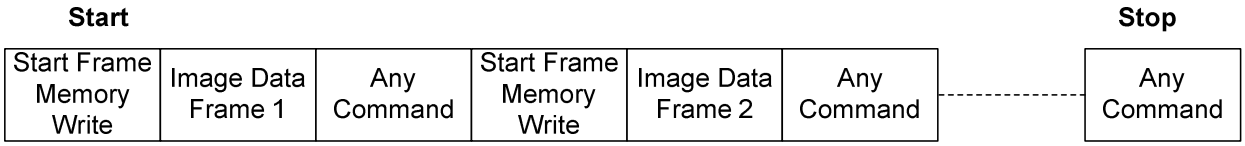


Figure 21: Data Transfer Mode - Method 2

Note 1. These apply to all five kinds of color depth on both serial and parallel interfaces.

2. The Frame Memory can contain both odd and even number of pixels for both methods. Only complete pixel data will be stored in the Frame Memory.

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3.11. DPI (RGB) Interface

The DPI can display moving pictures by two ways: rewrite into the GRAM and transmit directly to the shift register. The selection is set by the register BPGRAM and RM bit. The RM bit selects an interface for the access operation of the Frame Memory. For the DPI, RM should be set as 1.

BPGRAM	Display data path
0	Write into Memory

RM	Interface for RAM access
0	System interface
1	RGB interface

The DM bit selects the clock operation mode. It allows switching between display operations in synchronization with the internal oscillation clock. If DM = 1, the external DCK cannot be stopped unless it enters the Sleep In mode.

DM	RGB interface operating clock selection
0	Internal system clock
1	RGB interface DCK (Dot clock)

3.11.1. DPI Interface Selection

The DPI interface is operated with VSYNC, HSYNC, ENABLE, DCK, and DB [23:0] lines. It supports several pixel formats that can be selected by DPI [2:0] bits in Pixel Format Set (R3Ah) command. The selection of a given interface is done by DPI [2:0], as shown in the Table 10 and Figure 22.

Table 10: DPI Interface Selection

DPI [2:0]			DPI (RGB) Interface Mode	Used Pins
1	0	1	16-bit RGB interface	VSYNC, HSYNC, ENABLE, DCK, DB [20:16] , DB [13:8], DB [4:0]
1	1	0	18-bit RGB interface	VSYNC, HSYNC, ENABLE, DCK, DB [21:16] , DB [13:8], DB [5:0]
1	1	1	24-bit RGB interface	VSYNC, HSYNC, ENABLE, DCK, DB [23:0]
Other			Setting prohibited	

16-bit DPI interface connection:set pixel format DPI[2:0]=3'h5

DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
			R[4]	R[3]	R[2]	R[1]	R[0]			G[5]	G[4]	G[3]	G[2]	G[1]	G[0]				B[4]	B[3]	B[2]	B[1]	B[0]

18-bit DPI interface connection:set pixel format DPI[2:0]=3'h6

DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
		R[5]	R4]	R[3]	R[2]	R[1]	R[0]			G[5]	G[4]	G[3]	G[2]	G[1]	G[0]			B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

24-bit DPI interface connection:set pixel format DPI[2:0]=3'h7

DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
	R[6]	R[5]	R4]	R[3]	R[2]	R[1]	R[0]	G[7]	G[6]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[7]	B[6]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

Figure 22: DPI Interface 16/18/24-bit Pixel Format Selection

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The Pixel clock (DCK) runs all the time without stop, and it is used to enter VSYNC, HSYNC, ENABLE, and DB [23:0] states when there is a rising edge of the DCK. The DCK cannot be used as the internal clock for other functions of the display module.

Vertical synchronization (VSYNC) is used to indicate when a new frame of the display is received. This is a low enable and its state is read to the display module by the rising edge of the DCK signal.

Horizontal synchronization (HSYNC) is used to indicate when a new line of the frame is received. This is a low enable and its state is read to the display module by the rising edge of the DCK signal.

Data Enable (ENABLE) is used to indicate when the RGB information that should be transferred to the display is received. This is a high enable and its state is read to the display module by the rising edge of the DCK signal. DB [23:0] are used to indicate what is the information of the image that is transferred to the display (when ENABLE = 0 (low) and there is a rising edge of DCK). DB [23:0] can be 0 (low) or 1 (high). These lines are read by the rising edge of the DCK signal. In RGB interface modes, the input display data is written to the GRAM first then outputs the corresponding source voltage according to the gray data from the GRAM.

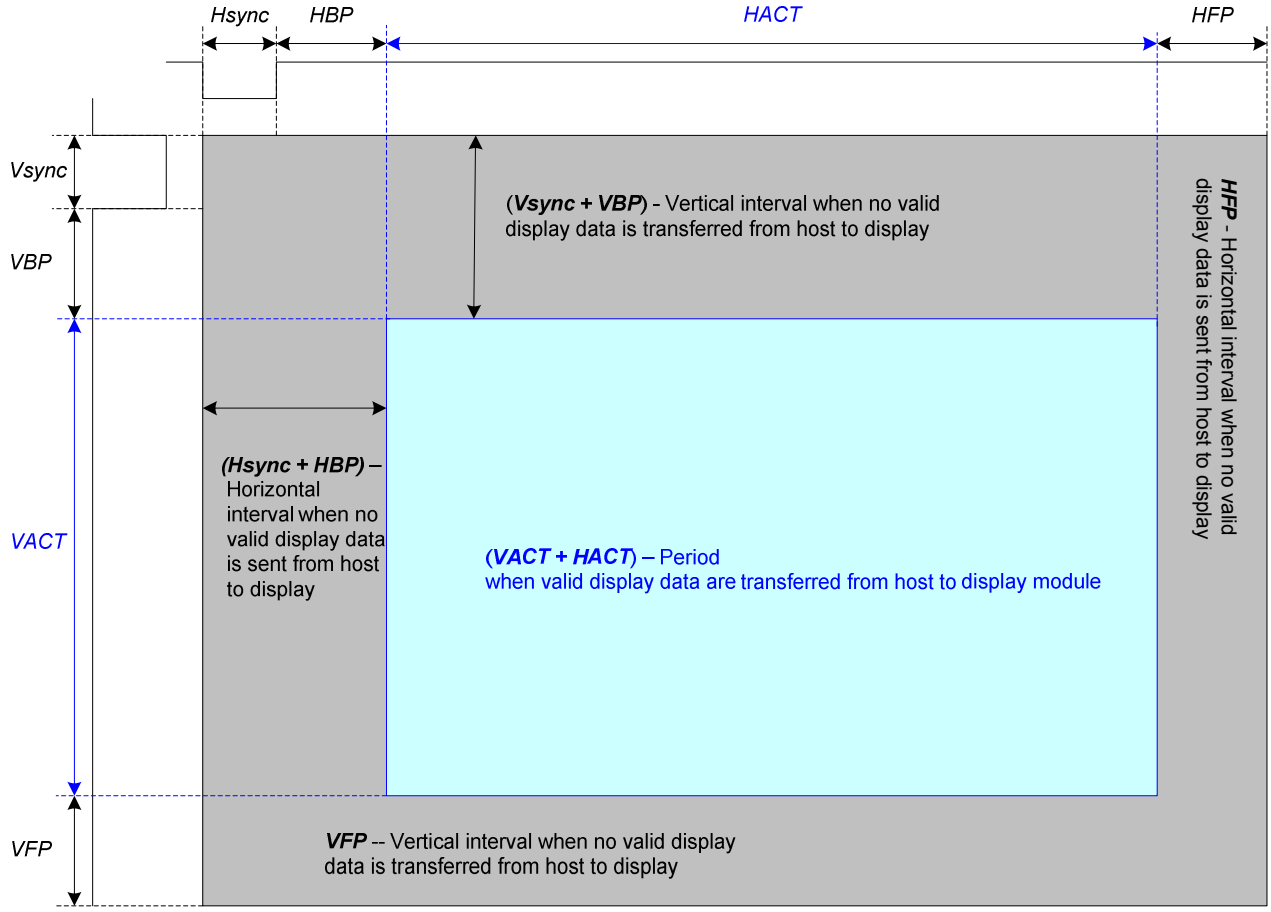


Figure 23: General DPI Timing Diagram

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3.11.2. DPI Interface Timing

The timing chart of 24-/18-/16-bit DPI interface mode is illustrated in Figure 24.

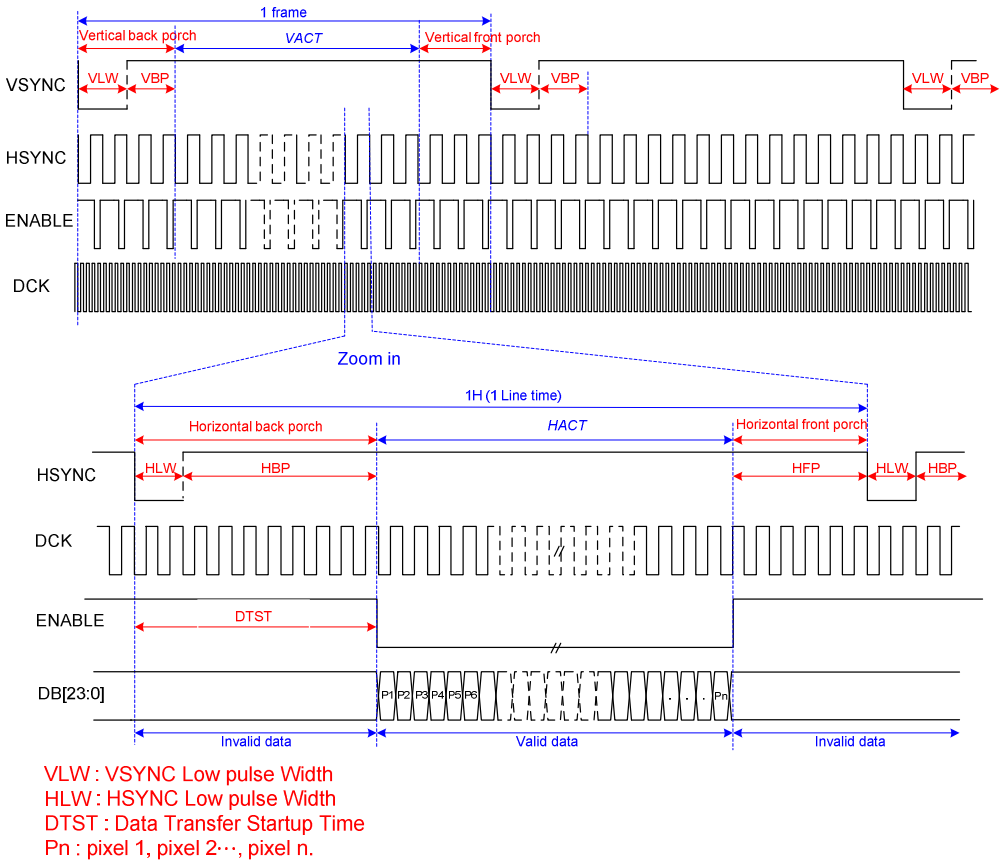
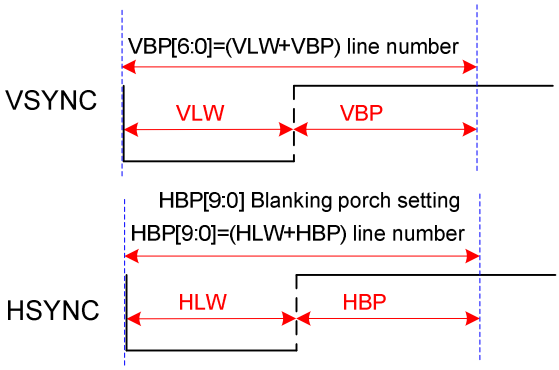


Figure 24: DPI Interface Timing diagram[Note]



Constraint1：Vertical back porch: Minimum VLW >=2 H , VBP >=8, VFP>=8

Constraint2：Horizontal porch: Minimum HLW >=2 clocks , HBP>=20 clocks, HFP>=20 clocks

Note *VSPL = 0, HSPL = 0, DPL = 0, and EPL = 0 of Interface Mode Control B0h command.*

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3.12. DSI System Interface

3.12.1. General Description

The MIPI DSI is enabled or disabled by the external IM [3:0] pin.

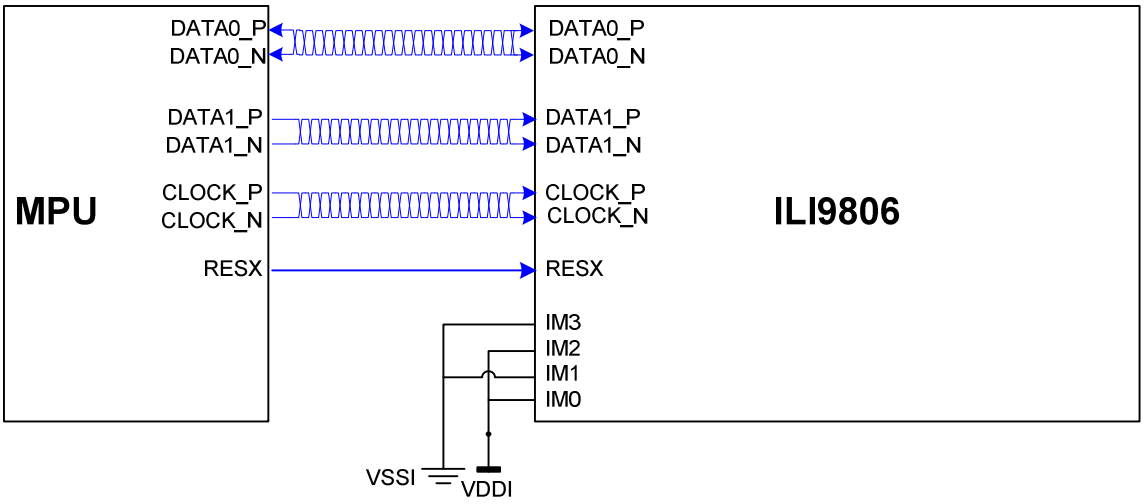


Figure 25: DSI System Interface Diagram

IM3	IM2	IM1	IM0	MPU Interface	Data Pin in Use
0	1	0	1	DSI interface	DSI_CP, DSI_CN DSI_D0P, DSI_D0N DSI_D1P, DSI_D1N

The communication is separated into two different levels between the MPU and the display module:

- Low level communication is done on the interface level.
- High level communication is done on the packet level.

3.12.2. Interface Level Communication

3.12.3. General

The display module uses data and clock lane differential pairs for DSI (DSI-2M). Both differential lane pairs can be driven to the Low Power (LP) or High Speed (HS) mode.

Low Power mode means that each line of the differential pair is used in the single ended mode, a differential receiver is disable (a termination resistor of the receiver is disable), and it can be driven into a low power mode.

High Speed mode means that differential pairs (the termination resistor of the receiver is enable) are not used in the single ended mode.

Different modes and protocols are used in each mode when information is to be transferred from the MPU to the display module and vice versa.

The State Codes of the High Speed (HS) and Low Power (LP) lane pair are defined below.

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Table 11: High Speed and Low-Power Lane Pair State Codes

Lane Pair State Code	Line DC Voltage Levels		High Speed (HS)	Low Power	
	DATA_P	DATA_N	Burst Mode	CLOCK_P	CLOCK_N
HS-0	Low (HS)	High (HS)	Differential – 0	Note 1	Note 1
HS-1	High (HS)	Low (HS)	Differential – 1	Note 1	Note 1
LP-00	Low (LP)	Low (LP)	Not Defined	Bridge	Space
LP-01	Low (LP)	High (LP)	Not Defined	HS – Request	Mark - 0
LP-10	High (LP)	Low (LP)	Not Defined	LP - Request	Mark - 1
LP-11	High (LP)	High (LP)	Not Defined	Stop	Note 2

3.12.4. DSI-CLK Lanes

DSI-CLK+/- lanes can be driven into three different power modes: Low Power Mode (LPM), Ultra Low Power Mode (ULPM), or High Speed Clock Mode (HSCM). Clock lanes are in the single ended mode (LP = Low Power) when entering or leaving the Low Power Mode (LPM) or Ultra Low Power Mode (ULPM). Likewise, clock lanes are in the single ended mode (LP = Low Power) when entering in or leaving the High Speed Clock Mode (HSCM). These entering and leaving protocols use clock lanes in the single ended mode to generate an entering or leaving sequence.

The principal flow chart of the different power modes of clock lanes is illustrated below.

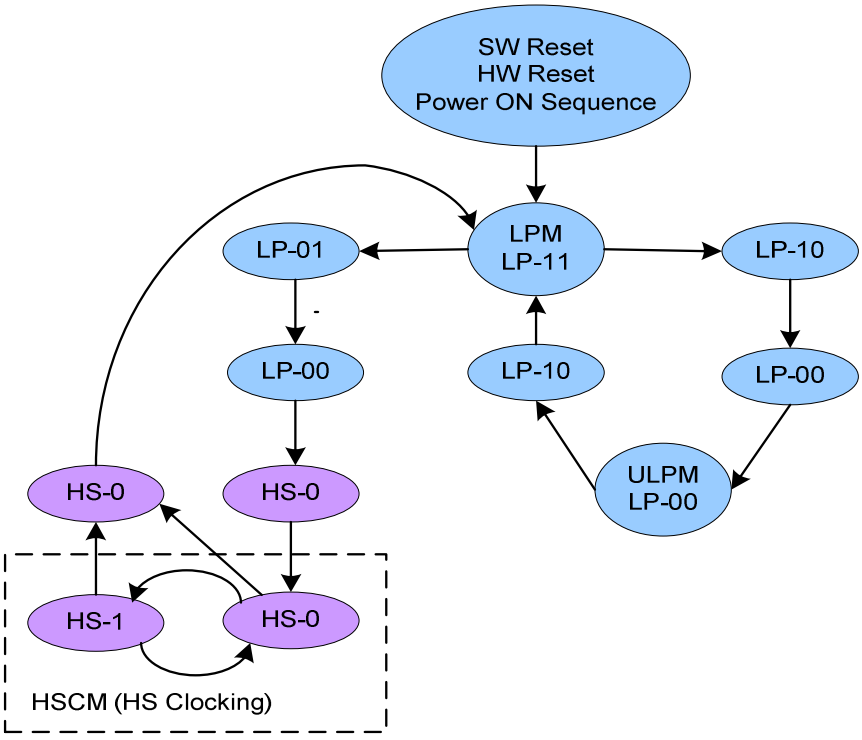


Figure 26: Power Modes of Clock Lanes

Note 1 Low-Power Receivers (LP-Rx) of the lane pair will check the LP-00 state code when the Lane Pair is in the High Speed (HS) mode.

Note 2 If Low-Power Receivers (LP-Rx) of the lane pair recognizes the LP-11 state code, the lane pair returns to the LP-11 of the Control Mode.

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3.12.5. Low Power Mode (LPM)

When DSI-CLK lanes enter the LP-11 State Code, DSI-CLK+/- lanes can be driven to the Low Power Mode (LPM) in three different ways:

1) After SW Reset, HW Reset or Power On Sequence => LP-11

2) After DSI-CLK+/- lanes leave the Ultra Low Power Mode (ULPM, LP-00 State Code) => LP-10 => LP-11 (LPM)

The sequence is illustrated below.

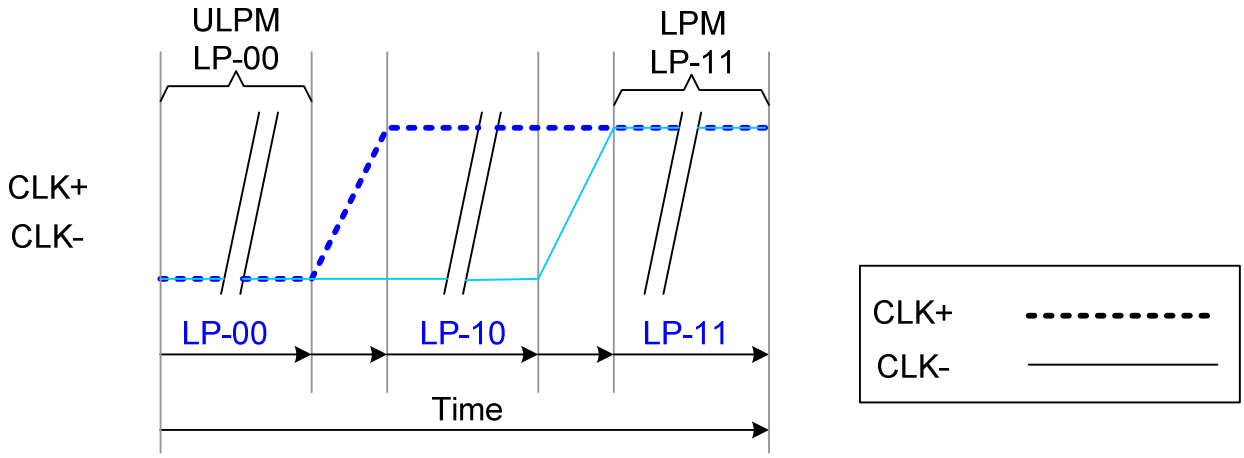


Figure 27: From ULPM to LPM

3) After DSI-CLK+/- lanes leave the High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) => HS-0 => LP-11 (LPM). The sequence is illustrated below.

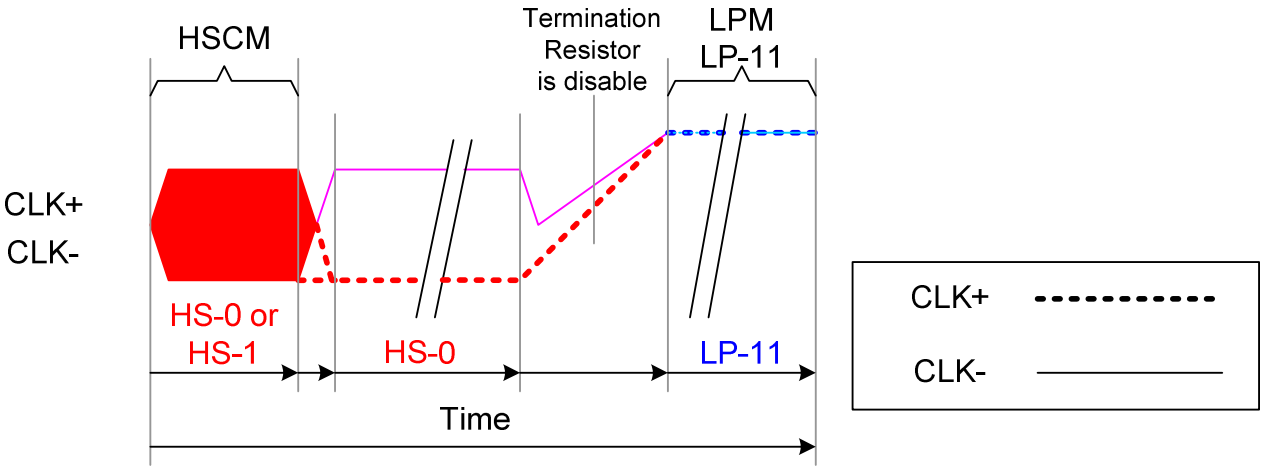


Figure 28: From High Speed Clock Mode (HSCM) to LPM

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All the changes of the three modes are illustrated in the flow chart below.

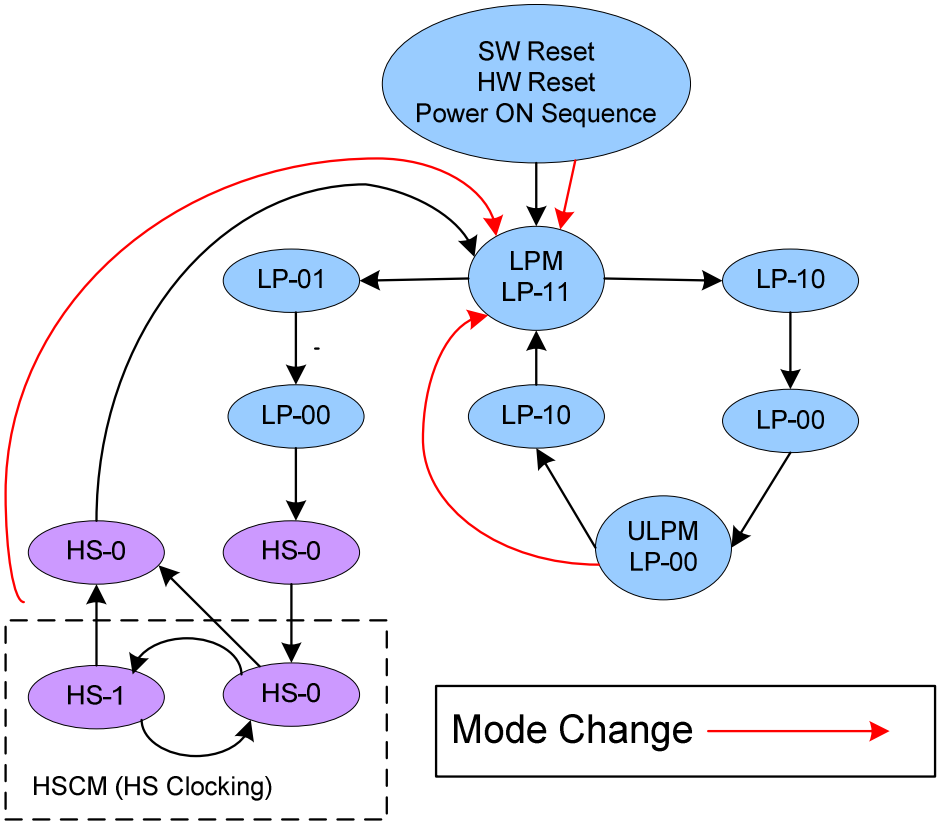


Figure 29: All the Changes of the Three Modes to the LPM

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3.12.6. Ultra Low Power Mode (ULPM)

DSI-CLK+/- lanes can be driven to the Ultra Low power Mode (ULPM) when DSI-CLK lanes enter the LP-00 State Code. The only possibility is from the Low Power Mode (LPM, LP-11 State Code) => LP-10 => LP-00 (ULPM).

The sequence is illustrated below.

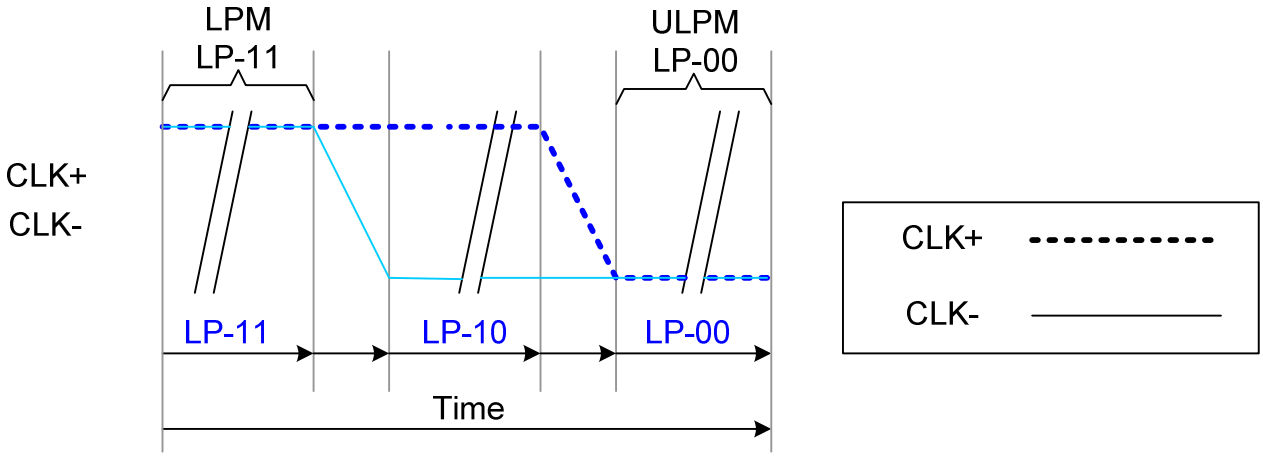


Figure 30: From LPM to ULPM

The mode change is illustrated below.

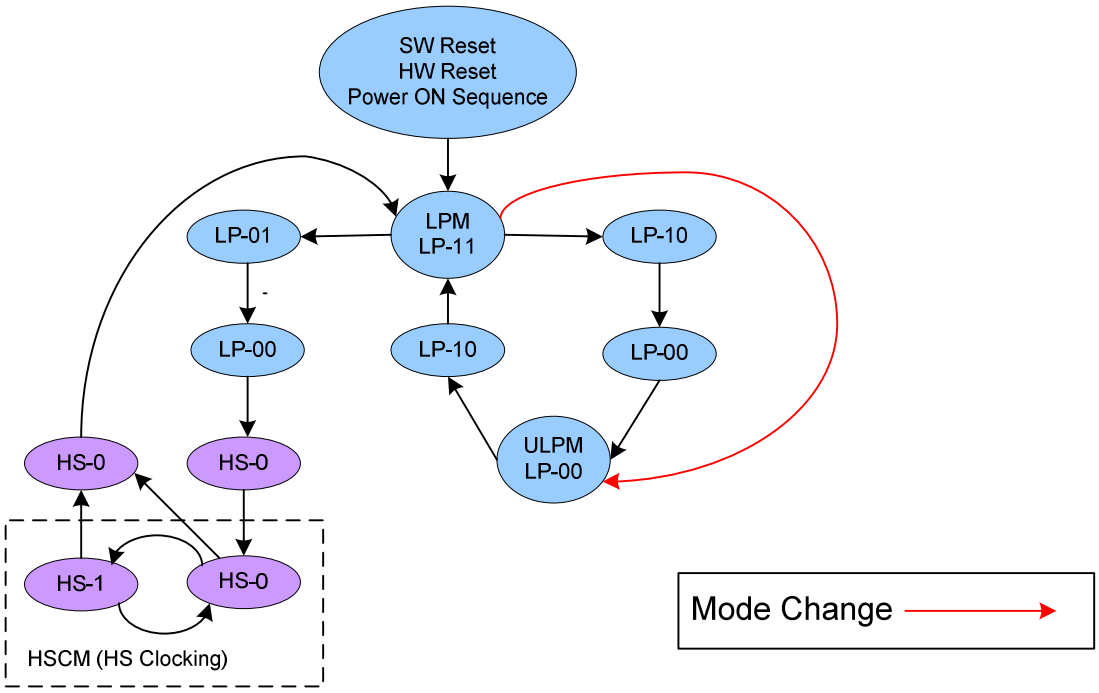


Figure 31: Mode Change from LPM to ULPM

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3.12.7. High-Speed Clock Mode (HSCM)

DSI-CLK+/- lanes can be driven to the High Speed Clock Mode (HSCM) when DSI-CLK lanes start to work between HS-0 and HS-1 State Codes.

The only entering possibility is from the Low Power Mode (LPM, LP-11 State Code) =>LP-01 => LP-00 => HS-0 => HS-0/1 (HSCM). The sequence is illustrated below.

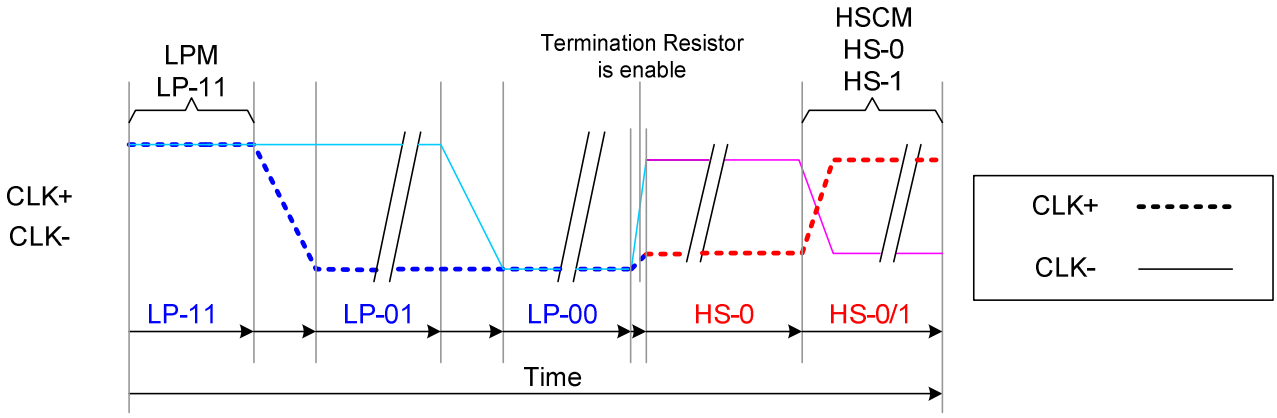


Figure 32: From LPM to HSCM

The mode change is illustrated below.

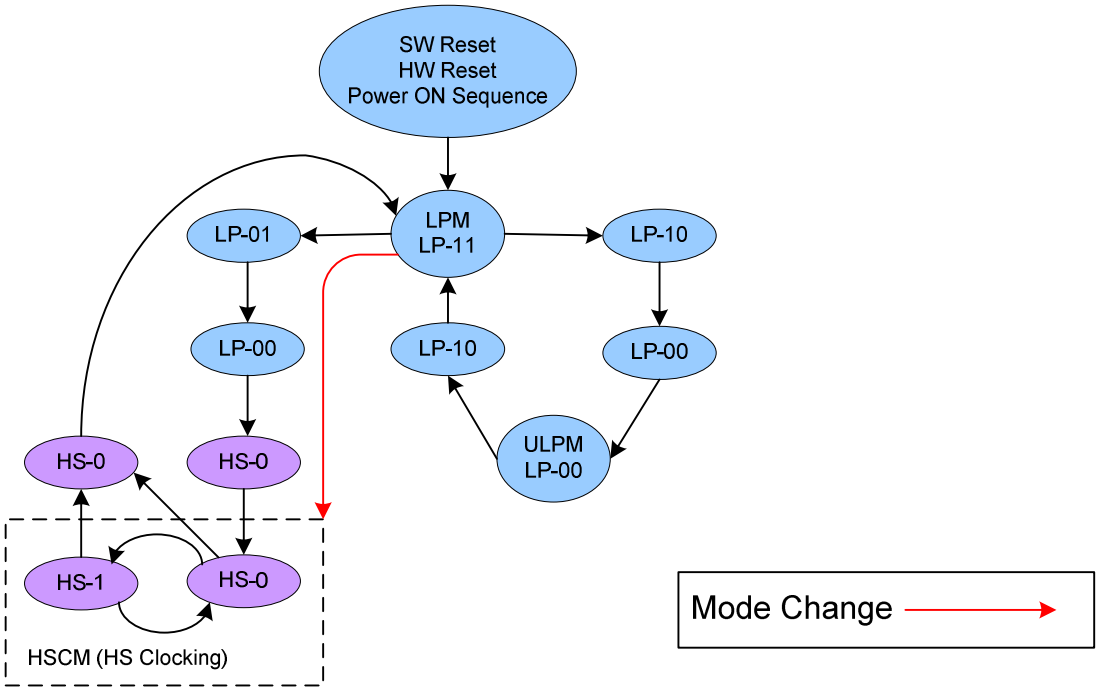


Figure 33: Mode Change from LPM to HSCM

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The high speed clock (DSI-CLK+/-) starts before high speed data is sent via DSI-D1+/- or DSI-D0+/- lanes. The high speed clock continues clocking after the high speed data sending is stopped.

The burst of the high speed clock consists of:

- Even number of transitions
- Start state is HS-0
- End state is HS-0

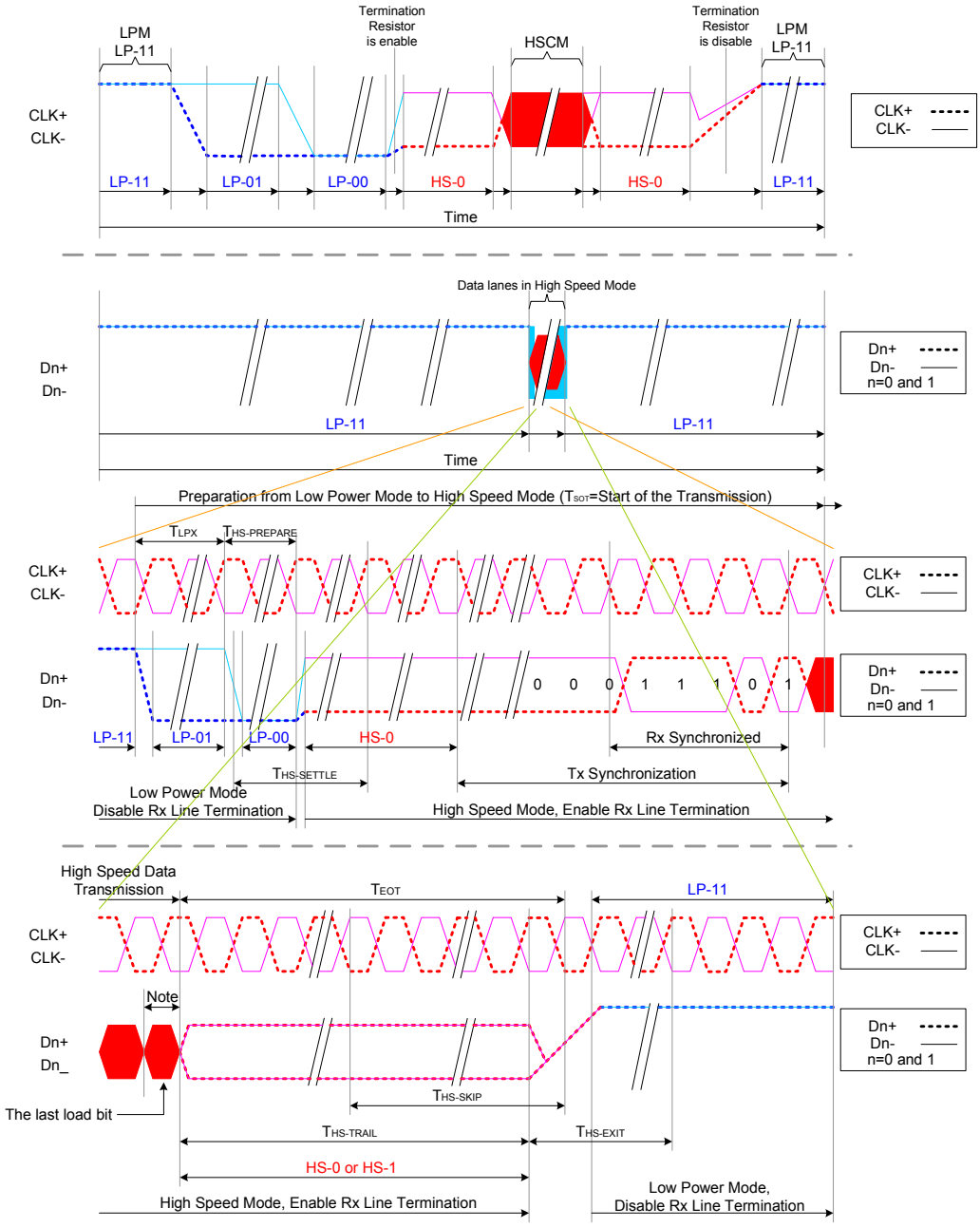


Figure 34: High Speed Clock Burst[Note]

Note 1. If the last load bit is HS-0, the transmitter changes from HS-0 to HS-1.
2. If the last load bit is HS-1, the transmitter changes from HS-1 to HS-0.

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3.12.8. DSI-D1 and DSI-D0 Data Lanes

3.12.9. General

DSI-D1+/- and DSI-D0+/- Data Lanes can be driven to different modes:

- Escape Mode (only DSI-D0+/- data lanes are used)
- High-Speed Data Transmission (DSI-D1+/- and DSI-D0+/- data lanes are used)
- Bus Turnaround Request (only DSI-D0+/- data lanes are used)

These modes and their entering codes are defined in the following table.

Table 12: Entering and Leaving Sequences[Note]

Mode	Entering Mode Sequence	Leaving Mode Sequence
Escape Mode	LP-11 → LP-10 → LP-00 → LP-01 → LP-00	LP-00 → LP-10 → LP-11 (Mark-1)
High-Speed Data Transmission	LP-11 → LP-01 → LP-00 → HS-0	(HS-0 or HS-1) → LP-11
Bus Turnaround Request	LP-11 → LP-10 → LP-00 → LP-10 → LP-00	Hi-Z

3.12.10. Escape Modes

DSI-D0+/- data lanes can be used in different Escape Modes when data lanes are in the Low Power (LP) mode.

These Escape Modes are used to:

- Send “Low-Power Data Transmission” (LPDT) from the MPU to the display module,
- Drive data lanes to “Ultra-Low Power State” (ULPS),
- Indicate “Remote Application Reset” (RAR), which can reset the display module,
- Indicate “Tearing Effect” (TEE), which is used to transmit a TE line event from the display module to the MPU,
- Indicate “Acknowledge” (ACK), which is used to transmit a non-error event from the display module to the MPU.

The basic sequence of the Escape Mode is as follows:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Escape Command (EC), which is coded when one of the data lanes changes from low-to-high-to-low, then this changed data lane presents the value of the current data bit (DSI-D0+ = 1, DSI-D0- = 0). For example, when DSI-D0- changes from low-to-high-to-low, the receiver latches a data bit, which value is the logical 0. The receiver uses this low-to-high-to-low transition for its internal clock.
- A load if necessary
- Exit Escape (Mark-1) LP-00 => LP-10 => LP-11
- End: LP-11

Note 1. DSI-D1+/- and DSI-D0+/- data lanes are used.
2. See more information in the chapter of Bus Turnaround.

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This basic construction is illustrated below:

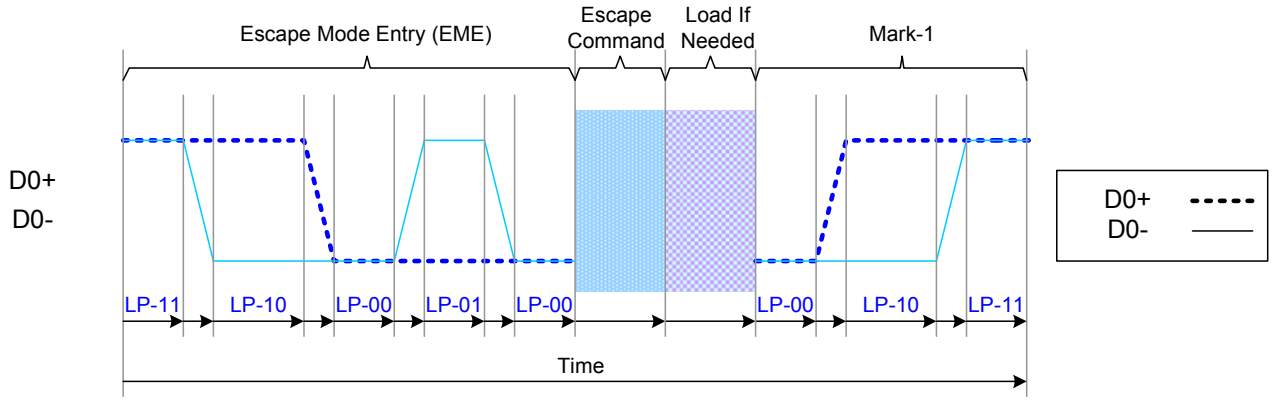


Figure 35: General Escape Mode Sequence

A total of eight Escape Commands (EC) are divided into two types: Mode and Trigger, as described in Table 13: Escape Commands.

An example of a Mode type Escape Command is Ultra-Low Power Mode, where the MPU instructs the display module to enter its Ultra-Low Power Mode.

An example of a Trigger type Escape Command is Tearing Effect. In this case, the MPU has already instructed the display module to provide this trigger and is waiting for a response. The display module then sends a TE trigger (TEE) on the next V-sync event.

Escape commands are defined in the table below.

Table 13: Escape Commands[Note]

Escape command	Command Type Mode/Trigger	Entry command Pattern (First Bit ➔ Last Bit Transmitted)	Dn	D0
Low-Power Data Transmission	Mode	1110 0001 b	-	X
Ultra-Low Power Mode	Mode	0001 1110 b	X	X
Undefined-1, [Note 1]	Mode	1001 1111 b	-	-
Undefined-2, [Note 1]	Mode	1101 1110 b	-	-
Remote Application Reset	Trigger	0110 0010 b	-	X
Tearing Effect	Trigger	0101 1101 b	-	X
Acknowledge	Trigger	0010 0001 b	-	X
Uknown-5, [Note 1]	Trigger	1010 0000 b	-	-

Note 1. This Escape command is not implemented in the display module.
2. n = 1
3. x = Supported
4. - = Not Supported

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3.12.11. Low-Power Data Transmission (LPDT)

The MPU can send data to the display module in the Low-Power Data Transmission (LPDT) mode when data lanes enter the Escape Mode and Low-Power Data Transmission (LPDT) command is sent to the display module. The display module uses the same sequence as which it sends data to the MPU.

The Low Power Data Transmission (LPDT) uses the following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Low-Power Data Transmission (LPDT) command in the Escape Mode: 1110 0001 (first to last bit)
- Load (Data):
  - One or more bytes (8 bit)
  - Data lanes are in the pause mode when data lanes are stopped (both lanes are low) between bytes
- Mark-1: LP-00 => LP-10 => LP-11
- End: LP-11

This sequence is illustrated below for reference purpose:

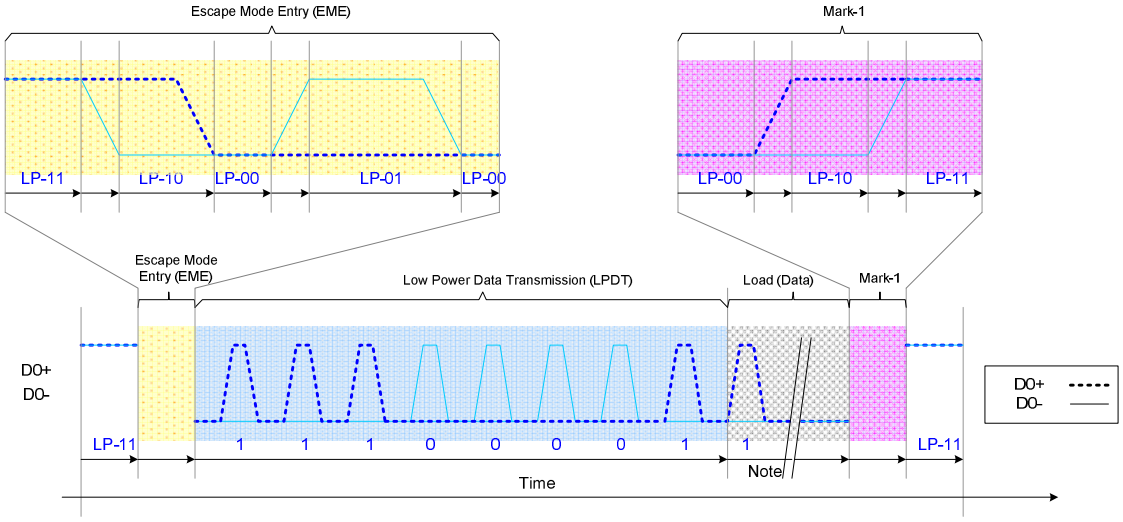


Figure 36: Low-Power Data Transmission (LPDT)[Note]

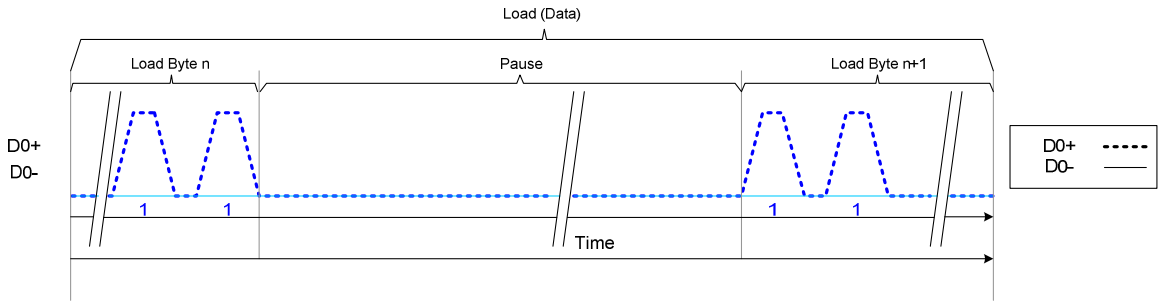


Figure 37: Pause (Example)

Note In this example, load (Data) presents that the first bit is the logical 1.

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3.12.12. Ultra-Low Power State (ULPS)

The MPU can force data lanes get into the Ultra-Low Power State (ULPS) mode when data lanes enter the Escape Mode.

The Ultra-Low Power State (ULPS) uses the following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Ultra-Low Power State (ULPS) command in the Escape Mode: 0001 1110 (first to last bit)
- Ultra-Low Power State (ULPS) when the MPU keeps data lanes low
- Mark-1: LP-00 => LP-10 => LP-11
- End: LP-11

This sequence is illustrated below for reference purpose:

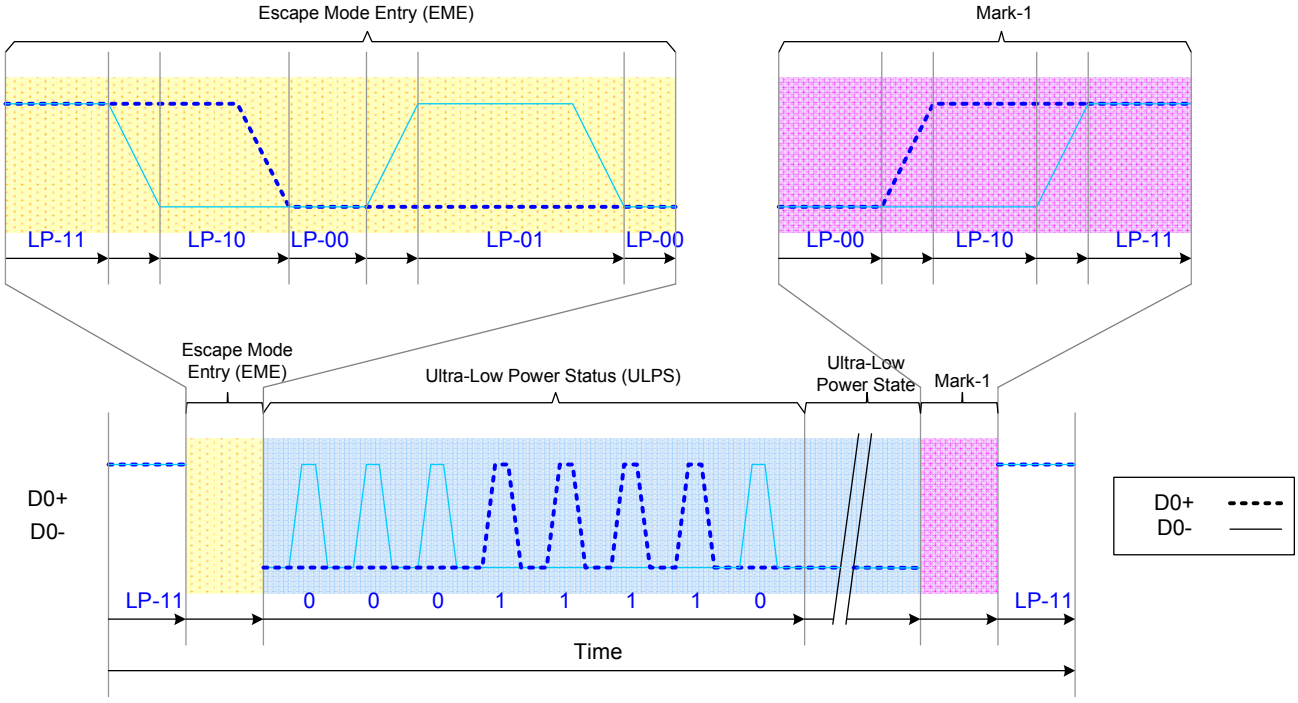


Figure 38: Ultra-Low Power State (ULPS)

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3.12.13. Remote Application Reset (RAR)

The MPU can inform the display module that it should be reset in Remote Application Reset (RAR) trigger when data lanes enter the Escape Mode.

The Remote Application Reset (RAR) uses the following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Remote Application Reset (RAR) command in the Escape Mode: 0110 0010 (first to last bit)
- Mark-1: LP-00 => LP-10 => LP-11
- End: LP-11

This sequence is illustrated below for reference purpose:

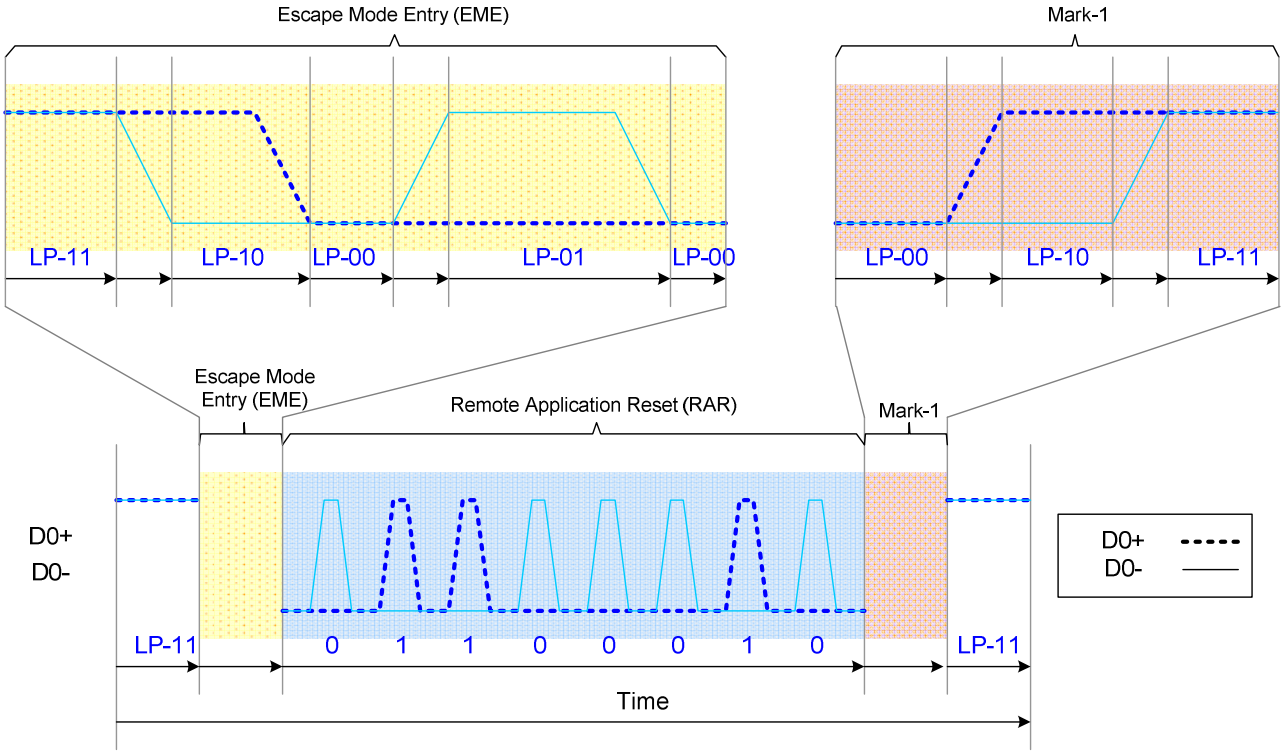


Figure 39: Remote Application Reset (RAR)

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3.12.14. Tearing Effect (TEE)

The display module can inform the MPU when a tearing effect event (New V-synch) has happened in the display module by the Tearing Effect (TEE).

The display module sends the Tearing Effect (TEE) with the following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Tearing Effect (TEE) trigger in the Escape Mode: 0101 1101 (first to last bit)
- Mark-1: LP-00 => LP-10 => LP-11
- End: LP-11

This sequence is illustrated below for reference purpose:

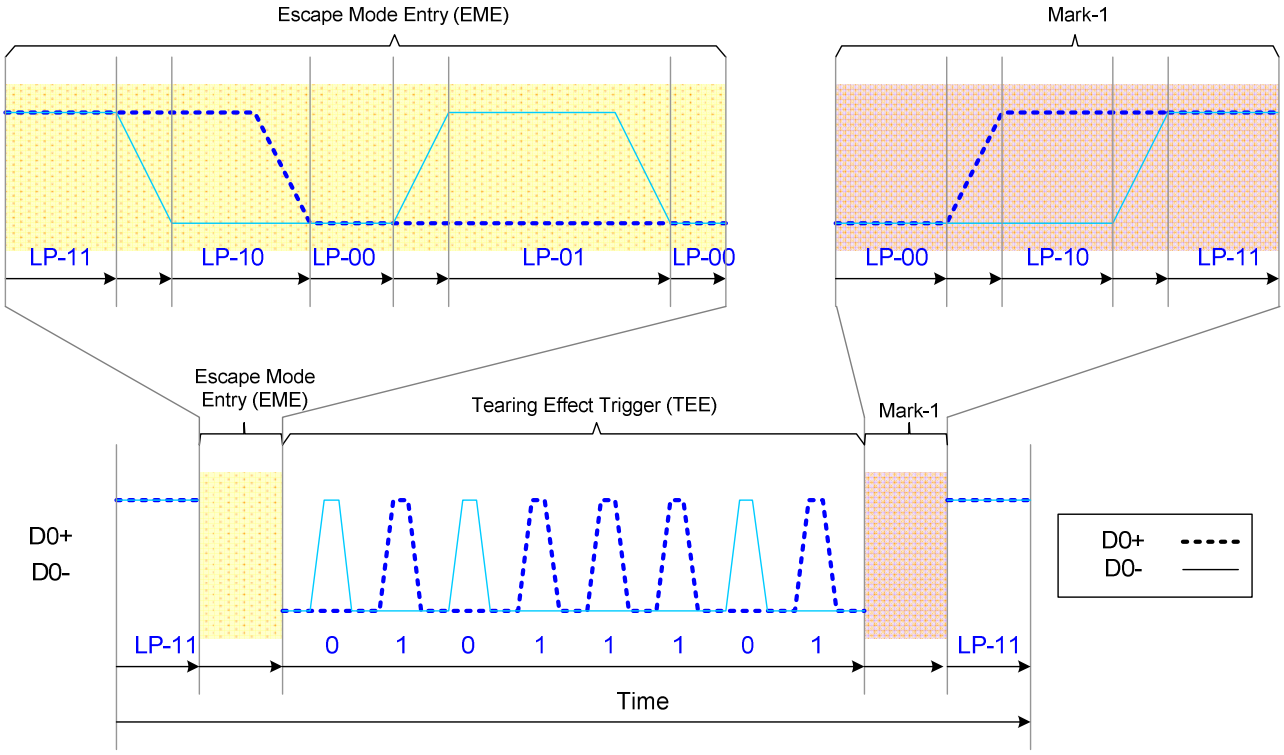


Figure 40: Tearing Effect (TEE)

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3.12.15. Acknowledge (ACK)

The display module can inform the MPU that no errors are found by the Acknowledge (ACK).

The display module sends the Acknowledge (ACK) with the following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 => LP-10 => LP-00 => LP-01 => LP-00
- Acknowledge (ACK) command in the Escape Mode: 0010 0001 (first to last bit)
- Mark-1: LP-00 => LP-10 => LP-11
- End: LP-11

This sequence is illustrated below for reference purpose:

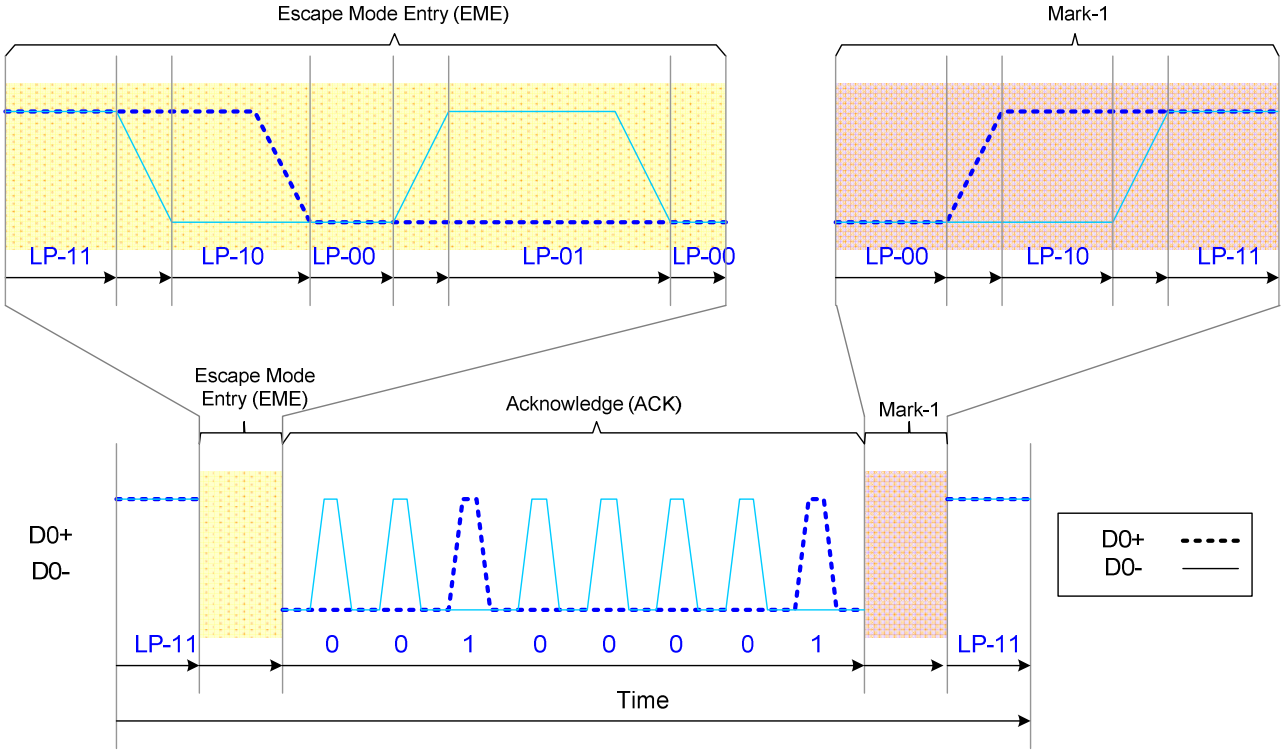


Figure 41: Acknowledge (ACK)

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3.12.16. High-Speed Data Transmission (HSDT)

3.12.17. Entering High-Speed Data Transmission (TSOT of HSDT)

The display module enters the High-Speed Data Transmission (HSDT) when Clock lanes DSI-CLK+/- have already entered the High-Speed Clock Mode (HSCM) through the MPU. See more information in the chapter “High-Speed Clock Mode (HSCM)”.

Data lanes DSI-D1+/- and DSI-D0+/- of the display module enter the High-Speed Data Transmission (TSOT of HSDT) with the following sequence:

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= MPU) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This sequence of entering High-Speed Data Transmission (TSOT of HSDT) is illustrated below:

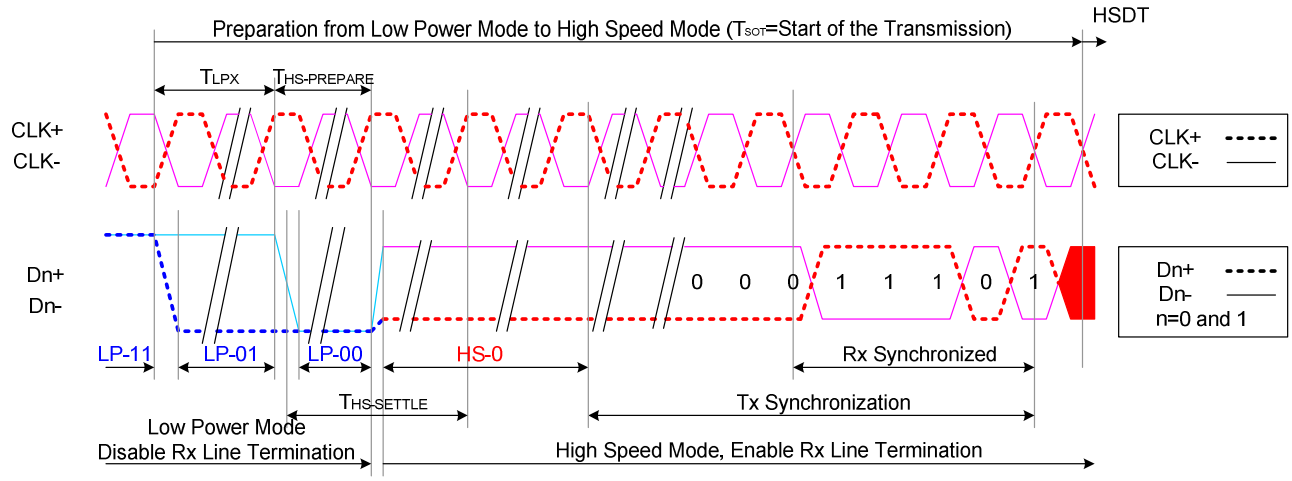


Figure 42: Entering High-Speed Data Transmission (TSOT of HSDT)

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3.12.18. Leaving High-Speed Data Transmission (TEOT of HSDT)

The display module leaves the High-Speed Data Transmission (TEOT of HSDT) when Clock lanes DSICLK+/- are in the High-Speed Clock Mode (HSCM) through the MPU. This HSCM is kept until data lanes DSI-D1+/- and DSI-D0+/- are in the LP-11 mode. See more information in the chapter “High-Speed Clock Mode (HSCM)”.

Data lanes DSI-D1+/- and DSI-D0+/- of the display module leave the High-Speed Data Transmission (TEOT of HSDT) with the following sequence:

- Start: High-Speed Data Transmission (HSDT)
- Stops High-Speed Data Transmission
  - o MPU changes to HS-1, if the last load bit is HS-0
  - o MPU changes to HS-0, if the last load bit is HS-1
- End: LP-11 (Rx: Lane Termination Disable)

The sequence of leaving High-Speed Data Transmission (TEOT of HSDT) is illustrated below:

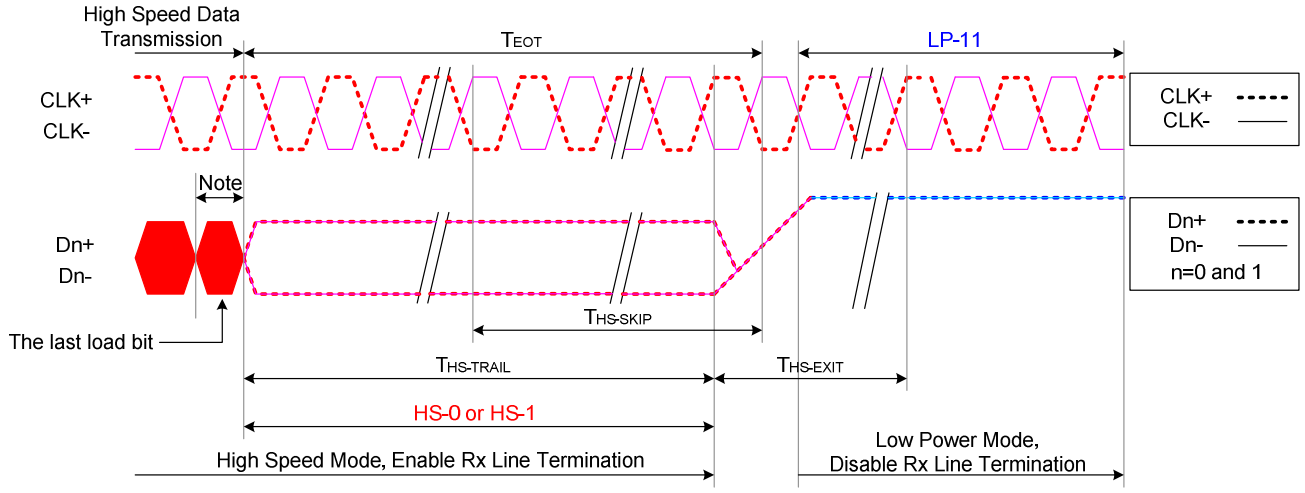


Figure 43: Leaving High-Speed Data Transmission (TEOT of HSDT)[Note]

Note 1. If the last load bit is HS-0, the transmitter changes from HS-0 to HS-1.
2. If the last load bit is HS-1, the transmitter changes from HS-1 to HS-0.

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3.12.19. Burst of the High-Speed Data Transmission (HSDT)

The burst of the “High-Speed Data Transmission” (HSDT) can consist of one data packet or several data packets. These data packets can be Long (LPa) or Short (SPa) packets. These packets are defined in the chapter “Short Packet (SPa) and Long Packet (LPa) Structures”.

These different burst of the High-Speed Data Transmission (HSDT) cases are illustrated below for reference purpose.

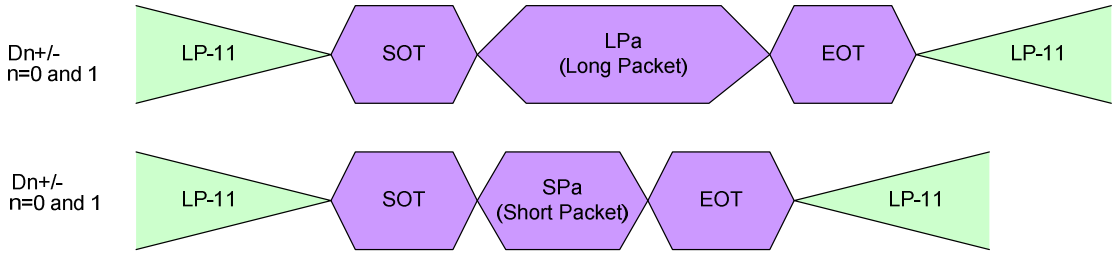


Figure 44: Single Packet in High-Speed Data Transmissions

The multiple packets in High-Speed Data Transmission are illustrated below for reference purposes:

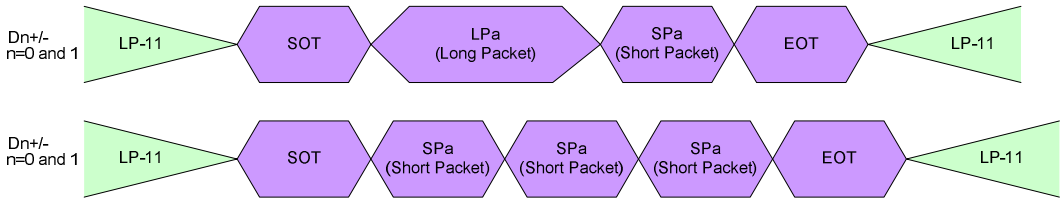


Figure 45: Multiple Packets in High-Speed Data Transmission – Examples

Table 14: Abbreviations

Abbreviation	Explanation
EOT	End of the Transmission
LPa	Long Packet
LP-11	Low Power Mode, Both of Data lanes are 1 (Stop Mode)
SPa	Short Packet
SOT	Start of the Transmission

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Byte orders of the sent packet in High-Speed Data Transmission (HSDT) are as follows.

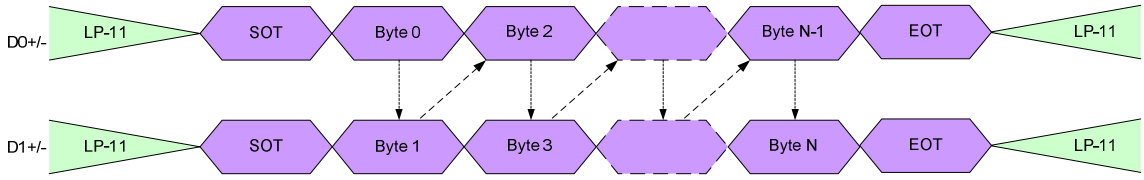


Figure 46: Single Packet in HSDT – Even Number of Bytes

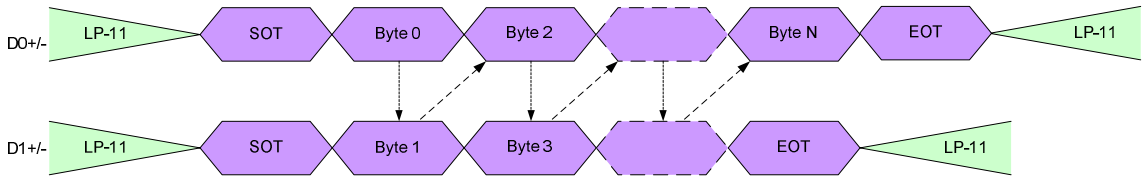


Figure 47: Single Packet in HSDT – Odd Number of Bytes

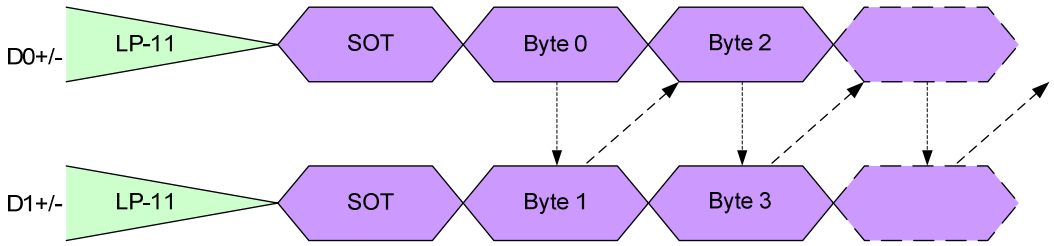


Figure 48: Start of Transmission (SoT) in HSDT for Multiple Packets

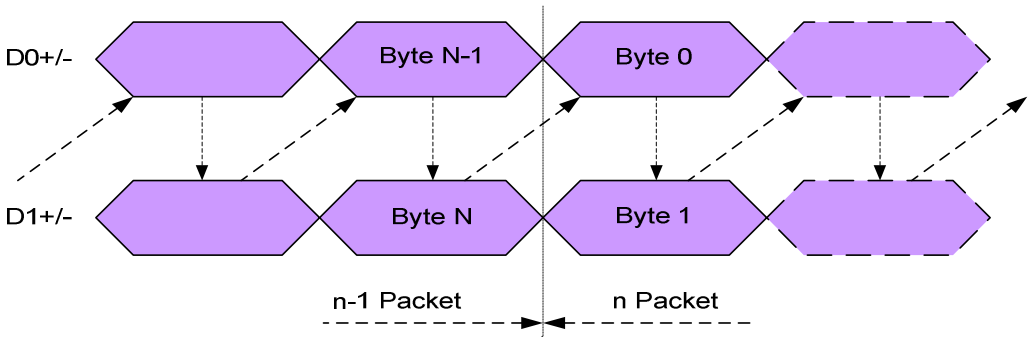


Figure 49: Continuous Multiple Packets in HSDT when Number of Bytes is Equal on Both Data Lanes at the End of the Packet

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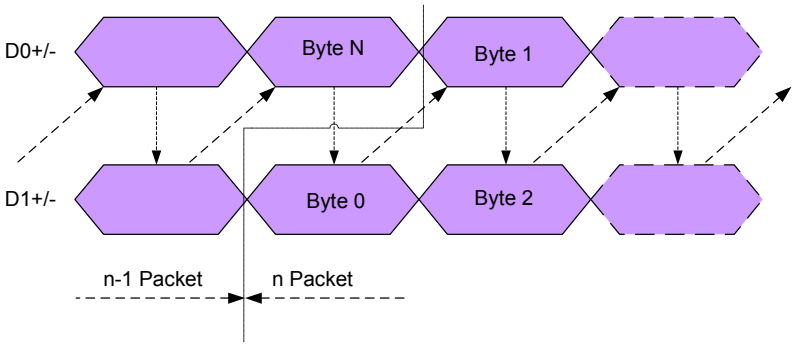


Figure 50: Continuous Multiple Packets in HSDT when Number of Bytes is not Equal on Both Data Lanes at the End of the Packet

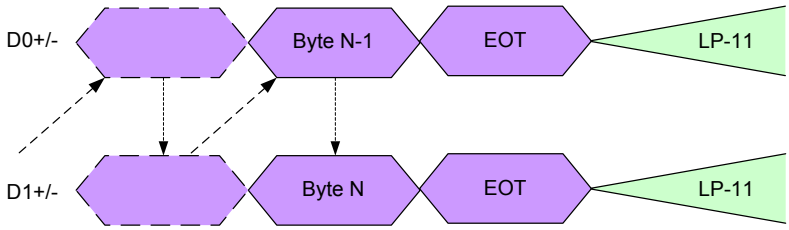


Figure 51: End of Transmission (EoT) in HSDT when Number of Bytes is Equal on Both Data Lanes at the End of the Packet

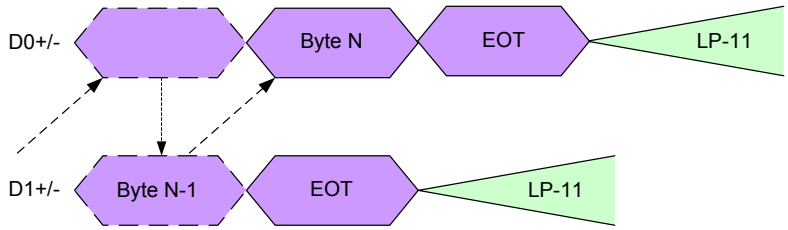


Figure 52: End of Transmission (EoT) in HSDT when Number of Bytes is not Equal on Both Data Lanes at the End of the Packet

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3.12.20. Bus Turnaround (BTA)

The MPU or the display module, which controls DSI-D0+/- Data Lanes, can start a bus turnaround procedure when it requires information from a receiver, which can be the MPU or the display module.

The MPU and the display module use the same sequence when this bus turnaround procedure is applied. The sequence when the MPU wants to perform the bus turnaround procedure to the display module is described below for reference purpose:

- Start (MPU): LP-11
- Turnaround Request (MPU): LP-11 => LP-10 => LP-00 => LP-10 => LP-00
- The MPU waits until the display module starts to control DSI-D0+/- data lanes and the MPU stops to control DSI-D0+/- data lanes (= High-Z)
- The display module changes to the stop mode: LP-00 => LP-10 => LP-11

The same bus turnaround procedure (from the MPU to the display module) is illustrated below:

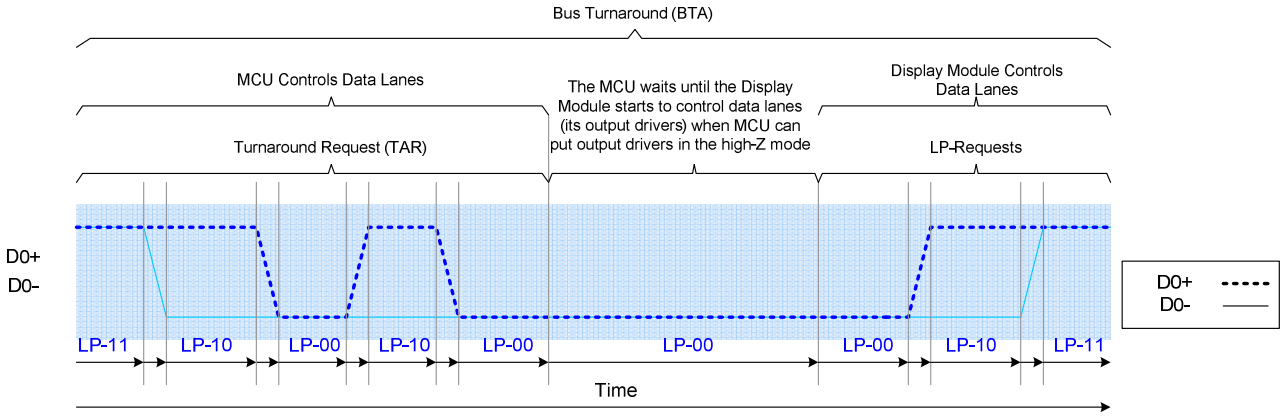


Figure 53: Bus Turnaround Procedure

The MPU and the display module can be switched in Figure 53: Bus Turnaround Procedure, if the Bus Turnaround (BTA) is from the display module to the MPU.

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3.12.21. Packet Level Communication

3.12.22. Short Packet (SPa) and Long Packet (LPa) Structures

Short Packets (SPa) and Long Packets (LPa) can always be used when data transmission is done in Low Power Data Transmission (LPDT) or High-Speed Data Transmission (HSDT) modes[Note].

The lengths of the packets are:

• Short Packet (SPa): 4 bytes

• Long Packet (LPa): 6 to 65,541 bytes

The type of the packet (SPa or LPa) can be recognized from their packet headers (PH).

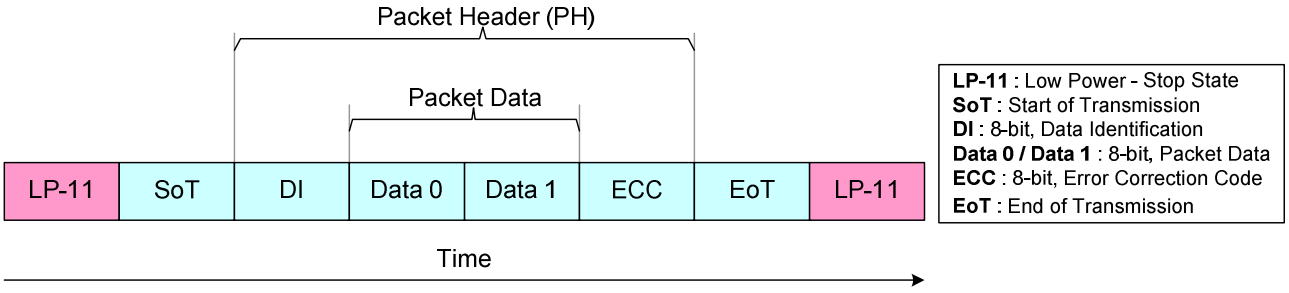


Figure 54: Short Packet (SPa) Structure

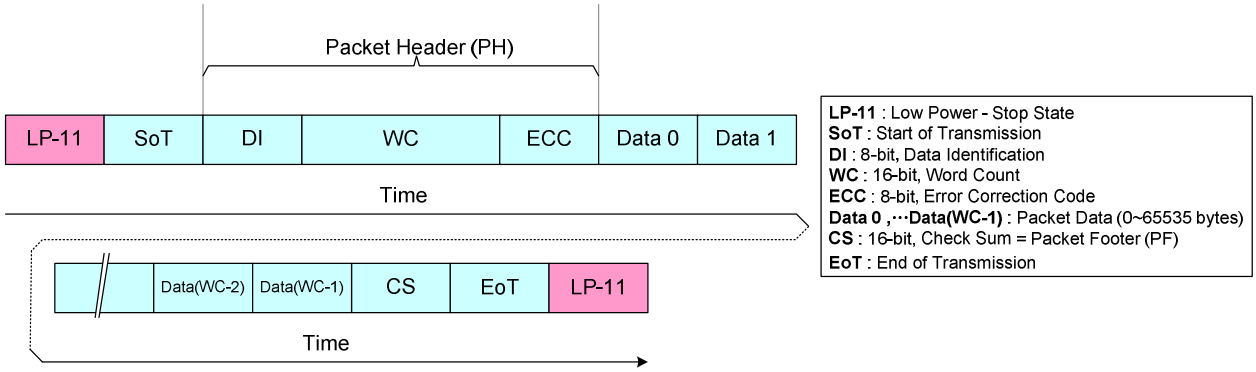


Figure 55: Long Packet (LPa) Structure

Note *Short Packet (SPa) and Long Packet (LPa) present a single packet sending (= including LP-11, SoT, and EoT for each packet sending).*
The other possibility is that SoT, EoT and LP-11 between packets are not necessary if packets are sent in multiple packet formats, for example:

- *LP-11 => SoT => SPa => LPa => SPa => SPa => EoT => LP-11*
- *LP-11 => SoT => SPa => SPa => SPa => EoT => LP-11*
- *LP-11 => SoT => LPa => LPa => LPa => EoT => LP-11*

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3.12.23. Bit Transmission Order of Bytes in Packets

The bit transmission order in a byte, which is used in packets, is that the Least Significant Bit (LSB) of a byte is sent first and the Most Significant Bit (MSB) of a byte is sent last.

This transmission order is illustrated below for reference purpose.

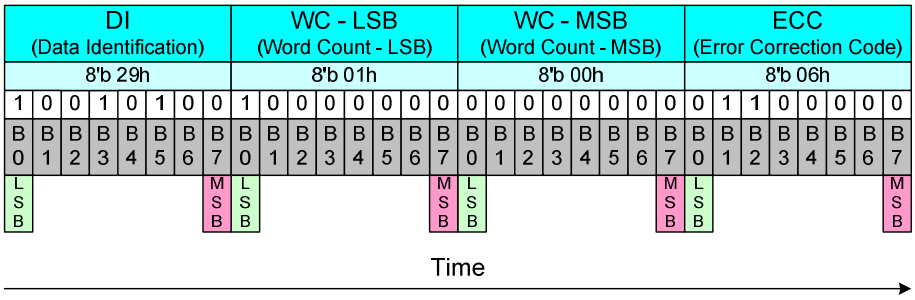


Figure 56: Bit Transmission Order of Bytes in Packets

3.12.24. Byte Transmission Order of Multiple-Byte Information in Packets

The byte transmission order of the multiple-byte information, what is used in packets, is that the Least Significant (LS) Byte of the information is sent first and the Most Significant (MS) Byte of the information is sent last. For example, Word Count (WC) consists of 2 bytes (16 bits); the LS byte is sent first and the MS byte is sent last.

This transmission order is illustrated below for reference purpose.

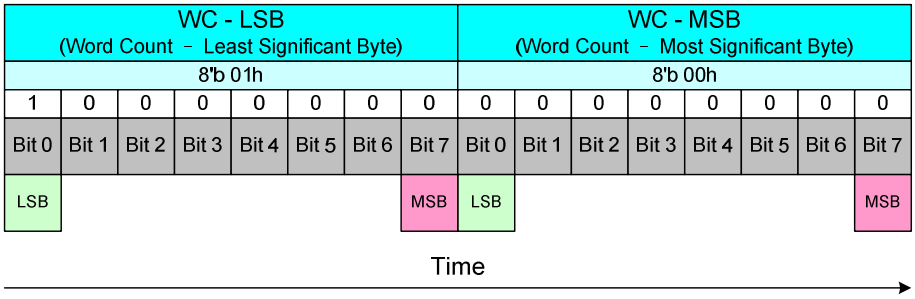


Figure 57: Byte Transmission Order of the Multiple-Byte Information in Packets

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3.12.25. Packet Header (PH)

The packet header always consists of 4 bytes. The content of these 4 bytes are different for Short Packets (SPa) and Long Packets (LPa).

Short Packet (SPa):

• 1st byte: Data Identification (DI) => identify that this is a Short Packet (SPa)

• 2nd and 3rd bytes: Packet Data (PD), Data 0 and 1

• 4th byte: Error Correction Code (ECC)

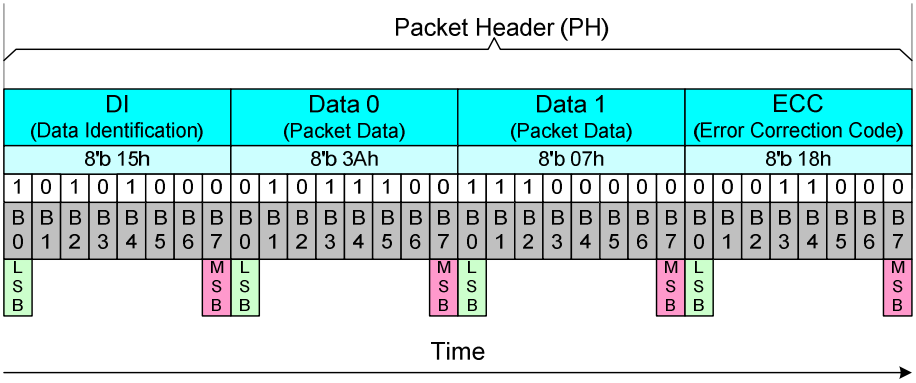


Figure 58: Packet Header (PH) of a Short Packet (SPa)

Long Packet (LPa):

• 1st byte: Data Identification (DI) => identify that this is a Long Packet (LPa)

• 2nd and 3rd bytes: Word Count (WC)

• 4th byte: Error Correction Code (ECC)

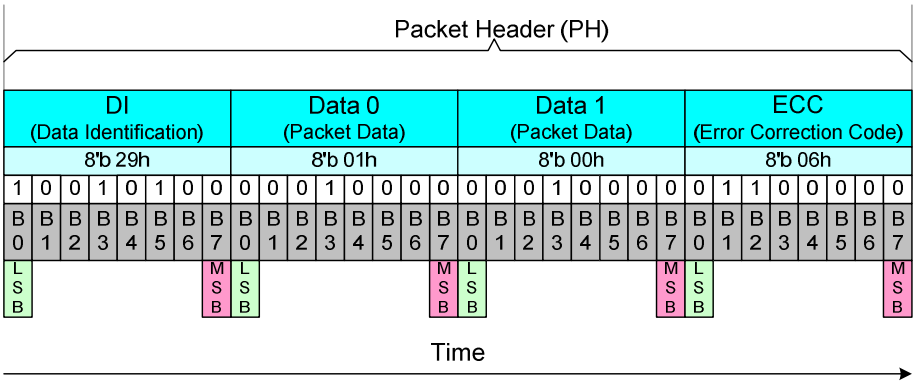


Figure 59: Packet Header (PH) on Long Packet (LPa)

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3.12.26. Data Identification (DI)

Data Identification (DI) is a part of the Packet Header (PH) and it consists of 2 parts:

- Virtual Channel (VC), 2 bits, DI [7...6]
- Data Type (DT), 6 bits, DI [5…0]

The Data Identification (DI) structure is illustrated in the figure below.

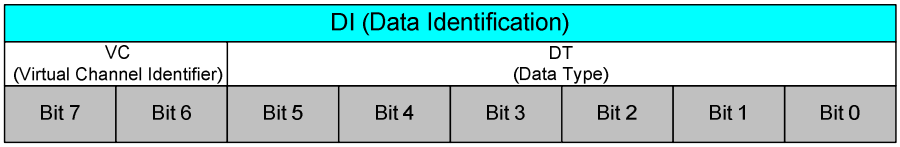


Figure 60: Data Identification (DI) Structure

Data Identification (DI) is illustrated in the Packet Header (PH) below for reference purpose.

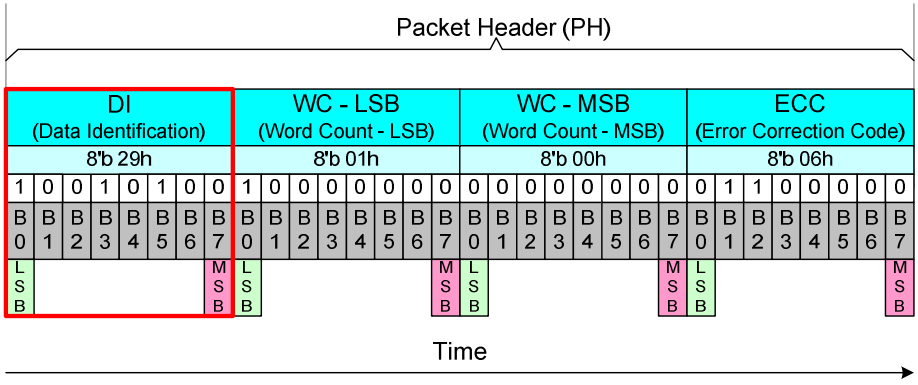


Figure 61: Data Identification (DI) in the Packet Header (PH)

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3.12.27. Virtual Channel (VC)

Virtual Channel (VC) is a part of the Data Identification (DI [7…6]) structure and it is used to indicate where a packet is to be sent from the MPU.

Bits of the Virtual Channel (VC) are illustrated below for reference purpose.

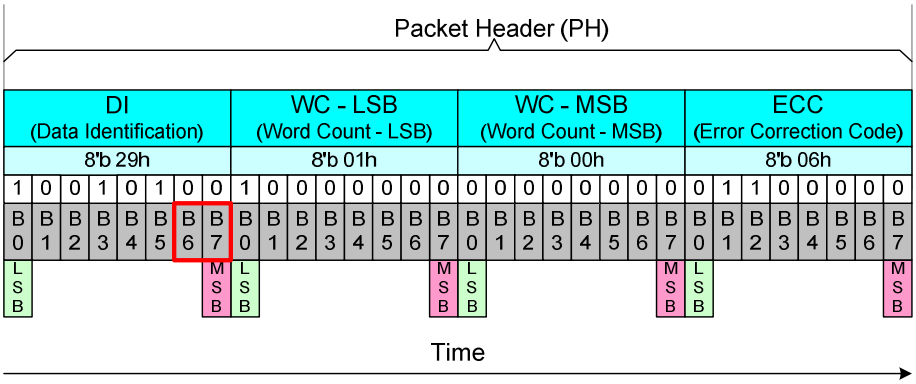


Figure 62: Virtual Channel (VC) on the Packet Header (PH)

Virtual Channel (VC) can assign 4 different channels to 4 different display modules. Devices use the same virtual channel as which the MPU uses to send packets to them, for example,

- The MPU uses the virtual channel 0 when it sends packets to ILI9806G
- ILI9806G also uses the virtual channel 0 when it sends packets to the MPU

This functionality is illustrated below.

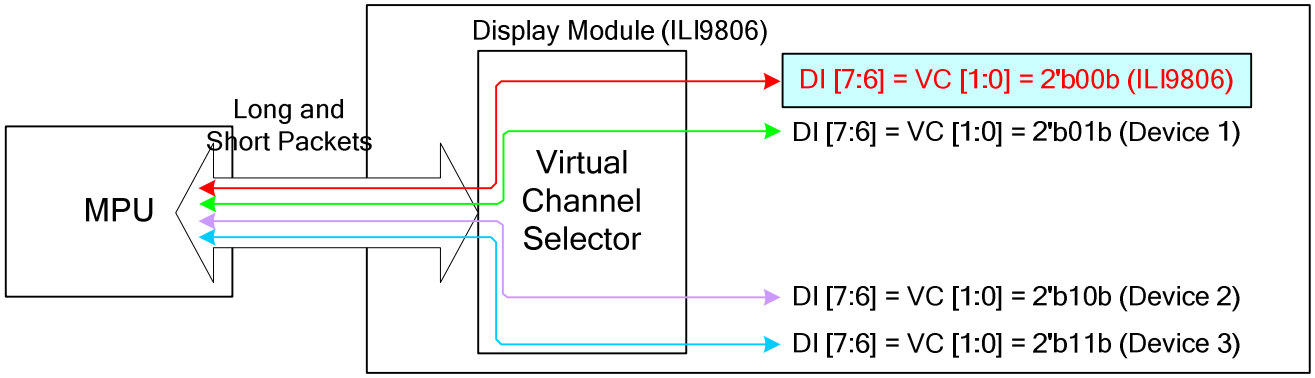


Figure 63: Virtual Channel (VC) Configuration

Virtual Channel (VC) is always 0 (DI [7..6] = VC [1..0] = 00_b) when the MPU sends the End of Transmission Packet to the display module. See the chapter “End of Transmission Packet (EoTP)”.

This display module does not support the virtual channel selector for other devices (1 to 3) when the only possible virtual channel (VC [1..0]) is 00b for this display module.

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3.12.28. Data Type (DT)

Data Type (DT) is a part of the Data Identification (DI [5…0]) structure and it is used to define the type of the used data in a packet.

Bits of the Data Type (DT) are illustrated below for reference purpose.

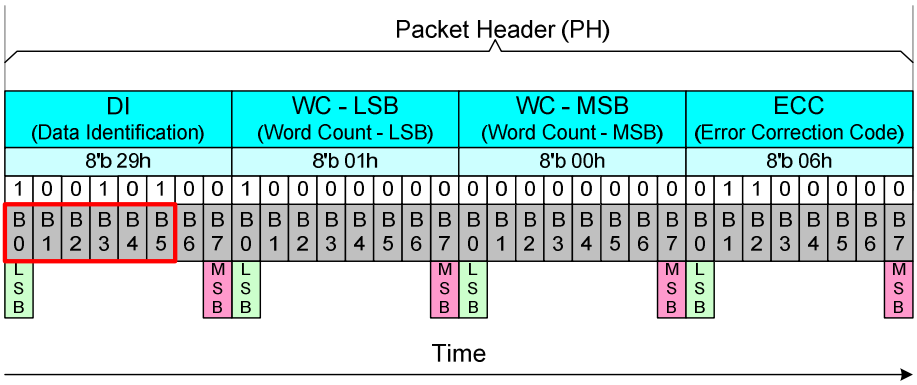


Figure 64: Data Type (DT) in the Packet Header (PH)

This Data Type (DT) also defines the used packet is Short Packet (SPa) or Long Packet (LPa). Data Types (DT) are different from the MPU to the display module (or other devices) and vice versa.

These Data Types (DT) are defined in the tables below.

Table 15: Data Types (DT) from the MPU to the Display Module (ILI9806G)

From the MPU to the Display Module (ILI9806G)									
Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Hex	Description	Short/Long Packet	Abbreviation
0	0	1	0	0	0	08	End of Transmission Packet, Note 1	SPa (Short Packet)	EoTP
0	0	0	1	0	1	05	DCS Write, No Parameter	SPa (Short Packet)	DCSWN-S
0	1	0	1	0	1	15	DCS Write, 1 Parameter	SPa (Short Packet)	DCSW1-S
0	0	0	1	1	0	06	DCS Read, No Parameter	SPa (Short Packet)	DCSRN-S
1	1	0	1	1	1	37	Set Maximum Return Packet Size	SPa (Short Packet)	SMRPS-S
0	0	1	0	0	1	09	Null Packet, No Data, Note 2	LPa (Long Packet)	NP-L
1	1	1	0	0	1	39	DCS Write Long	LPa (Long Packet)	DCSW-L

Note 1 *This can be used when the MPU wants to make sure that it is the end of the transmission in the High Speed Data Transferring (HSDT) mode.*

Note 2 *This can be used when data lanes are to be kept in the High Speed Data Transferring (HSDT) Mode.*

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Table 16: Data Types (DT) from the Display Module (ILI9806G) to the MPU

From the Display Module (ILI9806G) to the MPU									
Bit5	Bit4	Bit3	Bit2	Bit1	Bit0	Hex	Description	Short/Long Packet	Abbreviation
0	0	0	0	1	0	02	Acknowledge with Error Report	SPa (Short Packet)	AwER
0	1	1	1	0	0	1C	DCS Read Long Response	LPa (Long Packet)	DCSRR-L
1	0	0	0	0	1	21	DCS Read Short Response, 1 byte returned	SPa (Short Packet)	DCSRR1-S
1	0	0	0	1	0	22	DCS Read Short Response, 2 byte returned	SPa (Short Packet)	DCSRR2-S

The receiver will ignore other Data Types (DT) if they are not defined in the “Table 15: Data Type (DT) from the MPU to the Display Module (or Other Devices)” or “Table 16: Data Type (DT) from the Display Module (or Other Devices) to the MPU”.

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3.12.29. Packet Data (PD) of a Short Packet (SPa)

Packet Data (PD) of a Short Packet (SPa) is defined after Data Type (DT) of the Data Identification (DI), which indicates that a Short Packet (SPa) is to be sent.

Packet Data (PD) of a Short Packet (SPa) consists of 2 data bytes: Data 0 and Data 1.

The sending order of Packet Data (PD) is that Data 0 is sent first and the Data 1 is sent last.

Bits of Data 1 are set to 0 if the information length is 1 byte.

Packet Data (PD) of a Short Packet (SPa), when the length of the information is 1 or 2 bytes and Virtual Channel (VC) is 0, are illustrated below for reference purpose.

Packet Data (PD) information:

• Data 0: 35hex (Display Command Set (DCS) with 1 Parameter => DI (Data Type (DT)) = 15hex)

• Data 1: 01hex (DCS's parameter)

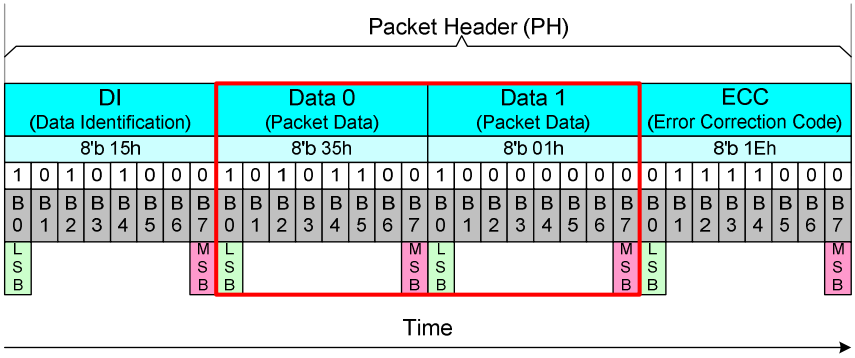


Figure 65: Packet Data (PD) of a Short Packet (SPa), 2 Bytes Information

Packet Data (PD) information:

• Data 0: 10hex (DCS without parameter => DI (Data Type (DT)) = 05hex)

• Data 1: 00hex (Null)

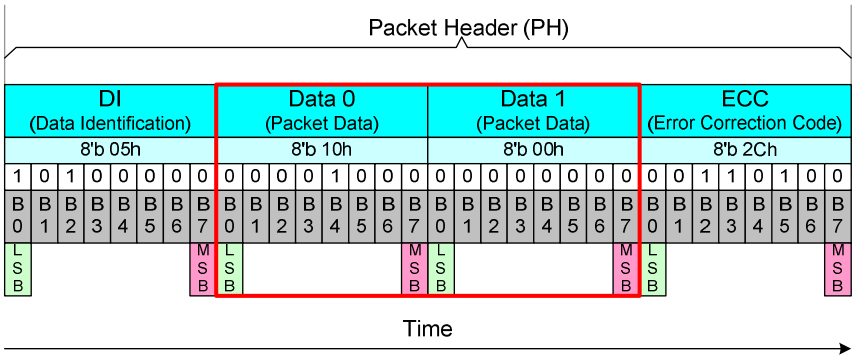


Figure 66: Packet Data (PD) of a Short Packet (SPa), 1 Byte Information

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3.12.30. Word Count (WC) of a Long Packet (LPa)

Word Count (WC) of a Long Packet (LPa) is defined after Data Type (DT) of the Data Identification (DI), which indicates that Long Packet (LPa) is to be sent.

Word Count (WC) indicates the amount of data bytes of the Packet Data (PD) which are to be sent after the Packet Header (PH). The location of Word Count (WC) in a Long Packet is the same as which of Packet Data (PD) in a Short Packet.

Word Count (WC) of a Long Packet (LPa) consists of 2 bytes.

The sending order of these 2 bytes of the Word Count (WC) is that the Least Significant (LS) Byte is sent first and the Most Significant (MS) Byte is sent last.

Word Count (WC) of a Long Packet (LPa) is illustrated below for reference purposes.

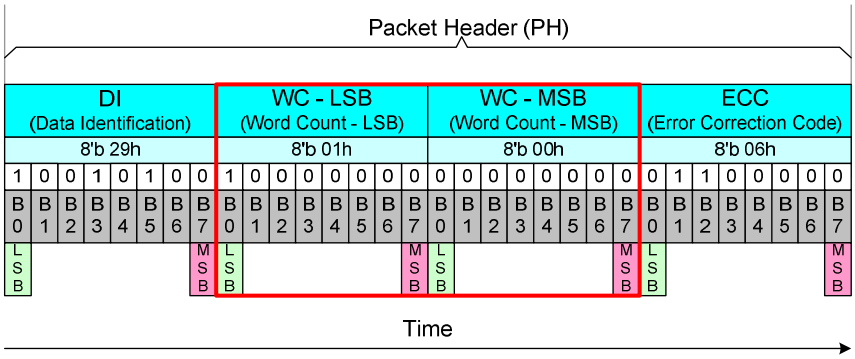


Figure 67: Word Count (WC) of a Long Packet (LPa)

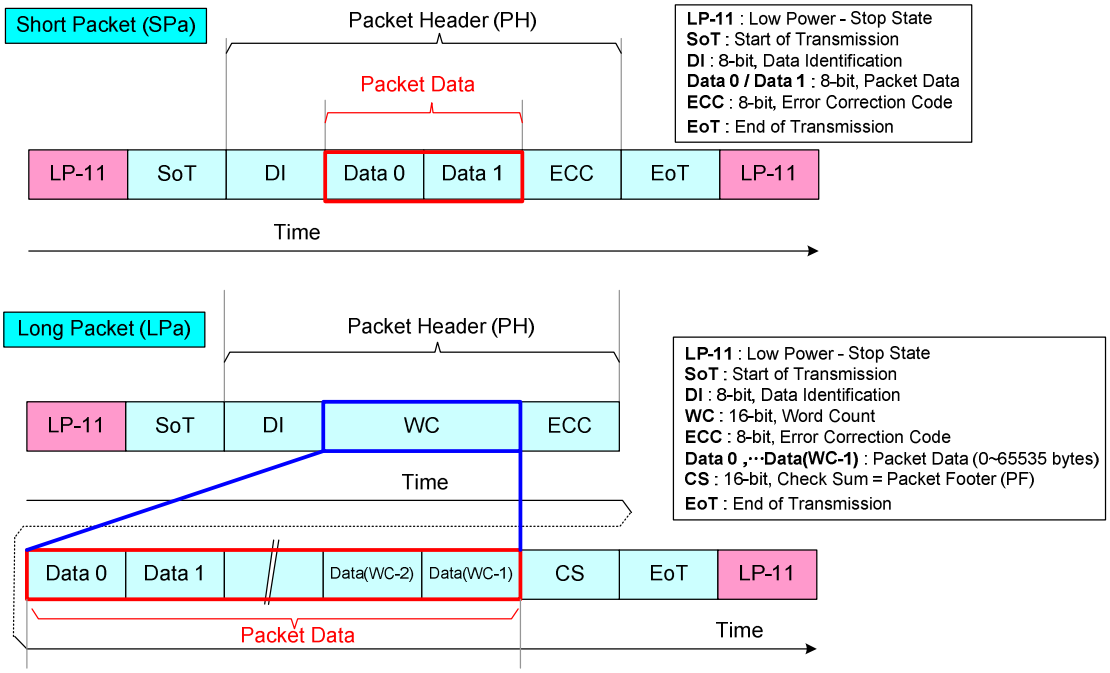


Figure 68: Packet Data in Short and Long Packets

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3.12.31. Error Correction Code (ECC)

Error Correction Code (ECC) is a part of the Packet Header (PH), and its purpose is to identify an error or errors.

The ECC protects the following fields:

- Short Packet (SPa): Data Identification (DI) byte (8 bits: D [0…7]), Packet Data (PD) bytes (16 bits: D [8…23]) and ECC (8 bits: P [0…7])
- Long Packet (LPa): Data Identification (DI) byte (8 bits: D [0…7]), Word Count (WC) bytes (16 bits: D [8…23]) and ECC (8 bits: P [0…7])

D [23…0] and P [7…0] are illustrated below for reference purpose.

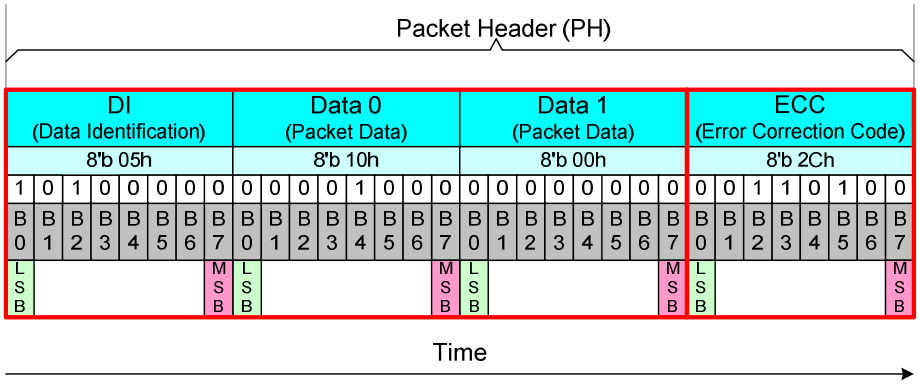


Figure 69: D [23…0] and P [7…0] of a Short Packet (SPa)

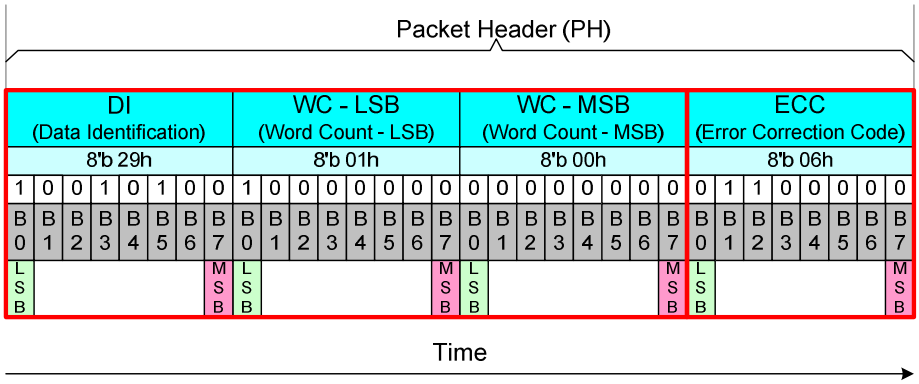


Figure 70: D [23…0] and P [7…0] of a Long Packet (LPa)

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Error Correction Code (ECC) can recognize one error or several errors, but only a one-bit error will be corrected. Bits (P [7…0]) of the Error Correction Code (ECC) are defined, where the symbol '^' presents the XOR function (Pn is 1 if there is odd number of 1, and Pn is 0 if there is even number of 1), as below.

• P7 = 0

• P6 = 0

• P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23

• P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23

• P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23

• P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22

• P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23

• P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

P7 and P6 are set to 0 because the Error Correction Code (ECC) is based on 64 bit value (D [63…0]), but this implementation is based on 24 bit value (D [23…0]). Therefore, only 6 bits (P [5…0]) for Error Correction Code (ECC) are needed.

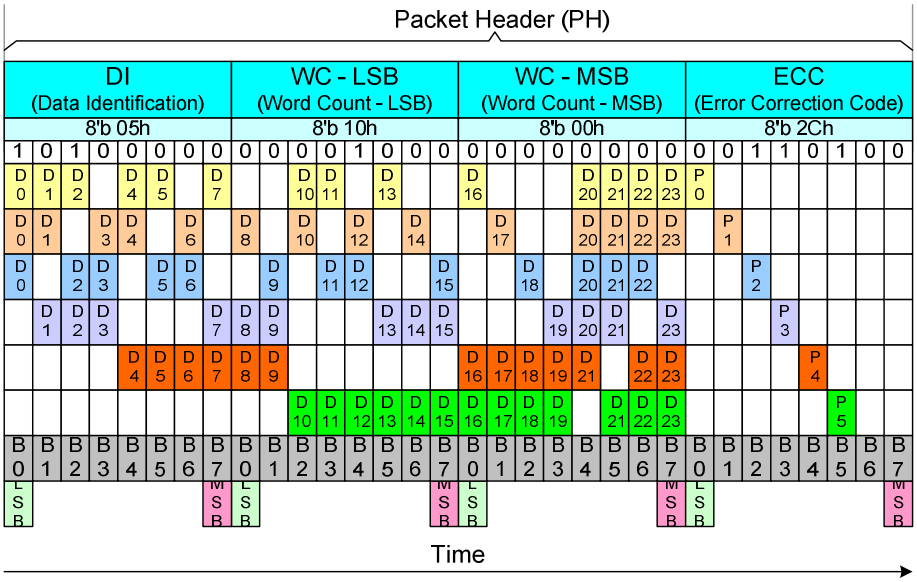


Figure 71: XOR Functionality in a Short Packet (SPa)

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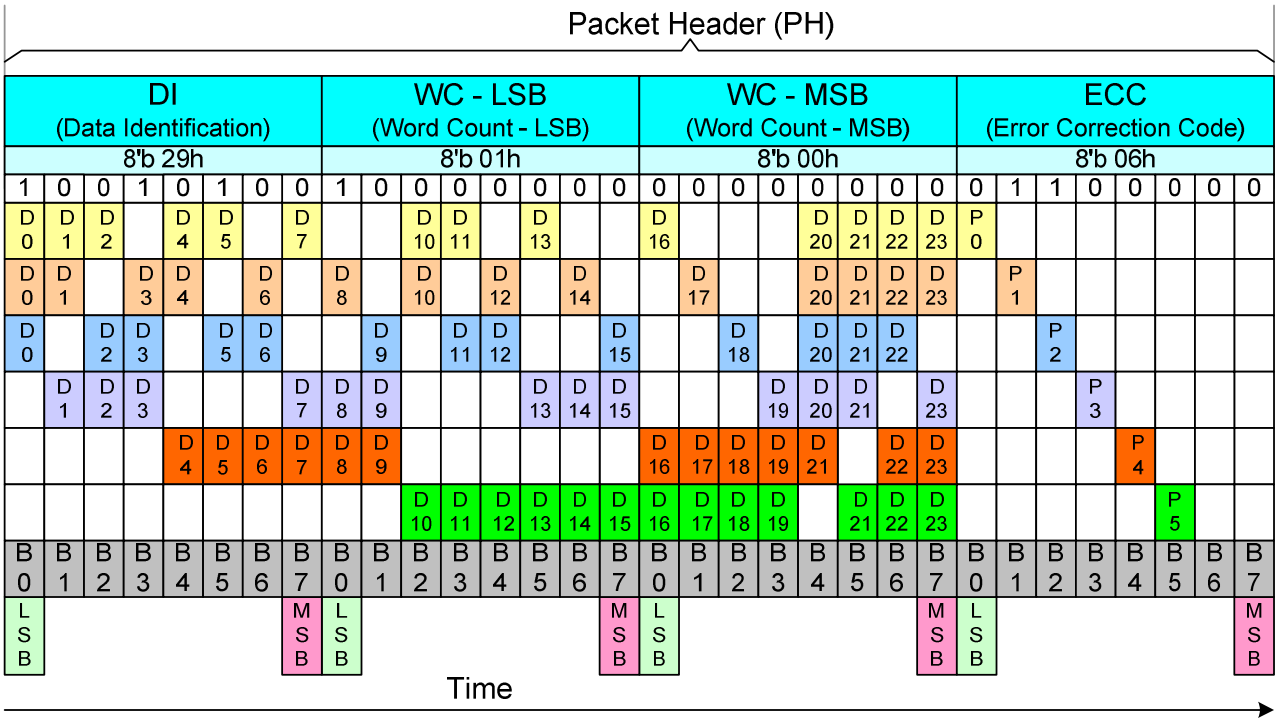


Figure 72: XOR Functionality in a Long Packet (LPa)

The transmitter (the MPU or the Display Module) sends data bits D [23…0] and Error Correction Code (ECC) P [7…0]. The receiver (the Display module or the MPU) calculates an Internal Error Correction Code (IECC) and compares the received Error Correction Code (ECC) and the Internal Error Correction Code (IECC). This comparison is done when each power bit of ECC and IECC have performed the XOR function. The result of this function is PO [7…0]. This functionality, where the transmitter is the MPU and the receiver is the display module, is illustrated below for reference purposes.

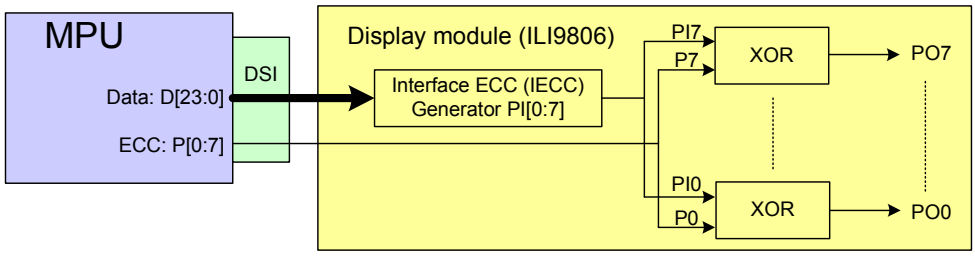


Figure 73: Internal Error Correction Code (IECC) on the Display Module (the Receiver)

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The sent data bits (D [23…0]) and ECC (P [7…0]) are received correctly, if the value of the PO [7…0]) is 00h.

The sent data bits (D [23…0]) and ECC (P [7…0]) are not received correctly, if the value of the PO [7…0]) is not 00h.

ECC P[7…0]	1	1	0	0	0	0	0	0	03h
IECC PI[7…0]	1	1	0	0	0	0	0	0	03h
XOR(ECC, IECC) => PO[7…0]	0	0	0	0	0	0	0	0	= 00h => No Error
	L							M	
	S							S	
	B							B	

Figure 74: Internal XOR Calculation between ECC and IECC Values – No Error

ECC P[7…0]	1	1	0	0	0	0	0	0	03h
IECC PI[7…0]	1	1	1	1	0	0	0	0	0Fh
XOR(ECC, IECC) => PO[7…0]	0	0	1	1	0	0	0	0	= 0Ch => Error
	L							M	
	S							S	
	B							B	

Figure 75: Internal XOR Calculation between ECC and IECC Values - Error

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The received Error Correction Code (ECC) can be 00h when the Error Correction Code (ECC) functionality is not used for data values D [23…0] on the transmitter side.

The amount of the errors (one or more) can be defined when the value of the PO [7…0] is compared to the values in the following table.

Table 17: One Bit Error Value of the Error Correction Code (ECC)

Data Bit	PO7	PO6	PO5	PO4	PO3	PO2	PO1	PO0	Hex
D [0]	0	0	0	0	0	1	1	1	07h
D [1]	0	0	0	0	1	0	1	1	0Bh
D [2]	0	0	0	0	1	1	0	1	0Dh
D [3]	0	0	0	0	1	1	1	0	0Eh
D [4]	0	0	0	1	0	0	1	1	13h
D [5]	0	0	0	1	0	1	0	1	15h
D [6]	0	0	0	1	0	1	1	0	16h
D [7]	0	0	0	1	1	0	0	1	19h
D [8]	0	0	0	1	1	0	1	0	1Ah
D [9]	0	0	0	1	1	1	0	0	1Ch
D [10]	0	0	1	0	0	0	1	1	23h
D [11]	0	0	1	0	0	1	0	1	25h
D [12]	0	0	1	0	0	1	1	0	26h
D [13]	0	0	1	0	1	0	0	1	29h
D [14]	0	0	1	0	1	0	1	0	2Ah
D [15]	0	0	1	0	1	1	0	0	2Ch
D [16]	0	0	1	1	0	0	0	1	31h
D [17]	0	0	1	1	0	0	1	0	32h
D [18]	0	0	1	1	0	1	0	0	34h
D [19]	0	0	1	1	1	0	0	0	38h
D [20]	0	0	0	1	1	1	1	1	1Fh
D [21]	0	0	1	0	1	1	1	1	2Fh
D [22]	0	0	1	1	0	1	1	1	37h
D [23]	0	0	1	1	1	0	1	1	3Bh

One error is detected if the value of the PO [7…0] is found in Table 17: One Bit Error Value of the Error Correction Code (ECC), and the receiver can correct this one-bit error because this found value also defines the location of the corrupt bit, for example,

• PO [7…0] = 0Eh

• The bit of the data (D [23…0]), which is not correct, is D [3]

More than one error is detected if the value of the PO [7…0] is not in Table 17: One Bit Error Value of the Error Correction Code (ECC), for example, PO [7…0] = 0Ch.

3.12.32. Packet Data (PD) of a Long Packet (LPa)

Packet Data (PD) of a Long Packet (LPa) is defined after the Packet Header (PH) of the Long Packet (LPa).

The amount of data bytes is defined in the chapter “Word Count (WC) of a Long Packet (LPa)”.

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3.12.33. Packet Footer (PF) of a Long Packet (LPa)

The Packet Footer (PF) of a Long Packet (LPa) is defined after the Packet Data (PD) of a Long Packet (LPa). The Packet Footer (PF) is the checksum value which is calculated from the Packet Data of the Long Packet (LPa).

The checksum uses a 16-bit Cyclic Redundancy Check (CRC) value which is generated by a polynomial X16+X12+X5+X0, as illustrated below.

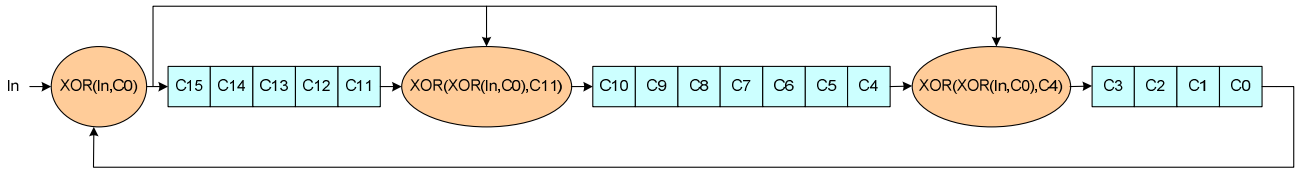
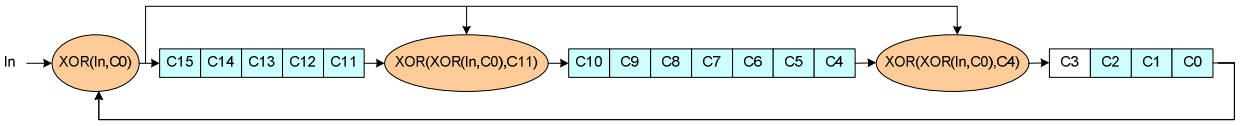


Figure 76: 16-bit Cyclic Redundancy Check (CRC) Calculation

The 16-bit Cyclic Redundancy Check (CRC) generator is initialized to FFFFh before calculations.

The Most Significant Bit (MSB) of the data byte of the Packet Data (PD) is the first bit to be inputted into the 16-bit Cyclic Redundancy Check (CRC).

An example of the 16-bit Cyclic Redundancy Check (CRC), where the Packet Data (PD) of the Long Packet (LPa) is 01h, is illustrated (step-by-step) below.



Step	In	XOR(In,C0)	C15	C14	C13	C12	C11	XOR(XOR(In,C0),C11(Step-1))	C10	C9	C8	C7	C6	C5	C4	XOR(XOR(In,C0),C4(Step-1))	C3	C2	C1	C0	C0
0	X	X	1	1	1	1	1	X	1	1	1	1	1	1	1	X	1	1	1	1	X
1	1(LSB)	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	0	1	1	0	1	1	1	0	0	1	1	1	1	1	1	0	0	1	1	1	1
3	0	1	1	1	0	1	1	0	0	0	1	1	1	1	1	0	0	0	1	1	1
4	0	1	1	1	1	0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1
5	0	1	1	1	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0
6	0	0	0	1	1	1	1	0	0	0	0	0	0	1	1	1	1	0	0	0	0
7	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0	0
8	0(MSB)	0	0	0	0	1	1	1	1	1	0	0	0	0	0	1	1	1	1	0	0
	1 byte	CRC result	0	0	0	1	1		1	1	0	0	0	0	0		1	1	1	0	
			MSB																	LSB	

Figure 77: CRC Calculation – Packet Data (PD) is 01h

The value of the Packet Footer (PF) is 1E0Eh in this example. This example (command 01h has been sent) is illustrated below.

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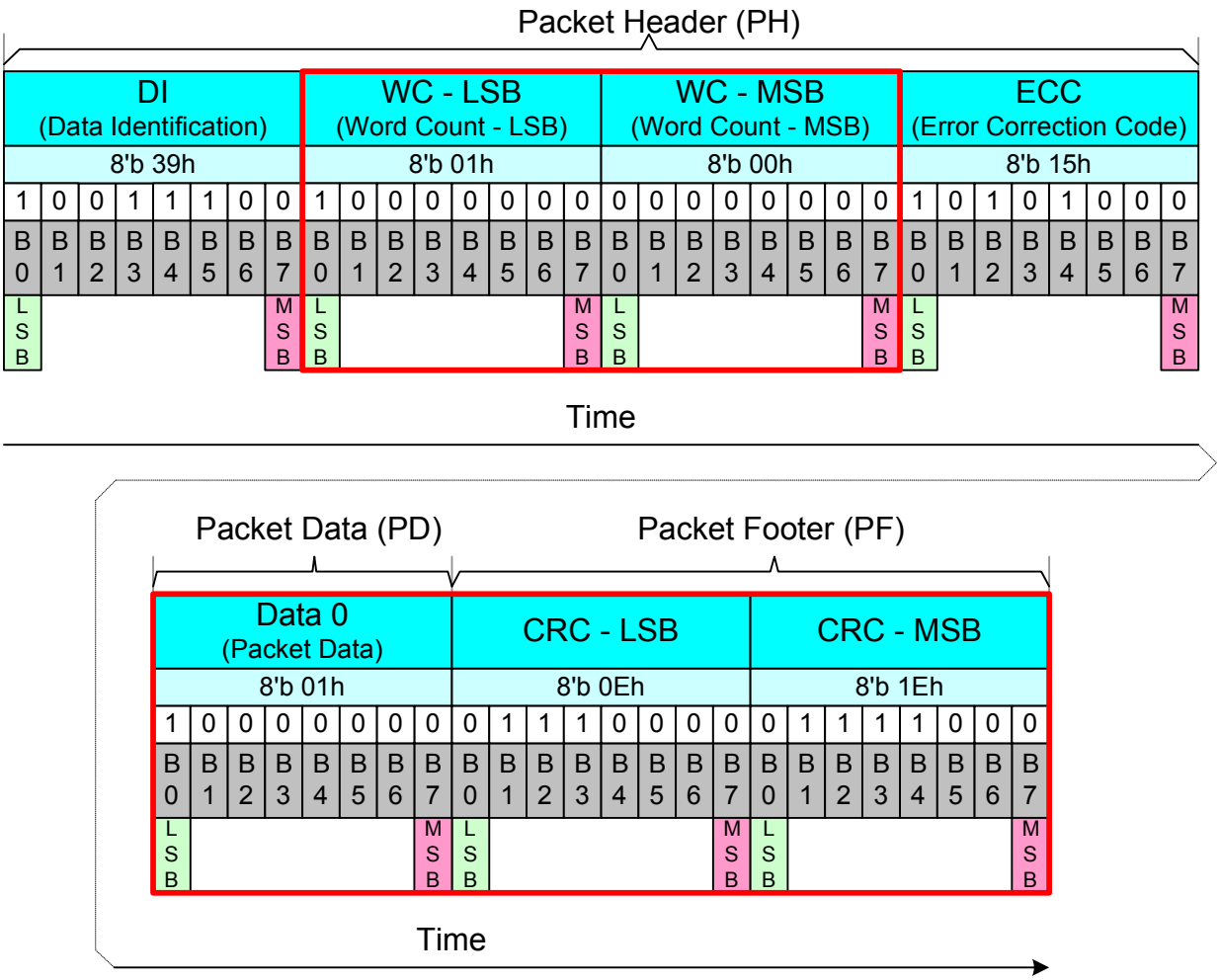


Figure 78: Packet Footer (PF) Example

The receiver calculates its own checksum value from the received Packet Data (PD). The receiver compares its own checksum with the Packet Footer (PF) which the transmitter has sent.

The received Packet Data (PD) and Packet Footer (PF) are correct if the checksums of the receiver and Packet Footer (PF) are equal. The received Packet Data (PD) and Packet Footer (PF) are not correct if the checksums of the receiver and Packet Footer (PF) are not equal.

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3.12.34. Packet Transmissions

3.12.35. Packet from the MPU to the Display Module

3.12.36. Display Command Set (DCS)

Display Command Set (DCS), which is defined in the Chapter 4.2 Command Description, is transmitted from the MPU to the display module. This Display Command Set (DCS) is always defined in the Data 0 of the Packet Data (PD), which is included in a Short Packet (SPa) and a Long packet (LPa), as illustrated below.

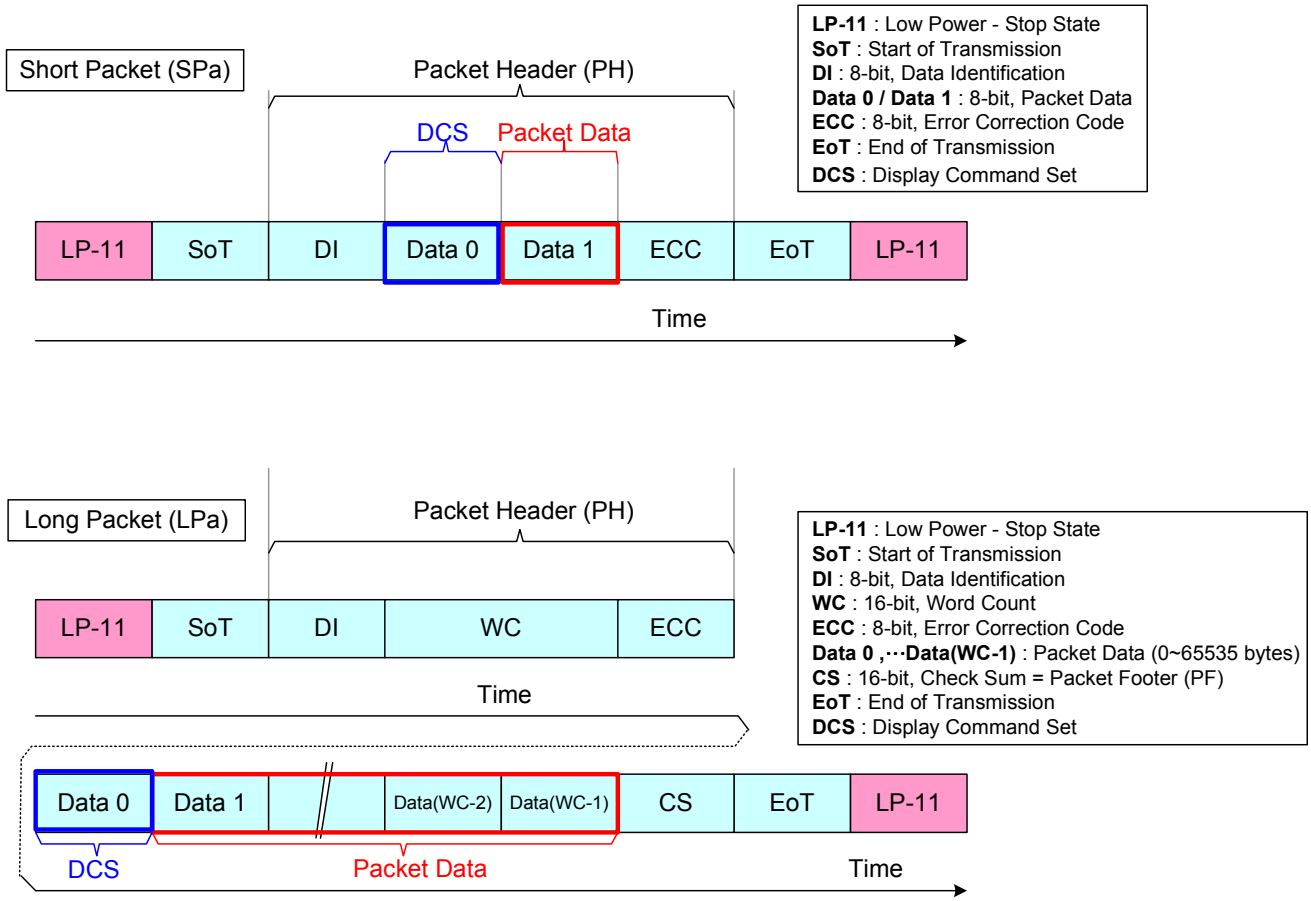


Figure 79: Display Command Set (DCS) in Short Packet (SPa) and Long Packet (LPa)

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3.12.37. Display Command Set (DCS) Write, No Parameter (DCSWN-S)

"Display Command Set (DCS) Write, No Parameter" is always used in a Short Packet (SPa), which is defined in the Data Type (DT, 00 0101b), from the MPU to the display module. These commands are defined in the table below. (See Chapter 5.2: Command Description.)

Table 18: Display Command Set (DCS) Write, No Parameters (DCSWN-S)

Command
NOP (00h)
Software Reset (01h)
Sleep In(10h)
Sleep Out (11h)
Partial Mode On (12h)
Normal Display Mode On (13h)
All Pixel Off (22h)
All Pixel On (23h)
Display Off (28h)
Display ON (29h)
Tearing Effect Line OFF (34h)
Idle Mode Off (38h)
Idle Mode On (39h)

A Short Packet (SPa) is defined as:

• Data Identification (DI)
  o Virtual Channel (VC, DI [7…6]): 00b
  o Data Type (DT, DI [5…0]): 00 0101b
• Packet Data (PD)
  o Data 0: "Sleep In (10h)", Display Command Set (DCS)
  o Data 1: Always 00hex
• Error Correction Code (ECC)

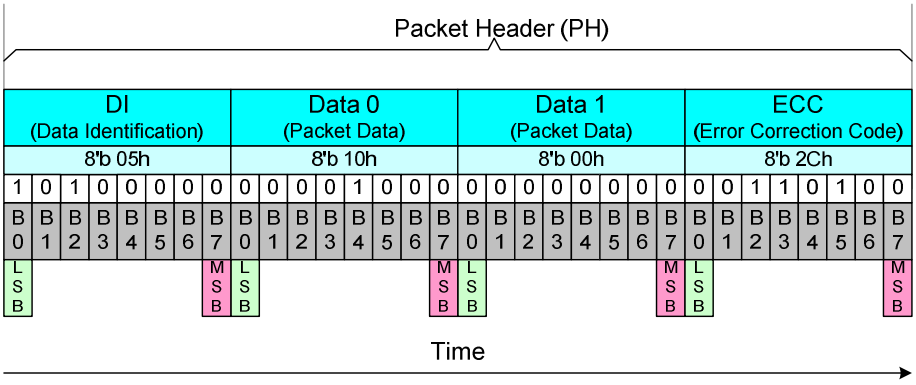


Figure 80: Display Command Set (DCS) Write, No Parameter (DCSWN-S) - Example

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3.12.38. Display Command Set (DCS) Write, 1 Parameter (DCSW1-S)

"Display Command Set (DCS) Write, 1 Parameter" (DCSW1-S) is always used in a Short Packet (SPa), which is defined in the Data Type (DT, 01 0101b), from the MPU to the display module. These commands are defined in the table below. (See chapter 5.2: Command Description.)

Table 19: Display Command Set (DCS) Write, 1 Parameter (DCSW1-S)

Command
Gamma Set (26h)
Memory Write (2Ch), Note
Tearing Effect Line ON(35h)
Memory Access Control (36h)
Interface Pixel Format (3Ah)
Memory Write Continue (3Ch), Note
Write Display Brightness (51h)
Write CTRL Display (53h)
Write Content Adaptive Brightness control (55h)
Write CABC Minimum Brightness (5Eh)

A Short Packet (SPa) is defined as:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 01 0101b
- Packet Data (PD)
  - o Data 0: "Gamma Set (26h)", Display Command Set (DCS)
  - o Data 1: 01hex, Parameter of the DCS
- Error Correction Code (ECC)

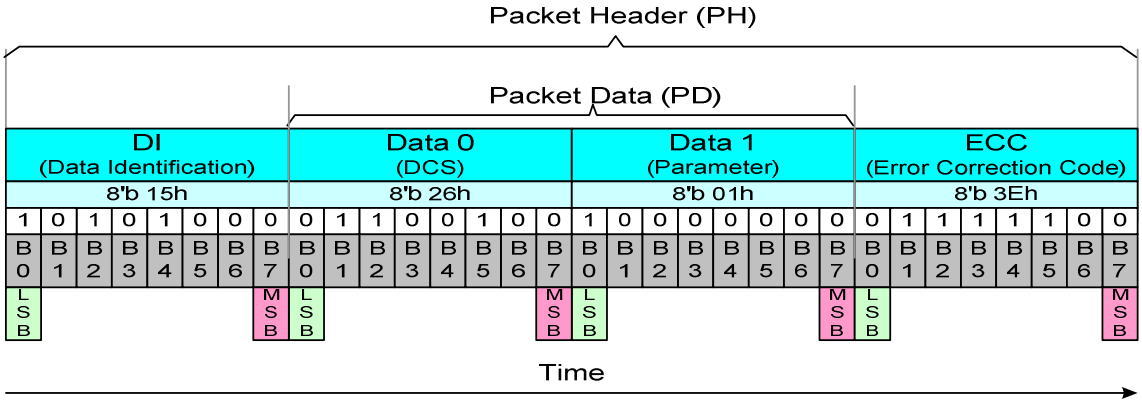


Figure 81: Display Command Set (DCS) Write, 1 Parameter (DCSW1-S) – Example

Note One subpixel has been written.

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3.12.39. Display Command Set (DCS) Write Long (DCSW-L)

"Display Command Set (DCS) Write Long" (DCSW-L) is always used in a Long Packet (LPa), which is defined in the Data Type (DT, 11 1001b), from the MPU to the display module. Command (No Parameters) and Write (1 or more parameters), are defined in the table below. (See chapter 5.2: Command Description.)

Table 20: Display Command Set (DCS) Write Long (DCSW-L)

Command
NOP (00h) , Note 1
Software Reset (01h) , Note 1
Sleep In(10h) , Note 1
Sleep Out (11h) , Note 1
Partial Mode On (12h) , Note 1
Normal Display Mode On (13h) , Note 1
All Pixel Off (22h)
All Pixel On (23h)
Gamma Set (26h), Note 2
Display Off (28h) , Note 1
Display ON (29h) , Note 1
Column Address Set (2Ah)
Page Address Set (2Bh)
Memory Write (2Ch), Note 2
Partial Area (30h)
Tearing Effect Line OFF (34h), Note 1
Tearing Effect Line ON (35h), Note 2
Memory Access Control (36h), Note 2
Idle Mode Off (38h) , Note 1
Idle Mode On (39h) , Note 1
Interface Pixel Format (3Ah)
Memory Write Continue (3Ch), Note 2
Write Display Brightness (51h) , Note 2
Write CTRL Display (53h) , Note 2
Write Content Adaptive Brightness control (55h) , Note 2
Write CABC Minimum Brightness (5Eh)

A Long Packet (LPa), when one command (no Parameter) was sent, is defined as:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 11 1001b
- Word Count (WC)
  - o Word Count (WC): 0001h
- Error Correction Code (ECC)
- Packet Data (PD): Data 0: "Sleep In (10h)", Display Command Set (DCS)
- Packet Footer (PF)

Note 1 *It can also be used in a Short Packet (SPa); See chapter "Display Command Set (DCS) Write, No Parameter".*

Note 2 *It can also be used in a Short Packet (SPa); See chapter "Display Command Set (DCS) Write, 1 Parameter".*

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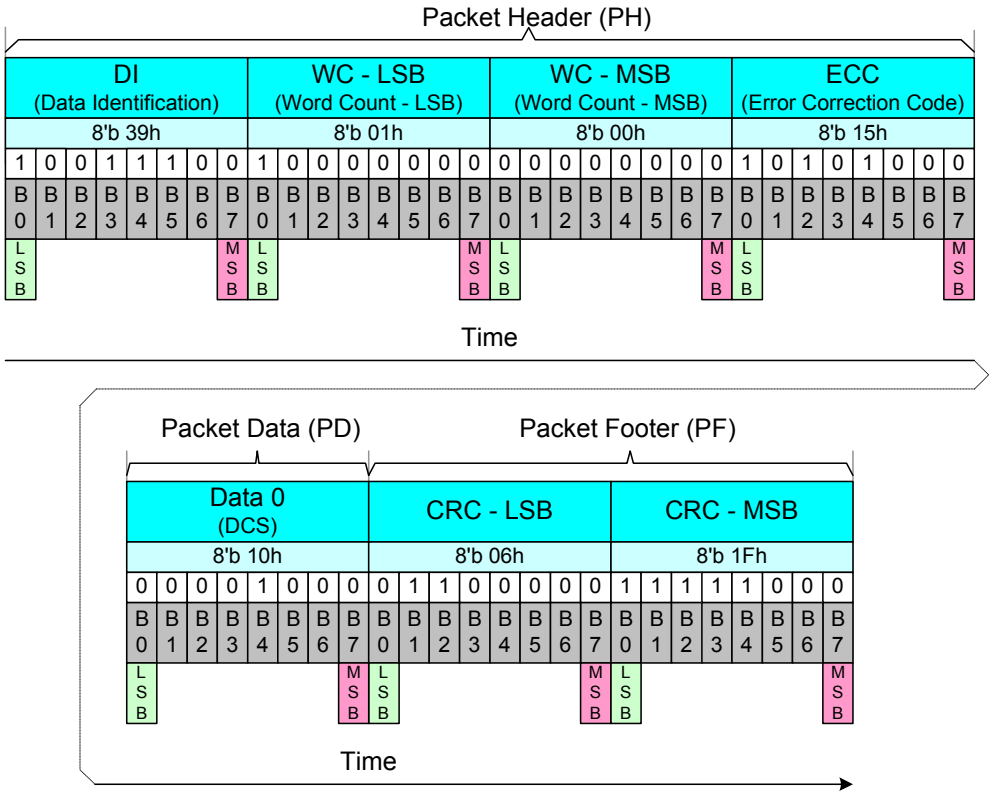


Figure 82: Display Command Set (DCS) Write Long (DCSW-L) with DCS Only - Example

A Long Packet (LPa), when one Write (1 parameter) was sent, is defined as:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 11 1001b
- Word Count (WC)
  - o Word Count (WC): 0002h
- Error Correction Code (ECC)
- Packet Data (PD):
  - o Data 0: “Gamma Set (26h)”, Display Command Set (DCS)
  - o Data 1: 01hex, Parameter of the DCS
- Packet Footer (PF)

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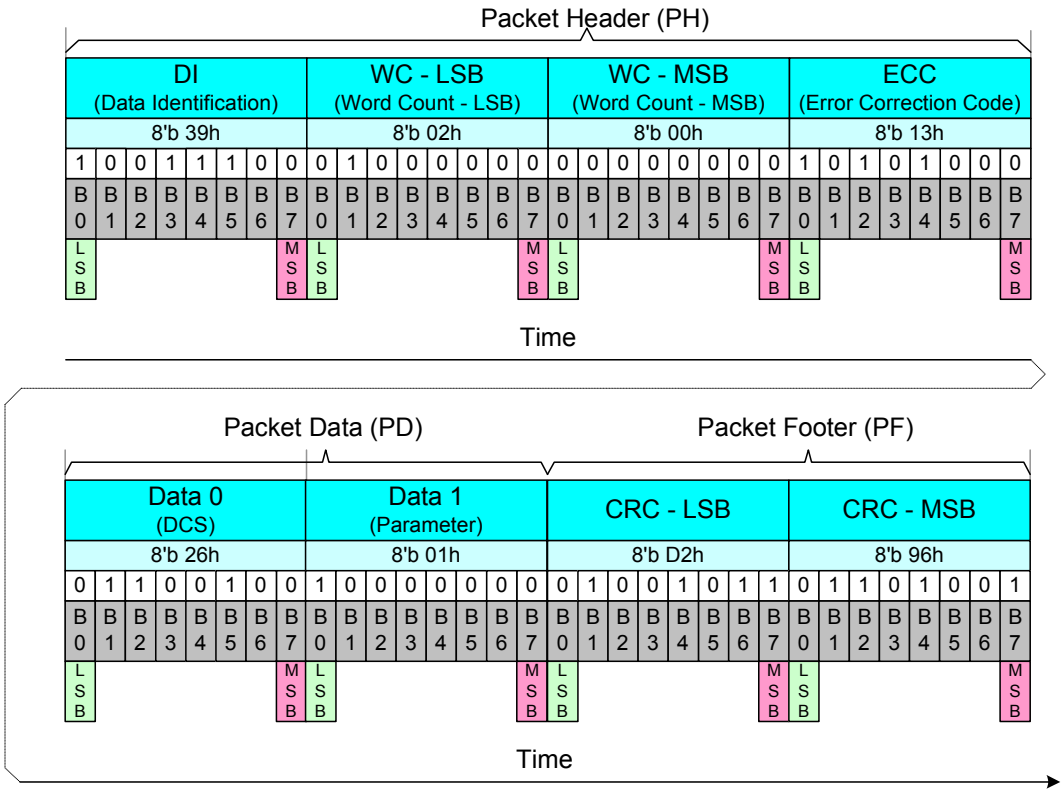


Figure 83: Display Command Set (DCS) Write Long with DCS and 1 Parameter - Example

A Long Packet (LPa), when one Write (4 parameters) was sent, is defined as:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 11 1001b
- Word Count (WC)
  - o Word Count (WC): 0005h
- Error Correction Code (ECC)
- Packet Data (PD):
  - o Data 0: “Column Address Set (2Ah)”, Display Command Set (DCS)
  - o Data 1: 00hex, 1st Parameter of the DCS, Start Column SC [15…8]
  - o Data 2: 12hex, 2nd Parameter of the DCS, Start Column SC [7…0]
  - o Data 3: 01hex, 3rd Parameter of the DCS, End Column EC [15…8]
  - o Data 4: EFhex, 4th Parameter of the DCS, End Column EC [7…0]
- Packet Footer (PF)

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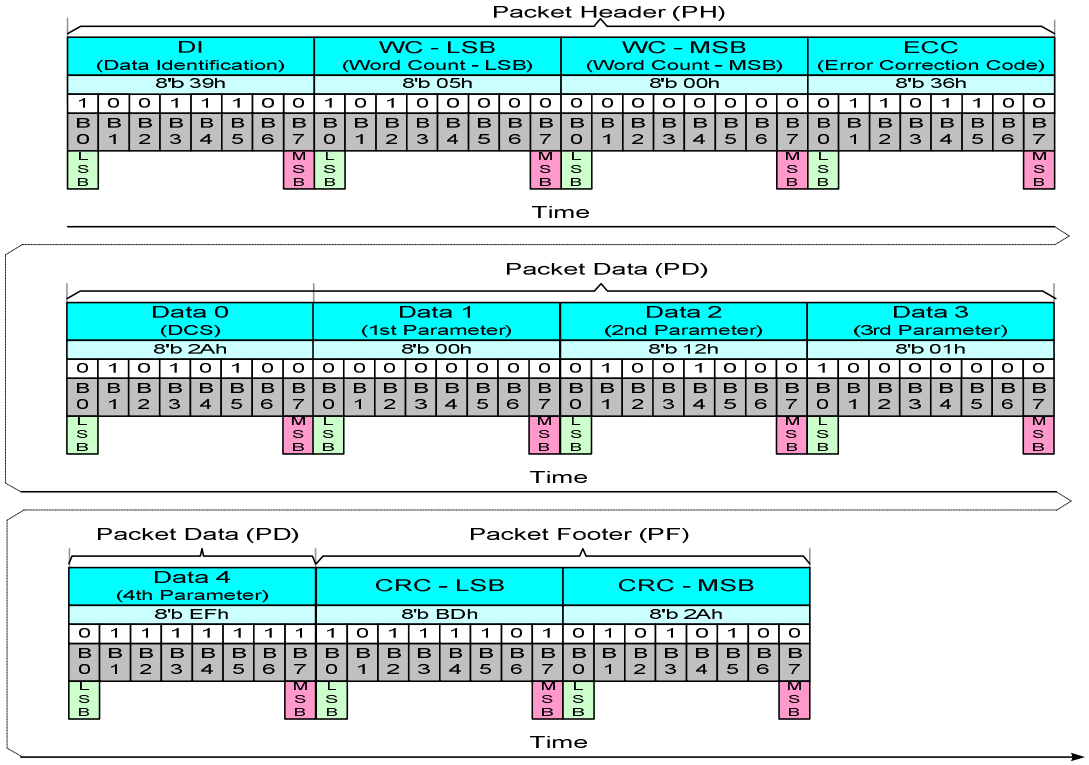


Figure 84: Display Command Set (DCS) Write Long with DCS and 4 Parameters - Example

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3.12.40. Display Command Set (DCS) Read, No Parameter (DCSRN-S)

"Display Command Set (DCS) Read, No Parameter" (DCSRN-S) is always used in a Short Packet (SPa), which is defined in the Data Type (DT, 00 0110b), from the MPU to the display module. These commands are defined in the table below. (See Chapter 5.2: Command Description.)

Table 21: Display Command Set (DCS) Read, No Parameter (DCSRN-S)

Command
Read Number of the Errors on DSI (05h)
Read Display Power Mode (0Ah)
Read Display MADCTL (0Bh)
Read Display Pixel Format (0Ch)
Read Display Image Mode (0Dh)
Read Display Signal Mode (0Eh)
Read Display Self-Diagnostic Result (0Fh)
Memory Read (2Eh)
Memory Read Continue (3Eh)
Read Display Brightness Value (52h)
Read CTRL Value Display (54h)
Read Content Adaptive Brightness Control (56h)
Read CABC Minimum Brightness (5Fh)
Read Black/White Low Bits (70h)
Read Bkx (71h)
Read Bky(72h)
Read Wx (73h)
Read Wy (74h)
Read Red/Green Low Bits (75h)
Read Rx (76h)
Read Ry (77h)
Read Gx (78h)
Read Gy (79h)
Read Blue/A Color Low Bits (7Ah)
Read Bx (7Bh)
Read By (7Ch)
Read Ax (7Dh)
Read Ay (7Eh)
Read DDB Start (A1h)
Read DDB Continue (A8h)
First Checksum (AAh)
Read Continue Checksum (AFh)
Read ID1 (DAh)
Read ID2 (DBh)
Read ID3 (DCh)

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The MPU has to define to the display module the maximum size of the returned packet. A command used for this purpose is “Set Maximum Return Packet Size” (SMRPS-S), which Data Type (DT) is 11 0111b and which uses Short Packets (SPa) before the MPU can send “Display Command Set (DCS) Read, No Parameter” to the display module. This sequence is illustrated below for reference purposes.

Step 1:

- The MPU sends “Set Maximum Return Packet Size” (Short Packet (SPa)) (SMRPS-S) to the display module when it wants to return one byte from the display module
- Data Identification (DI)
    - o Virtual Channel (VC, DI [7…6]): 00b
    - o Data Type (DT, DI [5…0]): 11 0111b
- Maximum Return Packet Size (MRPS)
    - o Data 0: 01hex
    - o Data 1: 00hex
- Error Correction Code (ECC)

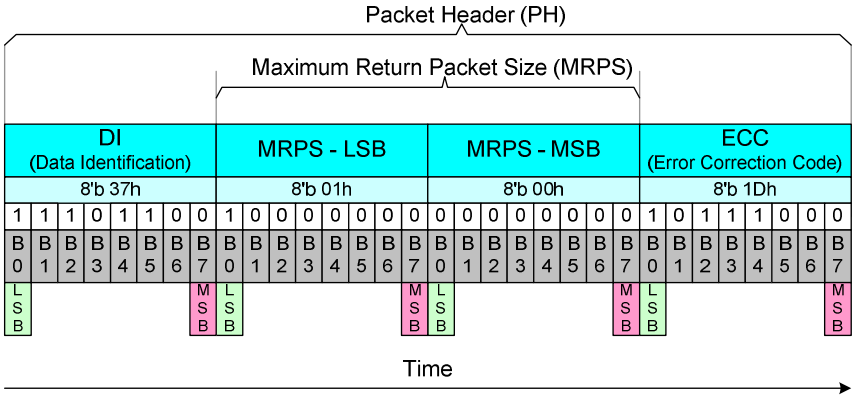


Figure 85 Set Maximum Return Packet Size (SMRPS-S) - Example

Step 2:

- The MPU wants to receive the value of the “Read ID1 (DAh)” from the display module when the MPU sends “Display Command Set (DCS) Read, No Parameter” to the display module.
- Data Identification (DI)
    - o Virtual Channel (VC, DI [7…6]): 00b
    - o Data Type (DT, DI [5…0]): 00 0110b
- Packet Data (PD)
    - o Data 0: “Read ID1 (DAh)”, Display Command Set (DCS)
    - o Data 1: Always 00hex
- Error Correction Code (ECC)

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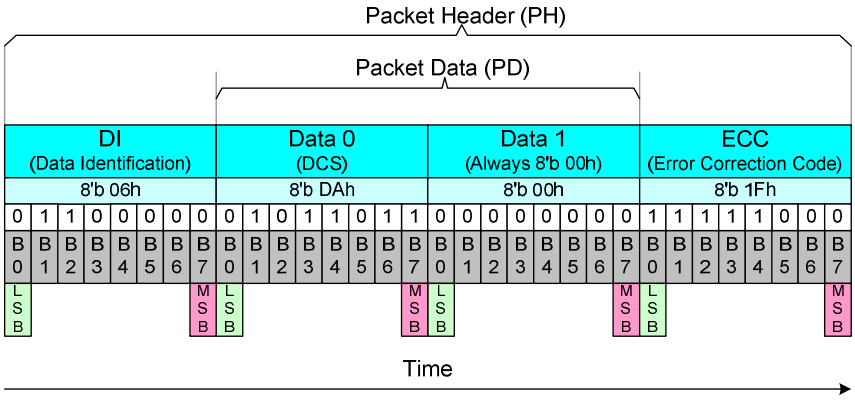


Figure 86 Display Command Set (DCS) Read, No Parameter (DCSRN-S) - Example

Step 3:

The display module can send 2 different information to the MPU after the Bus Turnaround (BTA).

1. An acknowledge with Error Report (AwER): it is used in a Short Packet (SPa) if an error is found while receiving a command. See the chapter “Acknowledge with Error Report (AwER)”.
2. Information of the received command: it can be transmitted in a Short Packet (SPa) or Long Packet (LPa).

3.12.41. Null Packet, No Data (NP-L)

“Null Packet, No Data” (NP-L) is always used in a Long Packet (LPa), which is defined in the Data Type (DT, 001001b), from the MPU to the display module. The purpose of this command is to keep data lanes in the high speed mode (HSDT), if necessary.

The display module can ignore the Packet Data (PD), which is sent by the MPU.

A Long Packet (LPa), when 5 random data bytes of the Packet Data (PD) are sent, is defined as:

• Data Identification (DI)
  o Virtual Channel (VC, DI [7…6]): 00b
  o Data Type (DT, DI [5…0]): 00 1001b
• Word Count (WC)
  o Word Count (WC): 0005hex
• Error Correction Code (ECC)
• Packet Data (PD):
  o Data 0: 89hex (Random data)
  o Data 1: 23hex (Random data)
  o Data 2: 12hex (Random data)
  o Data 3: A2hex (Random data)
  o Data 4: E2hex (Random data)

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• Packet Footer (PF)

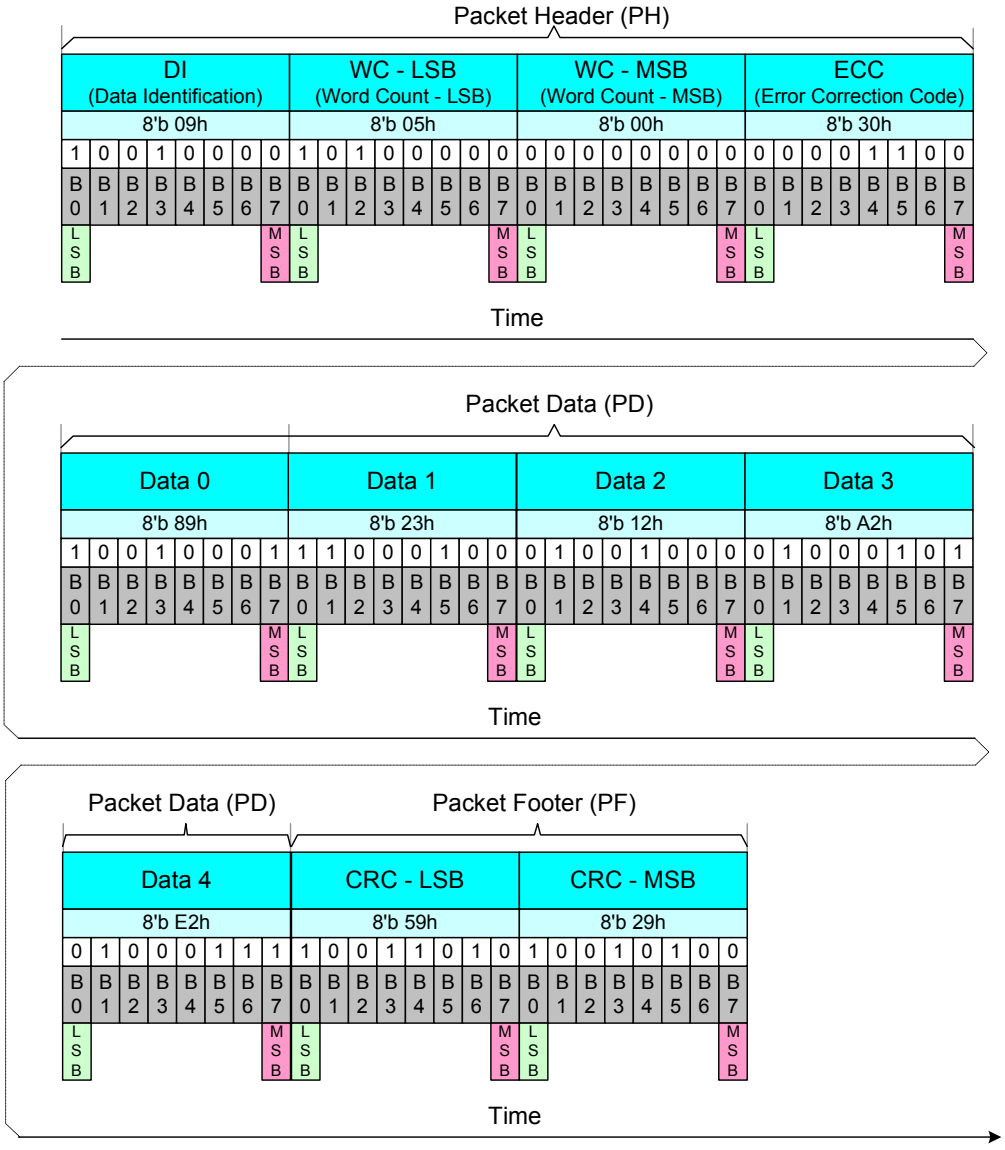


Figure 87: Null Packet, No Data (NP-L) - Example

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3.12.42. End of Transmission Packet (EoTP)

"End of Transmission Packet" (EoTP), which is defined in the Data Type (DT, 00 1000b) and optional in the interface level, is always used in a Short Packet (SPa) from the MPU to the display module. The purpose of this command is to terminate the high Speed Data Transmission (HSDT) mode properly when the EoTP is added after the last payload packet before "End of Transmission" (EoT).

The MPU can decide if it wants to use the "End of Transmission Packet" (EoTP) or not. The display shall have the capability to support both. That is, if the MPU applies the EoTP, it shall report the "DSI Protocol Violation Error" when the EoTP is not detected in the High-Speed (HS). The display module error reporting shall be enabled/disabled statistically, according to the module application.

The display module will/will not receive "End of Transmission Packet" (EoTP) from the MPU during the Low Power Data Transmission (LPDT) mode before "Mark-1" (= leaving Escape mode), which ends the Low Power Data Transmission (LPDT) mode.

The display module is not allowed to send "End of Transmission Packet" (EoTP) to the MPU during the Low Power Data Transmission (LPDT) mode.

The summary of the receiving and transmitting EoTP is listed below.

Table 22: Receiving and Transmitting EoTP during LPDT

Direction	Display Module (DM) in High Speed Data Transmission (HSDT)	Display Module (DM) in Low Power Data Transmission (LPDT)
MPU => Display Module	With or Without EoTP is Supported	With or Without EoTP is Supported
Display Module => MPU	HS Mode is not available (EoTP is not available)	EoTP cannot be sent by the Display Module (DM)

Short Packet (SPa) uses a fixed format as follows:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 00 1000b
- Packet Data (PD)
  - o Data 0: 0Fhex
  - o Data 1: 0Fhex
- Error Correction Code (ECC)
  - o ECC: 01hex

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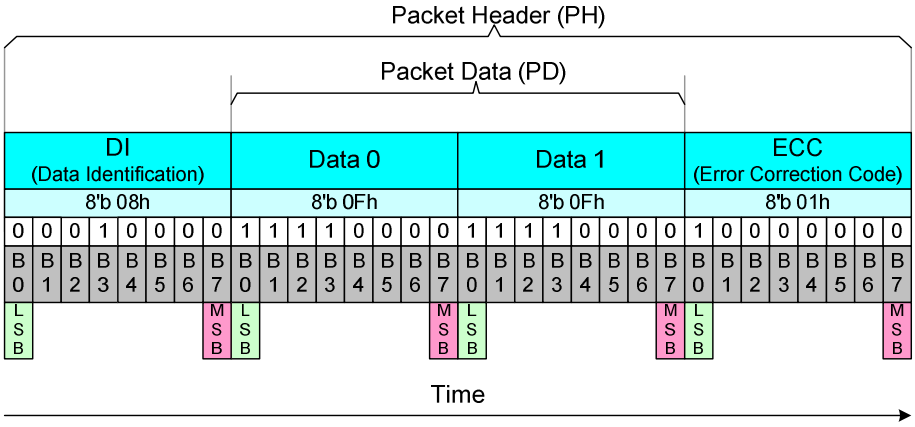


Figure 88: End of Transmission Packet (EoTP)

Some cases of the “End of Transmission Packet” (EoTP) are illustrated below for reference purpose only.

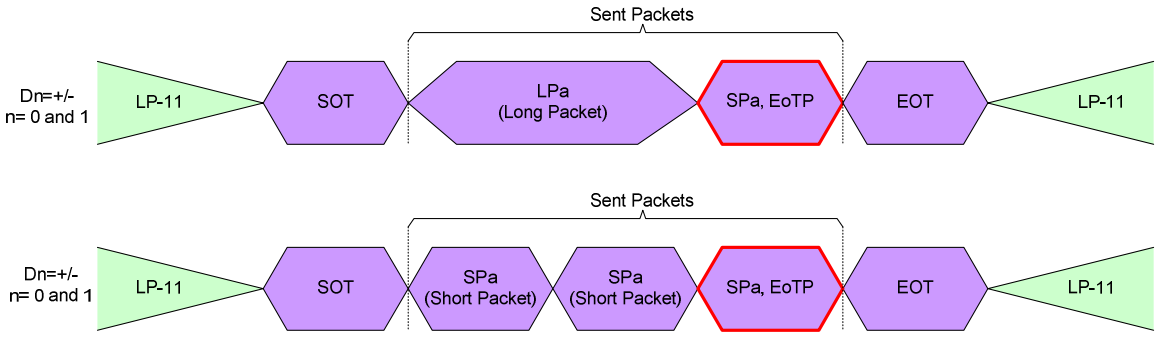


Figure 89: End of Transmission Packet (EoTP) - Examples

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3.12.43. Packet from the Display Module to the MPU

3.12.44. Used Packet Types

The display module can use Short Packets (SPa) or Long Packets (LPa) when it returns information to the MPU after the MPU requests information from the Display Module. This information can be a response of the Display Command Set (DCS) (See the chapter “Display Command Set (DCS) Read, No Parameter” (DCSRN-S)) or an Acknowledge with Error Report (See the chapter: “Acknowledge with Error Report (AwER)” (AwER)).

The used packet type is defined in the Data Type (DT). See the chapter “Data Type (DT)”. If the maximum size of the Packet Data (PD) could be sent in one packet, then the display module cannot separate returned bytes into several packets.

Both cases are illustrated below for reference purpose.

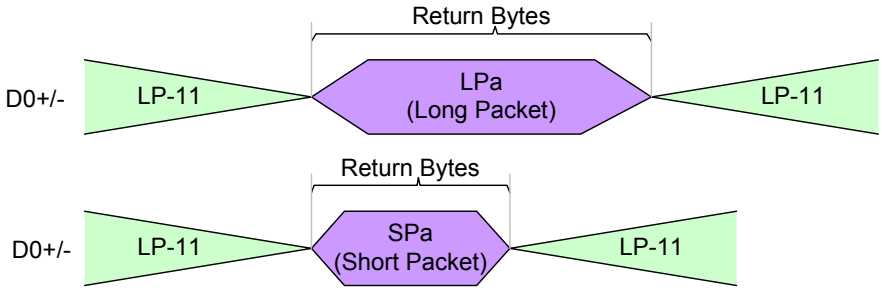


Figure 90: Return Bytes in a Single Packet

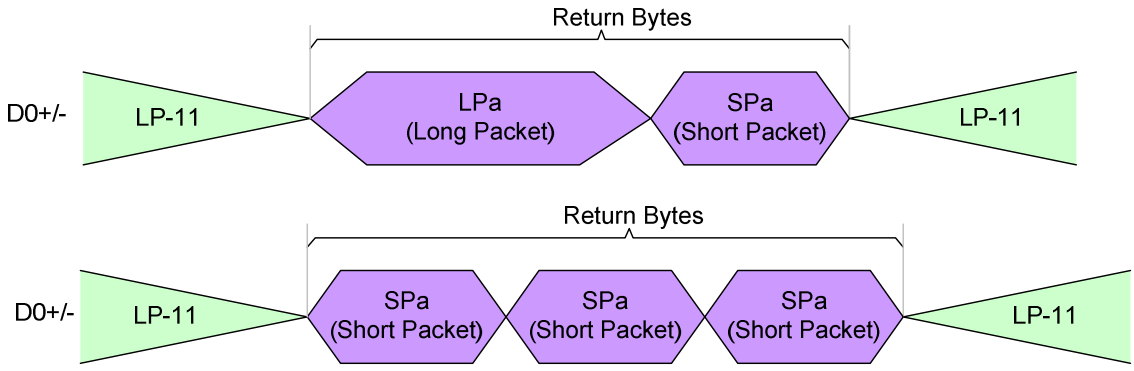


Figure 91: Return Bytes in Several Packets – Not Allowed

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Exception:

The display module returns 2 packets (1st packet: Data, 2nd Packet: Acknowledge with Error Report) to the MPU when the display module receives a read command (see the chapter "Display Command Set (DCS) Read, No Parameter (DCSRN-S)"), in which a single bit error is detected and corrected by the EEC (see bit 8 in "Table 24: Acknowledge with Error Report (AwER) for Short Packet (SPa) Response").

These returned packets are illustrated below for reference purpose.

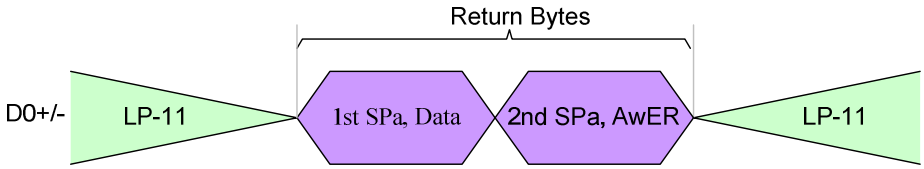


Figure 92: Exception when Return Bytes in Several Packets

AwER = Acknowledge with Error Report

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3.12.45. Acknowledge with Error Report (AwER)

"Acknowledge with Error Report" (AwER), which is defined in the Data Type (DT, 00 0010b), is always transmitted through a Short Packet (SPa) from the display module to the MPU. The 16 bits in the Packet Data (PD) can indicate the current error(s) if one or more than one bit(s) is/are set to 1, as defined in the following table.

Table 23: Acknowledge with Error Report (AwER) for Long Packet (LPa) Response

Bit	Description
0	SoT Error
1	SoT Sync Error
2	EoT Sync Error
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	Any Protocol Timer Time-Out
6	False Control Error
7	Contention is Detected on the Display Module
8	ECC Error, single-bit (detected and corrected)
9	ECC Error, multi-bit (detected, not corrected)
10	Checksum Error
11	DSI Data Type (DT) Not Recognized
12	DSI Virtual Channel (VC) ID Invalid
13	Invalid Transmission Length
14	Reserved, Set to 0 internally
15	DSI Protocol Violation

Table 24: Acknowledge with Error Report (AwER) for Short Packet (SPa) Response

Bit	Description
0	SoT Error
1	SoT Sync Error
2	EoT Sync Error
3	Escape Mode Entry Command Error
4	Low-Power Transmit Sync Error
5	Any Protocol Timer Time-Out
6	False Control Error
7	Contention is Detected on the Display Module
8	ECC Error, single-bit (detected and corrected)
9	ECC Error, multi-bit (detected, not corrected)
10	Reserved, Set to 0 internally (Only for Long Packet (LP)
11	DSI Data Type (DT) Not Recognized
12	DSI Virtual Channel (VC) ID Invalid
13	Invalid Transmission Length
14	Reserved, Set to 0 internally
15	DSI Protocol Violation

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These errors are included in all packages, which are received from the MPU to the display module, before the Bus Turnaround (BTA).

The display module ignores the received packet which includes an error or errors.

Acknowledge with Error Report (AwER) in a Short Packet (SPa) is defined as:

• Data Identification (DI)
  o Virtual Channel (VC, DI [7…6]): 00b
  o Data Type (DT, DI [5…0]): 00 0010b
• Packet Data (PD)
  o Bit 8: ECC Error, single-bit (detected and corrected)
  o AwER: 0100h
• Error Correction Code (ECC)

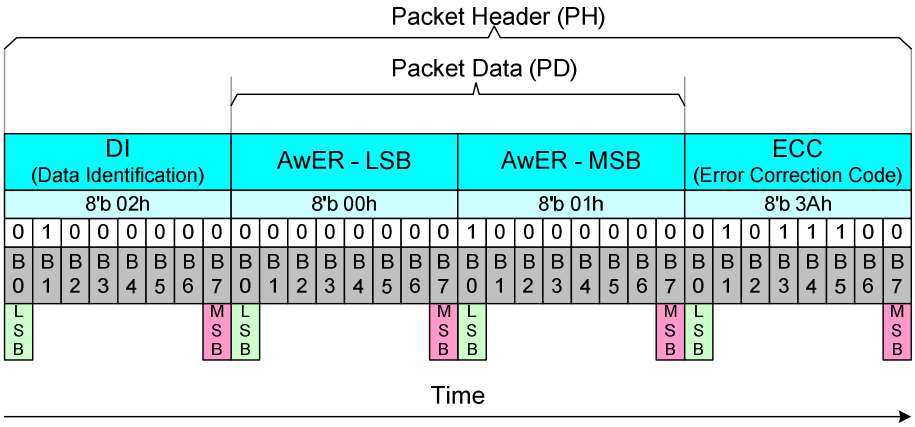


Figure 93: Acknowledge with Error Report (AwER) – Example

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It is possible that the display module receives several packets, which include errors, from the MPU before the MPU performs the Bus Turnaround (BTA). Some examples are illustrated below for reference purpose.

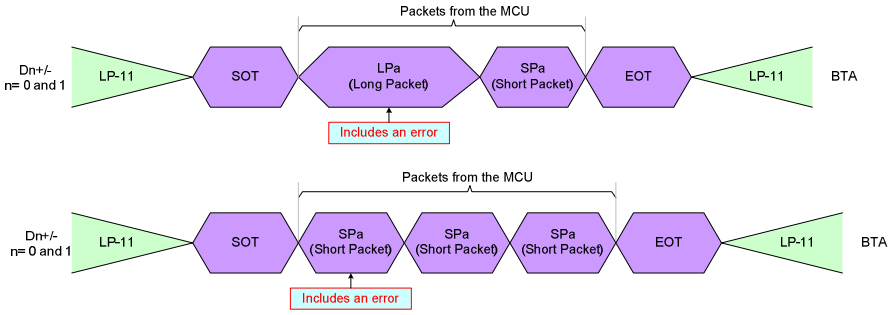


Figure 94: Errors Packets

Therefore, a method is needed to check if there are errors in the previous packets. These errors of the previous packets are indicated by “Read Display Signal Mode (0Eh)” and “Read Number of the Errors on DSI (05h)” commands. If a received packet includes an error, the bit D0 of the “Read Display Signal Mode (0Eh)” command will be set to 1.

The numbers of the packets, including an ***ECC or CRC*** error, are calculated in the RDNUMED register, which can read the “Read Number of the Errors on DSI (05h)” command. This command also sets the RDNUMED register to 00h and the bit D0 of the “Read Display Signal Mode (0Eh)” command to 0 after the MPU has read the RDNUMED register from the display module.

The functionality of the RDNUMED register is illustrated below for reference purpose.

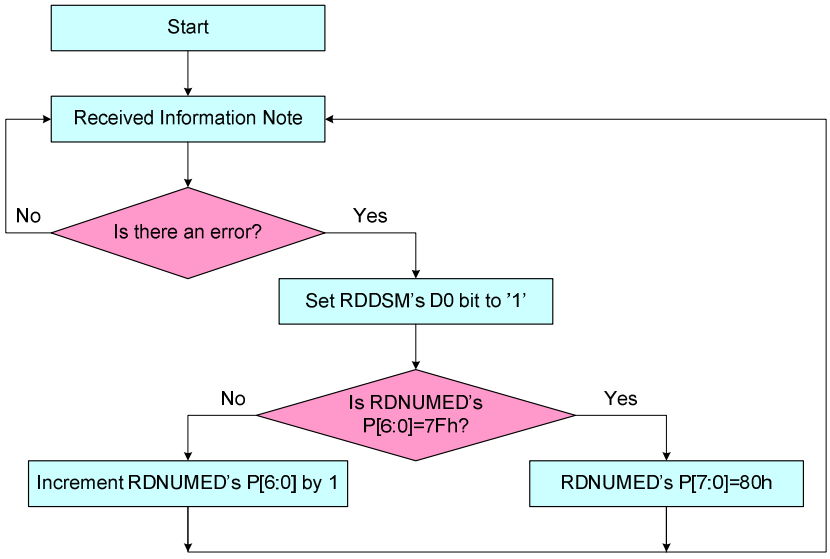


Figure 95: Flow Chart for Errors on DSI[Note]

Note 1. This information can be Interface or Packet Level Communication but it is always from the MPU to the display module in this case.

2. CRC or ECC error

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3.12.46. DCS Read Long Response (DCSRR-L)

"DCS Read Long Response" (DCSRR-L), which is defined in the Data Type (DT, 011100b), is always used in a Long Packet (LPa) from the display module to the MPU. "DCS Read Long Response" (DCSRR-L) is used when the display module wants to respond to a DCS Read command, which the MPU has sent to the display module.

A Long Packet (LPa), which includes 5 data bytes of the Packet Data (PD), is defined as:

- Data Identification (DI)
  - o Virtual Channel (VC, DI [7…6]): 00b
  - o Data Type (DT, DI [5…0]): 01 1100b
- Word Count (WC)
  - o Word Count (WC): 0005hex
- Error Correction Code (ECC)
- Packet Data (PD):
  - o Data 0: 89hex
  - o Data 1: 23hex
  - o Data 2: 12hex
  - o Data 3: A2hex
  - o Data 4: E2hex
- Packet Footer (PF)

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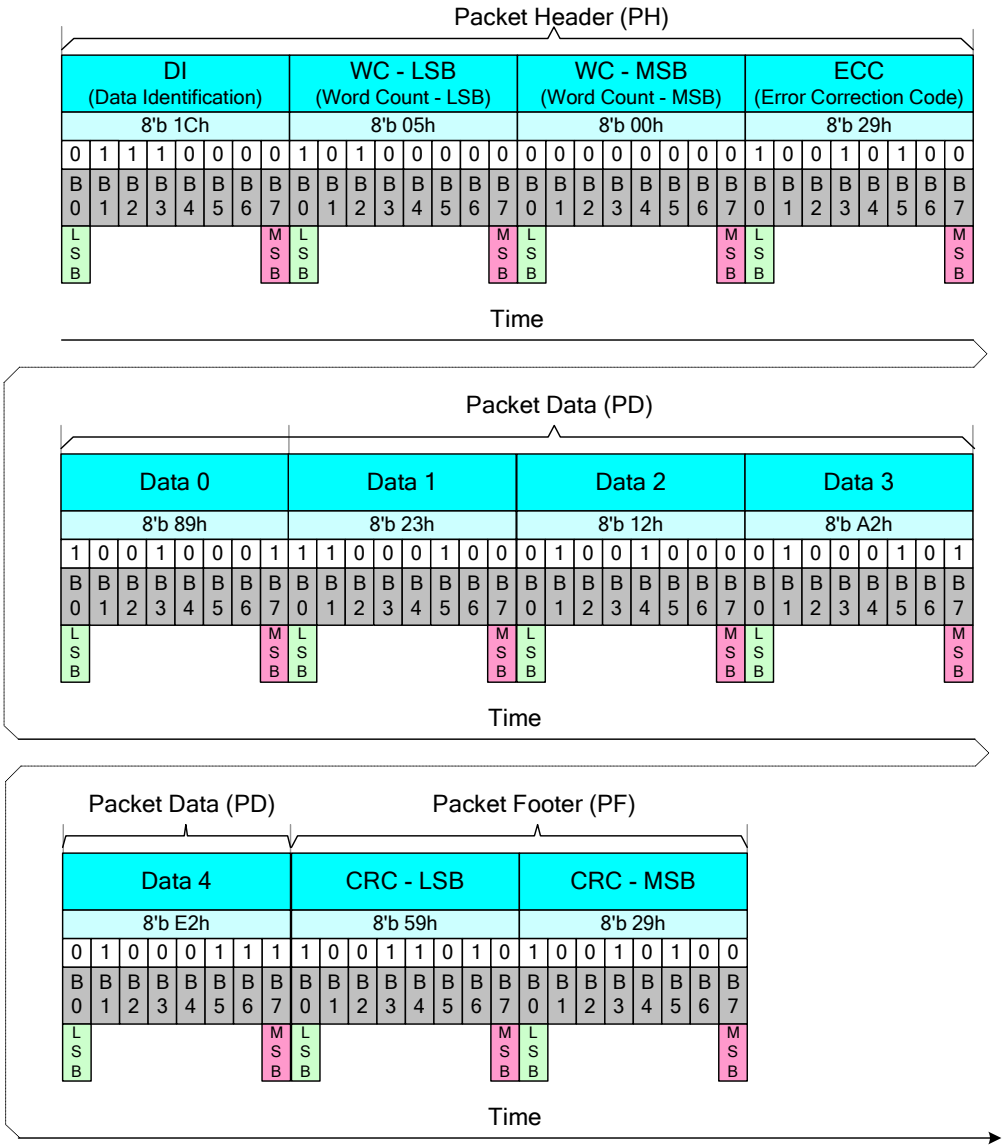


Figure 96: DCS Read Long Response (DCSRR-L) - Example

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3.12.47. DCS Read Short Response, 1 Byte Returned (DCSRR1-S)

"DCS Read Short Response, 1 Byte Returned", which is defined in the Data Type (DT, 10 0001b), (DCSRR1-S) is always used in a Short Packet (SPa) from the display module to the MPU. "DCS Read Short Response, 1 Byte Returned (DCSRR1-S) is used when the display module wants to respond to a DCS Read command, which the MPU has sent to the display module.

A Short Packet (SPa) is defined as:

• Data Identification (DI)
  o Virtual Channel (VC, DI [7…6]): 00b
  o Data Type (DT, DI [5…0]): 10 0001b
• Packet Data (PD)
  o Data 0: 45hex
  o Data 1: 00hex (Always)
• Error Correction Code (ECC)

This is defined on the Short Packet (SP) as follows.

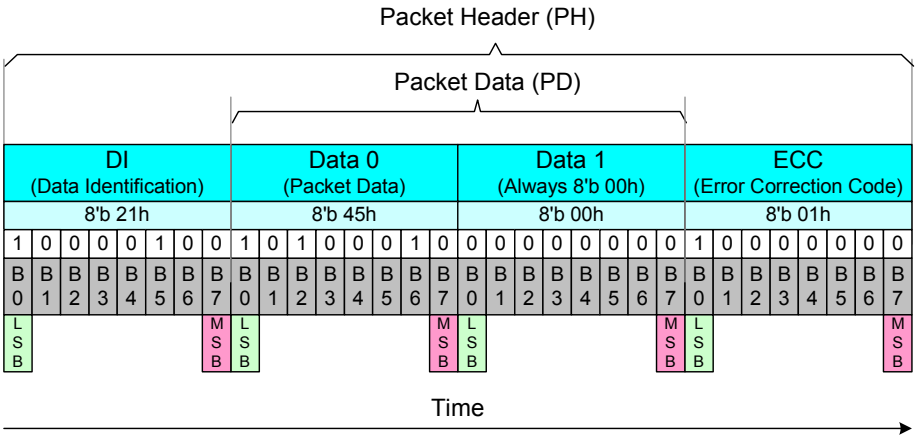


Figure 97: DCS Read Short Response, 1 Byte Returned (DCSRR1-S) - Example

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3.12.48. DCS Read Short Response, 2 Bytes Returned (DCSRR2-S)

"DCS Read Short Response, 2 Bytes Returned" (DCSRR2-S) is always used in a Short Packet (SPa), which is defined in the Data Type (DT, 10 0010b), from the display module to the MPU. "DCS Read Short Response, 2 Bytes Returned" (DCSRR2-S) is used when the display module wants to respond to a DCS Read command, which the MPU has sent to the display module.

A Short Packet (SPa) is defined as:

• Data Identification (DI)
  o Virtual Channel (VC, DI [7…6]): 00b
  o Data Type (DT, DI [5…0]): 10 0010b
• Packet Data (PD)
  o Data 0: 45hex
  o Data 1: 32hex
• Error Correction Code (ECC)

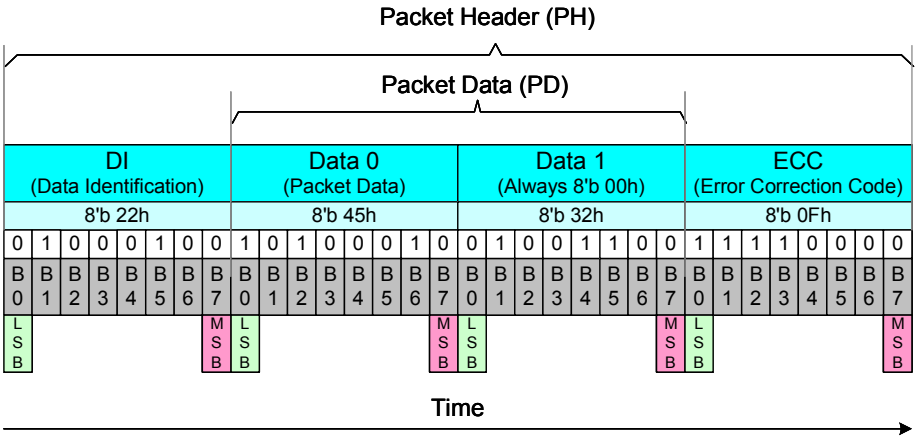


Figure 98: DCS Read Short Response, 2 Bytes Returned (DCSRR2-S) - Example

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3.12.49. Communication Sequences

3.12.50. General

The communication sequences can be done on interface or packet levels between the MPU and the display module. See chapters “Interface Level Communication” and “Packet Level Communication”.

This communication sequence description is for DSI data lanes (DSI-D0+/- and DSI-D1+/-), and it is assumed that the needed low level communication is done on DSI clock lanes (DSI-CLK+/-) automatically. See the chapter “DSI-CLK Lanes”.

Functions of the interface level communication are described in the following table.

Table 25: Interface Level Communication

Interface Mode	Abbreviation	Interface Action Description
Low Power	LP-11	Stop State
	LPDT	Low Power Data Transmission
	ULPS	Ultra-Low Power State
	RAR	Remote Application Reset
	TEE	Tearing Effect Event
	ACK	Acknowledge (No Error)
	BTA	Bus Turnaround
High Speed	HSDT	High speed Data Transmission

Functions of the packet level communication are described in the following table.

Table 26: Packet Level Communication

Interface Mode	Abbreviation	Packet Size	Interface Action Description
MPU	DCSW1-S	Short Packet	DCS Write, 1 Parameter
	DCSWN-S	Short Packet	DCS Write, No Parameter
	DCSW-L	Long Packet	DCS Write Long
	DCSRN-S	Short Packet	DCS Read, No Parameter
	SMRPS-S	Short Packet	Set Maximum Return Packet Size
	NP-L	Long Packet	Null Packet, No Data
	EoTP	Short Packet	End of Transmission Packet
Display Module (ILI9806G)	AwER	Short Packet	Acknowledge with Error Packet
	DCSRR-L	Long Packet	DCS Read Long Response
	DCSRR1-S	Short Packet	DCS Read Short Response
	DCSRR2-S	Short Packet	DCS Read Short Response

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3.12.51. Sequences

3.12.52. DCS Write, 1 Parameter Sequence

A Short Packet (SPa) of "Display Command Set (DCS) Write, 1 Parameter (DCSW1-S)" is defined in the chapter "Display Command Set (DCS) Write, 1 Parameter (DCSW1-S)", and examples of sequences on how this packet is used is described in the following tables.

Table 27: DCS Write, 1 Parameter Sequence – Example 1

DCS Write, 1 Parameter Sequence – Example 1						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSW1-S	LPDT	→	--	--	
3	--	LP-11	→	--	--	End

Table 28: DCS Write, 1 Parameter Sequence – Example 2

DCS Write, 1 Parameter Sequence – Example 2						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSW1-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	End

Table 29: DCS Write, 1 Parameter Sequence – Example 3

DCS Write, 1 Parameter Sequence – Example 3						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSW1-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	
5	--	BTA	⇔	BTA	--	Interface Control Change from MPU to the display module
6	--	--	←	LP-11	--	If No Error → Go to Line 8 If Error Occurs → Go to Line 13
7						
8	--	--	←	ACK	--	No Error
9	--	--	←	LP-11	--	
10	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
11	--	LP-11	→	--	--	End
12						
13	--	--	←	LPDT	AwER	Error Report
14	--	--	←	LP-11	--	
15	--	BTA	⇔	BTA	--	
16	--	LP-11	→	--	--	End

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3.12.53. DCS Write, No Parameter Sequence

A Short Packet (SPa) of "Display Command Set (DCS) Write, No Parameter (DCSWN-S)" is defined in the chapter "Display Command Set (DCS) Write, No Parameter (DCSWN-S)", and examples of sequences on how this packet is used are described in the following tables.

Table 30: DCS Write, No Parameter Sequence – Example 1

DCS Write, No Parameter Sequence – Example 1						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSWN-S	LPDT	→	--	--	
3	--	LP-11	→	--	--	End

Table 31: DCS Write, No Parameter Sequence – Example 2

DCS Write, No Parameter Sequence – Example 2						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSWN-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	End

Table 32: DCS Write, No Parameter Sequence – Example 3

DCS Write, 1 Parameter Sequence – Example 3						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSWN-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	
5	--	BTA	⇔	BTA	--	Interface Control Change from MPU to the display module
6	--	--	←	LP-11	--	If No Error → Go to Line 8 If Error Occurs → Go to Line 13
7						
8	--	--	←	ACK	--	No Error
9	--	--	←	LP-11	--	
10	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
11	--	LP-11	→	--	--	End
12						
13	--	--	←	LPDT	AwER	Error Report
14	--	--	←	LP-11	--	
15	--	BTA	⇔	BTA	--	
16	--	LP-11	→	--	--	End

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3.12.54. DCS Write Long Sequence

A Long Packet (LPa) of "Display Command Set (DCS) Write Long (DCSW-L)" is defined in the chapter "Display Command Set (DCS) Write Long (DCSW-L)", and examples of sequences on how this packet is used are described in the following tables.

Table 33: DCS Write Long Sequence – Example 1

DCS Write Long Sequence – Example 1						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSW-L	LPDT	→	--	--	
3	--	LP-11	→	--	--	End

Table 34: DCS Write Long Sequence – Example 2

DCS Write Long Sequence – Example 2						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSRN-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	End

Table 35: DCS Write Long Sequence – Example 3

DCS Write Long Sequence – Example 3						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSRN-S	HSDT	→	--	--	
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	
5	--	BTA	⇔	BTA	--	Interface Control Change from MPU to the display module
6	--	--	←	LP-11	--	If No Error → Go to Line 8 If Error Occurs → Go to Line 13
7						
8	--	--	←	ACK	--	No Error
9	--	--	←	LP-11	--	
10	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
11	--	LP-11	→	--	--	End
12						
13	--	--	←	LPDT	AwER	Error Report
14	--	--	←	LP-11	--	
15	--	BTA	⇔	BTA	--	

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16	--	LP-11	➔	--	--	End

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Table 36: DCS Write Long Sequence – Example 4[Note]

DCS Write Long Sequence – Example 4						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	DCSW-L	HSDT	→	--	--	Memory Write (2Ch)
3	DCSW-L	HSDT	→	--	--	Memory Write Continue (3Ch)
4	DCSW-L	HSDT	→	--	--	Memory Write Continue (3Ch)
5	DCSW1-S	HSDT	→	--	--	Memory Write Continue (3Ch) with 1 Parameter
6	EoTP	HSDT	→	--	--	End of Transmission Packet
7	--	LP-11	→	--	--	End

3.12.55. DCS Read, No Parameter Sequence

A Short Packet (SPa) of "Display Command Set (DCS) Read, No Parameter (DCSRN-S)" is defined in the chapter "Display Command Set (DCS) Read, No Parameter (DCSRN-S)", and examples of sequences on how this packet is used are described in the following tables.

Table 37: DCS Read, No Parameter Sequence – Example 1

DCS Read, No Parameter Sequence – Example 1						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	SMRPS-S	HSDT	→	--	--	Defined how many data byte is wanted to read : 1 byte
3	DCSRN-S	HSDT	→	--	--	Wanted to get a response ID1 (DAh)
4	EoTP	HSDT	→	--	--	End of Transmission Packet
5	--	LP-11	→	--	--	
6	--	BTA	⇔	BTA	--	Interface Control Change from MPU to the display module
7	--	--	←	LP-11	--	If No Error → Go to Line 9 If Error Occurs → Go to Line 14 If Error is Corrected by ECC → Go to Line 19
8						
9	--	--	←	LPDT	DCSRR1-S	Response 1 byte return
10	--	--	←	LP-11	--	
11	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
12	--	LP-11	→	--	--	End
13						
14	--	--	←	LPDT	AwER	Error Report
15	--	--	←	LP-11	--	
16	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
17	--	LP-11	→	--	--	End
18						

Note This is an example that image data are sent in 4 packets.

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19	--	--	←	LPDT	DCSRR1-S	Response 1 byte return
20	--	--	←	LPDT	AwER	Error Report (Error is corrected by ECC)
21			←	LP-11	--	
22	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
23	--	LP-11	→	--	--	End

Table 38: DCS Read, No Parameter Sequence – Example 2

DCS Read, No Parameter Sequence – Example 2						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	SMRPS-S	HSDT	→	--	--	Defined how many data byte is wanted to read : 200 bytes
3	DCSRN-S	HSDT	→	--	--	Wanted to get a response "Memory Read" (2Eh)
4	EoTP	HSDT	→	--	--	End of Transmission Packet
5	--	LP-11	→	--	--	
6	--	BTA	⇔	BTA	--	Interface Control Change from MPU to the display module
7	--	--	←	LP-11	--	If No Error → Go to Line 9 If Error Occurs → Go to Line 14 If Error is Corrected by ECC → Go to Line 19
8						
9	--	--	←	LPDT	DCSRR-L	Response 200 byte return
10	--	--	←	LP-11	--	
11	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
12	--	LP-11	→	--	--	End
13						
14	--	--	←	LPDT	AwER	Error Report
15	--	--	←	LP-11	--	
16	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
17	--	LP-11	→	--	--	End
18						
19	--	--	←	LPDT	DCSRR-S	Response 200 byte return
20	--	--	←	LPDT	AwER	Error Report (Error is corrected by ECC)
21			←	LP-11	--	
22	--	BTA	⇔	BTA	--	Interface Control Change from the display module to MPU
23	--	LP-11	→	--	--	End

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3.12.56. Null Packet, No Data Sequence

A Long Packet (LPa) of "Null Packet, No Data (NP-L)" is defined in the chapter "Null Packet, No Data (NP-L)", and an example of sequence on how this packet is used is described in the following table.

Table 39: Null Packet, No Data Sequence - Example

Null Packet, No Data Sequence – Example						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	NP-L	HSDT	→	--	--	Only High Speed Data Transmission is used
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	End

3.12.57. End of Transmission Packet

A Short Packet (SPa) of "End of Transmission (EoTP)" is defined in the chapter "End of Transmission Packet (EoTP)", and an example of sequence on how this packet is used is described in the following table.

Table 40: End of Transmission Packet – Example

End of Transmission Packet – Example						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	--	LP-11	→	--	--	Start
2	NP-L	HSDT	→	--	--	Only High Speed Data Transmission is used
3	EoTP	HSDT	→	--	--	End of Transmission Packet
4	--	LP-11	→	--	--	End

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3.12.58. 16 Bit/Pixel Writing

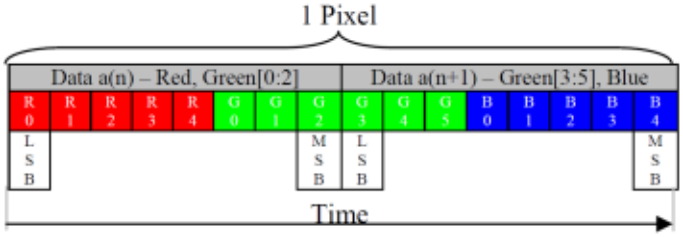


Figure 99: One Pixel Bit and Write Color Orders

The MPU can send the following packet to the display module.

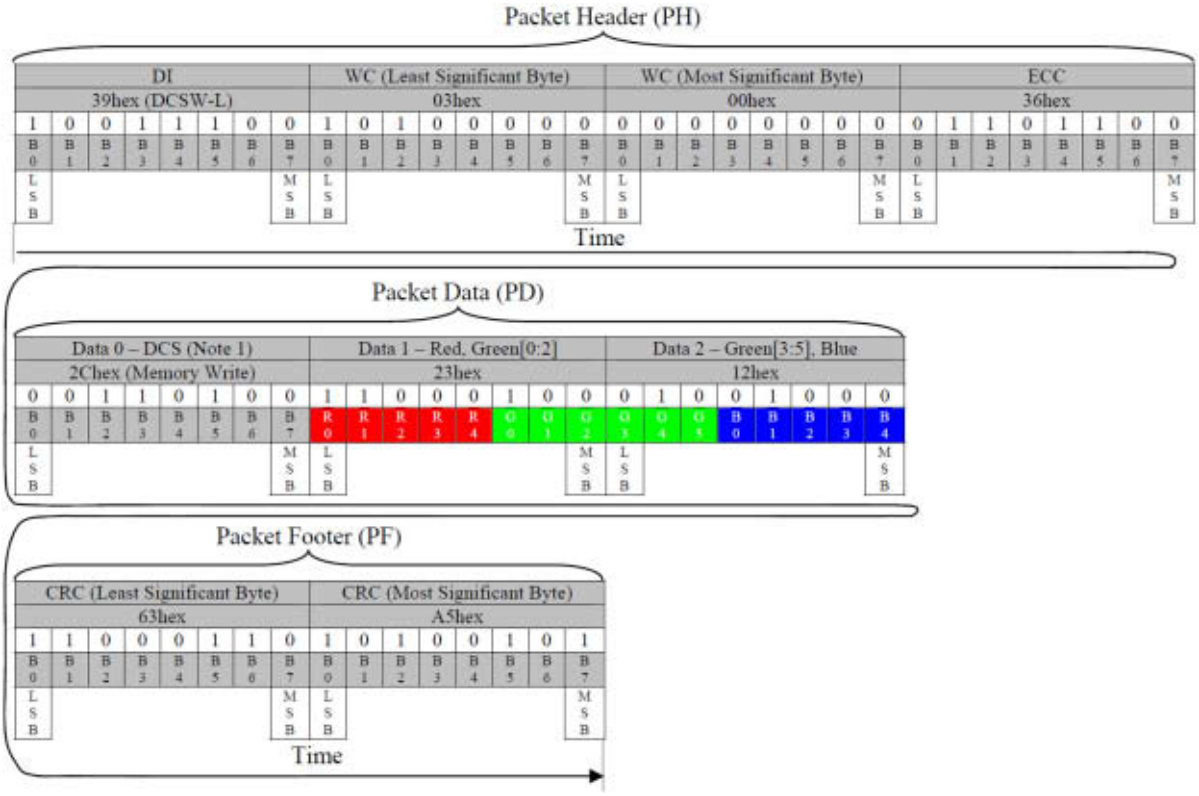


Figure 100: One Pixel Write (DCSW-L) – Example 1[Note]

Note 1. DCS (Data 0) can be Memory Write (2Ch) or Memory Write Continue (3Ch) command.
2. It is possible that one pixel information is split into different packets which end and star as follows: RG – GB (2 packets).
3. A packet can include several pixels (not just one pixel as in this example).

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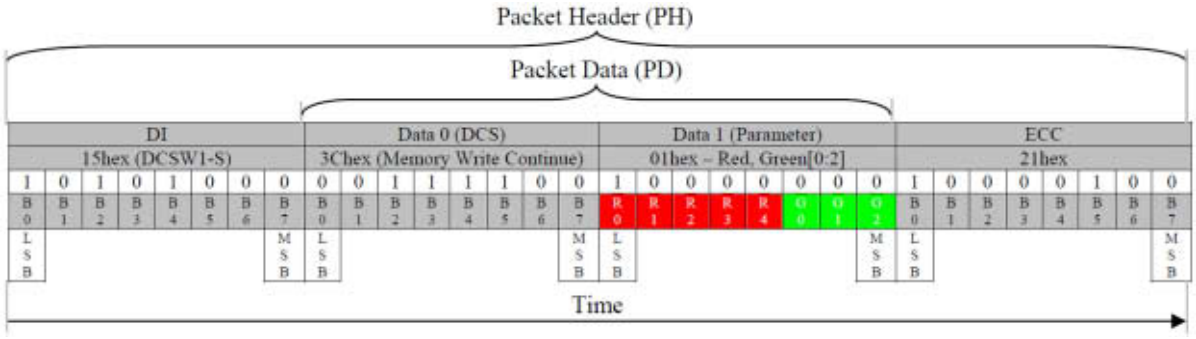


Figure 101: Red/Green [0:2] Subpixel Write (DCSW1-S) – Example 2[Note 1]

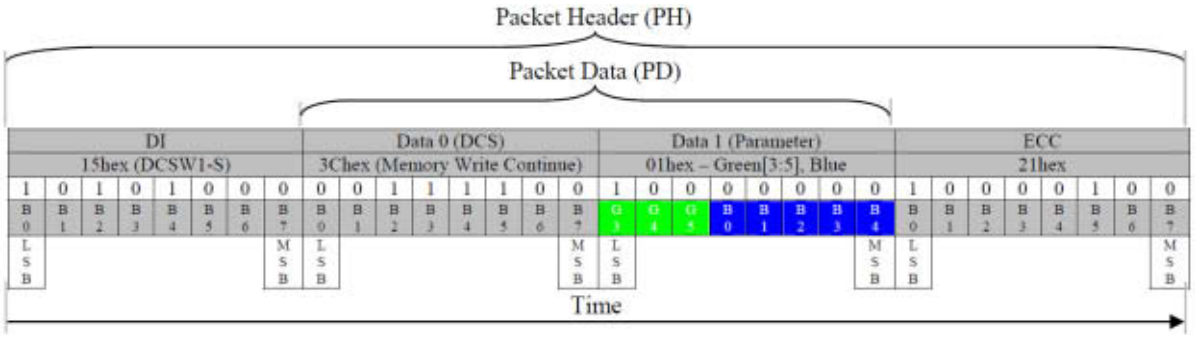


Figure 102: Green [3:5]/Blue Subpixel Write (DCSW1-S) – Example 3[Note 2]

3.12.59. 24 Bit/Pixel Writing



Figure 103: One Pixel Bit and Color Write Orders

The MPU can send the following packet to the display module.

Note 1 DCS (Data 0) can also be "Memory Write" (2Ch) command.

Note 2 1. DCS (Data 0) cannot be "Memory Write" (2Ch) command. It must always be "Memory Write Continue" (3Ch).
2. Previous data byte is R [0:4] G [0:2].

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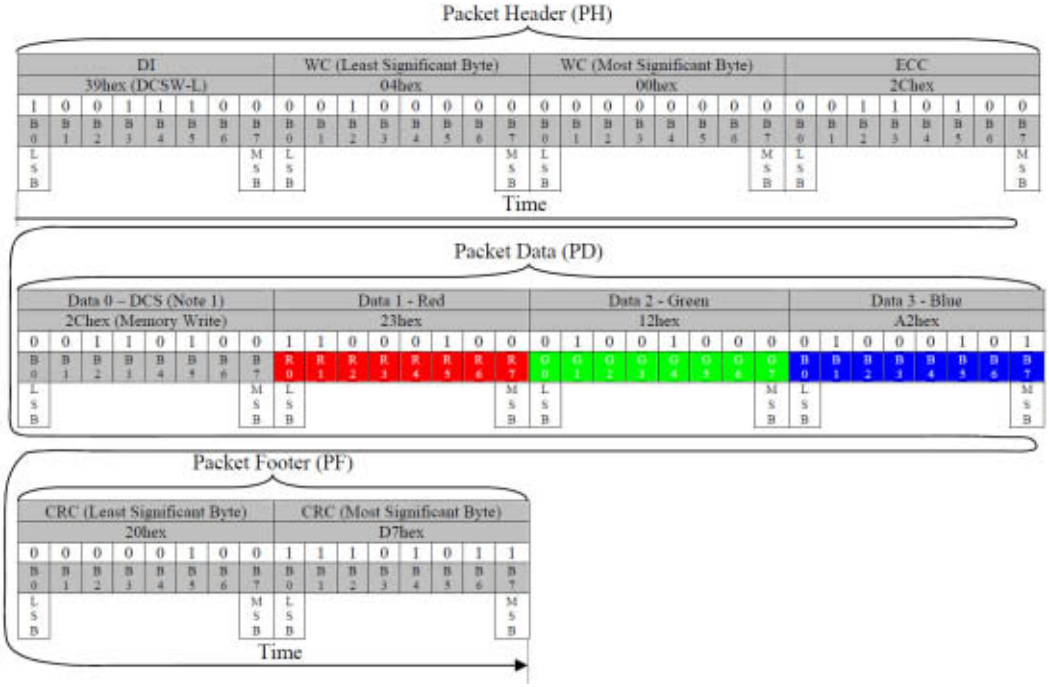


Figure 104: One Pixel Write (DCSW-L) – Example[Note]

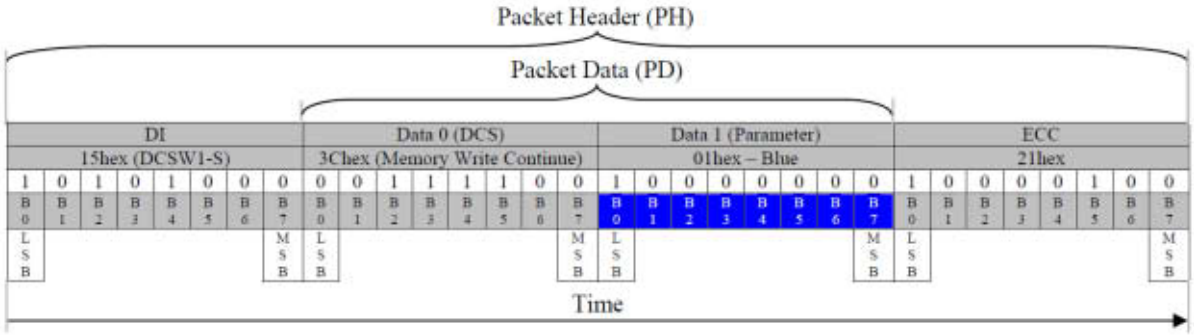


Figure 105: Blue Subpixel Write (DCSW1-S) – Example 2

Notes:

1. DCS (Data 0) cannot be “Memory Write” (2Ch) command. It must always be “Memory Write Continue” (3Ch)
2. Previous data byte is G [0:7].

Note 1. DCS (Data 0) can be Memory Write (2Ch) or Memory Write Continue (3Ch) command.

2. It is possible that one pixel information is split in two or three different packets which end and start as follows:

- *R – GB (2 packets)*
- *RG – B (2 packets)*
- *R – G – B (3 packets)*

3. A packet can include several pixels (not just one pixel as in this example).

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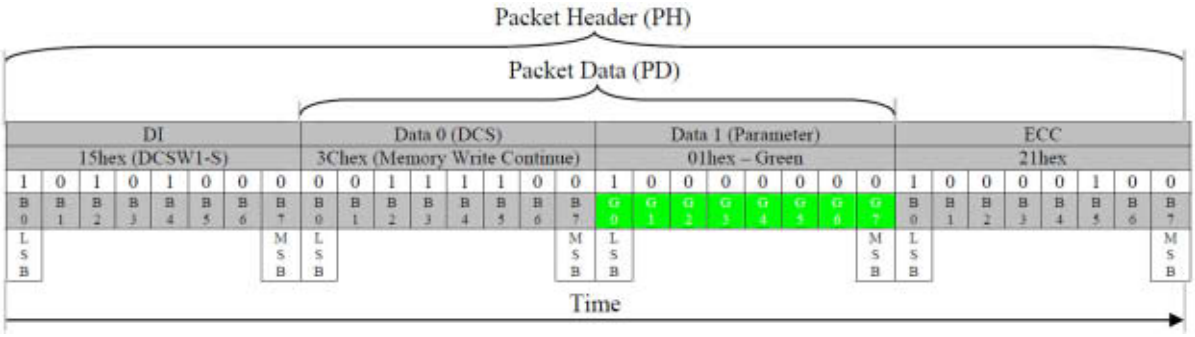


Figure 106: Green Subpixel Write (DCSW1-S) – Example 3

Notes:

1. DCS (Data 0) cannot be “Memory Write” (2Ch) command. It must always be “Memory Write Continue” (3Ch)
2. Previous data byte is R [0:7].

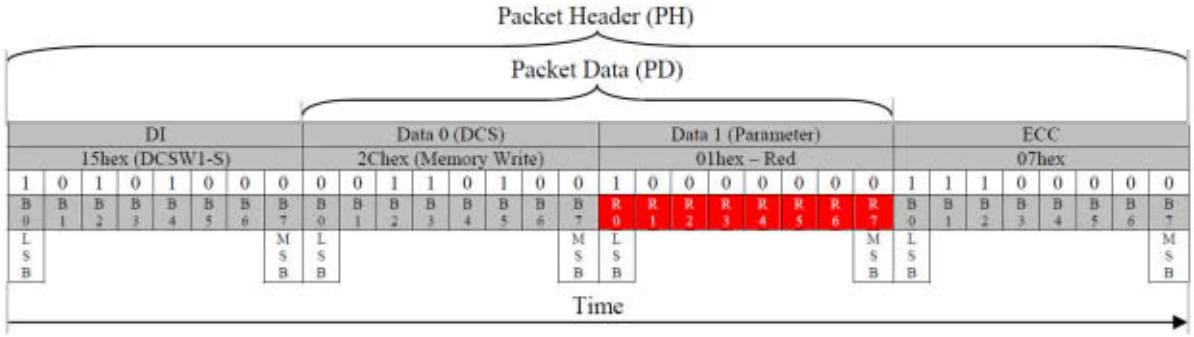


Figure 107: Red Subpixel Write (DCSW1-S) – Example 4

Notes:

1. DCS (Data 0) can also be “Memory Write Continue” (3Ch) command.
2. Previous data byte is B [0:7].

3.12.60. 24 bit/pixel Reading



Figure 108: One Pixel Bit and Color Read Order

The display module can send following packets to the MPU after the MPU has sent a read command “Memory Read (2Eh)” or “Memory Read Continue (3Eh)”.

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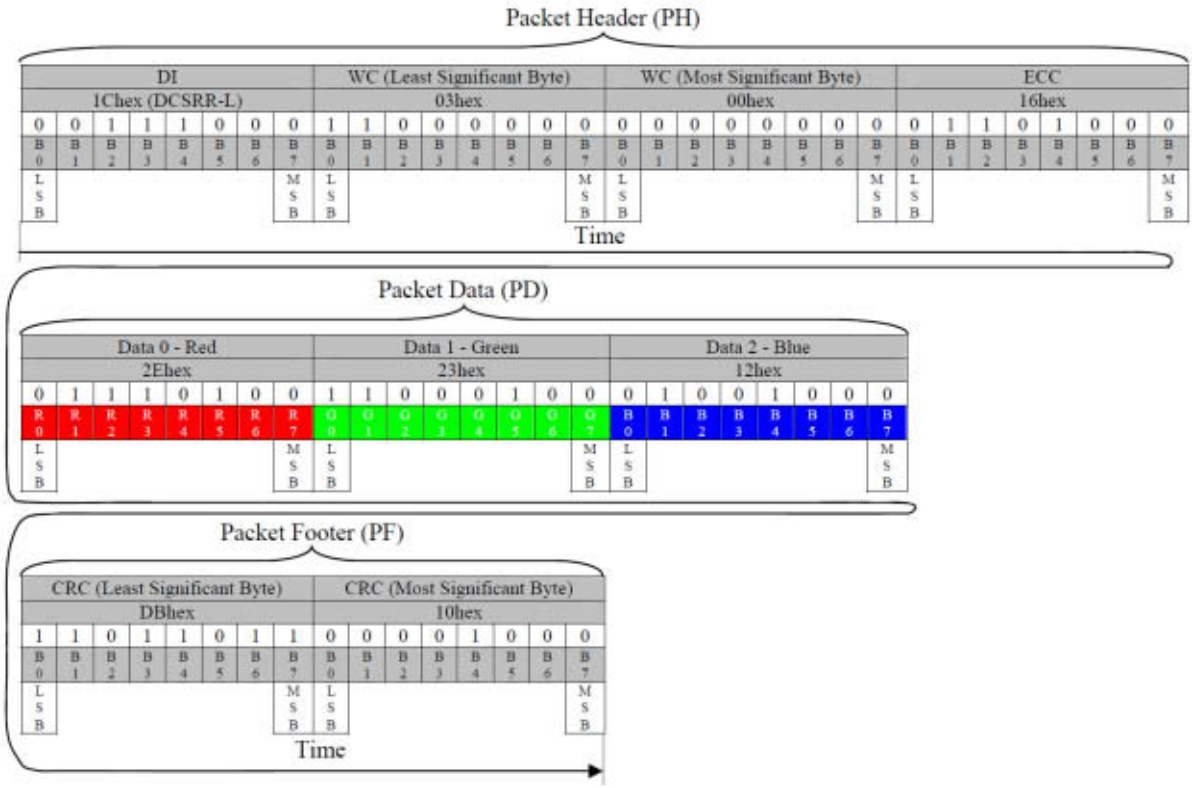


Figure 109: One Pixel Read Response (DCSRR-L) – Example 1

Note: It is possible that one pixel information is split in two or three different packets:

- R – GB (2 packets)
- RG – B (2 packets)
- R – G – B (3 packets)

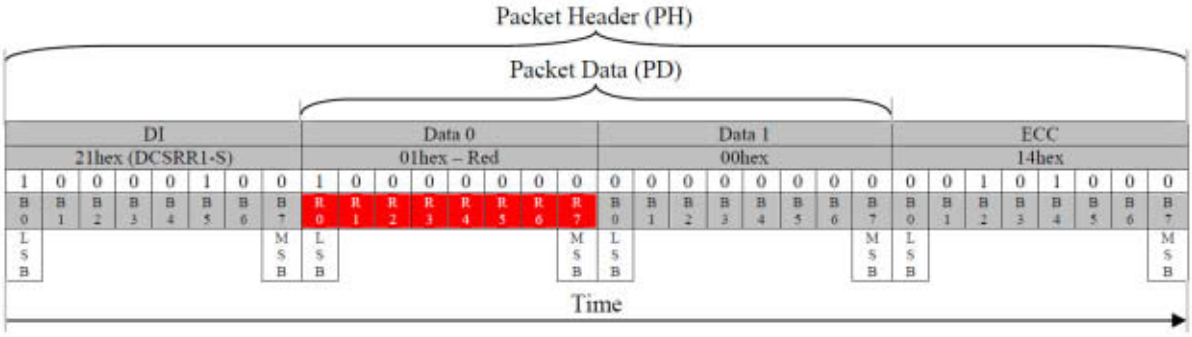


Figure 110: Red Subpixel Response (DCSRR1-S) – Example 2

Notes:

1. Data 1 is always 00h.
2. Previous data byte is B [0:7].

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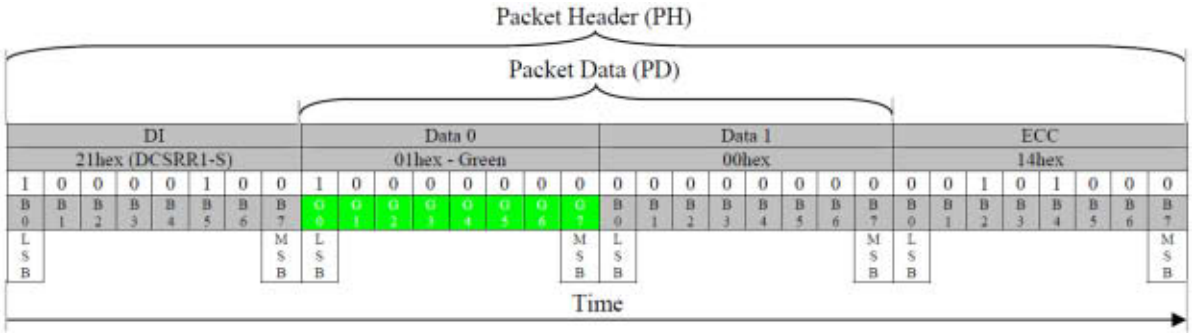


Figure 111: Green Subpixel Response (DCSRR1-S) – Example 3

Notes:

1. Data 1 is always 00h.
2. Previous data byte is R [0:7].

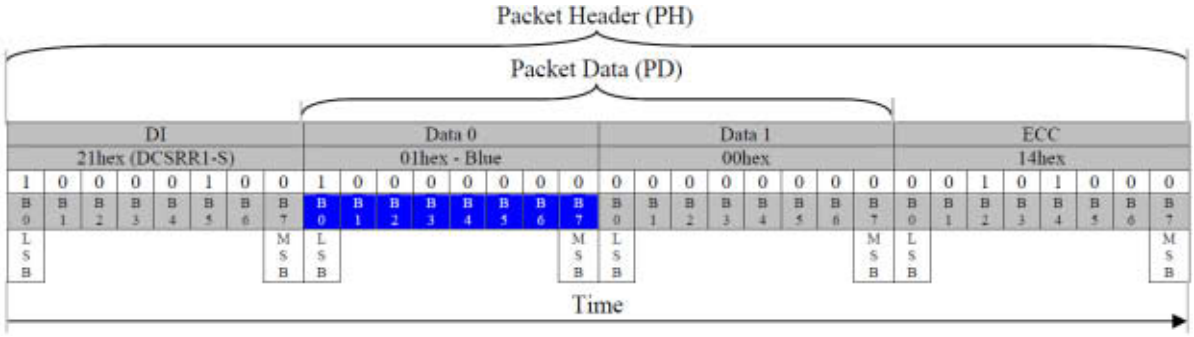


Figure 112: Blue Subpixel Response (DCSRR1-S) – Example 4

Notes:

1. Data 1 is always 00h.
2. Previous data byte is G [0:7].

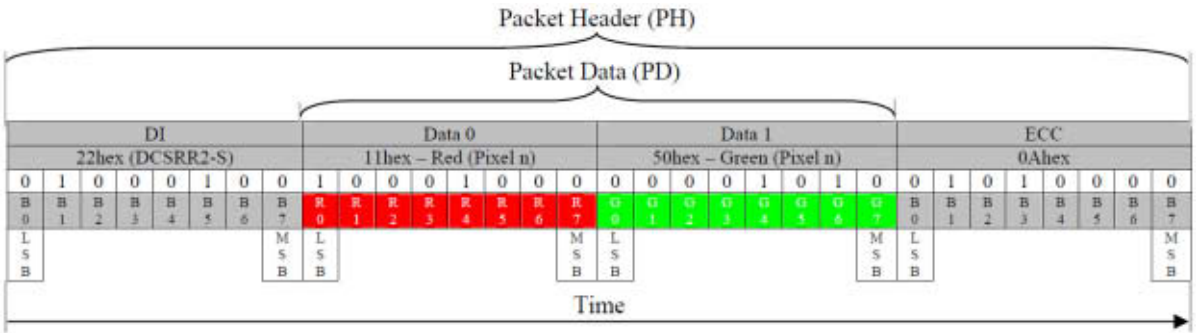


Figure 113: Red and Green Subpixels Response (DCSRR2-S) – Example 5

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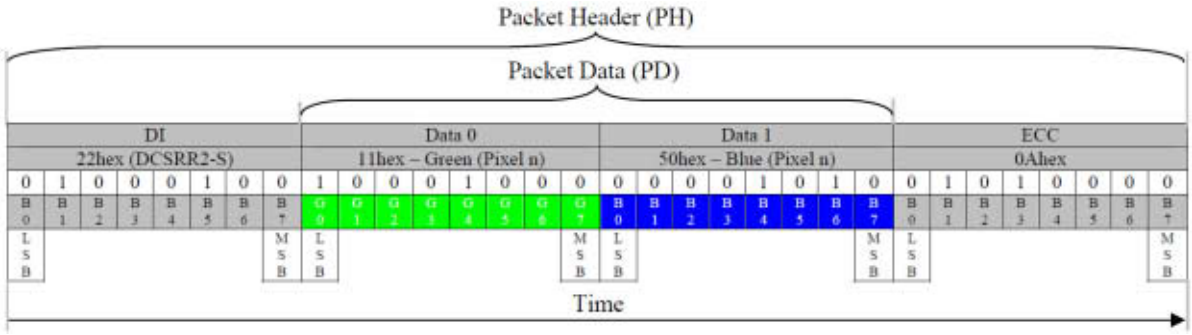


Figure 114: Green and Blue Subpixels Response (DCSRR2-S) – Example 6

Note: Previous data byte is R [0:7].

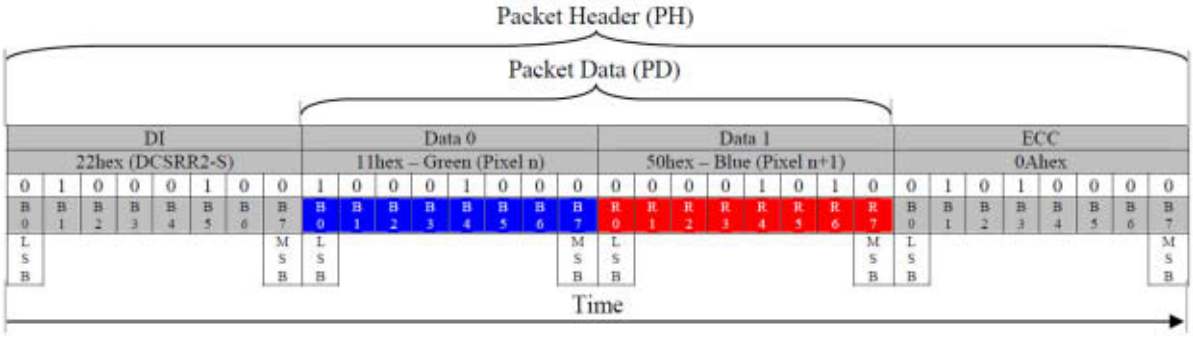


Figure 115: Blue and Red Subpixels Response (DCSRR2-S) – Example 7

Note: Previous data byte is G [0:7].

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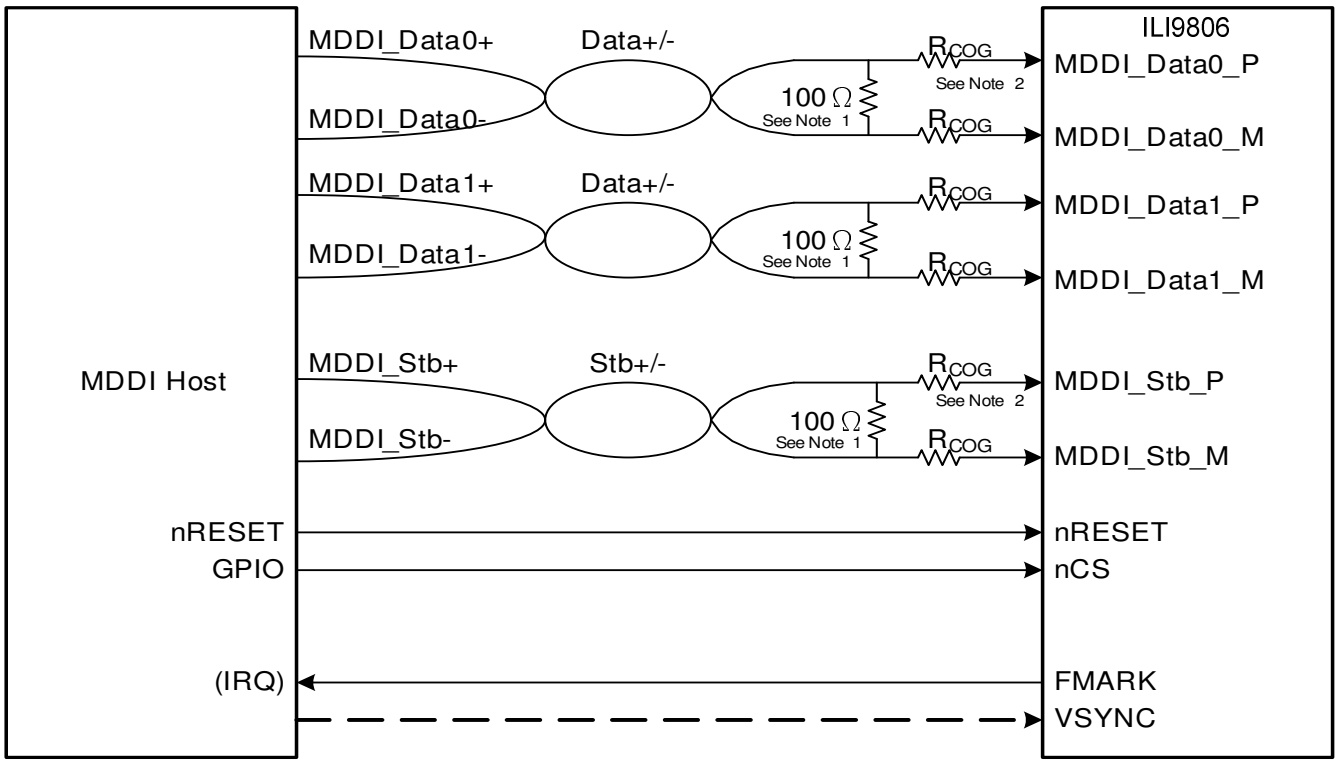
3.13. MDDI (Mobile Display Digital Interface)

MDDI (Mobile display digital interface) is a differential small amplitude serial interface for high-speed data transfer via following 6 lines: Stb+/- (MDDI_STB_P, MDDI_STB_M), Data+/- (MDDI_DATA0_P/ MDDI_DATA0_M, MDDI_DATA1_P/ MDDI_DATA1_M),.

The specifications of MDDI supported by the ILI9806G are compatible to the MDDI specifications disclosed by VESA, Video Electronics Standards Association. The following are the specifications particular to the ILI9806G's MDDI.

ILI9806G MDDI Specifications

- MDDI specification version V1.2 , support Type- Ⅱ (2 Data Lanes)
- High-speed, differential, small-amplitude data transfer via Stb+/-, Data0+/-, Data1+/- lines
- MDDI client: the ILI9806G enables direct connection to the base band (BB) chip without bridge chip
- Cost-performance optimized interface for mobile display systems
  1. Only internal mode (one client) and Forward Link are supported
  2. Hibernation mode to save power consumption
  3. Tearing-free moving picture display via TE/VSYNC interface
  4. Moving picture display with low power consumption, realized by the features 2 ~ 3
  5. Shutdown mode for saving power consumption in the standby state



Notes:

1. An external end resistor of 100 ohm is necessary between Data+ and Data- lines

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2. Make the COG wiring resistances of Data+/-, Stb+/- lines as small as possible (RCOG < 10 ohm).

MDDI Link Protocol (Packets Supported by the ILI9806G)

The MDDI Link Protocol of the ILI9806G is compliant with the MDDI specifications disclosed by VESA. See the MDDI specifications by VESA for details on the MDDI Link Protocol.

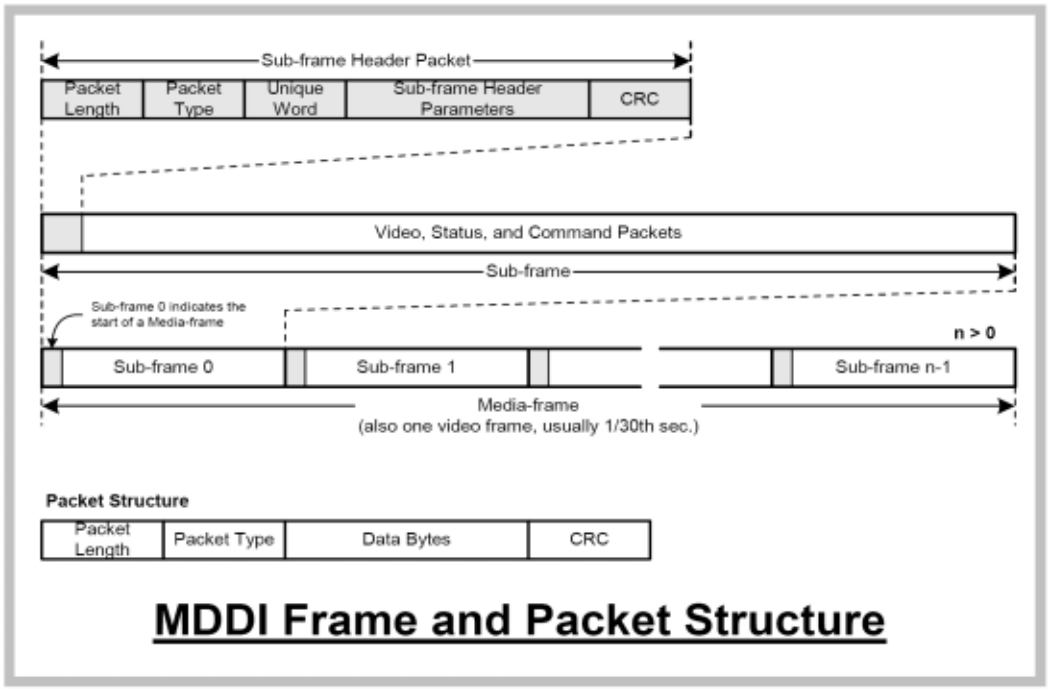
The MDDI packets supported by the ILI9806G are listed in Table 41.

The MDDI packets supported by the ILI9806G are as follows. Do not send packets not supported by the ILI9806G in the system incorporating the ILI9806G.

Refer to MDDI packet structure, sub-frame header packet is placed in front of a sub-frame and some sub-frame construct media-frame together. The following table describes 10 types of packet which is supported in ILI9806G.

Table 41: List of Supported MDDI Packets

Packet	Function	Direction
Sub-frame header packet	Header of each sub frame	Forward
Register access packet	Register setting	Forward
Video stream packet	Video data transfer	Forward
Filler packet	Fill empty packet space	Forward
Reverse link encapsulation packet	Reverse data packet	Reverse
Round-trip delay measurement packet	Host->client->host delay check	Forward/Reverse
Client capability packet	Capability of client check	Reverse
Client request and status packet	Information about client status	Reverse
Forward link skew calibration packet	Propagation delay calibration	Forward
Link shutdown packet	End of frame	Forward



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Figure 116: Forward Link Structure

Sub-frame Header Packet

The Sub-Frame Header Packet is the first packet of every sub-frame.

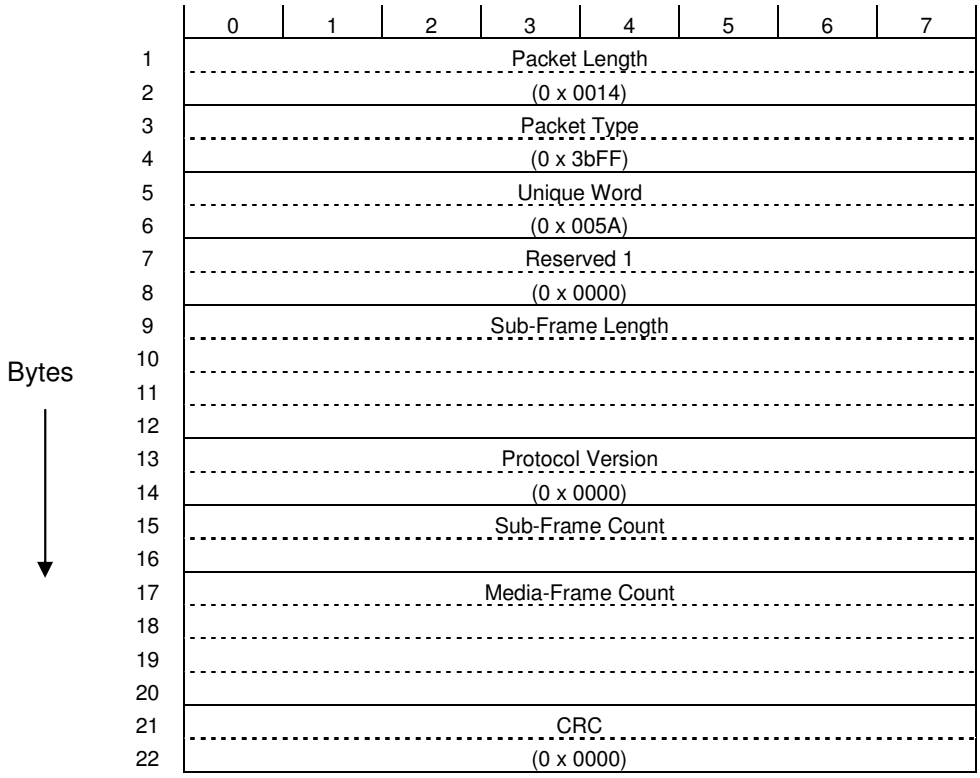


Figure 117: Sub-Frame Header Packet

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Video Stream Packet

The ILI9806G writes image data to the RAM via Video Stream Packet. The window and RAM addresses are set via Register Access Packet.

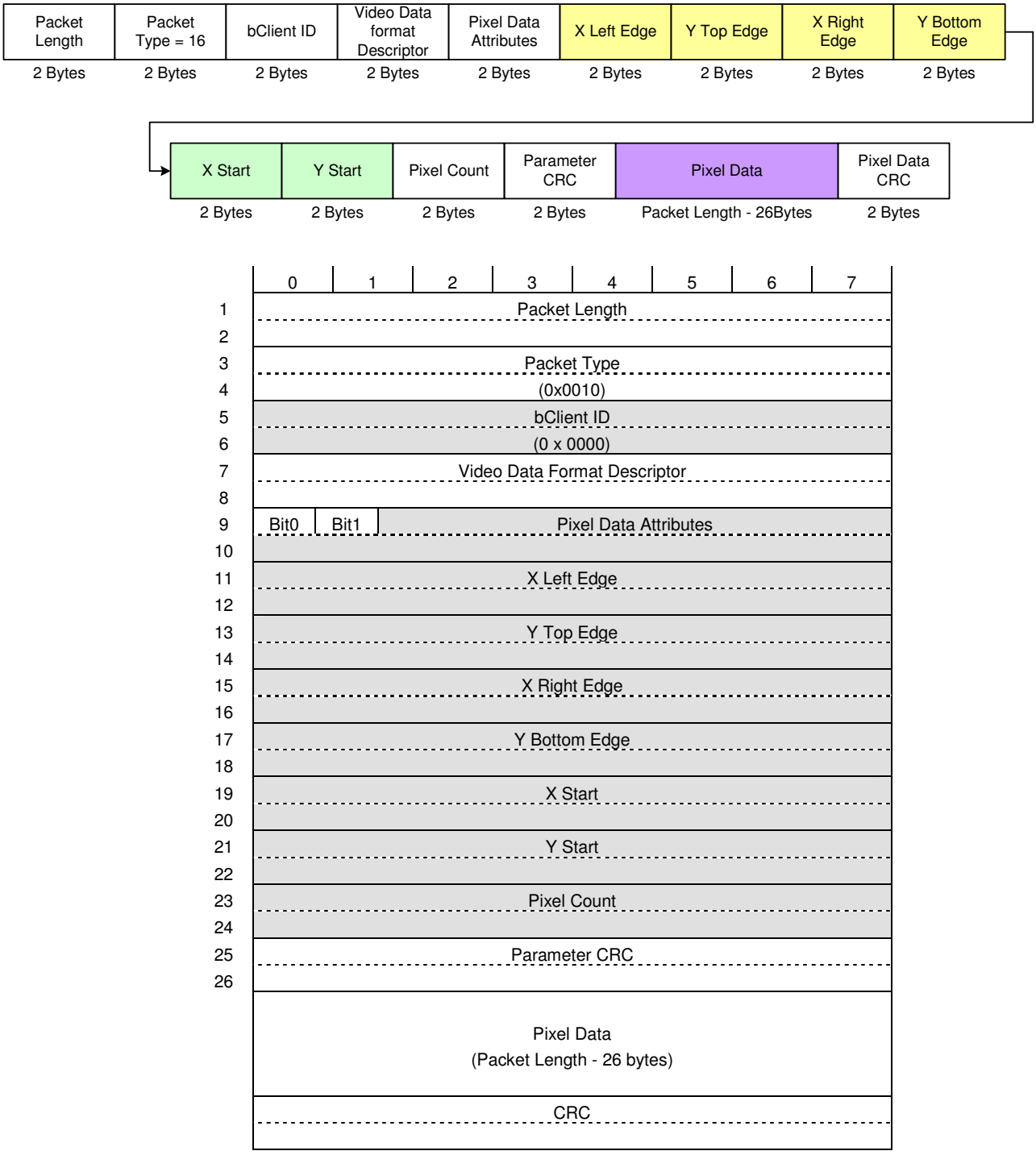


Figure 118: Example of Video Stream Packet Fields[Note]

Note *The parameters colored in gray are not supported by the ILI9806G.*

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Video Data Format Descriptor: This sets the pixel data format. 2 bytes that contain a 16-bit unsigned integer that specifies the format of each pixel in the Pixel Data in the present stream in the present packet.

The bits[15:13]=010: the video data consist of an array of color pixels in raw RGB format where the number of bits per pixel of red is defined by bits 11 through 8, the number of bits per pixel of green is defined by bits 7 through 4, and the number of bits per pixel of blue is defined by bits 3 through 0.

The bit 12=1: each pixel in Pixel Data is packed up against the previous pixel leaving no unused bits.

[15:13]	[12]	[11:8]	[7:4]	[3:0]	
010	1	0x5	0x6	0x5	Packed 16bpp RGB format (R:G:B = 5:6:5)
010	1	0x6	0x6	0x6	Packed 18bpp RGB format (R:G:B = 6:6:6)
010	1	0x8	0x8	0x8	Packed 24bpp RGB format (R:G:B = 8:8:8)
Others					Setting disabled

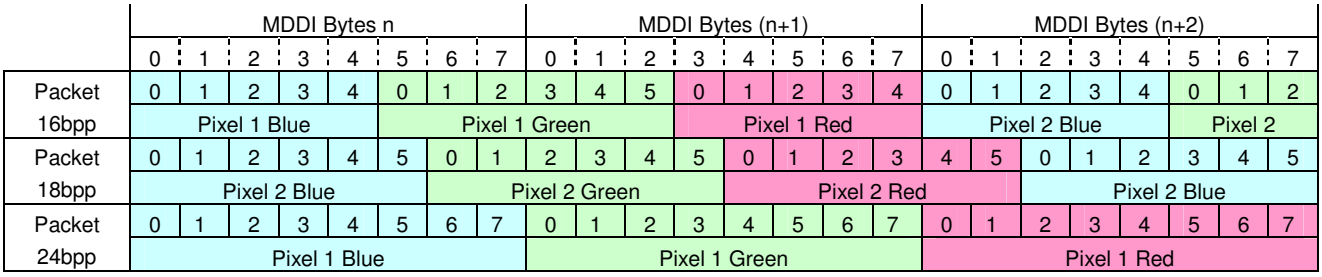


Figure 119: Video Data Format

Pixel Data Attributes: the image data sent via Video Stream Packet is recognized as the data written in the ILI9806G according to the setting in [1:0] bits in this field.

Table 42: Pixel Data Attributes

Pixel Data Attributes	Bits [1:0]	Description
0 x 0000	00	Setting disabled
0 x 0001	01	Setting disabled
0 x 0002	10	Setting disabled
0 x 0003	11	The Video Stream Packet data is recognized as the data written in the ILI9806G.

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Register Access Packet

The Register Access Packet is used when setting instructions to the ILI9806G. (detail see below example).

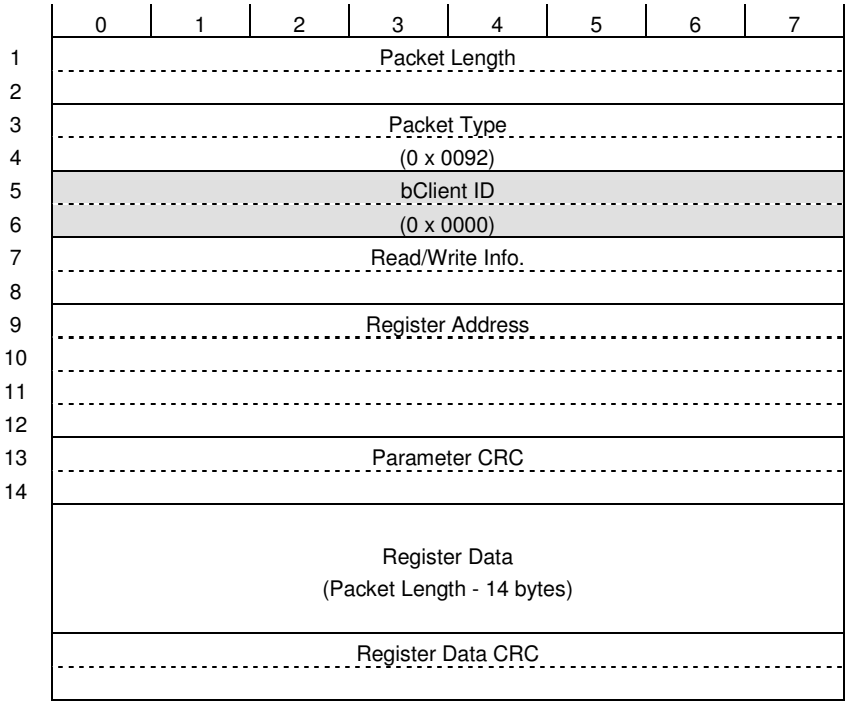


Figure 120: Register Access Packet[Note]

Read/Write Info: Read or Write information in register access. The ILI9806G supports the following access setting.

Bits [15:14]	Bits [13:00]	Description
2'b00	0 x n	Write one register by register access packet
2'b01	0 x n	Reserved
2'b10	0 x n	Read one register by register access packet
2'b11	0 x n	Response to read

Register Address: The index of the register to be accessed is set in the Register Address area, and the Register Address Packet is directed to the ILI9806G, or the sub display as determined by the setting in the Register Address area.

Bits [31:16]	Description
16'h0000	The Register Access Packet is directed to the ILI9806G via the main-display interface.
16'h0001 ~ 16'h7FFF	Setting disabled

Bits [15:0]	Description
16'h0000~16'h FFFF	Bits [15:0] are used as index [15:0].

Register Data: The data for register access is written in the Register Data area. The length of the Register Data Packet depends on the parameter length of the command.

Note *The parameters colored in gray are not supported by the ILI9806G.*

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	0	1	2	3	4	5	6	7
1	Packet Length					**(0 x 16)**		
2						(0 x 00)		
3	Packet Type					(0 x 92)		
4						(0 x 00)		
5	bClient ID					(0 x 00)		
6						(0 x 00)		
7	Read/Write Info.					(0 x 01)		
8						(0 x 00)		
9	Register Address					(index ID [7:0])		
10						(index ID [15:8])		
11						**(0 x 00) → Main Panel (ILI9806G)**		
12						(0 x 00)		
13	Parameter CRC							
14								
15	Register Data List **(Various Length)**					1st Parameter		
16						0 x 00		
17						0 x 00		
18						0 x 00		
19						2nd Parameter		
20						0 x 00		
21						0 x 00		
22						0 x 00		
23	Parameter CRC							
24								

Figure 121: Example of Register Access Packet (write to the ILI9806G)[Note]

Register Access Packet Restrictions

The internal RAM of the ILI9806G is accessible via the Video Stream Packet. RAM access data is not included in the Register Access Packet.

Note *The parameters colored in gray are not supported by the ILI9806G.*

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Link Shutdown Packet

The Link Shutdown Packet is sent from the host to the client to indicate that the MDDI data transmission and strobe will be shut down and go into a low-power hibernation state.

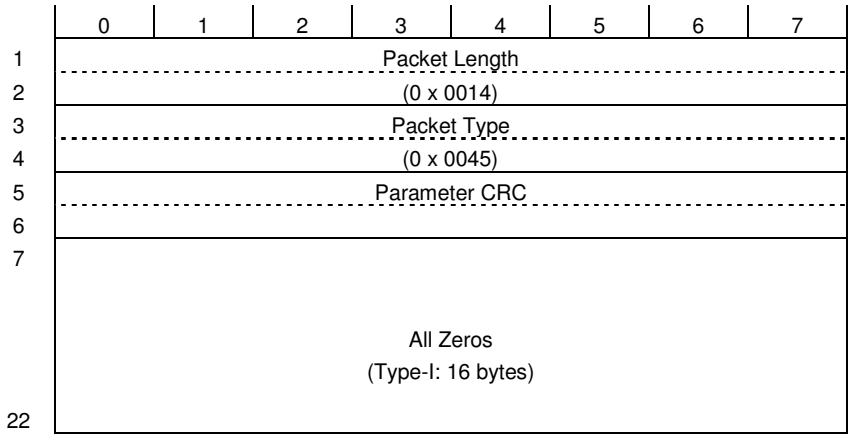


Figure 122: Link Shutdown Packet

Filler Packet

A Filler Packet is sent when no other information is available to be sent on the forward or reverse link.

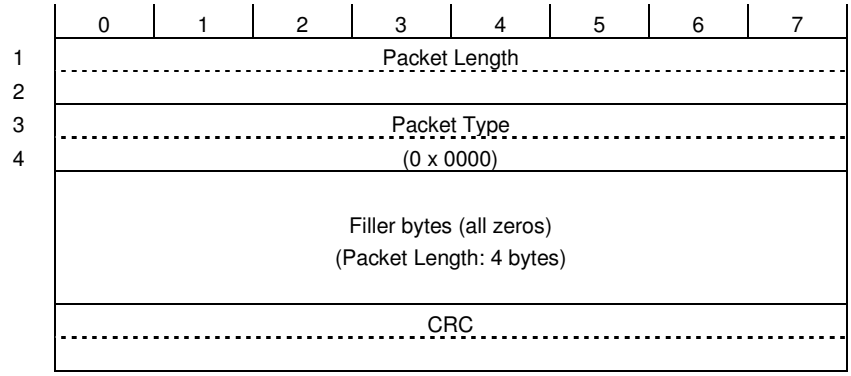


Figure 123: Filler Packet

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Hibernation Setting

The client MDDI supports the Hibernation setting. There are two ways to cancel the Hibernation setting selected according to the condition of use. The two cancellation methods are listed below and described in detail in the following page.

Hibernation Cancellation	
Method	Description
Host-initiated wake up	In power-saving mode such as standby
TE-initiated wake up	Reduced power consumption in transferring moving picture data Host-initiated wake up triggered by the output from TE.

The Hibernation setting and cancellation sequence must be compatible with the VESA-MDDI specifications.

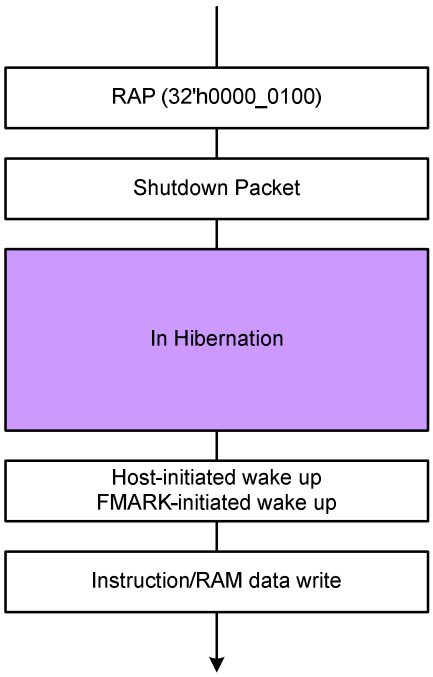


Figure 124: Hibernation Mode Sequence

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The host initiated wake up is described below without contention from the client trying to wake up at the same time. The following sequence of events is illustrated in the figure below.

A. The host sends a Link Shutdown Packet to inform the client that the link will transition to the low power hibernation state.
B. Following the CRC of the Link Shutdown Packet, the host toggles MDDI_Stb for 64 cycles to allow processing in the client to finish before it stops MDDI_Stb from toggling which stops the recovered clock in the client device. During the interval, the host initially sets MDDI_Data0 to the logical zero level, and then disables the MDDI_Data0 output in the range of 16 to 48 MDDI_Stb cycles (including output disable propagation delays) after the CRC. It may be desirable for the client to place its high-speed receivers for MDDI_Data0 and MDDI_Stb into a low power state any time after 48 MDDI_Stb cycles after the CRC and before point C.
C. The host enters the low power hibernation state by disabling the MDDI_Data0 and MDDI_Stb drivers and by placing the host controller into a low power hibernation state. It is also allowable for MDDI_Stb to be driven to the logical zero level or to continue toggling during hibernation. The client is also in the low power hibernation state.
D. After a while, the host begins the line restart sequence by enabling the MDDI_Data0 and MDDI_Stb driver outputs. The host drivers MDDI_Data0 to the logical one level and MDDI_Stb to the logical zero level for at least 200nsec after MDDI_Data0 reaches the valid logical one level and MDDI_Stb reaches the valid logical zero level before driving pulses on MDDI_Stb. This gives the client sufficient time to prepare to receive high speed pulses on MDDI_Stb. The client first detects the wake up pulse using a low power differential receiver having a +125mV input offset voltage.
E. The host drivers are fully enabled and MDDI_Data0 is being driven to the logical one level. The host begins to toggle MDDI_Stb in a manner consistent with having the logical zero level on MDDI_Data0 for a duration of 150 MDDI_Stb cycles.
F. The host drives MDDI_Data0 to the logical zero level for 50 MDDI_Stb cycles. The client begins to look for the Sub frame Header Packet after MDDI_Data0 is at the logical zero level for 40 MDDI_Stb cycles.
G. The host begins to transmit data on the forward link by sending a Sub-frame Header packet. Beginning at point G the MDDI host generates MDDI_Stb based on the logic level on MDDI_Data0 so that proper data-strobe encoding commences form point G.

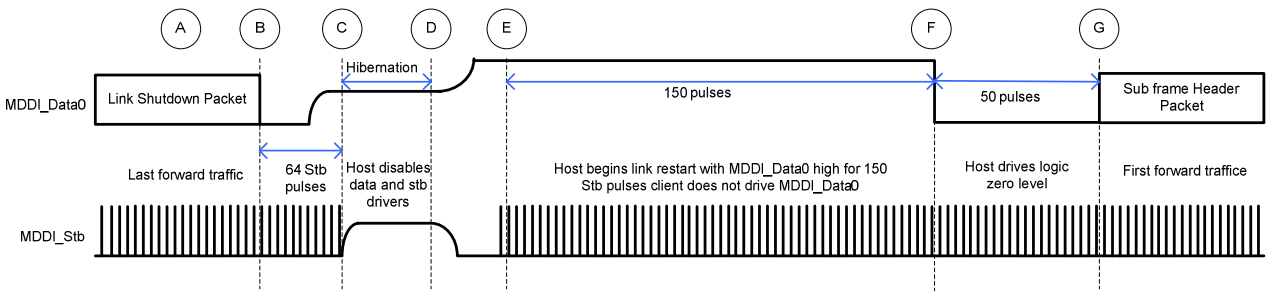


Figure 125: Host Initiated Wake-up without Contention

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3.14. Display Data Format

3.14.1. 3-line Serial Interface

The 3-line/9-bit serial bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to X011.

The Figure 126 is an interface with 8080 MPU system interface.

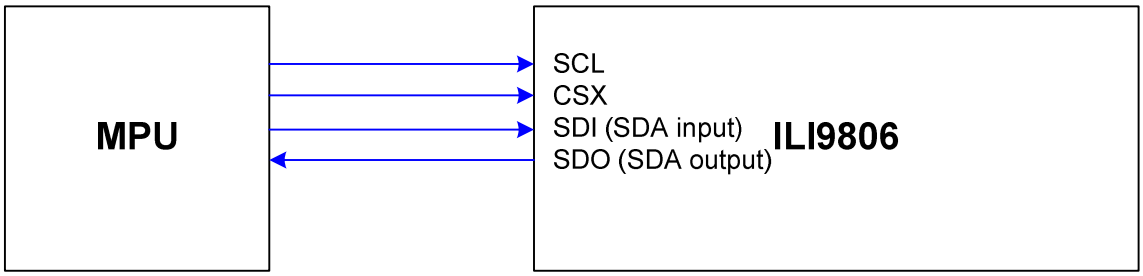


Figure 126: 3-line Serial Interface

The display data formats available for this type of color depth are:

-- 65K-Colors, RGB 5, 6, 5 bits input data (Set Standard Command 3Ah, DBI [2:0] = 101)

- 262K-Colors, RGB 6, 6, 6 bits input data (Set Standard Command 3Ah, DBI [2:0] = 110)

-16.7M colors, RGB 8, 8, 8 bits input (Set Standard Command 3Ah, DBI [2:0] = 111)

3.14.2. SPI Data for 3-bit/pixel (RGB 1-1-1 bits input), 8-color

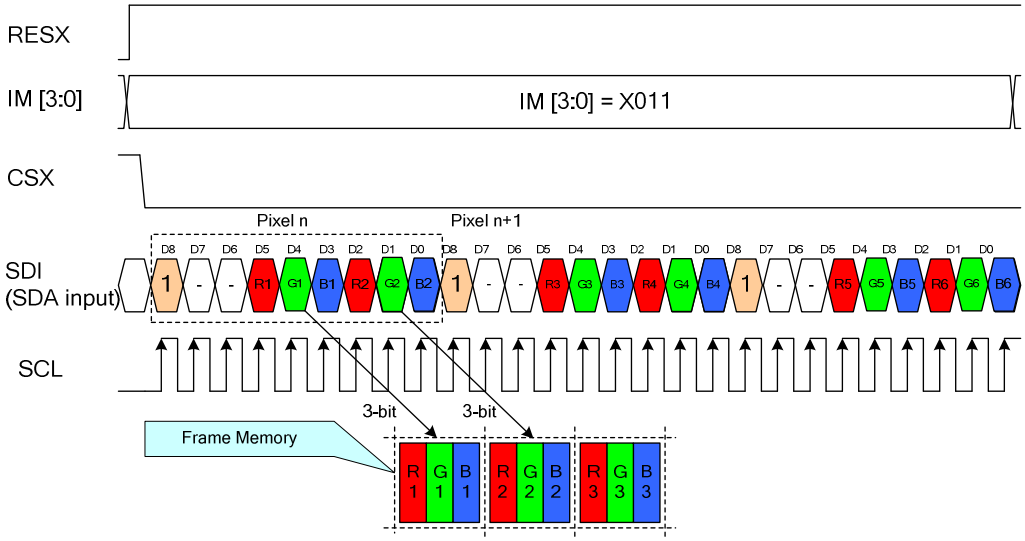


Figure 127: SPI Data for 3-bit/pixel (RGB 1-1-1 bits input), 8-color

Notes:

1. One pixel data contains 3-bit color depth information.
2. '-' = void

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3.14.3. SPI Data for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

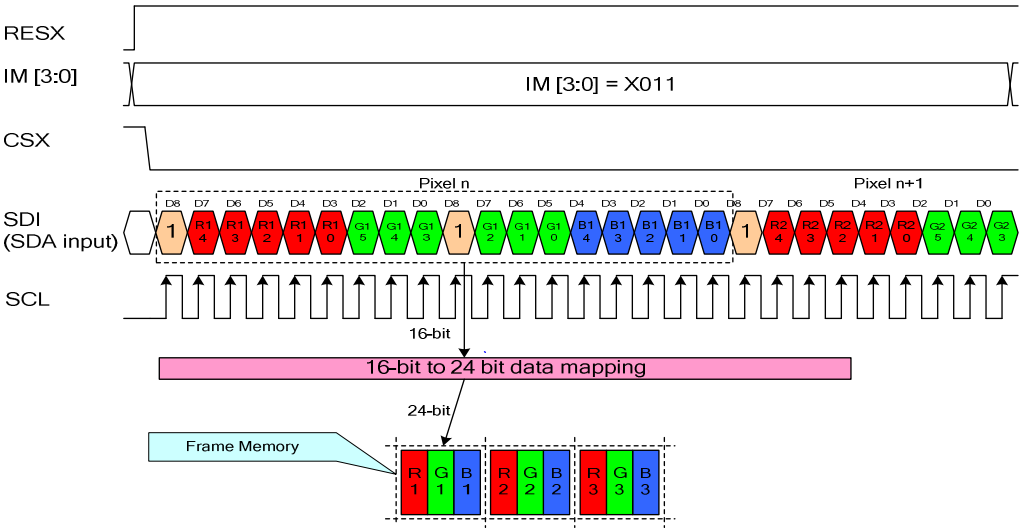


Figure 128: SPI Data for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

Notes:

1. One pixel data contains 16-bit color depth information.
2. The most significant bits are: R x 4, G x 5 and B x 4.
3. The least significant bits are: R x 0, G x 0 and B x 0.

3.14.4. SPI Data for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

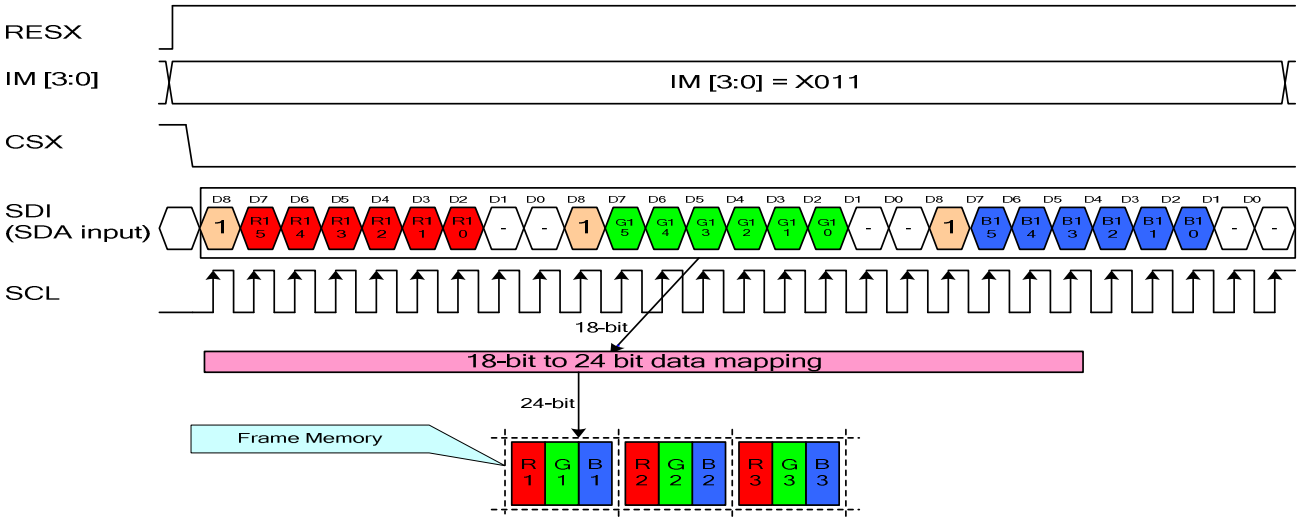


Figure 129: SPI Data for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

Notes:

1. One pixel data contains 18-bit color depth information.
2. The most significant bits are: R x 5, G x 5 and B x 5.
3. The least significant bits are: R x 0, G x 0 and B x 0.
4: '-' = void

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3.14.5. SPI Data for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

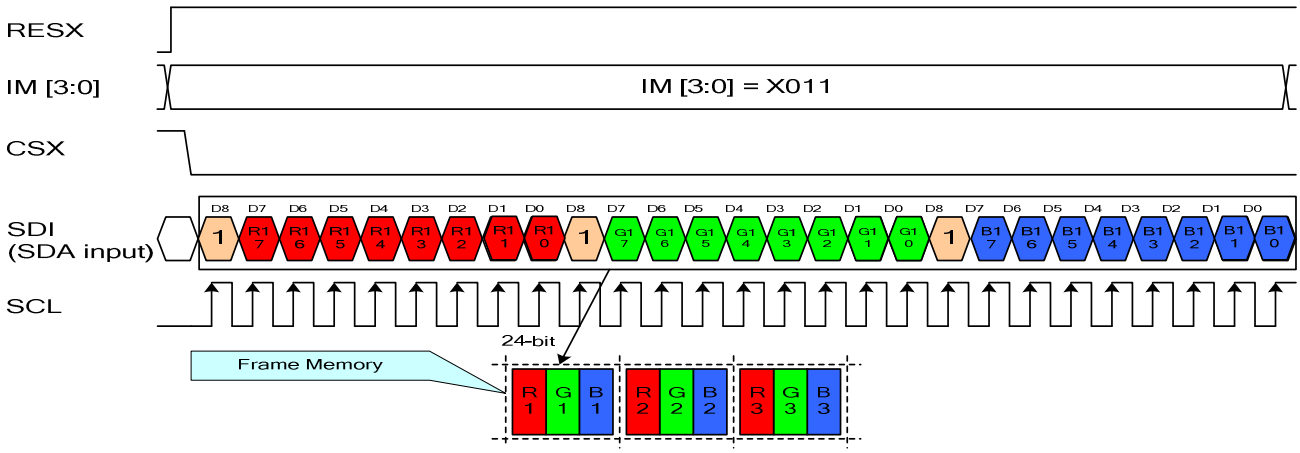


Figure 130: SPI Data for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. One pixel data contains 24-bit color depth information.
2. The most significant bits are: R x 7, G x 7 and B x 7.
3. The least significant bits are: R x 0, G x 0 and B x 0.

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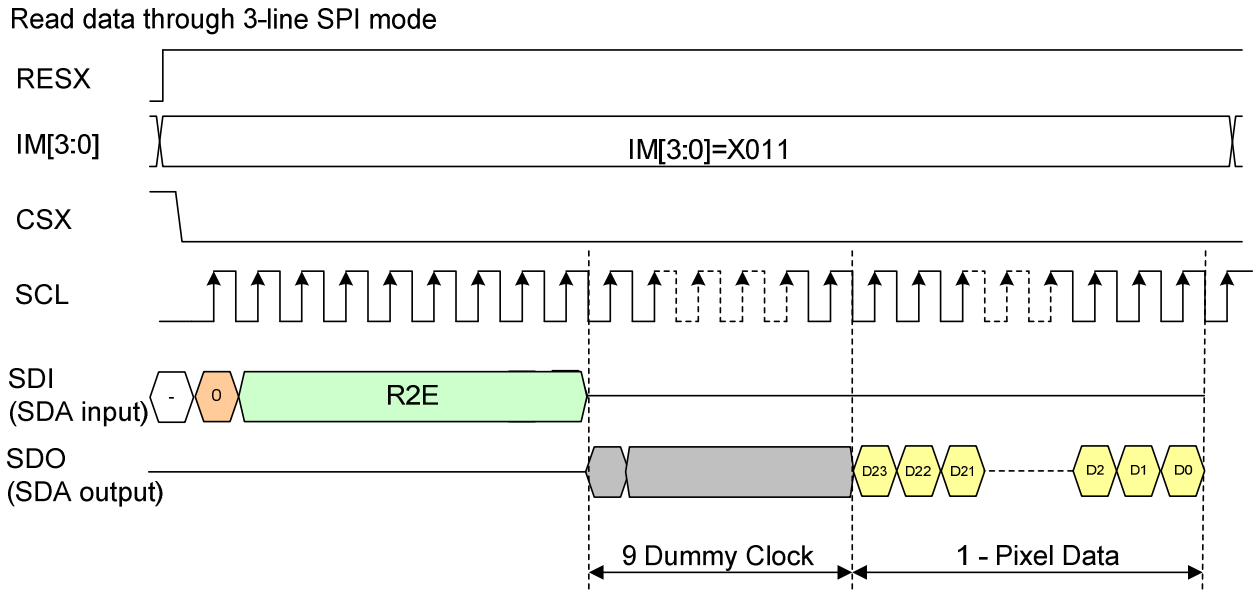


Figure 131: 3-line SPI Mode Read Data

Note: '-' = void

3.14.6. 8-bit Parallel MPU Interface

The DBI TYPE B 8-bit parallel bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to 0000.

The Figure 132 shows this system interface.

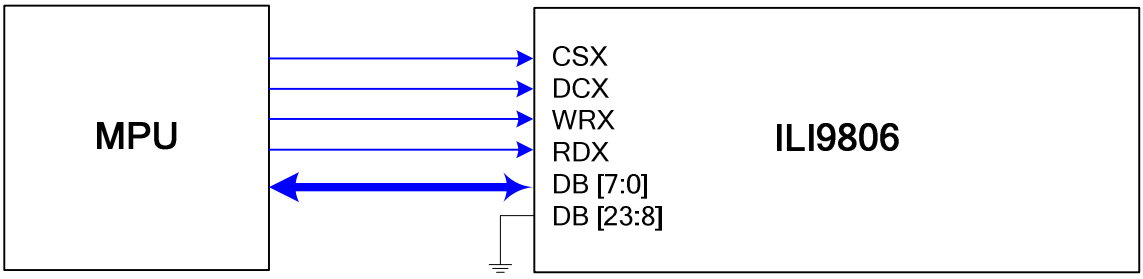


Figure 132: 8-bit Parallel MPU Interface

The display data formats available for this type of color depth are:

- 65K-Colors, RGB 5, 6, 5 bits input data (Set Standard Command 3Ah, DBI [2:0] = 101)
- 262K-Colors, RGB 6, 6, 6 bits input data (Set Standard Command 3Ah, DBI [2:0] = 110)
- 16.7M-Colors, RGB 8, 8, 8 bits input data (Set Standard Command 3Ah, DBI [2:0] = 111)

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3.14.7. 8-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

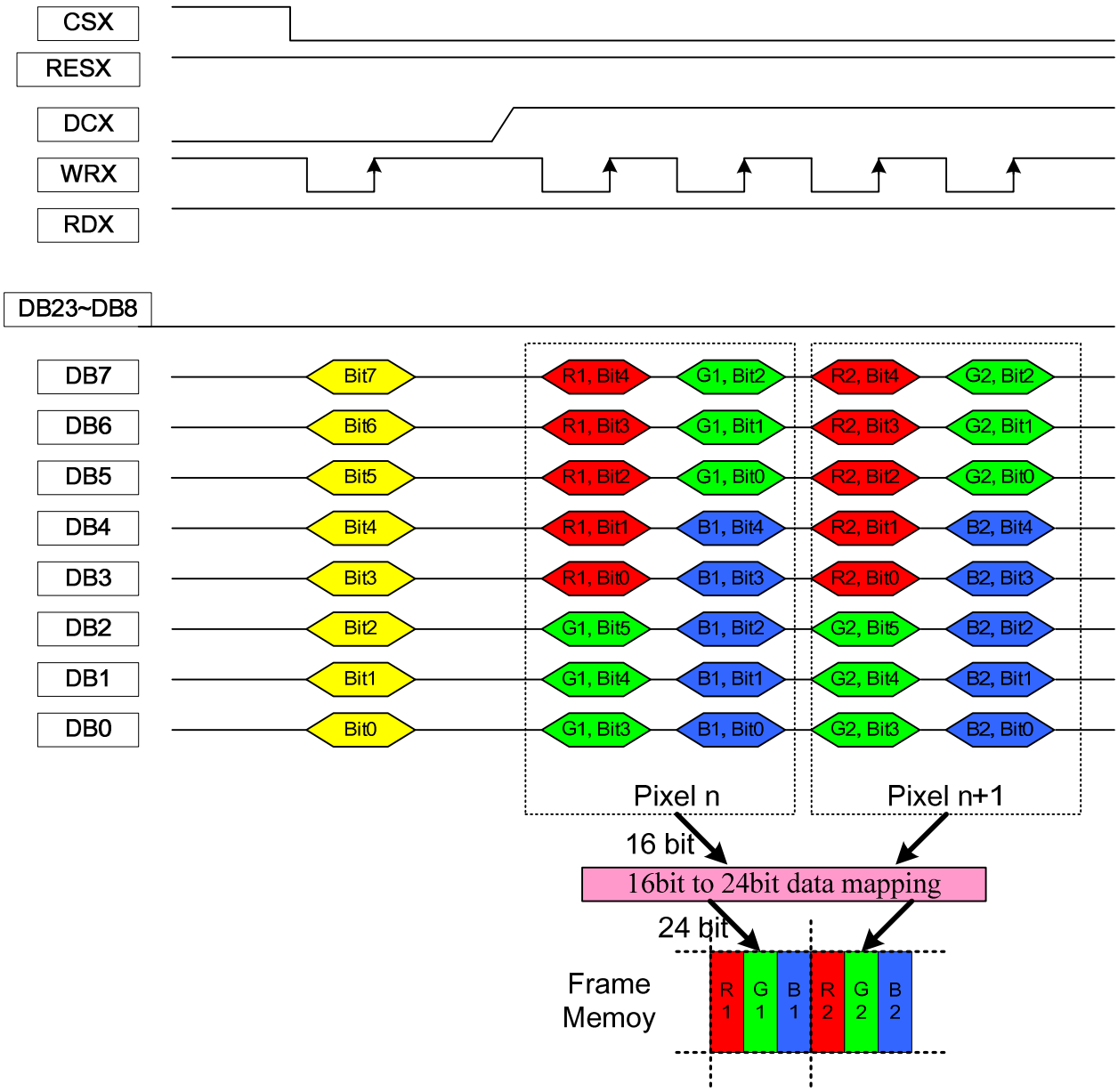


Figure 133: 8-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

Notes:

1. The data order is as follows: MSB = DB7, LSB = DB0 and picture data is MSB = Bit 5, LSB = Bit 0 for Green data and MSB = Bit 4, LSB = Bit 0 for Red and Blue data.
2. 2-times transfer is used to transmit 1 pixel data to the 16-bit color depth information.
3. '-' = void

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3.14.8. 8-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

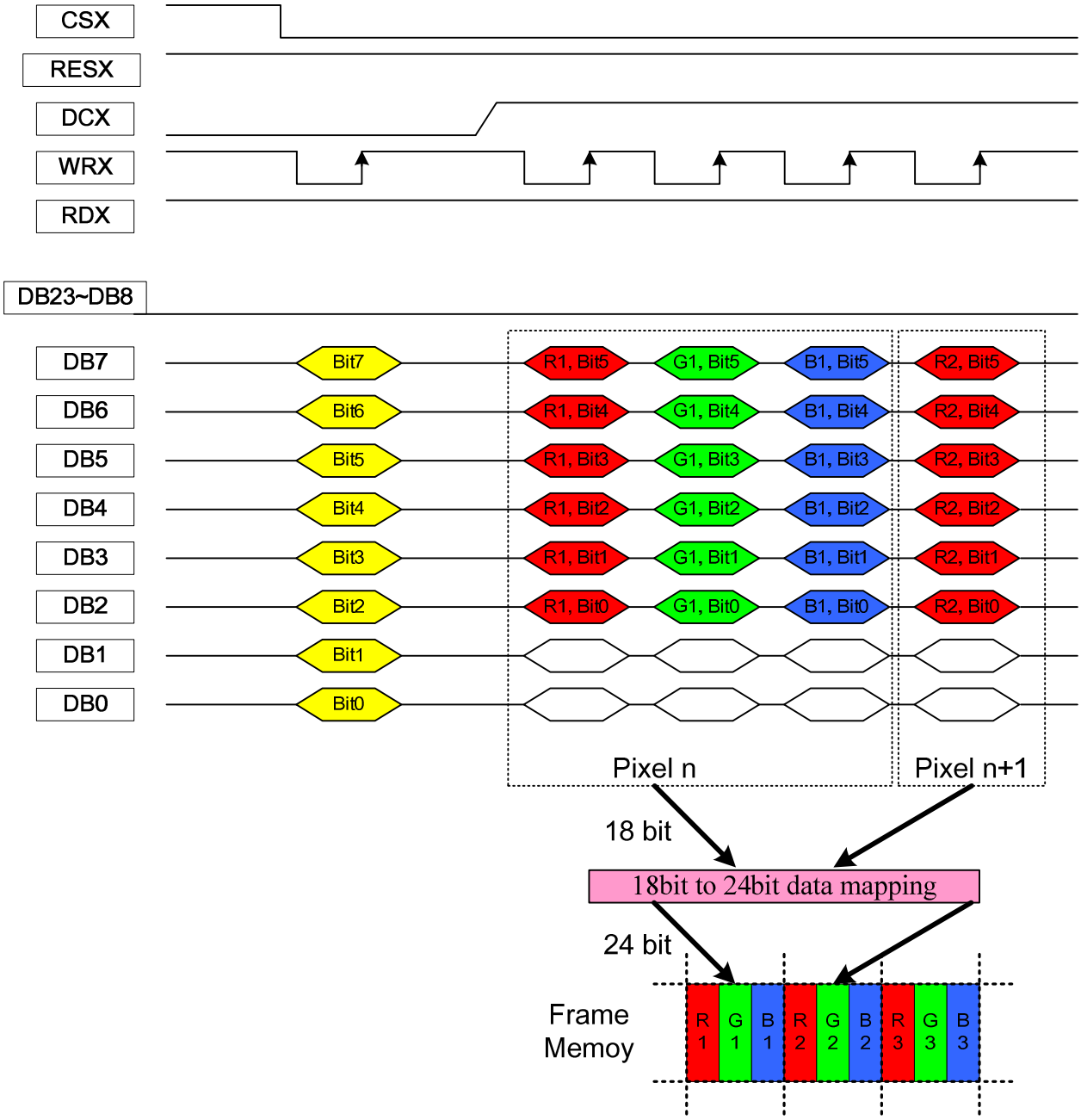


Figure 134: 8-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

Notes:

1. The data order is as follows: MSB = DB7, LSB = DB0 and picture data is MSB = Bit 5, LSB = Bit 0 for Green, Red and Blue data.
2. 3-times transfer is used to transmit 1 pixel data to the 18-bit color depth information.
3. '-' = void

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3.14.9. 8-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

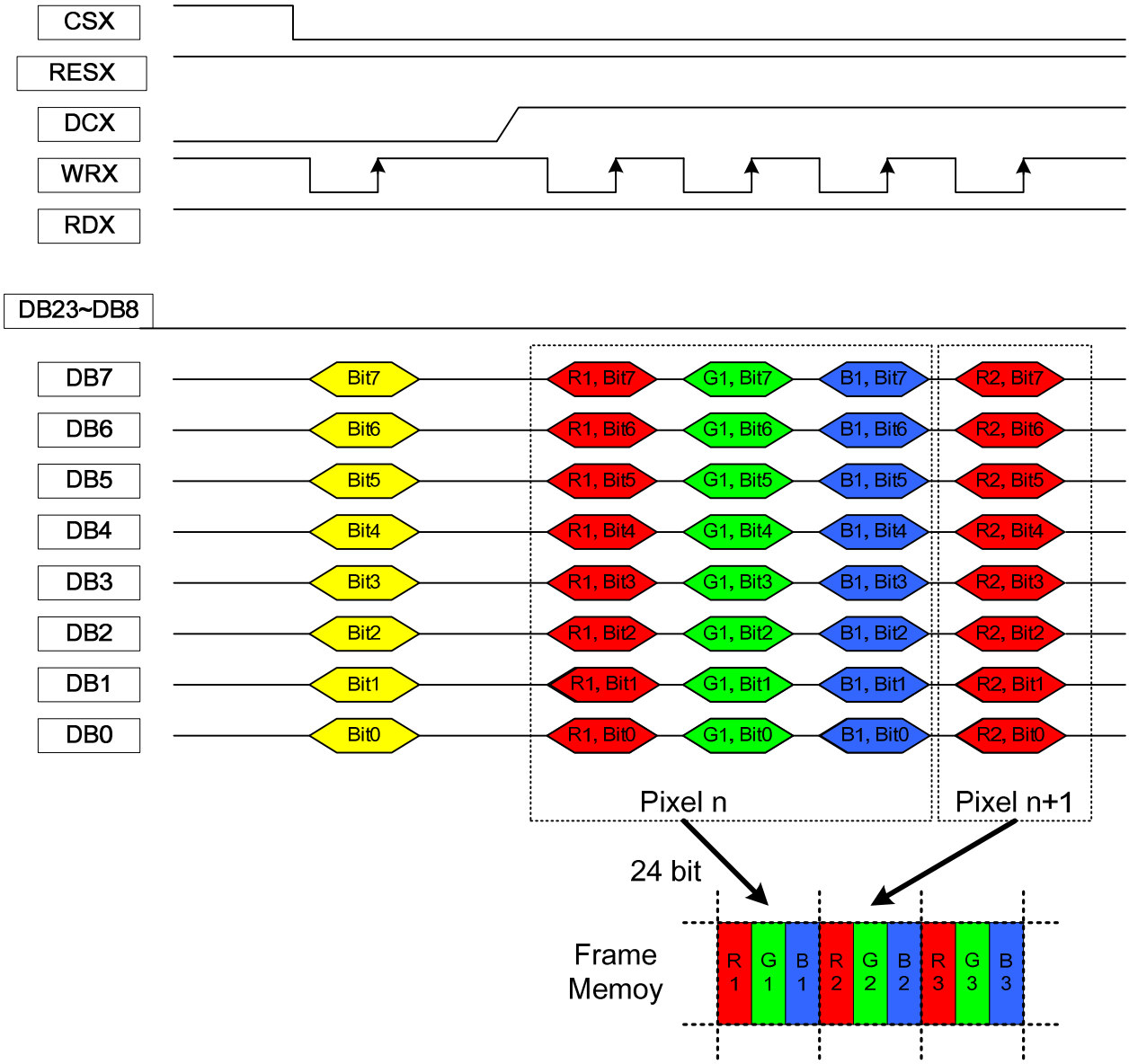


Figure 135: 8-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. The data order is as follows: MSB = DB7, LSB = DB0 and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 3-times transfer is used to transmit 1 pixel data to the 24-bit color depth information.
3. '-' = void

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3.14.10. 9-bit Parallel MPU Interface

The DBI TYPE B 9-bit parallel bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to 1100.

The Figure 136 shows this system interface.

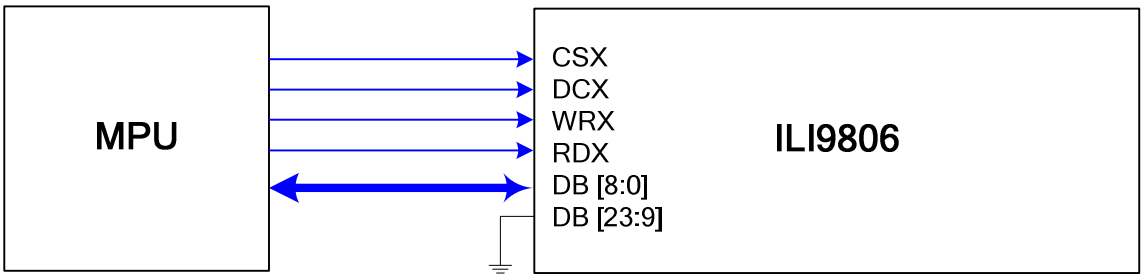


Figure 136: 9-bit Parallel MPU Interface

The display data formats available for this type of color depth are:

- 65K-Colors, RGB 5, 6, 5 bits input data (Set Standard Command 3Ah, DBI [2:0] = 101)
- 262K-Colors, RGB 6, 6, 6 bits input data (Set Standard Command 3Ah, DBI [2:0] = 110)
- 16.7M-Colors, RGB 8, 8, 8 bits input data (Set Standard Command 3Ah, DBI [2:0] = 111)

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3.14.11. 9-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

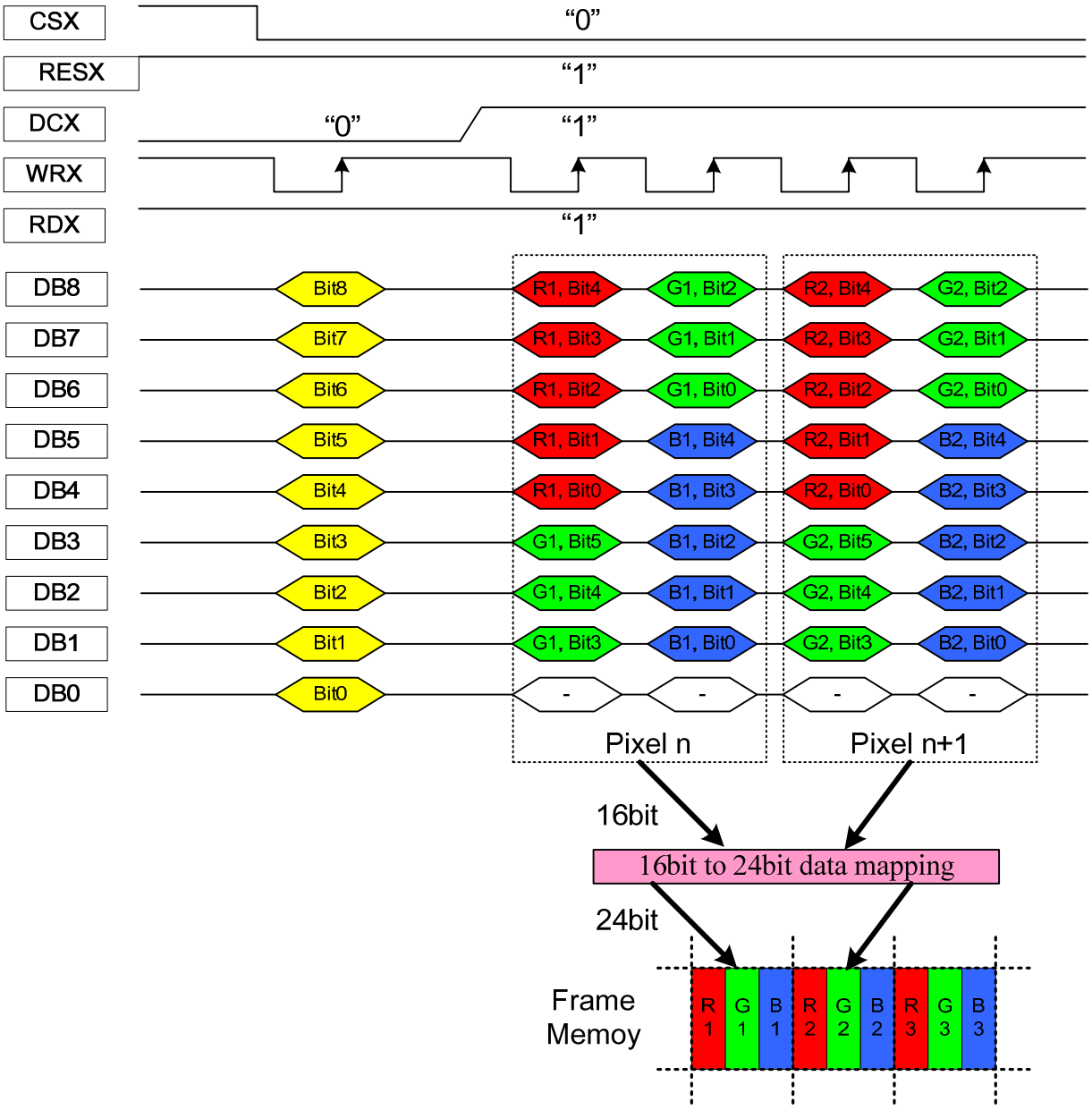


Figure 137: 9-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

Notes:

1. The data order is as follows: MSB = DB8, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, MSB = Bit 4, LSB = Bit 0 for Red and Blue data.
2. 2-times transfer is used to transmit 1 pixel data to the 16-bit color depth information.
3. '-' = void

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3.14.12. 9-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

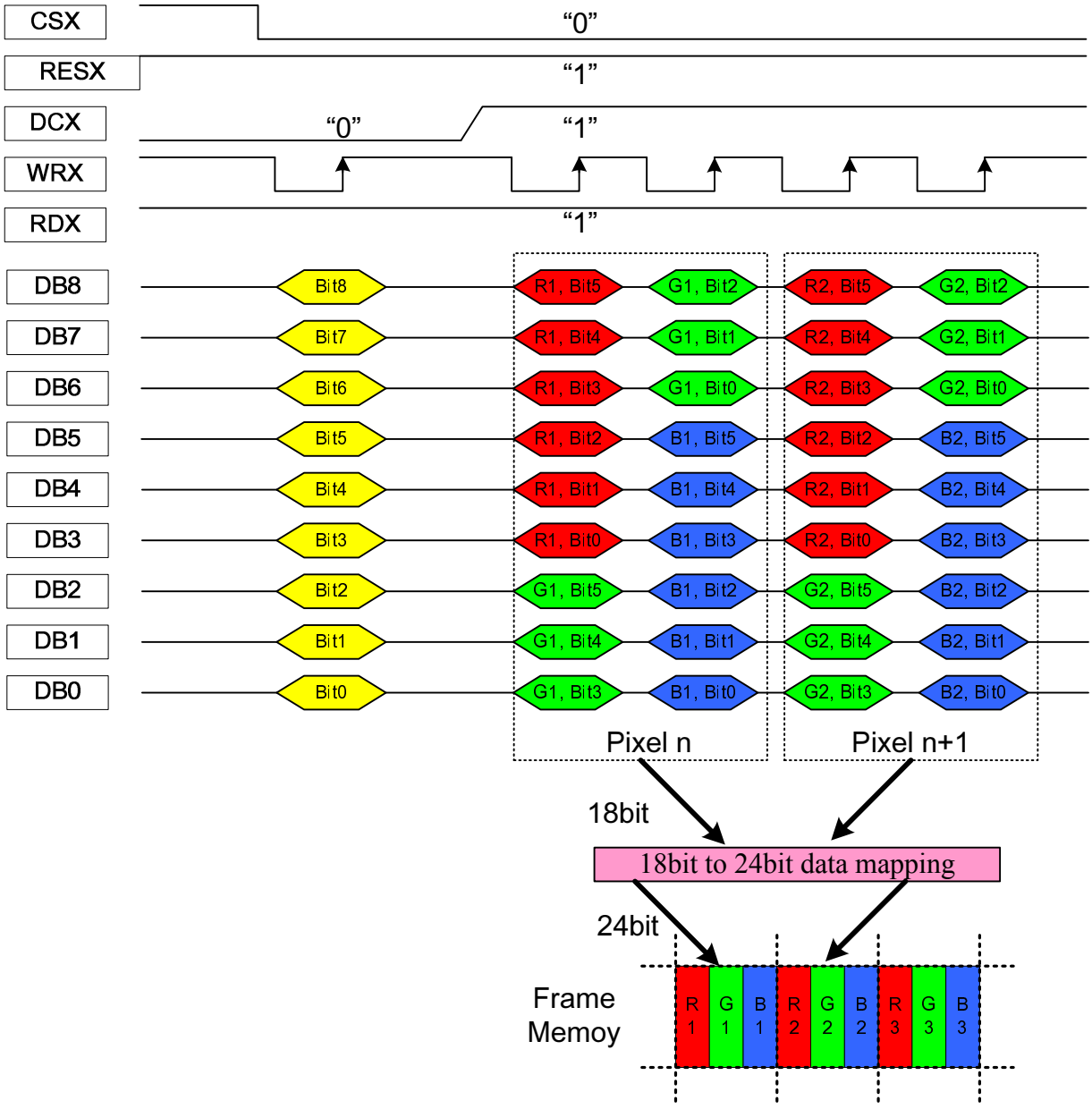


Figure 138: 9-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

Notes:

1. The data order is as follows: MSB = DB8, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, Red and Blue data.
2. 2-times transfer is used to transmit 1 pixel data to the 18-bit color depth information.
3. '-' = void

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3.14.13. 9-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

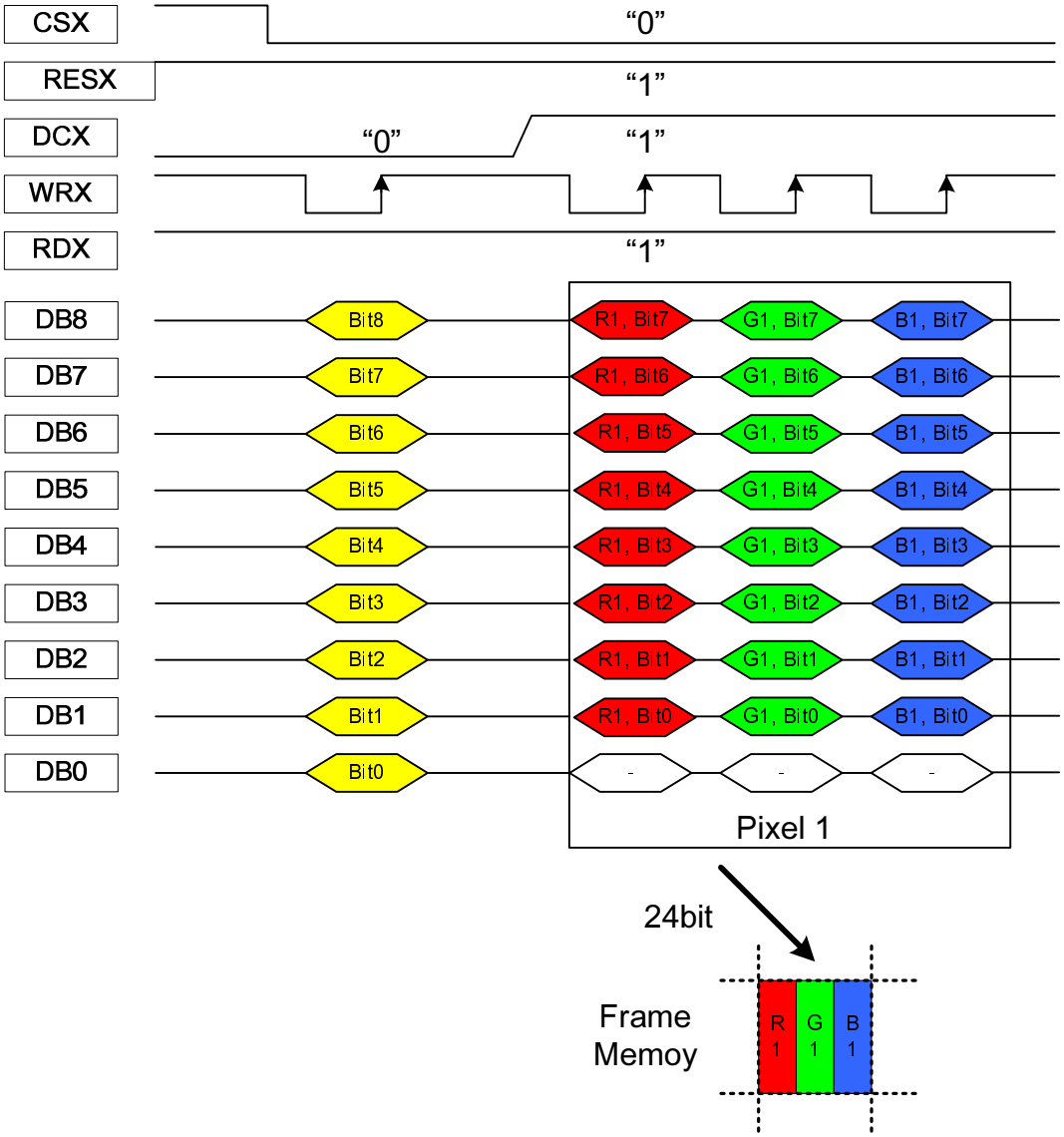


Figure 139: 9-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. The data order is as follows: MSB = DB8, LSB = DB0, and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 3-times transfer is used to transmit 1 pixel data to the 24-bit color depth information.
3. '-' = void

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3.14.14. 16-bit Parallel MPU Interface

The DBI TYPE B 16-bit parallel bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to 0001. The Figure 140 shows this system interface.

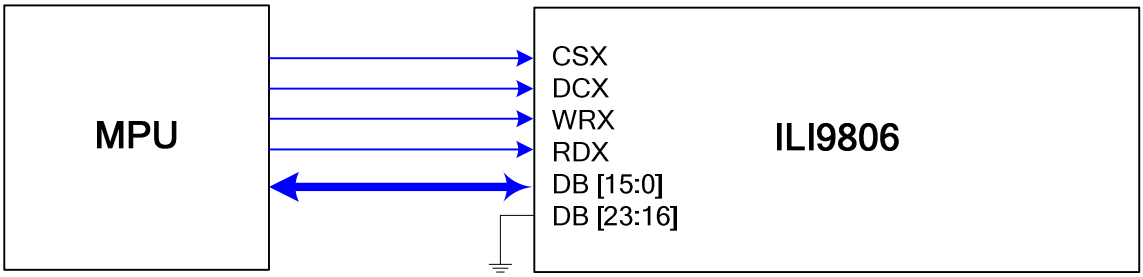


Figure 140: 16-bit Parallel MPU Interface

The display data formats available for this type of color depth are:

- 65K-Colors, RGB 5, 6, 5 bits input data (Set Standard Command 3Ah, DBI [2:0] = 101)
- 262K-Colors, RGB 6, 6, 6 bits input data (Set Standard Command 3Ah, DBI [2:0] = 110)
- 16.7M-Colors, RGB 8, 8, 8 bits input data (Set Standard Command 3Ah, DBI [2:0] = 111)

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3.14.15. 16-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

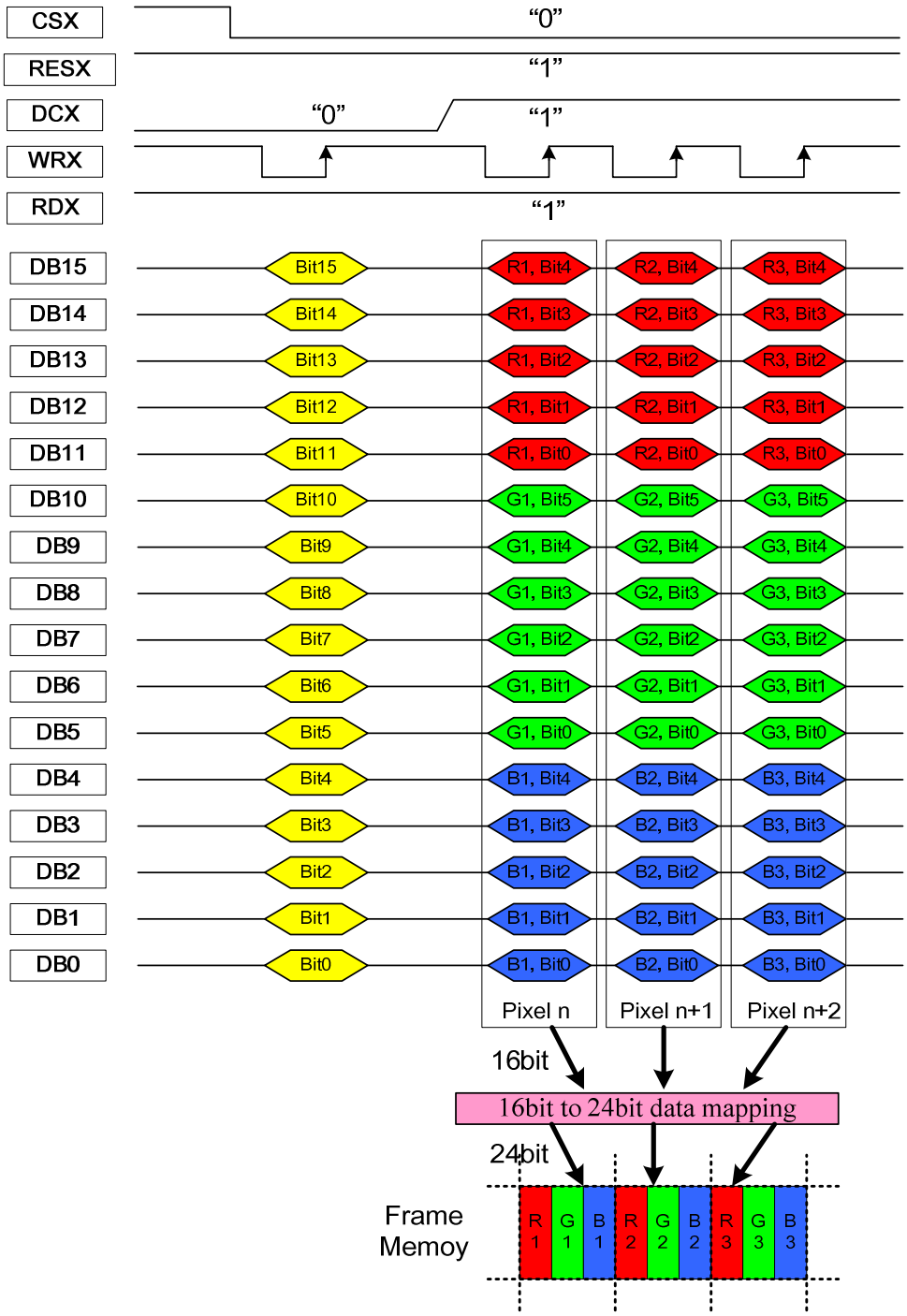


Figure 141: 16-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

Notes:

1. The data order is as follows: MSB = DB15, LSB = DB0 and picture data is MSB = Bit 5, LSB = Bit 0 for Green data and MSB = Bit 4, LSB = Bit0 for Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 16-bit color depth information.
3. '-' = void

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3.14.16. 16-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

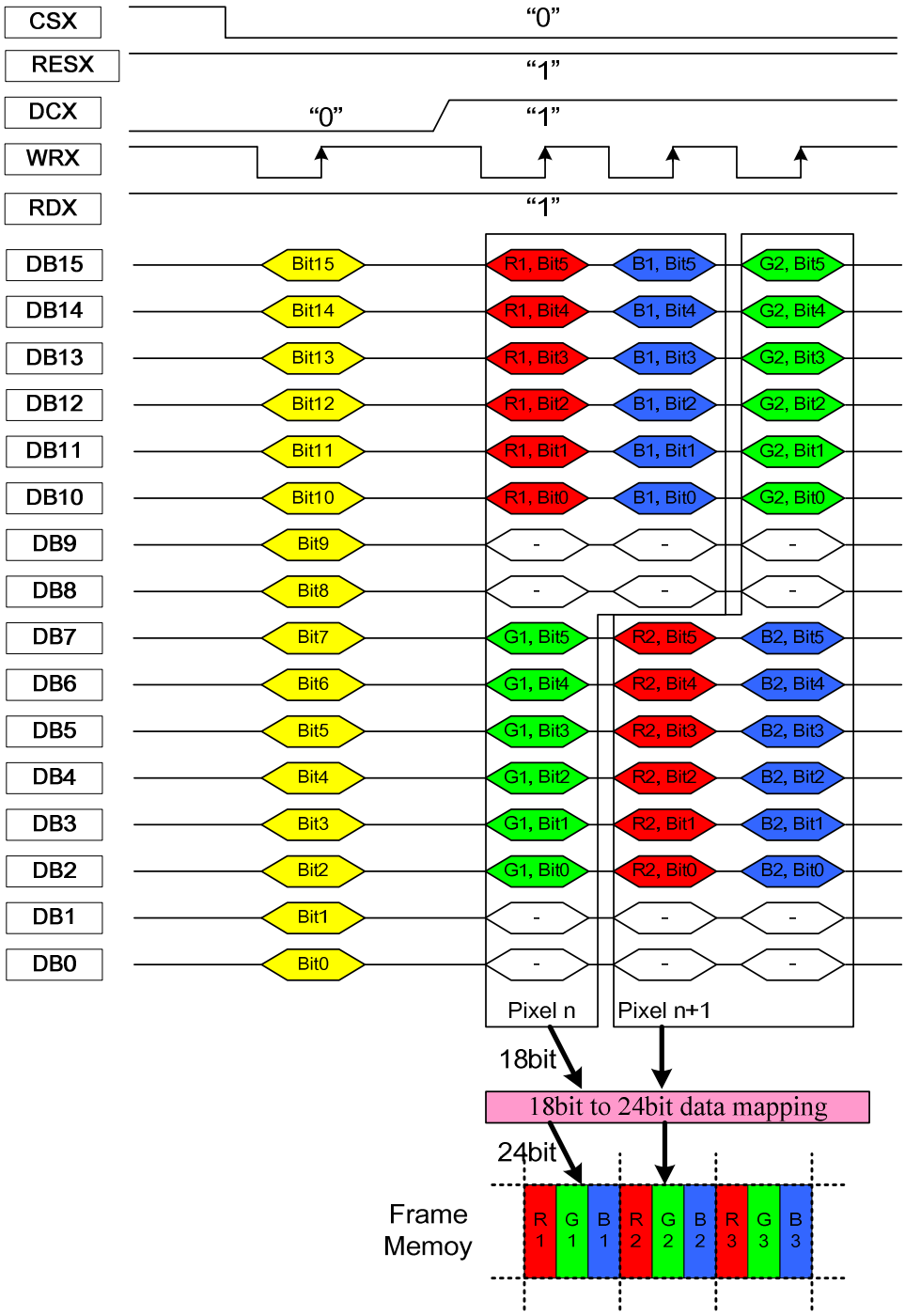


Figure 142: 16-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

Notes:

1. The data order is as follows: MSB = DB15, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, Red and Blue data.
2. 3-times transfer is used to transmit 2 pixel data to the 18-bit color depth information.
3. '-' = void

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3.14.17. 16-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

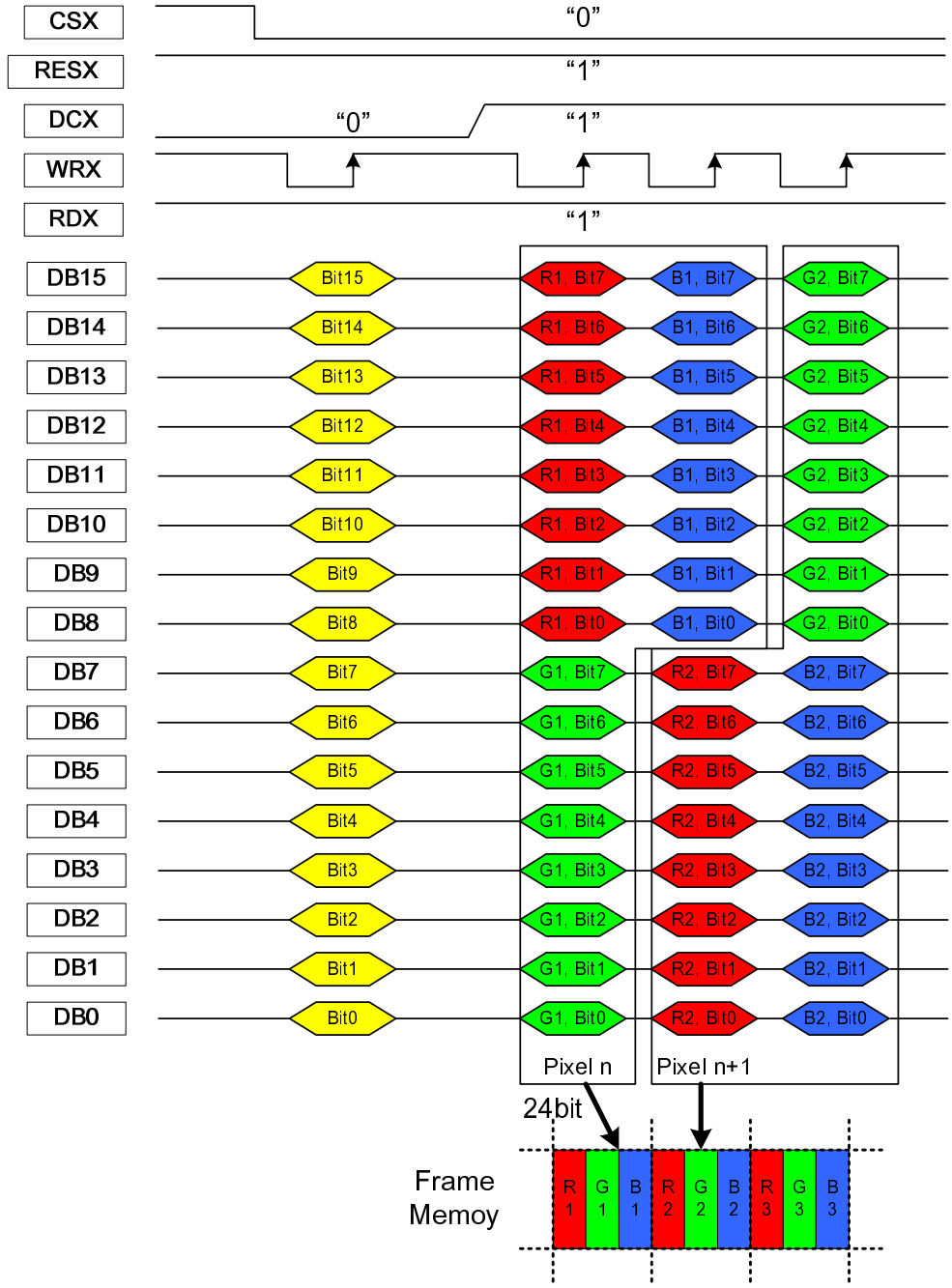


Figure 143: 16-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. The data order is as follows: MSB = DB15, LSB = DB0, and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 3-times transfer is used to transmit 2 pixel data to the 24-bit color depth information.
3. '-' = void

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3.14.18. 18-bit Parallel MPU Interface

The DBI TYPE B 18-bit parallel bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to 1101. The Figure 144 shows this system interface.

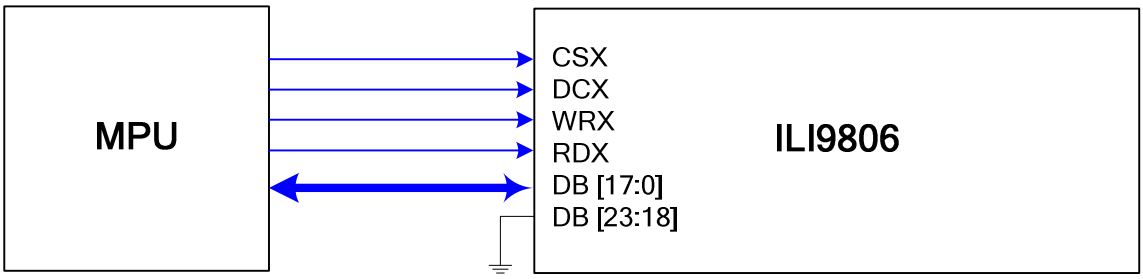


Figure 144: 18-bit Parallel MPU Interface

The display data formats available for this type of color depth are:

- 65K-Colors, RGB 5, 6, 5 bits input data (Set Standard Command 3Ah, DBI [2:0] = 101)
- 262K-Colors, RGB 6, 6, 6 bits input data (Set Standard Command 3Ah, DBI [2:0] = 110)
- 16.7M-Colors, RGB 8, 8, 8 bits input data (Set Standard Command 3Ah, DBI [2:0] = 111)

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3.14.19. 18-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

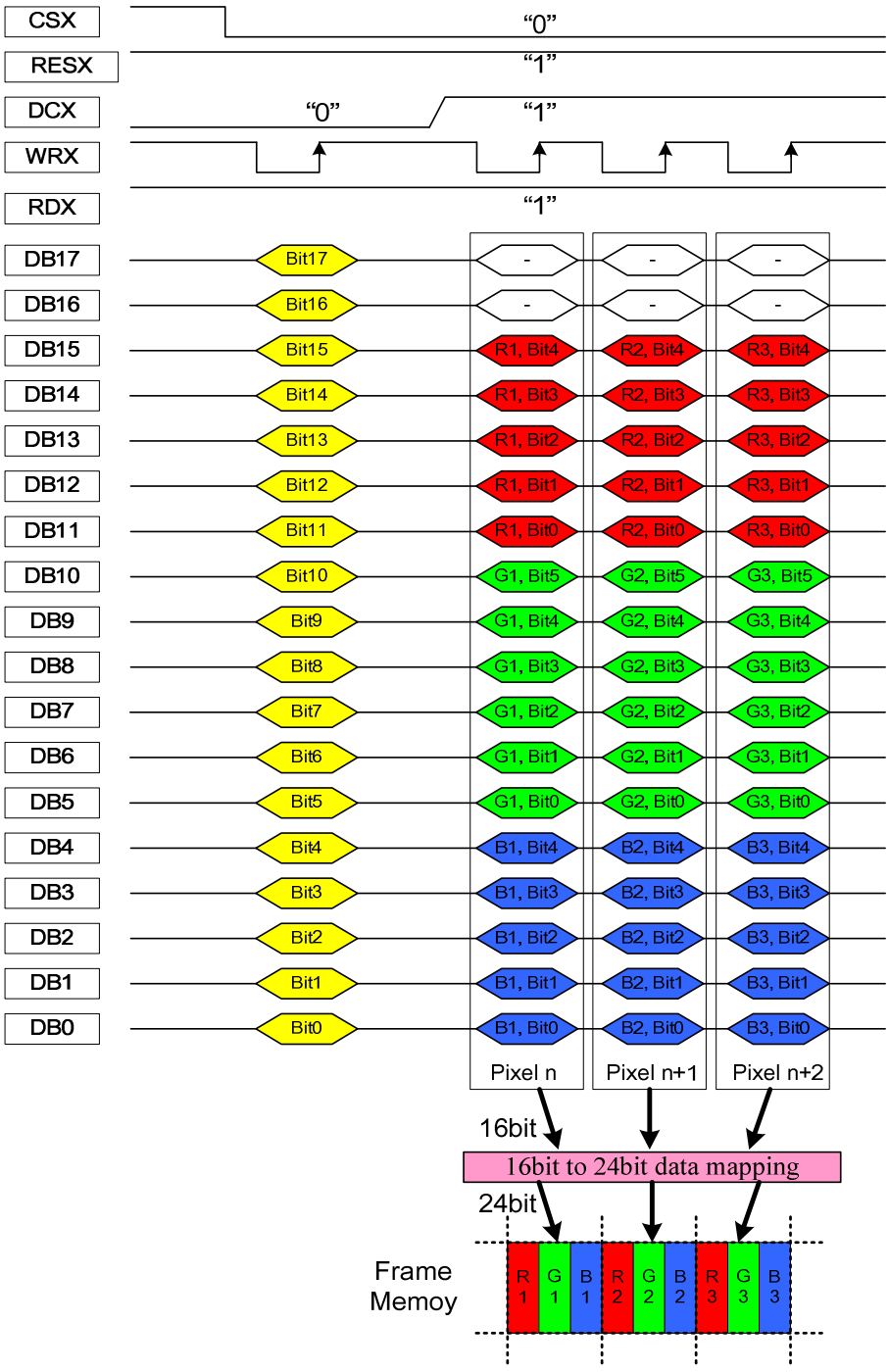


Figure 145: 18-bit Data Bus for 16-bit/pixel (RGB 5-6-5 bits input), 65K-color

Notes:

1. The data order is as follows: MSB = DB17, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green and MSB = Bit4, LSB = Bit0 for Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 16-bit color depth information.
3. '-' = void

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3.14.20. 18-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

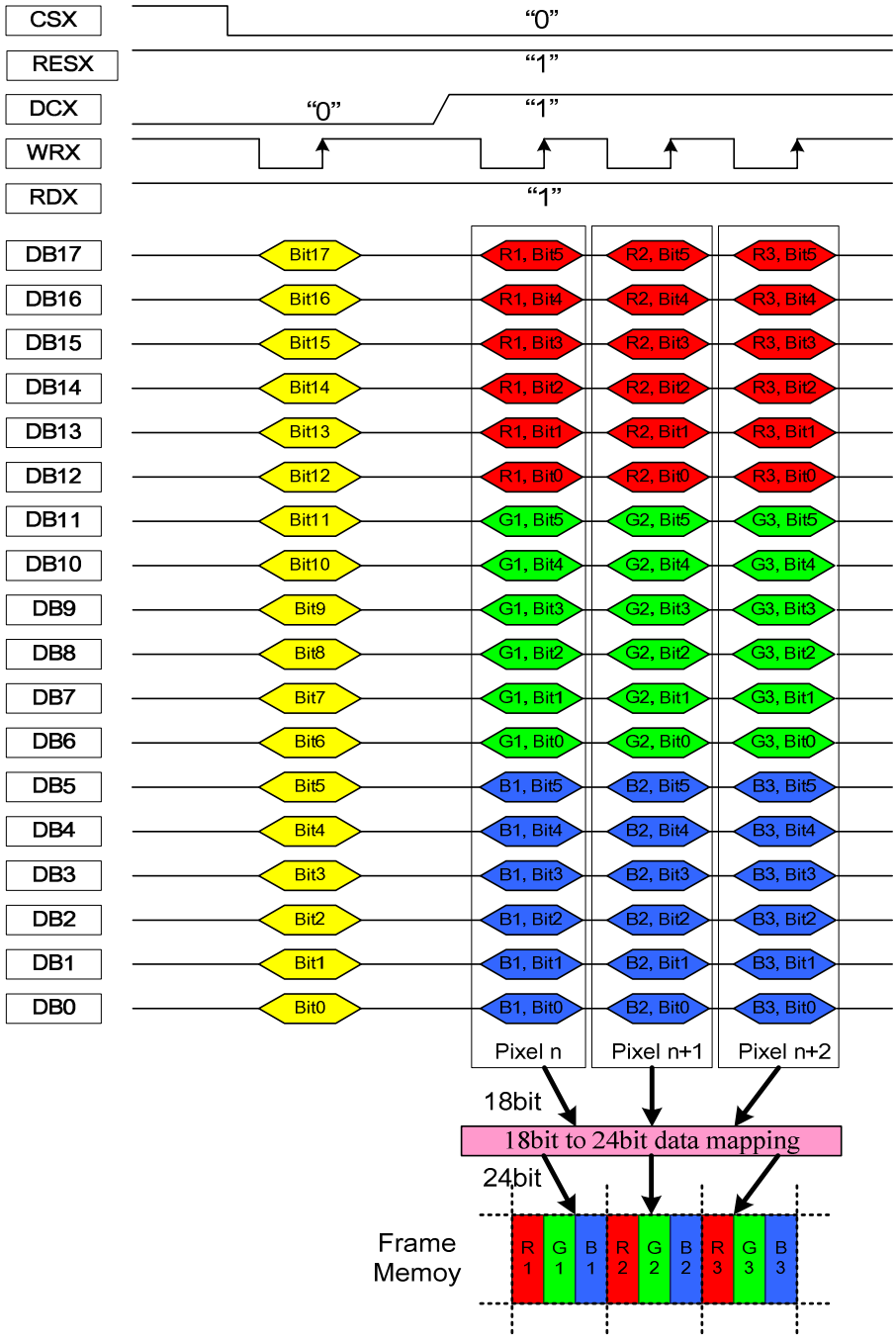


Figure 146: 18-bit Data Bus for 18-bit/pixel (RGB 6-6-6 bits input), 262K-color

Notes:

1. The data order is as follows: MSB = DB17, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 18-bit color depth information.
3. '-' = void

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3.14.21. 18-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

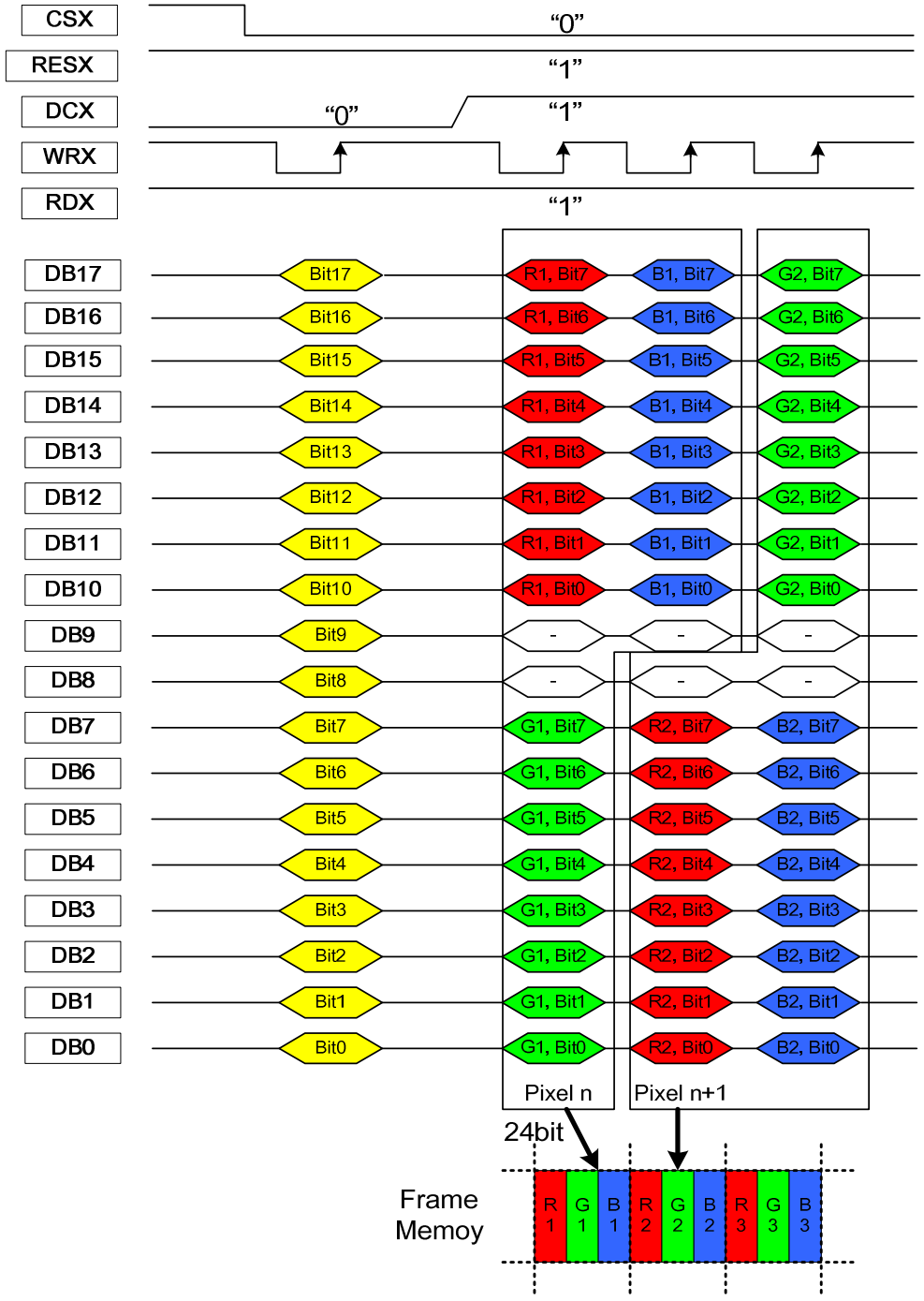


Figure 147: 18-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. The data order is as follows: MSB = DB17, LSB = DB0, and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 2-times transfer is used to transmit 1 pixel data to the 24-bit color depth information.
3. '-' = void

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3.14.22. 24-bit Parallel MPU Interface

The DBI TYPE B 24-bit parallel bus interface of the ILI9806G can be used by setting the external pin IM [3:0] to 0010. The Figure 148 shows this system interface.

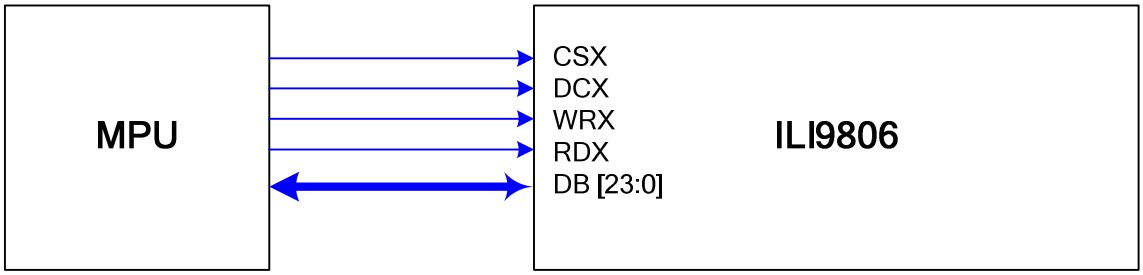


Figure 148: 24-bit Parallel MPU Interface

The display data format available for this type of color depth is:

- 16.7M-Colors, RGB 8, 8, 8 bits input data (Set Standard Command 3Ah, DBI [2:0] = 111)

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3.14.23. 24-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

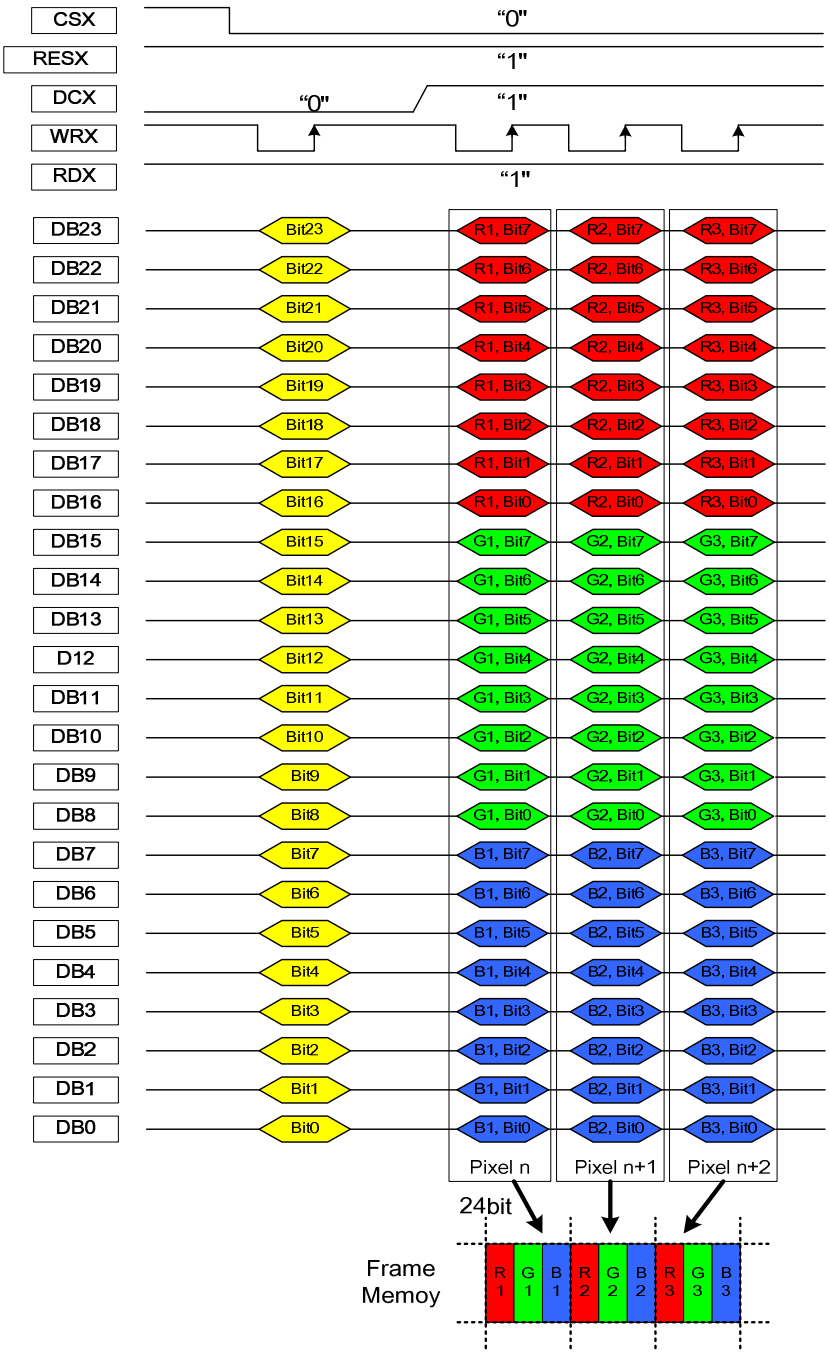


Figure 149: 24-bit Data Bus for 24-bit/pixel (RGB 8-8-8 bits input), 16.7M-color

Notes:

1. The data order is as follows: MSB = DB23, LSB = DB0 and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 24-bit color depth information.
3. '-' = void

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3.14.24. DPI (RGB) Interface

3.14.25. 16-bit/pixel 65K colors order on the DPI Interface

The 16-bit RGB interface is selected by setting the DPI [2:0] bits to 101. The display operation is synchronized with VSYNC, HSYNC and DCK signals. Display data is transferred to the internal GRAM through the 16-bit RGB data bus in synchronization with the display operation.

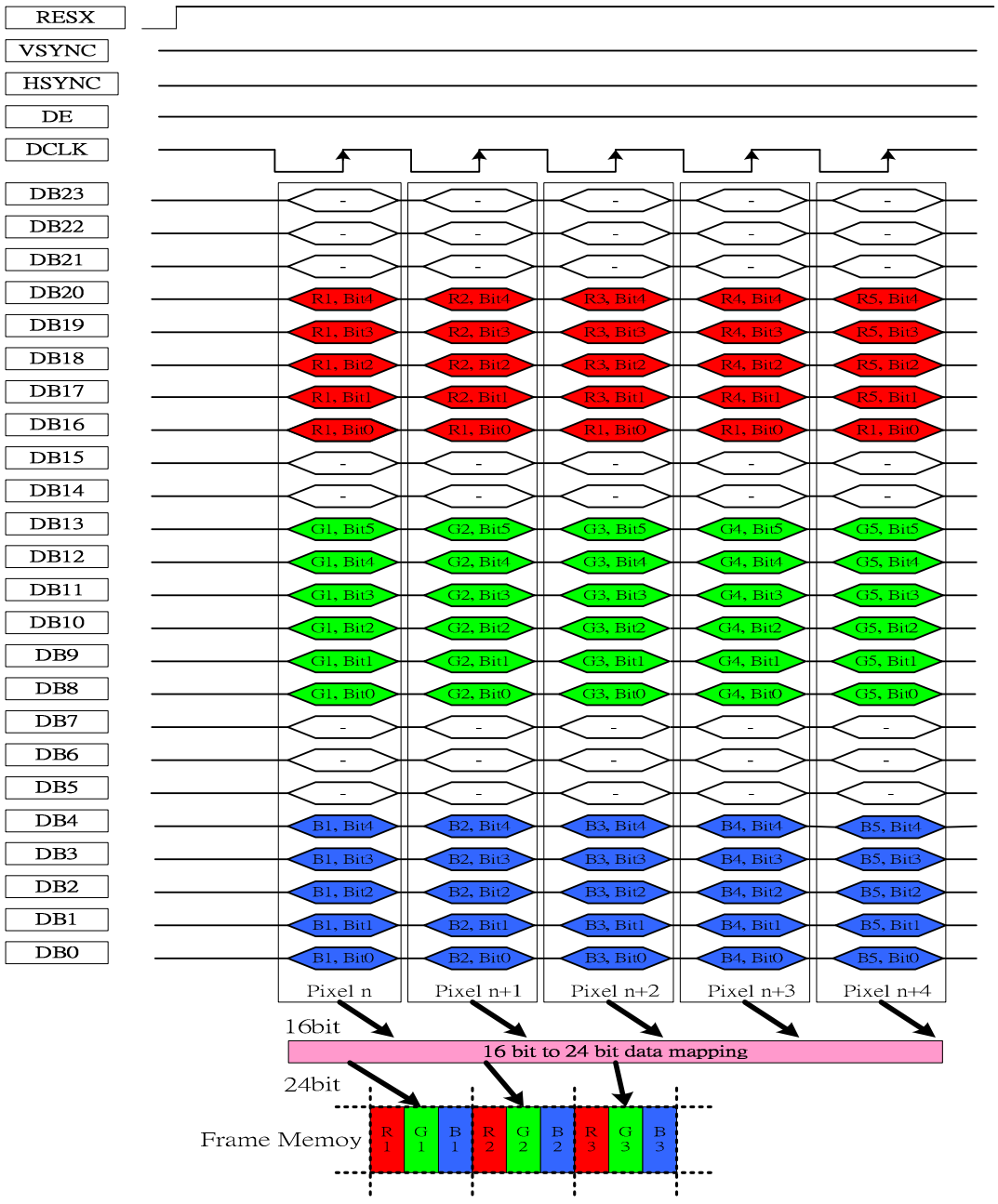


Figure 150: 16-bit/pixel 65K Colors Order on the DPI Interface

Notes:

1. The data order is as follows: MSB = DB23, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, MSB = Bit 4, LSB = Bit 0 for Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 16-bit color depth information.
3. '-' = void

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3.14.26. 18-bit/pixel 262K Colors Order on the DPI Interface

The 18-bit RGB interface is selected by setting the DPI [2:0] bits to 110. The display operation is synchronized with VSYNC, HSYNC and DCK signals. Display data is transferred to the internal GRAM through the 18-bit RGB data bus in synchronization with the display operation

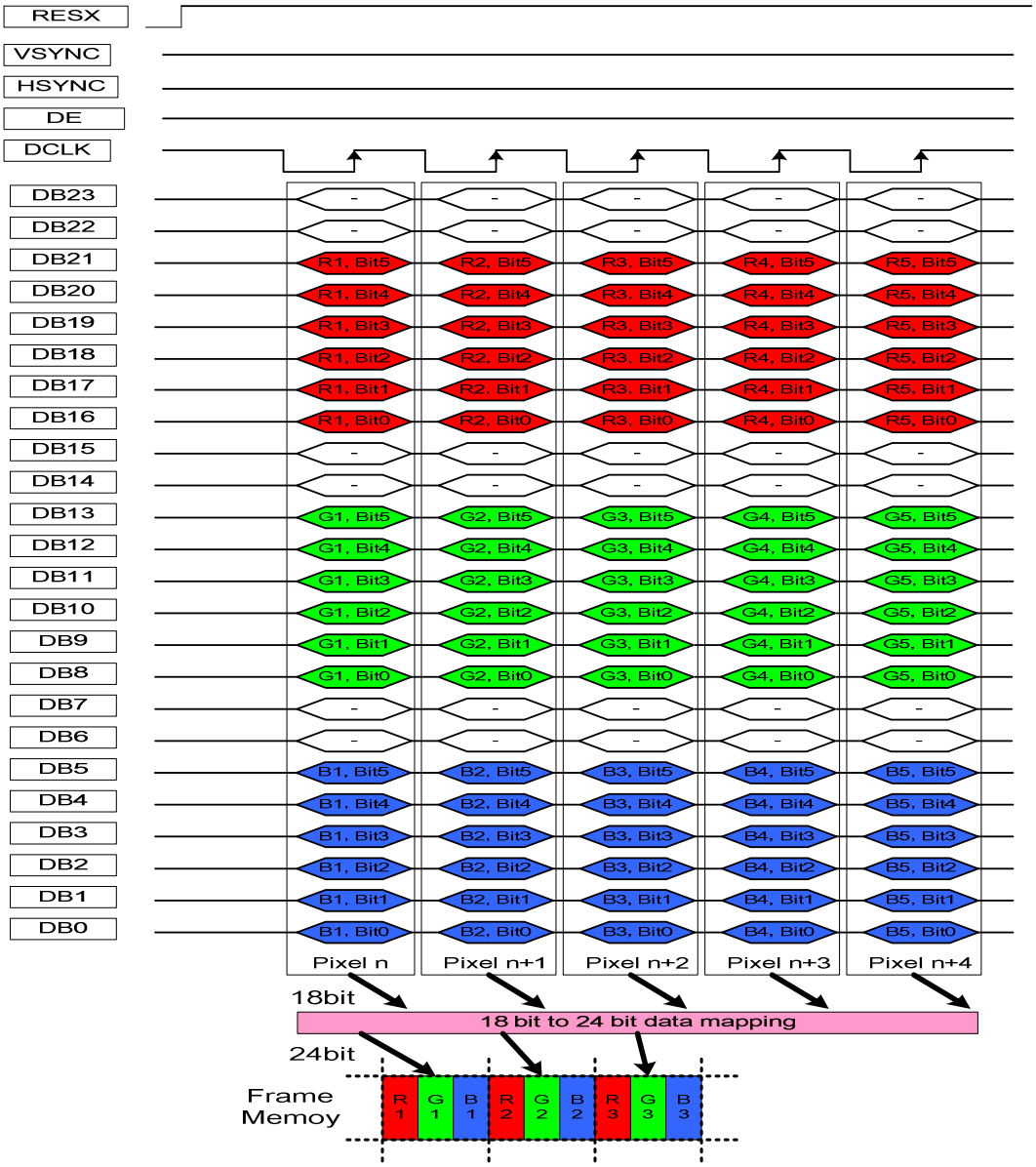


Figure 151: 18-bit/pixel 262K colors order on the DPI Interface

Notes:

1. The data order is as follows: MSB = DB23, LSB = DB0, and picture data is MSB = Bit 5, LSB = Bit 0 for Green, Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 18-bit color depth information.
3. '-' = void

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3.14.27. 24-bit/pixel 16.7M colors order on the DPI Interface

The 24-bit RGB interface is selected by setting the DPI [2:0] bits to 111. The display operation is synchronized with VSYNC, HSYNC and DCK signals. Display data is transferred to the internal GRAM through the 24-bit RGB data bus in synchronization with the display operation.

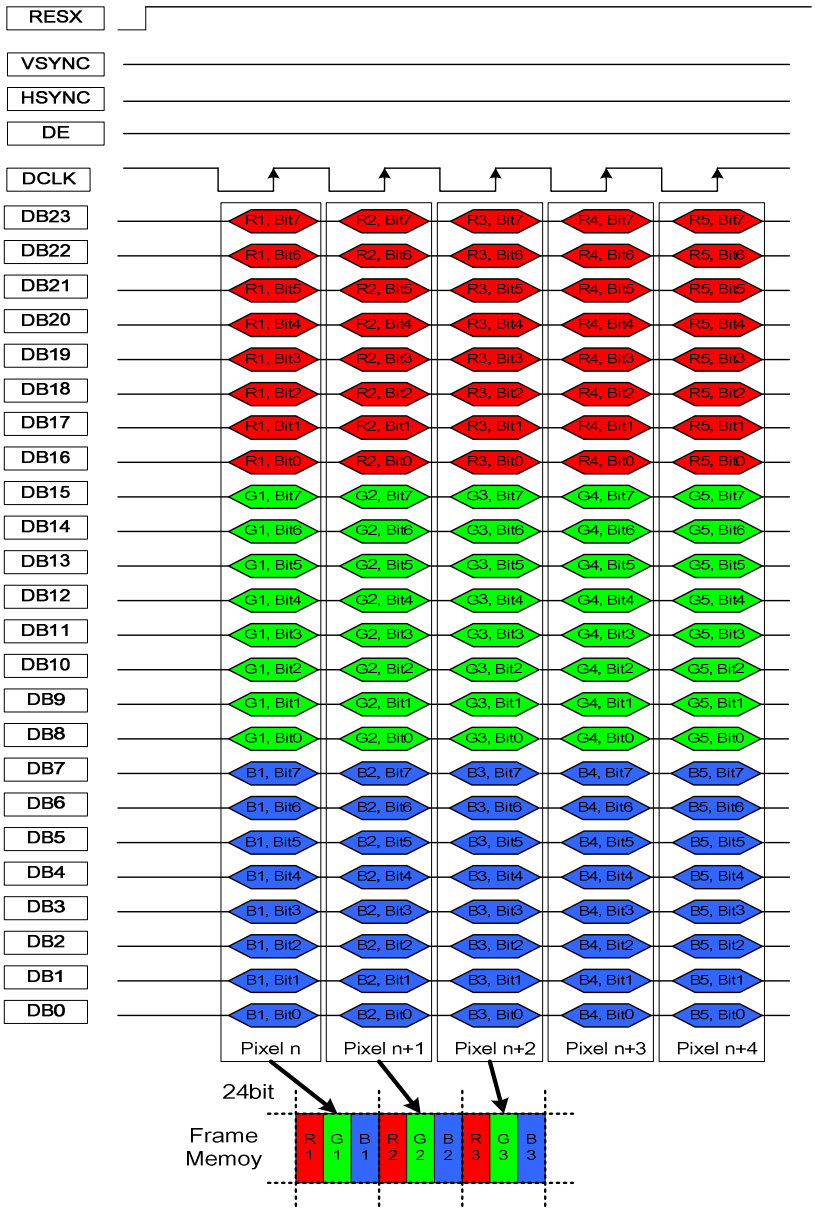


Figure 152: 24-bit/pixel 16.7M Colors Order on the DPI Interface

Notes:

1. The data order is as follows: MSB = DB23, LSB = DB0, and picture data is MSB = Bit 7, LSB = Bit 0 for Green, Red and Blue data.
2. 1-time transfer is used to transmit 1 pixel data to the 24-bit color depth information.

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3.14.28. 16/18-bit Color Data Mapping to 24-bit Pixel Data Operation

Table 43 below lists settings for 24-bit data mapping. Set the EPF [1:0] bits of the register “Interface Setting (B8H)” that define three types of data formats for 24-bit data (pixel data r, g, b) mapping that pixel data is stored in the internal GRAM.

Table 43: 16/18-bit Color Data Mapping to 24-bit Pixel Data Operation

EPF [1:0]	Expand 16-bit color data (R,G,B) to 24-bit subpixel data (r, g, b)	Expand 18-bit color data (R,G,B) to 24-bit subpixel data (r, g, b)
00	“0” is written to the LSB. 8 bits subpixel, data r [7:0] = {16-bit color data R [4:0], 3’h0} 8 bits subpixel, data g [7:0] = {16-bit color data G [5:0], 2’h0} 8 bits subpixel, data b [7:0] = {16-bit color data B [4:0], 3’h0} (Note3): the data are converted as follows. 16-bit color data R [4:0] = 5’h1F, G [5:0] = 6’h3F, B [4:0] = 5’h1F → 24-bit pixel data r, g, b [7:0] = 24’hFFFFFF	“0” is written to the LSB. 8 bits subpixel, data r [7:0] = {18-bit color data R [5:0], 2’h0} 8 bits subpixel, data g [7:0] = {18-bit color data G [5:0], 2’h0} 8 bits subpixel, data b [7:0] = {18-bit color data B [5:0], 2’h0} (Note1): the data are converted as follows. 18-bit color data R [5:0] = 6’h3F, G [5:0] = 6’h3F ,B [5:0] = 6’h3F → 24-bit pixel data r, g, b [7:0] = 24’hFFFFFF
01	“1” is written to the LSB. 8 bits subpixel, data r [7:0] = {16-bit color data R [4:0], 3’h7} 8 bits subpixel, data g [7:0] = {16-bit color data G [5:0], 2’h3} 8 bits subpixel, data b [7:0] = {16-bit color data B [4:0], 3’h7} (Note4): the data are converted as follows. 16-bit color data R [4:0] = 5’h0, G [5:0] = 6’h0, B [4:0] = 5’h0 → 24-bit pixel data r, g, b [7:0] = 24’h000000	“1” is written to the LSB. 8 bits subpixel, data r [7:0] = {18-bit color data R [5:0], 2’h3} 8 bits subpixel, data g [7:0] = {18-bit color data G [5:0], 2’h3} 8 bits subpixel, data b [7:0] = {18-bit color data B [5:0], 2’h3} (Note2): the data are converted as follows. 18-bit color data R [5:0] = 6’h0, G [5:0] = 6’h0, B [5:0] = 6’h0 → 24-bit pixel data r, g, b [7:0] = 24’h000000
10	The MSB value is written to the LSB. 8 bits subpixel, data r [7:0] = {16-bit color data R [4:0], R [4:2]} 8 bits subpixel, data g [7:0] = {16-bit color data G [5:0], G [5:4]} 8 bits subpixel, data b [7:0] = {16-bit color data B [4:0], B [4:2]}	The MSB value is written to the LSB. 8 bits subpixel, data r [7:0] = {18-bit color data R [5:0], R [5:4]} 8 bits subpixel, data g [7:0] = {18-bit color data G [5:0], G [5:4]} 8 bits subpixel, data b [7:0] = {18-bit color data B [5:0], B [5:4]}
11	Setting disabled	Setting disabled

	Frame memory data (24 bits)																							
	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
24-bit	R[7]	R[6]	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	G[7]	G[6]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[7]	B[6]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]
18-bit EPF[1:0]=00 (Note 1)	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	0	0	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	0	0	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]	0	0
18-bit EPF[1:0]=01 (Note 2)	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	1	1	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	1	1	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]	1	1
18-bit EPF[1:0]=10	R[5]	R[4]	R[3]	R[2]	R[1]	R[0]	R[5]	R[4]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	G[5]	G[4]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]	B[5]	B[4]
16-bit EPF[1:0]=00 (Note 3)	R[4]	R[3]	R[2]	R[1]	R[0]	0	0	0	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	0	0	B[4]	B[3]	B[2]	B[1]	B[0]	0	0	0
16-bit EPF[1:0]=01 (Note 4)	R[4]	R[3]	R[2]	R[1]	R[0]	1	1	1	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	1	1	B[4]	B[3]	B[2]	B[1]	B[0]	1	1	1
16-bit EPF[1:0]=10	R[4]	R[3]	R[2]	R[1]	R[0]	R[4]	R[3]	R[2]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	G[5]	G[4]	B[4]	B[3]	B[2]	B[1]	B[0]	B[4]	B[3]	B[2]

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For example:

16-bit data mapping to 24-bit, EPF [1:0] = 10

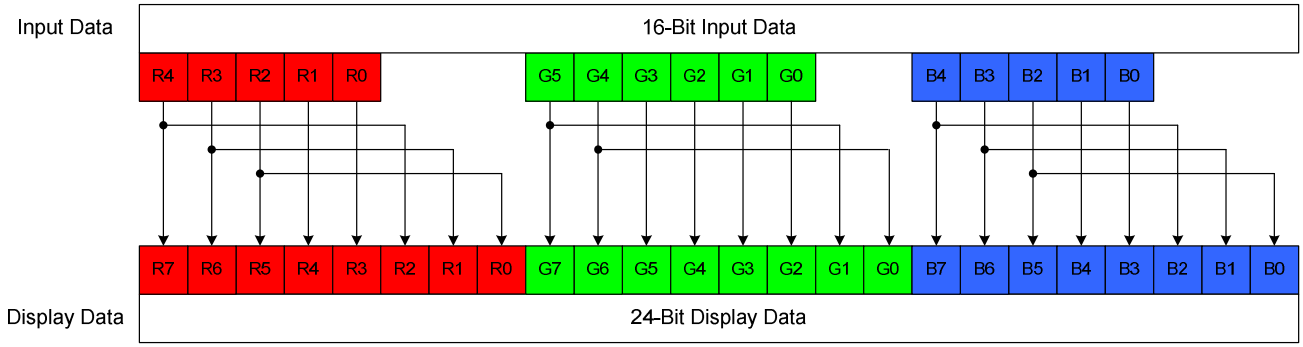


Figure 153: EPF [1:0] = 10, 16-bit data mapping to 24-bit

18-bit data mapping to 24-bit, EPF [1:0] = 10

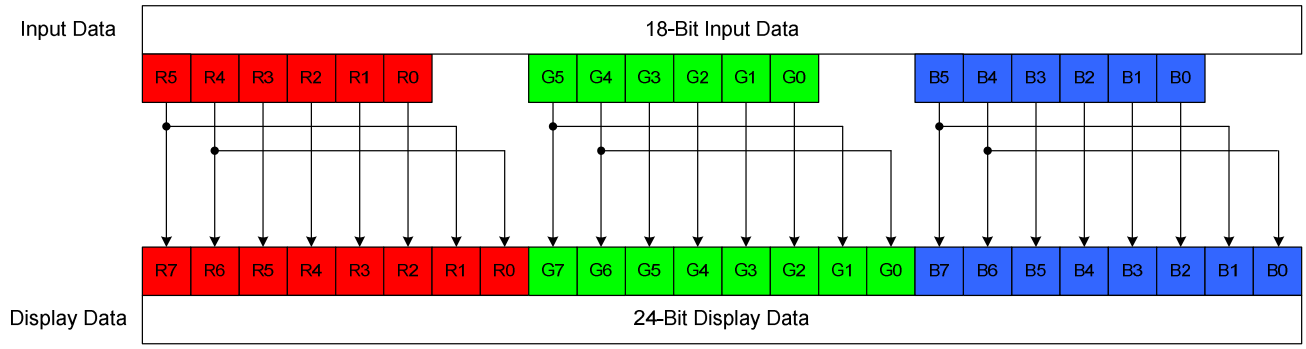


Figure 154: EPF [1:0] = 10, 18-bit data mapping to 24-bit

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3.14.29. DSI Transmission Data Format

3.14.30. 16-bit per Pixel, Long Packet, Data Type 00 1110 (0Eh)

Packed Pixel Stream 16-Bit Format is a Long Packet used to transmit image data formatted as 16-bit pixels to a Video Mode display module. The packet consists of the DI byte, a two-byte WC, an ECC byte, a payload of length WC bytes and a two-byte checksum. Pixel format is red (5 bits), green (6 bits), and blue (5 bits), in that order. Note that the Green component is split across two bytes. Within a color component, the LSB is sent first, the MSB last. With this format, pixel boundaries align with byte boundaries every two bytes. The total line width (displayed plus non-displayed pixels) should be a multiple of two bytes.

Normally, the ILI9806G has no frame buffer of its own, so all image data shall be supplied by the host processor at a sufficiently high rate to avoid flicker or other visible artifice.

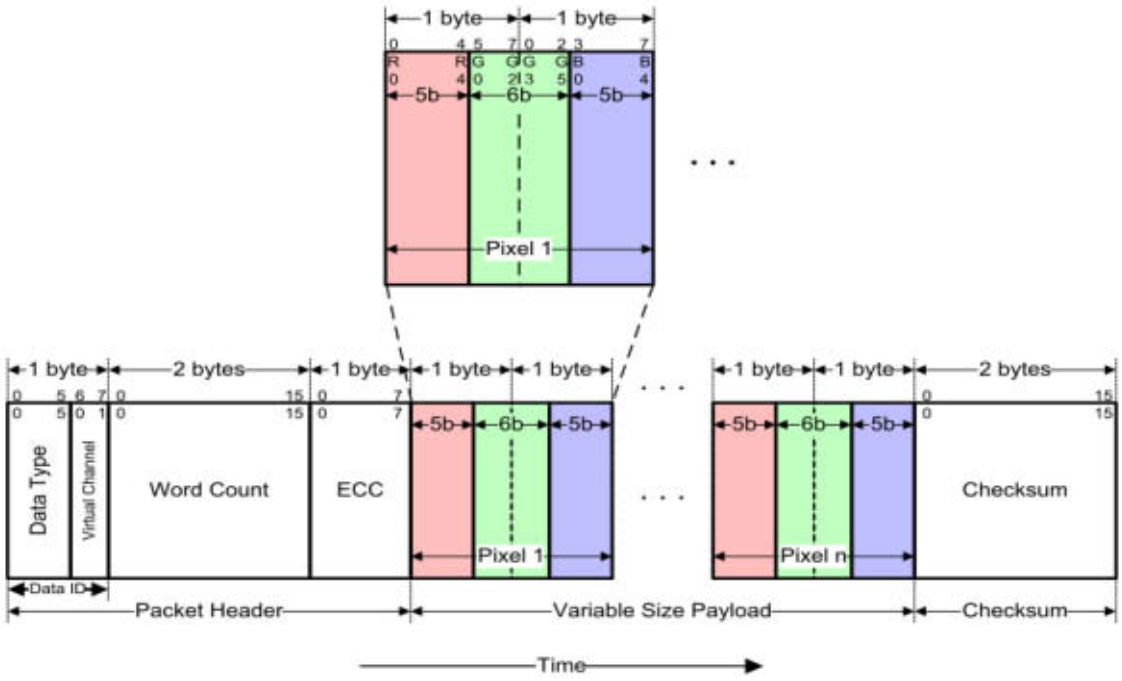


Figure 155: 16-bit per Pixel, Data Type 00 1110 (0Eh)

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3.14.31. 18-bit per Pixel, Long Packet, Data Type = 01 1110 (1Eh)

Packed Pixel Stream 18-Bit Format (Packed) is a Long packet. It is used to transmit RGB image data formatted as pixels to a Video Mode display module that displays 18-bit pixels. The packet consists of the DI byte, a two-byte WC, an ECC byte, a payload of length WC bytes, and a two-byte Checksum. Pixel format is red (6 bits), green (6 bits) and blue (6 bits), in that order. Within a color component, the LSB is sent first, the MSB last.

Note that pixel boundaries only align with byte boundaries every four pixels (nine bytes). Preferably, display modules employing this format have a horizontal extent (width in pixels) evenly divisible by four, so no partial bytes remain at the end of the display line data. If the active (displayed) horizontal width is not a multiple of four pixels, the transmitter shall send additional filled pixels at the end of the display line to make the transmitted width a multiple of four pixels. The receiving peripheral shall not display the filled pixels when refreshing the display device. For example, if a display device has an active display width of 399 pixels, the transmitter should send 400 pixels in one or more packets. The receiver should display the first 399 pixels and discard the last pixel of the transmission. With this format, the total line width (displayed and non-displayed pixels) should be a multiple of four 1246 pixels (nine bytes).

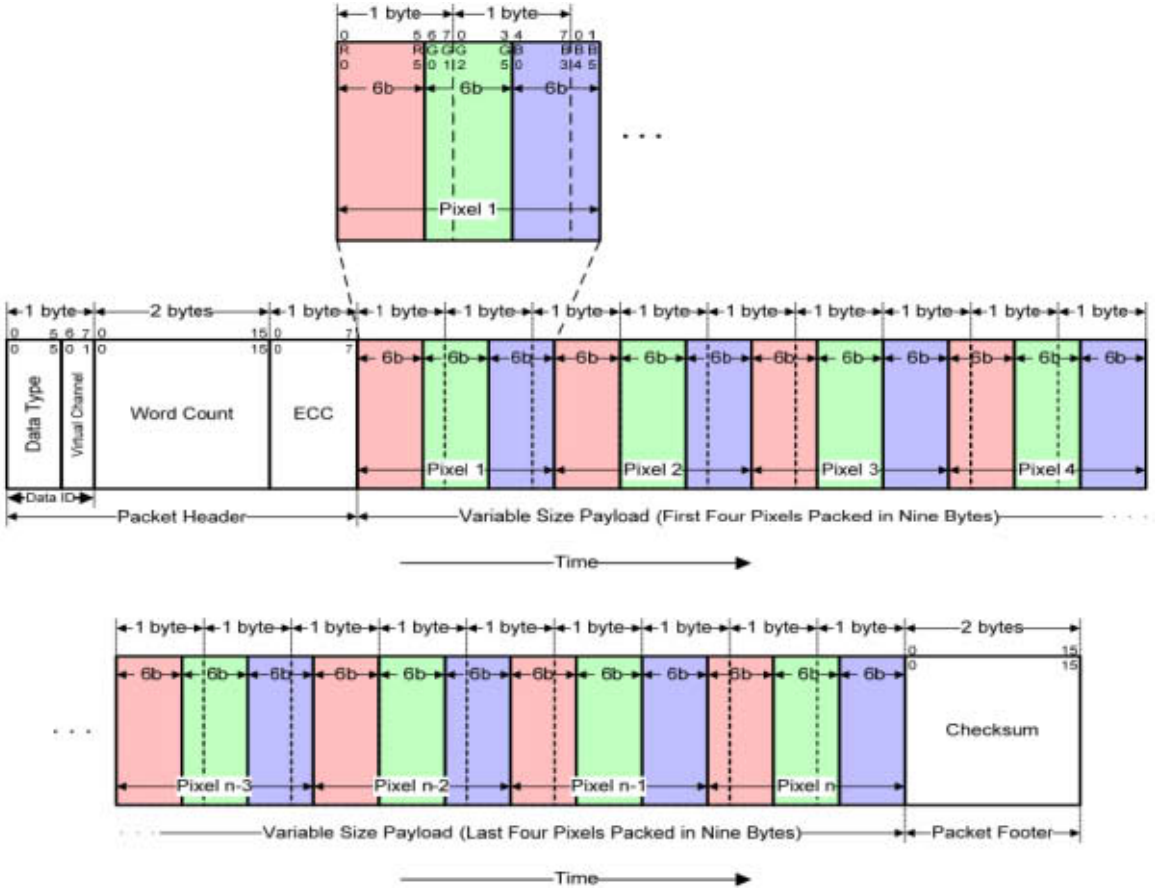


Figure 156: 18-bit per Pixel, Data Type = 01 1110 (1Eh)

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3.14.32. 18-bit per Pixel, Long Packet, Data Type = 10 1110 (2Eh)

In the 18-bit Pixel Loosely Packed format, each R, G, or B color component is six bits but is shifted to the upper bits of the byte, such that the valid pixel bits occupy bits [7:2] of each byte. Bits [1:0] of each payload byte representing active pixels are ignored. As a result, each pixel requires three bytes as it is transmitted across the link. This requires more bandwidth than the “packed” format, but requires less shifting and multiplexing logic in the packing and unpacking functions on each end of the Link.

This format is used to transmit RGB image data formatted as pixels to a Video Mode display module that displays 18-bit pixels. The packet consists of the DI byte, a two-byte WC, an ECC byte, a payload of length WC bytes and a two-byte Checksum. The pixel format is red (6 bits), green (6 bits) and blue (6 bits) in that order. Within a color component, the LSB is sent first, the MSB last.

With this format, pixel boundaries align with byte boundaries every three bytes. The total line width (displayed and non-displayed pixels) should be a multiple of three bytes.

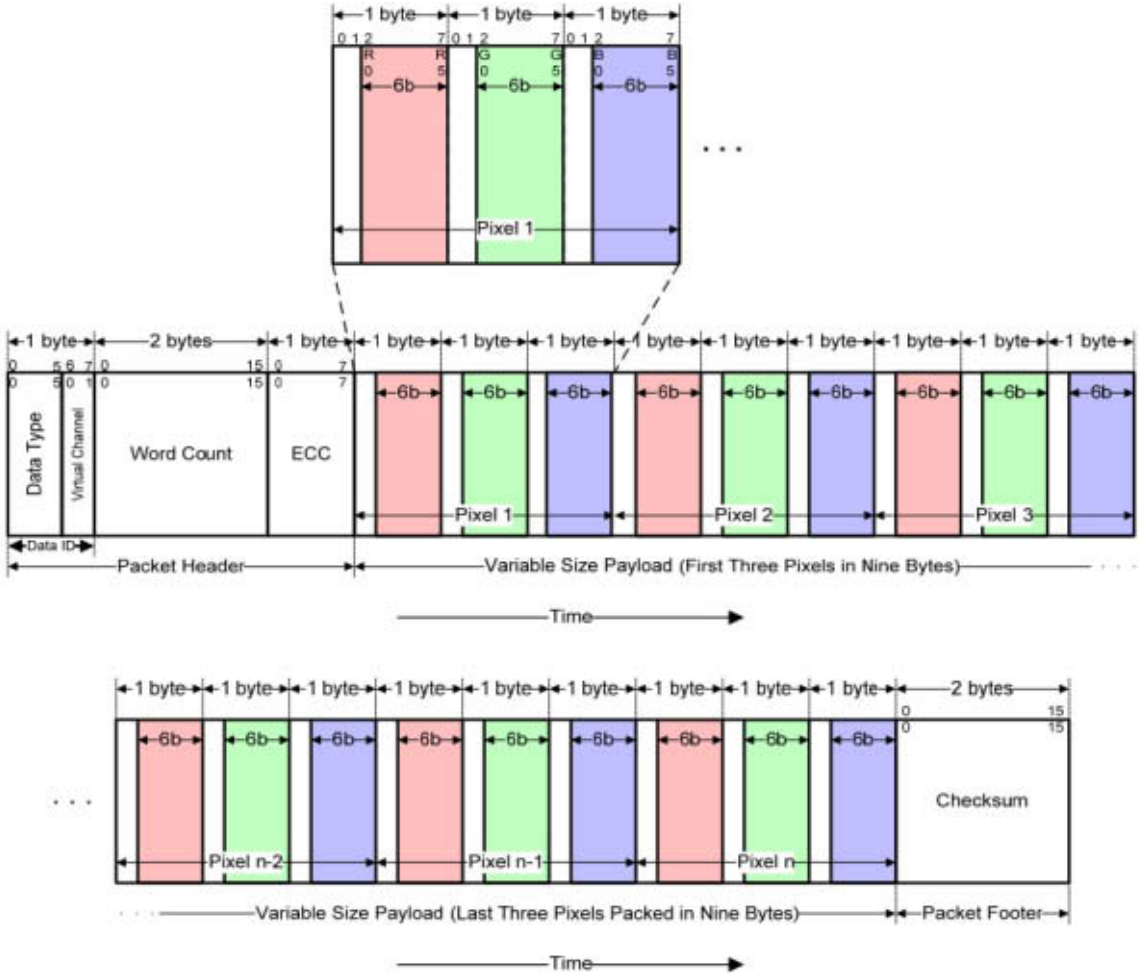


Figure 157: 18-bit per Pixel, Data Type = 10 1110 (2Eh)

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3.14.33. 24-bit per Pixel, Long Packet, Data Type = 11 1110 (3Eh)

Packed Pixel Stream 24-Bit Format is a Long packet. It is used to transmit image data formatted as 24-bit pixels to a Video Mode display module. The packet consists of the DI byte, a two-byte WC, an ECC byte, a payload of length WC bytes and a two-byte Checksum. The pixel format is red (8 bits), green (8 bits) and blue (8 bits), in that order. Each color component occupies one byte in the pixel stream; no components are split across byte boundaries. Within a color component, the LSB is sent first, the MSB last.

With this format, pixel boundaries align with byte boundaries every three bytes. The total line width (displayed and non-displayed pixels) should be a multiple of three bytes.

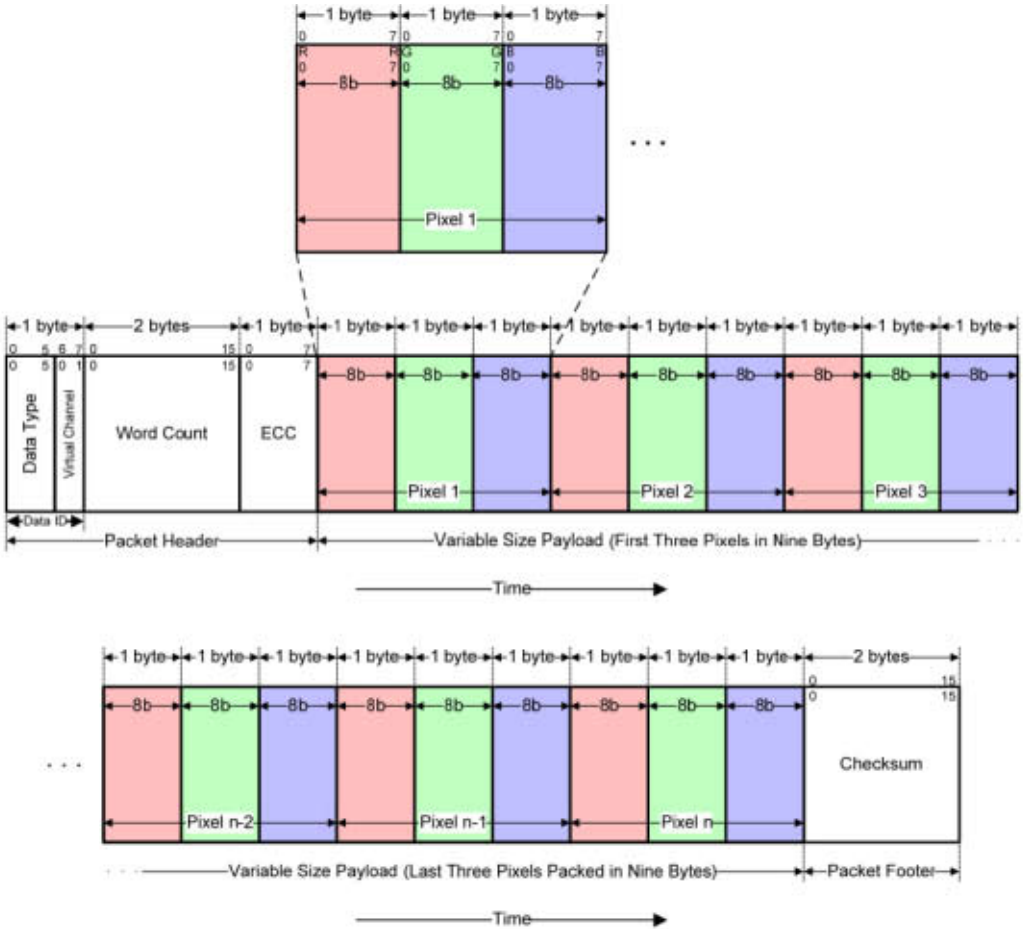


Figure 158: 24-bit per Pixel, Data Type = 11 1110 (3Eh)

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4. Command

4.1. Command List

4.1.1. Standard Command List

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
NOP	0	1	↑	xx	0	0	0	0	0	0	0	0	00h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Software Reset	0	1	↑	xx	0	0	0	0	0	0	0	1	01h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display Identification Information	0	1	↑	xx	0	0	0	0	0	1	0	0	04h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	ID1 [7:0]								xx
	1	↑	1	xx	ID2 [7:0]								xx
	1	↑	1	xx	ID3 [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Number of the Errors on DSI	0	1	↑	xx	0	0	0	0	0	1	0	1	05h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	P [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Red color	0	1	↑	xx	0	0	0	0	0	1	1	0	06h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	R [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Green color	0	1	↑	xx	0	0	0	0	0	1	1	1	07h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	G [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Blue color	0	1	↑	xx	0	0	0	0	1	0	0	0	08h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	B [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display Status	0	1	↑	xx	0	0	0	0	1	0	0	1	09h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [31:24]								xx
	1	↑	1	xx	D [23:16]								xx
	1	↑	1	xx	D [15:8]								xx
	1	↑	1	xx	D [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Power Mode	0	1	↑	xx	0	0	0	0	1	0	1	0	0Ah
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [7:2]						0	0	xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Address Mode	0	1	↑	xx	0	0	0	0	1	0	1	1	0Bh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Pixel Format	0	1	↑	xx	0	0	0	0	1	1	0	0	0Ch
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	0	DPI [2:0]			0	DBI [2:0]			xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display Mode	0	1	↑	xx	0	0	0	0	1	1	0	1	0Dh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display signal Mode	0	1	↑	xx	0	0	0	0	1	1	1	0	0Eh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D7	D6	D5	D4	D3	D2	D1	D0	xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display Self-Diagnostic Result	0	1	↑	xx	0	0	0	0	1	1	1	1	0Fh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Sleep In	0	1	↑	xx	0	0	0	1	0	0	0	0	10h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Sleep Out	0	1	↑	xx	0	0	0	1	0	0	0	1	11h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Partial Mode On	0	1	↑	xx	0	0	0	1	0	0	1	0	12h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Normal Display Mode On	0	1	↑	xx	0	0	0	1	0	0	1	1	13h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display Inversion Off	0	1	↑	xx	0	0	1	0	0	0	0	0	20h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display Inversion ON	0	1	↑	xx	0	0	1	0	0	0	0	1	21h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
All Pixel Off	0	1	↑	xx	0	0	1	0	0	0	1	0	22h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
All Pixel On	0	1	↑	xx	0	0	1	0	0	0	1	1	23h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Gamma Curve Set	0	1	↑	xx	0	0	1	0	0	1	1	0	26h
	1	1	↑	xx	GC7	GC6	GC5	GC4	GC3	GC2	GC1	GC0	xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display Off	0	1	↑	xx	0	0	1	0	1	0	0	0	28h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display ON	0	1	↑	xx	0	0	1	0	1	0	0	1	29h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Set Column Address	0	1	↑	xx	0	0	1	0	1	0	1	0	2Ah
	1	1	↑	xx	SC [15:8]								xx
	1	1	↑	xx	SC [7:0]								xx
	1	1	↑	xx	EC [15:8]								xx
	1	1	↑	xx	EC [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Set Page Address	0	1	↑	xx	0	0	1	0	1	0	1	1	2Bh
	1	1	↑	xx	SP [15:8]								xx
	1	1	↑	xx	SP [7:0]								xx
	1	1	↑	xx	EP [15:8]								xx
	1	1	↑	xx	EP [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write Memory Start	0	1	↑	xx	0	0	1	0	1	1	0	0	2Ch
	1	1	↑	Image Data 1 [D23:D0]									xx
	1	1	↑	Image Data x [D23:D0]									xx
	1	1	↑	Image Data n [D23:D0]									xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Memory Read	0	1	↑	xx	0	0	1	0	1	1	1	0	2Eh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	Image Data 1 [D23:D0]									xx
	1	↑	1	Image Data x [D23:D0]									xx
	1	↑	1	Image Data n [D23:D0]									xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Partial Area	0	1	↑	xx	0	0	1	1	0	0	0	0	30h
	1	1	↑	xx	SR [15:8]								xx
	1	1	↑	xx	SR [7:0]								xx
	1	1	↑	xx	ER [15:8]								xx
	1	1	↑	xx	ER [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Vertical Scrolling Definition	0	1	↑	xx	0	0	1	1	0	0	1	1	33h
	1	1	↑	xx	TFA [15:8]								xx
	1	1	↑	xx	TFA [7:0]								xx
	1	1	↑	xx	VSA [15:8]								xx
	1	1	↑	xx	VSA [7:0]								xx
	1	1	↑	xx	BFA [15:8]								xx
	1	1	↑	xx	BFA [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Tearing Effect Line OFF	0	1	↑	xx	0	0	1	1	0	1	0	0	34h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Tearing Effect Line ON	0	1	↑	xx	0	0	1	1	0	1	0	1	35h
	1	1	↑	xx	0	0	0	0	0	0	0	M	xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Memory Access Control	0	1	↑	xx	0	0	1	1	0	1	1	0	36h
	1	1	↑	xx	MY	MX	MV	ML	BGR	MH	SS	GS	xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Vertical Scrolling Start Address	0	1	↑	xx	0	0	1	1	0	1	1	1	37h
	1	1	↑	xx	VSP [15:8]								xx
	1	1	↑	xx	VSP [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Idle Mode Off	0	1	↑	xx	0	0	1	1	1	0	0	0	38h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Idle Mode On	0	1	↑	xx	0	0	1	1	1	0	0	1	39h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Interface Pixel Format	0	1	↑	xx	0	0	1	1	1	0	1	0	3Ah
	1	1	↑	xx	0	DPI [2:0]			0	DBI [2:0]			xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Memory Write Continue	0	1	↑	xx	0	0	1	1	1	1	0	0	3Ch
	1	1	↑	Image Data 1 [D23:D0]									xx
	1	1	↑	Image Data x [D23:D0]									xx
	1	1	↑	Image Data n [D23:D0]									xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Memory Read Continue	0	1	↑	xx	0	0	1	1	1	1	1	0	3Eh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	Image Data 1 [D23:D0]									xx
	1	↑	1	Image Data x [D23:D0]									xx
	1	↑	1	Image Data n [D23:D0]									xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write Tear Scan line	0	1	↑	xx	0	1	0	0	0	1	0	0	44h
	1	1	↑	xx	N [15:8]								xx
	1	1	↑	xx	N [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Tear Line	0	1	↑	xx	0	1	0	0	0	1	0	1	45h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	N [15:8]								xx
	1	↑	1	xx	N [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write Display Brightness	0	1	↑	xx	0	1	0	1	0	0	0	1	51h
	1	1	↑	xx	DBV [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Display Brightness Value	0	1	↑	xx	0	1	0	1	0	0	1	0	52h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	DBV [7:0]								xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write CTRL Display	0	1	↑	xx	0	1	0	1	0	0	1	1	53h
	1	1	↑	xx	0	0	BCTRL	0	DD	BL	0	0	xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read CTRL Display	0	1	↑	xx	0	1	0	1	0	1	0	0	54h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	0	0	BCTRL	0	DD	BL	0	0	xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write Content Adaptive Brightness Control	0	1	↑	xx	0	1	0	1	0	1	0	1	55h
	1	1	↑	xx	C [7:4]				C [3:0]				xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Content Adaptive Brightness Control	0	1	↑	xx	0	1	0	1	0	1	1	0	56h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	C [7:4]				C [3:0]				xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Write CABC Minimum Brightness	0	1	↑	xx	0	1	0	1	1	1	1	0	5Eh
	1	1	↑	xx	CMB [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read CABC Minimum Brightness	0	1	↑	xx	0	1	0	1	1	1	1	1	5Fh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	CMB [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Automatic Brightness Control Self-Diagnostic Result	0	1	↑	xx	0	1	1	0	1	0	0	0	68h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D [7:6]		0	0	0	0	0	0	xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Black/White Low Bits	0	1	↑	xx	0	1	1	1	0	0	0	0	70h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	BKx [1:0]		BKy [1:0]		Wx [1:0]		Wy [1:0]		xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Bkx	0	1	↑	xx	0	1	1	1	0	0	0	1	71h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	BKx [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Bky	0	1	↑	xx	0	1	1	1	0	0	1	0	72h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	BKy [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Wx	0	1	↑	xx	0	1	1	1	0	0	1	1	73h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Wx [9:2]								xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Wy	0	1	↑	xx	0	1	1	1	0	1	0	0	74h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Wy [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Red/Green Low bits	0	1	↑	xx	0	1	1	1	0	1	0	1	75h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Rx [1:0]		Ry [1:0]		Gx [1:0]		Gy [1:0]		xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Rx	0	1	↑	xx	0	1	1	1	0	1	1	0	76h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Rx [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Ry	0	1	↑	xx	0	1	1	1	0	1	1	1	77h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Ry [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Gx	0	1	↑	xx	0	1	1	1	1	0	0	0	78h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Gx [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Gy	0	1	↑	xx	0	1	1	1	1	0	0	1	79h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Gy [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Blue/AColour Low Bits	0	1	↑	xx	0	1	1	1	1	0	1	0	7Ah
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Bx [1:0]		By [1:0]		Ax [1:0]		Ay [1:0]		xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Bx	0	1	↑	xx	0	1	1	1	1	0	1	1	7Bh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Bx [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read By	0	1	↑	xx	0	1	1	1	1	1	0	0	7Ch
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	By [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Ax	0	1	↑	xx	0	1	1	1	1	1	0	1	7Dh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Ax [9:2]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Ay	0	1	↑	xx	0	1	1	1	1	1	1	0	7Eh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	Ay [9:2]								xx

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Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read DDB Start	0	1	↑	xx	1	0	1	0	0	0	0	1	A1h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	SID [7:0]								xx
	1	↑	1	xx	SID [15:8]								xx
	1	↑	1	xx	MRID [7:0]								xx
	1	↑	1	xx	MRID [15:8]								xx
	1	↑	1	xx	1	1	1	1	1	1	1	1	FFh

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read DDB Continue	0	1	↑	xx	1	0	1	0	1	0	0	0	A8h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	D1 [7:0]								xx
	1	↑	1	xx	Dx [7:0]								xx
	1	↑	1	xx	Dn [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read First Checksum	0	1	↑	xx	1	0	1	0	1	0	1	0	AAh
	1	↑	1	xx	FCS [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Continue Checksum	0	1	↑	xx	1	0	1	0	1	1	1	1	AFh
	1	↑	1	xx	CCS [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read Device Code	0	1	↑	xx	1	1	0	1	1	0	1	0	D3h
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	0	0	0	0	0	0	0	0	00h
	1	↑	1	xx	1	0	0	1	1	0	0	0	98h
	1	↑	1	xx	0	0	0	0	0	1	1	0	06h

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read ID1	0	1	↑	xx	1	1	0	1	1	0	1	0	DAh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	ID1 [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read ID2	0	1	↑	xx	1	1	0	1	1	0	1	1	DBh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	ID2 [7:0]								xx

Command Function	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Read ID3	0	1	↑	xx	1	1	0	1	1	1	0	0	DCh
	1	↑	1	xx	x	x	x	x	x	x	x	x	xx
	1	↑	1	xx	ID3 [7:0]								xx

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4.1.2. Extended Command List

Extended Command Set												
Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Interface Mode Control	0	1	↑	1	0	1	1	0	0	0	0	B0h
	1	1	↑	0	0	0	0	VSPL	HSPL	DPL	EPL	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Frame Rate Control 1	0	1	↑	1	0	1	1	0	0	0	1	B1h
	1	1	↑	0	0	0	0	0	0	DIVA [1:0]		XX
	1	1	↑	0	0	0	RTNA [4:0]					XX
	1	1	↑	0	0	FRSA [5:0]						XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Frame Rate Control 2	0	1	↑	1	0	1	1	0	0	1	0	B2h
	1	1	↑	0	0	0	0	0	0	DIVB [1:0]		XX
	1	1	↑	0	0	0	RTNB [4:0]					XX
	1	1	↑	0	0	FRSB [5:0]						XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Frame Rate Control 3	0	1	↑	1	0	1	1	0	0	1	1	B3h
	1	1	↑	0	0	0	0	0	0	DIVC[1:0]		XX
	1	1	↑	0	0	0	RTNC [4:0]					XX
	1	1	↑	0	0	FRSC [5:0]						XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display Inversion Control	0	1	↑	1	0	1	1	0	1	0	0	B4h
	1	1	↑	0	0	0	0	NLA [3:0]				XX
	1	1	↑	0	0	0	0	NLB [3:0]				XX
	1	1	↑	0	0	0	0	NLC [3:0]				XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Blanking Porch Control	0	1	↑	1	0	1	1	0	1	0	1	B5h
	1	1	↑	0	VFP [6:0]							XX
	1	1	↑	0	VBP [6:0]							XX
	1	1	↑	HBP [7:0]								XX
	1	1	↑	0	0	0	0	0	0	HBP [9:8]		XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Display Function Control	0	1	↑	1	0	1	1	0	1	1	0	B6h
	1	1	↑	SYNC mode	0	RM	0	0	0	1	PT	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Entry Mode Set	0	1	↑	1	0	1	1	0	1	1	1	B7h
	1	1	↑	0	0	0	0	DSTB	1	1	0	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Interface Setting	0	1	↑	1	0	1	1	1	0	0	0	B8h
	1	1	↑	0	WEMODE	EPF [1:0]		0	0	0	0	XX

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Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Panel Control	0	1	↑	1	0	1	1	1	0	0	1	B9h
	1	1	↑	MY_ Panel	MX_ Panel	MV_ Panel	ML_ Panel	BGR_ Panel	MH_ Panel	SS_ Panel	GS_ Panel	XX
	1	1	↑	0	0	0	0	0	0	REV_ Panel	0	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
SPI Interface Setting	0	1	↑	1	0	1	1	1	0	1	0	BAh
	1	1	↑	Spitype	1	1	0	0	0r	0	0	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Power Control 1	0	1	↑	1	1	0	0	0	0	0	0	C0h
	1	1	↑	VD[2:0]			VE[2:0]			1	1	03
	1	1	↑	0	0	0	0	1	0	1	1	0B
	1	1	↑	0	0	0	0	BL [1:0]		1	0	0A

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Power Control 2	0	1	↑	1	1	0	0	0	0	0	1	C1h
	1	1	↑	0	0	BG_ISC [1:0]		0	1	1	1	17
	1	1	↑	VRH1 [7:0]								90
	1	1	↑	VRH2 [7:0]								90
	1	1	↑	0	0	EXT_CPCK_SEL[5:4]		0	0	0	0	20

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Power Control 3	0	1	↑	1	1	0	0	0	0	1	0	C2h
	1	1	↑	0	1	0	0	0	0	0	1	41
	1	1	↑	0	1	0	0	0	1	0	0	44
	1	1	↑	0	0	0	0	0	1	0	0	04
	1	1	↑	VGSP [7:0]								00
	1	1	↑	VGSN [7:0]								00

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
VCOM Control 1	0	1	↑	1	1	0	0	0	1	1	1	C7h
	1	1	↑	VCM [7:0]								8F
	1	1	↑	nVM	0	0	0	0	0	0	0	80

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 1	0	1	↑	1	1	0	0	1	0	0	0	C8h
	1	1	↑	PWM_DIV [7:0]								XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 2	0	1	↑	1	1	0	0	1	0	0	1	C9h
	1	1	↑	THRES_MOV [3:0]				THRES_STILL [3:0]				XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 3	0	1	↑	1	1	0	0	1	0	1	0	CAh
	1	1	↑	0	0	0	0	THRES_UI [3:0]				XX

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Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 4	0	1	↑	1	1	0	0	1	0	1	1	CBh
	1	1	↑	DTH_MOV [3:0]				DTH_STILL [3:0]				XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 5	0	1	↑	1	1	0	0	1	1	0	0	CCh
	1	1	↑	0	0	0	0	DTH_UI [3:0]				XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 6	0	1	↑	1	1	0	0	1	1	0	1	CDh
	1	1	↑	0	DIM_MOV [2:0]			0	DIM_STILL [2:0]			XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 7	0	1	↑	1	1	0	0	1	1	1	0	CEh
	1	1	↑	DIM_MIN [3:0]				0	DIM_UI [2:0]			XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Backlight Control 8	0	1	↑	1	1	0	0	1	1	1	1	CFh
	1	1	↑	0	0	0	0	0	LEDONR	LEDON POL	PWM POL	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
NV Memory Write	0	1	↑	1	1	0	1	0	0	0	0	D0h
	1	1	↑	PGM_DATA [7:0]								XX
	1	1	↑	PGM_ADR [7:0]								XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
NV Memory Protection Key	0	1	↑	1	1	0	1	0	0	0	1	D1h
	1	1	↑	KEY [23:16]								XX
	1	1	↑	KEY [15:8]								XX
	1	1	↑	KEY [7:0]								XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
NV Memory Status Read	0	1	↑	1	1	0	1	0	0	1	0	D2h
	1	↑	1	Dummy read								XX
	1	↑	1	0	ID2_Mk [2:0]			0	ID1_ Mk [2:0]			XX
	1	↑	1	OTP Busy	VCM_Mk [2:0]			0	ID3_ Mk [2:0]			XX
	1	↑	1	0	0	0	0	RB9_P2_Mk		RB9_P1_Mk		XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Engineering Setting	0	1	↑	1	1	0	1	1	1	1	1	DFh
	1	1	↑	0	0	0	0	0	0	0	0	00
	1	1	↑	0	0	0	0	0	0	0	0	00
	1	1	↑	0	0	0	0	0	0	0	0	00
	1	1	↑	0	0	0	0	0	0	0	0	00
	1	1	↑	0	0	0	0	0	0	0	0	00
	1	1	↑	0	0	dsi_pclk _div	0	0	0	0	0	00

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Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Positive Gamma Control	0	1	↑	1	1	1	0	0	0	0	0	E0h
	1	1	↑	0	0	VP0 [5:0]						XX
	1	1	↑	0	0	VP4 [5:0]						XX
	1	1	↑	0	0	VP8 [5:0]						XX
	1	1	↑	0	0	0	VP16 [4:0]					XX
	1	1	↑	0	0	0	VP32 [4:0]					XX
	1	1	↑	0	0	0	VP52 [4:0]					XX
	1	1	↑	0	0	0	0	VP80 [3:0]				XX
	1	1	↑	0	0	0	0	VP108 [3:0]				XX
	1	1	↑	0	0	0	0	VP147 [3:0]				XX
	1	1	↑	0	0	0	0	VP175 [3:0]				XX
	1	1	↑	0	0	0	VP203 [4:0]					XX
	1	1	↑	0	0	0	VP223 [4:0]					XX
	1	1	↑	0	0	0	VP239 [4:0]					XX
	1	1	↑	0	0	VP247 [5:0]						XX
	1	1	↑	0	0	VP251 [5:0]						XX
	1	1	↑	0	0	VP255 [5:0]						XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Negative Gamma Control	0	1	↑	1	1	1	0	0	0	0	1	E1h
	1	1	↑	0	0	VN0 [5:0]						XX
	1	1	↑	0	0	VN4 [5:0]						XX
	1	1	↑	0	0	VN8 [5:0]						XX
	1	1	↑	0	0	0	VN16 [4:0]					XX
	1	1	↑	0	0	0	VN32 [4:0]					XX
	1	1	↑	0	0	0	VN52 [4:0]					XX
	1	1	↑	0	0	0	0	VN80 [3:0]				XX
	1	1	↑	0	0	0	0	VN108 [3:0]				XX
	1	1	↑	0	0	0	0	VN147 [3:0]				XX
	1	1	↑	0	0	0	0	VN175 [3:0]				XX
	1	1	↑	0	0	0	VN203 [4:0]					XX
	1	1	↑	0	0	0	VN223 [4:0]					XX
	1	1	↑	0	0	0	VN239 [4:0]					XX
	1	1	↑	0	0	VN247 [5:0]						XX
	1	1	↑	0	0	VN251 [5:0]						XX
	1	1	↑	0	0	VN255 [5:0]						XX

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Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Digital Gamma Control 1	0	1	↑	1	1	1	0	0	0	1	0	E2h
	1	1	↑	RCA0 [3:0]				BCA0 [3:0]				XX
	1	1	↑	RCA1 [3:0]				BCA1 [3:0]				XX
	1	1	↑	:				:				XX
	1	1	↑	RCAx [3:0]				BCAx [3:0]				XX
	1	1	↑	:				:				XX
	1	1	↑	RCA62 [3:0]				BCA62 [3:0]				XX
	1	1	↑	RCA63 [3:0]				BCA63 [3:0]				XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Digital Gamma Control 2	0	1	↑	1	1	1	0	0	0	1	1	E3h
	1	1	↑	RFA0 [3:0]				BFA0 [3:0]				XX
				:				:				XX
	1	1	↑	RFAx [3:0]				BFAx [3:0]				XX
				:				:				XX
	1	1	↑	RFA256 [3:0]				BFA256 [3:0]				XX
	1	1	↑	0	0	0	0	0	0	0	0	00

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
D3GE	0	1	↑	1	1	1	0	1	0	1	0	EAh
	1	1	↑	0	0	0	0	0	0	En_3G	Dith_en	XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
VGMP / VGMN / VGSP / VGSN Voltage Measurement Set	0	1	↑	1	1	1	0	1	1	0	1	EDh
	1	1	↑	0	1	1	1	1	1	1	1	7F
	1	1	↑	0	0	0	0	0	1	1	1	07

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
PTCTRL1	0	1	↑	1	1	1	1	0	0	0	1	F1h
	1	1	↑	0	0	1	0	1	0	0	1	29
	1	1	↑	Chopper_opt	Chopper_sel[1:0]		0	1	0	1	0	CA
	1	1	↑	0	0	Outsre	0	0	1	1	1	07

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
PTCTRL2	0	1	↑	1	1	1	1	0	0	1	0	F2h
	1	1	↑	SDTI [1:0]		0	0	0	CRTI [2:0]			40
	1	1	↑	1	1	0	1	0	EQTI [2:0]			D2
	1	1	↑	0	creqpc	0	1	0	PCTI [2:0]			52
	1	1	↑	0	0	1	0	1	0	SDTI[2]	0	2A

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
DVDD Voltage Setting	0	1	↑	1	1	1	1	0	0	1	1	F3h
	1	1	↑	0	1	1	1	0	0	0	0	70

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Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
PWCTRL 5	0	1	↑	1	1	1	1	0	1	0	1	F5h
	1	1	↑	0	1	0	0	0	0	1	0	42
	1	1	↑	0	0	0	0	0	1	0	0	04
	1	1	↑	1	0	VRGH[5:0]						80

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Panel Resolution Selection Set	0	1	↑	1	1	1	1	0	1	1	1	F7h
	1	1	↑	1	0	0	0	0	rso_in_test_mode[2:0]			80

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
RDEXTCSPI	0	1	↑	1	1	1	1	1	0	1	1	FBh
	1	1	↑	ext_spi_read_en	ext_spi_cnt[6:0]							XX
	1	1	↑	0	0	0	0	0	0	ext_spi_cnt[8:7]		XX

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
LVGL Voltage Setting	0	1	↑	1	1	1	1	1	1	0	0	FCh
	1	1	↑	0	0	0	EXB1T_IN	LVGL_SEL[3:0]				04

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
External Power Selection Set	0	1	↑	1	1	1	1	1	1	0	1	FDh
	1	1	↑	0	0	0	0	1	0	1	0	0A
	1	1	↑	0	0	pccs_reg[1:0]		0	0	0	0	00

Command Function	DCX	RDX	WRX	D7	D6	D5	D4	D3	D2	D1	D0	Hex
EXTC Command Set enable register	0	1	↑	1	1	1	1	1	1	1	1	FFh
	1	1	↑	1	1	1	1	1	1	1	1	FF
	1	1	↑	1	0	0	1	1	0	0	0	98
	1	1	↑	0	0	0	0	0	1	1	0	06

Notes:

1. Undefined commands are treated as NOP (00h) command.
2. B0 to D9 and DE to FF are for factory use of display supplier. Users can decide if these commands are available or they are treated as NOP (00h) commands before shipping to Users. The default value is NOP (00h).
3. Commands 10h, 12h, 13h, 26h, 28h, 29h, 30h, 36h (Bit D4 only), 38h and 39h are updated during V-SYNC. When the module is in the Sleep Out mode to avoid abnormal visual effects. During the Sleep In mode, these commands are updated immediately. Read status (09h), Read display power mode (0Ah), Read display MADCTL (0Bh), Read display pixel format (0Ch), Read display image mode (0Dh), Read display signal mode (0Eh) and Read display self-diagnostic result (0Fh) are updated immediately in both Sleep In mode and Sleep Out mode.

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4.2. Command Description

4.2.1. NOP (00h)

<table>
<tr><th>00h</th><th colspan="12">NOP (No Operation)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>00h</td></tr>
<tr><td>Parameter</td><td colspan="13">No parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command is an empty command. It does not have any effect on the ILI9806G. However, it can be used to terminate the Frame Memory Write or Read as described in RAMWR (Memory Write) and RAMRD (Memory Read) Commands.
X = void</td></tr>
<tr><td>Restriction</td><td colspan="13">None</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="13"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>S/W Reset</td><td>N/A</td></tr><tr><td>H/W Reset</td><td>N/A</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">None</td></tr>
</table>

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4.2.2. Software Reset (01h)

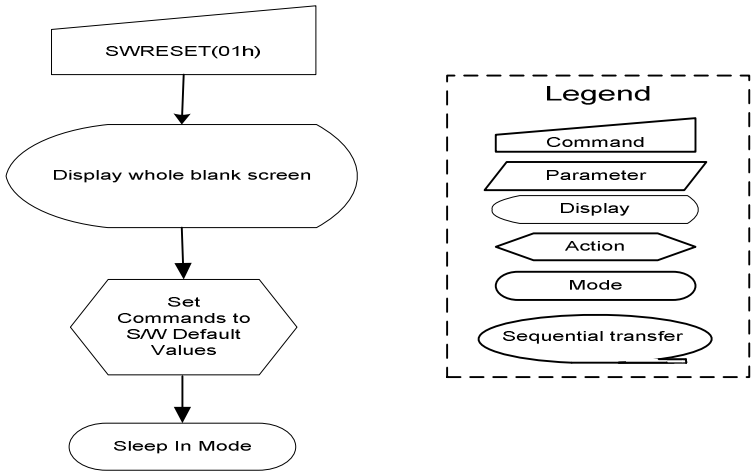
01h	SWRESET (Software Reset)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	0	0	1	01h
Parameter	No parameter												
Description	When the Software Reset command is written, it causes software reset. It resets the commands and parameters to their S/W Reset default values. (See default tables in each command description.) Note: The Frame Memory contents are not affected by this command. X = void												
Restriction	It is necessary to wait 5msec before sending a new command after software reset. The display module loads all factory default values of the display supplier to the registers during this 5msec. If Software Reset is applied during the Sleep Out mode, it will be necessary to wait 120msec before sending the Sleep Out command. The Software Reset Command cannot be sent during the Sleep Out sequence.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart



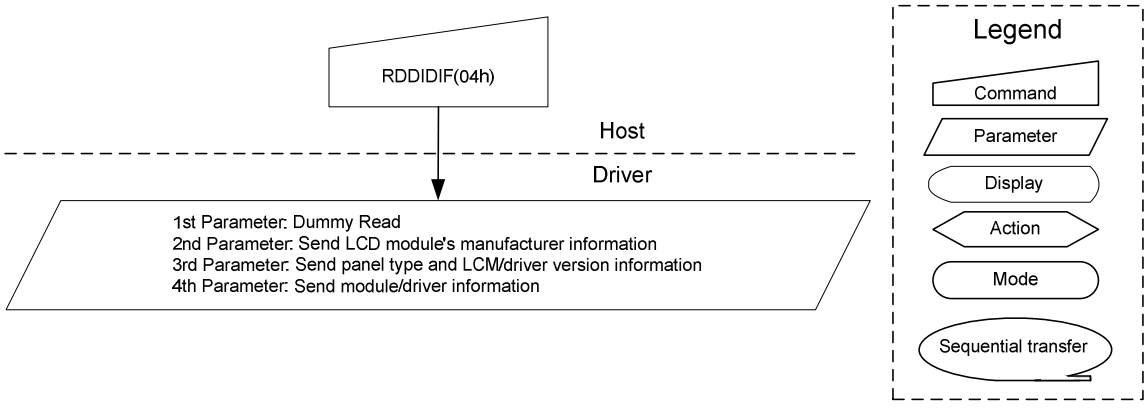
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4.2.3. Read display identification information (04h)

04h	RDDIDIF (Read Display Identification Information)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	1	0	0	04h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	ID1 [7:0]								XX
3rd Parameter	1	↑	1	XX	ID2 [7:0]								XX
4th Parameter	1	↑	1	XX	ID3 [7:0]								XX
Description	This read byte returns 24 bits of display identification information. The 1st parameter is a dummy data. The 2nd parameter (ID1 [7:0]): LCD module's manufacturer ID. The 3rd parameter (ID2 [7:0]): LCD module/driver version ID. The 4th parameter (ID3 [7:0]): LCD module/driver ID. X = void												
Restriction													
Register Availability													
Default													
Flow Chart													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
Power On Sequence	See description
S/W Reset	See description
H/W Reset	See description



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4.2.4. Read Number of the Errors on DSI (05h)

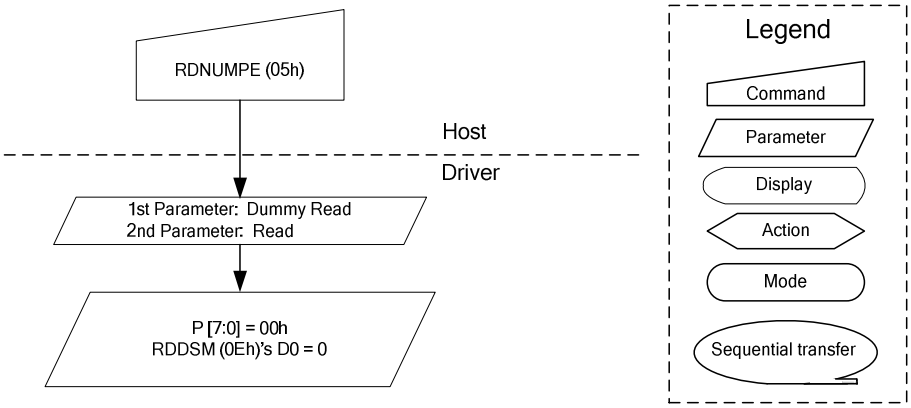
05h	RDNUMED (Read Number of the Errors on DSI)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	1	0	1	05h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	P [7:0]								XX
Description	The 1st parameter is a dummy data. The 2nd parameter indicates the amount of errors on the DSI. The more detailed description of the bits is below. P [6..0] bits indicate the amount of the error. P [7] is set to 1 if there is overflow with P [6..0] bits. P [7...0] bits are set to 0 (and RDDSM (0Eh)'s D0 is set 0 at the same time) after the second parameter information is sent (= the read function is completed). This function always returns P [7...0] = 00h if the parallel MPU interface is selected. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on DSI; the 1st parameter is not sent.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00_{HEX}
S/W Reset	00_{HEX}
H/W Reset	00_{HEX}

Flow Chart



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4.2.5. Get Red Channel (06h)

06h	RDRED (Read Red Color)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	1	1	0	06h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	R [7:0]								XX
Description	This command returns the red component value of the first pixel in the active frame. Only the relevant bits are used according to the pixel format. Unused bits are set to 0. 16-bit format R5 is MSB and R1 is LSB; R7, R6 and R0 are set to 0. 18-bit format R5 is MSB and R0 is LSB; R7, R6 are set to 0. 24-bit format R7 is MSB and R0 is LSB. X = void												
Restriction													

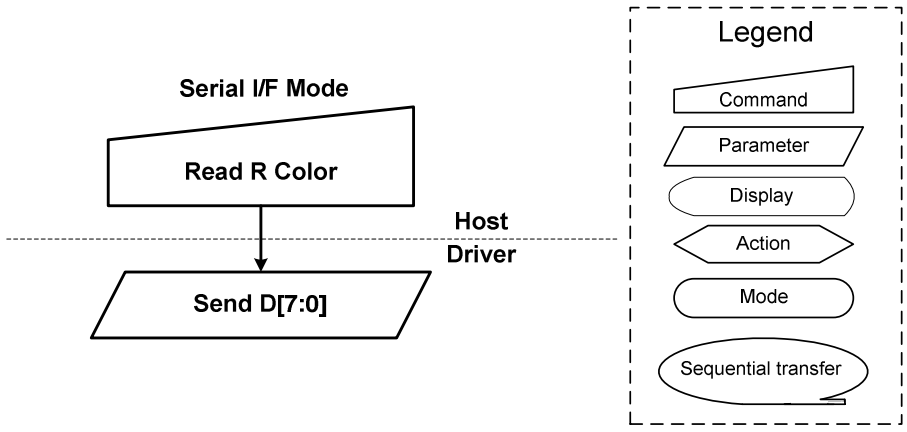
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart



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4.2.6. Get Green Channel (07h)

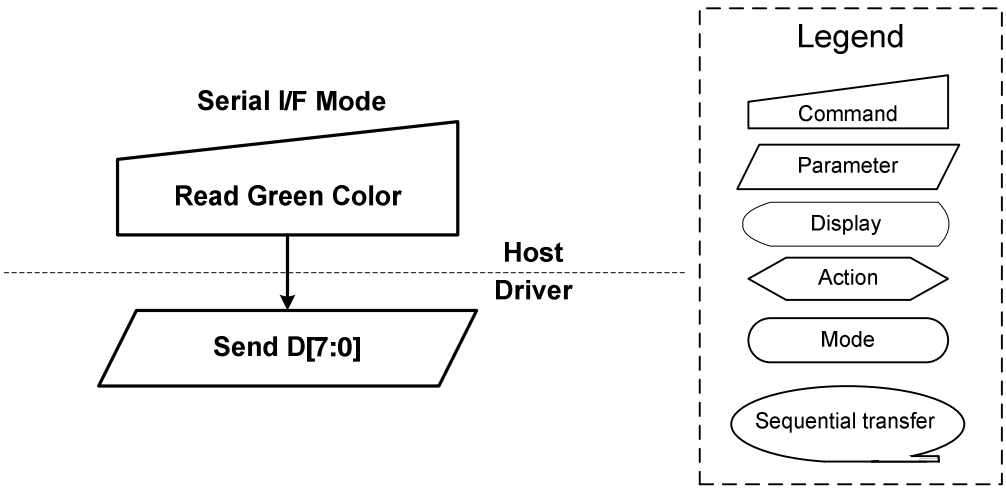
07h	RDGREEN (Read Green Color)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	0	1	1	1	07h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	G [7:0]								XX
Description	This command returns the green component value of the first pixel in the active frame. Only the relevant bits are used according to the pixel format. Unused bits are set to 0. 16-bit format G5 is MSB and G0 is LSB; G7, G6 are set to 0. 18-bit format G5 is MSB and G0 is LSB; G7, G6 are set to 0. 24-bit format G7 is MSB and G0 is LSB. X = void												
Restriction													
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart



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4.2.7. Get Blue Channel (08h)

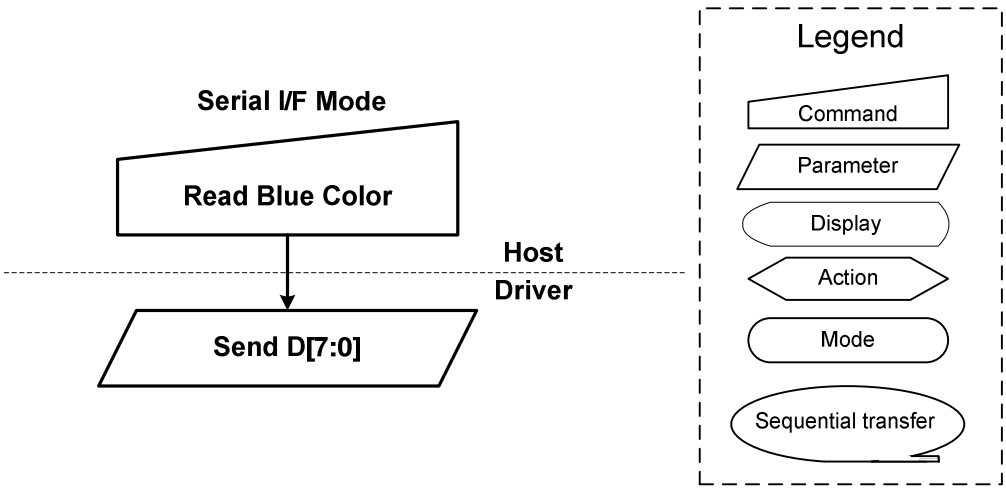
08h	RDBLUE (Read Blue Color)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	0	0	0	08h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	B [7:0]								XX
Description	This command returns the blue component value of the first pixel in the active frame. Only the relevant bits are used according to the pixel format. Unused bits are set to 0. 16-bit format B5 is MSB and B1 is LSB; B7, B6 and B0 are set to 0. 18-bit format B5 is MSB and B0 is LSB; B7, B6 are set to 0. 24-bit format B7 is MSB and B0 is LSB. X = void												
Restriction													
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	N/A
S/W Reset	N/A
H/W Reset	N/A

Flow Chart



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4.2.8. Read Display Status (09h)

09h	RDDST (Read Display Status)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	0	0	1	09h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	D [31:24]								XX
3rdParameter	1	↑	1	XX	D [23:16]								XX
4thParameter	1	↑	1	XX	D [15:8]								XX
5thParameter	1	↑	1	XX	D [7:0]								XX

Description

This command indicates the current status of the display, as described in the table below:

Bit	Description	Value	Status
D31	Booster voltage status	0	Booster Off
		1	Booster On
D30	Row address order	0	Top to Bottom (When MADCTL D7 = 0)
		1	Bottom to Top (When MADCTL D7 = 1)
D29	Column address order	0	Left to Right (When MADCTL D6 = 0)
		1	Right to Left (When MADCTL D6 = 1)
D28	Row/column exchange	0	Normal Mode (When MADCTL D5 = 0)
		1	Reverse Mode (When MADCTL D5 = 1)
D27	Vertical refresh	0	LCD Refresh Top to Bottom (When MADCTL D4 = 0)
		1	LCD Refresh Bottom to Top (When MADCTL D4 = 1)
D26	RGB/BGR order	0	RGB (When MADCTL D3 = 0)
		1	BGR (When MADCTL D3= 1)
D25	Horizontal refresh order	0	LCD Refresh Left to Right (When MADCTL D2 = 0)
		1	LCD Refresh Right to Left (When MADCTL D2 = 1)
D24	Source scan sequence	0	Source output Left to Right (When MADCTL D1 = 0)
		1	Source output Right to Left (When MADCTL D1 = 1)
D23	Gate scan sequence	0	Gate output Top to Bottom (When MADCTL D0 = 0)
		1	Gate output Bottom to Top (When MADCTL D0 = 1)
D [22:20]	Interface color pixel format definition	101	16-bit/pixel
		110	18-bit/pixel
		111	24-bit/pixel
		others	Not defined
D19	Idle Mode On/Off	0	Idle Mode Off
		1	Idle Mode On
D18	Partial Mode On/Off	0	Partial Mode Off
		1	Partial Mode On
D17	Sleep In/Out	0	Sleep In Mode
		1	Sleep Out Mode
D16	Display Normal Mode On/Off	0	Display Normal Mode Off (Partial or Scrolling Mode)
		1	Display Normal Mode On
D15	Vertical scrolling status	0	Vertical Scroll Off
		1	Vertical Scroll On
D14	Horizontal Scrolling Status	0	This bit is not applicable for this project, so it is set to 0
D13	Inversion status	0	Inversion Off
		1	Inversion On

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D12	All Pixel On	0	Normal mode
		1	All Pixels On
D11	All Pixel Off	0	Normal mode
		1	All Pixels Off
D10	Display ON/OFF	0	Display is OFF
		1	Display is ON
D9	Tearing effect line ON/OFF	0	Tearing Effect Line OFF
		1	Tearing Effect ON
D [8:6]	Gamma curve selection	000	Gamma Curve 1
		001	Gamma Curve 2
		010	Gamma Curve 3
		011	Gamma Curve 4
		others	Not defined
D5	Tearing Effect Line mode	0	Mode 1, V-Blanking only
		1	Mode 2, both H-Blanking and V-Blanking.
D4	Horizontal Sync. (HSYNC, DPI I/F) (Note)	0	Horizontal Sync. line is Off (Low)
		1	Horizontal Sync. line is On (High)
D3	Vertical Sync. (VSYNC, DPI I/F) (Note)	0	Vertical Sync. line is Off (Low)
		1	Vertical Sync. line is On (High)
D2	DOT Clock (DCK, DPI I/F) (Note)	0	DCK line is Off (Low)
		1	DCK line is On (High)
D1	Data Enable (ENABLE, DPI I/F) (Note)	0	DE line is Off (Low)
		1	DE line is On (High)
D0	Parity Error on DSI	0	No Parity Error
		1	Parity Error

Note: This bit indicates the current status of the line when this command is sent.

X = void

Restriction

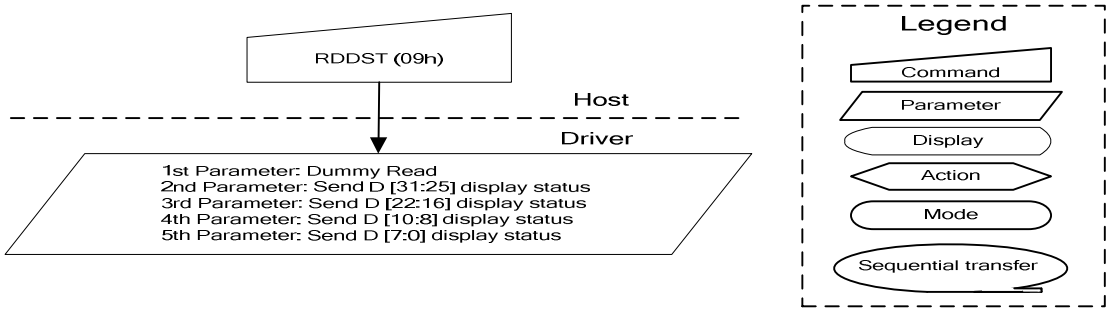
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	32'h00710000h
S/W Reset	D20~D30 no change, keep the current status before S/W Reset
H/W Reset	32'h00710000h

Flow Chart

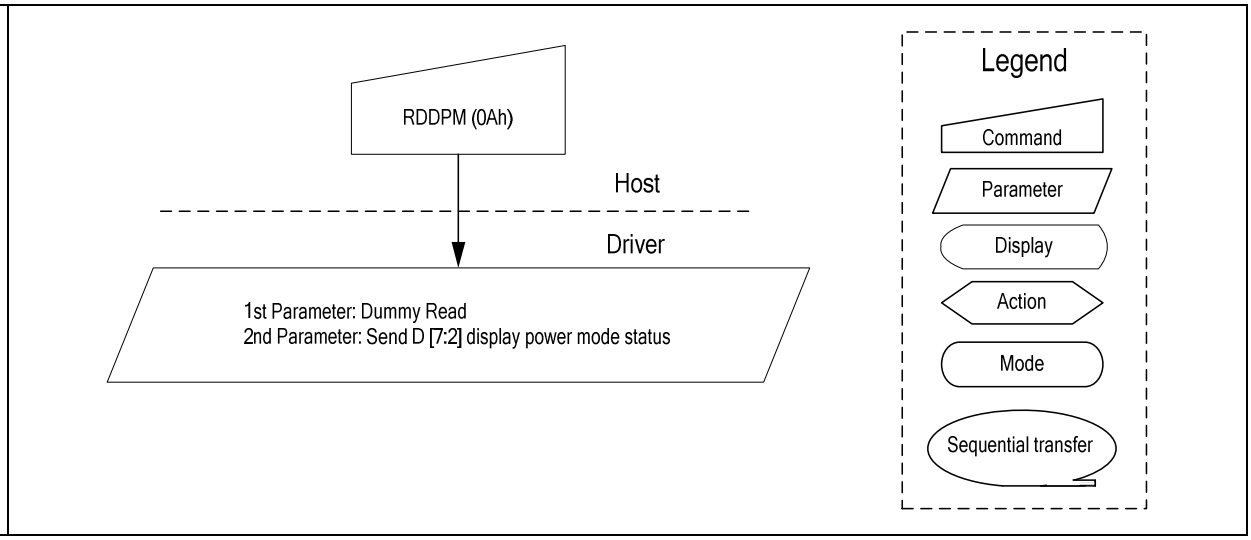


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4.2.9. Read Display Power Mode (0Ah)

<table>
<tr><th>0Ah</th><th colspan="12">RDDPM (Read Display Power Mode)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0Ah</td></tr>
<tr><td>1st Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>XX</td></tr>
<tr><td>2nd Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td colspan="7">D [7:2]</td><td>0</td><td>0</td><td>XX</td></tr>
<tr><td>Description</td><td colspan="13">This command indicates the current status of the display, as described in the table below.
<table>
<tr><th>Bit</th><th>Description</th><th>Value</th><th>Status</th></tr>
<tr><td rowspan="2">D7</td><td rowspan="2">Booster Voltage Status</td><td>0</td><td>Booster Off or has a fault</td></tr>
<tr><td>1</td><td>Booster On and working OK</td></tr>
<tr><td rowspan="2">D6</td><td rowspan="2">Idle Mode On/Off</td><td>0</td><td>Idle Mode Off</td></tr>
<tr><td>1</td><td>Idle Mode On</td></tr>
<tr><td rowspan="2">D5</td><td rowspan="2">Partial Mode On/Off</td><td>0</td><td>Partial Mode Off</td></tr>
<tr><td>1</td><td>Partial Mode On</td></tr>
<tr><td rowspan="2">D4</td><td rowspan="2">Sleep In/Out</td><td>0</td><td>Sleep In Mode</td></tr>
<tr><td>1</td><td>Sleep Out Mode</td></tr>
<tr><td rowspan="2">D3</td><td rowspan="2">Display Normal Mode On/Off</td><td>0</td><td>Display Normal Mode Off</td></tr>
<tr><td>1</td><td>Display Normal Mode On</td></tr>
<tr><td rowspan="2">D2</td><td rowspan="2">Display On/Off</td><td>0</td><td>Display is Off</td></tr>
<tr><td>1</td><td>Display is On</td></tr>
<tr><td>D1</td><td>Not Defined</td><td>--</td><td>Set to 0</td></tr>
<tr><td>D0</td><td>Not Defined</td><td>--</td><td>Set to 0</td></tr>
</table>
X = void</td></tr>
<tr><td>Restriction</td><td colspan="13">The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.
Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.</td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>08_{HEX}</td></tr>
<tr><td>S/W Reset</td><td>08_{HEX}</td></tr>
<tr><td>H/W Reset</td><td>08_{HEX}</td></tr>
</table>
</td></tr>
</table>

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Flow Chart	

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4.2.10. Read Display MADCTL (0Bh)

0Bh	RDDMADCTL (Read Display MADCTL)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	0	1	1	0Bh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	D [7:0]								XX

Description

This command indicates the current status of the display, as described in the table below.

Bit	Description	Value	Status
D7	Page Address Order	0	Top to Bottom (When MADCTL D7= 0)
		1	Bottom to Top (When MADCTL D7= 1)
D6	Column Address Order	0	Left to Right (When MADCTL D6= 0)
		1	Right to Left (When MADCTL D6= 1)
D5	Page/Column Order	0	Normal Mode (When MADCTL D5= 0)
		1	Reverse Mode (When MADCTL D5= 1)
D4	Line Address Order	0	LCD Refresh Top to Bottom (When MADCTL D4= 0)
		1	LCD Refresh Bottom to Top (When MADCTL D4= 1)
D3	RGB/BGR Order	0	RGB (When MADCTL D3= 0)
		1	BGR (When MADCTL D3= 1)
D2	Display Data Latch Data Order	0	LCD Refresh Left to Right (When MADCTL D2= 0)
		1	LCD Refresh Right to Left (When MADCTL D2= 1)
D1	Source scan sequence	0	Source output Left to Right
		1	Source output Right to Left
D0	Gate scan sequence	0	Gate output Top to Bottom
		1	Gate output Bottom to Top

X = void

Restriction

The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.

Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.

Register Availability

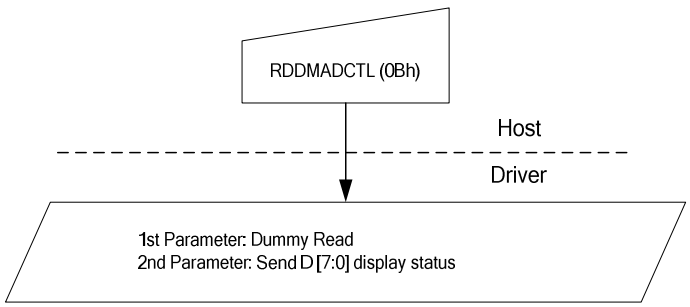
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

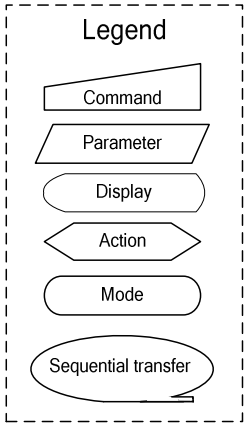
Default

Status	Default Value
Power On Sequence	00_{HEX}
S/W Reset	No Change
H/W Reset	00_{HEX}

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Flow Chart





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4.2.11. Read Display Pixel Format (0Ch)

0Ch	RDDCOLMOD (Read Display COLMOD)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	1	0	0	0Ch
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	0	DPI [2:0]			0	DBI [2:0]			XX

Description

This command indicates the current status of the display, as described in the table below:

DPI [2:0]			RGB Interface Format
1	0	1	16-bit/pixel
1	1	0	18-bit/pixel
1	1	1	24-bit/pixel
Others			Reserved

DBI [2:0]			CPU Interface Format
1	0	1	16-bit/pixel
1	1	0	18-bit/pixel
1	1	1	24-bit/pixel
Others			Reserved

X = void

Restriction

The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.

Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.

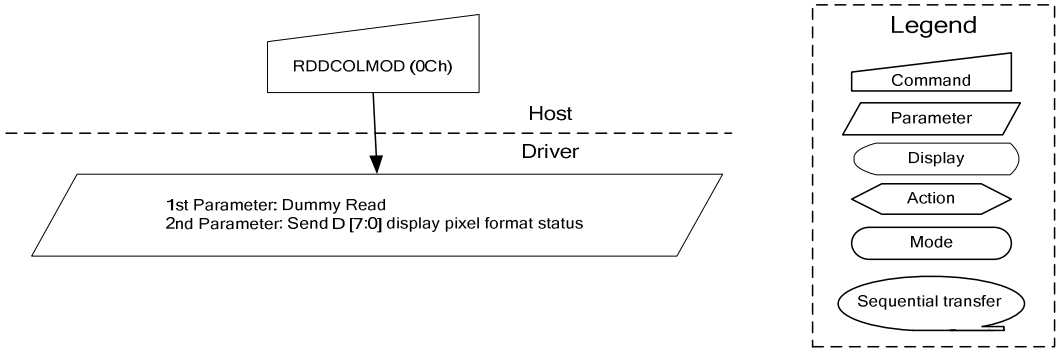
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	07_{HEX}
S/W Reset	07_{HEX}
H/W Reset	07_{HEX}

Flow Chart



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4.2.12. Read Display Image Mode (0Dh)

0Dh	RDDIM (Read Display Image Mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	1	0	1	0Dh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	D [7:0]								XX

Description

This command indicates the Image Mode status of the display, as described in the Tables below:

Bit	Description	Value	Status
D7	Vertical Scrolling Status	0	Vertical Scrolling is Off
		1	Vertical Scrolling is On
D6	Horizontal Scrolling Status (Reserved)	0	It is not applicable for this project, so set to 0
D5	Inversion On/Off	0	Inversion is Off
		1	Inversion is On
D4	All Pixels On	0	Normal Display
		1	White Display
D3	All Pixels Off	0	Normal Display
		1	Black Display

Bit D [2:0]	Gamma Cure Selection	Gamma Set (26h) Parameter
000	Gamma curve 1	GC0
001	Gamma curve 2	GC1
010	Gamma curve 3	GC2
011	Gamma curve 4	GC3
Others	Not defined	Not defined

X = void

Restriction

The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.

Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.

Register Availability

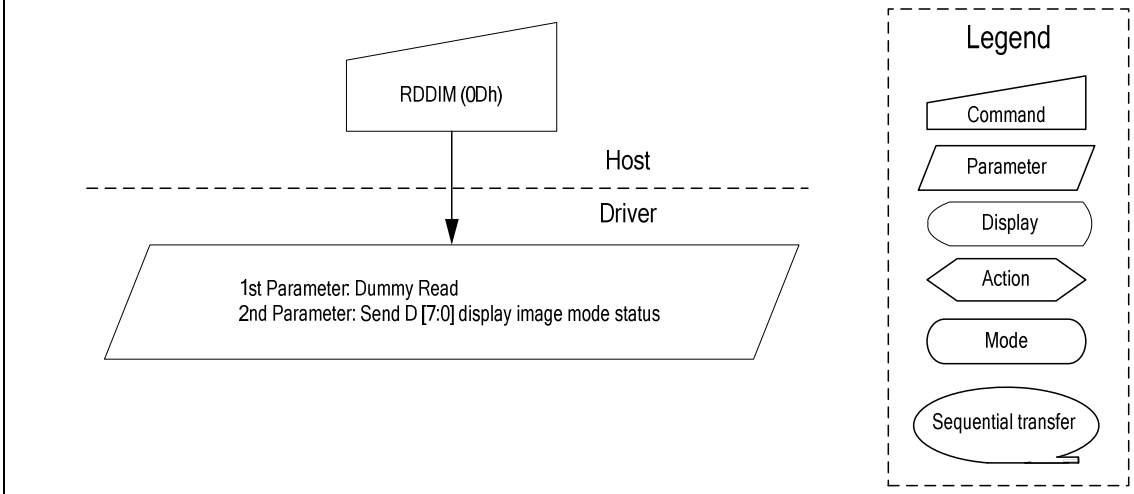
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00_{HEX}
S/W Reset	00_{HEX}
H/W Reset	00_{HEX}

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Flow Chart



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4.2.13. Read Display Signal Mode (0Eh)

0Eh	RDDSM (Read Display Signal Mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	1	1	0	0Eh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	D7	D6	D5	D4	D3	D2	D1	D0	XX

Description

This command indicates the current status of the display, as described in the table below:

Bit	Description	Value	Status
D7	Tearing Effect Line On/Off	0	Tearing Effect Line OFF
		1	Tearing Effect Line ON
D6	Tearing Effect Line Output Mode,	0	Tearing Effect Line Mode 1
		1	Tearing Effect Line Mode 2
D5	Horizontal Sync. (RGB I/F) On/Off	0	HSYNC line is OFF (low)
		1	HSYNC line is ON (high)
D4	Vertical Sync. (RGB I/F) On/Off	0	VSYNC line is OFF (low)
		1	VSYNC line is ON (high)
D3	Pixel Clock (DCK, RGB I/F) On/Off	0	DCK line is OFF (low)
		1	DCK line is ON (high)
D2	Data Enable (DE, RGB I/F) On/Off	0	Enable Line is OFF (low)
		1	Enable Line is ON (high)
D1	Reserved	0	Reserved, so it is set to 0
D0	Error on DSI,	0	No Error on DSI
		1	Error on DSI

X = void

Restriction

The ILI9487 sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.

Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.

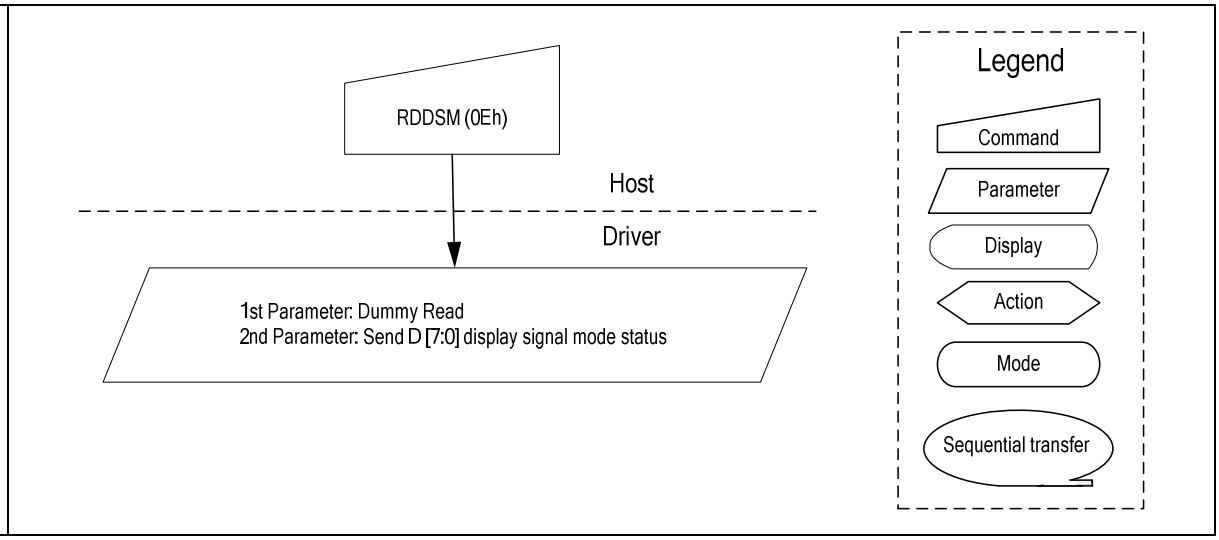
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00_{HEX}
S/W Reset	00_{HEX}
H/W Reset	00_{HEX}

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Flow Chart	

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4.2.14. Read Display Self-Diagnostic Result (0Fh)

0Fh	RDDSDR (Read Display Self-Diagnostic Result)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	0	1	1	1	1	0Fh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	D7	D6	D5	D4	0	0	0	D0	XX

Description

This command indicates the status of the display self-diagnostic results after the Sleep Out command, as described in the table below:

Bit	Description	Action
D7	Register Loading Detection	Invert the D7 bit when the EEPROM and register values are the same.
D6	Functionality Detection	Invert the D6 bit when the chip meets user's functionality requirements
D5	Chip Attachment Detection	Set bit D5 to 0, if this function is not implemented.
D4	Display Glass Break Detection	Set bit D4 to 0, if this function is not implemented.
D3	Not Used	Set to 0
D2	Not Used	Set to 0
D1	Not Used	Set to 0
D0	Checksums Comparison	0 = Checksums are the same 1 = Checksums are not the same

X = void

Restriction

It is necessary to wait 300ms after the last write access to registers on the User area before the Bit D0 value can be read.

The ILI9487 sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface.

Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.

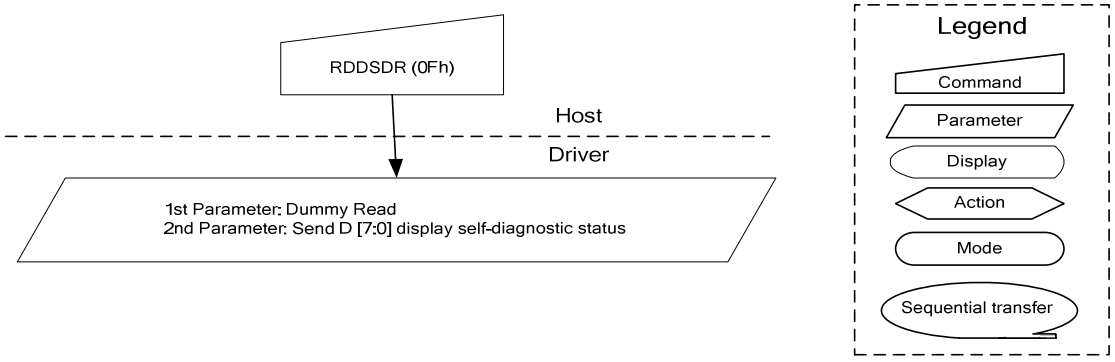
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

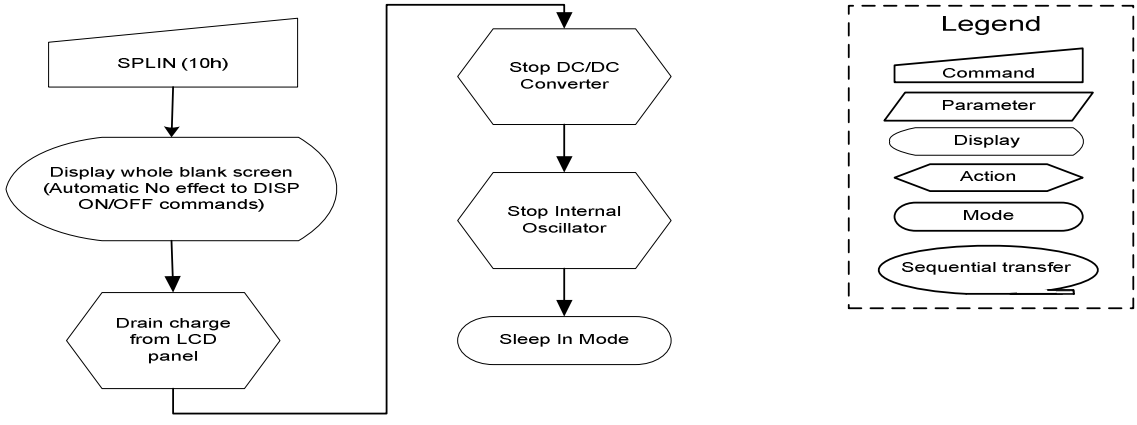
Status	Default Value
Power On Sequence	00_{HEX}
S/W Reset	00_{HEX}
H/W Reset	00_{HEX}

Flow Chart



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4.2.15. Sleep In (10h)

10h	SLPIN (Sleep In)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	0	0	10h
Parameter	No parameter												
Description	This command causes the ILI9806G to enter the minimum power consumption mode. In this mode, the DC/DC converter, Internal oscillator, and panel scanning are all stopped.  The MPU interface and memory are still working, and the memory keeps its contents. X = Void												
Restriction	This command has no effect when the module is already in the Sleep In mode. To exit the Sleep In mode, only the Sleep Out Command (11h) is workable. It is necessary to wait 5msec before sending the next command; this is to allow time for the supply voltages and clock circuits to become stable. It is necessary to wait 120msec after sending the Sleep Out command (when in the Sleep In Mode) before the Sleep In command can be sent.												
Register Availability													
Default													
Flow Chart													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
Power On Sequence	Sleep In Mode
S/W Reset	Sleep In Mode
H/W Reset	Sleep In Mode

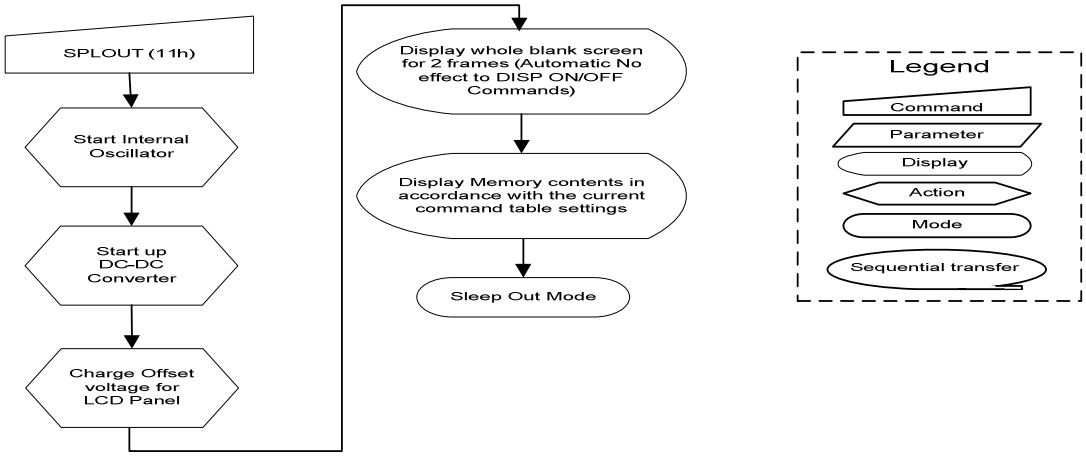
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4.2.16. Sleep Out (11h)

11h	SLPOUT (Sleep Out)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	0	1	11h
Parameter	No parameter												
Description	This command turns off the sleep mode. In this mode, the DC/DC converter is enabled, the Internal oscillator is started, and the panel scanning is started.  X = Void												
Restriction	This command has no effect when the module is already in the Sleep Out mode. To exit the Sleep Out Mode, Sleep In command (10h), S/W reset command (01h), and H/W reset are workable. It is necessary to wait 5msec before sending the next command; this is to allow time for the supply voltages and clock circuits to become stable. The ILI9806G loads all factory default values of the display supplier to the registers during this 5msec. There cannot be any abnormal visual effect on the display image if factory defaults and register values are the same when this load is done and when the ILI9806G is already in the Sleep Out mode. The ILI9806G performs self-diagnostic functions during this 5msec. It is necessary to wait 120msec after sending the Sleep In command (when in the Sleep Out mode) before the Sleep Out command can be sent.												
Register Availability													
Default													
Flow Chart													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
Power On Sequence	Sleep In Mode
S/W Reset	Sleep In Mode
H/W Reset	Sleep In Mode



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4.2.17. Partial Mode On (12h)

<table>
<tr><th>12h</th><th colspan="12">PTLON (Partial Mode On)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>12h</td></tr>
<tr><td>Parameter</td><td colspan="13">No parameter</td></tr>
<tr><td>Description</td><td colspan="13">This command turns on the Partial Mode. The Partial Mode window is described in the Partial Area command (30H).
To leave the Partial Mode, the Normal Display Mode On command (13H) should be written.</td></tr>
<tr><td>Restriction</td><td colspan="13">This command has no effect when the Partial Display Mode is already active.</td></tr>
<tr><td>Register Availability</td><td colspan="13"><table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table></td></tr>
<tr><td>Default</td><td colspan="13"><table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal Display Mode</td></tr><tr><td>S/W Reset</td><td>Normal Display Mode</td></tr><tr><td>H/W Reset</td><td>Normal Display Mode</td></tr></table></td></tr>
<tr><td>Flow Chart</td><td colspan="13">See Partial Area (30h)</td></tr>
</table>

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4.2.18. Normal Display Mode On (13h)

13h	NORON (Normal Display Mode On)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	0	1	0	0	1	1	13h
Parameter	No parameter												
Description	This command returns the display to the Normal Display Mode. Normal Display Mode On means Partial Mode Off and Scroll mode off. X = Void												
Restriction	This command has no effect when the Normal Display Mode is active.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Normal Display Mode On
S/W Reset	Normal Display Mode On
H/W Reset	Normal Display Mode On

Flow Chart	See Partial Area Descriptions for details of when to use this command.

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4.2.19. Display Inversion Off (20h)

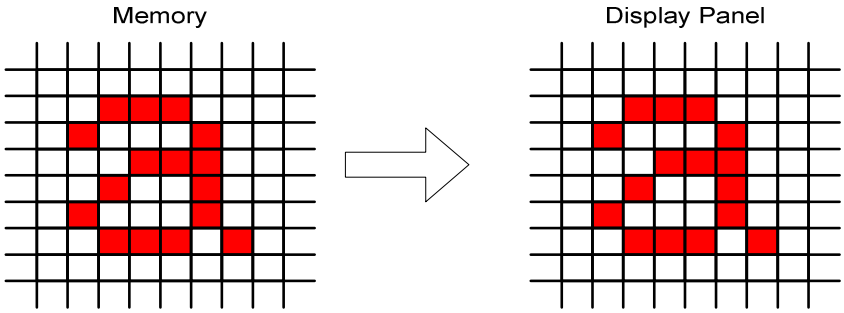
20h	INVOFF (Display Inversion Off)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	0	0	0	0	20h
Parameter	No parameter												

Description

This command is used to recover from the Display Inversion On mode.

This command makes no change of the content of Frame Memory.

This command does not change any other status.



X = Void

Restriction

This command has no effect when the module is already in the Display Inversion Off mode.

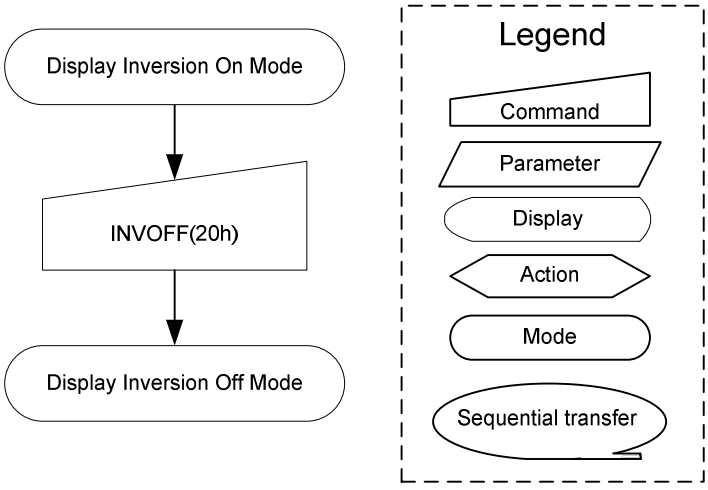
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display Inversion Off
S/W Reset	Display Inversion Off
H/W Reset	Display Inversion Off

Flow Chart



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4.2.20. fDisplay Inversion On (21h)

21h	INVON (Display Inversion ON)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	0	0	0	1	21h
Parameter	No parameter												

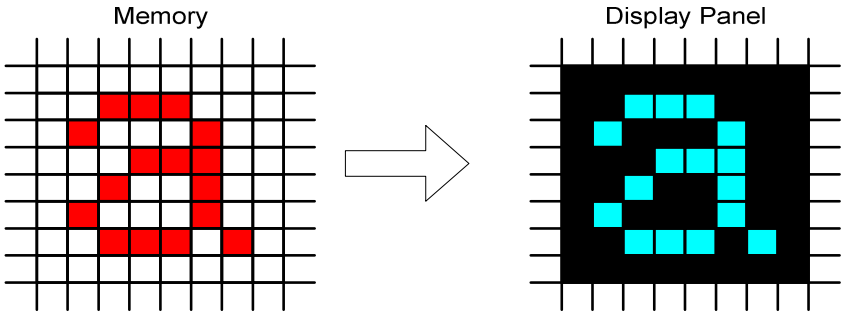
Description

This command is used to enter the Display Inversion On mode.

This command makes no change of the content of the Frame Memory. Every bit is inverted from the Frame Memory to the display.

This command does not change any other status.

To exit the Display Inversion On mode, the Display Inversion Off command (20h) should be written.



X = Void

Restriction

This command has no effect when the ILI9806G is already in the Inversion On mode.

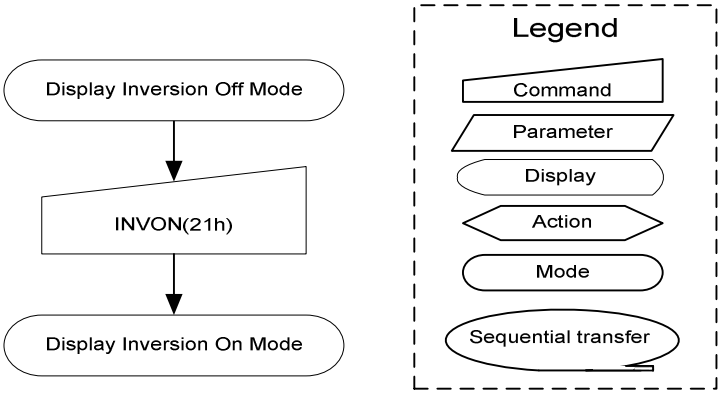
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

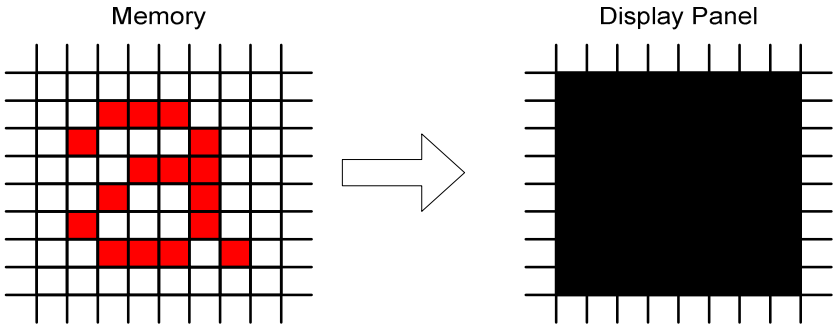
Status	Default Value
Power On Sequence	Display Inversion Off
S/W Reset	Display Inversion Off
H/W Reset	Display Inversion Off

Flow Chart



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4.2.21. All Pixel Off (22h)

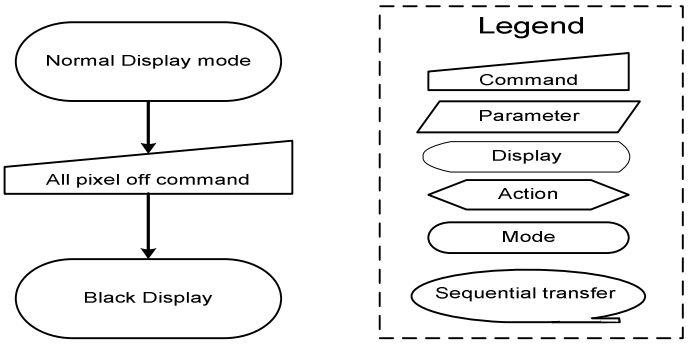
22h	INVON (Display Inversion ON)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	0	0	1	0	22h
Parameter	No parameter												
Description	This command turns the display panel black in the Sleep Out mode and the status of the Display On/Off register can be ON or OFF. This command makes no change to the contents of the Frame Memory. This command does not change any other status.  To exit this mode, All Pixels On, Normal Display Mode On, or Partial Mode On commands can be used. The display panel shows the content of the Frame Memory after Normal Display Mode On and Partial Mode On commands.												
Restriction	This command has no effect when the ILI9806G is already in the Display Inversion On mode.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

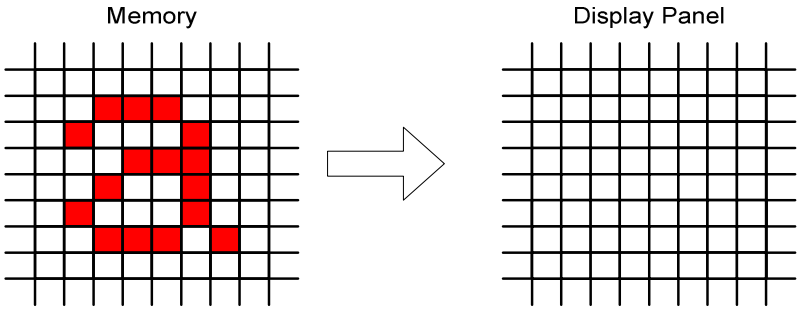
Status	Default Value
Power On Sequence	OFF
S/W Reset	OFF
H/W Reset	OFF

Flow Chart



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4.2.22. All Pixel On (23h)

23h	INVON (Display Inversion ON)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	0	0	1	1	23h
Parameter	No parameter												
Description	This command turns the display panel white in the Sleep Out mode and the status of the Display On/Off register can be ON or OFF. This command makes no change of contents of the Frame Memory. This command does not change any other status.  To exit this mode, All Pixels Off, Normal Display Mode On or Partial Mode On commands can be used. The display shows the content of the Frame Memory after Normal Display Mode On and Partial Mode On commands.												
Restriction	This command has no effect when the ILI9806G is already in the Inversion on mode.												

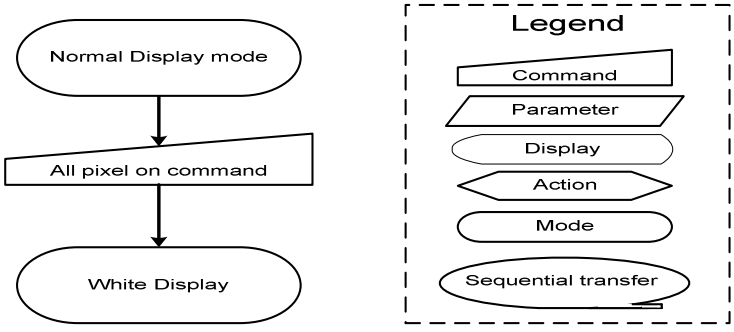
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

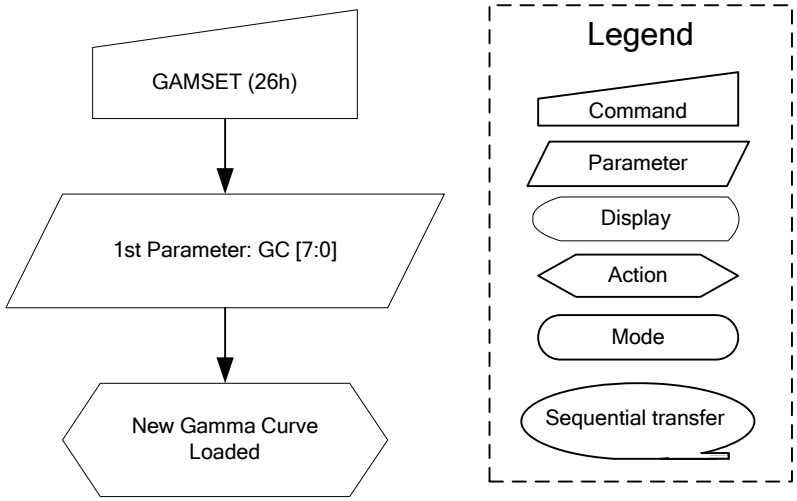
Status	Default Value
Power On Sequence	OFF
S/W Reset	OFF
H/W Reset	OFF

Flow Chart



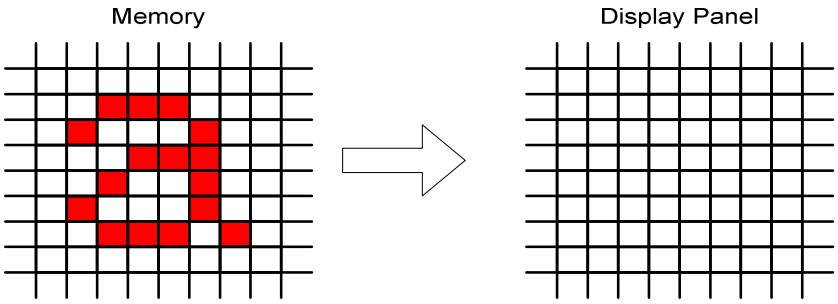
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4.2.23. Gamma Set (26h)

<table>
<tr><th>26h</th><th colspan="13">GAMSET (Gamma Set)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>26h</td></tr>
<tr><td>Parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td colspan="8">GC [7:0]</td><td>01</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to select the desired Gamma curve for the current display. A maximum of 4 fixed Gamma curves can be selected. The curve is selected by setting the appropriate bit in the parameter as described in the Table below:
<table>
<tr><th>GC [7:0]</th><th>Parameter</th><th>Curve Selected</th></tr>
<tr><td>01h</td><td>GC0</td><td>Gamma curve 1 (G2.2)</td></tr>
<tr><td>02h</td><td>GC1</td><td>Gamma curve 2 (G1.8)</td></tr>
<tr><td>04h</td><td>GC2</td><td>Gamma curve 3 (G2.5)</td></tr>
<tr><td>08h</td><td>GC3</td><td>Gamma curve 4 (G1)</td></tr>
</table>
Note: All others value are undefined.
X = Void</td></tr>
<tr><td>Restriction</td><td colspan="13">Values of GC [7:0] not shown in the table above are invalid and will not change the current selected Gamma curve until a valid value is received.</td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>8’h01h</td></tr>
<tr><td>S/W Reset</td><td>8’h01h</td></tr>
<tr><td>H/W Reset</td><td>8’h01h</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td colspan="13"> 
</td></tr>
</table>

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4.2.24. Display Off (28h)

28h	DISOFF (Display Off)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	0	0	28h
Parameter	No parameter												
Description	This command is used to enter the Display Off mode. In this mode, the output from the Frame Memory is disabled and a blank page inserted. This command makes no change of contents of the Frame Memory and does not change any other status. There will be no abnormal visible effect on the display.  X = Void												
Restriction	This command has no effect when the module is already in the Display Off mode.												

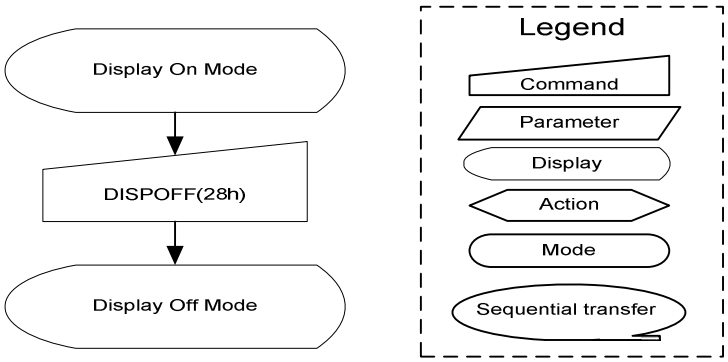
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display Off
S/W Reset	Display Off
H/W Reset	Display Off

Flow Chart



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4.2.25. Display ON (29h)

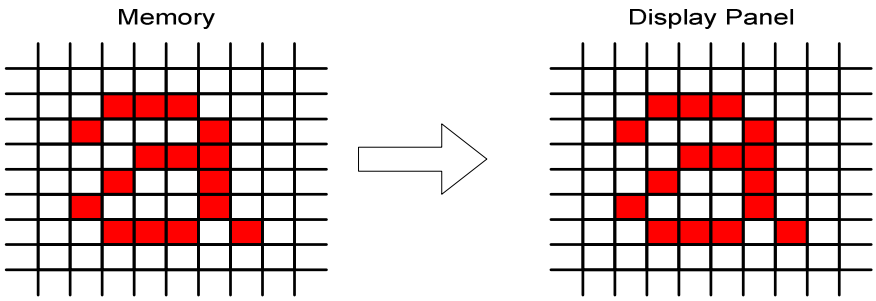
29h	DISON (Display ON)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	0	1	29h
Parameter	No parameter												

Description

This command is used to recover from the Display Off mode. Output from the Frame Memory is enabled.

This command makes no change to the contents of the Frame Memory.

This command does not change any other status.



X = Void

Restriction

This command has no effect when the ILI9806G is already in the Display On mode.

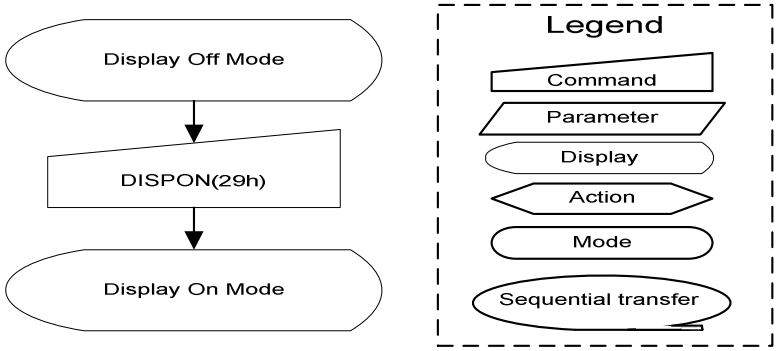
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Display Off
S/W Reset	Display Off
H/W Reset	Display Off

Flow Chart



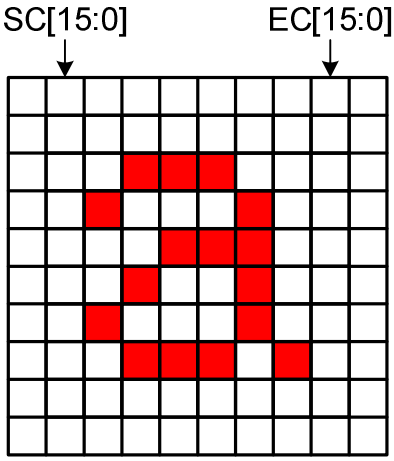
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4.2.26. Column Address Set (2Ah)

2Ah	CASET (Column Address Set)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	1	0	2Ah
1st Parameter	1	1	↑	XX	SC [15:8]								XX
2nd Parameter	1	1	↑	XX	SC [7:0]								XX
3rd Parameter	1	1	↑	XX	EC [15:8]								XX
4th Parameter	1	1	↑	XX	EC [7:0]								XX

Description

This command is used to define the area of the frame memory the MPU can access. This command makes no change on the other status of the driver. When the RAMWR command is applied, the values of SC [15:0] and EC [15:0] are referred. Each value represents one column line in the Frame Memory.



Restriction

SC [15:0] must always be equal to or less than EC [15:0]. For example (480 (RGB) x 864 Resolution):

When SC [15:0] or EC [15:0] is greater than 01DFh (when MADCTL's D5 = 0) or 035Fh (when MADCTL's D5 = 1), data out of range will be ignored.

Register Availability

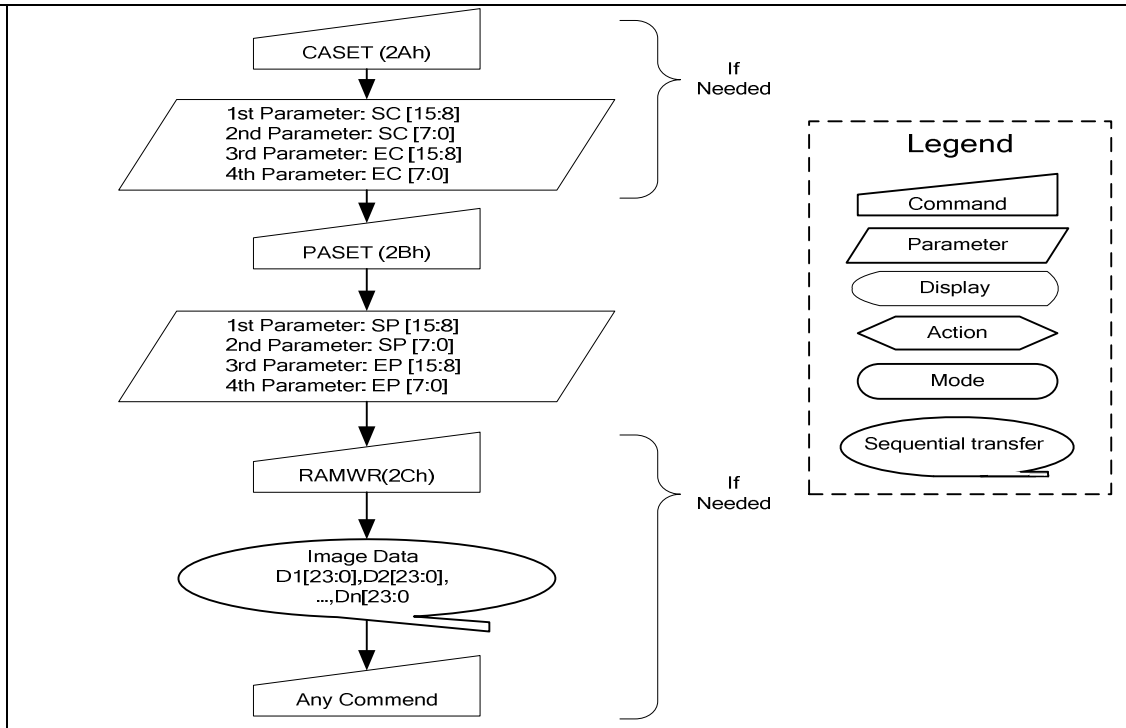
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

RSO [2:0]	Resolution	SC [15:0]	EC [15:0]
000	480 (RGB) x 864	0 x 0000h	0 x 01DFh
001	480 (RGB) x 854	0 x 0000h	0 x 01DFh
010	480 (RGB) x 800	0 x 0000h	0 x 01DFh
011	480 (RGB) x 640	0 x 0000h	0 x 01DFh

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Flow Chart



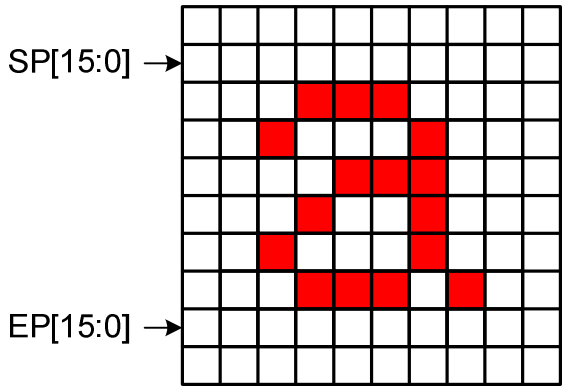
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4.2.27. Page Address Set (2Bh)

2Bh	PASET (Page Address Set)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	0	1	1	2Bh
1st Parameter	1	1	↑	XX	SP [15:8]								XX
2nd Parameter	1	1	↑	XX	SP [7:0]								XX
3rd Parameter	1	1	↑	XX	EP [15:8]								XX
4th Parameter	1	1	↑	XX	EP [7:0]								XX

Description

This command is used to define the area of the frame memory the MPU can access. This command makes no change on the other status of the driver. The values of SP [15:0] and EP [15:0] are referred when RAMWR command is applied. Each value represents one Page line in the Frame Memory.



X = void

Restriction

SP [15:0] must always be equal to or less than EP [15:0]. For example (480 (RGB) x 864 Resolution):

When SP [15:0] or EP [15:0] is greater than 035Fh (when MADCTL's D5 = 0) or 01DFh (when MADCTL's D5 = 1), data out of range will be ignored.

Register Availability

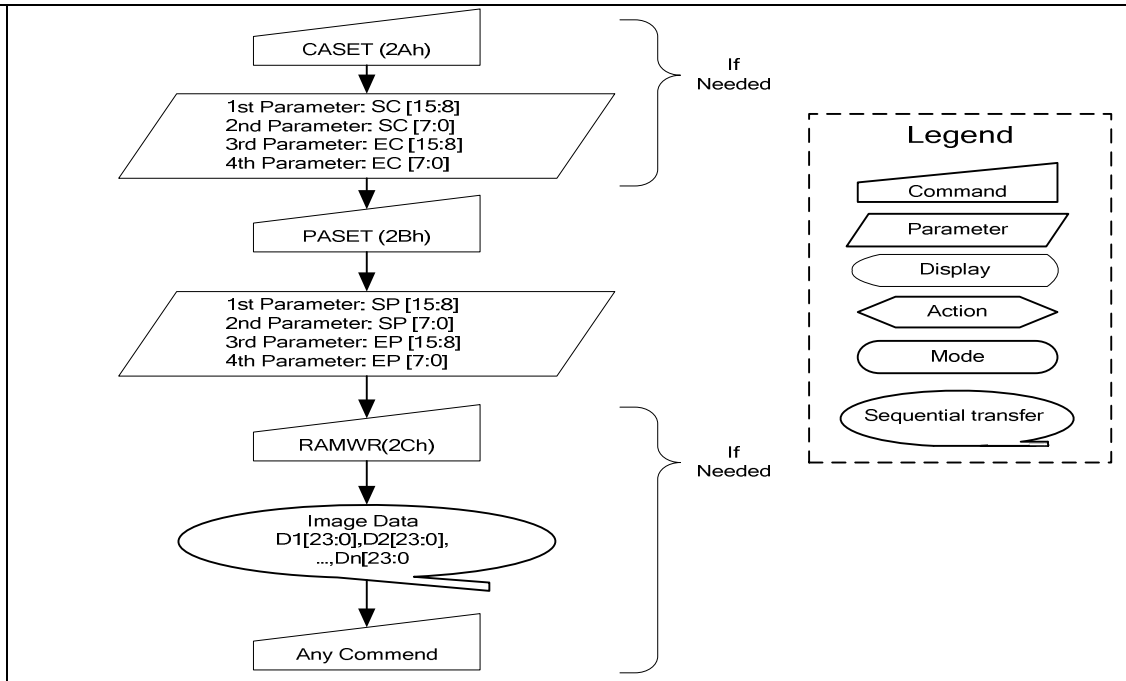
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

RSO [2:0]	Resolution	SP [15:0]	EP [15:0]
000	480 (RGB) x 864	0 x 0000h	0 x 035Fh
001	480 (RGB) x 854	0 x 0000h	0 x 0355h
010	480 (RGB) x 800	0 x 0000h	0 x 031Fh
011	480 (RGB) x 640	0 x 0000h	0 x 027Fh

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Flow Chart



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4.2.28. Memory Write (2Ch)

2Ch	RAMWR (Memory Write)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	1	0	0	2Ch
1st Parameter	1	1	↑	D1 [23:0]									XX
:	1	1	↑	Dx [23:0]									XX
Nth Parameter	1	1	↑	Dn [23:0]									XX
Description	This command transfers data from the MPU to the Frame Memory. This command makes no change to the other status of the driver. When this command is accepted, the column register and the page register are reset to the Start Column (SC) and the Start Page (SP) positions. The Start Column and Start Page positions are different in accordance with the MADCTL setting. Then D [23:0] is stored in the Frame Memory, and the column register and the page register are incremented at the same time. Sending any other commands can stop the frame Write. X = void.												
Restriction	There is no restriction on the length of parameters.												

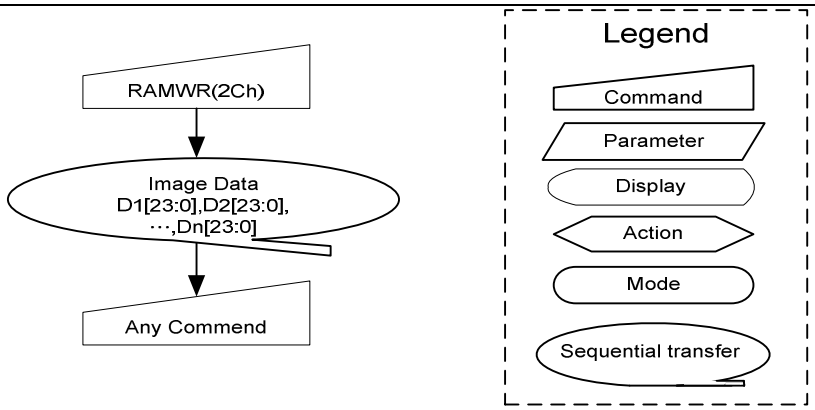
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is set randomly
H/W Reset	Contents of memory is set randomly

Flow Chart



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4.2.29. Memory Read (2Eh)

2Eh	RAMRD (Memory Read)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	0	1	1	1	0	2Eh
1st Parameter	1	1	↑	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	1	↑	D1 [23:0]									XX
:	1	1	↑	Dx [23:0]									XX
(N+1)th Parameter	1	1	↑	Dn [23:0]									XX

Description

The command transfers image data from the Frame Memory to the MPU. This command makes no change to the other status of the driver. When this command is accepted, the column register and the page register are reset to the Start Column (SC) and the Start Page (SP) positions.

The Start Column and Start Page positions are different in accordance with the MADCTL setting.

Then D [23:0] is read back from the Frame Memory, and the column register and the page register are incremented at the same time. Sending any other command can stop the frame read.

X = void

Restriction

There is no restriction on the length of parameters.

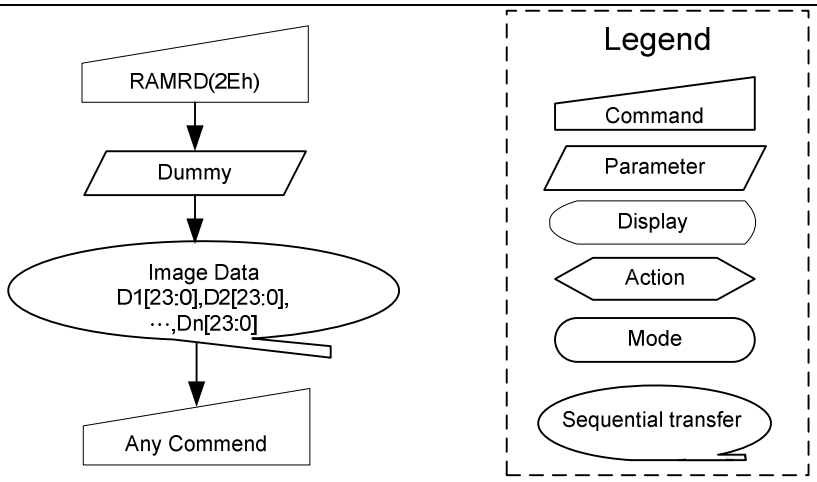
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is set randomly
H/W Reset	Contents of memory is set randomly

Flow Chart



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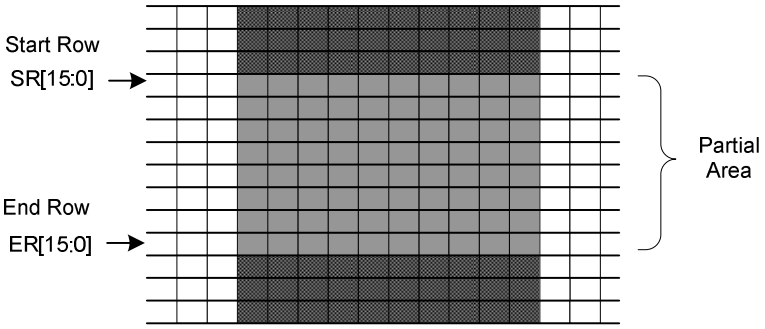
4.2.30. Partial Area (30h)

30h	PLTAR (Partial Area)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	0	0	0	30h
1st Parameter	1	1	↑	XX	SR [15:8]								XX
2nd Parameter	1	1	↑	XX	SR [7:0]								XX
3rd Parameter	1	1	↑	XX	ER [15:8]								XX
4th Parameter	1	1	↑	XX	ER [7:0]								XX

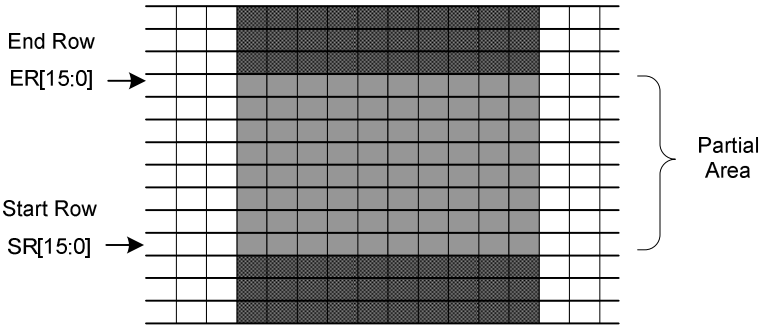
Description

This command defines the module's partial display area. There are two parameters associated with this command. The first defines the Start Row (SR) and the second the End Row (ER), as illustrated in the following figure. SR and ER refer to the Frame Memory line pointer.

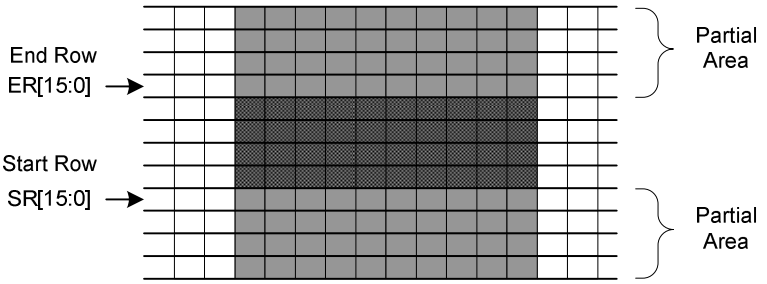
If End Row > Start Row when MADCTL D4 = 0.



If End Row > Start Row when MADCTL D4 = 1.



If End Row < Start Row when MADCTL D4 = 0.



If End Row = Start Row, then the Partial Area will be one row deep.

X = void

Restriction

SR [15..0] and ER [15..0] cannot be greater than the horizontal line number.

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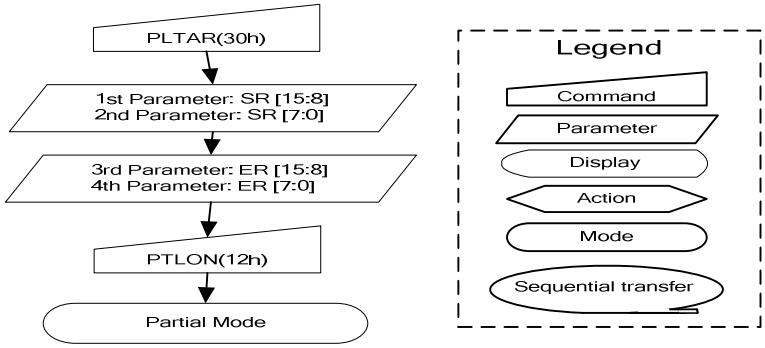
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

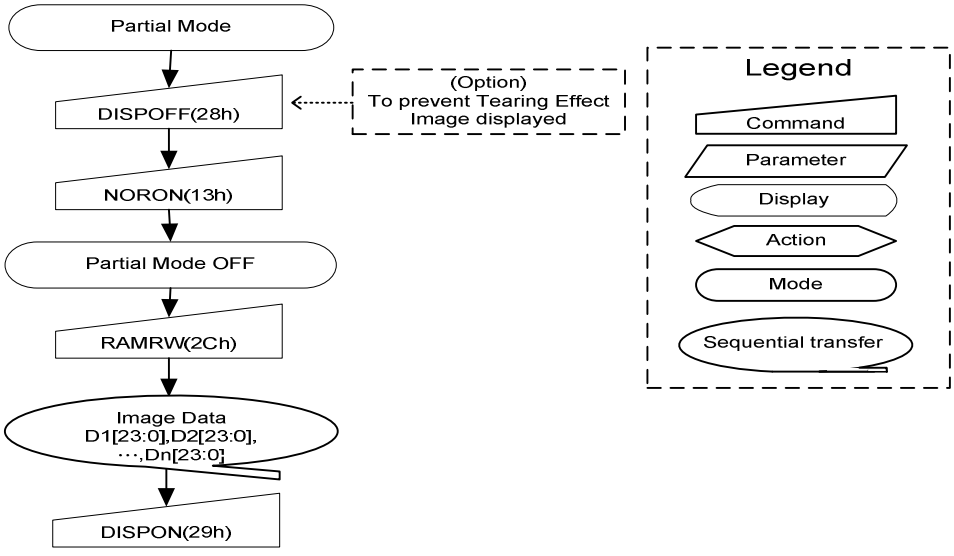
Default

Flow Chart

1. To Enter the Partial Mode



2. To Leave the Partial Mode



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4.2.31. Vertical Scrolling Definition (33h)

33h	VSCRDEF (Vertical Scrolling Definition)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	0	1	1	33h
1st Parameter	1	1	↑	XX	TFA [15:8]								XX
2nd Parameter	1	1	↑	XX	TFA [7:0]								XX
3rd Parameter	1	1	↑	XX	VSA [15:8]								XX
4th Parameter	1	1	↑	XX	VSA [7:0]								XX
5th Parameter	1	1	↑	XX	BFA [15:8]								XX
6th Parameter	1	1	↑	XX	BFA [7:0]								XX

Description

This command defines the display vertical scrolling area.

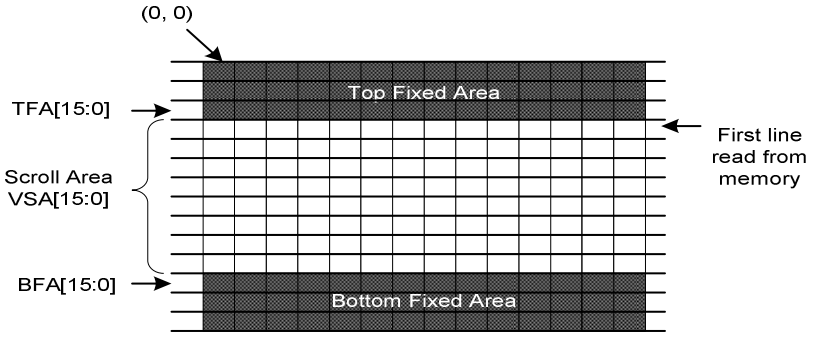
Memory Access Control (36h) D4 = 0:

The 1st & 2nd parameters, TFA [15:0], describe the Top Fixed Area in number of lines from the top of the Frame Memory. The top of the Frame Memory and top of the display device are aligned.

The 3rd & 4th parameters, VSA [15:0], describe the height of the Vertical Scrolling Area in number of lines of the Frame Memory from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the bottom most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the top most line of the Bottom Fixed Area.

The 5th & 6th parameters, BFA [15:0], describe the Bottom Fixed Area in number of lines from the bottom of the Frame Memory. The bottom of the Frame Memory and bottom of the display device are aligned.

TFA, VSA, and BFA refer to the Frame Memory Line Pointer.



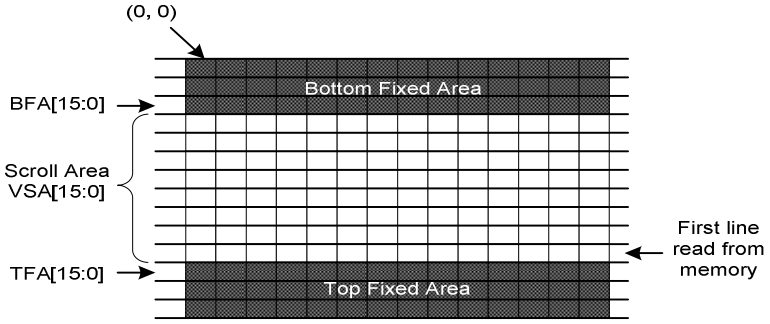
Memory Access Control (36h) D4 = 1:

The 1st & 2nd parameters, TFA [15:0], describe the Top Fixed Area in number of lines from the bottom of the Frame Memory. The bottom of the Frame Memory and bottom of the display device are aligned.

The 3rd & 4th parameters, VSA [15:0], describe the height of the Vertical Scrolling Area in number of lines of the Frame Memory from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the top most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the bottom most line of the Bottom Fixed Area.

The 5th & 6th parameters, BFA [15:0], describe the Bottom Fixed Area in number of lines from the top of the Frame Memory. The top of the Frame Memory and top of the display device are aligned.

TFA, VSA and BFA refer to the Frame Memory Line Pointer.



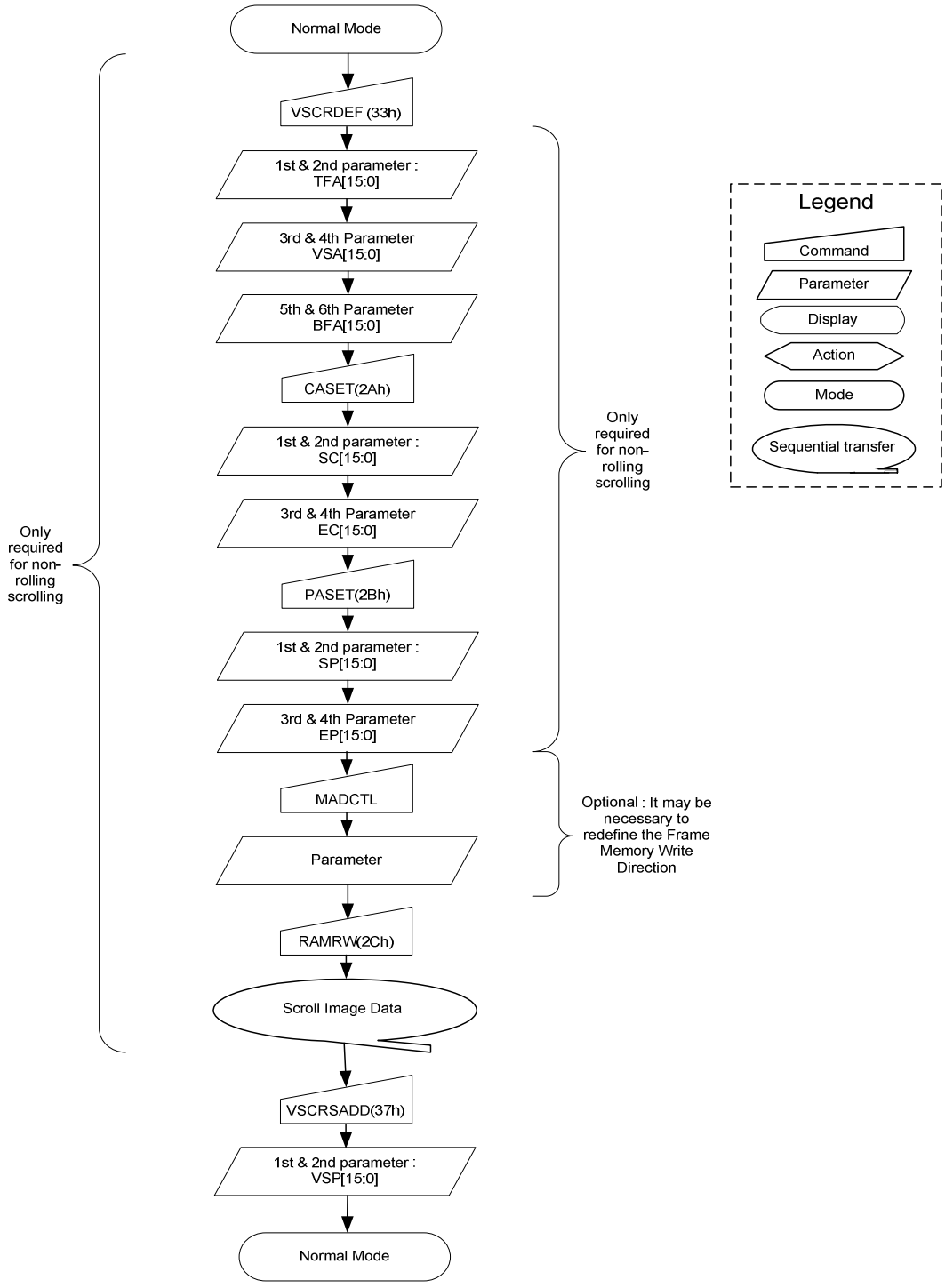
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<table>
<tr><td>Restriction</td><td>The sum of TFA, VSA and BFA must equal the number of the display device's horizontal lines; otherwise, the Scrolling mode is undefined. In the Vertical Scroll mode, Memory Access Control (36h) D5 should be set to 0 – this only affects the Frame Memory write.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Resolution</th><th>TFA [15..0]</th><th>VSA [15..0]</th><th>BFA [15..0]</th></tr>
<tr><td>480 (RGB) x 864</td><td>0 x 0000h</td><td>0 x 0360h</td><td>0 x 0000h</td></tr>
<tr><td>480 (RGB) x 854</td><td>0 x 0000h</td><td>0 x 0356h</td><td>0 x 0000h</td></tr>
<tr><td>480 (RGB) x 800</td><td>0 x 0000h</td><td>0 x 0320h</td><td>0 x 0000h</td></tr>
<tr><td>480 (RGB) x 640</td><td>0 x 0000h</td><td>0 x 0280h</td><td>0 x 0000h</td></tr>
</table>
</td></tr>
</table>

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Flow Chart

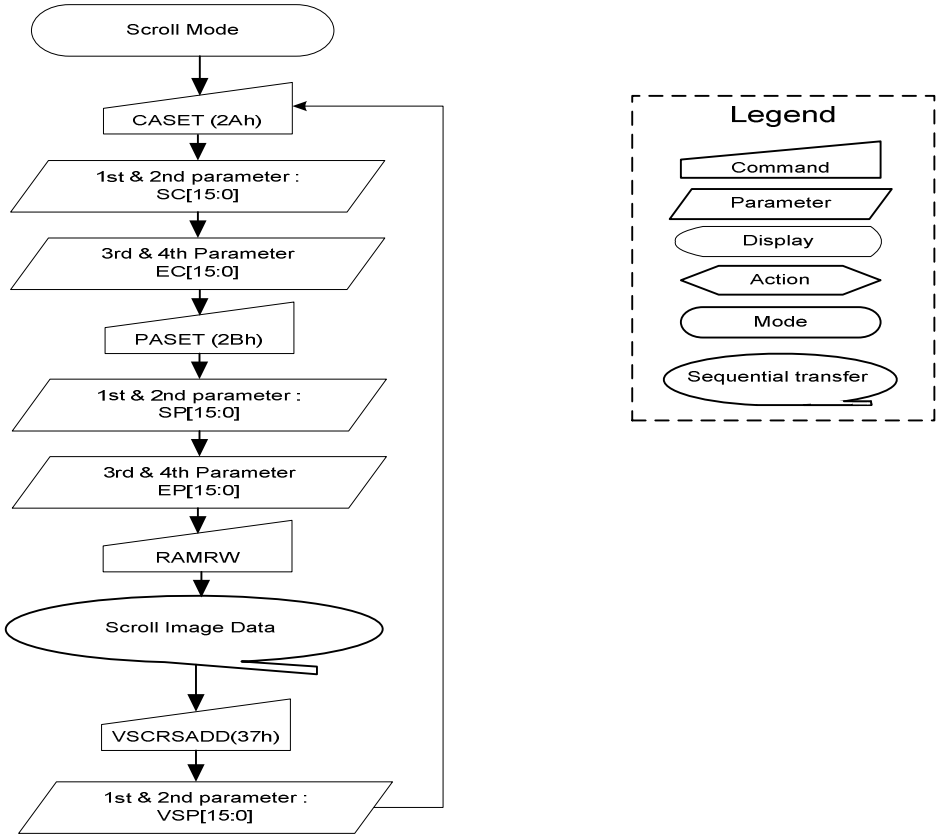
1. To enter the Vertical Scroll Mode:



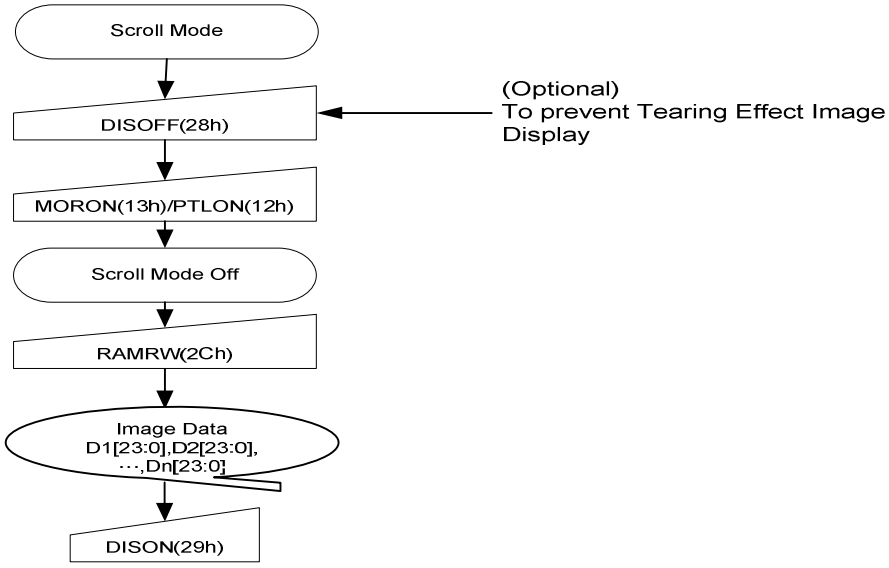
Since the value of the Vertical Scrolling Start Address is absolute with reference to the Frame Memory, it must not enter the fixed area; otherwise, an undesirable image may appear on the Display Panel.

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2. Continuous Scroll :



3. To Leave Vertical Scroll Mode:



Note: To exit the Scroll Mode, both Normal Display Mode On (13h) and Partial Mode On (12h) commands can be used.

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4.2.32. Tearing Effect Line OFF (34h)

34h	TEOFF (Tearing Effect Line OFF)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	0	0	34h
Parameter	No parameter												
Description	This command is used to turn off the Display module's Tearing Effect output signal (Active Low) on the TE signal line.												
Restriction	This command has no effect when the Tearing Effect output is already OFF.												

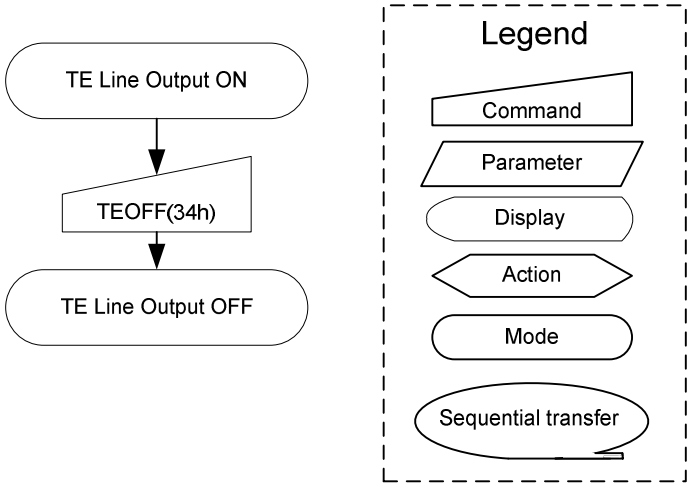
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

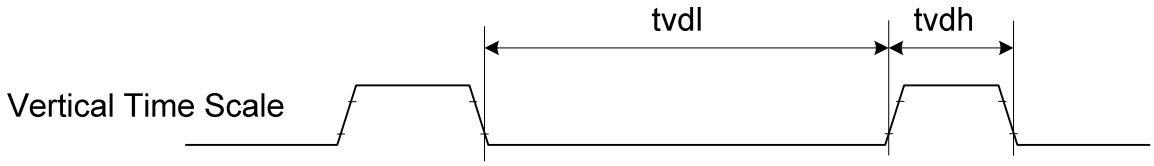
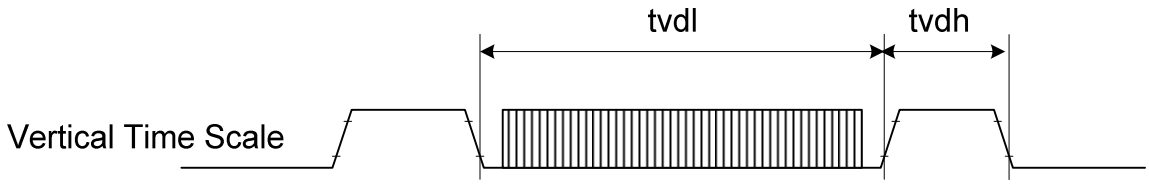
Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

Flow Chart



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4.2.33. Tearing Effect Line ON (35h)

35h	TEON (Tearing Effect Line ON)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	0	1	35h
Parameter	1	1	↑	XX	X	X	X	X	X	X	X	M	XX
Parameter	No parameter												
Description	This command is used to turn ON the Tearing Effect output signal on the TE signal line. Changing the MADCTL bit B4 will not affect this output. The Tearing Effect Line On has one parameter, which describes the mode of the Tearing Effect Output Line. When **M = 0**: The Tearing Effect Output line consists of V-Blanking information only:  When **M = 1**: The Tearing Effect Output Line consists of both V-Blanking and H-Blanking information:  Note: During the Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low. X = void												
Restriction	This command has no effect when the Tearing Effect output is already OFF.												

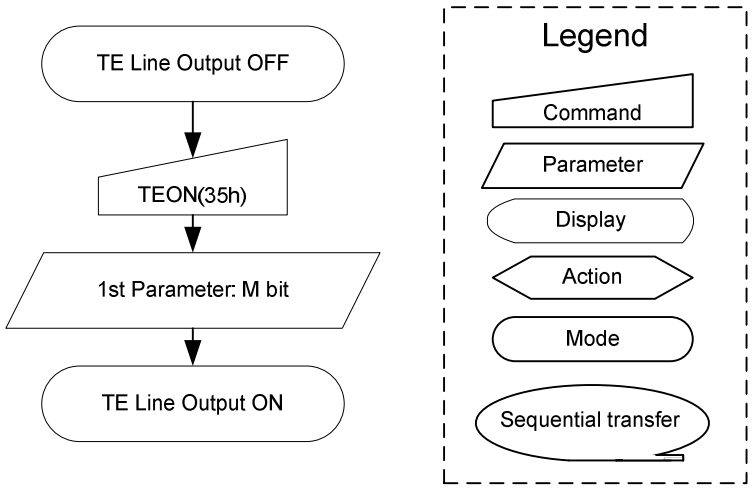
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Off
S/W Reset	Off
H/W Reset	Off

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Flow Chart	

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4.2.34. Memory Access Control (36h)

36h	MADCTL (Memory Access Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	1	0	36h
Parameter	1	1	↑	XX	MY	MX	MV	ML	BGR	MH	SS	GS	XX

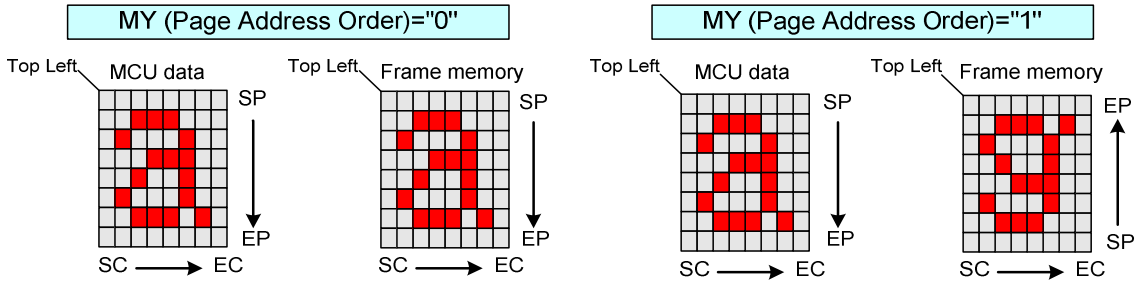
Description

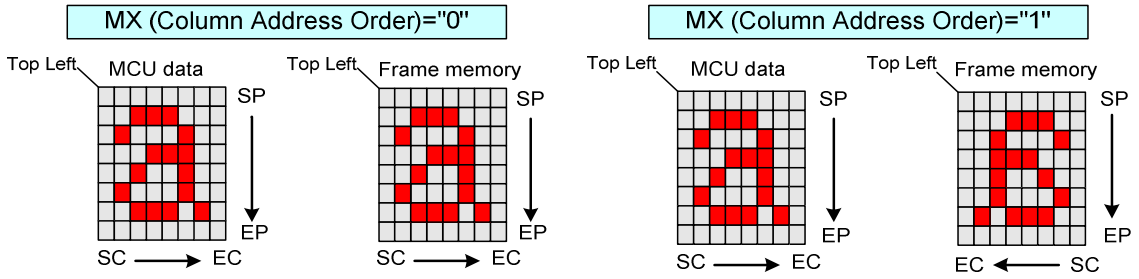
This command defines read/write scanning direction of the Frame Memory.

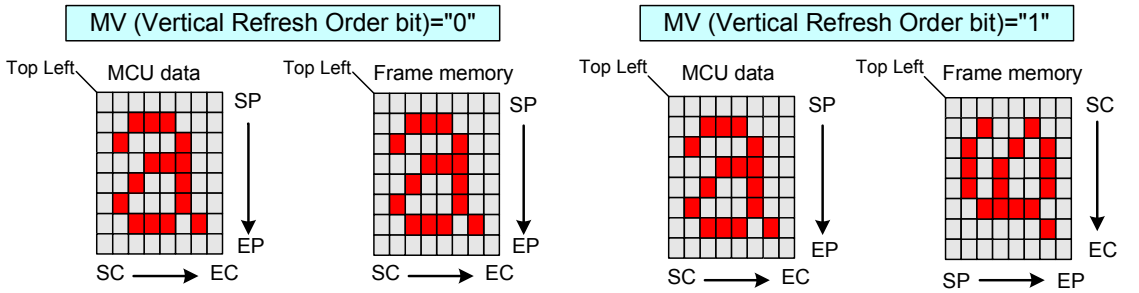
This command makes no change on the other status of the driver.

Bit	Symbol	Name	Description
D7	MY	Row Address Order	These 3 bits control the write/read direction from the MPU to memory
D6	MX	Column Address Order	
D5	MV	Row/Column Exchange	
D4	ML	Vertical Refresh Order	LCD vertical refresh direction control
D3	BGR	RGB-BGR Order	Color selector switch control (0 = RGB color filter panel, 1 = BGR color filter panel)
D2	MH	Horizontal Refresh Order	LCD horizontal refreshing direction control
D1	SS	Flip Horizontal(SS)	Select the Source driver scan direction on the panel module
D0	GS	Flip Vertical (GS)	Select the Gate driver scan direction on the panel module

Top-Left (0,0) means the physical memory location

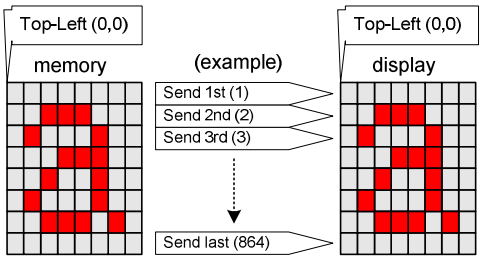




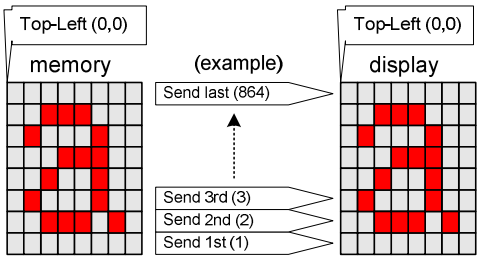


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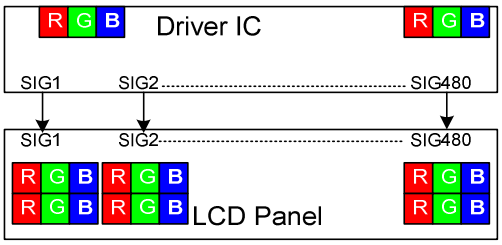
ML (Vertical refresh order bit)="0"



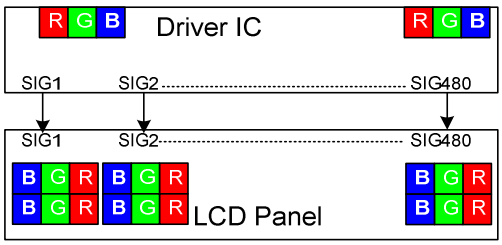
ML (Vertical refresh order bit)="1"



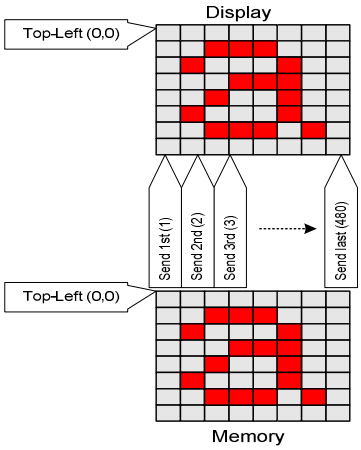
BGR (RGB-BGR Order control bit)="0"



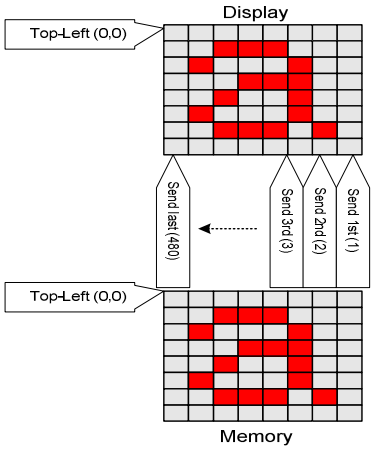
BGR (RGB-BGR Order control bit)="1"



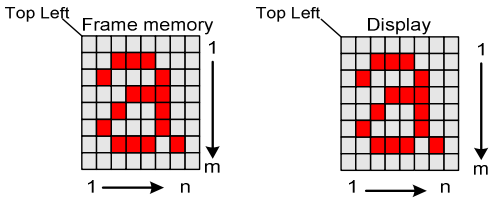
MH (Horizontal refresh order control bit)="0"



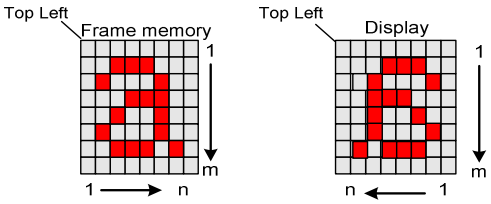
MH (Horizontal refresh order control bit)="1"



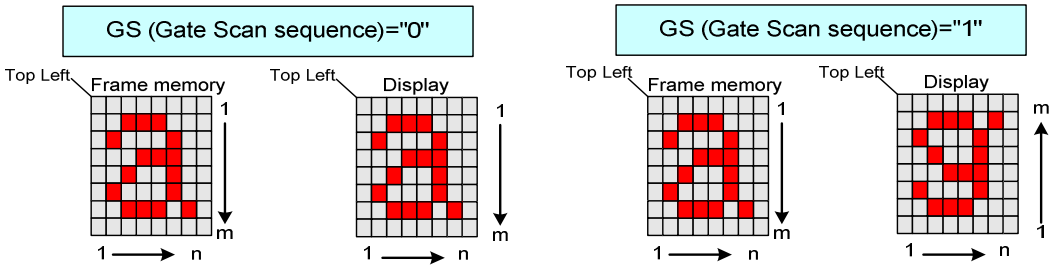
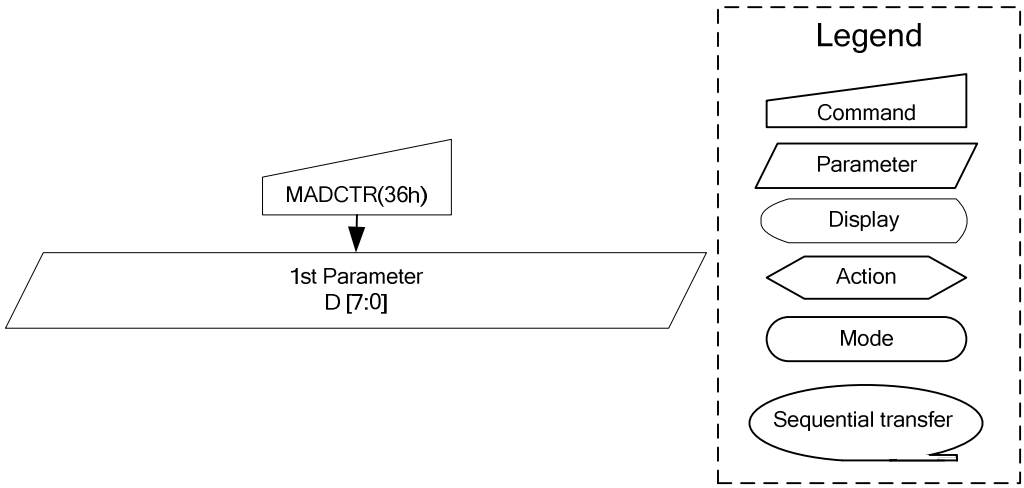
SS (Source Scan sequence)="0"



SS (Source Scan sequence)="1"

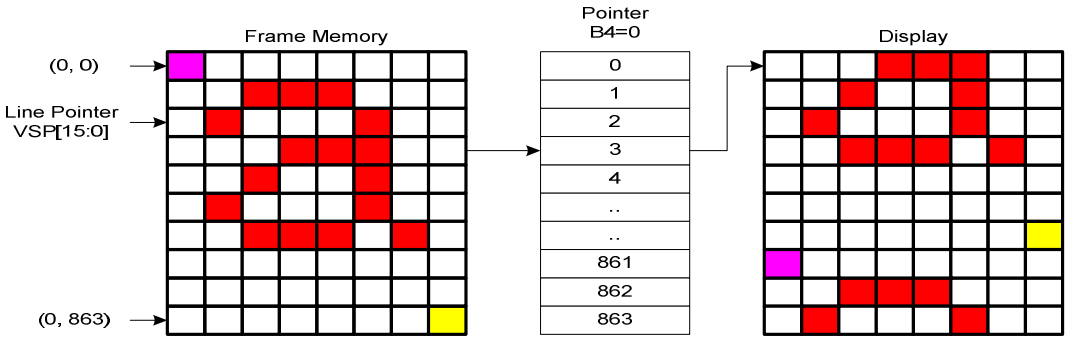
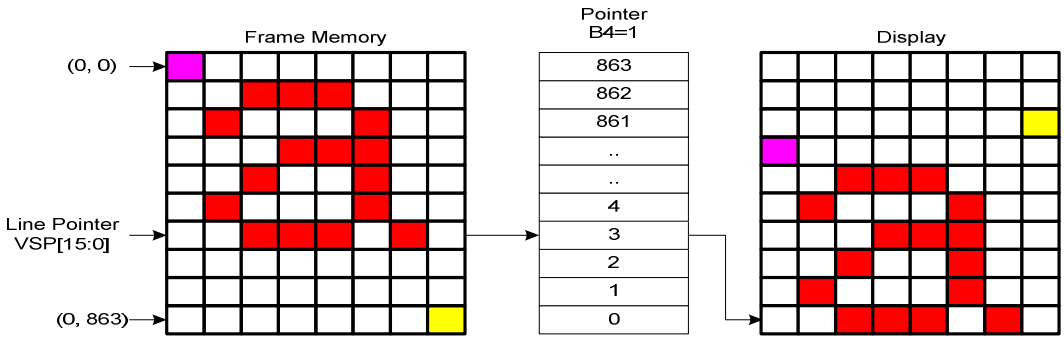


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<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Restriction</td><td></td></tr>
<tr><td>Register
Availability</td><td><table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td><table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h</td></tr>
<tr><td>S/W Reset</td><td>No change</td></tr>
<tr><td>H/W Reset</td><td>00h</td></tr>
</table></td></tr>
<tr><td>Flow Chart</td><td> 
</td></tr>
</table>

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4.2.35. Vertical Scrolling Start Address (37h)

37h	VSCRSADD (Vertical Scrolling Start Address)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	0	1	1	1	37h
1st Parameter	1	1	↑	XX	VSP [15:8]								XX
2nd Parameter	1	1	↑	XX	VSP [7:0]								XX
Description	This command is used together with the Vertical Scrolling Definition (33h). These two commands describe the scrolling area and the scrolling mode. The Vertical Scrolling Start Address command has one parameter which describes the address of the line in the Frame Memory that will be written as the first line after the last line of the Top Fixed Area on the display, as illustrated below. When MADCTL D4 = 0 Example: When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 864 and VSP= 3  When MADCTL D4=1 Example: When Top Fixed Area = Bottom Fixed Area = 00, Vertical Scrolling Area = 864 and VSP= 3  Note: When the new Pointer position and Picture Data are sent, the result on the display will appear at the next Panel Scan to avoid tearing effect. VSP refers to the Frame Memory line Pointer. X = void												
Restriction	Since the value of the Vertical Scrolling Start Address is absolute (with reference to the Frame Memory), it must not enter the fixed area (defined by Vertical Scrolling Definition (33h). Otherwise, undesirable image will occur on the Panel.												

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<table>
<tr><td>Register
Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>00h</td></tr>
<tr><td>S/W Reset</td><td>00h</td></tr>
<tr><td>H/W Reset</td><td>00h</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td>See Vertical Scrolling Definition (33h) description.</td></tr>
</table>

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4.2.36. Idle Mode Off (38h)

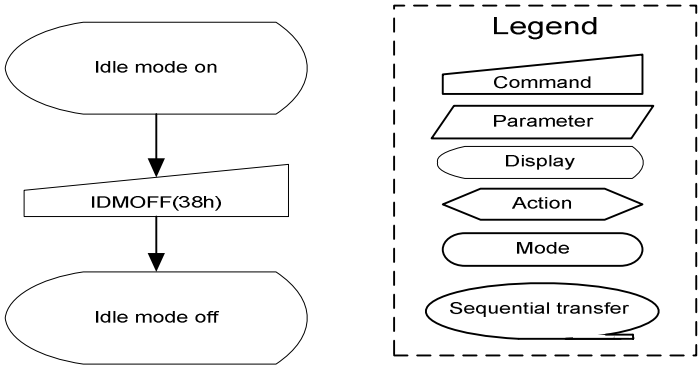
38h	IDMOFF (Idle Mode Off)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	0	0	0	38h
Parameter	No parameter												
Description	This command causes the Display module to exit the Idle mode. In the Idle Mode Off, the display panel can display a maximum of 16.7M colors.												
Restriction	This command has no effect when the module is already in the Idle Mode Off.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Idle Mode Off
S/W Reset	Idle Mode Off
H/W Reset	Idle Mode Off

Flow Chart



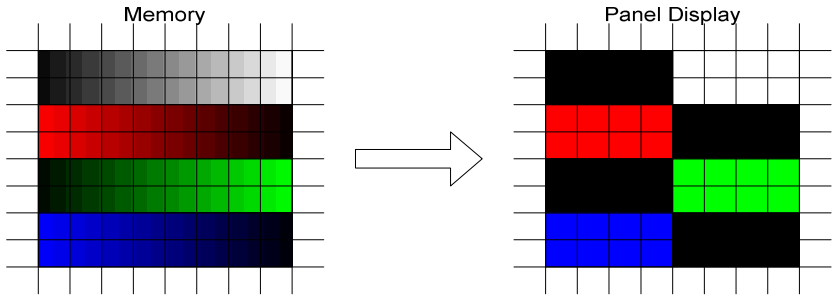
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4.2.37. Idle Mode On (39h)

39h	IDMON (Idle Mode On)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	0	0	1	39h
Parameter	No parameter												

Description

This command is used to enter the Idle Mode On. In the Idle Mode On, color expression is reduced. The first bits of R, G, and B in the Frame Memory will determine the display color, as shown in the table below.



Memory Contents vs. Display Color			
	R_7 - R_0	G_7 - G_0	B_7 - B_0
Black	0XXXXXXX	0XXXXXXX	0XXXXXXX
Blue	0XXXXXXX	0XXXXXXX	1XXXXXXX
Red	1XXXXXXX	0XXXXXXX	0XXXXXXX
Magenta	1XXXXXXX	0XXXXXXX	1XXXXXXX
Green	0XXXXXXX	1XXXXXXX	0XXXXXXX
Cyan	0XXXXXXX	1XXXXXXX	1XXXXXXX
Yellow	1XXXXXXX	1XXXXXXX	0XXXXXXX
White	1XXXXXXX	1XXXXXXX	1XXXXXXX

X = void

Restriction

This command has no effect when the module is already in the Idle Mode Off.

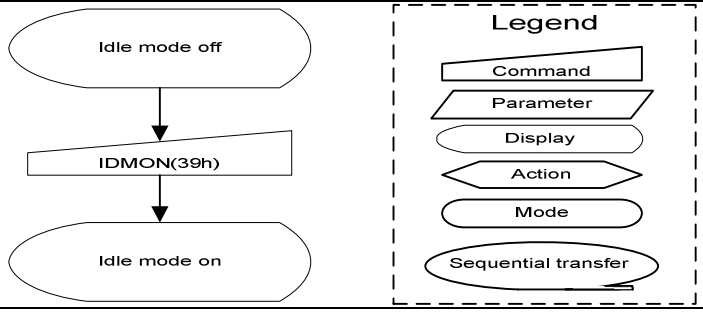
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

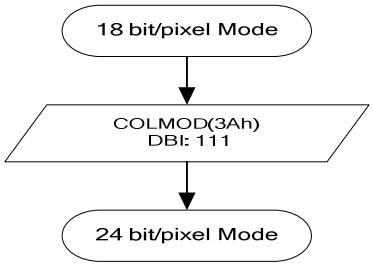
Status	Default Value
Power On Sequence	Idle Mode Off
S/W Reset	Idle Mode Off
H/W Reset	Idle Mode Off

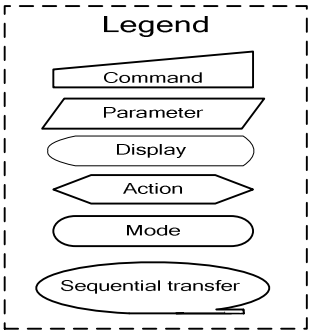
Flow Chart



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4.2.38. Interface Pixel Format (3Ah)

<table>
<tr><th>3Ah</th><th colspan="13">COLMOD (Interface Pixel Format)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>3Ah</td></tr>
<tr><td>Parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td colspan="3">DPI [2:0]</td><td>X</td><td colspan="3">DBI [2:0]</td><td>XX</td></tr>
<tr><td>Description</td><td colspan="13">This command sets the pixel format. DPI [3:0] selects the pixel format of the RGB interface, and DBI [2:0] select the pixel format of the MPU interface.
<table>
<tr><th colspan="3">DPI [2:0]</th><th>RGB Interface Format</th></tr>
<tr><td>1</td><td>0</td><td>1</td><td>16-bit/pixel</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>18-bit/pixel</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>24-bit/pixel</td></tr>
</table>
<table>
<tr><th colspan="3">DBI [2:0]</th><th>MPU Interface Format</th></tr>
<tr><td>1</td><td>0</td><td>1</td><td>16-bit/pixel</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>18-bit/pixel</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>24-bit/pixel</td></tr>
</table>
X = void</td></tr>
<tr><td>Restriction</td><td colspan="13"></td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial M de On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power On Sequence</td><td>07h</td></tr>
<tr><td>S/W Reset</td><td>07h</td></tr>
<tr><td>H/W Reset</td><td>07h</td></tr>
</table>
</td></tr>
<tr><td>Flow Chart</td><td colspan="13">Example :
 


</td></tr>
</table>

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4.2.39. Memory Write Continue (3Ch)

3Ch	RAMWRC (Memory Write Continue)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	1	0	0	3Ch
1st Parameter	1	1	↑	D1 [23:0]									XX
:	1	1	↑	Dx [23:0]									XX
Nth Parameter	1	1	↑	Dn [23:0]									XX

Description

This command is used to transfer data from the MPU to the frame memory if the frame memory wants to continue memory write after the "Memory Write (2Ch)" command. This command makes no change to the other status of the driver.

When this command is accepted, the column register and the page register will not reset to the Start Column/Start Page positions after it is done on the "Memory Write (2Ch)" command.

Then D [23:0] is stored in the frame memory and the column register and the page register are incremented, as the table below:

Column and Page Counter Control.

Condition	Column counter	Page Counter
When RAMWR/RAMRD command is accepted	Return to "Start Column"	Return to "Start Page"
Complete Pixel Read/Write action	Increment by 1	No change
The Column counter value is large than "End Column"	Return to "Start Column"	Increment by 1
The Page counter value is large than "End Page"	Return to "Start Column"	Return to "Start Page"

Sending any other command can stop the Memory Write Continue command.

X = void

Restriction

There is no restriction on the length of parameters. No access to the frame memory in the Sleep In mode.

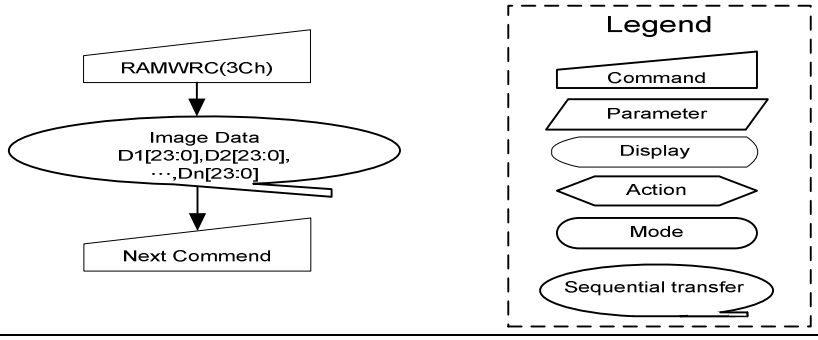
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is set randomly
H/W Reset	Contents of memory is set randomly

Flow Chart



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4.2.40. Memory Read Continue (3Eh)

3Eh	RAMRDRC (Memory Read Continue)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	0	1	1	1	1	1	0	3Eh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	D1 [15:0]									XX
:	1	↑	1	Dx [15:0]									XX
Nth Parameter	1	↑	1	Dn [15:0]									XX

Description

This command is used to transfer data from the Frame Memory to the MPU, if the MPU wants to continue memory read after "Memory Read (2Eh)" command. This command makes no change to the other status of the driver.

When this command is accepted, the column register and the page register will not reset to the Start Column/Start Page positions since it is done on the "Memory Read (2Eh)" command.

Then D [15:0] is read back from the frame memory, and the column register and the page register are incremented, as the table below: Column and Page Counter Control.

Condition	Column counter	Page Counter
When RAMWR/RAMRD command is accepted	Return to "Start Column"	Return to "Start Page"
Complete Pixel Read/Write action	Increment by 1	No change
The Column counter value is large than "End Column"	Return to "Start Column"	Increment by 1
The Page counter value is large than "End Page"	Return to "Start Column"	Return to "Start Page"

Sending any other command can stop the Memory Read Continue command.

X = void.

Restriction

There is no restriction on the length of parameters.

No access to the frame memory in the Sleep In mode.

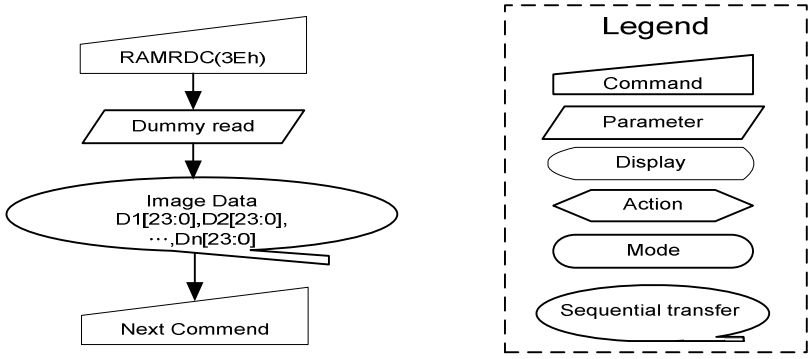
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

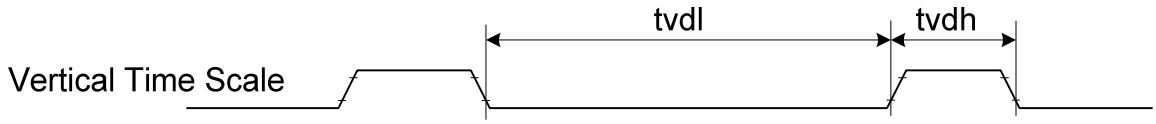
Status	Default Value
Power On Sequence	Contents of memory is set randomly
S/W Reset	Contents of memory is set randomly
H/W Reset	Contents of memory is set randomly

Flow Chart



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4.2.41. Write Tear Scan Line (44h)

44h	TESLWR (Write Tear Scan Line)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	0	0	1	0	0	44h
1st Parameter	1	1	↑	XX	N [15:8]								XX
2nd Parameter	1	1	↑	XX	N [7:0]								XX
Description	This command turns on the display Tearing Effect output signal on the TE signal line when the display reaches line N. Changing Memory Access Control (36h) D4 will not affect the TE signal. The Tearing Effect Line On has one parameter that describes the Tearing Effect Output Line mode. The Tearing Effect Output line consists of V-Blanking information only.  Note that set Tear Scan Line with N = 0 is equivalent to Tearing Effect Line ON with M = 0. The Tearing Effect Output line shall be active low when the ILI9806G is in the Sleep mode.												
Restriction	This command has no effect when the Tearing Effect output is already ON.												

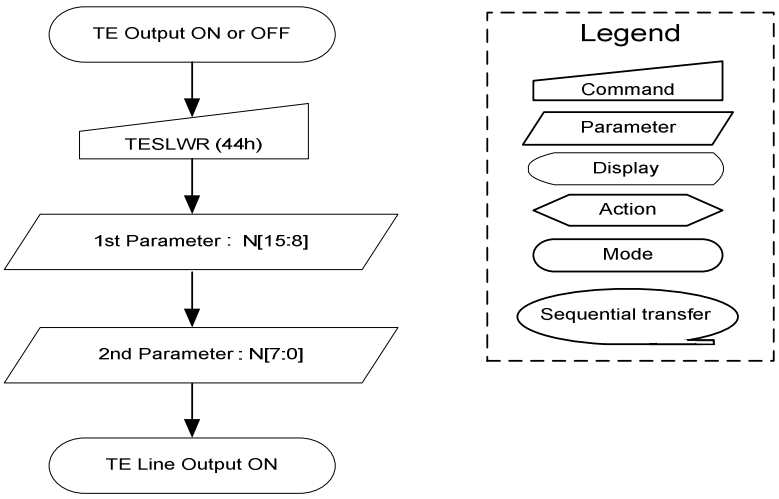
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.42. Read Scan Line (45h)

45h	TESLRD (Read Tear Scan Line)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	0	0	1	0	1	45h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	N [15:8]								XX
3rd Parameter	1	↑	1	XX	N [7:0]								XX
Description	The display reads the current scan line N, used to update the display device. The total number of scan lines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scan line is defined as the first line of V-Sync and is denoted as Line 0. When in the Sleep Mode, the value returned by the Read Scan Line command is undefined.												
Restriction													

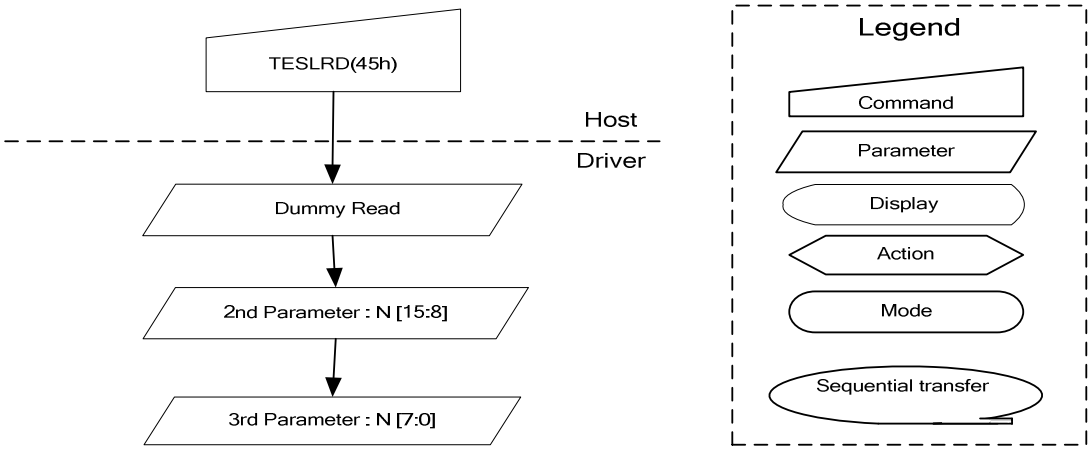
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.43. Write Display Brightness Value (51h)

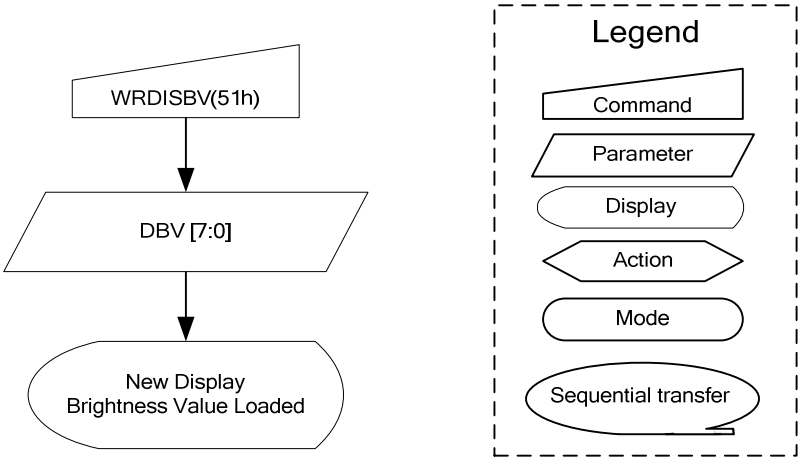
51h	WRDISBV (Write Display Brightness)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	0	0	1	51h
1st Parameter	1	1	↑	XX	DBV [7:0]								XX
Description	This command is used to adjust the brightness value of the display. **DBV [7:0]**: 8 bit, for display brightness of manual brightness setting and the CABC in the ILI9806G. PWM output signal and LEDPWM pin will control the LED driver IC in order to control the display brightness.												
Restriction													
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.44. Read Display Brightness Value (52h)

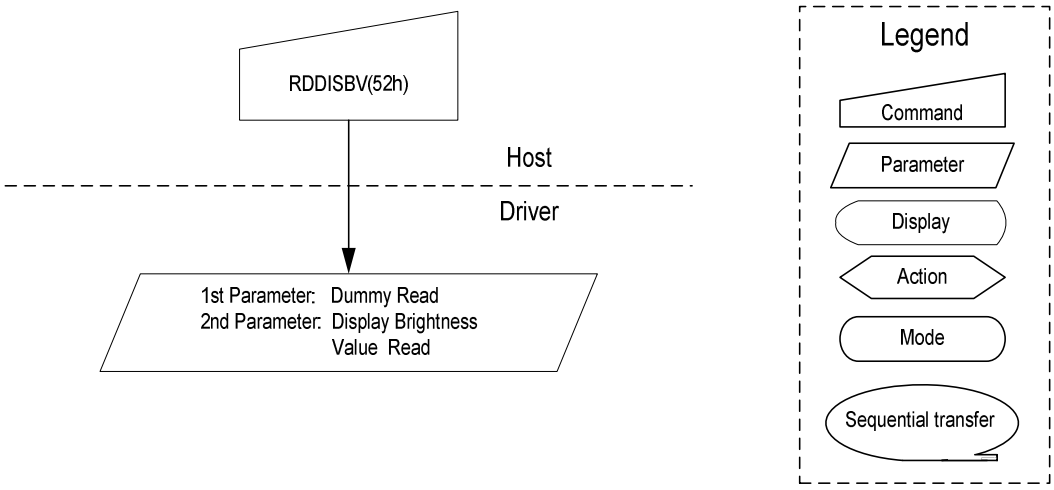
52h	RDDISBV (Read Display Brightness Value)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	0	1	0	52h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	DBV [7:0]								XX
Description	This command is used to return the brightness value of the display. DBV [7:0] is reset when display is in the Sleep In mode. DBV [7:0] is 0 when the bit BCTRL of "Write CTRL Display (53h)" command is 0. DBV [7:0] is the manual set brightness specified by the Write CTRL Display (53h) command when BCTRL bit is 1. When the bit BCTRL of the Write CTRL Display (53h) command is 1 and C1/C0 bit of the Write Content Adaptive Brightness Control (55h) command are 0, DBV [7:0] output is the brightness value specified by the Write Display Brightness (51h) command.												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h



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4.2.45. Write CTRL Display Value (53h)

53h	WRCTRLD (Write Control Display)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	0	1	1	53h
1st Parameter	1	1	↑	XX	X	X	BCTRL	X	DD	BL	X	X	XX

Description

This command is used to control the display brightness.

BCTRL: Brightness Control Block On/Off. This bit is always used to switch brightness for display.

BCTRL	Description
0	Brightness Control Block Off (DBV [7:0] = 00h)
1	Brightness Control Block On (DBV [7:0] is active)

DD: Display Dimming Control. This function is only for manual brightness setting.

DD	Description
0	Display Dimming Off
1	Display Dimming On

BL: Backlight Control On/Off

BL	Description
0	Backlight Control Off
1	Backlight Control On

Dimming function is adapted to the brightness registers for display when the bit BCTRL is changed at DD = 1, e.g. BCTRL: 0-> 1 or 1-> 0.

When the BL bit changes from ON to OFF, backlight is turned off without gradual dimming, even if Display Dimming On (DD = 1) are selected.

X = void

Restriction

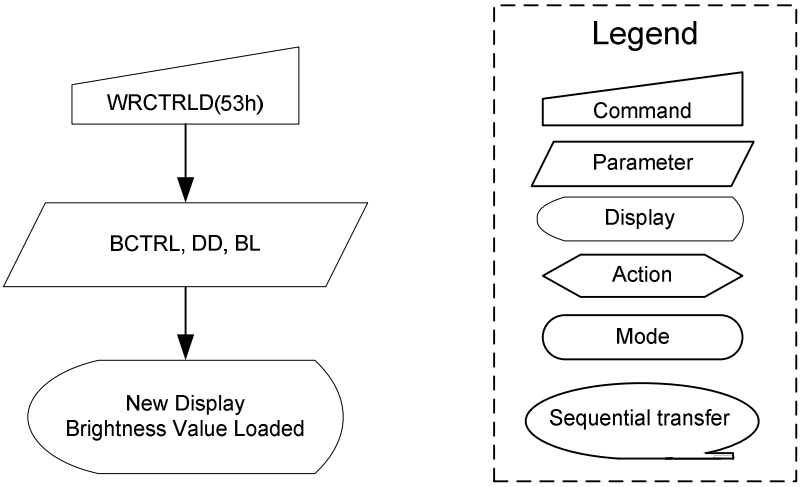
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

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Flow Chart	

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4.2.46. Read CTRL Display Value (54h)

54h	RDCTRLD (Read Control Display Value)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	1	0	0	54h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	X	X	BCTRL	X	DD	BL	X	X	XX

Description

This command is used to control the display brightness.

BCTRL: Brightness Control Block On/Off. This bit is always used to switch brightness for display.

BCTRL	Description
0	Brightness Control Block Off (DBV [7:0] = 00h)
1	Brightness Control Block On (DBV [7:0] is active)

DD: Display Dimming Control. This function is only for manual brightness setting.

DD	Description
0	Display Dimming Off
1	Display Dimming On

BL: Backlight Control On/Off

BL	Description
0	Backlight Control Off
1	Backlight Control On

X = Void

Restriction

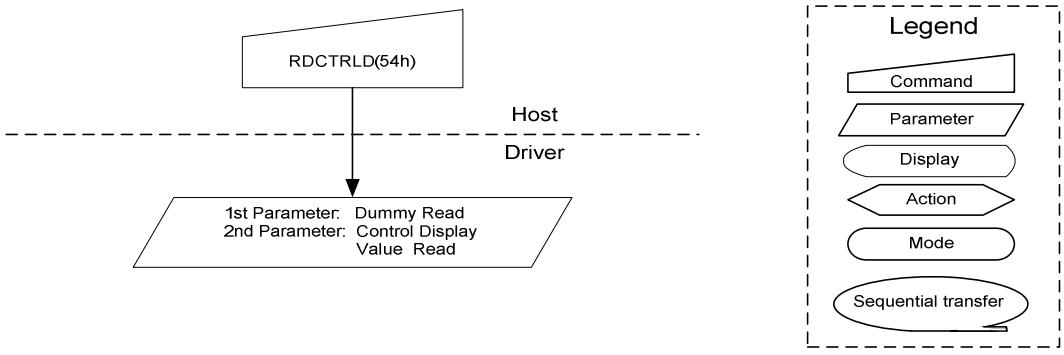
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

Flow Chart



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4.2.47. Write Content Adaptive Brightness Control Value (55h)

<table>
<tr><th>55h</th><th colspan="13">WRCABC (Write Content Adaptive Brightness Control)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>55h</td></tr>
<tr><td>1st Parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td colspan="4">C[7:4]</td><td colspan="4">C [3:0]</td><td>XX</td></tr>
<tr><td>Description</td><td colspan="13">This command is used to set parameters for image content based on the adaptive brightness control functionality.

There are 4 different modes for content adaptive image functionality, which are defined in the table below.

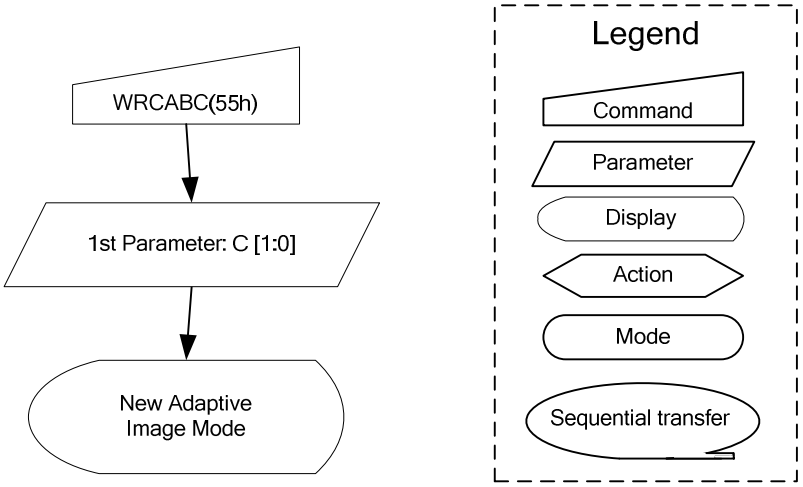
C [3:0] | Description
0 0 0 0 | CABC Off
0 0 0 1 | User Interface Image mode
0 0 1 0 | Still Picture mode
0 0 1 1 | Moving Image mode

Color enhancement level selection.

C [7:4] | Description
0 0 0 0 | Enhancement Disable
1 0 0 0 | Low Enhancement
1 0 0 1 | Medium Enhancement
1 0 1 1 | High Enhancement

X = Void</td></tr>
<tr><td>Restriction</td><td colspan="13"></td></tr>
<tr><td>Register Availability</td><td colspan="13">Status | Availability
Normal Mode On, Idle Mode Off, Sleep Out | Yes
Normal Mode On, Idle Mode On, Sleep Out | Yes
Partial Mode On, Idle Mode Off, Sleep Out | Yes
Partial Mode On, Idle Mode On, Sleep Out | Yes
Sleep In | Yes</td></tr>
<tr><td>Default</td><td colspan="13">Status | Default Value
Power On Sequence | 00h
S/W Reset | 00h
H/W Reset | 00h</td></tr>
</table>

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Flow Chart	

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4.2.48. Read Content Adaptive Brightness Control Value (56h)

56h	RDCABC (Read Content Adaptive Brightness Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	0	1	1	0	56h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	C[7:4]				C [3:0]				XX

Description

This command is used to read the settings for image content based on the adaptive brightness control functionality. There are 4 different modes for content adaptive image functionality, which are defined in the table below.

C [3:0]				Description
0	0	0	0	CABC Off
0	0	0	1	User Interface Image mode
0	0	1	0	Still Picture mode
0	0	1	1	Moving Image mode

Color enhancement level selection.

C [7:4]				Description
0	0	0	0	Enhancement Disable
1	0	0	0	Low Enhancement
1	0	0	1	Medium Enhancement
1	0	1	1	High Enhancement

X = Void

Restriction

Register Availability

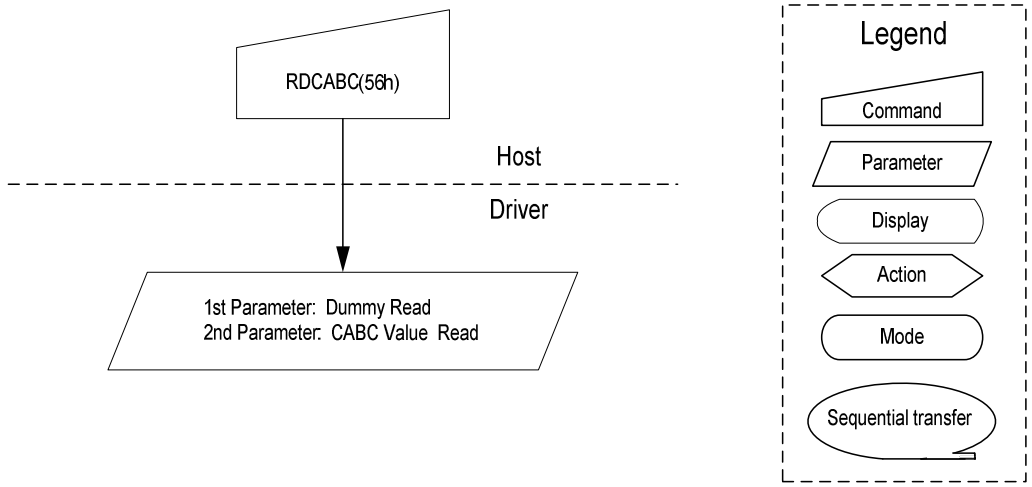
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

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Flow Chart



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4.2.49. Write CABC Minimum Brightness (5Eh)

5Eh	WRCABCMB (Write CABC Minimum Brightness)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	1	1	1	0	5Eh
1st Parameter	1	1	↑	XX	CMB [7:0]								XX
Description	This command is used to set the minimum brightness value of the display for the CABC function. **CMB [7:0]:** CABC minimum brightness control. This parameter is used to set a limit to the amount of brightness reduction allowed. When the CABC is active, it cannot reduce the display brightness to less than the CABC minimum brightness setting. Image processing function works normally, even if the brightness cannot be changed. This function does not affect the manual brightness setting. Manual brightness setting does not have a limit on allowable brightness reduction; display brightness can be set less than the CABC minimum brightness. Smooth transition and dimming function work normally. When the display brightness is turned off (BCTRL = 0 of Write CTRL Display (53h)), the CABC minimum brightness setting is ignored. The principle relationship is such that 00h value means the lowest brightness for the CABC, and FFh value means the highest brightness for the CABC.												
Restriction													

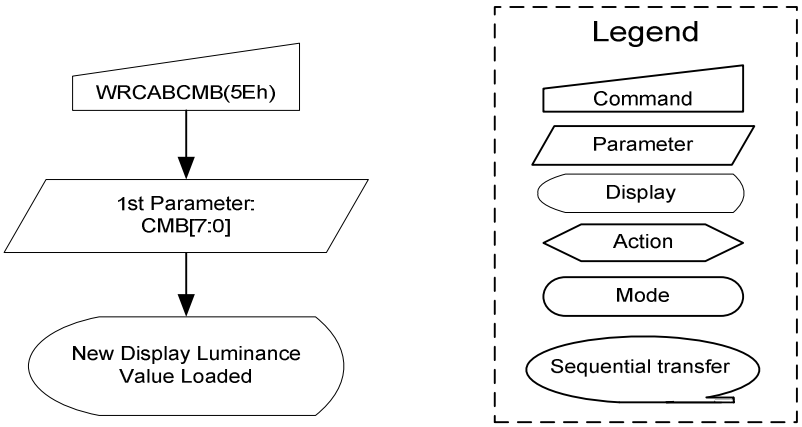
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

Flow Chart



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4.2.50. Read CABC Minimum Brightness (5Fh)

5Fh	RDCABCMB (Read CABC Minimum Brightness)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	0	1	1	1	1	1	5Fh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	CMB [7:0]								XX
Description	This command reads the minimum brightness value of the CABC function. The principle relationship is such that 00h value means the lowest brightness, and FFh value means the highest brightness. CMB [7:0] is the CABC minimum brightness specified by the Write CABC minimum brightness (5Eh) command.												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

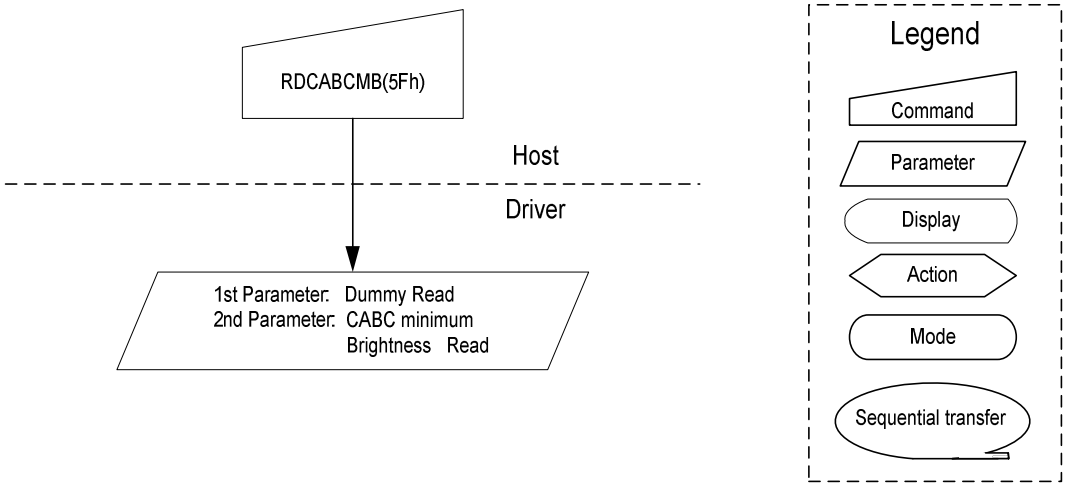
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	No Change
H/W Reset	00h

Flow Chart



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4.2.51. Read Automatic Brightness Control Self-diagnostic Result (68h)

68h	RDABCSDR (Read automatic brightness control self-diagnostic result)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	0	1	0	0	0	68h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	D [7:6]		0	0	0	0	0	0	XX
Description	This command indicates the status of the display self-diagnostic results for automatic brightness control after the Sleep Out command, as described in the table below: Bit D7 – Register Loading Detection, see the section “Register loading Detection”. Bit D6 – Functionality Detection, see the section “Functionality Detection”. Bits D5, D4, D3, D2, D1 and D0 are for future use and are set to 0.												
Restriction													

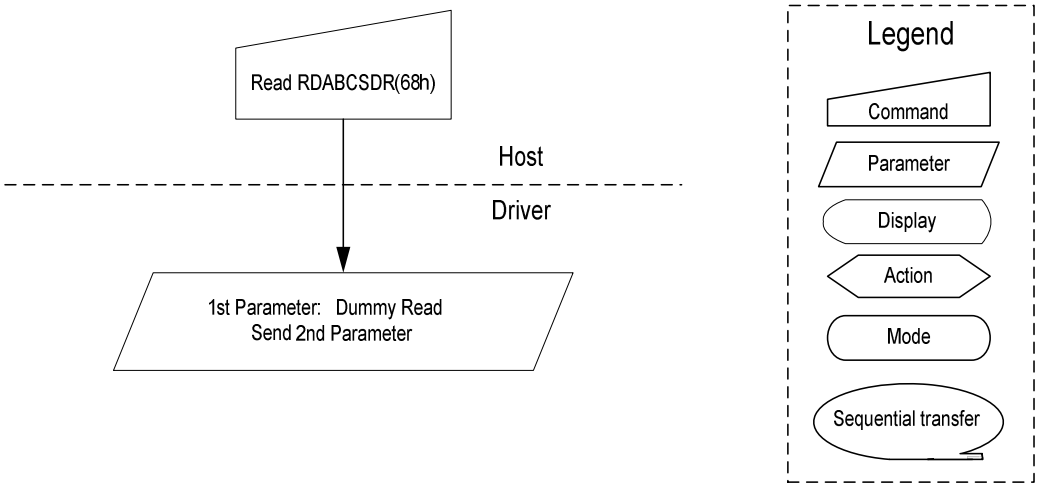
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.52. Read Black/White Low Bits (70h)

70h	RDBWLB (Read Black/White Low Bits)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	0	0	0	70h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Bkx [1:0]		Bky [1:0]		Wx [1:0]		Wy [1:0]		XX
Description	This command reads the lowest bits of black and white color characteristics. Black: Bkx and Bky White: Wx and Wy X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

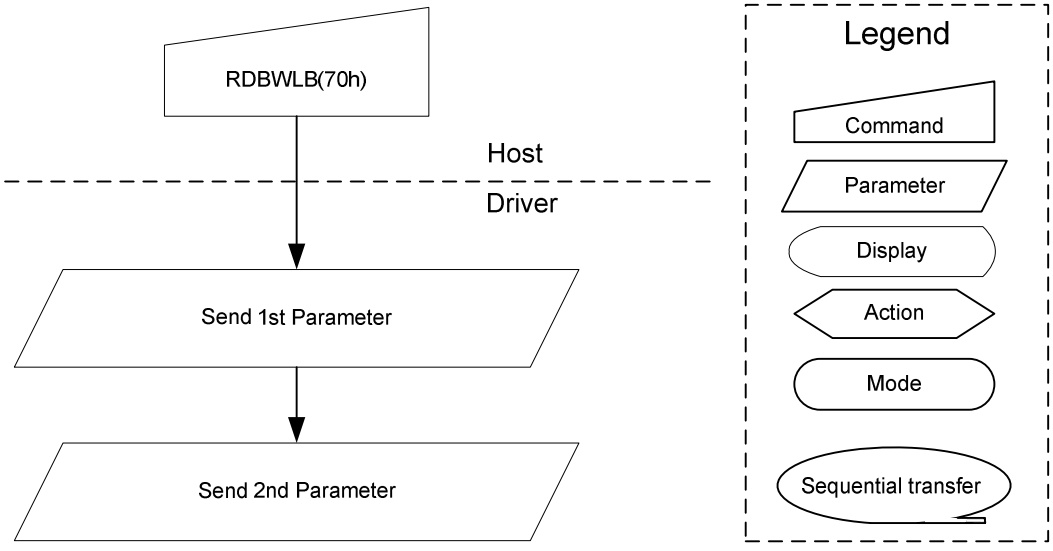
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.53. Read Bkx (71h)

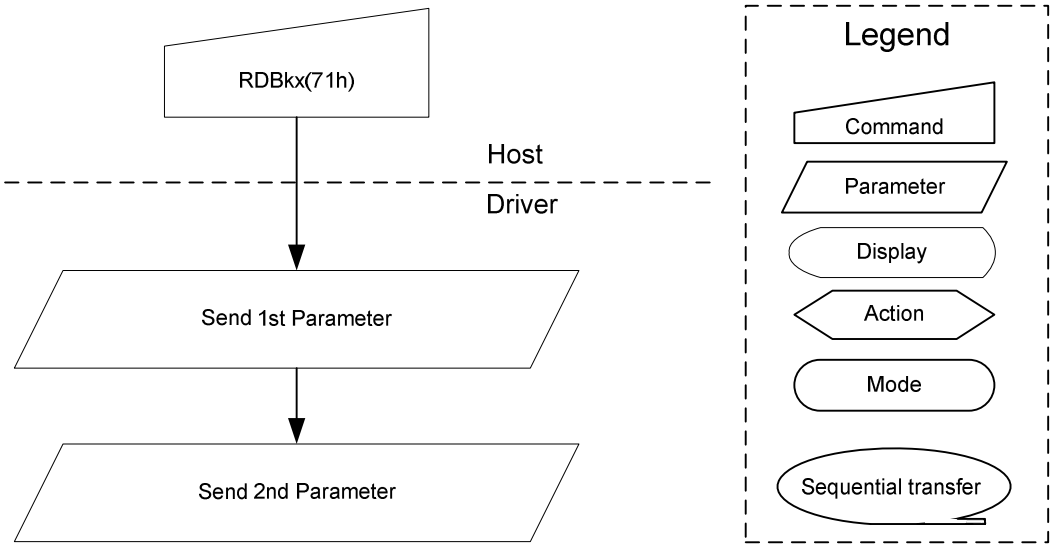
71h	RDBkx (Read Bkx)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	0	0	1	71h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Bkx [9:2]								XX
Description	This command reads the Bkx bits (Bkx [9:2]) of black color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.54. Read Bky (72h)

72h	RDBky (Read Bky)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	0	1	0	72h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Bky [9:2]								XX
Description	This command reads the Bky bits (Bky [9:2]) of black color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

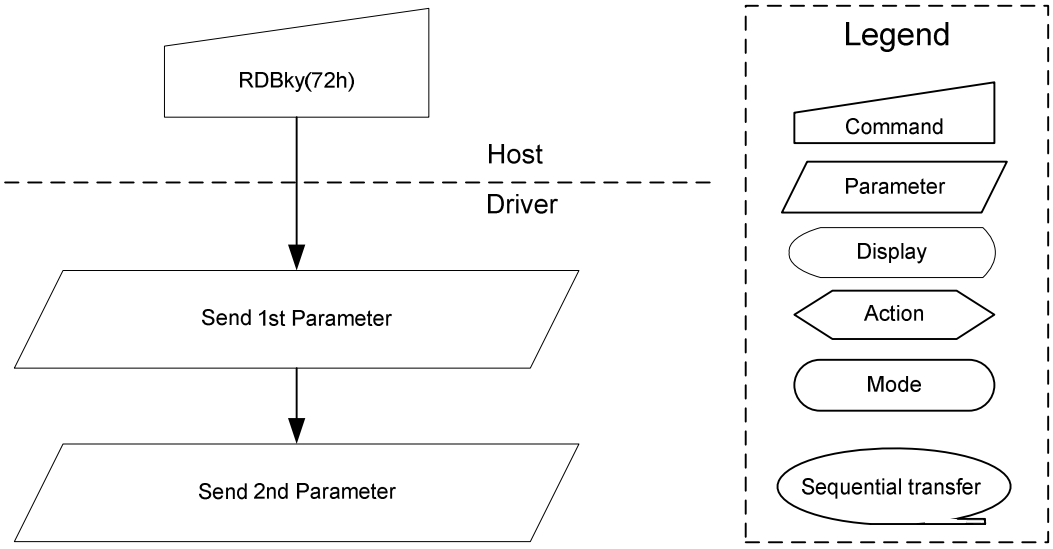
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.55. Read Wx (73h)

73h	RDWx (Read Wx)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	0	1	1	73h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Wx [9:2]								XX
Description	This command reads the Wx bits (Wx [9:2]) of white color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

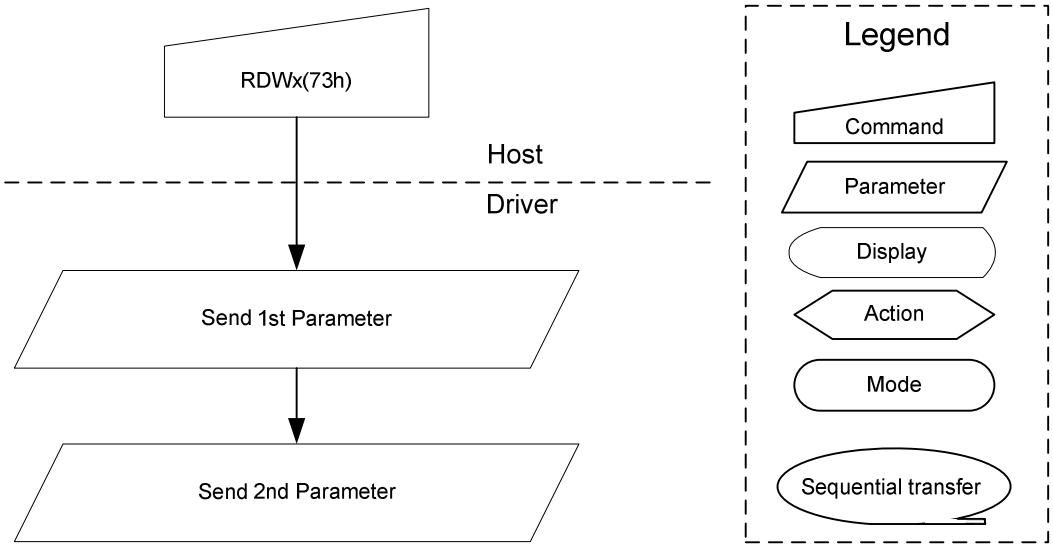
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.56. Read Wy (74h)

74h	RDWy (Read Wy)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	1	0	0	74h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Wy [9:2]								XX
Description	This command returns the Wy bits (Wy [9:2]) of white color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

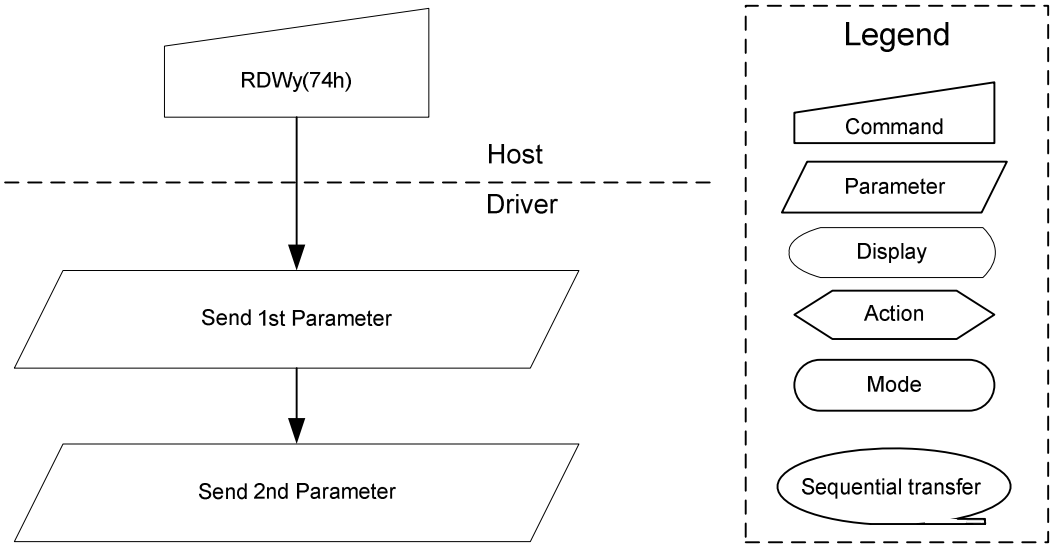
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.57. Read Red/Green Low Bits (75h)

75h	RDRGLB (Read Red / Green Low Bits)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	1	0	1	75h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Rx [1:0]		Ry [1:0]		Gx [1:0]		Gy [1:0]		XX
Description	This command reads the lowest bits of red and green color characteristics. Red: Rx and Ry Green: Gx and Gy X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

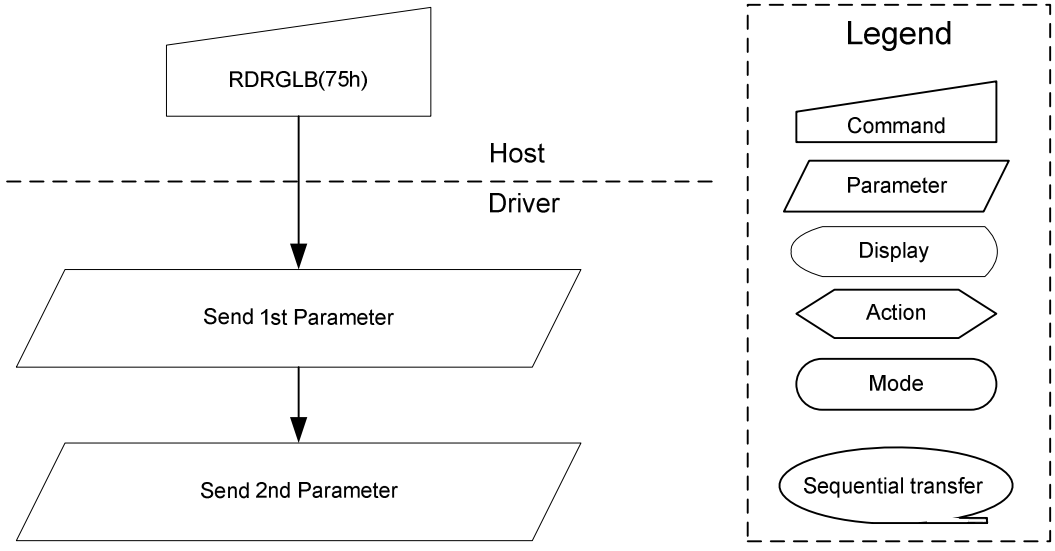
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.58. Read Rx (76h)

76h	RDRx (Read Rx)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	1	1	0	76h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Rx [9:2]								XX
Description	This command reads the Rx bits (Rx [9:2]) of red color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

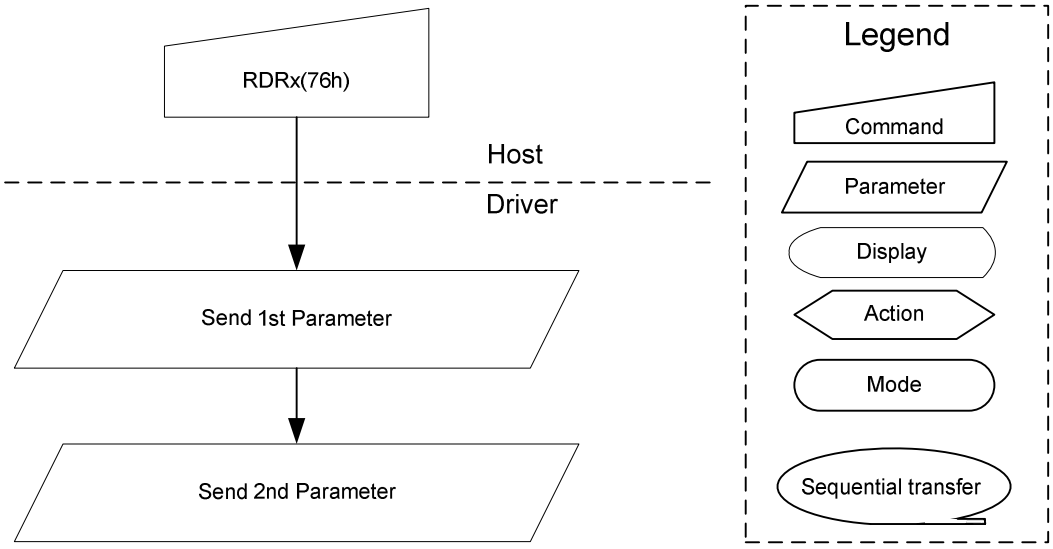
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.59. Read Ry (77h)

77h	RDRy (Read Ry)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	0	1	1	1	77h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Ry [9:2]								XX
Description	This command reads the Ry bits (Ry [9:2]) of red color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

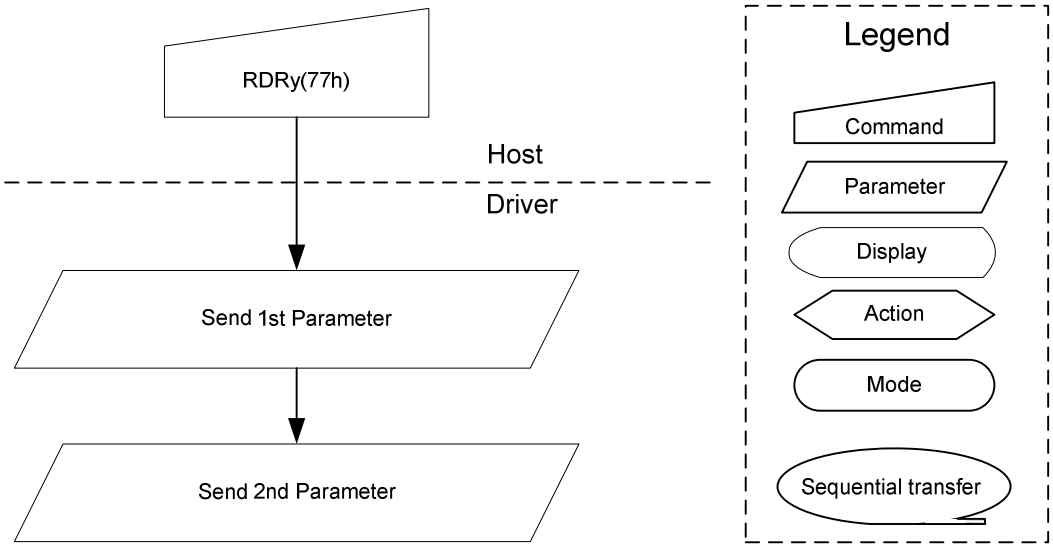
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.60. Read Gx (78h)

78h	RDGx (Read Gx)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	0	0	0	78h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	Gx [9:2]								XX
Description	This command reads the Gx bits (Gx [9:2]) of green color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

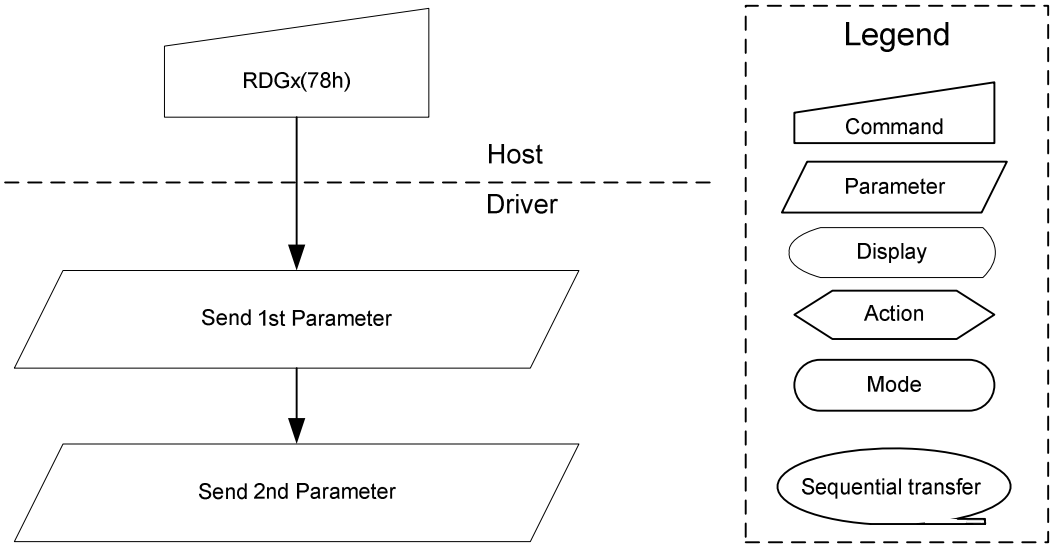
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.61. Read Gy (79h)

79h	RDGy (Read Gy)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	0	0	1	79h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Gy [9:2]								XX
Description	This command reads the Gy bits (Gy [9:2]) of green color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

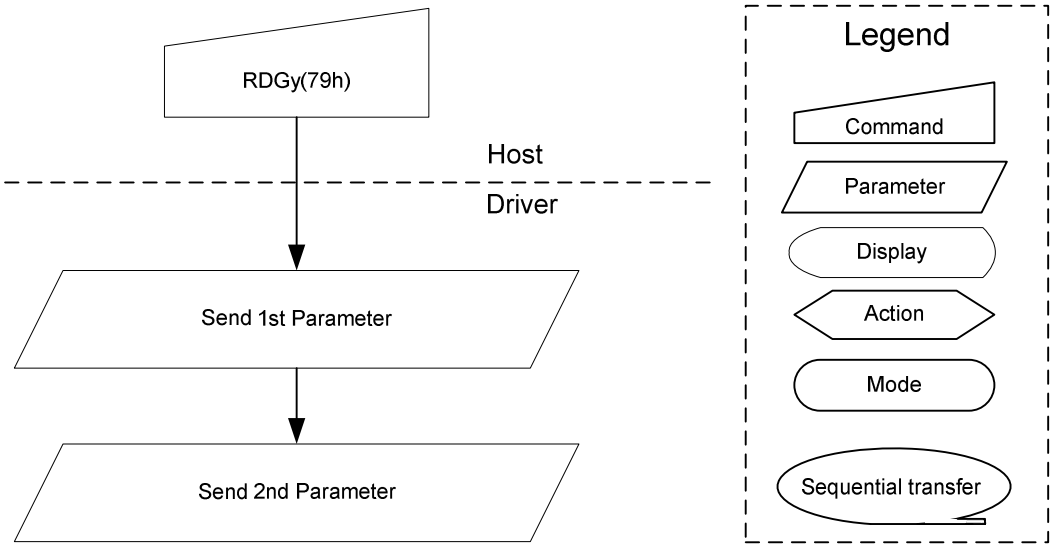
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.62. Read Blue/A Color Low Bits (7Ah)

7Ah	RDBALB (Read Blue / A Color Low Bits)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	0	1	0	7Ah
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Bx [1:0]		By [1:0]		Ax [1:0]		Ay [1:0]		XX
Description	This command reads the lowest bits of blue and A color color characteristics. Blue: Bx and By A color: Ax and Ay X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

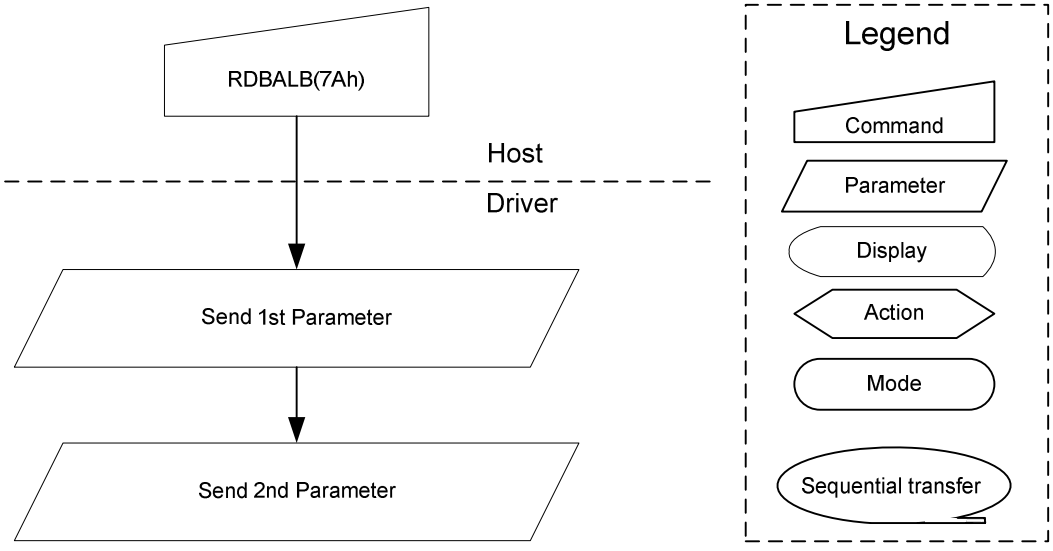
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.63. Read Bx (7Bh)

7Bh	RDBx (Read Bx)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	0	1	1	7Bh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Bx [9:2]								XX
Description	This command reads the Bx bits (Bx [9:2]) of blue color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

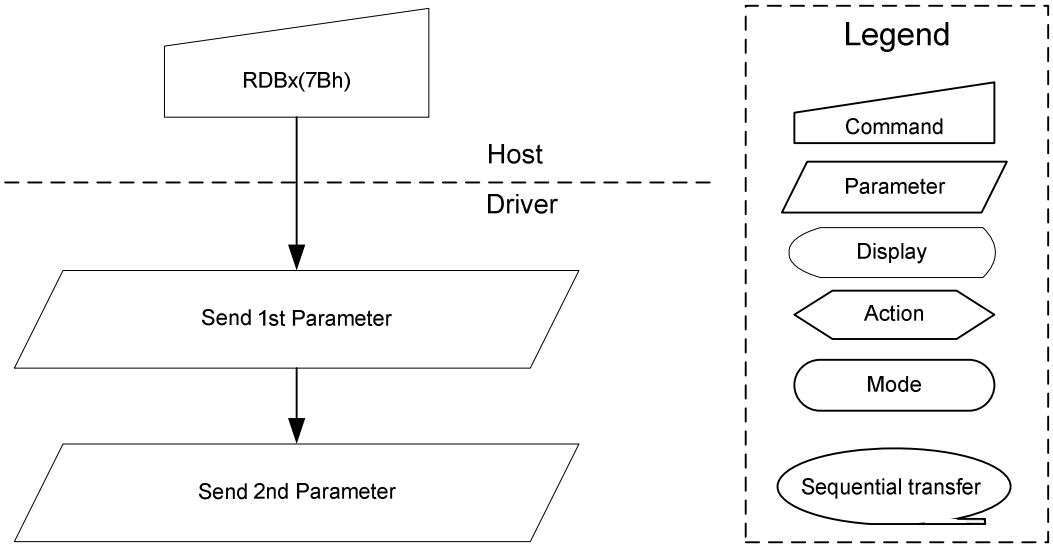
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.64. Read By (7Ch)

7Ch	RDBy (Read By)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	1	0	0	7Ch
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	By [9:2]								XX
Description	This command reads the By bits (By [9:2]) of blue color characteristics. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

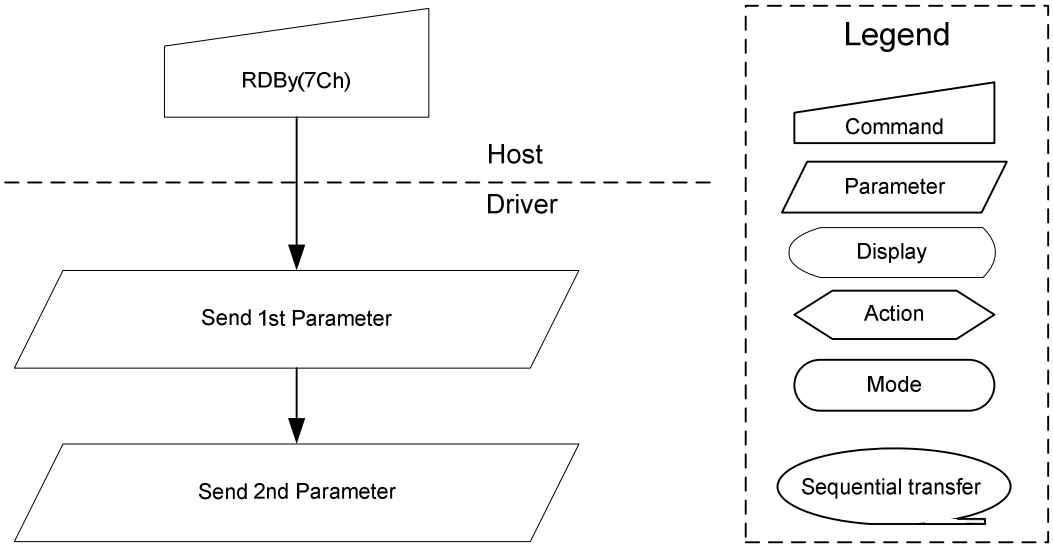
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.65. Read Ax (7Dh)

7Dh	RDAx (Read Ax)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	1	0	1	7Dh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Ax [9:2]								XX
Description	This command returns the Ax bits (Ax [9:2]) of A color characteristics. Ax [9:2] are set to 0 if they are not used. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

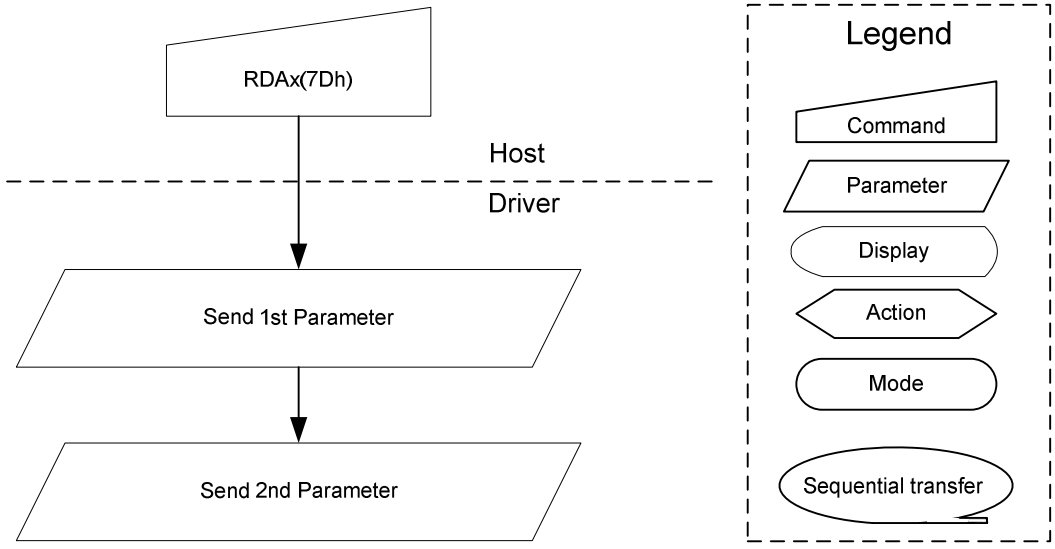
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.66. Read Ay (7Eh)

7Eh	RDAy (Read Ay)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	0	1	1	1	1	1	1	0	7Eh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	Ay [9:2]								XX
Description	This command reads the Ay bits (Ay [9:2]) of A color characteristics. Ay [9:2] are set to 0 if they are not used. X = void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

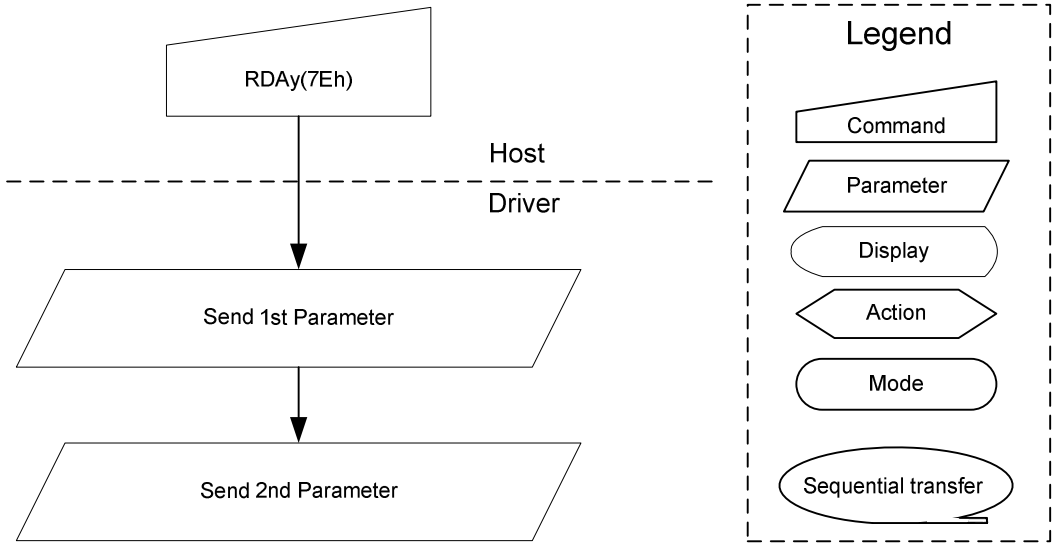
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.67. Read DDB Start (A1h)

A1h	RDDDBS (Read DDB Start)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	0	0	0	0	1	A1h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd Parameter	1	↑	1	XX	SID [7:0]								XX
3rd Parameter	1	↑	1	XX	SID [15:8]								XX
4th Parameter	1	↑	1	XX	MRID [7:0]								XX
5th Parameter	1	↑	1	XX	MRID [15:8]								XX
6th Parameter	1	↑	1	XX	1	1	1	1	1	1	1	1	FF

Description

This command reads the supplier identification and display module mode/revision information.

Note: This information is not the same as which "Read ID1 (DAh)", "Read ID2 (DBh)" and "Read ID3 (DCh)" commands return.

Parameter 2: SID [7:0] LCD module's manufacturer ID.

Parameter 3: SID [15:8] LCD module/driver version ID.

Parameter 4: MRID [7:0] LCD module/driver ID.

Parameter 5: MRID [15:8] IC version code.

Parameter 6: FFh - Exit code – there is no more data in the Descriptor Block

This read sequence can be interrupted by any command and it can be continued by the Read DDB Continue (A8h) command.

For example, RDDDBS => 1st parameter has been sent => 2nd parameter has been sent => interrupt => RDDDBC => 3rd parameter of the RDDDBS has been sent.

Restriction

Register Availability

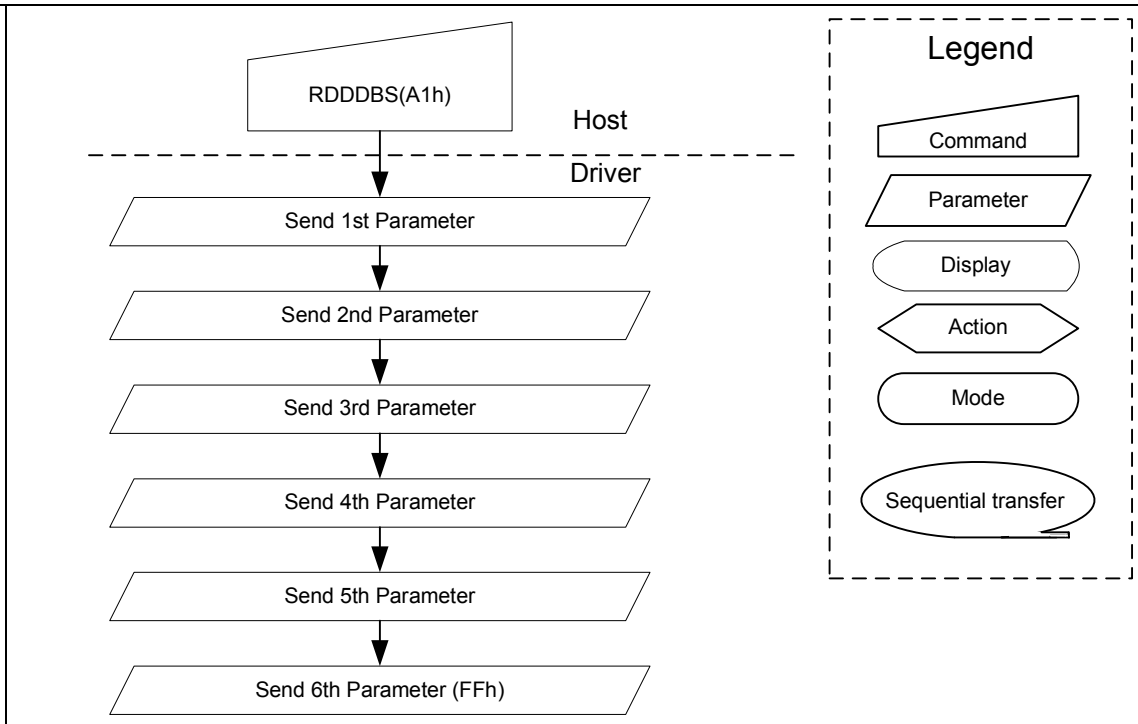
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value			
	SID [7:0]	SID [15:8]	MRID [7:0]	MRID [15:8]
Power On Sequence	00h	00h	00h	00h
S/W Reset	00h	00h	00h	00h
H/W Reset	00h	00h	00h	00h

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Flow Chart



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4.2.68. Read DDB Continue (A8h)

A8h	RDDDBC (Read DDB Continue)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	0	1	0	0	0	A8h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	D1 [7:0]								XX
:	1	↑	1	XX	Dx [7:0]								XX
NthParameter	1	↑	1	XX	Dn [7:0]								XX
Description	This command is used to read the supplier's identification and revision information from the point where RDDDBS (A1h) was interrupted by another command.												
Restriction	-												

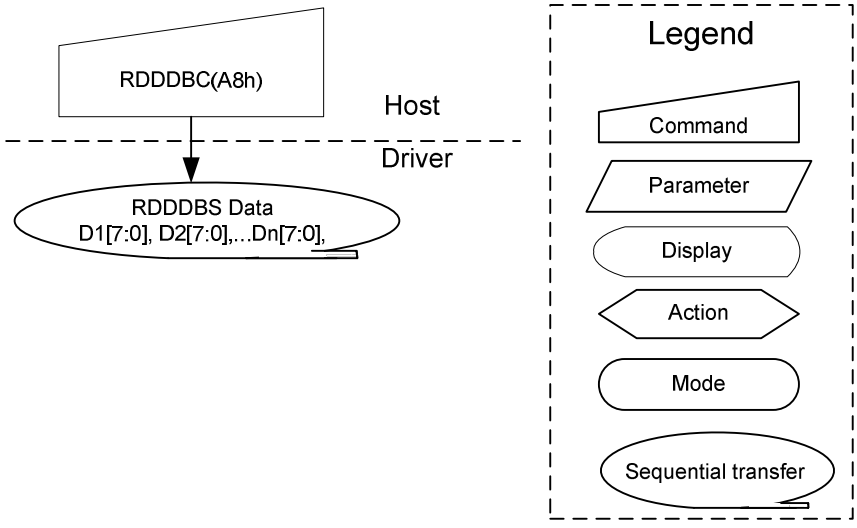
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	XXh
S/W Reset	XXh
H/W Reset	XXh

Flow Chart



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4.2.69. Read First Checksum (AAh)

AAh	RDFCS (Read First Checksum)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	0	1	0	1	0	AAh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	FCS [7:0]								XX
Description	This command reads the first checksum calculated from registers of the User's area and the Frame Memory after the write access to those registers and/or Frame Memory has been done. X = void												
Restriction	It is necessary to wait 150ms after the last write access to registers on the User area before this checksum value can be read. The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

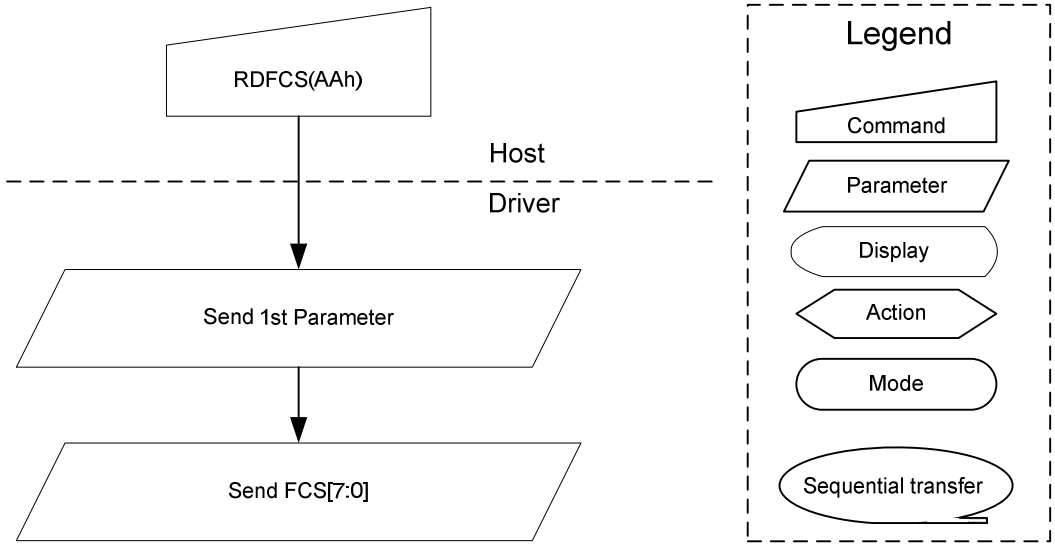
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.70. Read Continue Checksum (AFh)

AFh	RDCFCS (Read Continue Checksum)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	0	1	1	1	1	AFh
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	CCS [7:0]								XX
Description	This command reads the following checksum that is calculated continuously after the first checksum from registers of the User's area and the Frame Memory after the write access to those registers and/or Frame Memory is done. X = void												
Restriction	It is necessary to wait 300ms after the last write access to registers of the User's area before this checksum value can be read the first time. The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

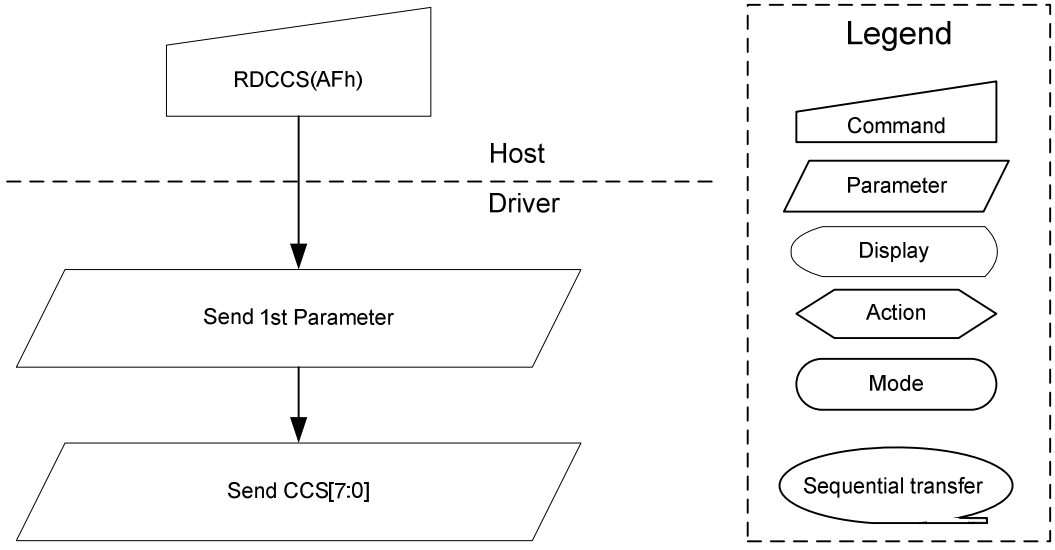
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	00h
S/W Reset	00h
H/W Reset	00h

Flow Chart



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4.2.71. Read Device Code (D3h)

D3h	Read Device Code												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	0	0	1	1	D3h
1st Parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2ndParameter	1	↑	1	XX	0	0	0	0	0	0	0	0	00h
3rdParameter	1	↑	1	XX	1	0	0	1	1	0	0	0	98h
4th Parameter	1	↑	1	XX	0	0	0	0	0	1	1	0	06h
Description	Read IC device code. The 1st parameter is a dummy data The 2nd parameter is reserved for future use. The 3rd and 4th parameters indicate the IC model name. X=void												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	24'h009806
S/W Reset	24'h009806
H/W Reset	24'h009806

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4.2.72. Read ID1 (DAh)

DBh	RDID1 (Read ID1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	0	1	0	DAh
1st parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd parameter	1	↑	1	XX	ID1 [7:0]								XX
Description	This read byte is used to track the LCD module/driver version. It is defined by the display supplier (with User's agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is a dummy data. The 2nd parameter is the LCD module/driver's version ID. The ID parameter range is from 80h to FFh. The ID2 is programmed by the OTP function. X = Void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

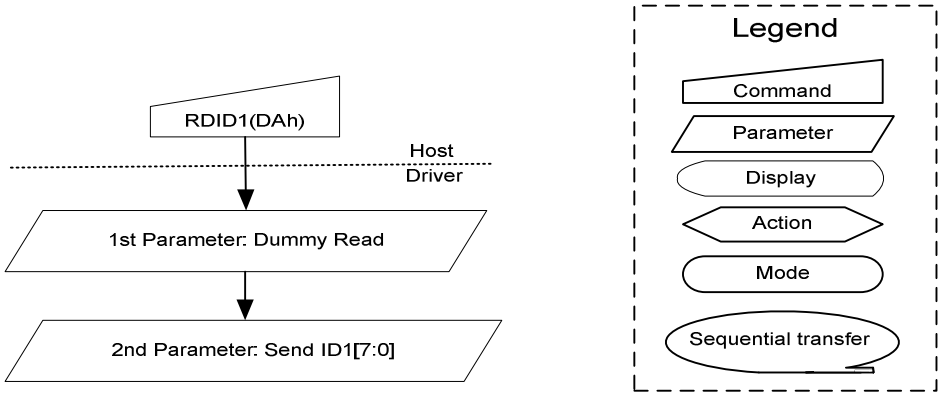
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (Before OTP program)	Default Value (After OTP program)
Power On Sequence	00h	OTP value
H/W Reset	00h	OTP value

Flow Chart



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4.2.73. Read ID2 (DBh)

DBh	RDID2 (Read ID2)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	0	1	1	DBh
1st parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd parameter	1	↑	1	XX	ID2 [7:0]								XX
Description	This read byte is used to track the LCD module/driver version. It is defined by the display supplier (with User's agreement) and changes each time a revision is made to the display, material or construction specifications. The 1st parameter is a dummy data. The 2nd parameter is the LCD module/driver's version ID. The ID parameter range is from 80h to FFh. The ID2 is programmed by the OTP function. X = Void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

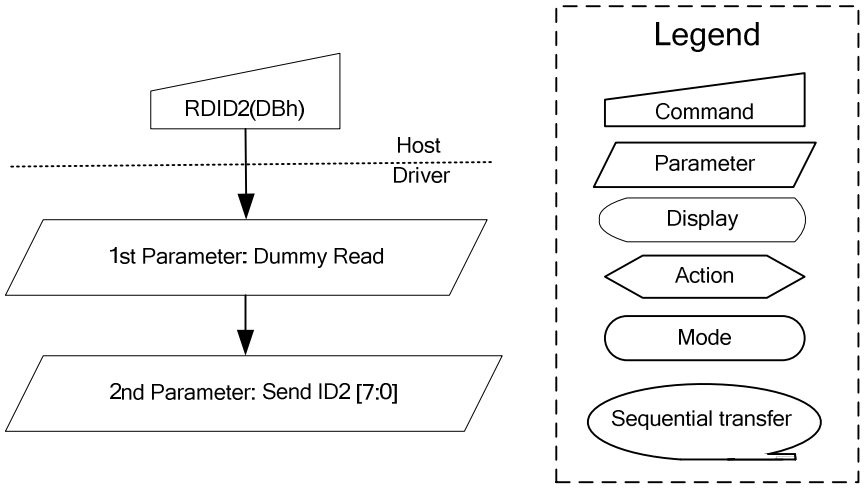
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (Before OTP program)	Default Value (After OTP program)
Power On Sequence	80h	OTP value
H/W Reset	80h	OTP value

Flow Chart



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4.2.74. Read ID3 (DCh)

DCh	RDID3 (Read ID3)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	1	0	0	DCh
1st parameter	1	↑	1	XX	X	X	X	X	X	X	X	X	XX
2nd parameter	1	↑	1	XX	ID3 [7:0]								XX
Description	This read byte identifies the LCD module/driver. It is specified by users. The 1st parameter is a dummy data. The 2nd parameter is the LCD module/driver's ID. The ID3 is programmed by the OTP function. X = Void												
Restriction	The ILI9806G sends the 2nd parameter value to the data lines if the MPU wants to read more than one parameter (= more than 2 RDX cycle) on the parallel MPU interface. Only the 2nd parameter is sent on the DSI; the 1st parameter is not sent.												

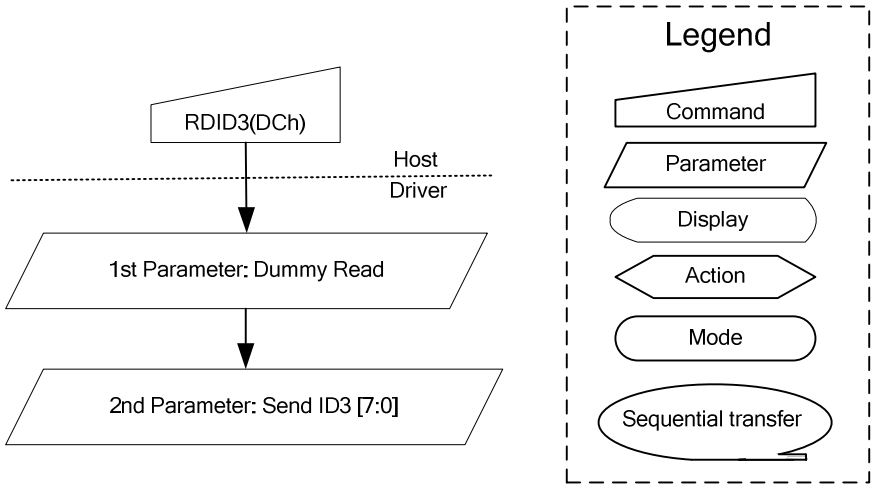
Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value (Before OTP program)	Default Value (After OTP program)
Power On Sequence	00h	OTP value
H/W Reset	00h	OTP value

Flow Chart



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4.3. Extend Command (EXTC) Description

4.3.1. Interface Mode Control (B0h)

B0h	IFMODE (Interface Mode Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	0	0	0	B0h
1st Parameter	1	1	↑	XX	0	0	0	0	VSPL	HSPL	DPL	EPL	00
Description	Sets the operation status of the RGB interface. The setting becomes effective as soon as the command is received. **EPL**: Enable polarity (0 = Low level enable, 1 = High level enable) **DPL**: DCK polarity set (0 = data latched at the rising edge, 1 = data latched at the falling edge) **HSPL**: HSYNC polarity (0 = Low pulse active, 1 = High pulse active) **VSPL**: VSYNC polarity (0 = Low pulse active, 1 = High pulse active)												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													
Default													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value			
	EPL	DPL	HSPL	VSPL
Power ON Sequence	1'b0	1'b0	1'b0	1'b0
S/W Reset	1'b0	1'b0	1'b0	1'b0
H/W Reset	1'b0	1'b0	1'b0	1'b0

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4.3.2. Frame Rate Control 1 (In Normal Mode/Full Colors) (B1h)

B1h	FRMCTR1 (Frame Rate Control In Normal Mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	0	0	1	B1h
1st parameter	1	1	↑	XX	0	0	0	0	0	0	DIVA[1:0]		00
2nd parameter	1	1	↑	XX	0	0	0	RTNA [4:0]					10
3rd parameter	1	1	↑	XX	0	0	FRSA [5:0]						14

Description

Sets the division ratio for internal clocks of the Normal Display Mode On

$$\text{Frame Rate} = \frac{\text{Fosc}}{(\text{DIVAx}(\text{FRSA}+1)) \times \text{RTNA} \times (\text{Display Line}+\text{VBP}+\text{VFP})}$$

Fosc: Internal oscillator frequency

DIVA: Division Ratio

FRSA: Frame rate setting

RTNA: Clocks per line

Display Line: Total driving line number

VBP: Back porch line number

VFP: Front porch line number

Note: If "VBP+VFP" < 60 or = 60, The "VBP+VFP" data of the frame rate calculation formula = 60.

If "VBP+VFP" > 60 , The "VBP+VFP" data of the frame rate calculation formula = "VBP+VFP" real data.

FRSA [5:0]						Decimal	Frame rate (Hz)
000000~010001						0~17	Setting prohibited
0	1	0	0	1	0	18	75.19
0	1	0	0	1	1	19	71.43
0	1	0	1	0	0	20	68.03
0	1	0	1	0	1	21	64.94
0	1	0	1	1	0	22	62.11
0	1	0	1	1	1	23	59.52
0	1	1	0	0	0	24	57.14
0	1	1	0	0	1	25	54.95
0	1	1	0	1	0	26	52.91
0	1	1	0	1	1	27	51.02
0	1	1	1	0	0	28	49.26
0	1	1	1	0	1	29	47.62
0	1	1	1	1	0	30	46.08
0	1	1	1	1	1	31	44.64
1	0	0	0	0	0	32	43.29
100001~111111						33~63	Setting prohibited

Frame rate calculation example:

Fosc = 18MHz, DIVA = 1, FRSA = 20, RTNA = 15 clock, Display Line = 800lines, VBP = VFP = 20lines.

Following the formula, the frame rate = 67.04Hz.

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DIVA [1:0]: division ratio for internal clocks for the Normal Display Mode On.

DIVA [1:0]		Division Ratio
0	0	fosc
0	1	fosc/2
1	0	fosc/4
1	1	fosc/8

RTNA [4:0]: RTNA [4:0] is used to set 1H (line) period for the Normal Display Mode On.

RTNA [4:0]					Clock per Line
0	0	0	0	0	Setting prohibited
0	0	0	0	1	Setting prohibited
0	0	0	1	0	Setting prohibited
0	0	0	1	1	Setting prohibited
0	0	1	0	0	Setting prohibited
0	0	1	0	1	Setting prohibited
0	0	1	1	0	Setting prohibited
0	0	1	1	1	Setting prohibited
0	1	0	0	0	Setting prohibited
0	1	0	0	1	Setting prohibited
0	1	0	1	0	Setting prohibited

RTNA [4:0]					Clock per Line
0	1	0	1	1	Setting prohibited
0	1	1	0	0	Setting prohibited
0	1	1	0	1	12 clocks
0	1	1	1	0	13 clocks
0	1	1	1	1	14 clocks
1	0	0	0	0	15 clocks
1	0	0	0	1	16 clocks
1	0	0	1	0	17 clocks
1	0	0	1	1	18 clocks
1	0	1	0	0	19 clocks
1	0	1	0	1	20 clocks

RTNA [4:0]					Clock per Line
1	0	1	1	0	21 clocks
1	0	1	1	1	22 clocks
1	1	0	0	0	23 clocks
1	1	0	0	1	24 clocks
1	1	0	1	0	25 clocks
1	1	0	1	1	26 clocks
1	1	1	0	0	27 clocks
1	1	1	0	1	28 clocks
1	1	1	1	0	29 clocks
1	1	1	1	1	30 clocks

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	DIVA [1:0]	RTNA [4:0]	FRSA [5:0]
Power ON Sequence	2'b00	5'b10000	6'b010100
S/W Reset	2'b00	5'b10000	6'b010100
H/W Reset	2'b00	5'b10000	6'b010100

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4.3.3. Frame Rate Control 2 (In Idle Mode/8 colors) (B2h)

B2h	FRMCTR2 (Frame Rate Control In Idle Mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	0	1	0	B2h
1st parameter	1	1	↑	XX	0	0	0	0	0	0	DIVB [1:0]		00
2nd parameter	1	1	↑	XX	0	0	0	RTNB [4:0]					10
3rd parameter	1	1	↑	XX	0	0	FRSB [5:0]						14

Description

Sets the division ratio for internal clocks of the Idle Mode On

$$\text{Frame Rate} = \frac{\text{Fosc}}{(\text{DIVBx}(\text{FRSB}+1)) \times \text{RTNB} \times (\text{Display Line}+\text{VBP}+\text{VFP})}$$

Fosc: Internal oscillator frequency

DIVB: Division Ratio

FRSB: Frame rate setting

RTNB: Clocks per line

Display Line: Total driving line number

VBP: Back porch line number

VFP: Front porch line number

Note: If “VBP+VFP” < 60 or = 60, The “VBP+VFP” data of the frame rate calculation formula = 60.

If “VBP+VFP” > 60 , The “VBP+VFP” data of the frame rate calculation formula = “VBP+VFP” real data.

FRSB [5:0]						Decimal	Frame rate (Hz)
000000~010001						0~17	Setting prohibited
0	1	0	0	1	0	18	75.19
0	1	0	0	1	1	19	71.43
0	1	0	1	0	0	20	68.03
0	1	0	1	0	1	21	64.94
0	1	0	1	1	0	22	62.11
0	1	0	1	1	1	23	59.52
0	1	1	0	0	0	24	57.14
0	1	1	0	0	1	25	54.95
0	1	1	0	1	0	26	52.91
0	1	1	0	1	1	27	51.02
0	1	1	1	0	0	28	49.26
0	1	1	1	0	1	29	47.62
0	1	1	1	1	0	30	46.08
0	1	1	1	1	1	31	44.64
1	0	0	0	0	0	32	43.29
100001~111111						33~63	Setting prohibited

Frame rate calculation example:

Fosc = 18MHz, DIVB = 1, FRSB = 20, RTNB = 15 clock, Display Line = 800lines, VBP = VFP = 20lines.

Following the formula, the frame rate = 67.04Hz.

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DIVB [1:0]: division ratio for internal clocks for the Idle Mode On.

DIVB [1:0]		Division Ratio
0	0	fosc
0	1	fosc/2
1	0	fosc/4
1	1	fosc/8

RTNB [4:0]: RTNB [4:0] is used to set 1H (line) period for the Idle Mode On.

RTNB [4:0]					Clock per Line
0	0	0	0	0	Setting prohibited
0	0	0	0	1	Setting prohibited
0	0	0	1	0	Setting prohibited
0	0	0	1	1	Setting prohibited
0	0	1	0	0	Setting prohibited
0	0	1	0	1	Setting prohibited
0	0	1	1	0	Setting prohibited
0	0	1	1	1	Setting prohibited
0	1	0	0	0	Setting prohibited
0	1	0	0	1	Setting prohibited
0	1	0	1	0	Setting prohibited

RTNB [4:0]					Clock per Line
0	1	0	1	1	Setting prohibited
0	1	1	0	0	Setting prohibited
0	1	1	0	1	12 clocks
0	1	1	1	0	13 clocks
0	1	1	1	1	14 clocks
1	0	0	0	0	15 clocks
1	0	0	0	1	16 clocks
1	0	0	1	0	17 clocks
1	0	0	1	1	18 clocks
1	0	1	0	0	19 clocks
1	0	1	0	1	20 clocks

RTNB [4:0]					Clock per Line
1	0	1	1	0	21 clocks
1	0	1	1	1	22 clocks
1	1	0	0	0	23 clocks
1	1	0	0	1	24 clocks
1	1	0	1	0	25 clocks
1	1	0	1	1	26 clocks
1	1	1	0	0	27 clocks
1	1	1	0	1	28 clocks
1	1	1	1	0	29 clocks
1	1	1	1	1	30 clocks

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	DIVB [1:0]	RTNB [4:0]	FRSB [5:0]
Power ON Sequence	2'b00	5'b10000	6'b010100
S/W Reset	2'b00	5'b10000	6'b010100
H/W Reset	2'b00	5'b10000	6'b010100

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4.3.4. Frame Rate control 3 (In Partial Mode/Full Colors) (B3h)

B3h	FRMCTR3 (Frame Rate Control In Partial Mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	0	1	1	B3h
1st parameter	1	1	↑	XX	0	0	0	0	0	0	DIVC[1:0]		00
2nd parameter	1	1	↑	XX	0	0	0	RTNC [4:0]					10
3rd parameter	1	1	↑	XX	0	0	FRSC [5:0]						14

Description

Sets the division ratio for internal clocks of Partial Mode On.

$$\text{Frame Rate} = \frac{\text{Fosc}}{(\text{DIVCx}(\text{FRSC}+1)) \times \text{RTNC} \times (\text{Display Line}+\text{VBP}+\text{VFP})}$$

Fosc: Internal oscillator frequency

DIVC: Division Ratio

FRSC: Frame rate setting

RTNC: Clocks per line

Display Line: Total driving line number

VBP: Back porch line number

VFP: Front porch line number

Note: If "VBP+VFP" < 60 or = 60, The "VBP+VFP" data of the frame rate calculation formula = 60.

If "VBP+VFP" > 60 , The "VBP+VFP" data of the frame rate calculation formula = "VBP+VFP" real data.

FRSC [5:0]						Decimal	Frame rate (Hz)
000000~010001						0~17	Setting prohibited
0	1	0	0	1	0	18	75.19
0	1	0	0	1	1	19	71.43
0	1	0	1	0	0	20	68.03
0	1	0	1	0	1	21	64.94
0	1	0	1	1	0	22	62.11
0	1	0	1	1	1	23	59.52
0	1	1	0	0	0	24	57.14
0	1	1	0	0	1	25	54.95
0	1	1	0	1	0	26	52.91
0	1	1	0	1	1	27	51.02
0	1	1	1	0	0	28	49.26
0	1	1	1	0	1	29	47.62
0	1	1	1	1	0	30	46.08
0	1	1	1	1	1	31	44.64
1	0	0	0	0	0	32	43.29
100001~111111						33~63	Setting prohibited

Frame rate calculation example:

Fosc = 18MHz, DIVC = 1, FRSC = 20, RTNC = 15 clock, Display Line = 800lines, VBP = VFP = 20lines.

Following the formula, the frame rate = 67.04Hz.

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DIVC [1:0]: division ratio for internal clocks for the Partial Mode On

DIVC [1:0]		Division Ratio
0	0	fosc
0	1	fosc/2
1	0	fosc/4
1	1	fosc/8

RTNC [4:0]: RTNC [4:0] is used to set 1H (line) period for the Partial Mode On.

RTNC [4:0]					Clock per Line
0	0	0	0	0	Setting prohibited
0	0	0	0	1	Setting prohibited
0	0	0	1	0	Setting prohibited
0	0	0	1	1	Setting prohibited
0	0	1	0	0	Setting prohibited
0	0	1	0	1	Setting prohibited
0	0	1	1	0	Setting prohibited
0	0	1	1	1	Setting prohibited
0	1	0	0	0	Setting prohibited
0	1	0	0	1	Setting prohibited
0	1	0	1	0	Setting prohibited

RTNC [4:0]					Clock per Line
0	1	0	1	1	Setting prohibited
0	1	1	0	0	Setting prohibited
0	1	1	0	1	12 clocks
0	1	1	1	0	13 clocks
0	1	1	1	1	14 clocks
1	0	0	0	0	15 clocks
1	0	0	0	1	16 clocks
1	0	0	1	0	17 clocks
1	0	0	1	1	18 clocks
1	0	1	0	0	19 clocks
1	0	1	0	1	20 clocks

RTNC [4:0]					Clock per Line
1	0	1	1	0	21 clocks
1	0	1	1	1	22 clocks
1	1	0	0	0	23 clocks
1	1	0	0	1	24 clocks
1	1	0	1	0	25 clocks
1	1	0	1	1	26 clocks
1	1	1	0	0	27 clocks
1	1	1	0	1	28 clocks
1	1	1	1	0	29 clocks
1	1	1	1	1	30 clocks

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	DIVC [1:0]	RTNC [4:0]	FRSC [5:0]
Power ON Sequence	2'b00	5'b10000	6'b010100
S/W Reset	2'b00	5'b10000	6'b010100
H/W Reset	2'b00	5'b10000	6'b010100

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4.3.5. Display Inversion Control (B4h)

B4h	INVTR (Display Inversion Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	0	0	B4h
1st parameter	1	1	↑	XX	0	0	0	0	NLA [3:0]				00
2nd parameter	1	1	↑	XX	0	0	0	0	NLB [3:0]				00
3rd parameter	1	1	↑	XX	0	0	0	0	NLC [3:0]				00

Description

Set Display inversion mode

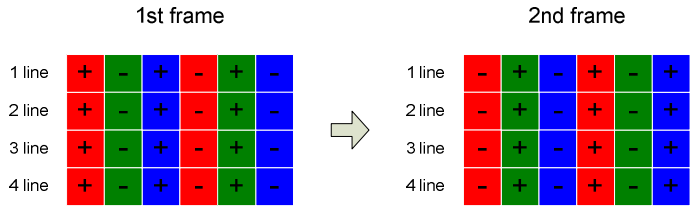
NLA [3:0]: Inversion setting in full colors normal mode (Normal Mode On)

NLB [3:0]: Inversion setting in Idle mode (Idle Mode On)

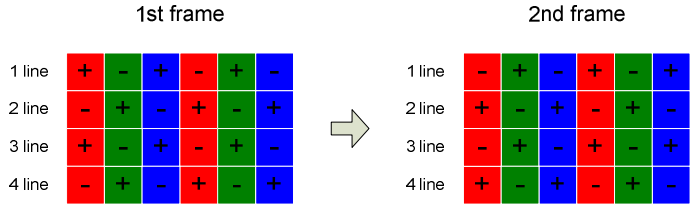
NLC [3:0]: Inversion setting in full colors partial mode (Partial Mode On/Idle Mode Off)

NLA/NLB/NLC [3:0]				Inversion
0	0	0	0	Column inversion
0	0	0	1	1 dot inversion
0	0	1	0	2 dot inversion
0	0	1	1	4 dot inversion
0	1	1	1	3 dot inversion

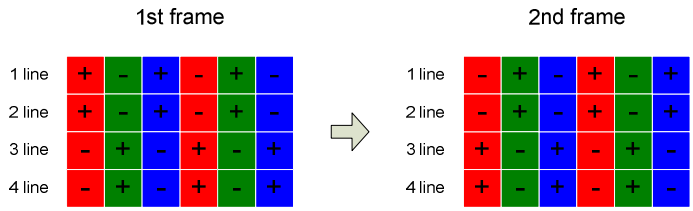
Column Inversion



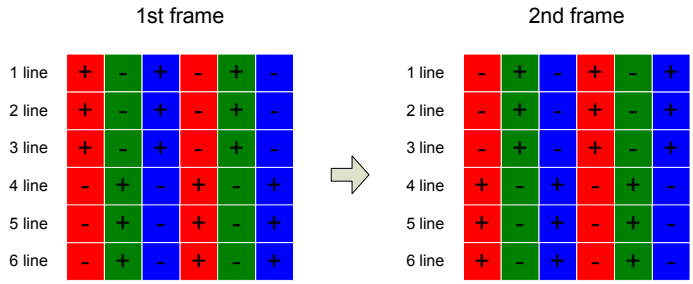
1-Dot Inversion

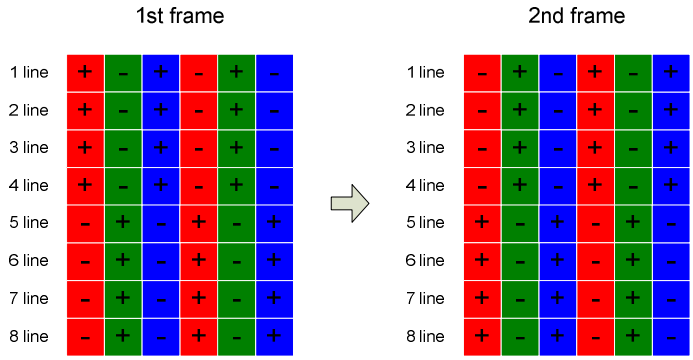


2-Dot Inversion



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<table>
<tr><td></td><td>
3-Dot Inversion


4-Dot Inversion


</td></tr>
<tr><td>Restriction</td><td>To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th>Default Value</th></tr>
<tr><th>NLA/NLB/NLC [3:0]</th></tr>
<tr><td>Power ON Sequence</td><td>4’b0000</td></tr>
<tr><td>S/W Reset</td><td>4’b0000</td></tr>
<tr><td>H/W Reset</td><td>4’b0000</td></tr>
</table>
</td></tr>
</table>

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4.3.6. Blanking Porch Control (B5h)

B5h	BLKPRH (Blanking Porch)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	0	1	B5h
1st Parameter	1	1	↑	XX	0	VFP [6:0]							14
2nd Parameter	1	1	↑	XX	0	VBP [6:0]							14
3rd Parameter	1	1	↑	XX	HBP [7:0]								04
4th Parameter	1	1	↑	XX	0	0	0	0	0	0	HBP [9:8]		00

Description

VFP [6:0]/VBP [6:0]: The VFP [6:0] and VBP [6:0] bits specify the line number of vertical front and back porch period respectively.

VFP [6:0] VBP [6:0]	Number of HSYNC of front/back porch
0000000	Setting prohibited
0000001	Setting prohibited
0000010	2
0000011	3
: :	: :
1111101	125
1111110	126
1111111	127

HBP [9:0]: The HBP [9:0] bits specify the line number of horizontal back porch period.

HBP [9:0]	Number of DCK of the back porch
0000000000	Setting prohibited
0000000001	Setting prohibited
0000000010	2
0000000011	3
0000000100	4 (HBP [9:0] default)
: :	: :
1111111110	1022
1111111111	1023

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	VFP [6:0]	VBP [6:0]	HBP [9:0]
Power ON Sequence	7'h14	7'h14	10'h04
S/W Reset	7'h14	7'h14	10'h04
H/W Reset	7'h14	7'h14	10'h04

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4.3.7. Display Function Control (B6h)

B6h	DISCTRL (Display Function Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	1	0	B6h
1st parameter	1	1	↑	XX	SYNC mode	0	RM	0	0	0	1	PT	02

Description

SYNC mode: Select the operation mode of the RGB interface

SYNC mode	RGB interface selection
0	DE mode
1	SYNC mode

RM: Select the interface to access the GRAM. When RM = 0, the driver will write display data to the GRAM via the system interface, and the driver will write display data to the GRAM via RGB interface when RM = 1.

RM	Interface for RAM access
0	System interface
1	RGB interface

PT : Determine source/VCOM output in a non-display area in the partial display mode.

PT	Source output on non-display area
0	V255
1	V0

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	PT	RM	SYNC Mode
Power ON Sequence	1'b0	1'b0	1'b0
S/W Reset	1'b0	1'b0	1'b0
H/W Reset	1'b0	1'b0	1'b0

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4.3.8. Entry Mode Set (B7h)

B7h	ETMOD (Entry Mode Set)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	0	1	1	1	B7h
1st Parameter	1	1	↑	XX	0	0	0	0	DSTB	1	1	0	06

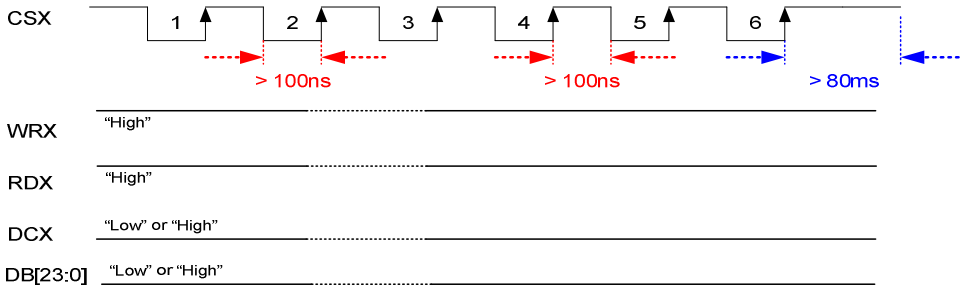
Description

DSTB: The ILI9806G driver enters the Deep Standby Mode when DSTB is set to high (= 1). In the Deep Standby mode, both internal logic power and SRAM power are turned off, and the display data stored in the Frame Memory and the instructions are not saved. Rewrite Frame Memory content and instructions after exiting the Deep Standby Mode.

The ILI9806G provides two ways to exit the Deep Standby Mode:

(1) Exit the Deep Standby Mode by pulling down the CSX to low (= 0) 6 times.

(2) Input a RESX pulse with effective low level duration to start up the inner logic regulator and make a transition to the initial state.



Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	DSTB
Power ON Sequence	1'b0
S/W Reset	1'b0
H/W Reset	1'b0

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4.3.9. DBI Type B Interface Setting (B8h)

B8H	Frame Memory Access and Interface Setting												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	1	0	0	0	B8h
1st parameter	1	1	↑	XX	0	WEMODE	EPF [1:0]		0	0	0	0	70
Description	**WEMODE:** Memory write control WEMODE = 0: When the transfer number of data exceeds (EC-SC+1)*(EP-SP+1), the exceeding data is ignored. WEMODE = 1: When the transfer number of data exceeds (EC-SC+1)*(EP-SP+1), the column and page number are reset, and the exceeding data is written into the following column and page. **EPF [1:0]:** Set the data format from 16/18-bit (R, G, B) to 24-bit (r, g, b) that is stored in the internal GRAM. See chapter 3.14.28 “16/18-bit color data mapping to 24-bit pixel data operation” for detail description.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	WEMODE	EPF [1:0]
Power ON Sequence	1b’1	2b’11
H/W Reset	1b’1	2b’11

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4.3.10. Panel Control (B9h)

B9h	PANELCTRL (Set panel operation mode)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	1	0	0	1	B9h
1st parameter	1	1	↑	XX	MY_ Panel	MX_ Panel	MV_ Panel	ML_ Panel	BGR_ Panel	MH_ Panel	SS_ Panel	GS_ Panel	00
2nd parameter	1	1	↑	XX	0	0	0	0	0	0	REV_ Panel	0	00

Description

This command defines the panel operation mode.

Symbol	Name	Description
MY_ Panel	Panel Row Address Order	These 3 bits control the write/read direction from the MPU to the memory.
MX_ Panel	Panel Column Address Order	
MV_ Panel	Panel Row/Column Exchange	
ML_ Panel	Panel Vertical Refresh Order	LCD vertical refresh direction control
BGR_ Panel	Panel RGB-BGR Order	Color selector switch control (0 = RGB color filter panel, 1 = BGR color filter panel)
MH_ Panel	Panel Horizontal Refresh Order	LCD horizontal refreshing direction control
SS_ Panel	Panel Flip Horizontal	Select the Source driver scan direction on the panel module
GS_ Panel	Panel Flip Vertical	Select the Gate driver scan direction on the panel module
REV_ Panel	Panel inversion mode	Panel display inversion on the panel module

REV_ Panel: Select normally white or normally black panel.

REV	Panel	Data	Color	Source
0	Normally white	0 x 00	Black	V0+/V0-
		0 x FF	White	V255+/V255-
1	normally black	0 x 00	Black	V255+/V255-
		0 x FF	White	V0+/V0-

SS_ Panel**:** Select the shift direction of outputs from the source driver.

SS_ Panel	Source Output Scan Direction
0	S1 → S1440
1	S1440 → S1

GS_ Panel**:** Set the direction of the gate driver scan in the display range.

Note: Its usage depends on the design of the Panel.

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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Default

Status	Default Value				
	MY_ Panel	MX_ Panel	MV_ Panel	ML_ Panel	BGR_ Panel
Power On Sequence	1b'0	1b'0	1b'0	1b'0	1b'0
S/W Reset	1b'0	1b'0	1b'0	1b'0	1b'0
H/W Reset	1b'0	1b'0	1b'0	1b'0	1b'0

Status	Default Value			
	MH_ Panel	SS_ Panel	GS_ Panel	REV_ Panel
Power On Sequence	1b'0	1b'0	1b'0	1b'0
S/W Reset	1b'0	1b'0	1b'0	1b'0
H/W Reset	1b'0	1b'0	1b'0	1b'0

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4.3.11. SPI Interface Setting (BAh)

BAH	Frame Memory Access and Interface Setting												
	DCX	RDX	WRX	D[23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	0	1	1	1	0	1	0	BAh
1st parameter	1	1	↑	XX	Spitype	1	1	0	0	0	0	0	XX
Description	**Spitype:** Spitype = 0: SPI interface transfers data through SDI and SDO pins. Spitype = 1: SPI interface transfers data through SDI (SDA) pin.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	Spitype
Power ON Sequence	1b'1
S/W Reset	1b'1
H/W Reset	1b'1

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4.3.12. Power Control 1 (C0h)

C0h	PWCTRL 1 (Power Control 1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	0	0	0	0	C0h
1st parameter	1	1	↑	XX	VD[2:0]			VE[2:0]			1	1	03
2nd parameter	1	1	↑	XX	0	0	0	0	1	0	1	1	0B
3rd parameter	1	1	↑	XX	0	BH	0	0	BL [1:0]		1	0	0A
4th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
5th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
6th parameter	1	1	↑	XX	1	0	0	0	1	0	0	0	88
7th parameter	1	1	↑	XX	0	0	1	0	1	1	0	0	2C
8th parameter	1	1	↑	XX	0	1	0	1	0	0	0	0	50
9th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
10th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
11th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
12th parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
13th parameter	1	1	↑	XX	PCK_1[1:0]		PCK_2[1:0]		PCK_3[1:0]		PCK_4[1:0]		FF

Description

VD [2:0]: Set the AVDD clamp level.

AVDD Cla p Level	
VD[2:0]	Voltage
000	5.2
001	5.3
010	5.4
011	5.5
100	5.6
101	5.1
110	5.0
111	5.7

VE [2:0]: Set the AVEE clamp level.

AVEE Clamp Level	
VE[2:0]	Voltage
000	-4.9
001	-5.0
010	-5.1
011	-5.2
100	-5.2
101	-5.3
110	-5.4
111	-5.5

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BH : Sets the factor used in the step-up circuits for VGH.

Select the optimal step-up factor for the operating voltage. To reduce power consumption, set a smaller factor.

VGH operation voltage	
BH	VGH
0	2AVDD – AVEE
1	AVDD – AVEE + VCI

BL [1:0]: Sets the factor used in the step-up circuits for VGL.

Select the optimal step-up factor for the operating voltage. To reduce power consumption, set a smaller factor.

VGH / VGL operation voltage			
BL [1:0]		VGH	VGL
0	0	2AVDD - AVEE	AVEE - AVDD
0	1	2AVDD - AVEE	2AVEE - AVDD
1	0	2AVDD - AVEE	AVEE – AVDD + VCL
1	1	2AVDD - AVEE	AVEE - VDDB + VCL

Note. The restriction setting VGH – VGL < 32 V.

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value		
	VD[2:0]	VE[2:0]	BL [1:0]
Power ON Sequence	3'b000	3'b000	2'b10
S/W Reset	3'b000	3'b000	2'b10
H/W Reset	3'b000	3'b000	2'b10

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4.3.13. Power Control 2 (C1h)

C1h	PWCTRL 2 (Power Control 2)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	0	0	0	1	C1h
1st parameter	1	1	↑	XX	0	0	BG_ISC [1:0]		0	1	1	1	17
2nd parameter	1	1	↑	XX	VRH1 [7:0]								90
3rd parameter	1	1	↑	XX	VRH2 [7:0]								90
4th parameter	1	1	↑	XX	0	0	EXT_CPCK_SEL[5:4]		0	0	0	0	20

Description

BG_ISC [1:0]: Source OP Amp bias control.

BG_ISC [1:0]		Source bias
0	0	0.5 x I
0	1	1.0 x I
1	0	1.5 x I
1	1	2.0 x I

VRH1 [7:0]: Sets the VGMP voltage for positive Gamma

VRH1 [7:0]	VGMP
8'h00	3.0000
8'h01	3.0125
8'h02	3.0250
:	:
8'h7F	4.5875
8'h80	4.6000
:	:
8'h8F	4.7875
8'h90	4.8000
:	:
8'hFE	6.1750
8'hFF	6.1875

Note: VGMP ≦ AVDD – 0.5V

VRH2 [7:0]: Sets the VGMN voltage for negative Gamma

VRH2 [7:0]	VGMN
8'h00	- 3.0000
8'h01	- 3.0125
8'h02	- 3.0250
:	:
8'h7F	- 4.5875
8'h80	- 4.6000
:	:
8'h8F	- 4.7875
8'h90	- 4.8000
:	:
8'hFE	- 6.1750
8'hFF	- 6.1875

Note: VGMN ≦ AVEE + 0.5V

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<table>
<tr><td></td><td>
EXT_CPCK_SEL [5:4]: Sets the ILI4002 AVDD / AVEE pumping ratio.
<table>
<tr><th colspan="2">EXT_CPCK_SEL [5:4]</th><th>Pumping Ratio</th></tr>
<tr><td>0</td><td>0</td><td>1.5 X</td></tr>
<tr><td>0</td><td>1</td><td>2.0 X</td></tr>
<tr><td>1</td><td>0</td><td>3.0 X</td></tr>
<tr><td>1</td><td>1</td><td>OFF</td></tr>
</table>
Note：If VCI>3V, ILI4002 AVDD/AVEE pumping ratio=3.0X setting is prohibited.
</td></tr>
<tr><td>Restriction</td><td>To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="4">Default Value</th></tr>
<tr><th>BG_ISC [1:0]</th><th>VRH1 [7:0]</th><th>VRH2 [7:0]</th><th>EXT_CPCK_SEL [1:0]</th></tr>
<tr><td>Power ON Sequence</td><td>2’b01</td><td>8’h90</td><td>8’h90</td><td>2’b10</td></tr>
<tr><td>S/W Reset</td><td>2’b01</td><td>No Change</td><td>No Change</td><td>2’b10</td></tr>
<tr><td>H/W Reset</td><td>2’b01</td><td>8’h90</td><td>8’h90</td><td>2’b10</td></tr>
</table>
</td></tr>
</table>

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4.3.14. Power Control 3 (C2h)

C2h	PWCTRL 3 (Power Control 3)									
	DCX	SCL	Parameter							HEX
			D7	D6	D5	D4	D3	D2	D1	D0	
Command	0	↑	1	1	0	0	0	0	1	0	C2h
1st parameter	1	↑	0	1	0	0	0	0	0	1	41h
2nd parameter	1	↑	0	1	0	0	0	1	0	0	44h
3rd parameter	1	↑	0	0	0	0	0	1	0	0	04h
4th parameter	1	↑	VGSP[7:0]								00h
5th parameter	1	↑	VGSN[7:0]								00h

Description

VGSP [7:0]: Selects positive gamma low voltage.

VGSP[7:0]	Spec (V)
8'h00	0
8'h01	0.3
8'h02	0.3125
8'h03	0.325
8'h04	0.3375
8'h05	0.35
8'h06	0.3625
8'h07	0.375
8'h08	0.3875
…	
8'hFA	3.4125
8'hFB	3.425
8'hFC	3.4375
8'hFD	3.45
8'hFE	3.4625
8'hFF	3.475

VGSN [7:0]: Selects negative gamma high voltage.

VGSN[7:0]	Spec(V)
8'h00	0
8'h01	-0.3
8'h02	-0.3125
8'h03	-0.325
8'h04	-0.3375
8'h05	-0.35
8'h06	-0.3625
8'h07	-0.375
8'h08	-0.3875
8'h09	-0.4
8'h0A	-0.4125
8'h0B	-0.425
…	
8'hFB	-3.425
8'hFC	-3.4375
8'hFD	-3.45
8'hFE	-3.4625
8'hFF	-3.475

Restriction	To enable this command, “EXTC Command Set enable register (FFh) ” must set first.

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Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Idle Mode Off, Sleep Out	Yes
Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	VGSP[7:0]	VGSN[7:0]
Power ON Sequence	8'h00	8'h00
S/W Reset	8'h00	8'h00
H/W Reset	8'h00	8'h00

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4.3.15. VCOM Control 1 (C7h)

C7h	VMCTRL1 (VCOM Control 1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	0	1	1	1	C7h
1st Parameter	1	1	↑	XX	VCM [7:0]								8F
2nd Parameter	1	1	↑	XX	nVM	0	0	0	0	0	0	0	80

Description

nVM: The source of selection of Vcom voltage setting either RC7h VCM[7:0] setting by user or NV memory programmed by user.
0 : Selection from NV memory.
1 : Selection from RC7h VCM[7:0].
If NV momory programmed before, the nVM bit set to "0" automatically, user sets nVM bit to "1", that can enable RC7 VCM[7:0] setting again.

VCM [7:0] is used to set factors to generate the VCOM voltage.

VCM [7:0]	VCOM
8'h00	0.0000
8'h01	- 0.0125
8'h02	- 0.0250
:	:
8'h7F	- 1.5875
8'h80	- 1.6000
:	:
8'h8F	- 1.7875
8'h90	- 1.8000
:	:
8'hFE	- 3.1750
8'hFF	- 3.1875

Note: VCOM ≧ VCL + 0.3V

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	VCM [7:0]	nVM
Power ON Sequence	8'h8F	1'b1
S/W Reset	No Change	1'b1
H/W Reset	8'h8F	1'b1

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4.3.16. Backlight Control 1 (C8h)

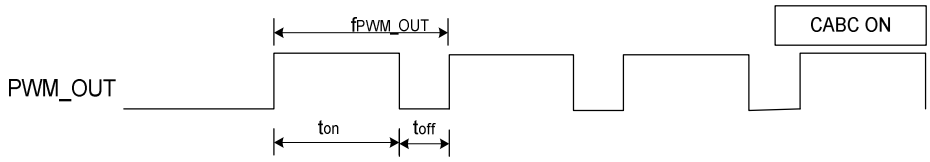
C8h	BLCTRL1 (Backlight Control 1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	0	0	C8h
1st Parameter	1	1	↑	XX	PWM_DIV [7:0]								XX

Description

PWM_DIV [7:0]: PWM_OUT output period control. This command is used to adjust the PWM waveform period of PWM_OUT. The PWM period is calculated by the following equation.

$$fpwm_out = \frac{18\ MHz}{(PWM_DIV[7:0]+1) \times 255}$$

PWM_DIV [7:0]								f_{PWM_OUT}
D7	D6	D	D4	D3	D2	D1	D0	
0	0	0	0	0	0	0	0	70.58 KHz
0	0	0	0	0	0	0	1	35.29 KHz
0	0	0	0	0	0	1	0	23.53 KHz
0	0	0	0	0	0	1	1	17.65 KHz
0	0	0	0	0	1	0	0	14.12 KHz
:								:
1	1	1	1	1	0	1	1	280.1 Hz
1	1	1	1	1	1	0	0	279.0 Hz
1	1	1	1	1	1	0	1	277.9 Hz
1	1	1	1	1	1	1	0	276.8 Hz
1	1	1	1	1	1	1	1	275.7 Hz



Note: The output frequency tolerance of internal frequency divider in the CABC is ±10%

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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Default	

Status	Default Value
Power ON Sequence	8'hE8
S/W Reset	No Change
H/W Reset	8'hE8

4.3.17. Backlight Control 2 (C9h)

C9h	BLCTRL2 (Backlight Control 2)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	0	1	C9h
1st Parameter	1	1	↑	XX	THRES_MOV [3:0]				THRES_STILL [3:0]				XX

Description

THRES_MOV [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data = 63) in the total pixels by image process in the Moving Image mode. After this parameter sets the number of pixels that makes display image white, then the threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter will not be changed.

THRES_MOV [3:0]				Description
D3	D2	D1	D0	
0	0	0	0	99 %
0	0	0	1	98 %
0	0	1	0	96 %
0	0	1	1	94 %
0	1	0	0	92 %
0	1	0	1	90 %
0	1	1	0	88 %
0	1	1	1	86 %

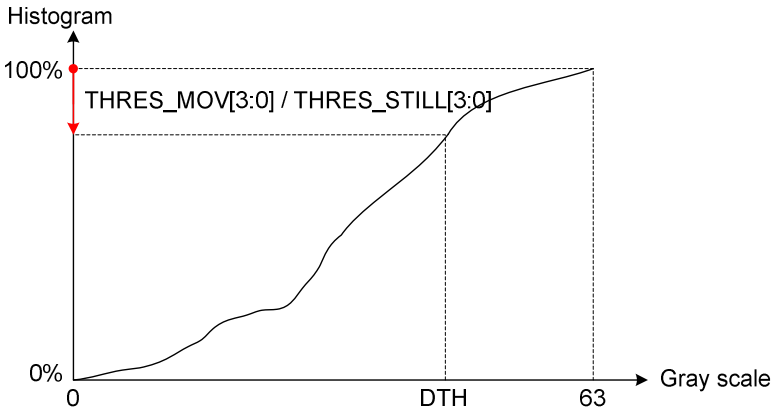
THRES_MOV [3:0]				Description
D3	D2	D1	D0	
1	0	0	0	84 %
1	0	0	1	82 %
1	0	1	0	80 %
1	0	1	1	78 %
1	1	0	0	76 %
1	1	0	1	74 %
1	1	1	0	72 %
1	1	1	1	70 %

THRES_STILL [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data = 63) in the total pixels by image process in the Still Picture mode. After this parameter sets the number of pixels that makes display image white, then the threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter will not be changed.

THRES_STILLI [3:0]				Description
D3	D2	D1	D0	
0	0	0	0	99 %
0	0	0	1	98 %
0	0	1	0	96 %
0	0	1	1	94 %
0	1	0	0	92 %
0	1	0	1	90 %
0	1	1	0	88 %
0	1	1	1	86 %

THRES_STILL [3:0]				Description
D3	D2	D1	D0	
1	0	0	0	84 %
1	0	0	1	82 %
1	0	1	0	80 %
1	0	1	1	78 %
1	1	0	0	76 %
1	1	0	1	74 %
1	1	1	0	72 %
1	1	1	1	70 %

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<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Restriction</td><td>To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
<tr><td>Register
Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
<tr><td>Default</td><td>
<table>
<tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr>
<tr><th>THRES_MOV [3:0]</th><th>THRES_STILL [3:0]</th></tr>
<tr><td>Power ON Sequence</td><td>4’b1011</td><td>4’b1011</td></tr>
<tr><td>S/W Reset</td><td>4’b1011</td><td>4’b1011</td></tr>
<tr><td>H/W Reset</td><td>4’b1011</td><td>4’b1011</td></tr>
</table>
</td></tr>
</table>

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4.3.18. Backlight Control 3 (CAh)

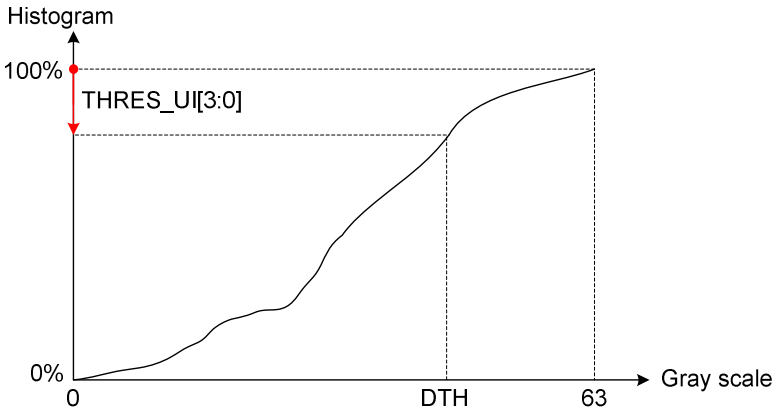
CAh	BLCTRL3 (Backlight Control 3)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	1	0	CAh
1st Parameter	1	1	↑	XX	0	0	0	0	THRES_UI [3:0]				XX

Description

THRES_UI [3:0]: This parameter is used to set the ratio (percentage) of the maximum number of pixels that makes display image white (data = 63) in the total pixels by image process in the UI (User Interface Image mode). After this parameter sets the number of pixels that makes display image white, then the threshold grayscale value (DTH) that makes display image white is set so that the number of the pixels set by this parameter will not be changed.

THRES_UI [3:0]				Description
D3	D2	D1	D0	
0	0	0	0	99 %
0	0	0	1	98 %
0	0	1	0	96 %
0	0	1	1	94 %
0	1	0	0	92 %
0	1	0	1	90 %
0	1	1	0	88 %
0	1	1	1	86 %

THRES_UI [3:0]				Description
D3	D2	D1	D0	
1	0	0	0	84 %
1	0	0	1	82 %
1	0	1	0	80 %
1	0	1	1	78 %
1	1	0	0	76 %
1	1	0	1	74 %
1	1	1	0	72 %
1	1	1	1	70 %



Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

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Register Availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	

Status	Default Value
Power ON Sequence	4'b1011
S/W Reset	4'b1011
H/W Reset	4'b1011

4.3.19. Backlight Control 4 (CBh)

CBh	BLCTRL4 (Backlight Control 4)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	0	1	1	CBh
1st Parameter	1	1	↑	XX	DTH_MOV [3:0]				DTH_STILL [3:0]				XX

Description

DTH_MOV [3:0]: This parameter is used to set the minimum limitation of the grayscale threshold value in the Moving Image mode.

DTH_MOV [3:0]				Description
D3	D2	D1	D0	
0	0	0	0	224
0	0	0	1	220
0	0	1	0	216
0	0	1	1	212
0	1	0	0	208
0	1	0	1	204
0	1	1	0	200
0	1	1	1	196

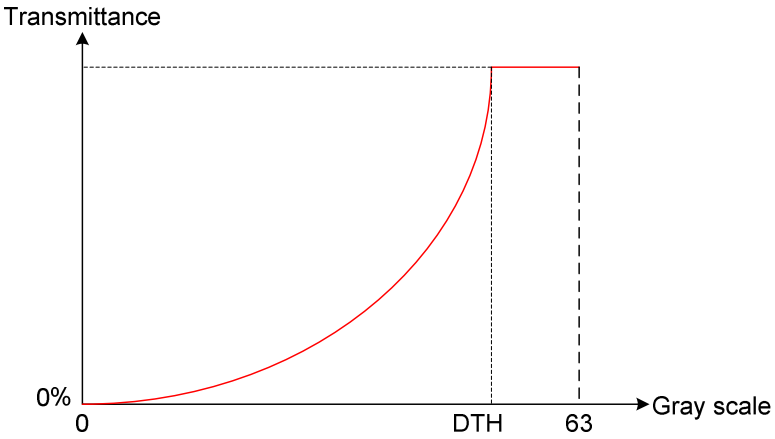
DTH_MOV [3:0]				Description
D3	D2	D1	D0	
1	0	0	0	192
1	0	0	1	188
1	0	1	0	184
1	0	1	1	180
1	1	0	0	176
1	1	0	1	172
1	1	1	0	168
1	1	1	1	164

DTH_STILL [2:0]: This parameter is used to set the minimum limitation of the grayscale threshold value in the Still Picture mode.

DTH_STILL [3:0]				Description
D3	D2	D1	D0	
0	0	0	0	224
0	0	0	1	220
0	0	1	0	216
0	0	1	1	212
0	1	0	0	208
0	1	0	1	204
0	1	1	0	200
0	1	1	1	196

DTH_STILL [3:0]				Description
D3	D2	D1	D0	
1	0	0	0	192
1	0	0	1	188
1	0	1	0	184
1	0	1	1	180
1	1	0	0	176
1	1	0	1	172
1	1	1	0	168
1	1	1	1	164

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Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

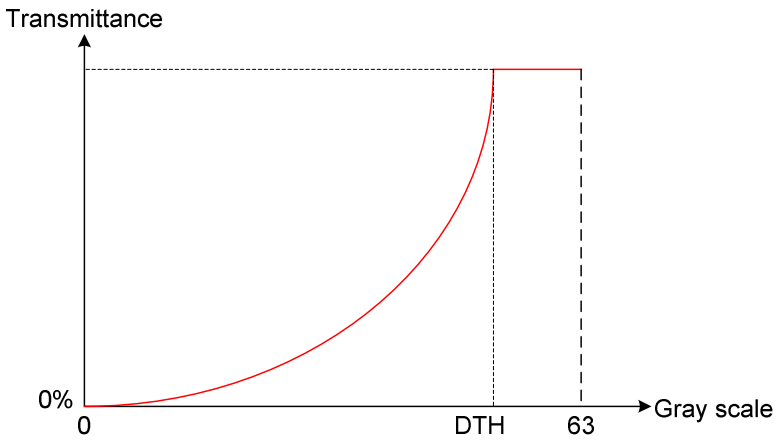
Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	DTH_MOV [3:0]	DTH_STILL [3:0]
Power ON Sequence	4'b1010	4'b1000
S/W Reset	4'b1010	4'b1000
H/W Reset	4'b1010	4'b1000

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4.3.20. Backlight Control 5 (CCh)

<table>
<tr><th>CCh</th><th colspan="13">BLCTRL5 (Backlight Control 5)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>CCh</td></tr>
<tr><td>1st Parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>0</td><td colspan="3">DTH_UI [3:0]</td><td>XX</td></tr>
<tr><td>Description</td><td colspan="13">DTH_UI [3:0]: This parameter is used to set the minimum limitation of the grayscale threshold value in the UI (User Interface Image mode).
<table>
<tr><th colspan="4">DTH_UI [3:0]</th><th rowspan="2">Description</th></tr>
<tr><th>D3</th><th>D</th><th>D1</th><th>D0</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>252</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>248</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>244</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>240</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>236</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>232</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>228</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>224</td></tr>
</table>
<table>
<tr><th colspan="4">DTH_UI [3:0]</th><th rowspan="2">Description</th></tr>
<tr><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>220</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>216</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>212</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>208</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>204</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>200</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>196</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>192</td></tr>
</table>

</td></tr>
<tr><td>Restriction</td><td colspan="13">To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
</table>

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Register Availability	

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default	

Status	Default Value
Power ON Sequence	4'b0100
S/W Reset	4'b0100
H/W Reset	4'b0100

4.3.21. Backlight Control 6 (CDh)

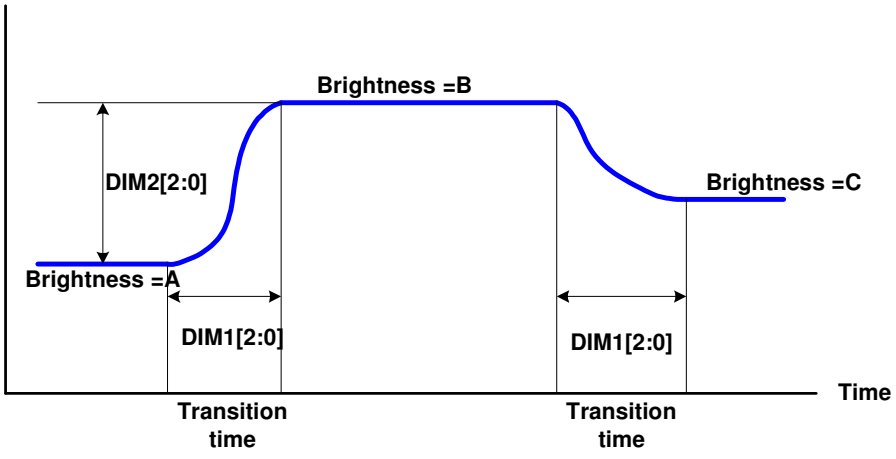
CDh	BLCTRL6 (Backlight Control 6)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	1	0	1	CDh
1st Parameter	1	1	↑	XX	0	DIM_MOV [2:0]			0	DIM_STILL [2:0]			XX

Description

DIM_STILL [2:0]: This parameter is used to set the transition time of the brightness level change to avoid the sharp brightness change in the Still Picture mode.

DIM_MOV [2:0]: This parameter is used to set the transition time of the brightness level change to avoid the sharp brightness change in the Moving Image mode.

DIM_MOV [2:0]/DIM_STILL [2:0]			Description
D2	D1	D0	
0	0	0	1 frame
0	0	1	1 frame
0	1	0	2 frames
0	1	1	4 frames
1	0	0	8 frames
1	0	1	16 frames
1	1	0	32 frames
1	1	1	64 frames



Note: In the above picture, **DIM1 [2:0]** means **DIM_MOV [2:0]**, **DIM_STILL [2:0],** and **DIM_UI [2:0]** are in different modes.

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Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Partial Mode On, Idle Mode Off, Sleep Out: Yes Partial Mode On, Idle Mode On, Sleep Out: Yes Sleep In: Yes
Default	Status / Default Value (DIM_MOV [2:0], DIM_STILL [2:0]) Power ON Sequence: 3'b100, 3'b011 S/W Reset: 3'b100, 3'b011 H/W Reset: 3'b100, 3'b011

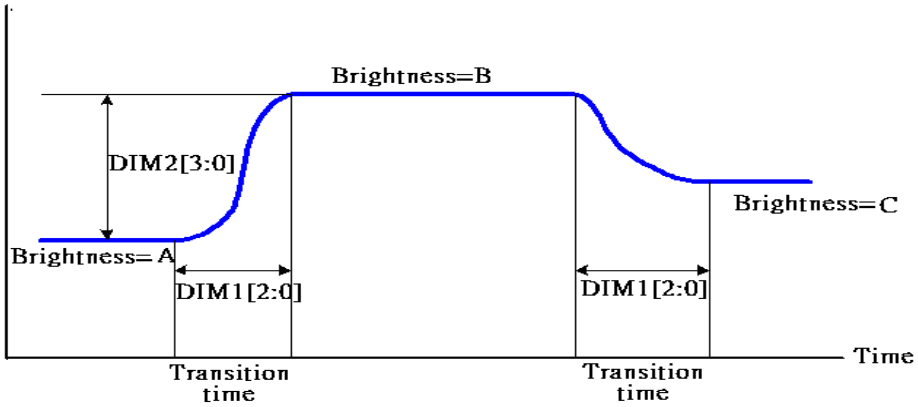
4.3.22. Backlight Control 7(CEh)

CEh	BLCTRL7 (Backlight Control 7)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	1	1	0	CEh
1st Parameter	1	1	↑	XX	DIM_MIN [3:0]				0	DIM_UI [2:0]			XX

Description

DIM_UI [2:0]: This parameter is used to set the transition time of the brightness level change to avoid the sharp brightness change in the UI mode (User Interface Image mode).

DIM_UI [2:0]			Description
D2	D1	D0	
0	0	0	1 frame
0	0	1	1 frame
0	1	0	2 frames
0	1	1	4 frames
1	0	0	8 frames
1	0	1	16 frames
1	1	0	32 frames
1	1	1	64 frames



Note1: In the above picture, **DIM1 [2:0]** means **DIM_MOV [2:0]**, **DIM_STILL [2:0]**, and **DIM_UI [2:0]** are in different modes.

Note2: In the above picture, **DIM2 [3:0]** means **DIM_MIN [3:0]**.

DIM_MIN [3:0]: This parameter is used to set the imitation of the minimum brightness change. If this parameter is larger than the difference between target brightness and current brightness, then the brightness will not change.

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Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Partial Mode On, Idle Mode Off, Sleep Out: Yes Partial Mode On, Idle Mode On, Sleep Out: Yes Sleep In: Yes
Default	Status / Default Value (DIM_MIN [3:0] / DIM_UI [2:0]) Power ON Sequence: 4'b0000 / 3'b010 S/W Reset: 4'b0000 / 3'b010 H/W Reset: 4'b0000 / 3'b010

4.3.23. Backlight Control 8 (CFh)

CFh	BLCTRL8 (Backlight Control 8)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	0	1	1	1	1	CFh
1st Parameter	1	1	↑	XX	0	0	0	0	0	LEDONR	LEDONPOL	PWMPOL	XX

Description

PWMPOL: The bit is used to define the polarity of the LEDPWM signal.

BL	LEDPWMPOL	LEDPWM pin
0	0	Always low
0	1	Always high
1	0	Original polarity of PWM signal
1	1	Inversed polarity of PWM signal

LEDONPOL: This bit is used to control the LEDON pin.

BL	LEDONPOL	LEDON pin
0	0	0
0	1	1
1	0	LEDONR
1	1	Inversed LEDONR

LEDONR: This bit is used to control the LEDON pin.

LEDONR	Description
0	Low
1	High

Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out: Yes Normal Mode On, Idle Mode On, Sleep Out: Yes Partial Mode On, Idle Mode Off, Sleep Out: Yes Partial Mode On, Idle Mode On, Sleep Out: Yes Sleep In: Yes

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Default	

Status	Default Value		
	LEDONR	LEDONPOL	PWMPOL
Power On Sequence	1'b0	1'b0	1'b0
H/W Reset	1'b0	1'b0	1'b0

4.3.24. NV Memory Write (D0h)

D0h	NVMWR (NV Memory Write)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	0	0	0	0	D0h
1st Parameter	1	1	↑	XX	PGM_DATA [7:0]								XX
2nd Parameter	1	1	↑	XX	PGM_ADR [7:0]								XX

Description

This command is used to program or read the NV memory data.

After a successful OTP operation, the information of PGM_DATA [7:0] will be programmed to the NV memory.

PGM_DATA [7:0]: The programmed data.

PGM_ADR [7:0]: Set the address of the NV memory for programming data. See "NV Memory Programming flow".

PGM_ADR [7:0]	Programming data
8'h00	ID1
8'h 01	ID2
8'h 02	ID3
8'h 03	VCM
8'h 3B	RB9h parameter 1
8'h 3C	RB9h parameter 2

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	
	PGM_ADR [7:0]/NVM_Read_ADR [7:0]	PGM_DATA [7:0]
Power ON Sequence	8'h00	8'h00
S/W Reset	No change	No change
H/W Reset	8'h00	8'h00

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4.3.25. NV Memory Protection Key (D1h)

D1h	NVMPKEY (NV Memory Protection Key)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	0	0	0	1	D1h
1st Parameter	1	1	↑	XX	KEY [23:16]								XX
2nd Parameter	1	1	↑	XX	KEY [15:8]								XX
3rd Parameter	1	1	↑	XX	KEY [7:0]								XX
Description	**KEY [23:0]:** NV memory programming protection key. Write an OTP data to D0h, this KEY [23:0] must set to 0x55AA66h to enable the OTP programming. If the KEY [23:0] is not 0 x 55AA66h, the NV Memory program will be aborted.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	24'h000000h
S/W Reset	24'h000000h

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4.3.26. NV Memory Status Read (D2h)

<table>
<tr><th>D2h</th><th colspan="13">RDNVM (NV Memory Status Read)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>D2h</td></tr>
<tr><td>1st Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>XX</td></tr>
<tr><td>2nd Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td>0</td><td colspan="4">ID2_Mk [2:0]</td><td>0</td><td colspan="3">ID1_ Mk [2:0]</td><td>XX</td></tr>
<tr><td>3rd Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td>OTP Busy</td><td colspan="4">VCM_Mk [2:0]</td><td>0</td><td colspan="3">ID3_ Mk [2:0]</td><td>XX</td></tr>
<tr><td>4th Parameter</td><td>1</td><td>↑</td><td>1</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>0</td><td colspan="2">RB9_P2_Mk</td><td colspan="2">RB9_P1_Mk</td><td>XX</td></tr>
<tr><td>Description</td><td colspan="13">
The register uses a mark to record the NV memory programmed time. The bits are increase “+1” automatically after writing the PGM_DATA [7:0] to the NV memory.
ID1_Mk [2:0]/ID2_Mk [2:0]/ID3_Mk [2:0]/VCM_Mk [2:0]:
<table>
<tr><th colspan="3">ID1_MK [2:0]/ID2_MK [2:0]/ID3_MK [2:0]/VCM_MK [2:0]</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>No Programmed</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>Programmed 1 time already</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>Programmed 2 times already</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>Programmed 3 times already</td></tr>
</table>
RB9_P1_Mk: Include MY_Panel, MX_Panel, MV_Panel, ML_Panel, BGR_Panel, MH_Panel,SS _Panel, GS_Panel bits
RB9_P2_Mk: Include REV_Panel, SM_Panel bits
<table>
<tr><th colspan="2">RB9_P1_Mk [2:0]/RB9_P1_Mk [2:0]</th><th>Description</th></tr>
<tr><td>0</td><td>0</td><td>No Programmed</td></tr>
<tr><td>0</td><td>1</td><td>Programmed 1 time already</td></tr>
<tr><td>1</td><td>1</td><td>Programmed 2 times already</td></tr>
</table>
OTP Busy: The status bit of the NV memory programming.
<table>
<tr><th>BUSY</th><th>The Status of NV Memory</th></tr>
<tr><td>0</td><td>Idle</td></tr>
<tr><td>1</td><td>Busy</td></tr>
</table>
</td></tr>
<tr><td>Restriction</td><td colspan="13">To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
</table>

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Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value						
	ID3_ Mk [2:0]	ID2_ Mk [2:0]	ID1_ Mk [2:0]	VCM_Mk [2:0]	OTP Busy	RB9_P1_MK	RB9_P2_MK
Power ON Sequence	3'b000	3'b000	3'b000	3'b000	0	2'b00	2'b00
S/W Reset	3'b000	3'b000	3'b000	3'b000	0	2'b00	2'b00

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4.3.27. Engineering Setting (DFh)

DFh	Engineering Setting												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	0	1	1	1	1	1	DFh
1st Parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
2nd Parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
3rd Parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
4th Parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
5th Parameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
6th Parameter	1	1	↑	XX	0	0	dsi_pclk_div	0	0	0	0	0	00
Description	MIPI internal PCLK setting. When dsi_pclk_div set 0 → Freq. = MIPI PCLK / 2 When dsi_pclk_div set 1 → Freq. = MIPI PCLK												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	dsi_pclk_div
Power ON Sequence	1'b0
S/W Reset	No Change
H/W Reset	1'b0

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4.3.28. Positive Gamma Control (E0h)

E0h	PGAMCTRL (Positive Gamma Control)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	0	0	0	0	E0h
1st Parameter	1	1	↑	XX	0	0	VP0 [5:0]						XX
2ndParameter	1	1	↑	XX	0	0	VP4 [5:0]						XX
3rdParameter	1	1	↑	XX	0	0	VP8 [5:0]						XX
4th Parameter	1	1	↑	XX	0	0	0	VP16 [4:0]					XX
5th Parameter	1	1	↑	XX	0	0	0	VP32 [4:0]					XX
6th Parameter	1	1	↑	XX	0	0	0	VP52 [4:0]					XX
7th Parameter	1	1	↑	XX	0	0	0	0	VP80 [3:0]				XX
8th Parameter	1	1	↑	XX	0	0	0	0	VP108 [3:0]				XX
9th Parameter	1	1	↑	XX	0	0	0	0	VP147 [3:0]				XX
10thParameter	1	1	↑	XX	0	0	0	0	VP175 [3:0]				XX
11thParameter	1	1	↑	XX	0	0	0	VP203 [4:0]					XX
12thParameter	1	1	↑	XX	0	0	0	VP223 [4:0]					XX
13thParameter	1	1	↑	XX	0	0	0	VP239 [4:0]					XX
14thParameter	1	1	↑	XX	0	0	VP247 [5:0]						XX
15thParameter	1	1	↑	XX	0	0	VP251 [5:0]						XX
16thParameter	1	1	↑	XX	0	0	VP255 [5:0]						XX
Description	Set the gray scale voltage to adjust the Gamma characteristics of the TFT panel.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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4.3.29. Negative Gamma Correction (E1h)

E1h	NGAMCTRL (Negative Gamma Correction)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	0	0	0	1	E1h
1st Parameter	1	1	↑	XX	0	0	VN0 [5:0]						XX
2ndParameter	1	1	↑	XX	0	0	VN4 [5:0]						XX
3rdParameter	1	1	↑	XX	0	0	VN8 [5:0]						XX
4th Parameter	1	1	↑	XX	0	0	0	VN16 [4:0]					XX
5th Parameter	1	1	↑	XX	0	0	0	VN32 [4:0]					XX
6th Parameter	1	1	↑	XX	0	0	0	VN52 [4:0]					XX
7th Parameter	1	1	↑	XX	0	0	0	0	VN80 [3:0]				XX
8th Parameter	1	1	↑	XX	0	0	0	0	VN108 [3:0]				XX
9th Parameter	1	1	↑	XX	0	0	0	0	VN147 [3:0]				XX
10thParameter	1	1	↑	XX	0	0	0	0	VN175 [3:0]				XX
11thParameter	1	1	↑	XX	0	0	0	VN203 [4:0]					XX
12thParameter	1	1	↑	XX	0	0	0	VN223 [4:0]					XX
13thParameter	1	1	↑	XX	0	0	0	VN239 [4:0]					XX
14thParameter	1	1	↑	XX	0	0	VN247 [5:0]						XX
15thParameter	1	1	↑	XX	0	0	VN251 [5:0]						XX
16thParameter	1	1	↑	XX	0	0	VN255 [5:0]						XX
Description	Set the gray scale voltage to adjust the Gamma characteristics of the TFT panel.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mo e OFF, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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4.3.30. Digital Gamma Control 1 (E2h)

E2h	DGAMCTRL (Digital Gamma Control 1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	0	0	1	0	E2h
1st Parameter	1	1	↑	XX	RCA0 [3:0]				BCA0 [3:0]				XX
:	1	1	↑	XX	RCAx [3:0]				BCAx [3:0]				XX
64th Parameter	1	1	↑	XX	RCA63 [3:0]				BCA63 [3:0]				XX
Description	**RCAx [3:0]:** Gamma Macro-adjustment registers for red Gamma curve. **BCAx [3:0]:** Gamma Macro-adjustment registers for blue Gamma curve. See the 3 Gamma setting flow in the chapter 16. Note：3 Gamma setting only support MIPI Low Power mode operation.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Partial Mode On, Idle Mode Off, Sleep Out / Yes Partial Mode On, Idle Mode On, Sleep Out / Yes Sleep In / Yes												

4.3.31. Digital Gamma Control 2 (E3h)

E3h	DGAMCTRL (Digital Gamma Control 2)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	0	0	1	1	E3h
1st Parameter	1	1	↑	XX	RFA0 [3:0]				BFA0 [3:0]				XX
:					:				:				XX
xthParameter	1	1	↑	XX	RFAx [3:0]				BFAx [3:0]				XX
:					:				:				XX
256thParameter	1	1	↑	XX	RFA256 [3:0]				BFA256 [3:0]				XX
257thParameter	1	1	↑	XX	0	0	0	0	0	0	0	0	00
Description	**RFAx [3:0]:** Gamma Micro-adjustment registers for red Gamma curve. **BFAx [3:0]:** Gamma Micro-adjustment registers for blue Gamma curve. 257th Parameter enables the register E3h setting value. See the 3 Gamma setting flow in the chapter 16. Note：3 Gamma setting only support MIPI Low Power mode operation.												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability	Status / Availability Normal Mode On, Idle Mode Off, Sleep Out / Yes Normal Mode On, Idle Mode On, Sleep Out / Yes Partial Mode On, Idle Mode Off, Sleep Out / Yes Partial Mode On, Idle Mode On, Sleep Out / Yes Sleep In / Yes												

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4.3.32. Digital 3 Gamma Enable (EAh)

EAh	D3GE (Digital 3 Gamma Enable)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	1	0	1	0	EAh
1st Parameter	1	1	↑	XX	0	0	0	0	0	0	En_3G	Dith_en	00
Description	**En_3G:** 1: digital 3 gamma enable 0: digital 3 gamma disable **Dith_en:** 1: dithering function enable 0: dithering function disable												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	8'h00
S/W Reset	No Change
H/W Reset	8'h00

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4.3.33. VGMP / VGMN /VGSP / VGSN Voltage Measurement Set (EDh)

EDh	VGMP / VGMN / VGSP / VGSN Voltage Measurement Set												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	0	1	1	0	1	EDh
1st parameter	1	1	↑	XX	0	1	1	1	1	1	1	1	7F
2nd parameter	1	1	↑	XX	0	0	0	0	en_volt_reg	1	1	1	07

Description

en_volt_reg:

1: Enable VGMP / VGMN /VGSP / VGSN voltage to output

0: disable VGMP / VGMN /VGSP / VGSN voltage to output

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	en_volt_reg
Power ON Sequence	1'b0
S/W Reset	No Change
H/W Reset	1'b0

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4.3.34. Panel Timing Control 1 (F1h)

F1h	PTCTRL1 (Panel Timing Control 1)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	0	0	0	1	F1h
1st Parameter	1	1	↑	XX	0	0	1	0	1	0	0	1	29
2nd Parameter	1	1	↑	XX	Chopper_opt	Chopper_sel [1:0]		0	1	0	1	0	CA
3rd Parameter	1	1	↑	XX	0	0	Outsre	0	0	1	1	1	07

Description

Chopper_opt: Source Op-amp chopper function option

Chopper_sel [1:0]: Source Op-amp chopper function option

Chopper_opt	Chopper_sel [1]	Chopper_sel [0]	Description
0	0	0	chopper disable
0	0	1	1 frame chopper
0	1	0	2 frame chopper
0	1	1	3 frame chopper
1	0	0	1 line chopper (1 frame chopper polarity change)
1	0	1	2 line chopper (1 frame chopper polarity change)
1	1	0	1 line chopper (2 frame chopper polarity change)
1	1	1	2 line chopper (2 frame chopper polarity change)

Outsre: Increase SRC Opamp output driving ability. 0:normal driving, 1:double driving

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	1st Parameter = 8'h29 2nd Parameter = 8'hCA 3rd Parameter = 8'h07
S/W Reset	No Change
H/W Reset	1st Parameter = 8'h29 2nd Parameter = 8'hCA 3rd Parameter = 8'h07

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4.3.35. Panel Timing Control 2 (F2h)

F2h	PTCTRL2 (Panel Timing Control 2)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	0	0	1	0	F2h
1st Parameter	1	1	↑	XX	SDTI [1:0]		0	0	0	CRTI [2:0]			40
2nd Parameter	1	1	↑	XX	1	1	0	1	0	EQTI [2:0]			D2
3rd Parameter	1	1	↑	XX	0	creqpc	0	1	0	PCTI [2:0]			52
4th Parameter	1	1	↑	XX	0	0	1	0	1	0	SDTI [2]	0	2A

Description

SDTI [2:0]: Source SDT timing adjustment (time scale: internal op_clk).

The timing can be adjusted 0 to 7 time scales.

CRTI [2:0]: Source CR timing adjustment (time scale: internal op_clk).

The timing can be adjusted 0 to 7 time scales.

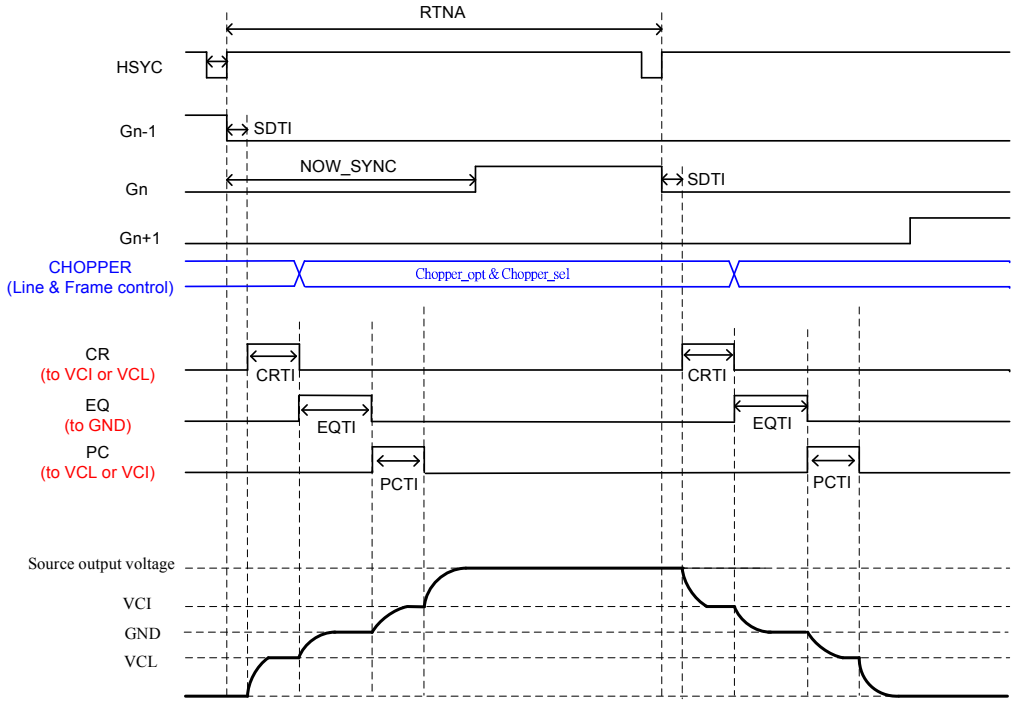
EQTI [2:0]: Source EQ timing adjustment (time scale: internal op_clk).

The timing can be adjusted 1 to 8 time scales.

PCTI [2:0]: Source PC timing adjustment (time scale: internal op_clk).

The timing can be adjusted 0 to 7 time scales.

creqpc: CR/EQ/PC function enable signal



Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

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<table>
<tr>
<td>Default</td>
<td>
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power ON Sequence</td><td>1st Parameter = 8'h40
2nd Parameter = 8'hD2
3rd Parameter = 8'h52
4th Parameter = 8'h2A</td></tr>
<tr><td>S/W Reset</td><td>No Change</td></tr>
<tr><td>H/W Reset</td><td>1st Parameter = 8'h40
2nd Parameter = 8'hD2
3rd Parameter = 8'h52
4th Parameter = 8'h2A</td></tr>
</table>
</td>
</tr>
</table>

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4.3.36. DVDD Voltage Setting (F3h)

F3h	DVDD Voltage Setting												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	0	0	1	1	F3h
1st parameter	1	1	↑	XX	0	1	1	reg_vd[2:0]			0	0	70
Description	reg_vd [2:0]: Select DVDD regulator voltage level												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												
Register Availability													
Default													

reg_vd[2:0]	DVDD Voltage(V)
3'h00	1.5 V
3'h01	setting prohibited
3'h02	setting prohibited
3'h03	setting prohibited
3'h04	1.6 V
3'h05	1.6 V
3'h06	1.7 V
3'h07	1.7 V

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Idle Mode Off, Sleep Out	Yes
Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Status	Default Value
	reg_vd[2:0]
Power ON Sequence	3'b100
S/W Reset	No Change
H/W Reset	3'b100

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4.3.37. Power Control 5 (F5h)

<table>
<tr><th>F5h</th><th colspan="12">PWCTRL 5 (Power Control 5)</th></tr>
<tr><td></td><td>DCX</td><td>RDX</td><td>WRX</td><td>D [23:8]</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>HEX</td></tr>
<tr><td>Command</td><td>0</td><td>1</td><td>↑</td><td>XX</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>F5h</td></tr>
<tr><td>1st parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>42h</td></tr>
<tr><td>2nd parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>04h</td></tr>
<tr><td>3rd parameter</td><td>1</td><td>1</td><td>↑</td><td>XX</td><td>1</td><td>0</td><td colspan="6">VRGH[5:0]</td><td>80h</td></tr>
<tr><td>Description</td><td colspan="13">VRGH [5:0]: Select AVDD regulator voltage level
<table>
<tr><th>VRGH[5:0]</th><th>Spec(V)</th></tr>
<tr><td>8'h00</td><td>1</td></tr>
<tr><td>8'h01</td><td>1.1</td></tr>
<tr><td>8'h02</td><td>1.2</td></tr>
<tr><td>8'h03</td><td>1.3</td></tr>
<tr><td>8'h04</td><td>1.4</td></tr>
<tr><td>8'h05</td><td>1.5</td></tr>
<tr><td>8'h06</td><td>1.6</td></tr>
<tr><td>8'h07</td><td>1.7</td></tr>
<tr><td>8'h08</td><td>1.8</td></tr>
<tr><td>8'h09</td><td>1.9</td></tr>
<tr><td>8'h0A</td><td>2</td></tr>
<tr><td colspan="2">…</td></tr>
<tr><td>8'h2F</td><td>5.7</td></tr>
<tr><td>8'h30</td><td>5.8</td></tr>
<tr><td>8'h31</td><td>5.9</td></tr>
<tr><td>8'h32</td><td>6</td></tr>
<tr><td>8'h33 → 8'h3F</td><td>6</td></tr>
</table></td></tr>
<tr><td>Restriction</td><td colspan="13">To enable this command, EXTC Command Set Enable Register (FFh) must be set first.</td></tr>
<tr><td>Register Availability</td><td colspan="13">
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Idle Mode Off, Sleep Out</td><td>Yes</td></tr>
<tr><td>Idle Mode On, Sleep Out</td><td>Yes</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table></td></tr>
<tr><td>Default</td><td colspan="13">
<table>
<tr><th>Status</th><th>Default Value</th></tr>
<tr><td>Power ON Sequence</td><td>3rd parameter = 8’h80</td></tr>
<tr><td>S/W Reset</td><td>No Change</td></tr>
<tr><td>H/W Reset</td><td>3rd parameter = 8’h80</td></tr>
</table></td></tr>
</table>

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4.3.38. Panel Resolution Selection Set (F7h)

F7h	Panel Resolution Selection Set												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	0	1	1	1	F7h
1st parameter	1	1	↑	XX	1	0	0	0	0	rso_in_test_mode[2:0]			80
Description	Rso_in_test_mode[2:0] : 000 : 480x864 resolution 001 : 480x854 resolution 010 : 480x800 resolution 011 : 480x640 resolution 100 : 480x720 resolution												
Restriction	To enable this command, EXTC Command Set Enable Register (FFh) must be set first.												

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power On Sequence	1st Parameter = 8'h80
S/W Reset	No Change
H/W Reset	1st Parameter = 8'h80

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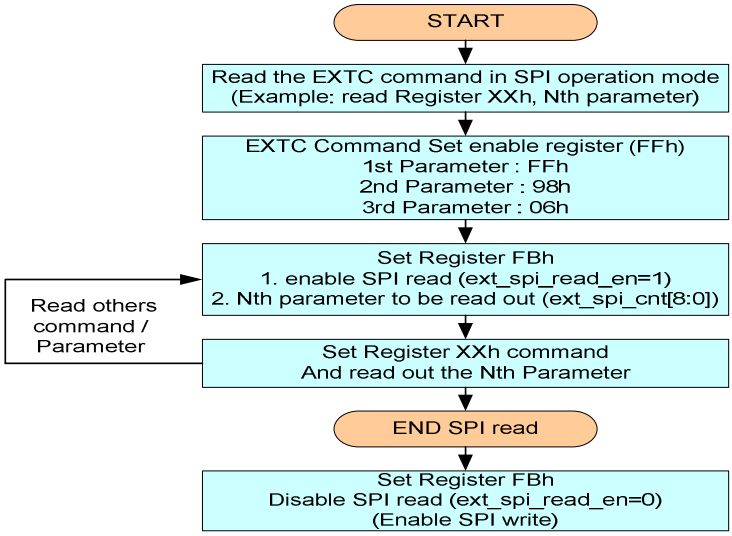
4.3.39. Read EXTC Command In SPI Mode (FBh)

FBh	RDEXTCSPI(Read EXTC command In SPI)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	1	0	1	1	FBh
1st Parameter	1	1	↑	XX	ext_spi_read_en	ext_spi_cnt [6:0]							XX
2nd Parameter	1	1	↑	XX	0	0	0	0	0	0	ext_spi_cnt [8:7]		XX

Description

ext_spi_read_en: enable the read function of the EXTC Command in the SPI operation mode

ext_spi_cnt [8:0]: the Nth parameter which wants to be read out



Restriction

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	1st Parameter = 8'h00 2nd Parameter = 8'h00
S/W Reset	1st Parameter = 8'h00 2nd Parameter = 8'h00
H/W Reset	1st Parameter = 8'h00 2nd Parameter = 8'h00

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4.3.40. LVGL Voltage Setting (FCh)

FCh	LVGL Voltage Setting												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	1	1	0	0	FCh
1st parameter	1	1	↑	XX	0	0	0	EXB1T_IN	LVGL_SEL[3:0]				04

Description

EXB1T_IN : Select the AVDD / AVEE power source.

EXB1T_IN bit = 0，AVDD / AVEE generated by internal pumping.

EXB1T_IN bit = 1，AVDD / AVEE generated by external pumping source(ex. ILI4002).

LVGL_SEL[3:0]: Select VGL regulator voltage level.

LVGL_SEL[3:0]	Spec (V)
4'h0	N/A
4'h1	N/A
4'h2	N/A
4'h3	N/A
4'h4	-6
4'h5	-7
4'h6	-8
4'h7	-9
4'h8	-10
4'h9	-11
4'hA	-12
4'hB	-13
4'hC	-14
4'hD	-15
4'hE	-16
4'hF	-17

LVGL>VGL+2V

Restriction: To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value	Default Value
	EXB1T_IN	LVGL_SEL[3:0]
Power On Sequence	1'b0	4'b0100
S/W Reset	No Change	No Change
H/W Reset	1'b0	4'b0100

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4.3.41. External Power Selection Set (FDh)

FCh	External Power Selection Set												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	1	1	0	1	FDh
1st parameter	1	1	↑	XX	0	0	0	0	1	0	1	0	0A
2nd parameter	1	1	↑	XX	0	0	pccs_reg[1:0]		0	0	0	0	00

Description

pccs_reg[1:0]: Select external driving mode.

pccs_reg[1]	pccs_reg[0]	Driving mode
0	0	Default
0	1	Prohibit
1	0	ILI4002
1	1	Prohibit

Restriction

To enable this command, EXTC Command Set Enable Register (FFh) must be set first.

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off, Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
	pccs_reg[1:0]
Power On Sequence	2'b00
S/W Reset	No Change
H/W Reset	2'b00

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4.3.42. EXTC Command Set enable register (FFh)

FFh	ENEXTC (EXTC command set enable register)												
	DCX	RDX	WRX	D [23:8]	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	0	1	↑	XX	1	1	1	1	1	1	1	1	FFh
1st Parameter	1	1	↑	XX	1	1	1	1	1	1	1	1	FFh
2nd Parameter	1	1	↑	XX	1	0	0	1	1	0	0	0	98h
3rd Parameter	1	1	↑	XX	0	0	0	0	0	1	1	0	06h
Description	Set the register, 1st Parameter = FFh, 2nd Parameter = 98h, 3rd Parameter = 06h to enable "EXTC command set". Set the register, 1st Parameter = FFh, 2nd Parameter = 98h, 3rd Parameter = any value except 06h to disable "EXTC command set".												
Restriction													

Register Availability

Status	Availability
Normal Mode On, Idle Mode Off Sleep Out	Yes
Normal Mode On, Idle Mode On, Sleep Out	Yes
Partial Mode On, Idle Mode Off, Sleep Out	Yes
Partial Mode On, Idle Mode On, Sleep Out	Yes
Sleep In	Yes

Default

Status	Default Value
Power ON Sequence	1st Parameter = 8'h00 2nd Parameter = 8'h00 3rd Parameter = 8'h00
S/W Reset	1st Parameter = 8'h00 2nd Parameter = 8'h00 3rd Parameter = 8'h00
H/W Reset	1st Parameter = 8'h00 2nd Parameter = 8'h00 3rd Parameter = 8'h00

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5. Source, Gate and Memory Map

5.1. Memory Map of the 480 (RGB) x 864 Resolution

For example: R36h MV bit (D5) = 0

Source Out		S1	S2	S3	S4	S5	S6	...	S1435	S1436	S1437	S1438	S1439	S1440		
		RGB=0	RGB=1		RGB=0	RGB=1		RGB Order	RGB=0	RGB=1		RGB=0	RGB=1			
RA															SA	
MY=0	MY=1														ML=0	ML=1
0	863	R0	G0	B0	R1	G1	B1	...	R478	G478	B478	R479	G479	B479	0	863
1	862							...							1	862
2	861							...							2	861
3	860							...							3	860
4	859							...							4	859
5	858							...							5	858
6	857							...							6	857
7	856							...							7	856
8	855							...							8	855
9	854							...							9	854
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
856	7							...							856	7
857	6							...							857	6
858	5							...							858	5
859	4							...							859	4
860	3							...							860	3
861	2							...							861	2
862	1							...							862	1
863	0							...							863	0
CA	MX=0	0			1			...	478			479				
	MX=1	479			478			...	1			0				

Figure 159: Memory Map of the 480 (RGB) x 864 Resolution, R36h MV bit D5 = 0

RA = Row Address

CA = Column Address

SA = Scan Address

MX = Colum Address direction parameter

MY = Row Address direction parameter

ML = Scan direction parameter

RGB = Red, Green and Blue pixel position change

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5.2. Memory Map of the 480 (RGB) x 854 Resolution

For example: R36h MV bit (D5) = 0

Source Out		S1	S2	S3	S4	S5	S6	...	S1435	S1436	S1437	S1438	S1439	S1440		
RA		RGB = 0 / RGB = 1						RGB Order							SA	
MY=0	MY=1														ML=0	ML=1
0	853	R0	G0	B0	R1	G1	B1	...	R478	G478	B478	R479	G479	B479	0	853
1	852							...							1	852
2	851							...							2	851
3	850							...							3	850
4	849							...							4	849
5	848							...							5	848
6	847							...							6	847
7	846							...							7	846
8	845							...							8	845
9	844							...							9	844
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
846	7							...							846	7
847	6							...							847	6
848	5							...							848	5
849	4							...							849	4
850	3							...							850	3
851	2							...							851	2
852	1							...							852	1
853	0							...							853	0
CA	MX=0	0			1			...	478			479				
	MX=1	479			478			...	1			0				

Figure 160: Memory Map of the 480 (RGB) x 854 Resolution, R36h MV bit D5 = 0

RA = Row Address

CA = Column Address

SA = Scan Address

MX = Colum Address direction parameter

MY = Row Address direction parameter

ML = Scan direction parameter

RGB = Red, Green and Blue pixel position change

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5.3. Memory Map of the 480 (RGB) x 800 Resolution

For example: R36h MV bit (D5) = 0

Source Out		S1	S2	S3	S4	S5	S6	...	S1435	S1436	S1437	S1438	S1439	S1440		
RA		RGB = 0		RGB = 1	RGB = 0		RGB = 1	RGB Order	RGB = 0		RGB = 1	RGB = 0		RGB = 1	SA	
MY=0	MY=1														ML=0	ML=1
0	799	R0	G0	B0	R1	G1	B1	...	R478	G478	B478	R479	G479	B479	0	799
1	798							...							1	798
2	797							...							2	797
3	796							...							3	796
4	795							...							4	795
5	794							...							5	794
6	793							...							6	793
7	792							...							7	792
8	791							...							8	791
9	790							...							9	790
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
792	7							...							792	7
793	6							...							793	6
794	5							...							794	5
795	4							...							795	4
796	3							...							796	3
797	2							...							797	2
798	1							...							798	1
799	0							...							799	0
CA	MX=0	0			1			...	478			479				
	MX=1	479			478			...	1			0				

Figure 161: Memory Map of the 480 (RGB) x 800 resolution, R36h MV bit D5 = 0

RA = Row Address

CA = Column Address

SA = Scan Address

MX = Colum Address direction parameter

MY = Row Address direction parameter

ML = Scan direction parameter

RGB = Red, Green and Blue pixel position change

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5.4. Memory Map of the 480 (RGB) x 640 Resolution

For example: R36h MV bit (D5) = 0

Source Out		S1	S2	S3	S4	S5	S6	...	S1435	S1436	S1437	S1438	S1439	S1440		
RA		RGB = 0		RGB = 1	RGB = 0		RGB = 1	RGB Order	RGB = 0		RGB = 1	RGB = 0		RGB = 1	SA	
MY=0	MY=1														ML=0	ML=1
0	639	R0	G0	B0	R1	G1	B1	...	R478	G478	B478	R479	G479	B479	0	639
1	638							...							1	638
2	637							...							2	637
3	636							...							3	636
4	635							...							4	635
5	634							...							5	634
6	633							...							6	633
7	632							...							7	632
8	631							...							8	631
9	630							...							9	630
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
632	7							...							632	7
633	6							...							633	6
634	5							...							634	5
635	4							...							635	4
636	3							...							636	3
637	2							...							637	2
638	1							...							638	1
639	0							...							639	0
CA	MX=0	0			1			...	478			479				
	MX=1	479			478			...	1			0				

Figure 162: Memory Map of the 480 (RGB) x 640 Resolution, R36h MV bit D5 = 0

RA = Row Address

CA = Column Address

SA = Scan Address

MX = Colum Address direction parameter

MY = Row Address direction parameter

ML = Scan direction parameter

RGB = Red, Green and Blue pixel position change

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6. Source Driver

The source driver uses 1440 channels (S1~S1440) are used for driving the source line of the TFT LCD panel. The source driver converts the digital data from the GRAM into the analog voltage and generates the corresponding gray scale voltage output, enabling up to 16.7M colors to be displayed simultaneously. The output circuit of this source driver incorporates with an operational amplifier, so that a positive and a negative voltage can be alternately outputted from each channel.

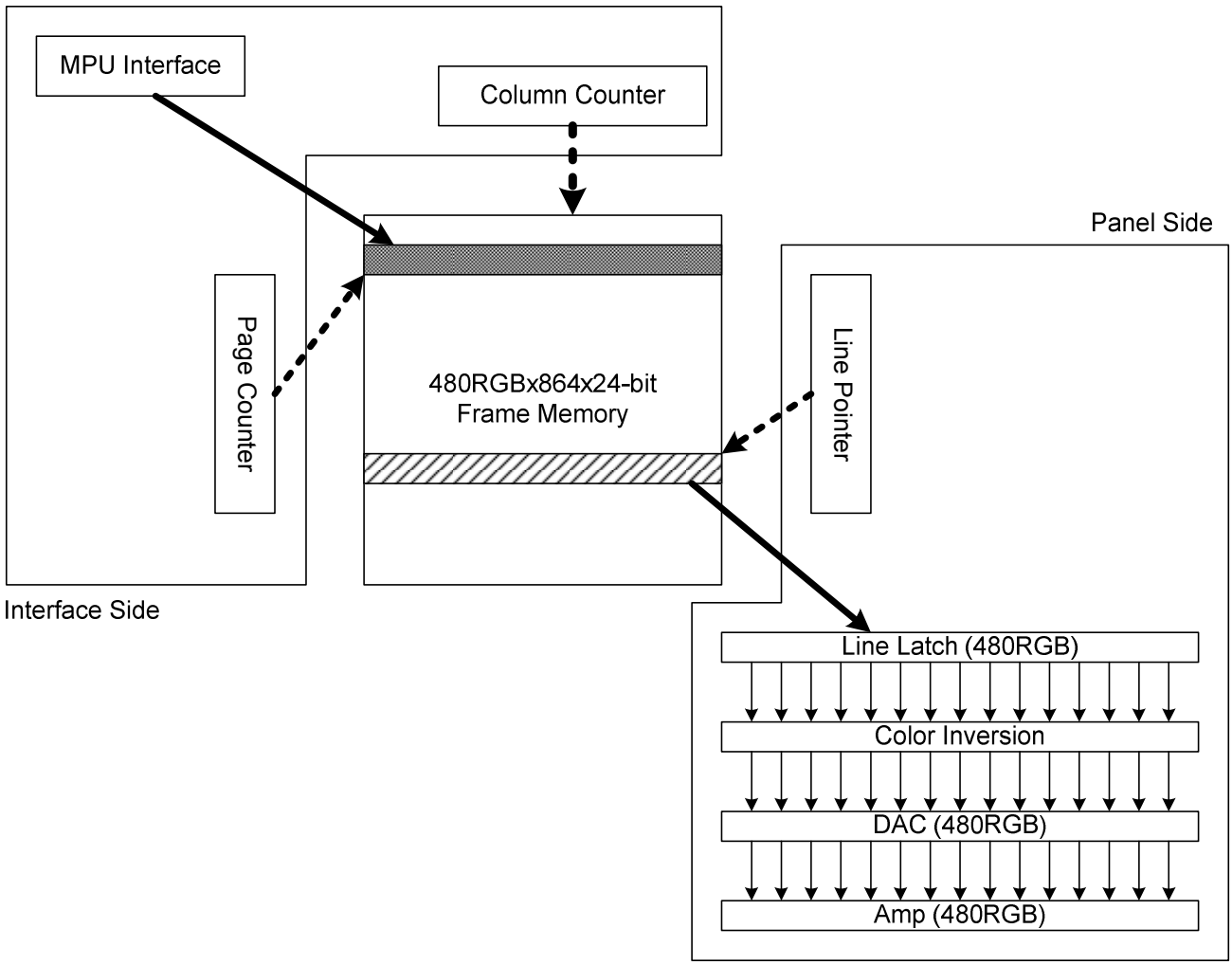
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7. Display Data RAM

7.1. Configuration

The 480 x 864 x 24-bit graphic type static RAM has a 1,244,160-byte memory, allowing the storage of a 480 (RGB)x 864 image with a 24-bit resolution.

Panel Read and Interface Read (or Interface Write) can occur simultaneously with the Frame Memory. This process will not cause any visible effects on the display.



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7.2. Memory to Display Address Mapping

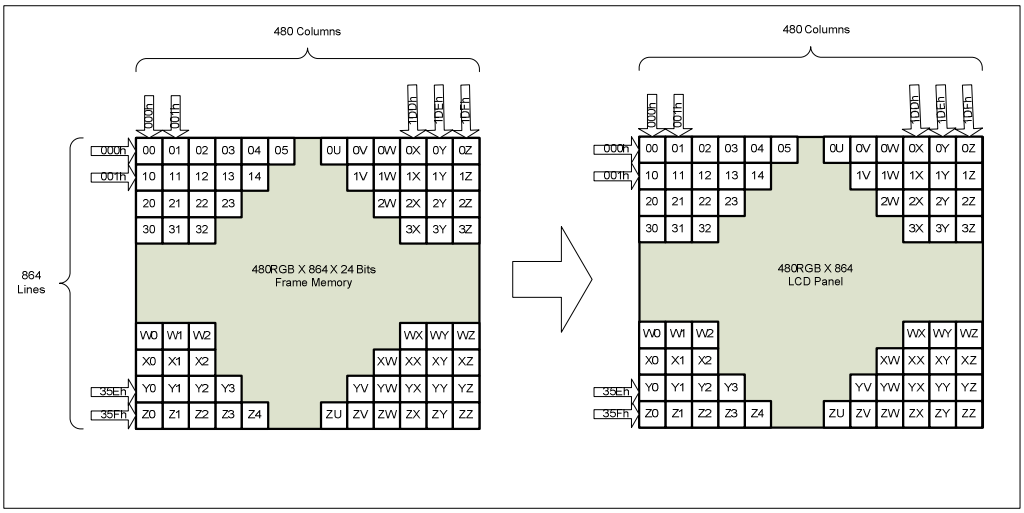
7.2.1. Fully Display

Example: 480 (RGB) x 864 dot display mode

(1) In this mode, the content of the Frame Memory within an area where the column pointer is 0000h to 01DFh and page pointer is 0000h to 035Fh is displayed.

(2) NORON (Normal Display Mode On) instruction (R13h)

(3) SC = 0 x 000h, EC = 0 x 1DFh (R2Ah) and SP = 0 x 000h, EP = 0 x 35Fh (R2Bh), ML = 0

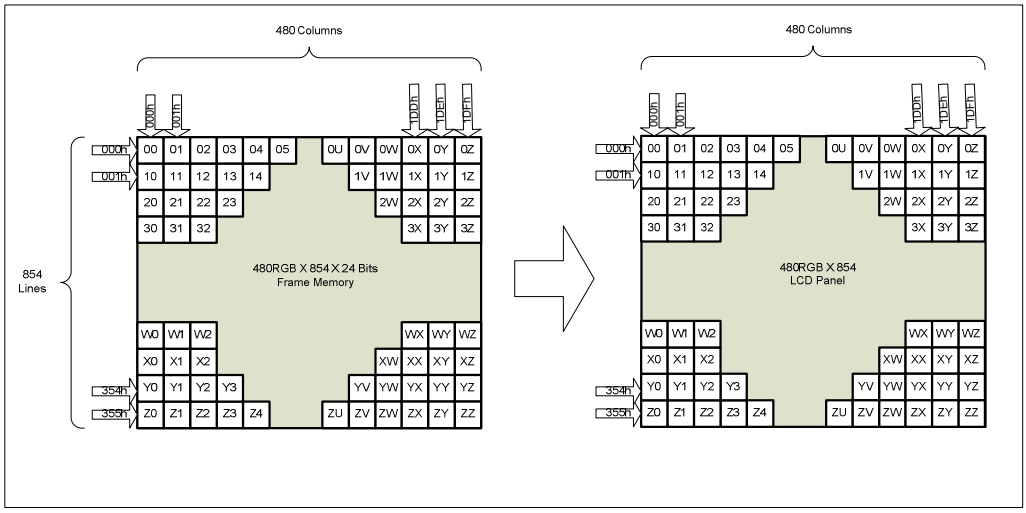


Example: 480 (RGB) x 854 dot display mode

(1) In this mode, the content of the Frame Memory within an area where the column pointer is 0000h to 01DFh and page pointer is 0000h to 0355h is displayed.

(2) NORON (Normal Display Mode On) instruction (R13h)

(3) SC = 0 x 000h, EC = 0 x 1DFh (R2Ah) and SP = 0 x 000h, EP = 0 x 355h (R2Bh), ML = 0



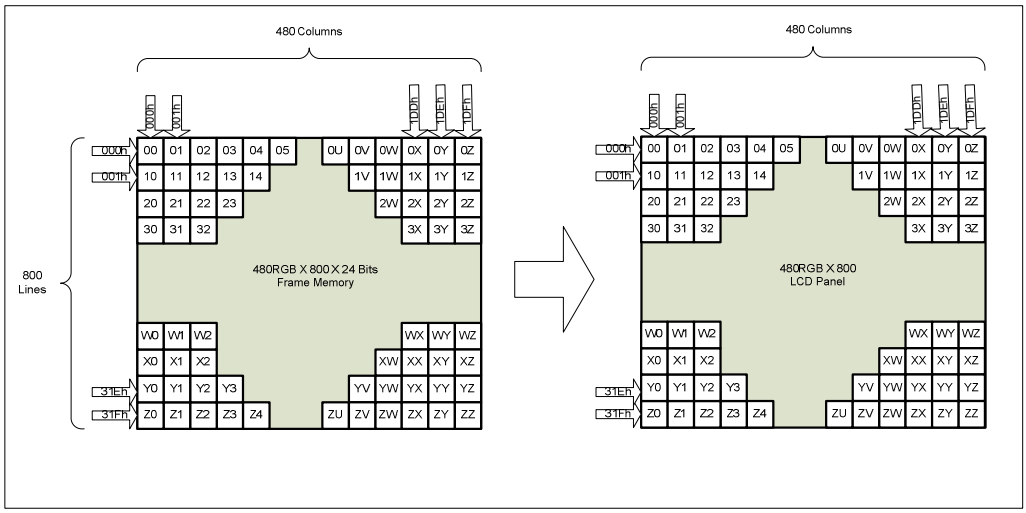
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Example: 480 (RGB) x 800 dot display mode

(1) In this mode, the content of the Frame Memory within an area where the column pointer is 0000h to 01DFh and page pointer is 0000h to 031Fh is displayed.

(2) NORON (Normal Display Mode On) instruction (R13h)

(3) SC = 0 x 000h, EC = 0 x 1DFh (R2Ah) and SP = 0 x 000h, EP = 0 x 31Fh (R2Bh), ML = 0

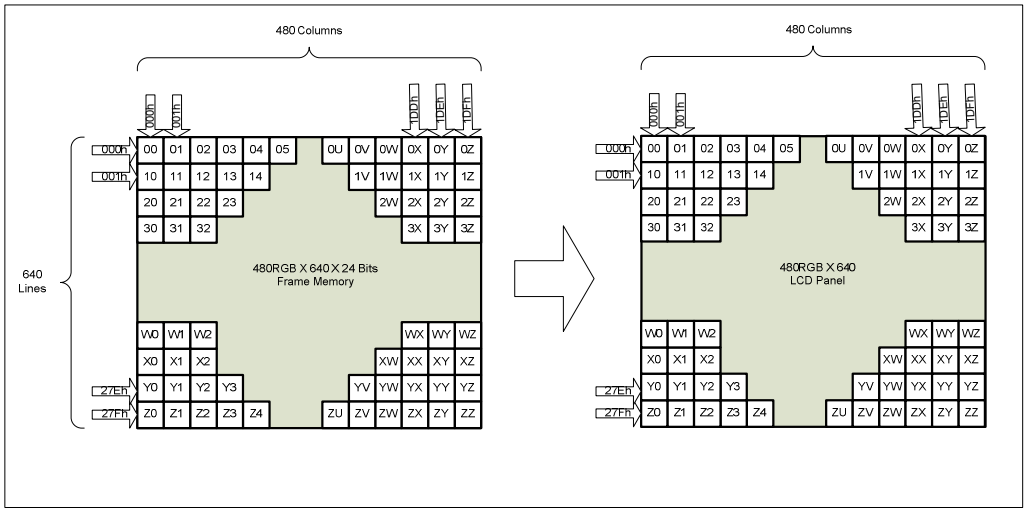


Example: 480 (RGB) x 640 dot display mode

(1) In this mode, the content of the Frame Memory within an area where the column pointer is 0000h to 01DFh and page pointer is 0000h to 027Fh is displayed.

(2) NORON (Normal Display Mode On) instruction (R13h)

(3) SC = 0 x 000h, EC = 0 x 1DFh (R2Ah) and SP = 0 x 000h, EP = 0 x 27Fh (R2Bh), ML = 0



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7.2.2. Vertical Scrolling Display

The vertical scrolling mode is determined by the commands “Vertical Scrolling Definition” (33h) and “Vertical Scrolling Start Address” (37h).

The Vertical Scroll Mode function is explained by the examples of (TFA+VSA+BFA) = 864 in Figure 163.

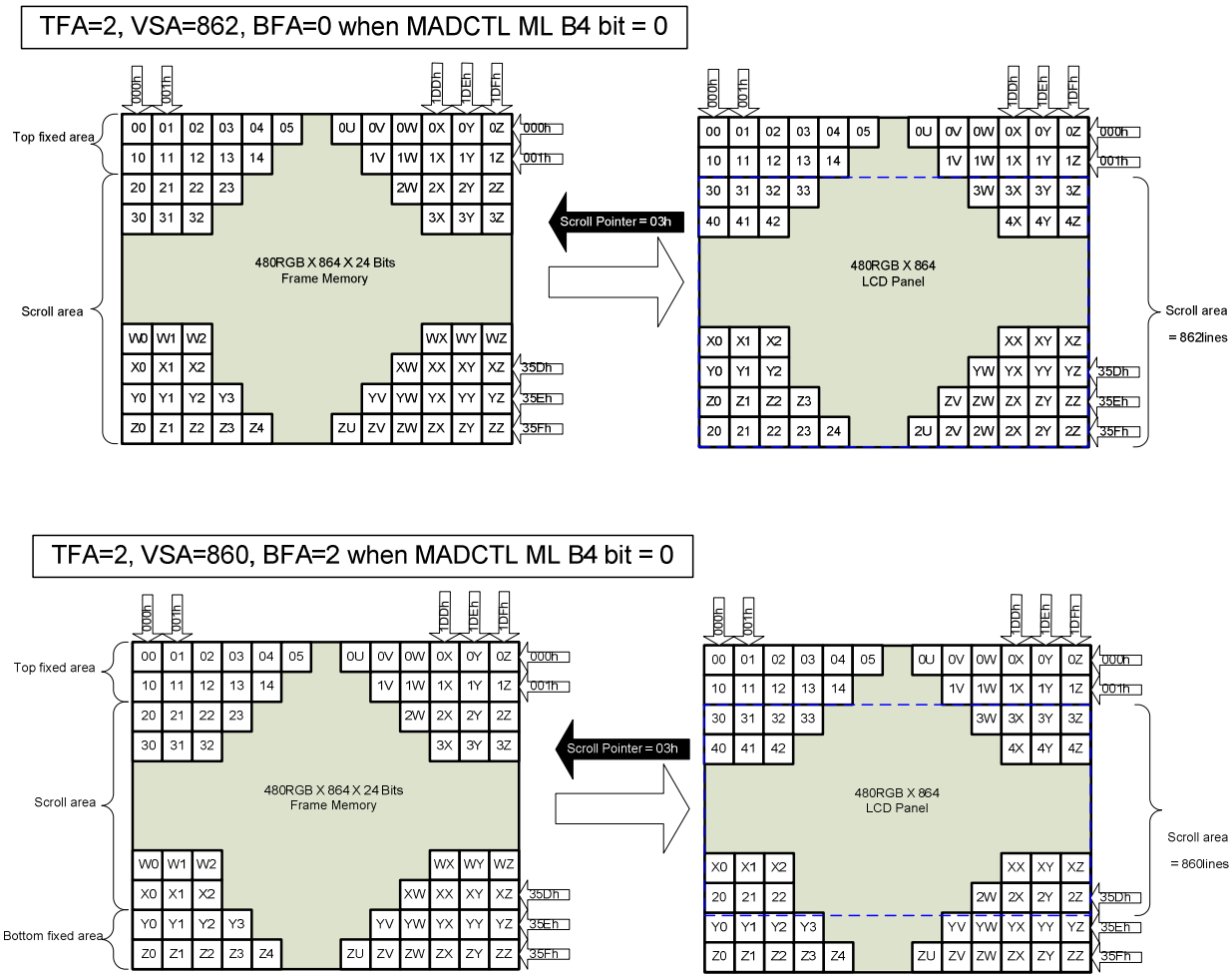


Figure 163: Vertical Scroll Mode Function

Note: When Vertical Scrolling Definition Parameters (TFA+VSA+BFA) ≠ 640, the Scrolling Mode is undefined.

7.2.3. Vertical Scroll 480(RGB) (H) X 864 (V) Example

7.2.4. Case1: TFA+VSA+BFA ≠ 864

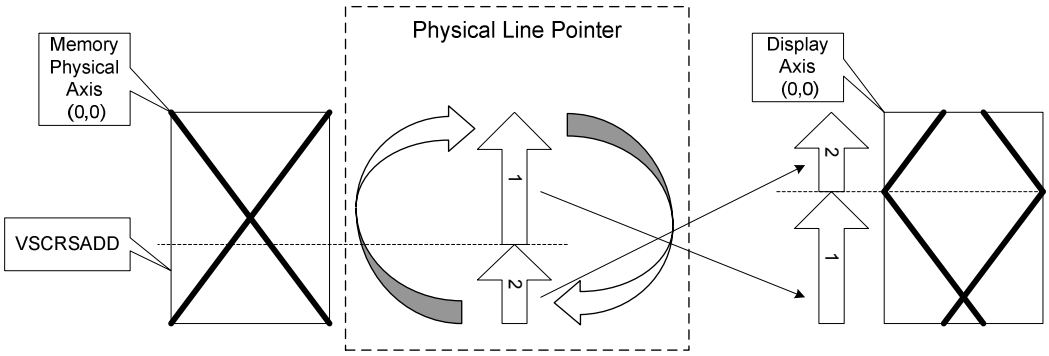
This setting is prohibited because it will cause a display output error.

7.2.5. Case2: TFA+VSA+BFA = 864 (Rolling Scrolling)

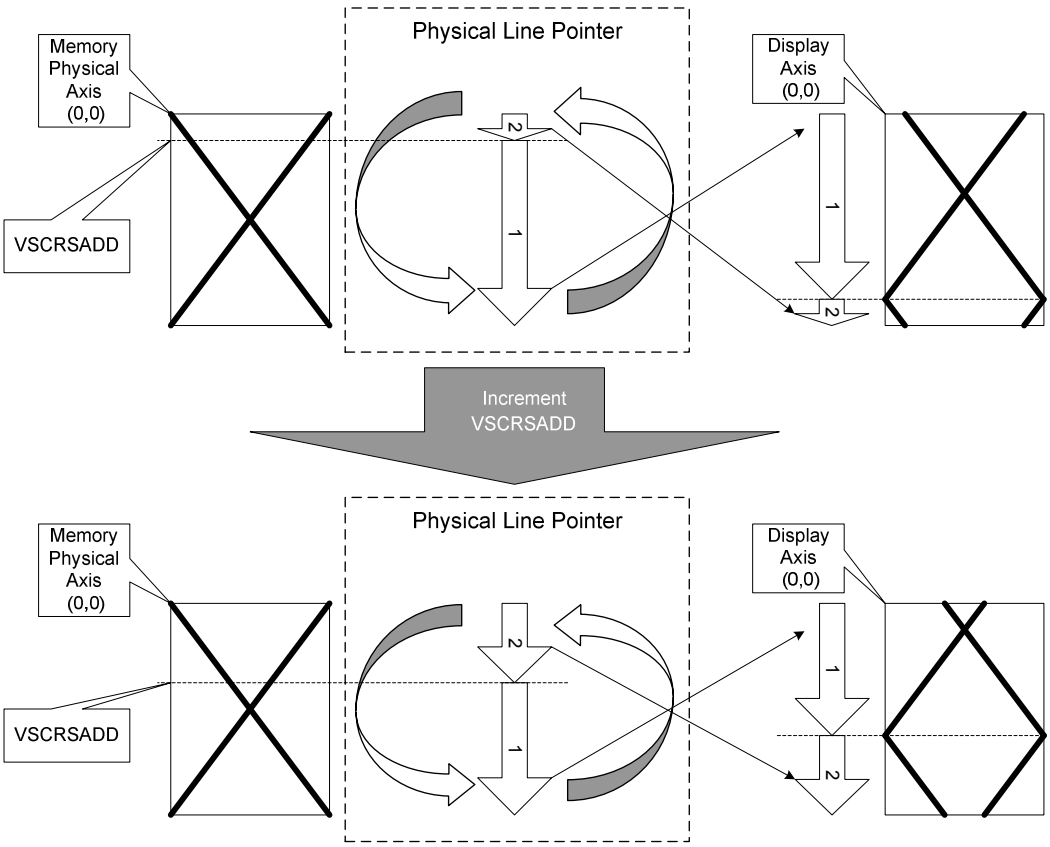
The operations of the Rolling Scrolling are explained by the examples below.

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When TFA=0, VSA=864, BFA=0, VSCRSADD=40 and MADCTL ML B4 bit = 1

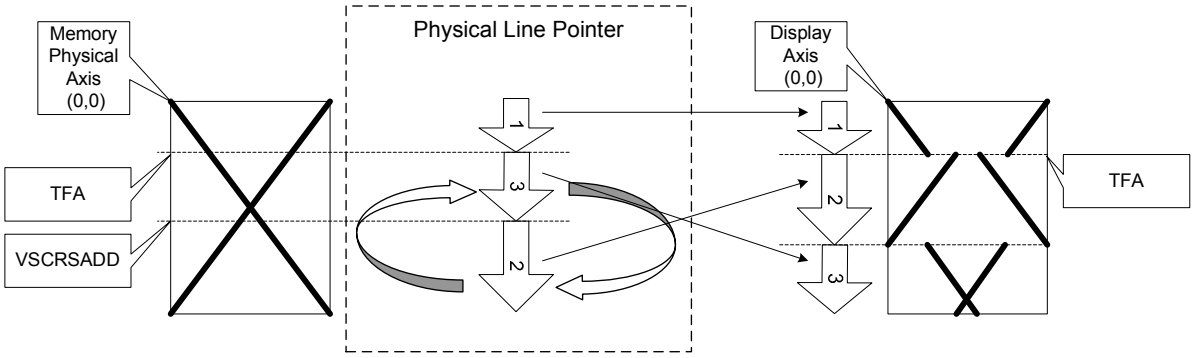


When TFA=0, VSA=864, BFA=0, VSCRSADD=40 and MADCTL ML bit = 0

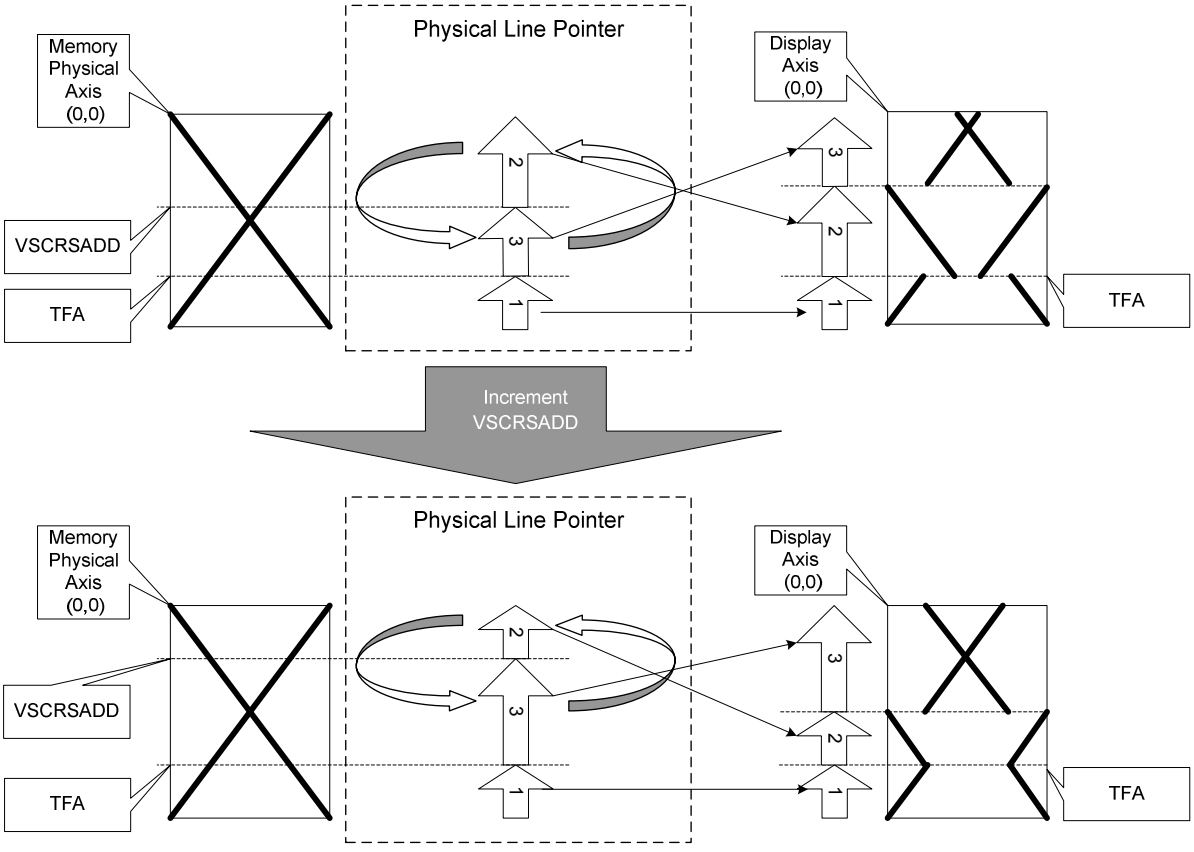


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When TFA=30, VSA=834, BFA=0, VSCRSADD=80 and MADCTL ML bit = 0

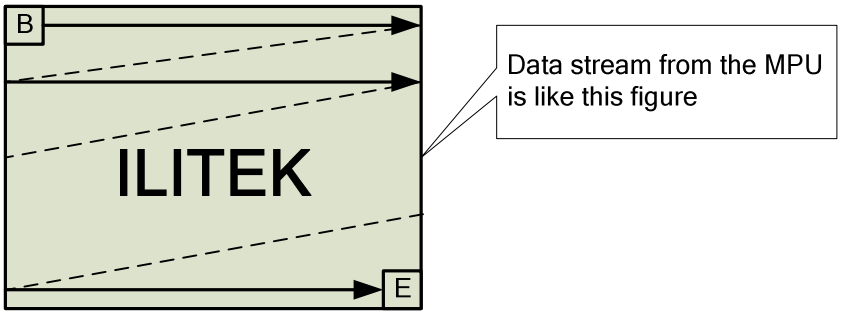


When TFA=30, VSA=834, BFA=0, VSCRSADD=80 and MADCTL ML bit = 1



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7.3. MPU to Memory Write/Read Direction



The data is written in the order illustrated above. Data is always written to the Frame Memory in the same order, regardless of the Memory Write Direction set by MADCTL bits D7, D6 and D5.

The write order for each pixel unit is shown below.

DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
R[7]	R[6]	R[5]	R4]	R[3]	R[2]	R[1]	R[0]	G[7]	G[6]	G[5]	G[4]	G[3]	G[2]	G[1]	G[0]	B[7]	B[6]	B[5]	B[4]	B[3]	B[2]	B[1]	B[0]

The Counter dictates where the data is to be written in the physical memory and is controlled by “Memory Data Access Control” Command, Bits D5, D6, and D7, as described in Figure 164.

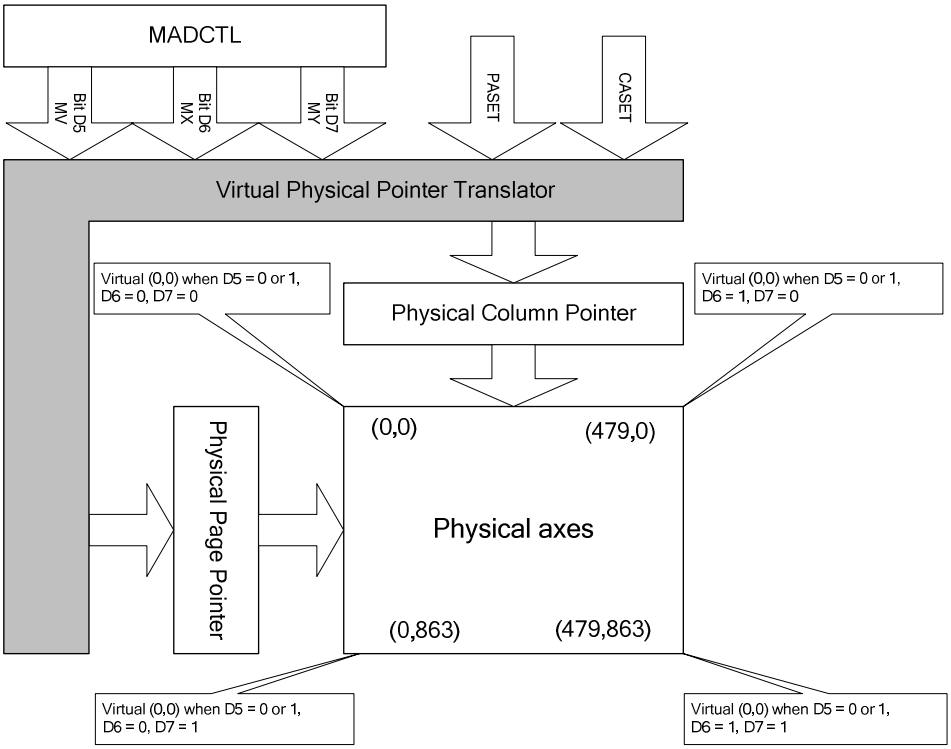


Figure 164: MV, MX, MY setting (480 (RGB) x 864)

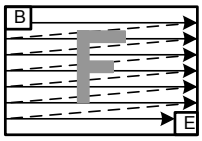
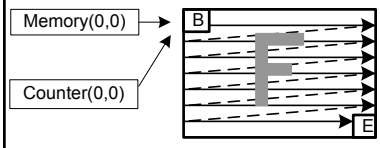
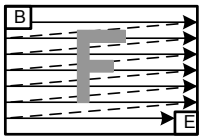
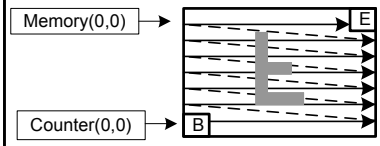
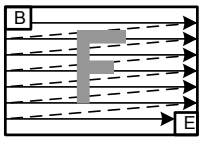
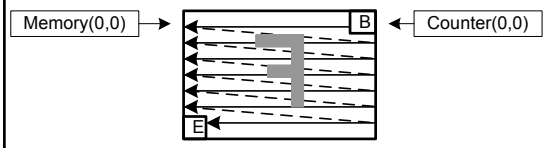
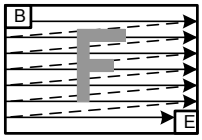
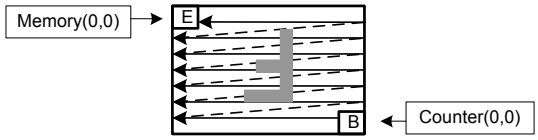
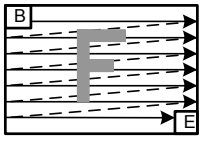
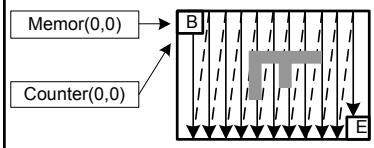
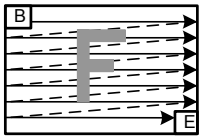
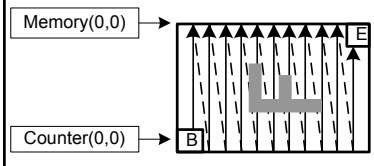
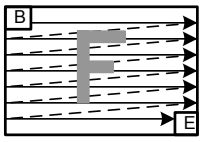
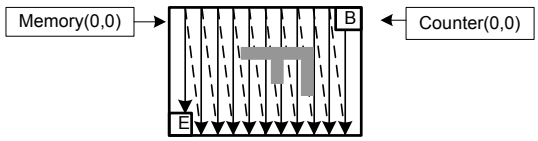
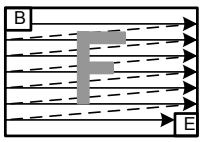
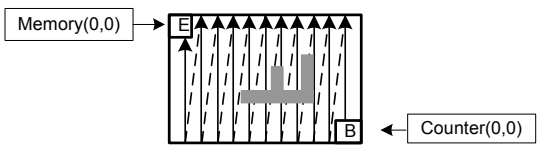
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D5 (MV)	D6 (MX)	D7 (MY)	CASET	PASET
0	0	0	Direct to Physical Column Pointer	Direct to Physical Page Pointer
0	0	1	Direct to Physical Column Pointer	Direct to (863-Physical Page Pointer)
0	1	0	Direct to (479-Physical Column Pointer)	Direct to Physical Page Pointer
0	1	1	Direct to (479-Physical Column Pointer)	Direct to (863-Physical Page Pointer)
1	0	0	Direct to Physical Page Pointer	Direct to Physical Column Pointer
1	0	1	Direct to (863-Physical Page Pointer)	Direct to Physical Column Pointer
1	1	0	Direct to Physical Page Pointer	Direct to (479-Physical Column Pointer)
1	1	1	Direct to (863-Physical Page Pointer)	Direct to (479-Physical Column Pointer)

Condition	Column Counter	Page counter
When RAMWR/RAMRD command is accepted	Return to “Start column”	Return to “Start Page”
Complete Pixel Read/Write action	Incremented by 1	No change
The Column values are larger than “End Column”	Return to “Start column”	Incremented by 1
The Page counter are larger than “End Page”	Return to “Start column”	Return to “Start Page”

One pixel unit represents 1 column and 1 page counter value on the Frame Memory.

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Display Data Direction	MADCTR Parameter			Image in the Memory (MPU)	Image in the Driver (Frame Memory)
	MV	MX	MY		
Normal	0	0	0		
Y-Mirror	0	0	1		
X-Mirror	0	1	0		
X-Mirror Y-Mirror	0	1	1		
X-Y Exchange	1	0	0		
X-Y Exchange Y-Mirror	1	0	1		
X-Y Exchange X-Mirror	1	1	0		
X-Y Exchange X-Mirror Y-Mirror	1	1	1		

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8. Tearing Effect Information

The Tearing Effect Line (TE) signal supplies panel synchronization information to the MPU. The TE signal determines the position for refreshing GRAM data for the display panel; the MPU can reference the TE to decide when to send image data in order to avoid abnormal visual effect on the display panel.

The TE signal is enabled or disabled with the Tearing Effect Line ON or OFF commands.

The TE Line output mode is defined by the parameter of the “Tearing Effect Line ON” command.

Tearing Effect information is sent in two different ways:

- Separated Line, which is the Tearing Effect Line (TE).
- Tearing Effect Bus Trigger (TEE), the ILI9806G sends a trigger to inform the MPU.

The TE is used in the MPU parallel interface.

The TE can also be used in the DSI case if the Tearing Effect Bus Trigger (TEE) is not possible to be used.

The Tearing Effect (TEE) Bus Trigger is only used in the DSI.

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8.1. Tearing Effect Line

8.1.1. Tearing Effect Line Modes

Mode 1: illustrated in Figure 165, the Tearing Effect output signal consists of V-Sync information only:

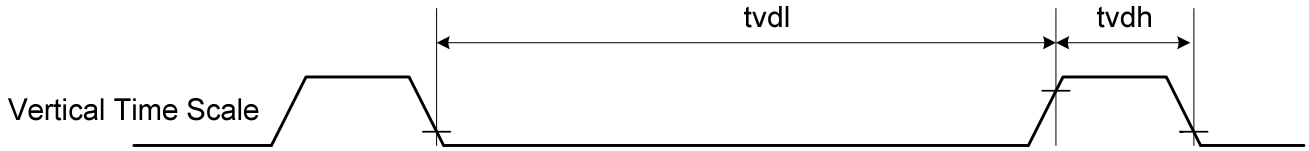


Figure 165: Tearing Effect Line Mode 1

tvdh = The LCD display is not updated from the Frame Memory.

tvdl = The LCD display is updated from the Frame Memory (except Invisible Line – see the figure above).

Mode 2: illustrated in Figure 166, the Tearing Effect output signal consists of V-Sync and H-Sync information; there is one V-sync and "N" H-sync pulses per field.

N: If RSO [2:0] = 3'b000, the resolution is 480 (RGB) X 864 and N = 864.

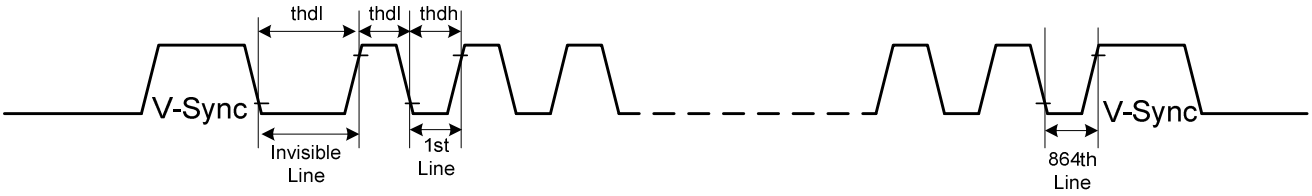
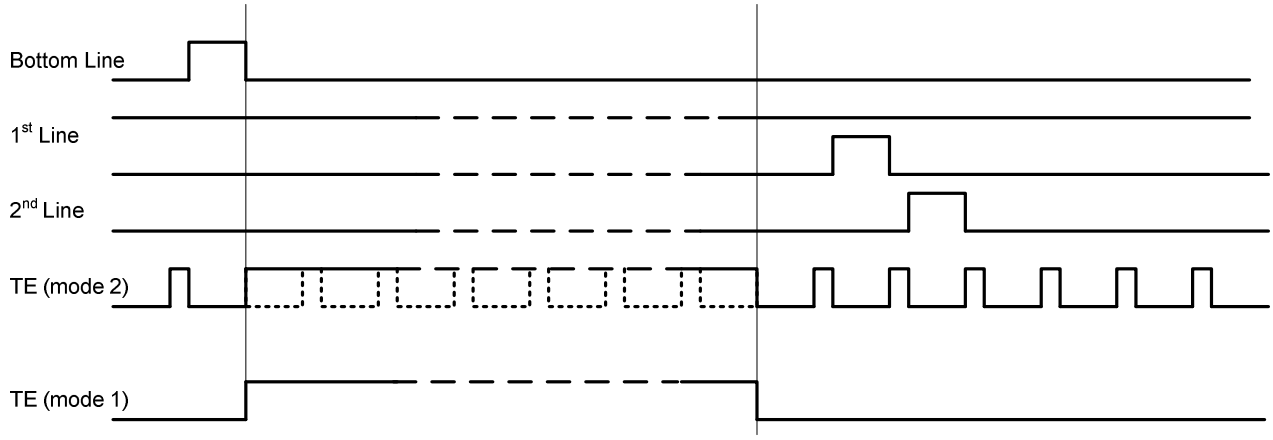


Figure 166: Tearing Effect Line Mode 2

thdh = The LCD display is not updated from the Frame Memory.

thdl = The LCD display is updated from the Frame Memory (except Invisible Line – see the figure above).

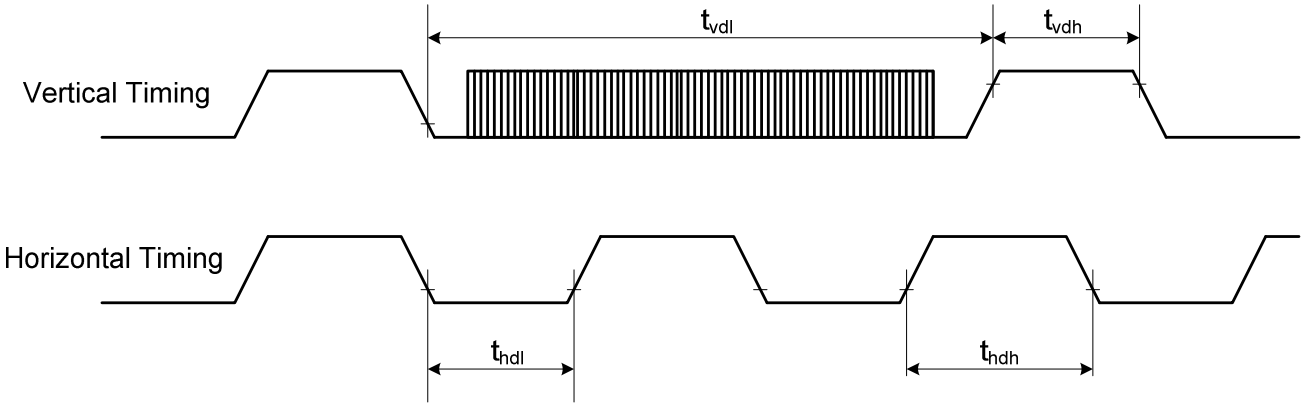


Note: During the Sleep In Mode, the Tearing Effect Output Pin is active Low.

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8.1.2. Tearing Effect Line Timing

The tearing effect signal is described below:



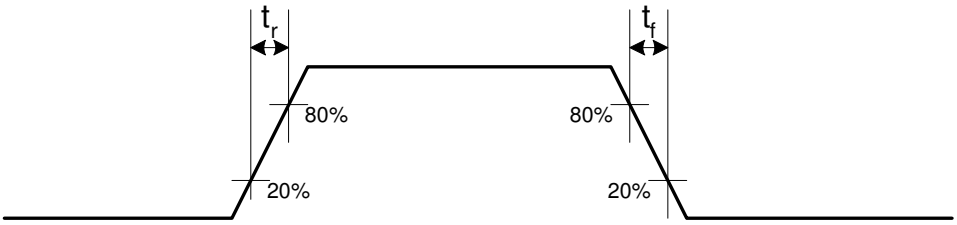
AC characteristics of the Tearing Effect Signal is: (Resolution 480(RGB)x800, Frame Rate = 60.5 Hz)

Table 44: Tearing Effect Line Timing

Symbol	Parameter	Min.	Typ.	Max.	Unit
t_{vdl}	Vertical timing low duration	15	-		ms
t_{vdh}	Vertical timing high duration	1000	-		us
t_{hdl}	Horizontal timing low duration	18	-		us
t_{hdh}	Horizontal timing high duration	0.13	500		us

Notes: 1. The timings in Table 44 are applied when MADCTL D4 = 0 and D4 = 1.

2. The signal's rising and falling times (tf, tr) are stipulated to be equal to or less than 15ns.



The Tearing Effect Output Line is fed back to the MPU and should be used to avoid the Tearing Effect.

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8.2. Tearing Effect Bus Trigger

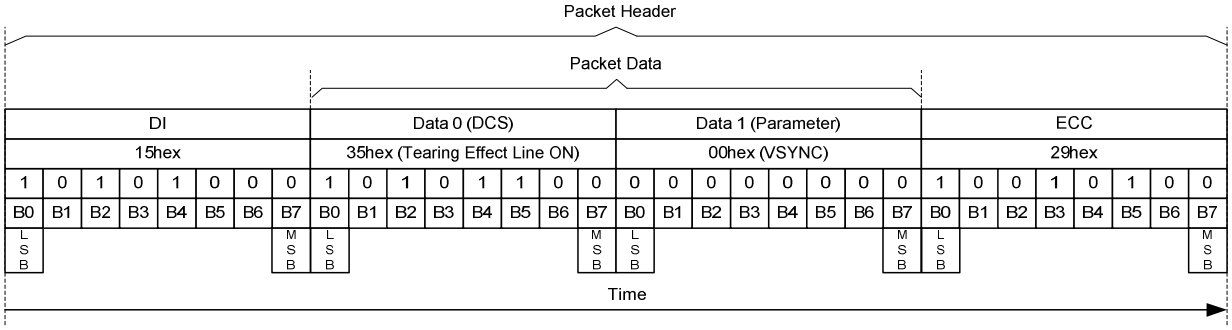
Tearing Effect Bus Trigger information supplies to the MPU a Panel synchronization trigger, and this Tearing Effect Bus Trigger information can be enabled or disabled by “Tearing Effect Line Off (34h)” and “Tearing Effect Line On (35h)” commands when the only mode of the Tearing Effect Signal is VSYNC information.

The ILI9806G sends this trigger information in the Escape Mode after the Bus Turnaround (BTA), and the Tearing Effect Bus Trigger can only be used in the DSI without the TE line.

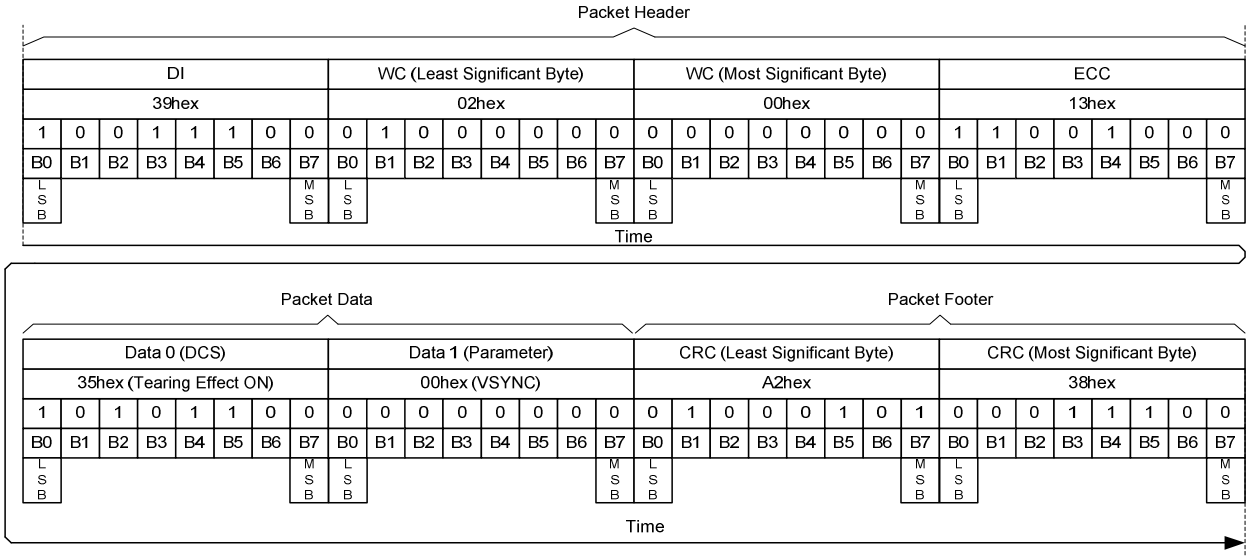
8.2.1. Tearing Effect Bus Trigger Enable

The MPU can enable the Tearing Effect Bus Trigger on the ILI9806G in 2 different ways: when a Short Packet (SPa) or a Long Packet (LPa) is used. These cases are illustrated below.

Tearing Effect Bus Trigger Enable - Short Packet



Tearing Effect Bus Trigger Enable - Long Packet

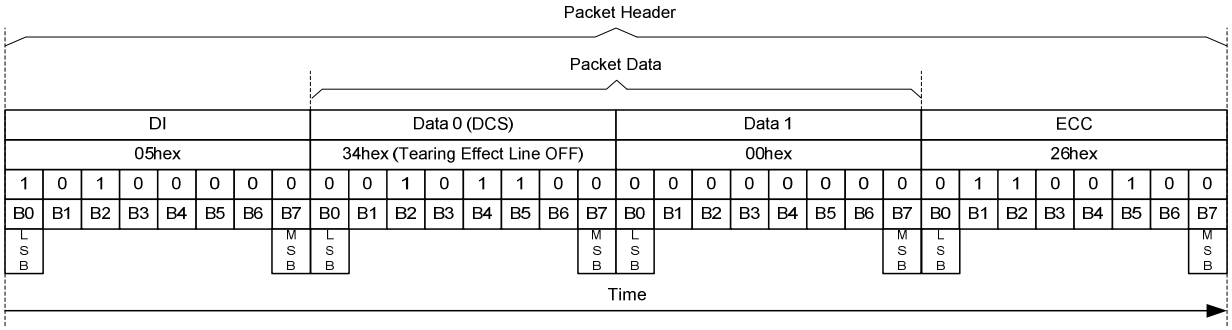


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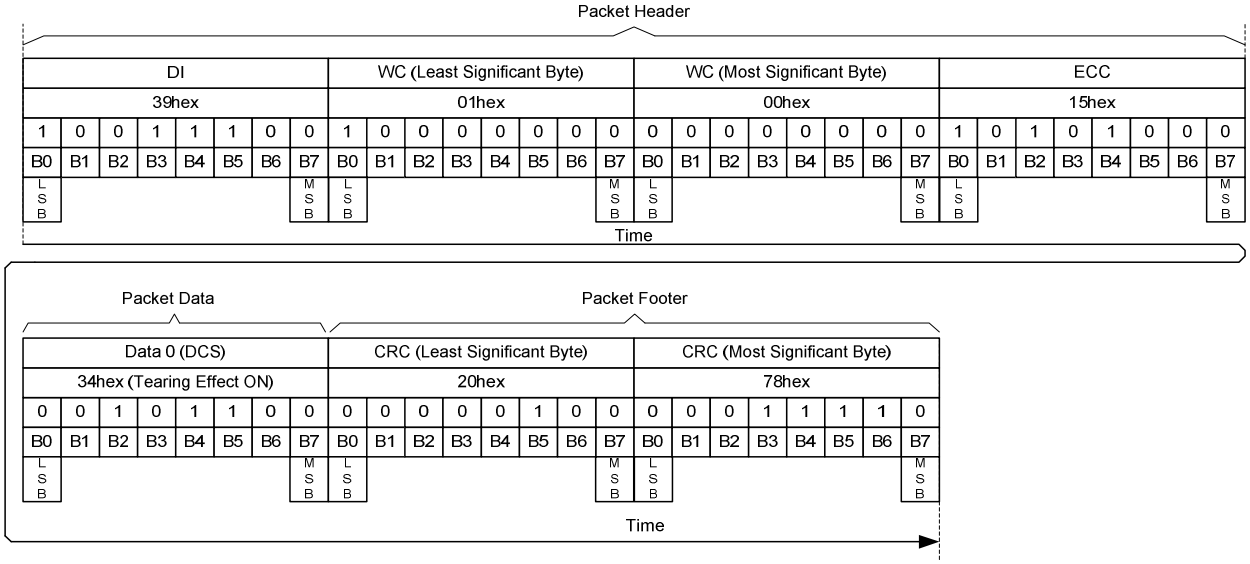
8.2.2. Tearing Effect Bus Trigger Disable

The MPU can enable the Tearing Effect Bus Trigger on the ILI9806G in 2 different ways: when a Short Packet (SPa) or a Long Packet (LPa) is used. These both possibilities are illustrated below.

Tearing Effect Bus Trigger Disable - Short Packet



Tearing Effect Bus Trigger Disable - Long Packet



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8.2.3. Tearing Effect Bus Trigger Sequences

Tearing Effect Trigger Enable Sequence – DCS Write (Long Packet) and HSDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCSW-L	HSDT	→	-	-	Tearing Effect Trigger Enable
3	EoTP	HSDT	→	-	-	End of Transmission Packet
4	-	LP-11	→	-	-	-
5	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
6	-	-	←	LP-11	-	If No Error => Goto Line 8 If Error is Corrected by ECC => Goto Line 19, If Error => Goto Line 30
7						
8	-	-	←	ACK	-	No Error
9	-	-	←	LP-11	-	-
10	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
11	-	LP-11	→	-	-	-
12	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
13	-	-	←	LP-11	-	-
14	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
15	-	-	←	LP-11	-	-
16	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
17	-	LP-11	→	-	-	End
18						
19	-	-	←	LPDT	AwER	Error Report (Error is Corrected by ECC)
20	-	-	←	LP-11	-	-
21	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
22	-	LP-11	→	-	-	-
23	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
24	-	-	←	LP-11	-	-
25	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
26	-	-	←	LP-11	-	-
27	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
28	-	LP-11	→	-	-	End
29						
30	-	-	←	LPDT	AwER	Error Report
31	-	-	←	LP-11	-	-
32	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
33	-	LP-11	→	-	-	If the MPU is not forcing BTA => Goto Line 34, If the MPU is forcing BTA => Goto Line36
34	-	LP-11	→	-	-	End
35						

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36	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
37	-	-	←	LP-11	-	Dead-Lock (No TE information) See Note 2
38	-	LP-11	→	-	-	The MPU is forced to start to control the interface. The display module detects, Bus Connection Error (BCE)
39	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
40	-	-	←	LP-11	-	-
41	-	-	←	LPDT	AwER	Error Report (Bus Connection Error (BCE) is reported) See Note 2
42	-	-	←	LP-11	-	-
43	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
44	-	LP-11	→	-	-	End

Notes:

1. Lines 1 – 17 are needed for every frame.

2. Bits 5 and 7 of the AwER are applied.

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Tearing Effect Trigger Enable Sequence – DCS Write (Long Packet) and LPDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCSW-L	LPDT	→	-	-	Tearing Effect Trigger Enable
3	-	LP-11	→	-	-	-
4	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
5	-	-	←	LP-11	-	If No Error => Goto Line 7 If Error is Corrected by ECC => Goto Line 18 , If Error => Goto Line 29
6						
7	-	-	←	ACK	-	No Error
8	-	-	←	LP-11	-	-
9	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
10	-	LP-11	→	-	-	-
11	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
12	-	-	←	LP-11	-	-
13	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
14	-	-	←	LP-11	-	-
15	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
16	-	LP-11	→	-	-	End
17						
18	-	-	←	LPDT	AwER	Error Report (Error is Corrected by ECC)
19	-	-	←	LP-11	-	-
20	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
21	-	LP-11	→	-	-	-
22	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
23	-	-	←	LP-11	-	-
24	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
25	-	-	←	LP-11	-	-
26	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
27	-	LP-11	→	-	-	End
28						
29	-	-	←	LPDT	AwER	Error Report
30	-	-	←	LP-11	-	-
31	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
32	-	LP-11	→	-	-	If the MPU is not forcing BTA => Goto Line 33 , If the MPU is forcing BTA => Goto Line 35
33	-	LP-11	→	-	-	End
34						
35	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module

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36	-	-	←	LP-11	-	Dead-Lock (No TE information) See Note 2
37	-	LP-11	→	-	-	The MPU is forced to start to control the interface. The display module detects Bus Connection Error (BCE)
38	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
39	-	-	←	LP-11	-	-
40	-	-	←	LPDT	AwER	Error Report (Bus Connection Error (BCE) is reported) See Note 2
41	-	-	←	LP-11	-	-
42	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
43	-	LP-11	→	-	-	End

Notes:

1. Lines 1 – 16 are needed for every frame.

2. Bits 5 and 7 of the AwER are applied.

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Tearing Effect Trigger Enable Sequence – DCS Write, 1 Parameter (Short Packet) and HSDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCSW1-S	HSDT	→	-	-	Tearing Effect Trigger Enable
3	EoTP	HSDT	→	-	-	End of Transmission Packet
4	-	LP-11	→	-	-	-
5	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
6	-	-	←	LP-11	-	If No Error => Goto Line 8 If Error is Corrected by ECC => Goto Line 19 , If Error => Goto Line 30
7						
8	-	-	←	ACK	-	No Error
9	-	-	←	LP-11	-	-
10	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
11	-	LP-11	→	-	-	-
12	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
13	-	-	←	LP-11	-	-
14	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
15	-	-	←	LP-11	-	-
16	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
17	-	LP-11	→	-	-	End
18						
19	-	-	←	LPDT	AwER	Error Report (Error is Corrected by ECC)
20	-	-	←	LP-11	-	-
21	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
22	-	LP-11	→	-	-	-
23	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
24	-	-	←	LP-11	-	-
25	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
26	-	-	←	LP-11	-	-
27	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
28	-	LP-11	→	-	-	End
29						
30	-	-	←	LPDT	AwER	Error Report
31	-	-	←	LP-11	-	-
32	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
33	-	LP-11	→	-	-	If the MPU is not forcing BTA => Goto Line 34 , If the MPU is forcing BTA => Goto Line 36
34	-	LP-11	→	-	-	End
35						
36	-	BTA	⇔	BTA	-	Interface Control Change from the

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						MPU to the display module
37	-	-	←	LP-11	-	Dead-Lock (No TE information) See Note 2
38	-	LP-11	→	-	-	The MPU is forced to start to control the interface. The display module detects Bus Connection Error (BCE)
39	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
40	-	-	←	LP-11	-	-
41	-	-	←	LPDT	AwER	Error Report (Bus Connection Error (BCE) is reported) See Note 2
42	-	-	←	LP-11	-	-
43	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
44	-	LP-11	→	-	-	End

Notes:

1. Lines 1 – 17 are needed for every frame.

2. Bits 5 and 7 of the AwER are applied.

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Tearing Effect Trigger Enable Sequence – DCS Write, 1 Parameter (Short Packet) and LPDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCSW1-S	LPDT	→	-	-	Tearing Effect Trigger Enable
3	-	LP-11	→	-	-	-
4	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
5	-	-	←	LP-11	-	If No Error => Goto Line 7 If Error is Corrected by ECC => Goto Line 18 , If Error => Goto Line 29
6						
7	-	-	←	ACK	-	No Error
8	-	-	←	LP-11	-	-
9	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
10	-	LP-11	→	-	-	-
11	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
12	-	-	←	LP-11	-	-
13	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
14	-	-	←	LP-11	-	-
15	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
16	-	LP-11	→	-	-	End
17						
18	-	-	←	LPDT	AwER	Error Report (Error is Corrected by ECC)
19	-	-	←	LP-11	-	-
20	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
21	-	LP-11	→	-	-	-
22	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
23	-	-	←	LP-11	-	-
24	-	-	←	TEE	-	TE (Escape Trigger) on the next V-Synch.
25	-	-	←	LP-11	-	-
26	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
27	-	LP-11	→	-	-	End
28						
29	-	-	←	LPDT	AwER	Error Report
30	-	-	←	LP-11	-	-
31	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
32	-	LP-11	→	-	-	If the MPU is not forcing BTA => Goto Line 33 , If the MPU is forcing BTA => Goto Line 35
33	-	LP-11	→	-	-	End
34						
35	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module

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36	-	-	←	LP-11	-	Dead-Lock (No TE information) See Note 2
37	-	LP-11	→	-	-	The MPU is forced to start to control the interface. The display module detects Bus Connection Error (BCE)
38	-	BTA	⇔	BTA	-	Interface Control Change from the MPU to the display module
39	-	-	←	LP-11	-	-
40	-	-	←	LPDT	AwER	Error Report (Bus Connection Error (BCE) is reported) See Note 2
41	-	-	←	LP-11	-	-
42	-	BTA	⇔	BTA	-	Interface Control Change from the display module to the MPU
43	-	LP-11	→	-	-	End

Notes:

1. Lines 1 – 16 are needed for every frame.

2. Bits 5 and 7 of the AwER are applied.

Tearing Effect Bus Trigger Disable Sequence – DCS Write, No Parameter (Short Packet) and LPDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCS Write (SPa)	LPDT	→	-	-	Tearing Effect Bus Trigger Disable
3	-	LP-11	→	-	-	End

Tearing Effect Bus Trigger Disable Sequence – DCS Write, No Parameter (Short Packet) and HSDT						
Line	MPU		Information Direction	Display Module (ILI9806G)		Comment
	Packet Sender	Interface Mode Control		Interface Mode Control	Packet Sender	
1	-	LP-11	→	-	-	Start
2	DCS Write (SPa)	HSDT	→	-	-	Tearing Effect Bus Trigger Disable
3	-	LP-11	→	-	-	End

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9. Content Adaptive Brightness Control (CABC) Function

The CABC, a dynamic backlight control function, drastically reduces power consumption of the luminance source. The ILI9806G will refer the gray scale content of the display image to output in the PWM waveform then to the LED driver for backlight brightness control. The content of gray scale can be increased while simultaneously lowering the brightness of the backlight to achieve the same perceived brightness. The adjusted gray level scale and the power consumption reduction depend on the content of the image.

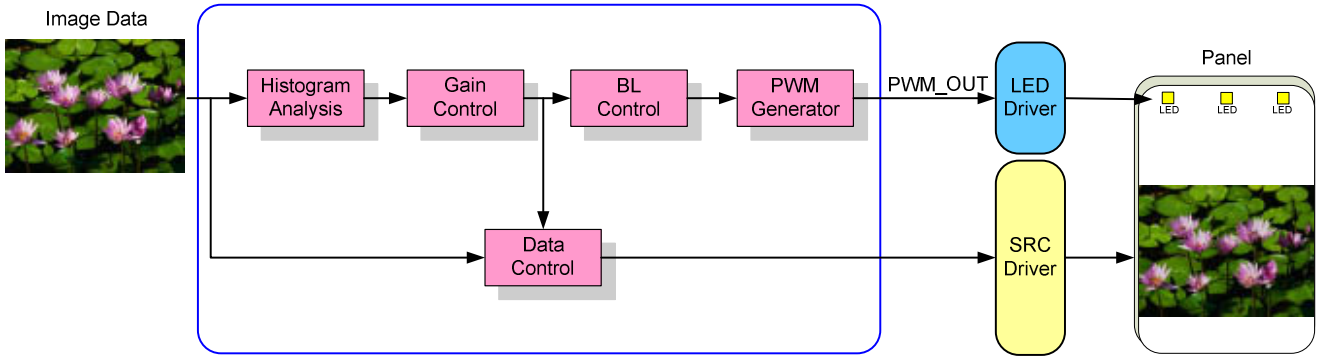


Figure 167: CABC Block Diagram

The ILI9806G can calculate the backlight brightness level and send a PWM_OUT pulse to the LED driver via PWM_OUT pin for backlight brightness control purposes. The PWM frequency can be adjusted by PWM_DIV parameters, and the calculating equation is shown below:

$$fpwm_out = \frac{18\ MHz}{(PWM_DIV[7:0]+1) \times 255}$$

Figure 168 is the basic timing diagram which is applied by the ILI9806G in order to control a LED driver.

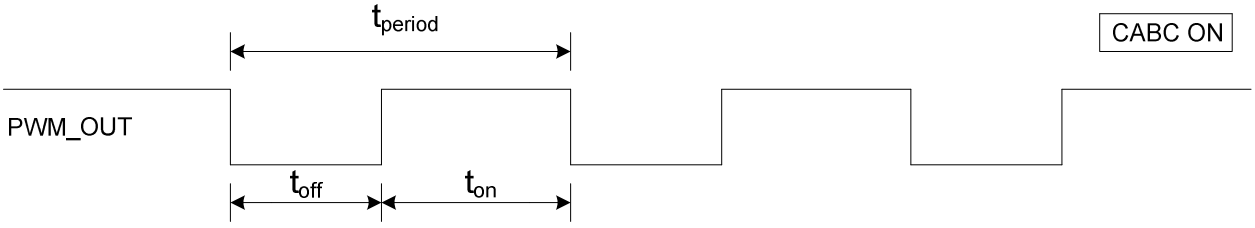


Figure 168: PWM OUT on/off period

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10. Color Enhancement Function

The ILI9806G supports color enhancement. The Color Enhancement Function enhances saturation by calculating image data of the display on the liquid crystal panel. The saturation enhancement coefficients of red, yellow, green, cyan, blue, magenta , are set independently. The function enhances color and makes pixel colors more vivid.

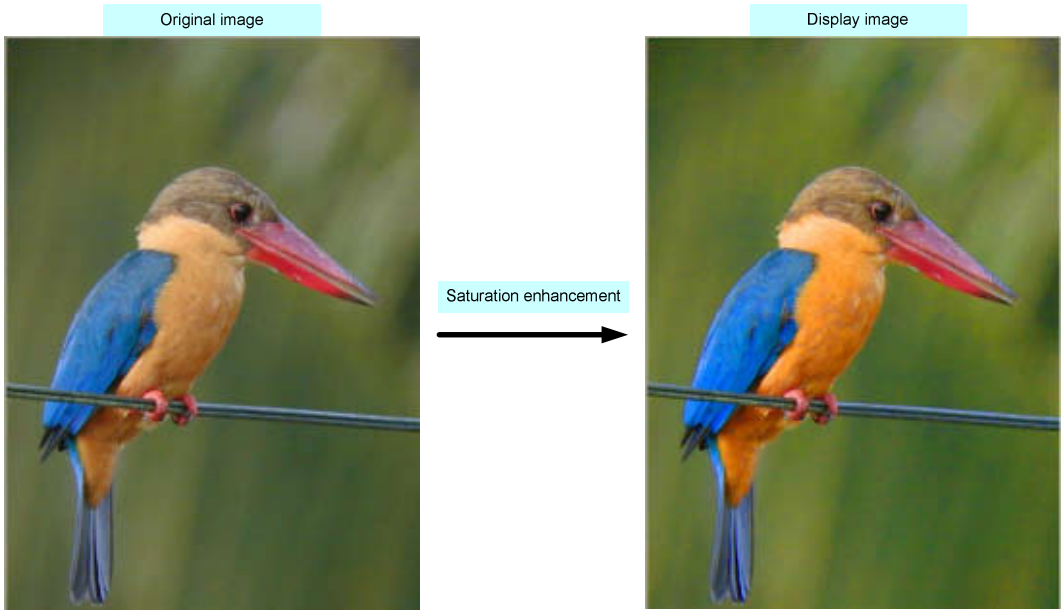


Figure 169: Saturation Enhancement Image

The display image with color enhanced is generated when the saturation enhancement coefficients of the input image are 1.0 or more. See the saturation diagram Figure 169, where the colors of the input image are enhanced. Both green and magenta colors keep the original gray value without enhanced.

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11. Sleep Out Command and Self-Diagnostic Functions

11.1. Register Loading Detection

The Sleep Out command (See “Sleep Out (11h)”) is a trigger for an internal function of the display module, which indicates if the display module loading function of factory default values from the EEPROM (or similar devices) to registers of the display controller works properly.

Factory values of the EEPROM and register values of the display controller are compared by the display controller (1st step: compare register and EEPROM values, 2nd step: loads EEPROM values to the register). If those values (EEPROM and register values) are the same, a bit is inverted (= increased by 1), which is defined in the command Read Display Self-Diagnostic Result (0Fh) (= RDDSDR). The used bit of this command is D7. If both values are not the same, this bit (D7) is not inverted (not increased by 1), and the used TE-line is set to low (registers, set by the “Tearing Effect Line On (35h)”command, keep their current values) and can only be reactivated by the “Tearing Effect Line On (35h)” command.

The flow chart for this internal function is:

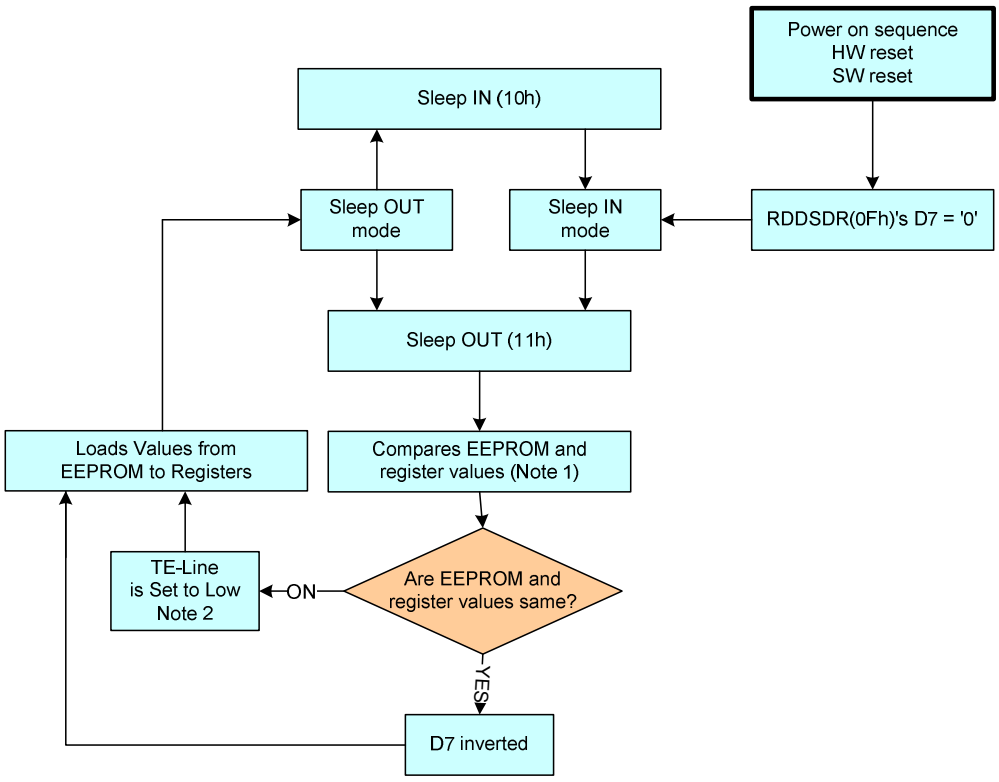


Figure 170: Register Loading Detection

Notes:

1. If the EEPROM and register values are not compared and loaded, they can be changed by commands 00h to AFh and DAh to DDh.
2. This information is only used when the TE line is used.

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11.2. Functionality Detection

The Sleep Out command (See “Sleep Out (11h)”) is a trigger for an internal function of the display module. It indicates if the display module is still running and meets functionality requirements.

The internal function (the display controller) is compared when the display module meets functionality requirements (e.g. booster voltage levels, timings, etc.). If functionality requirements are met, a bit is inverted (increased by 1), defined in the command “Read Display Self-Diagnostic Result (0Fh)” (RDDSDR). The used bit of this command is D6. If functionality requirements are not the same, this bit (D6) is not inverted (not increased by 1), and the used TE-line is set to low (registers keep their current values) and can only be reactivated by “Tearing Effect Line On (35h)” command.

The flow chart for this internal function is:

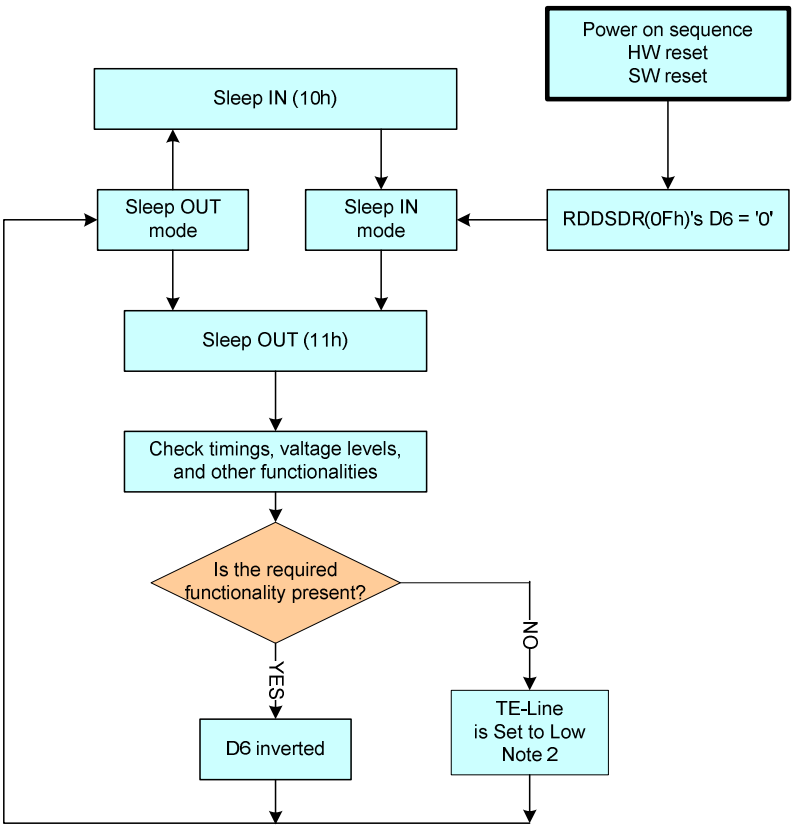


Figure 171: Functionality Detection

Notes:

1. When changing from the Sleep In mode to the Sleep Out mode, 120msec are needed after the Sleep Out command before it is able to check if functionality requirements are met and the value of RDDSDR’s D6 is valid. Otherwise, there will be 5msec delay for the D6’s value to be valid when the Sleep Out command is sent in the Sleep Out mode.
2. This function can only be used when the TE-line is used.

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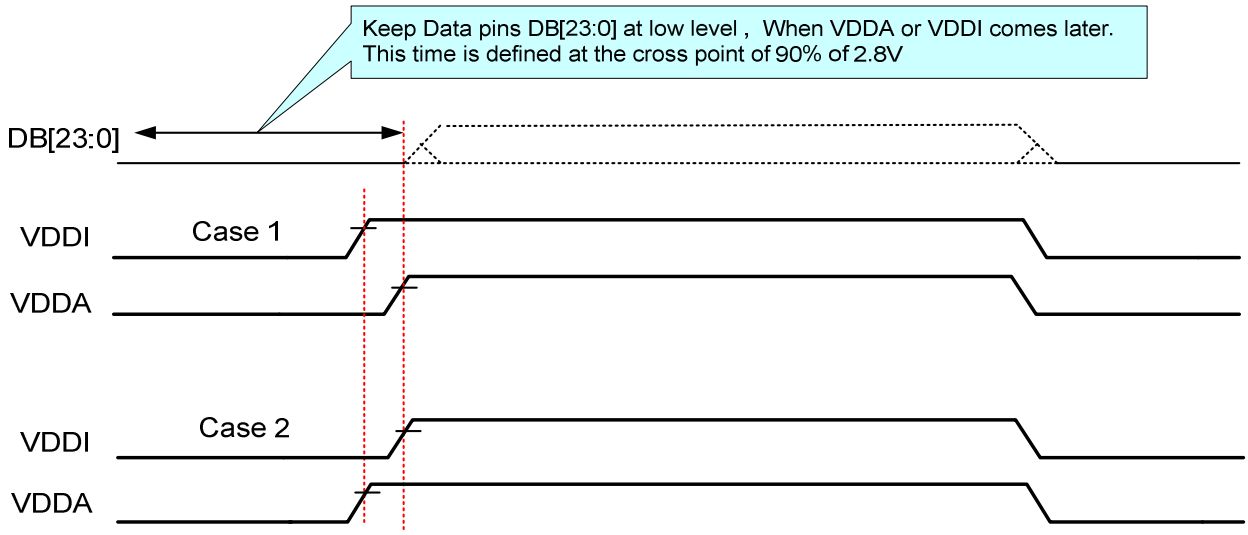
12. Power ON/OFF Sequence

VDDI and VDDA can be applied or powered down in any order. During the Power Off sequence, if the LCD is in the Sleep Out mode, VDDA and VDDI must be powered down with minimum 120msec. If the LCD is in the Sleep In mode, VDDA and VDDI can be powered down with minimum 0msec after the RESX is released.

CSX can be applied at any timing or can be permanently grounded. RESX has high priority over CSX.

Notes:

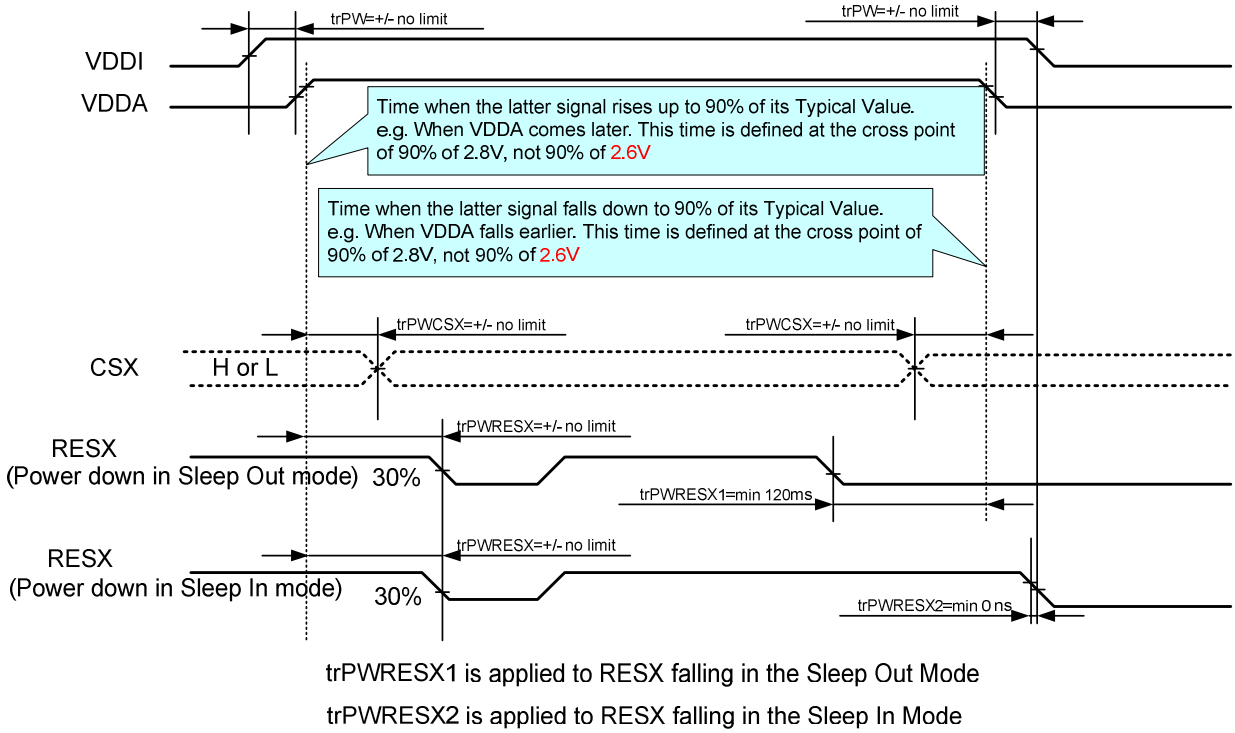
1. There will be no damage to the ILI9806G if the power sequences are not met.
2. There will be no abnormal visible effects on the display panel during the Power On/Off Sequences.
3. There will be no abnormal visible effects on the display between the end of Power On Sequence and before receiving the Sleep Out command, and also between receiving the Sleep In command and the Power Off Sequence.
4. If the RESX line is not steadily held by the host during the Power On Sequence as defined in Sections 12.1 and 12.2, then it will be necessary to apply the Hardware Reset (RESX) after the completion of the Host Power On Sequence to ensure correct operations. Otherwise, all the functions are not guaranteed.
5. Keep data pins DB[23:0] at low level, when VDDA or VDDI comes later.



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12.1. Case 1 – RESX Line is Held High or Unstable by Host at Power ON

If the RESX line is held high or unstable by the host during Power On, then the Hardware Reset must be applied after both VDDA and VDDI have been applied. Otherwise, the correct functionality is not guaranteed. There is no timing restriction upon this hardware reset.



trPWRESX1 is applied to RESX falling in the Sleep Out Mode

trPWRESX2 is applied to RESX falling in the Sleep In Mode

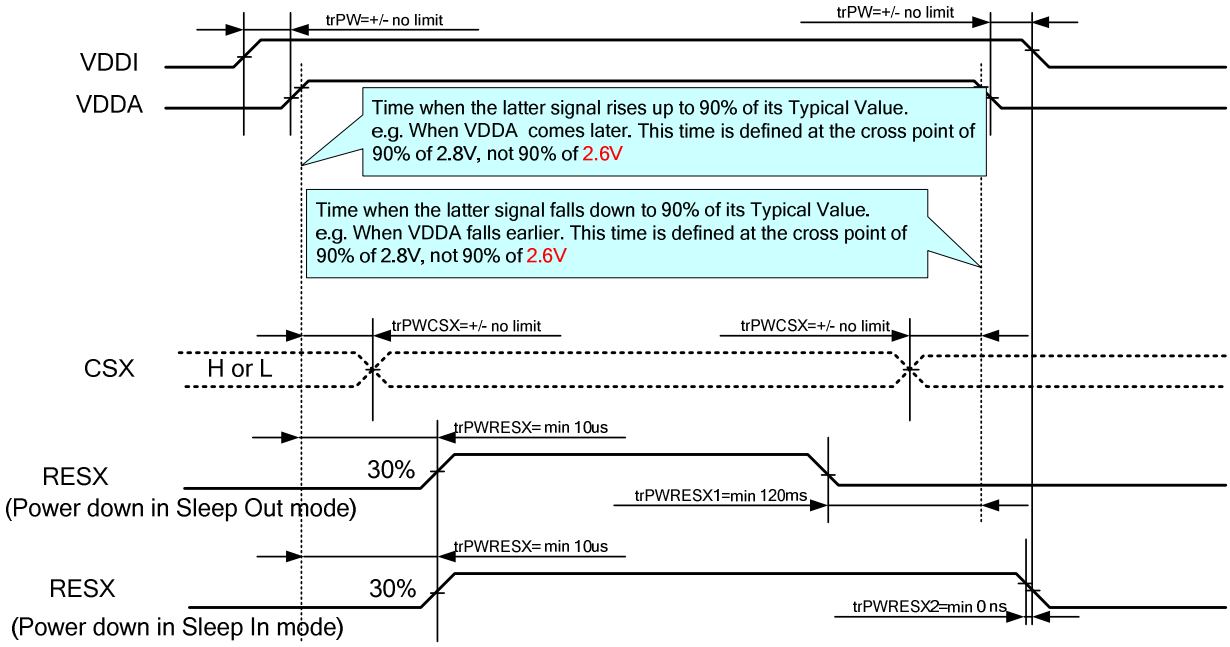
Figure 172: Case 1 – RESX Line is Held High or Unstable by Host at Power ON

Note: Unless otherwise specified, timings herein show the cross point at 50% of the signal power level.

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12.2. Case 2 – RESX Line is Held Low by Host at Power ON

If the RESX line is held Low (and stable) by the host during Power On, then the RESX must be held low for a minimum of 10µsec after both VDDA and VDDI have been applied.



trPWRESX1 is applied to RESX falling in the Sleep Out Mode
trPWRESX2 is applied to RESX falling in the Sleep In Mode

Figure 173: Case 2 – RESX Line is Held Low by Host at Power ON

Note: Unless otherwise specified, timings herein show the cross point at 50% of the signal power level.

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12.3. Abnormal Power Off

The Abnormal Power Off means a situation when a battery is removed without the controlled power off sequence. There will not be any damages on the display module, or the display module will not cause any damages on the host or lines of the interface. At an uncontrolled power off event, the ILI9806G will not cause any abnormal visible effects within 1 second on the display, and will force the display to become blank until the Power On Sequence powers it up.

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13. Power Level Definition

13.1. Power Levels

6 level modes are defined in order from Maximum to Minimum Power Consumption:

1. Normal Mode On (full display), Idle Mode Off, Sleep Out.
   In this mode, the display is able to show a maximum of 16.7M colors.
2. Partial Mode On, Idle Mode Off, Sleep Out.
   In this mode, part of the display is used with a maximum of 16.7M colors.
3. Normal Mode On (full display), Idle Mode On, Sleep Out.
   In this mode, the full display area is used but with 8 colors.
4. Partial Mode On, Idle Mode On, Sleep Out.
   In this mode, part of the display is used but with 8 colors.
5. Sleep In Mode.
   In this mode, the DC/DC converter, internal oscillator, and panel driver circuit are stopped. Only the MPU interface and memory work with VDDI power supply. Contents of the memory are safe and cannot be changed.
6. Power Off Mode.
   In this mode, both VDDA and VDDI are removed.

Transition between modes 1-5 is controllable by MPU commands. Only when both Power supplies are removed can Mode 6 be entered.

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13.2. Power Flow Chart

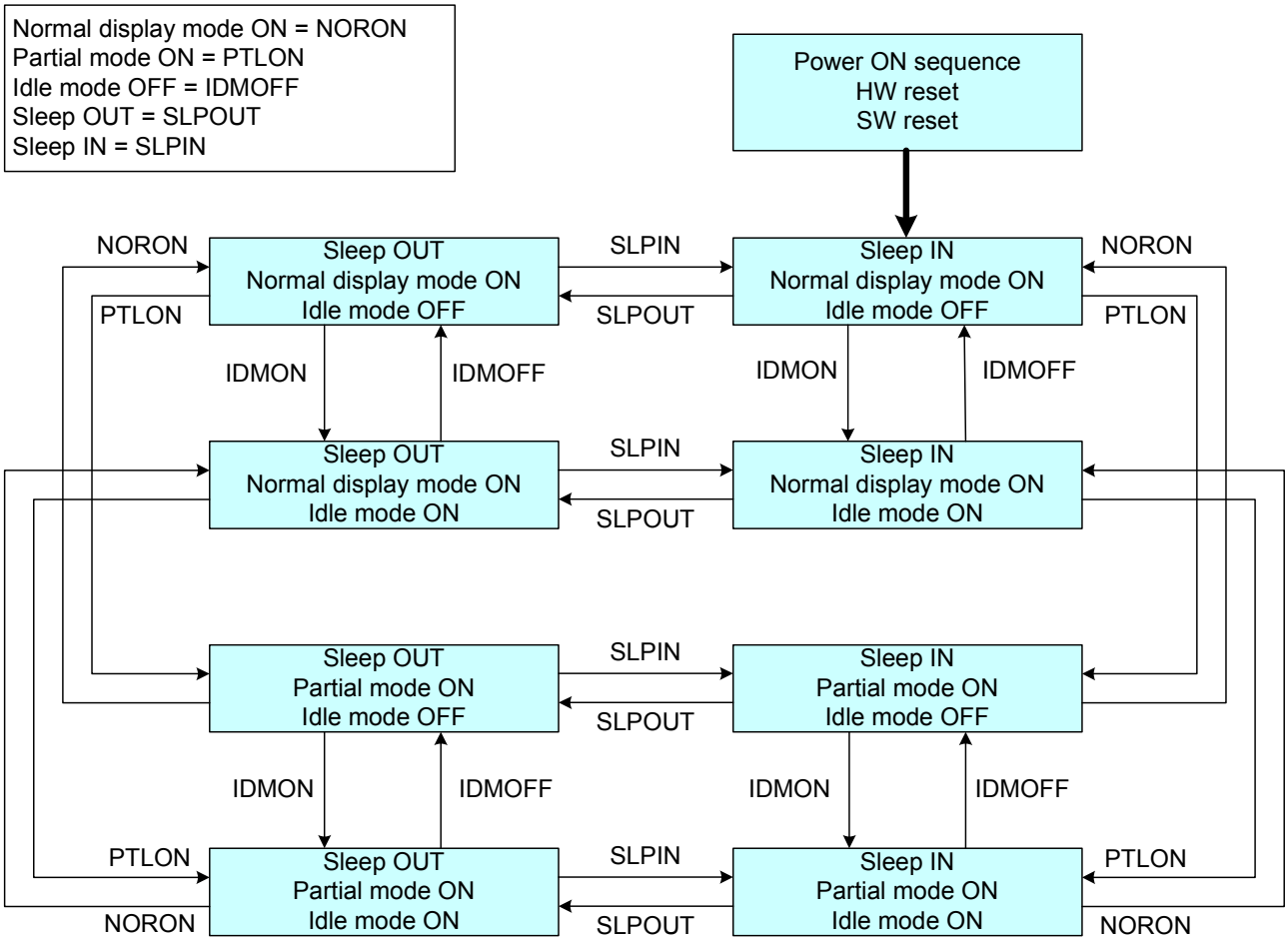


Figure 174: Power Mode Flow Chart

Notes:

1. There are no abnormal visual effects when one power mode changes to another power mode.
2. There is not any limitation, which is not specified by users, when one power mode changes to another power mode.

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13.3. LCM Voltage Generation

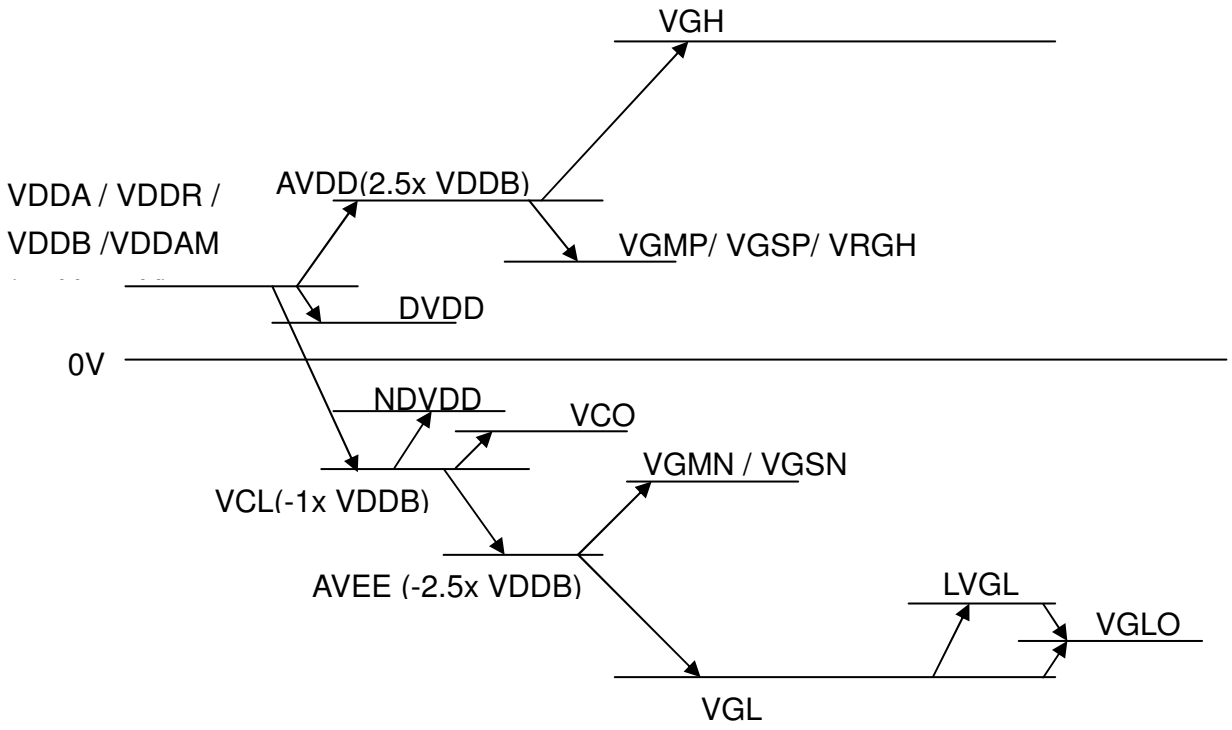


Figure 175: Power Stage Diagram

Power	VGL : BL[1:0]	
AVDD = 2.5x VDDB	00	AVEE-AVDD
AVEE = -2.5x VDDB	01	2AVEE-VDDB
VCL = -1x VDDB	10	AVEE-AVDD+VCL
VGH = 2AVDD-AVEE	11	AVEE-VDDB+VCL

Note: The AVDD, AVEE, VGMP, VGMN, VCOM, VGH, VGL and VCL output voltage levels are lower than their theoretical levels (ideal voltage levels) due to the current consumption at respective outputs.

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13.4. Gamma Curves

13.4.1. Gamma Curve 1 (GC0), applies the function y = x^2.2

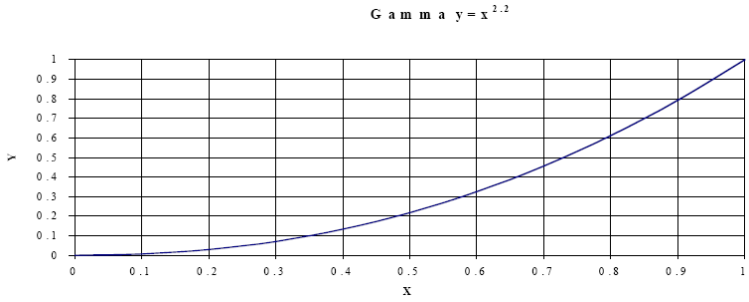


Figure 176: Gamma Curve 1 (GC0)

13.4.2. Gamma Curve 2 (GC1), applies the function y = x^1.8

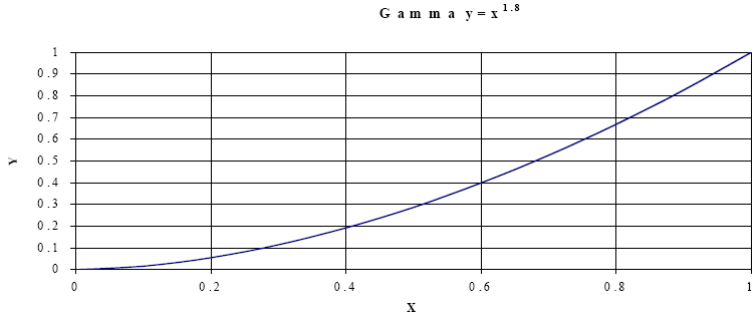


Figure 177: Gamma Curve 2 (GC1)

13.4.3. Gamma Curve 3 (GC2), applies the function y = x^2.5

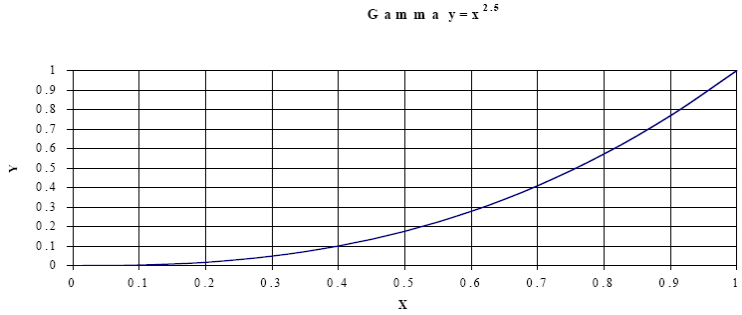


Figure 178: Gamma Curve 3 (GC2)

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13.4.4. Gamma Curve 4 (GC3), applies the function y = x^1.0

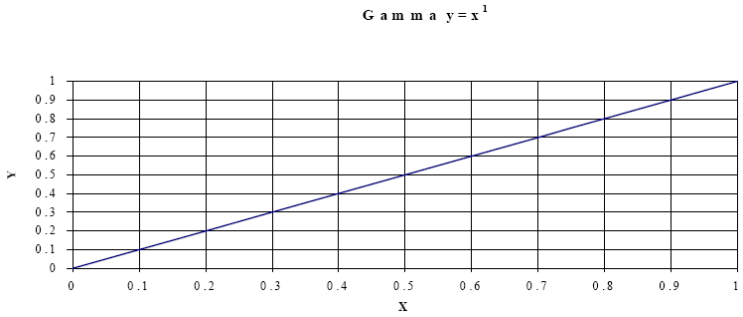


Figure 179: Gamma Curve 4 (GC3)

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14. Reset

14.1. Registers

The initialized values of registers are listed in the Table 45.

Table 45: Initial Values of Registers

Register	After Powered ON	After Hardware Reset	After Software Reset
Frame Memory	Random	Random	Random
Sleep	In	In	In
Display Mode	Normal	Normal	Normal
Display Status	Display Off	Display Off	Display Off
Idle Mode	Off	Off	Off
All Pixels Off	Off	Off	Off
All Pixels On	Off	Off	Off
Column Start Address (2Ah)	0000 h	0000 h	0000 h
Column End Address (2Ah)	01DF h	01DF h	01DF h
Page Start Address (2Bh)	0000 h	0000 h	0000 h
Page End Address (2Bh)	480(RGB)x864=035Fh	480(RGB)x864=035Fh	480(RGB)x864=035 h
Gamma Setting (26h)	GC0	GC0	GC0
Partial Area Start (30h)	0000 h	0000 h	0000 h
Partial Area End (30h)	480(RGB)x864=035Fh	480(RGB)x864=035Fh	480(RGB)x864=035Fh
MADCTL (36h)	00 h	00 h	00h
RDNUMED (05h)	00 h	00 h	00h
RDDPM (0Ah)	08 h	08 h	08 h
RDDMADCTL (0Bh)	00 h	00 h	no change
RDDCOLMOD (0Ch)	07 h	07 h	07 h
RDDIM (0Dh)	00 h	00 h	00 h
RDDSM (0Eh)	00 h	00 h	00 h
RDDSDR (0Fh)	00 h	00 h	00 h
Color Pixel Format (3Ah)	24 Bit/Pixel	24 Bit/Pixel	24 Bit/Pixel
TE Output Line (35h)	Off	Off	Off
TE Line Mode (35h)	Mode 1 (Note 3)	Mode 1 (Note 3)	Mode 1 (Note 3)
RDDISBV (52h)	00 h	00 h	00 h
RDCTRLD (54h)	00 h	00 h	00 h
RDCABC (56h)	00 h	00 h	00 h
RDCABCMB (5Fh)	00 h	00 h	00 h
RDFCS (AAh)	00 h	00 h	00 h
RDCCS (AFh)	00 h	00 h	00 h
MRPS (Data Type 37h)	01 h	01 h	01 h

Notes:

1. There will be no abnormal visible effects on the display when S/W or H/W Resets are applied.
2. After the Powered-On Reset finishes within 10µs after both VDDA & VDDI are applied.
3. Mode 1 means Tearing Effect Output Line consists of V-Blanking Information only.

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14.2. Driver IC Input and Output pins

14.2.1. Output Pins, I/O Pins

Table 46: Output and I/O Pins

Pin/Line	After Power ON	After Hardware Reset	After Software Reset
DB [23:0]	Hi-Z (Inactive)	Hi-Z (Inactive)	Hi-Z (Inactive)
SDA (Output direction), SDO	Hi-Z (Inactive)	Hi-Z (Inactive)	Hi-Z (Inactive)
HS_D0P	Hi-Z (Inactive)	Hi-Z (Inactive)	Hi-Z (Inactive)
HS_D0N	Hi-Z (Inactive)	Hi-Z (Inactive)	Hi-Z (Inactive)
LEDPWM	Low	Low	Low
TE_L / TE_R	Low	Low	Low

Note: There will be no output from DB [23:0], SDA, SDO, LEDPWM, TE_L / TE_R, HS_D0P, and HS_D0N, during the Power ON/OFF sequence, hardware reset and software reset.

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14.2.2. Reset Timing

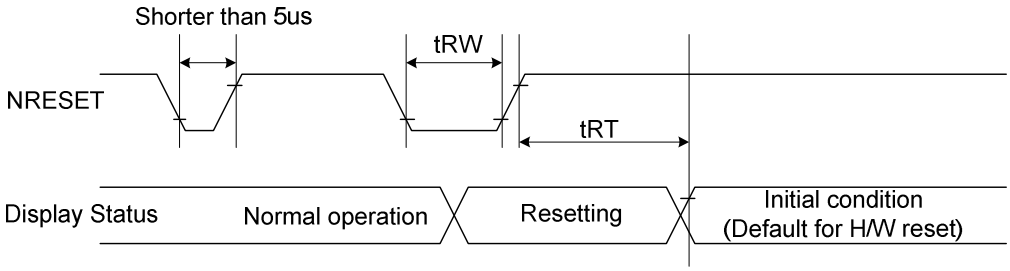


Figure 180: Reset Timing

Table 47: Reset Timing

Signal	Symbol	Parameter	Min	Max	Unit
RESX	tRW	Reset pulse duration	10		us
	tRT	Reset cancel		5 (note 1,5)	ms
				120 (note 1,6,7)	ms

Notes:

1. The reset cancel also includes the required time for loading ID bytes, VCOM setting and other settings from the EEPROM to registers. After a rising edge of RESX, this loading is done within 5 ms after the H/W reset cancel (tRT).
2. A spike due to an electrostatic discharge on the RESX line does not cause irregular system reset, according to the Table 48.

Table 48: Reset Description

RESX Pulse	Action
Shorter than 5us	Reset Rejected
Longer than 9us	Reset
Between 5us and 9us	Reset starts

3. During the Resetting period, the display will be blanked (When Reset starts in the Sleep Out mode, the display will enter the blanking sequence in at least 120 ms. The display remains blank in the Sleep In mode.), and then return to the default condition for the Hardware Reset.
4. Spike Rejection can also be applied during a valid reset pulse as shown below:

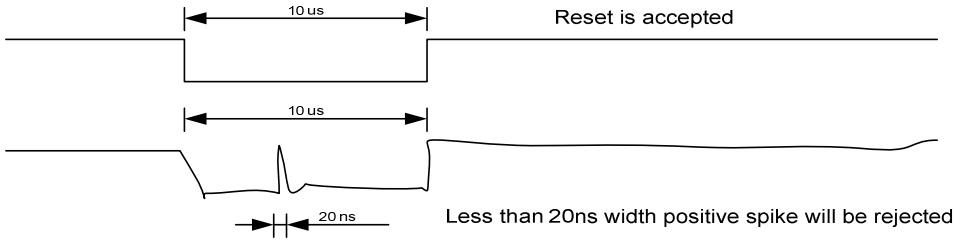


Figure 181: Positive Noise Pulse during Reset Low

5. When Reset applied during Sleep In Mode.
6. When Reset applied during Sleep Out Mode.
7. It is necessary to wait 5msec after releasing the RESX before sending commands. Moreover, the Sleep Out command cannot be sent in 120msec.

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15. NV Memory Programming

15.1. NV Memory Programming Flow

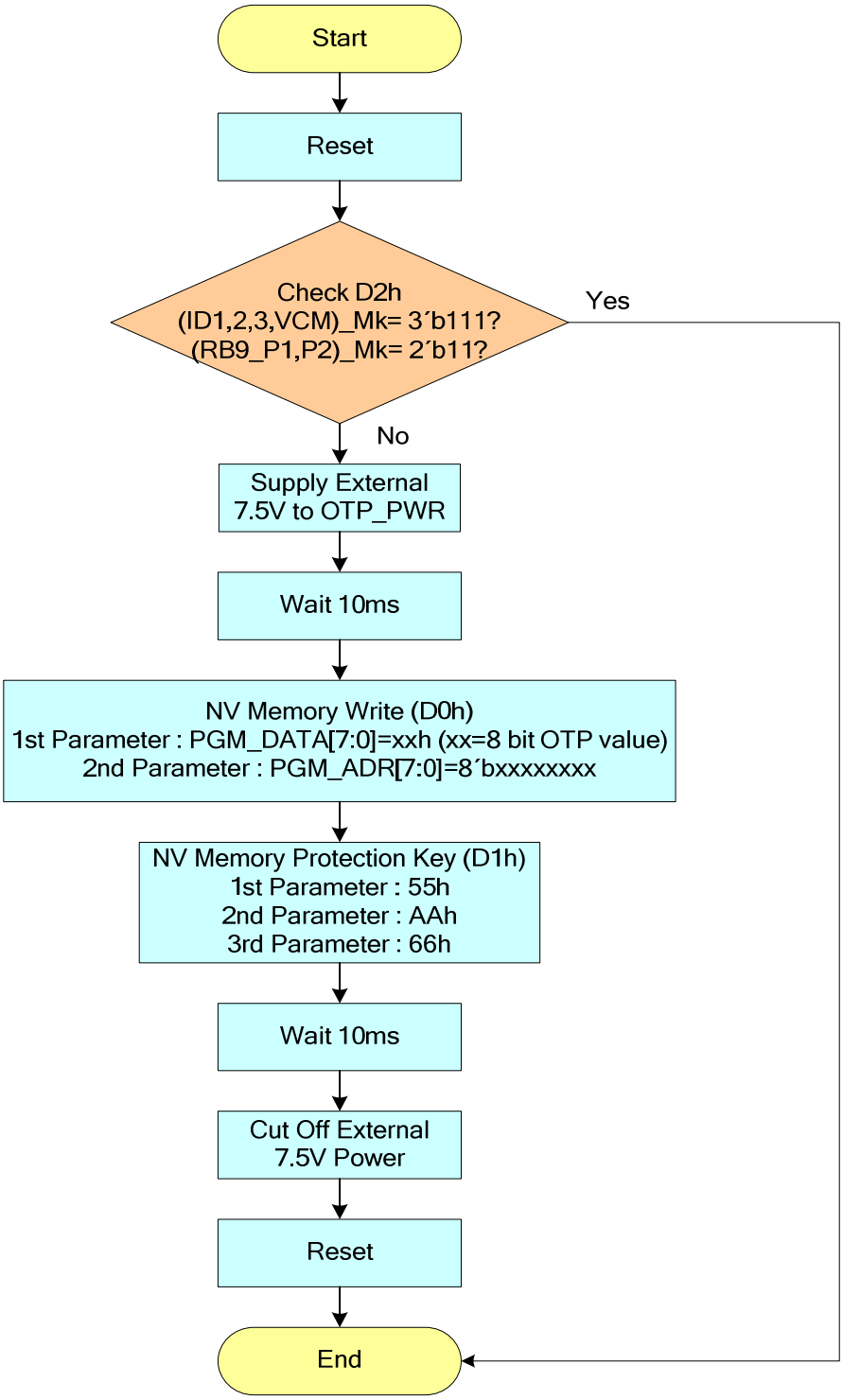


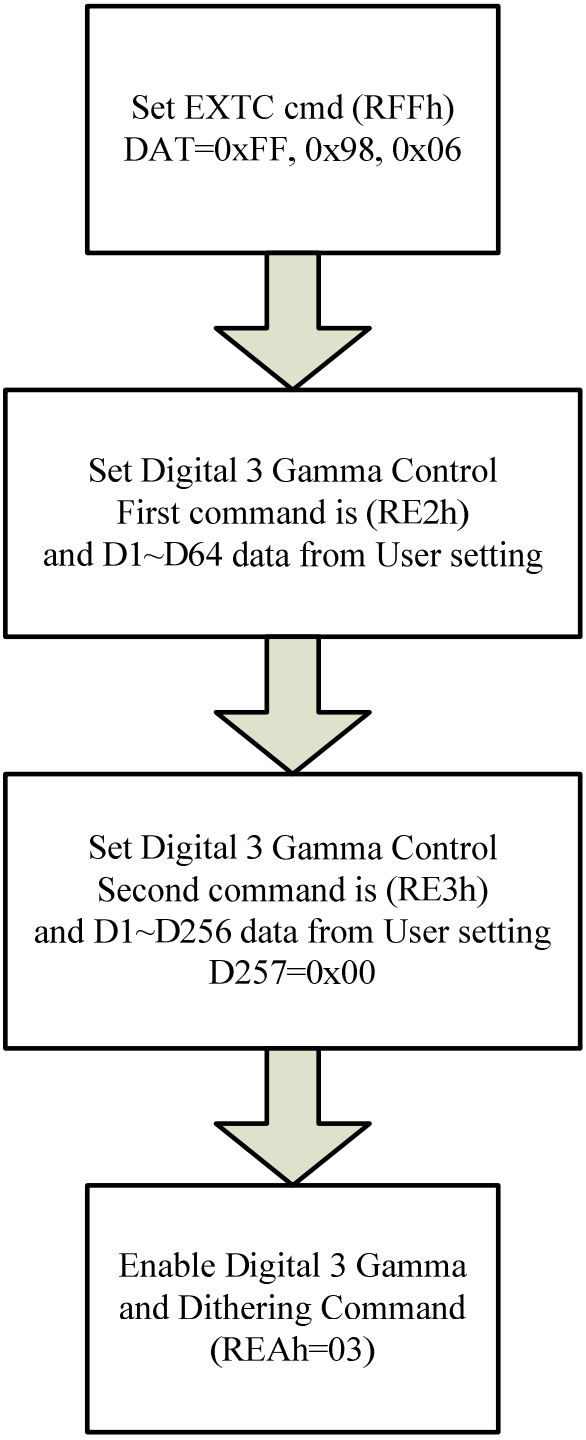
Figure 182: NV Memory Programming Flow

Note 1：In SPI operation mode, set register FBh “ext_spi_read_en”=1 to enable the read function of extend command.

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16. 3 Gamma Setting Flow

16.1. 3 Gamma Setting Flow



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17. Gamma Correction

Positive Gamma Control (E0h)

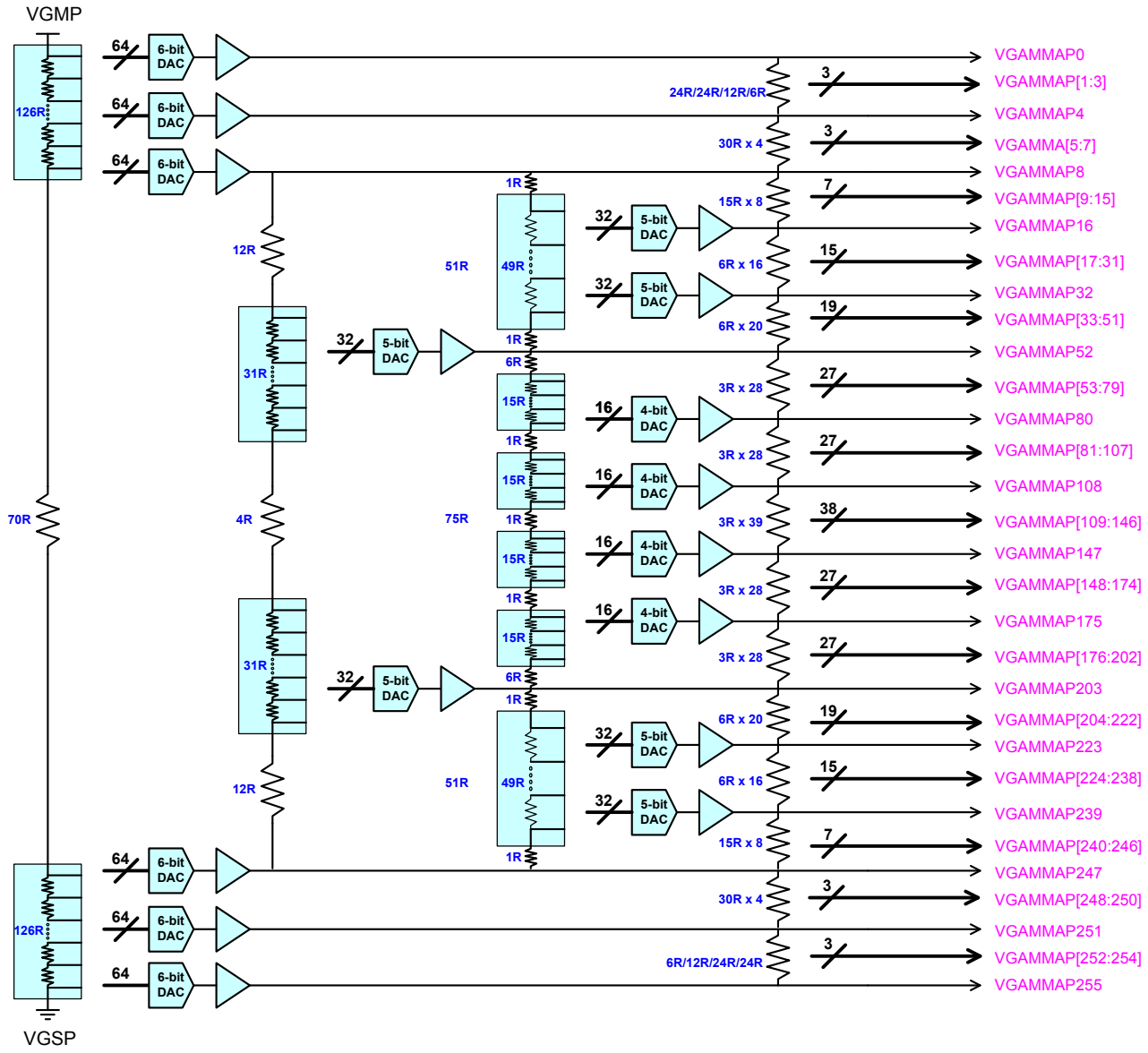


Figure 183: Positive Gamma Control (E0h)

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Negative Gamma Control (E1h)

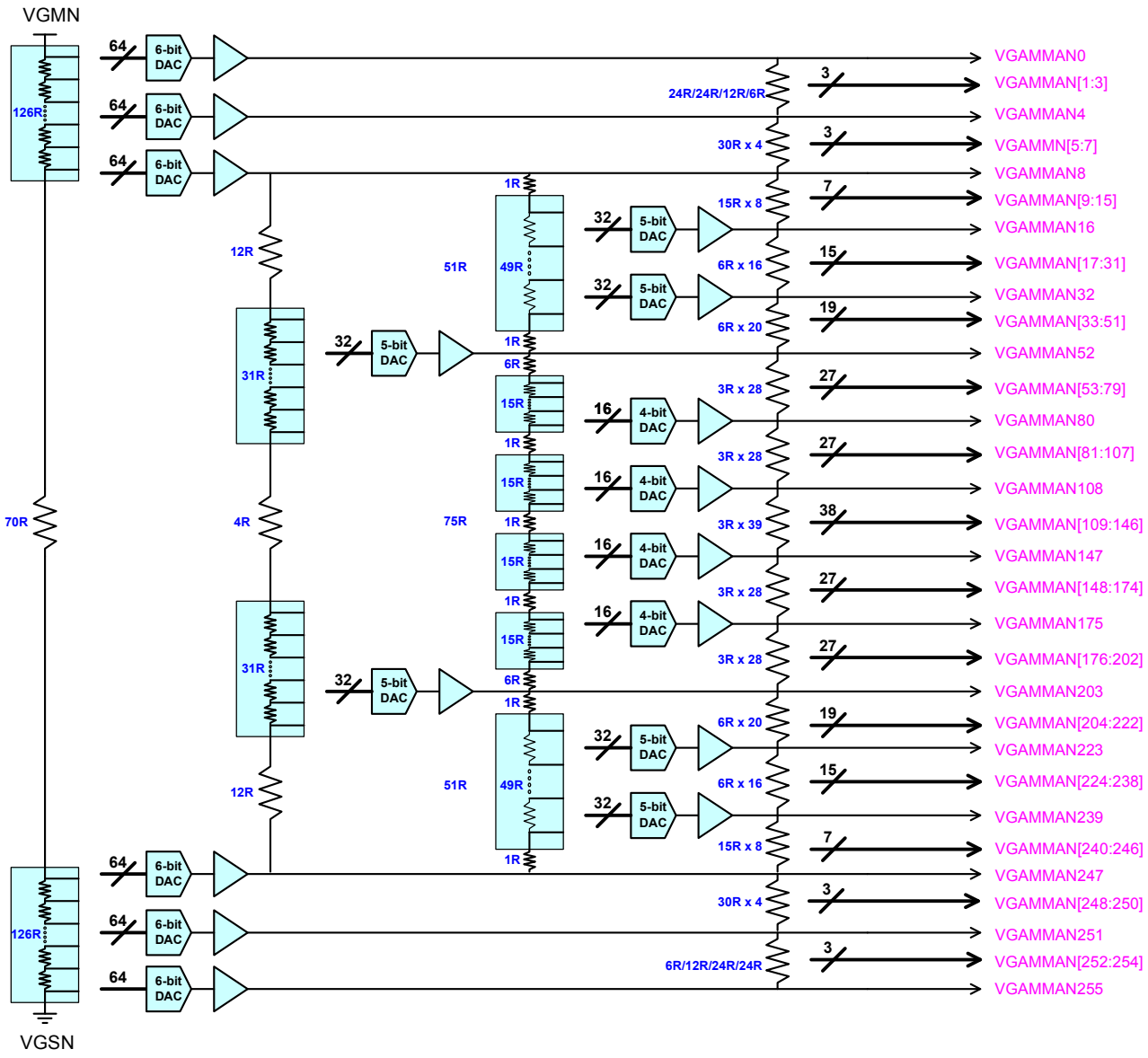


Figure 184: Negative Gamma Control (E1h)

Positive polarity	Resister stream	Gamma 256 grayscale voltage calculation formula
VGAMMAP0	24R	VGSP+ΔVDHP(322R-2R*VP0[5:0])/322R , △VDHP=(VGMP-VGSP)
VGAMMAP1	24R	VGAMMAP4+(VGAMMP0-VGAMMAP4)*(6R+12R+24R)/(42R+RCGPH+24R)
VGAMMAP2	12R	VGAMMAP4+(VGAMMP0-VGAMMAP4)*(6R+12R)/(42R+RCGPH+24R)
VGAMMAP3	6R	VGAMMAP4+(VGAMMP0-VGAMMAP4)*(6R)/(42R+RCGPH+24R)
VGAMMAP4	30R	VGSP+ΔVDHP(322R-2R*VP4[5:0])/322R , △VDHP=(VGMP-VGSP)
VGAMMAP5	30R	VGAMMAP8+((VGAMMAP4-VGAMMAP8)*(90R/120R)
VGAMMAP6	30R	VGAMMAP8+((VGAMMAP4-VGAMMAP8)*(60R/120R)
VGAMMAP7	30R	VGAMMAP8+((VGAMMAP4-VGAMMAP8)*(30R/120R)

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VGAMMAP8	15R	VGSP+ΔVDHP(322R-2R*VP8[5:0])/322R , △VDHP=(VGMP-VGSP)
VGAMMAP9	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(105R/120R)
VGAMMAP10	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(90R/120R)
VGAMMAP11	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(75R/120R)
VGAMMAP12	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(60R/120R)
VGAMMAP13	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(45R/120R)
VGAMMAP14	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(30R/120R)
VGAMMAP15	15R	VGAMMAP16+((VGAMMAP8-VGAMMAP16)*(15R/120R)
VGAMMAP16	6R	VGAMMAP52+(VGAMMAP8-VGAMMAP52)*((50R-1R*VP16[4:0])/51R)
VGAMMAP17	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(90R/96R)
VGAMMAP18	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(84R/96R)
VGAMMAP19	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(78R/96R)
VGAMMAP20	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(72R/96R)
VGAMMAP21	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(66R/96R)
VGAMMAP22	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(60R/96R)
VGAMMAP23	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(54R/96R)
VGAMMAP24	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(48R/96R)
VGAMMAP25	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(42R/96R)
VGAMMAP26	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(36R/96R)
VGAMMAP27	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(30R/96R)
VGAMMAP28	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(24R/96R)
VGAMMAP29	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(18R/96R)
VGAMMAP30	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(12R/96R)
VGAMMAP31	6R	VGAMMAP32+((VGAMMAP16-VGAMMAP32)*(6R/96R)
VGAMMAP32	6R	VGAMMAP52+(VGAMMAP8-VGAMMAP52)*((32R-1R*VP32[4:0])/51R)
VGAMMAP33	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(114R/120R)
VGAMMAP34	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(108R/120R)
VGAMMAP35	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(102R/120R)
VGAMMAP36	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(96R/120R)
VGAMMAP37	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(90R/120R)
VGAMMAP38	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(84R/120R)
VGAMMAP39	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(78R/120R)
VGAMMAP40	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(72R/120R)
VGAMMAP41	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(66R/120R)
VGAMMAP42	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(60R/120R)
VGAMMAP43	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(54R/120R)
VGAMMAP44	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(48R/120R)

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VGAMMAP45	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(42R/120R)
VGAMMAP46	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(36R/120R)
VGAMMAP47	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(30R/120R)
VGAMMAP48	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(24R/120R)
VGAMMAP49	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(18R/120R)
VGAMMAP50	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(12R/120R)
VGAMMAP51	6R	VGAMMAP52+((VGAMMAP32-VGAMMAP52)*(6R/120R)
VGAMMAP52	3R	VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((77R-0R)/90R)) , VP52[4:0] =0(Dec) VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((77R-2R)/90R)) , VP52[4:0] =1(Dec) VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((77R-4R)/90R)) , VP52[4:0] =2(Dec) VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((74R-1R*VP52[4:0])/90R)) , else
VGAMMAP53	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(81R/84R)
VGAMMAP54	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(78R/84R)
VGAMMAP55	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(75R/84R)
VGAMMAP56	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(72R/84R)
VGAMMAP57	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(69R/84R)
VGAMMAP58	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(66R/84R)
VGAMMAP59	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(63R/84R)
VGAMMAP60	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(60R/84R)
VGAMMAP61	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(57R/84R)
VGAMMAP62	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(54R/84R)
VGAMMAP63	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(51R/84R)
VGAMMAP64	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(48R/84R)
VGAMMAP65	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(45R/84R)
VGAMMAP66	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(42R/84R)
VGAMMAP67	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(39R/84R)
VGAMMAP68	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(36R/84R)
VGAMMAP69	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(33R/84R)
VGAMMAP70	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(30R/84R)
VGAMMAP71	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(27R/84R)
VGAMMAP72	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(24R/84R)
VGAMMAP73	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(21R/84R)
VGAMMAP74	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(18R/84R)
VGAMMAP75	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(15R/84R)
VGAMMAP76	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(12R/84R)
VGAMMAP77	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(9R/84R)
VGAMMAP78	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(6R/84R)

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VGAMMAP79	3R	VGAMMAP80+((VGAMMAP52-VGAMMAP80)*(3R/84R)
VGAMMAP80	3R	VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((68R-0R)/75R) , VP80[3:0] =0(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((68R-2R)/75R) , VP80[3:0] =1(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((68R-4R)/75R) , VP80[3:0] =2(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((65R-1R*VP80[3:0])/75R) , else
VGAMMAP81	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(81R+/84R)
VGAMMAP82	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(78R/84R)
VGAMMAP83	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(75R/84R)
VGAMMAP84	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(72R/84R)
VGAMMAP85	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(69R/84R)
VGAMMAP86	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(66R/84R)
VGAMMAP87	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(63R/84R)
VGAMMAP88	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(60R/84R)
VGAMMAP89	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(57R/84R)
VGAMMAP90	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(54R/84R)
VGAMMAP91	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(51R/84R)
VGAMMAP92	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(48R/84R)
VGAMMAP93	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(45R/84R)
VGAMMAP94	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(42R/84R)
VGAMMAP95	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(39R/84R)
VGAMMAP96	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(36R/84R)
VGAMMAP97	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(33R/84R)
VGAMMAP98	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(30R/84R)
VGAMMAP99	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(27R/84R)
VGAMMAP100	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(24R/84R)
VGAMMAP101	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(21R/84R)
VGAMMAP102	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(18R/84R)
VGAMMAP103	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(15R/84R)
VGAMMAP104	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(12R/84R)
VGAMMAP105	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(9R/84R)
VGAMMAP106	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(6R/84R)
VGAMMAP107	3R	VGAMMAP108+((VGAMMAP80-VGAMMAP108)*(3R/84R)
VGAMMAP108	3R	VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((52R-0R/75R) , VP108[3:0] =0(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((52R-2R/75R) , VP108[3:0] =1(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((52R-4R/75R) , VP108[3:0] =2(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((49R-1R*VP108[3:0])/75R) , else
VGAMMAP109	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(114R/117R)

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VGAMMAP110	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(111R/117R)
VGAMMAP111	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(108R/117R)
VGAMMAP112	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(105R/117R)
VGAMMAP113	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(102R/117R)
VGAMMAP114	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(99R/117R)
VGAMMAP115	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(96R/117R)
VGAMMAP116	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(93R/117R)
VGAMMAP117	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(90R/117R)
VGAMMAP118	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(87R/117R)
VGAMMAP119	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(84R/117R)
VGAMMAP120	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(81R/117R)
VGAMMAP121	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(78R/117R)
VGAMMAP122	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(75R/117R)
VGAMMAP123	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(72R/117R)
VGAMMAP124	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(69R/117R)
VGAMMAP125	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(66R/117R)
VGAMMAP126	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(63R/117R)
VGAMMAP127	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(60R/117R)
VGAMMAP128	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(57R/117R)
VGAMMAP129	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(54R/117R)
VGAMMAP130	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(51R/117R)
VGAMMAP131	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(48R/117R)
VGAMMAP132	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(45R/117R)
VGAMMAP133	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(42R/117R)
VGAMMAP134	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(39R/117R)
VGAMMAP135	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(36R/117R)
VGAMMAP136	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(33R/117R)
VGAMMAP137	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(30R/117R)
VGAMMAP138	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(27R/117R)
VGAMMAP139	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(24R/117R)
VGAMMAP140	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(21R/117R)
VGAMMAP141	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(18R/117R)
VGAMMAP142	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(15R/117R)
VGAMMAP143	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(12R/117R)
VGAMMAP144	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(9R/117R)
VGAMMAP145	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(6R/117R)
VGAMMAP146	3R	VGAMMAP147+((VGAMMAP108-VGAMMAP147)*(3R/117R)

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VGAMMAP147	3R	VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((17R+1R*VP147[3:0])/75R) , else VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((17R+31R/75R) , VP147[3:0] =13(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((17R+33R/75R) , VP147[3:0] =14(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((17R+35R/75R) , VP147[3:0] =15(Dec)
VGAMMAP148	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(81R/84R)
VGAMMAP149	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(78R/84R)
VGAMMAP150	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(75R/84R)
VGAMMAP151	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(72R/84R)
VGAMMAP152	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(69R/84R)
VGAMMAP153	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(66R/84R)
VGAMMAP154	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(63R/84R)
VGAMMAP155	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(60R/84R)
VGAMMAP156	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(57R/84R)
VGAMMAP157	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(54R/84R)
VGAMMAP158	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(51R/84R)
VGAMMAP159	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(48R/84R)
VGAMMAP160	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(45R/84R)
VGAMMAP161	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(42R/84R)
VGAMMAP162	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(39R/84R)
VGAMMAP163	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(36R/84R)
VGAMMAP164	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(33R/84R)
VGAMMAP165	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(30R/84R)
VGAMMAP166	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(27R/84R)
VGAMMAP167	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(24R/84R)
VGAMMAP168	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(21R/84R)
VGAMMAP169	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(18R/84R)
VGAMMAP170	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(15R/84R)
VGAMMAP171	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(12R/84R)
VGAMMAP172	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(9R/84R)
VGAMMAP173	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(6R/84R)
VGAMMAP174	3R	VGAMMAP175+((VGAMMAP147-VGAMMAP175)*(3R/84R)
VGAMMAP175	3R	VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((1R+1R*VP175[3:0])/75R) , else VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((1R+15R/75R) , VP175[3:0] =13(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((1R+17R/75R) , VP175[3:0] =14(Dec) VGAMMAP203+(VGAMMAP52-VGAMMAP203)*((1R+19R/75R) , VP175[3:0] =15(Dec)
VGAMMAP176	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(81R/84R)
VGAMMAP177	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(78R/84R)

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VGAMMAP178	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(75R/84R)
VGAMMAP179	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(72R/84R)
VGAMMAP180	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(69R/84R)
VGAMMAP181	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(66R/84R)
VGAMMAP182	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(63R/84R)
VGAMMAP183	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(60R/84R)
VGAMMAP184	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(57R/84R)
VGAMMAP185	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(54R/84R)
VGAMMAP186	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(51R/84R)
VGAMMAP187	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(48R/84R)
VGAMMAP188	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(45R/84R)
VGAMMAP189	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(42R/84R)
VGAMMAP190	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(39R/84R)
VGAMMAP191	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(36R/84R)
VGAMMAP192	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(33R/84R)
VGAMMAP193	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(30R/84R)
VGAMMAP194	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(27R/84R)
VGAMMAP195	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(24R/84R)
VGAMMAP196	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(21R/84R)
VGAMMAP197	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(18R/84R)
VGAMMAP198	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(15R/84R)
VGAMMAP199	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(12R/84R)
VGAMMAP200	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(9R/84R)
VGAMMAP201	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(6R/84R)
VGAMMAP202	3R	VGAMMAP203+((VGAMMAP175-VGAMMAP203)*(3R/84R)
VGAMMAP203	6R	VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((8R+1R*VP203[4:0])/90R)) , else VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((8R+38R/90R)) , VP203[4:0] =29(Dec) VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((8R+40R/90R)) , VP203[4:0] =30(Dec) VGAMMAP247+(VGAMMAP8-VGAMMAP247)*((8R+42R/90R)) , VP203[4:0] =31(Dec)
VGAMMAP204	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(114R/120R)
VGAMMAP205	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(108R/120R)
VGAMMAP206	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(102R/120R)
VGAMMAP207	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(96R/120R)
VGAMMAP208	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(90R/120R)
VGAMMAP209	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(84R/120R)
VGAMMAP210	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(78R/120R)
VGAMMAP211	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(72R/120R)

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VGAMMAP212	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(66R/120R)
VGAMMAP213	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(60R/120R)
VGAMMAP214	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(54R/120R)
VGAMMAP215	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(48R/120R)
VGAMMAP216	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(42R/120R)
VGAMMAP217	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(36R/120R)
VGAMMAP218	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(30R/120R)
VGAMMAP219	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(24R/120R)
VGAMMAP220	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(18R/120R)
VGAMMAP221	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(12R/120R)
VGAMMAP222	6R	VGAMMAP223+((VGAMMAP203-VGAMMAP223)*(6R/120R)
VGAMMAP223	6R	VGAMMAP247+(VGAMMAP203-VGAMMAP247)*((19R+1R*VP223[4:0])/51R)
VGAMMAP224	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(90R/96R)
VGAMMAP225	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(84R/96R)
VGAMMAP226	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(78R/96R)
VGAMMAP227	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(72R/96R)
VGAMMAP228	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(66R/96R)
VGAMMAP229	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(60R/96R)
VGAMMAP230	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(54R/96R)
VGAMMAP231	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(48R/96R)
VGAMMAP232	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(42R/96R)
VGAMMAP233	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(36R/96R)
VGAMMAP234	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(30R/96R)
VGAMMAP235	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(24R/96R)
VGAMMAP236	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(18R/96R)
VGAMMAP237	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(12R/96R)
VGAMMAP238	6R	VGAMMAP239+((VGAMMAP223-VGAMMAP239)*(6R/96R)
VGAMMAP239	15R	VGAMMAP247+(VGAMMAP203-VGAMMAP247)*((1R+1R*VP239[4:0])/51R)
VGAMMAP240	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(105R/120R)
VGAMMAP241	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(90R/120R)
VGAMMAP242	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(75R/120R)
VGAMMAP243	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(60R/120R)
VGAMMAP244	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(45R/120R)
VGAMMAP245	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(30R/120R)
VGAMMAP246	15R	VGAMMAP247+((VGAMMAP239-VGAMMAP247)*(15R/120R)
VGAMMAP247	30R	VGSP+ΔVDHP(3R*VP247[5:0])/322R , △VDHP=(VGMP-VGSP)
VGAMMAP248	30R	VGAMMAP251+((VGAMMAP247-VGAMMAP251)*(90R/120R)

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VGAMMAP249	30R	VGAMMAP251+((VGAMMAP247-VGAMMAP251)*(60R/120R)
VGAMMAP250	30R	VGAMMAP251+((VGAMMAP247-VGAMMAP251)*(30R/120R)
VGAMMAP251	6R	VGSP+ΔVDHP(3R*VP251[5:0])/322R , △VDHP=(VGMP-VGSP)
VGAMMAP252	12R	VGAMMAP255+(VGAMMP251-VGAMMAP255)*(RCGPL+6R+12R)/(42R+RCGNH)
VGAMMAP253	24R	VGAMMAP255+(VGAMMP251-VGAMMAP255)*(RCGPL+6R)/(42R+RCGNH)
VGAMMAP254	24R	VGAMMAP255+(VGAMMP251-VGAMMAP255)*(RCGPL)/(42R+RCGPL)
VGAMMAP255	0R	VGSP+ΔVDHP(3R*VP255[5:0])/322R , △VDHP=(VGMP-VGSP)

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Negative polarity	Resister stream	Gamma 256 grayscale voltage calculation formula
VGAMMAN255	24R	VGMN+ΔVDHN(322R-2R*VN255[5:0])/322R , △VDHN=(VGSN-VGMN)
VGAMMAN254	24R	VGAMMAN251+(VGAMMN255-VGAMMAN251)*(6R+12R+24R)/(42R+RCGNL+24R)
VGAMMAN253	12R	VGAMMAN251+(VGAMMN255-VGAMMAN251)*(6R+12R)/(42R+RCGNL+24R)
VGAMMAN252	6R	VGAMMAN251+(VGAMMN255-VGAMMAN251)*(6R)/(42R+RCGNL+24R)
VGAMMAN251	30R	VGMN+ΔVDHN(322R-2R*VN251[5:0])/322R , △VDHN=(VGSN-VGMN)
VGAMMAN250	30R	VGAMMAN247+((VGAMMAN251-VGAMMAN247)*(90R/120R)
VGAMMAN249	30R	VGAMMAN247+((VGAMMAN251-VGAMMAN247)*(60R/120R)
VGAMMAN248	30R	VGAMMAN247+((VGAMMAN251-VGAMMAN247)*(30R/120R)
VGAMMAN247	15R	VGMN+ΔVDHN(322R-2R*VN247[5:0])/322R , △VDHN=(VGSN-VGMN)
VGAMMAN246	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(105R/120R)
VGAMMAN245	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(90R/120R)
VGAMMAN244	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(75R/120R)
VGAMMAN243	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(60R/120R)
VGAMMAN242	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(45R/120R)
VGAMMAN241	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(30R/120R)
VGAMMAN240	15R	VGAMMAN239+((VGAMMAN247-VGAMMAN239)*(15R/120R)
VGAMMAN239	6R	VGAMMAN203+(VGAMMAN247-VGAMMAN203)*(50R-1R*VN239[4:0])/51R)
VGAMMAN238	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(90R/96R)
VGAMMAN237	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(84R/96R)
VGAMMAN236	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(78R/96R)
VGAMMAN235	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(72R/96R)
VGAMMAN234	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(66R/96R)
VGAMMAN233	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(60R/96R)
VGAMMAN232	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(54R/96R)
VGAMMAN231	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(48R/96R)
VGAMMAN230	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(42R/96R)
VGAMMAN229	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(36R/96R)
VGAMMAN228	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(30R/96R)
VGAMMAN227	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(24R/96R)
VGAMMAN226	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(18R/96R)
VGAMMAN225	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(12R/96R)
VGAMMAN224	6R	VGAMMAN232+((VGAMMAN239-VGAMMAN232)*(6R/96R)
VGAMMAN223	6R	VGAMMAN203+(VGAMMAN247-VGAMMAN203)*((32R-1R*VN223[4:0])/51R)
VGAMMAN222	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(114R/120R)
VGAMMAN221	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(108R/120R)

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VGAMMAN220	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(102R/120R)
VGAMMAN219	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(96R/120R)
VGAMMAN218	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(90R/120R)
VGAMMAN217	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(84R/120R)
VGAMMAN216	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(78R/120R)
VGAMMAN215	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(72R/120R)
VGAMMAN214	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(66R/120R)
VGAMMAN213	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(60R/120R)
VGAMMAN212	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(54R/120R)
VGAMMAN211	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(48R/120R)
VGAMMAN210	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(42R/120R)
VGAMMAN209	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(36R/120R)
VGAMMAN208	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(30R/120R)
VGAMMAN207	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(24R/120R)
VGAMMAN206	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(18R/120R)
VGAMMAN205	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(12R/120R)
VGAMMAN204	6R	VGAMMAN203+((VGAMMAN223-VGAMMAN203)*(6R/120R)
VGAMMAN203	3R	VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((77R-0R)/90R) , VP203[4:0] =0 (Dec) VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((77R-2R)/90R) , VP203[4:0] =1 (Dec) VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((77R-4R)/90R) , VP203[4:0] =2 (Dec) VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((74R-1R*VN203[4:0])/90R)) , else
VGAMMAN202	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(81R/84R)
VGAMMAN201	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(78R/84R)
VGAMMAN200	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(75R/84R)
VGAMMAN199	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(72R/84R)
VGAMMAN198	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(69R/84R)
VGAMMAN197	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(66R/84R)
VGAMMAN196	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(63R/84R)
VGAMMAN195	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(60R/84R)
VGAMMAN194	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(57R/84R)
VGAMMAN193	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(54R/84R)
VGAMMAN192	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(51R/84R)
VGAMMAN191	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(48R/84R)
VGAMMAN190	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(45R/84R)
VGAMMAN189	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(42R/84R)
VGAMMAN188	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(39R/84R)
VGAMMAN187	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(36R/84R)

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VGAMMAN186	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(33R/84R)
VGAMMAN185	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(30R/84R)
VGAMMAN184	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(27R/84R)
VGAMMAN183	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(24R/84R)
VGAMMAN182	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(21R/84R)
VGAMMAN181	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(18R/84R)
VGAMMAN180	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(15R/84R)
VGAMMAN179	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(12R/84R)
VGAMMAN178	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(9R/84R)
VGAMMAN177	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(6R/84R)
VGAMMAN176	3R	VGAMMAN175+((VGAMMAN203-VGAMMAN175)*(3R/84R)
VGAMMAN175	3R	VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((68R-0R)/76R) , VP175[3:0] =0 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((68R-2R)/76R) , VP175[3:0] =1 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((68R-4R)/76R) , VP175[3:0] =2 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((65R-1R*VN175[3:0])/76R) , else
VGAMMAN174	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(81R+/84R)
VGAMMAN173	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(78R/84R)
VGAMMAN172	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(75R/84R)
VGAMMAN171	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(72R/84R)
VGAMMAN170	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(69R/84R)
VGAMMAN169	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(66R/84R)
VGAMMAN168	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(63R/84R)
VGAMMAN167	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(60R/84R)
VGAMMAN166	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(57R/84R)
VGAMMAN165	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(54R/84R)
VGAMMAN164	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(51R/84R)
VGAMMAN163	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(48R/84R)
VGAMMAN162	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(45R/84R)
VGAMMAN161	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(42R/84R)
VGAMMAN160	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(39R/84R)
VGAMMAN159	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(36R/84R)
VGAMMAN158	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(33R/84R)
VGAMMAN157	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(30R/84R)
VGAMMAN156	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(27R/84R)
VGAMMAN155	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(24R/84R)
VGAMMAN154	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(21R/84R)
VGAMMAN153	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(18R/84R)

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VGAMMAN152	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(15R/84R)
VGAMMAN151	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(12R/84R)
VGAMMAN150	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(9R/84R)
VGAMMAN149	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(6R/84R)
VGAMMAN148	3R	VGAMMAN147+((VGAMMAN175-VGAMMAN147)*(3R/84R)
VGAMMAN147	3R	VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((52R-0R)/76R) , VP147[3:0] =0 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((52R-2R)/76R) , VP147[3:0] =1 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((52R-4R)/76R) , VP147[3:0] =2 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((49R-1R*VN147[3:0])/76R) , else
VGAMMAN146	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(114R/117R)
VGAMMAN145	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(111R/117R)
VGAMMAN144	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(108R/117R)
VGAMMAN143	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(105R/117R)
VGAMMAN142	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(102R/117R)
VGAMMAN141	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(99R/117R)
VGAMMAN140	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(96R/117R)
VGAMMAN139	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(93R/117R)
VGAMMAN138	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(90R/117R)
VGAMMAN137	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(87R/117R)
VGAMMAN136	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(84R/117R)
VGAMMAN135	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(81R/117R)
VGAMMAN134	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(78R/117R)
VGAMMAN133	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(75R/117R)
VGAMMAN132	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(72R/117R)
VGAMMAN131	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(69R/117R)
VGAMMAN130	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(66R/117R)
VGAMMAN129	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(63R/117R)
VGAMMAN128	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(60R/117R)
VGAMMAN127	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(57R/117R)
VGAMMAN126	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(54R/117R)
VGAMMAN125	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(51R/117R)
VGAMMAN124	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(48R/117R)
VGAMMAN123	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(45R/117R)
VGAMMAN122	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(42R/117R)
VGAMMAN121	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(39R/117R)
VGAMMAN120	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(36R/117R)
VGAMMAN119	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(33R/117R)

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VGAMMAN118	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(30R/117R)
VGAMMAN117	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(27R/117R)
VGAMMAN116	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(24R/117R)
VGAMMAN115	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(21R/117R)
VGAMMAN114	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(18R/117R)
VGAMMAN113	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(15R/117R)
VGAMMAN112	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(12R/117R)
VGAMMAN111	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(9R/117R)
VGAMMAN110	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(6R/117R)
VGAMMAN109	3R	VGAMMAN108+((VGAMMAN147-VGAMMAN108)*(3R/117R)
VGAMMAN108	3R	VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((17R+1R*VN108[3:0])/76R) , else VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((17R+31R)/76R) , VP108[3:0] =13 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((17R+33R)/76R) , VP108[3:0] =14 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((17R+35R)/76R) , VP108[3:0] =15 (Dec)
VGAMMAN107	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(81R/84R)
VGAMMAN106	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(78R/84R)
VGAMMAN105	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(75R/84R)
VGAMMAN104	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(72R/84R)
VGAMMAN103	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(69R/84R)
VGAMMAN102	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(66R/84R)
VGAMMAN101	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(63R/84R)
VGAMMAN100	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(60R/84R)
VGAMMAN99	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(57R/84R)
VGAMMAN98	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(54R/84R)
VGAMMAN97	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(51R/84R)
VGAMMAN96	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(48R/84R)
VGAMMAN95	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(45R/84R)
VGAMMAN94	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(42R/84R)
VGAMMAN93	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(39R/84R)
VGAMMAN92	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(36R/84R)
VGAMMAN91	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(33R/84R)
VGAMMAN90	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(30R/84R)
VGAMMAN89	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(27R/84R)
VGAMMAN88	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(24R/84R)
VGAMMAN87	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(21R/84R)
VGAMMAN86	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(18R/84R)
VGAMMAN85	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(15R/84R)

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VGAMMAN84	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(12R/84R)
VGAMMAN83	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(9R/84R)
VGAMMAN82	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(6R/84R)
VGAMMAN81	3R	VGAMMAN80+((VGAMMAN108-VGAMMAN80)*(3R/84R)
VGAMMAN80	3R	VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((1R+1R*VN80[3:0])/76R) , else VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((1R+15R)/76R) , VP80[3:0] =13 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((1R+17R)/76R) , VP80[3:0] =14 (Dec) VGAMMAN52+(VGAMMAN203-VGAMMAN52)*((1R+19R)/76R) , VP80[3:0] =15 (Dec)
VGAMMAN79	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(81R/84R)
VGAMMAN78	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(78R/84R)
VGAMMAN77	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(75R/84R)
VGAMMAN76	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(72R/84R)
VGAMMAN75	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(69R/84R)
VGAMMAN74	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(66R/84R)
VGAMMAN73	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(63R/84R)
VGAMMAN72	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(60R/84R)
VGAMMAN71	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(57R/84R)
VGAMMAN70	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(54R/84R)
VGAMMAN69	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(51R/84R)
VGAMMAN68	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(48R/84R)
VGAMMAN67	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(45R/84R)
VGAMMAN66	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(42R/84R)
VGAMMAN65	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(39R/84R)
VGAMMAN64	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(36R/84R)
VGAMMAN63	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(33R/84R)
VGAMMAN62	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(30R/84R)
VGAMMAN61	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(27R/84R)
VGAMMAN60	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(24R/84R)
VGAMMAN59	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(21R/84R)
VGAMMAN58	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(18R/84R)
VGAMMAN57	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(15R/84R)
VGAMMAN56	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(12R/84R)
VGAMMAN55	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(9R/84R)
VGAMMAN54	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(6R/84R)
VGAMMAN53	3R	VGAMMAN52+((VGAMMAN80-VGAMMAN52)*(3R/84R)
VGAMMAN52	6R	VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((8R+1R*VN52[4:0])/90R)) , else VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((8R+38R)/90R)) , VP52[4:0] =29 (Dec)

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		VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((8R+40R)/90R)) , VP52[4:0] =30 (Dec) VGAMMAN8+(VGAMMAN247-VGAMMAN8)*((8R+42R)/90R)) , VP52[4:0] =31 (Dec)
VGAMMAN51	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(114R/120R)
VGAMMAN50	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(108R/120R)
VGAMMAN49	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(102R/120R)
VGAMMAN48	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(96R/120R)
VGAMMAN47	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(90R/120R)
VGAMMAN46	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(84R/120R)
VGAMMAN45	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(78R/120R)
VGAMMAN44	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(72R/120R)
VGAMMAN43	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(66R/120R)
VGAMMAN42	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(60R/120R)
VGAMMAN41	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(54R/120R)
VGAMMAN40	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(48R/120R)
VGAMMAN39	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(42R/120R)
VGAMMAN38	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(36R/120R)
VGAMMAN37	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(30R/120R)
VGAMMAN36	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(24R/120R)
VGAMMAN35	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(18R/120R)
VGAMMAN34	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(12R/120R)
VGAMMAN33	6R	VGAMMAN32+((VGAMMAN52-VGAMMAN32)*(6R/120R)
VGAMMAN32	6R	VGAMMAN8+(VGAMMAN52-VGAMMAN8)*((19R+1R*VN32[4:0])/51R)
VGAMMAN31	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(90R/96R)
VGAMMAN30	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(84R/96R)
VGAMMAN29	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(78R/96R)
VGAMMAN28	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(72R/96R)
VGAMMAN27	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(66R/96R)
VGAMMAN26	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(60R/96R)
VGAMMAN25	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(54R/96R)
VGAMMAN24	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(48R/96R)
VGAMMAN23	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(42R/96R)
VGAMMAN22	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(36R/96R)
VGAMMAN21	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(30R/96R)
VGAMMAN20	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(24R/96R)
VGAMMAN19	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(18R/96R)
VGAMMAN18	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(12R/96R)
VGAMMAN17	6R	VGAMMAN16+((VGAMMAN32-VGAMMAN16)*(6R/96R)

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VGAMMAN16	15R	VGAMMAN8+(VGAMMAN52-VGAMMAN8)*((1R+1R*VN16[4:0])/51R)
VGAMMAN15	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(105R/120R)
VGAMMAN14	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(90R/120R)
VGAMMAN13	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(75R/120R)
VGAMMAN12	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(60R/120R)
VGAMMAN11	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(45R/120R)
VGAMMAN10	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(30R/120R)
VGAMMAN9	15R	VGAMMAN8+((VGAMMAN16-VGAMMAN8)*(15R/120R)
VGAMMAN8	30R	VGMN+ΔVDHN(3R*VN8[5:0])/322R , △VDHN=(VGSN-VGMN)
VGAMMAN7	30R	VGAMMAN4+((VGAMMAN8-VGAMMAN4)*(90R/120R)
VGAMMAN6	30R	VGAMMAN4+((VGAMMAN8-VGAMMAN4)*(60R/120R)
VGAMMAN5	30R	VGAMMAN4+((VGAMMAN8-VGAMMAN4)*(30R/120R)
VGAMMAN4	6R	VGMN+ΔVDHN(3R*VN4[5:0])/322R , △VDHN=(VGSN-VGMN)
VGAMMAN3	12R	VGAMMAN0+(VGAMMN4-VGAMMAN0)*(RCGNH+6R+12R)/(42R+RCGNH)
VGAMMAN2	24R	VGAMMAN0+(VGAMMN4-VGAMMAN0)*(RCGNH+6R)/(42R+RCGNH)
VGAMMAN1	24R	VGAMMAN0+(VGAMMN4-VGAMMAN0)*(RCGNH)/(42R+RCGNH)
VGAMMAN0		VGMN+ΔVDHN(3R*VN0[5:0])/322R , △VDHN=(VGSN-VGMN)

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18. Deep Standby Mode Setting

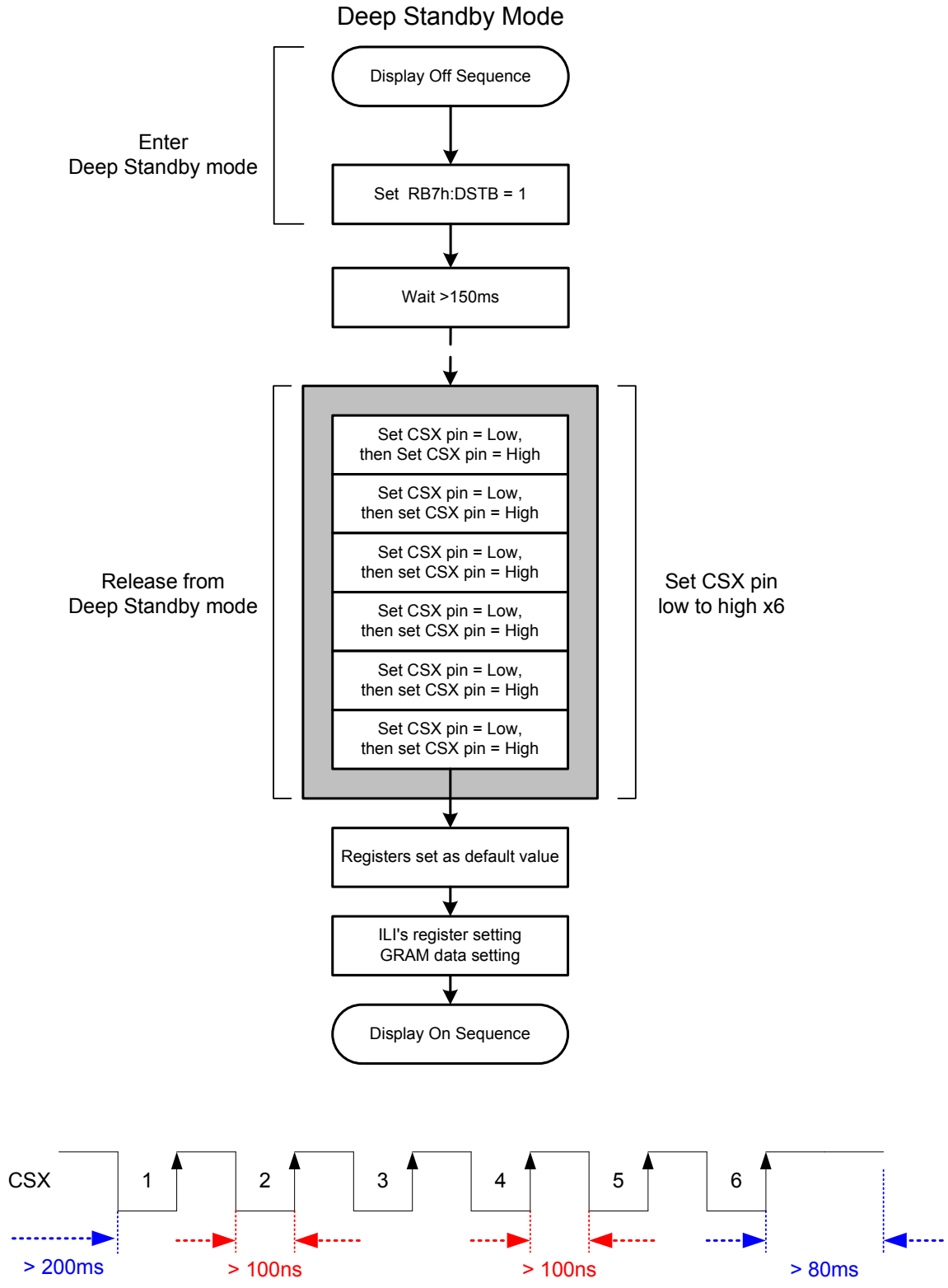


Figure 185: Deep Standby Mode Entry/Exit

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19. Electrical Characteristics

19.1. Absolute Maximum Ratings

The absolute maximum rating is listed in Table 49. When the ILI9806G is used beyond the absolute maximum ratings, it may be permanently damaged. It is strongly recommended that use the ILI9806G within the following electrical characteristics limit for normal operation. If these electrical characteristic conditions are exceeded during normal operation, the ILI9806G will malfunction and cause poor reliability.

Table 49: Absolute Maximum Ratings

Item	Symbol	Unit	Value
Supply voltage(Analog)	VDDA ~ VSSA	V	-0.3 ~ +4.6
Supply voltage (I/O)	VDDI ~ VSSI	V	-0.3 ~ +4.6
Supply voltage (DSI I/O)	VDDAM ~ VSSAM	V	-0.3 ~ +4.6
OTP Supply voltage	VPP ~ VSSA	V	7.5
Supply voltage	AVDD ~ DSSA	V	-0.3 ~ +6.6
Supply voltage	AVEE ~ DSSA	V	0.3 ~ -6.6
Supply voltage	VGH ~ DSSA	V	-0.3 ~ +25
Supply voltage	VGL ~ DSSA	V	0.3 ~ -16
Driver supply voltage	VDDA – VCL	V	≦ 6.0V
Driver supply voltage	VGH-VGL	V	≦ 32.0V
Input voltage	V_{IN}	V	-0.3 ~ VDDI + 0.3
HS Input voltage	V_{HSIN}	V	-0.3 ~ + 2
Operating temperature	Topr	℃	-30 ~ +70
Storage temperature	Tstg	℃	-55 ~ +110

Note:

Even if the absolute maximum rating of one of the above parameters is exceeded only for a short while, the quality of the product may be degraded. Therefore, be sure to use the product within the range of the absolute maximum ratings.

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19.2. DC Characteristics for Panel Driving

Item	Symbol	Condition	Min.	Typ.	Max.	Unit	Note
Power & Operation Voltage							
Analog operating voltage	VDDA	-	2.5	2.8	3.3	V	
Logic operating voltage	VDDAM	-	2.5	2.8	3.3	V	
Logic operating voltage	VDDI	-	1.65	1.8	3.3	V	Note1,2
OTP Supply voltage	VPP	-		7.5		V	Note1
Logic High level input voltage	V_{IH}	-	0.7*VDDI		VDDI	V	Note1
Logic Low level input voltage	V_{IL}	-	-0.3		0.3*VDDI	V	Note1
Logic High level output voltage TE, SDO (SDA) , LEDPWM	V_{OH}	IOH = -1.0mA	0.8*VDDI		VDDI	V	Note1
Logic Low level output voltage TE, SDO (SDA) , LEDPWM	V_{OL}	IOL = +1.0mA	0		0.2*VDDI	V	Note1
Gate Driver High Voltage	VGH	-	12.0	-	20	V	
Gate Driver Low Voltage	VGL	-	-15.0	-	-7.0	V	
Driver Supply Voltage	-	|VGH-VGL|	19	-	32	V	
VCOM Operation							
DC VCOM Amplitude Voltage	VCOM	-	-3.0	-	0	V	Note3
Source Driver							
Source Output Range	V_{SOUT}	-	0.1	-	VGMP-0.1	V	Note4
Positive Gamma Reference Voltage	VGMP	-	3.0	-	6.1875	V	
Negative Gamma Reference Voltage	VGMN	-	-6.1875	-	-3.0	V	
Source Output Setting Time	Tr	Below with 99% precision	-	15	20	us	Note3.4
Output Deviation Voltage (Source Output channel)	Vdev	Sout>=4.2V Sout<=0.8V	-	-	20	mV	Note3
		4.2V>Sout>0.8V	-	-	15	mV	-
Output Offset Voltage	V_{OFFSET}	-	-	-	35	mV	Note3
Booster Operation							
Booster (VDDBx2.5) Voltage	AVDD	-			6.5	V	
Booster (VDDBx-2.5) Voltage	AVEE	-	-6.5			V	
Booster (VDDBx2 Drop Voltage)	VDDBx2 drop	loading=1mA	-	-	5	%	
Gate Driver High Voltage	VGH	-	10.0	-	20	V	
Gate Driver Low Voltage	VGL	-	-15.0	-	-6.0	V	

Notes:

1. Ta = -30 to 70 ℃ (no damage up to 85 ℃), VDDI=1.65V to 3.3V, VDDA=2.5V to 3.3V, VSSAM=GND =0V.
2. Supply the digital VDDI voltage equal to or less than the analog VDDA voltage.
3. Source channel loading = 10pF/channel, Gate channel loading = 50pF/channel
4. The Max. Value is between with Note 3 measure point and the Gamma setting value

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19.3. DSI DC Characteristics

The DSI uses different state codes which depend on DC voltage levels of the clock and data lanes. The meaning of the state codes is defined on the following table.

State Code	Line DC Voltage Levels	
	CLOCK_P or DATA_P	CLOCK_N or DATA_N
HS-0	Low (HS)	High (HS)
HS-1	High (HS)	Low (HS)
LP-00	Low (LP)	Low (LP)
LP-01	Low (LP)	High (LP)
LP-10	High (LP)	Low (LP)
LP-11	High (LP)	Low (LP)

Note: Ta = -30℃ to 70℃ (no damage up to 85℃)

19.3.1. DC characteristics for Power Lines

Parameter	Symbol	Condition	Specification			Unit
			Min.	Typ.	Max.	
Analog power supply voltage	VDDA	Operating voltage	2.5	2.8	3.3	V
Digital power supply voltage	VDDI	I/O supply voltage	1.65	1.8	3.3	V
Analog power supply voltage noise	V_{VDDA_NOISE}	Noise Range, 0 to 100MHz, Sinusoidal Wave (peak-to-peak)	-	-	100	mV
		Noise Range, 0 to 30kHz, Pulse Wave with Duty Cycle (50%/50%)	-	-	500	mV
I/O power supply voltage noise	V_{VDDI_NOISE}	Noise Range, 0 to 100MHz, Sinusoidal Wave (peak-to-peak)	-	-	100	mV

Notes:

1. Ta = -30℃ to 70℃ (no damage up to 85℃)
2. These values are not symmetric amplitude, which center points are VDDI or VDDA. See examples, when V_{VDDA_NOISE} and V_{VDDI_NOISE} are maximums, below for reference.

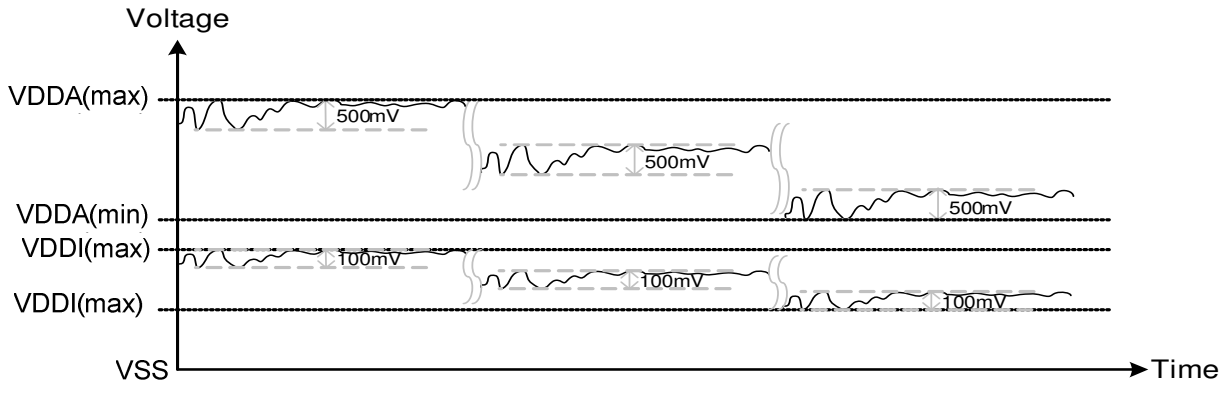


Figure 186: Noise on Power Supply Lines

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19.3.2. DC Characteristics for DSI LP Mode

DC levels of the LP-00, LP-01, LP-10, and LP-11 are defined in the table below: DC Characteristics for DSI LP mode when LP-RX, LP-CD, or LP-TX is mentioned in the condition column. Other logical levels of the table are for the MPU interface.

Parameter	Symbol	Condition	Specification			Unit
			Min.	Typ.	Max.	
Logic High level output voltage	V_{OH}	I_{OUT} = -1mA, Note 2	0.8 V_{VDDAM}	-	V_{VDDAM}	V
Logic Low level output voltage	V_{OL}	I_{OUT} = 1Ma, Note 2	0.0	-	0.2V_{VDDAM}	V
Logic High level input voltage	V_{IHLPCD}	LP-CD, Note 3	450	-	1350	mV
Logic Low level input voltage	V_{ILLPCD}	LP-CD, Note 3	0.0	-	200	mV
Logic High level input voltage	V_{IHLPRX}	LP-RX (CLK, D0 ,D1), Note 3	880	-	1350	mV
Logic Low level input voltage	V_{ILLPRX}	LP-RX (CLK, D0 ,D1), Note 3	0.0	-	550	mV
Logic Low level input voltage	$V_{ILLPRXULP}$	LP-RX (CLK ULP mode), Note 3	0.0	-	300	mV
Logic high level output voltage	V_{OHLPTX}	LP-TX (D0), Note 3	1.1	-	1.3	V
Logic Low level output voltage	V_{OLLPTX}	LP-TX (D0), Note 3	-50	-	50	mV
Logic High level input current	I_{IH}	LP-CD, LP-RX, Note 3	-	-	10	uA
Logic Low level input current	I_{IL}	LP-CD, LP-RX, Note 3	-10	-	-	uA

Notes:

1.Ta = -30℃ to 70℃ (no damage up to 85℃)

2. LEDPWM, TE_ENABLE

3. The DSI High Speed mode is Off.

19.3.3. Spike/Glitch Rejection

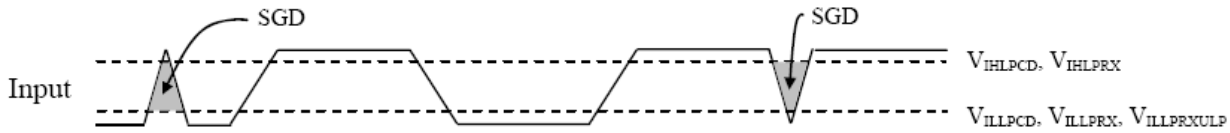


Figure 187: Spike/Glitch Rejection

Notes:

1. A spike/glitch can be rejected when the Peak Interference Amplitude is 200mV (at maximum) and the Interference Frequency is 450 MHz (at the very least).

2. n = 0 and 1.

Table 50: Spike/Glitch Rejection

Spike/Glitch Rejection – DSI					
Signal	Symbol	Parameter	Min	Max	Unit
DSI-CLK+/-, DSI-Dn+/-	SGD	Input pulse rejection for DSI	-	300	Vps

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19.3.4. DC Characteristics for DSI HS Mode

Parameter	Symbol	Condition	Specification			Unit
Input Common Mode Voltage for Clock	V_{CMCLK}	DSI-CLK+/- Note 2, Note 3	70	-	330	mV
Input Common Mode Voltage for Data	V_{CMDATA}	DSI-Dn+/- Note 2, Note 3, Note 5	70	-	330	mV
Common Mode Ripple for Clock Equal or Less than 450MHz	$V_{CMRCLKL450}$	DSI-CLK+/- Note 4	-50	-	50	mV
Common Mode Ripple for Data Equal or Less than 450MHz	$V_{CMRDATAL450}$	DSI-Dn+/- Note 4, Note 5	-50	-	50	mV
Common Mode Ripple for Clock More than 450MHz (peak sine wave)	$V_{CMRCLKM450}$	DSI-CLK+/-	-	-	100	mV
Common Mode Ripple for Data More than 450MHz (peak sine wave)	$V_{CMRDATAM450}$	DSI-Dn+/- Note 5	-	-	100	mV
Differential Input Low Level Threshold Voltage for Clock	$V_{THLCLK-}$	DSI-CLK+/-	-70	-	-	mV
Differential Input Low Level Threshold Voltage for Data	$V_{THLDATA-}$	DSI-Dn+/- Note 5	-70	-	-	mV
Differential Input High Level Threshold Voltage for Clock	$V_{THHCLK+}$	DSI-CLK+/-	-	-	70	mV
Differential Input High Level Threshold Voltage for Data	$V_{THHDATA+}$	DSI-Dn+/- Note 5	-	-	70	mV
Single-ended Input Low Voltage	V_{ILHS}	DSI-CLK+/-, DSI-Dn+/- Note 3, Note 5	-40	-	-	mV
Single-ended Input High Voltage	V_{IHHS}	DSI-CLK+/-, DSI-Dn+/- Note 3, Note 5	-	-	460	mV
Differential Termination Resistor	R_{TERM}	DSI-CLK+/-, DSI-Dn+/- Note 5	80	100	125	Ω
Single-ended Threshold Voltage for Termination Enable	$V_{TERM-EN}$	DSI-CLK+/-, DSI-Dn+/- Note 5	-	-	450	mV
Termination Capacitor	C_{TERM}	DSI-CLK+/-, DSI-Dn+/- Note 5, Note 6	-	-	60	pF

Notes:

1. Ta = -30℃ to 70℃ (no damage up to 85℃), VDDI = 1.65 to 1.95V, GND= 0V.
2. Includes 50mV (-50mV to 50mV) ground difference
3. Without VCMRCLKM450/VCMRDATAM450
4. Without 50mV (-50mV to 50mV) ground difference
5. n = 0 or 1
6. For higher bit rates, a 14pF capacitor will be needed to meet the common-mode return loss specification.

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The DSI receiver (HS mode) is understanding that there is logical '1' (HS-1) when a differential voltage is more than VTHH (CLK+/DATA+) and the DSI receiver (HS mode) is understanding that there is logical '0' (HS-0) when a differential voltage is more than VTHL (CLK-/DATA-). There is undefined state if the differential voltage is less than VTHH (CLK+/DATA+) and less than VTHL (CLK-/DATA-). A reference figure is below.

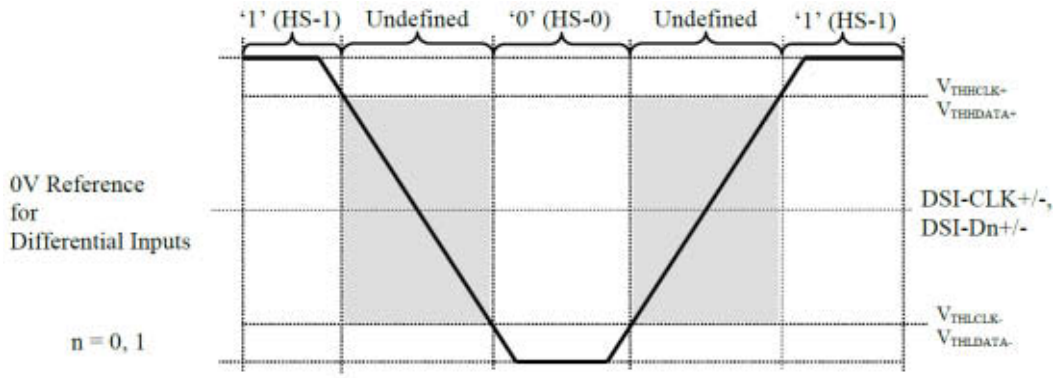
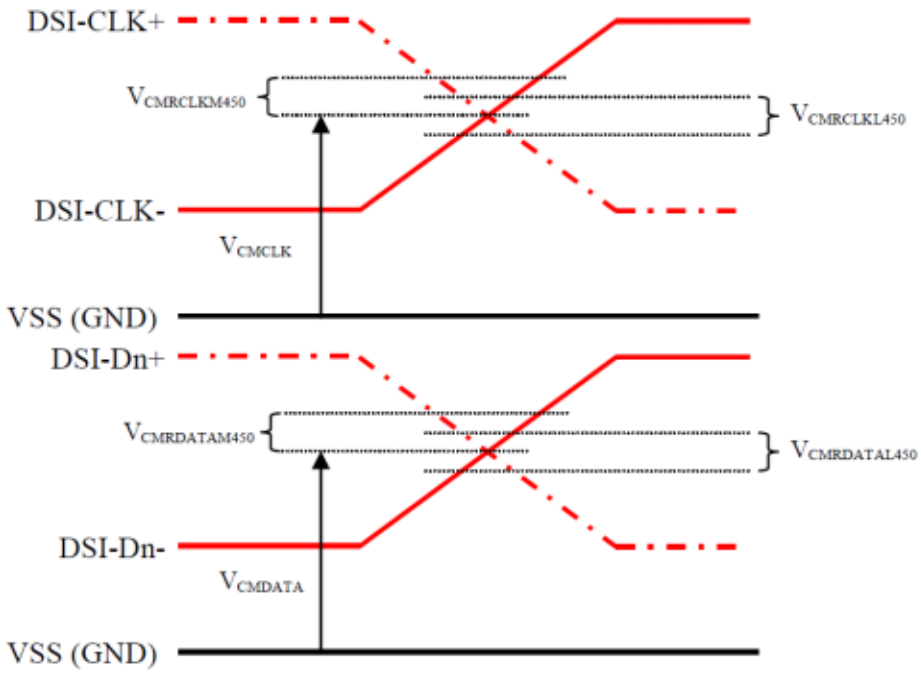


Figure 188: Differential Inputs Logical 0 and 1, Threshold High/Low, Differential Voltage Range



Note: n = 0 or 1

Figure 189: Common Mode Voltage on Clock and Data Channels

The termination resistor (RTERM) of the differential DSI receiver can be driven to two different states by the receiver:

- Low Power (LP) mode when the termination resistor is not connected between differential inputs (DSI-CLK+ <=> DSI-CLK- or DSI-D0+ <=> DSI-D0- or DSI-D1+ <=> DSI-D1-)
- High Speed (HS) mode when the termination resistor is connected between differential inputs (DSI-CLK+ <=> DSI-CLK- or DSI-D0+ <=> DSI-D0- or DSI-D1+ <=> DSI-D1-)

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The termination switch (HS/LP), when the termination resistor is not connected, is illustrated below.

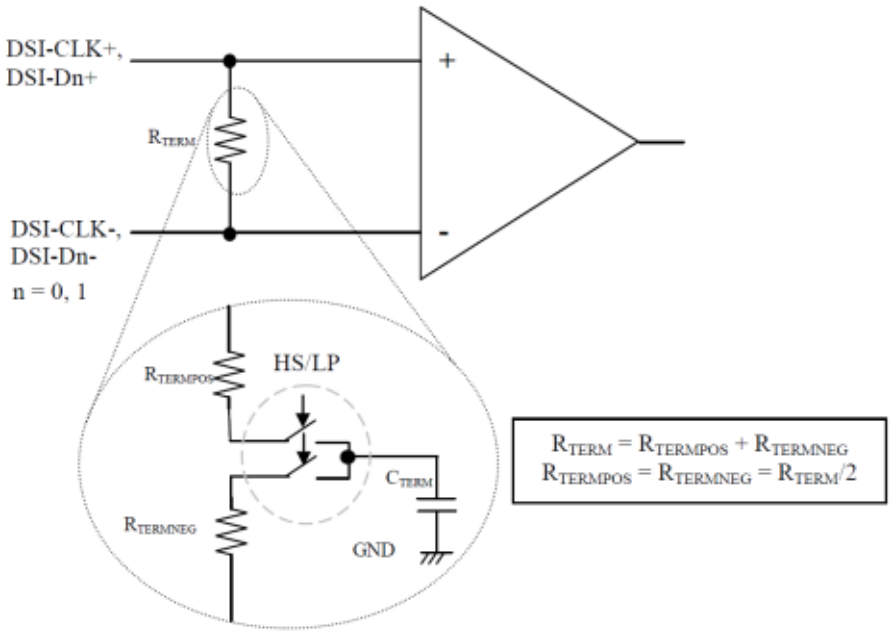


Figure 190: Differential Pair Termination Resistor on the Receiver Side

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19.3.5. DC Characteristics for MDDI

Client Receiver Electrical Specifications

Parameter	Description	Min	Typ	Max	Unit
V_{IT+}	Receiver differential input high threshold voltage. Above this differential voltage the input signal must be interpreted as a logic-one level.		0	50	mV
V_{IT-}	Receiver differential input low threshold voltage. Below this differential voltage the input signal must be interpreted as a logic-zero level.	-50	0		mV
V_{IT+}	Receiver differential input high threshold voltage (offset for hibernation wake-up). Above this differential voltage the input signal must be interpreted as a logic-one level.		100	125	mV
V_{IT-}	Receiver differential input low threshold voltage (offset for hibernation wake-up). Below this differential voltage the input signal must be interpreted as a logic-zero level.	75	100		mV
$V_{Input-Range}$	Allowable receiver input voltage range with respect to client ground.	0.5		1.2	V
R_{term}	Parallel termination resistance value	98	100	102	Ω

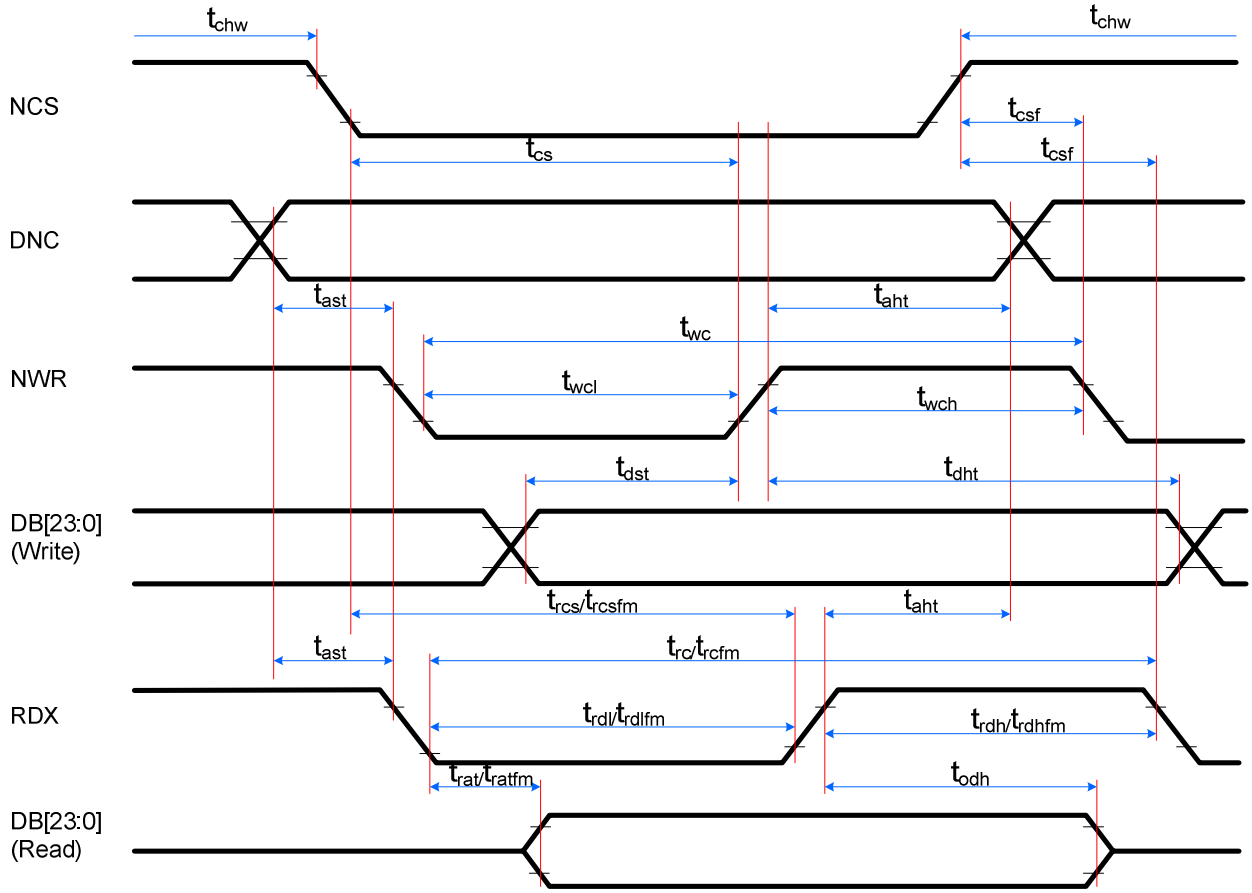
MDDI V1.2 Client Transmitter Electrical Specifications

Parameter	Description	Min	Typ	Max	Unit
$V_{output-Range-In}$	Allowable client driver output voltage range with respect to client ground.	0.6		1.1	V
I_{OD+}	Driver differential output high current corresponding to a logic-one level (while driving the equivalent of the pull-up and pull-down circuits that exist at the host and client)	1.5		2.5	mA
I_{OD-}	Driver differential output low current corresponding to a logic-zero level (while driving the equivalent of the pull-up and pull-down circuits that exist at the host and client)	-2.5		-1.5	mA

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AC Characteristics

19.3.6. Display Parallel 24/18/16/9/8-bit Interface Timing Characteristics (8080-series)

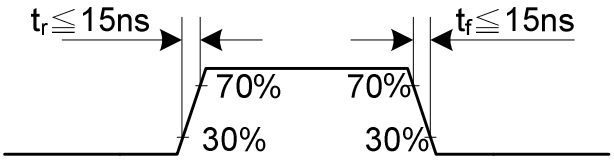


Signal	Symbol	Parameter	min	max	Unit	Description
DCX	tast	Address setup time	0	-	ns	-
	taht	Address hold time (Write/Read)	10	-	ns	-
CSX	tchw	CSX “H” pulse width	0	-	ns	-
	tcs	Chip Select setup time (Write)	10	-	ns	-
	trcs	Chip Select setup time (Read ID)	45	-	ns	-
	trcsfm	Chip Select setup time (Read FM)	355	-	ns	-
	tcsf	Chip Select Wait time (Write/Read)	10	-	ns	-
WRX	twc	Write cycle	30	-	ns	-
	twrh	Write Control pulse H duration	10	-	ns	-
	twrl	Write Control pulse L duration	10	-	ns	-
RDX (FM)	trcfm	Read Cycle (FM)	450	-	ns	When read from the Frame Memory
	trdhfm	Read Control H duration (FM)	90	-	ns	
	trdlfm	Read Control L duration (FM)	355	-	ns	
RDX (ID)	trc	Read cycle (ID)	160	-	ns	When read ID data
	trdh	Read Control pulse H duration	90	-	ns	
	trdl	Read Control pulse L duration	45	-	ns	
DB[17:0], DB[15:0], DB[8:0] DB[7:0]	tdst	Write data setup time	10	-	ns	CL = 30pF (maximum) CL = 8pF (minimum)
	tdht	Write data hold time	10	-	ns	
	trat	Read access time	-	40	ns	
	tratfm	Read access time	-	340	ns	
	trodh	Read output disable time	20	80	ns	

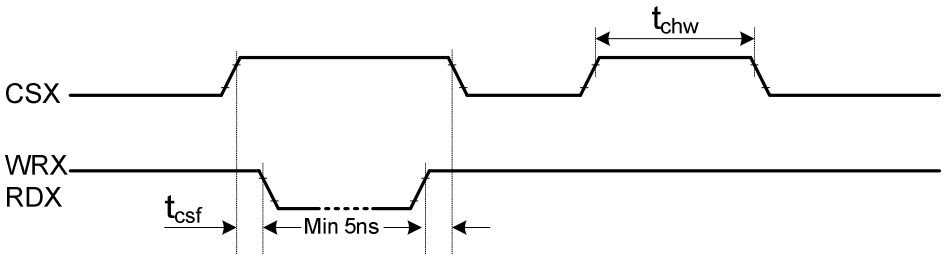
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Notes:

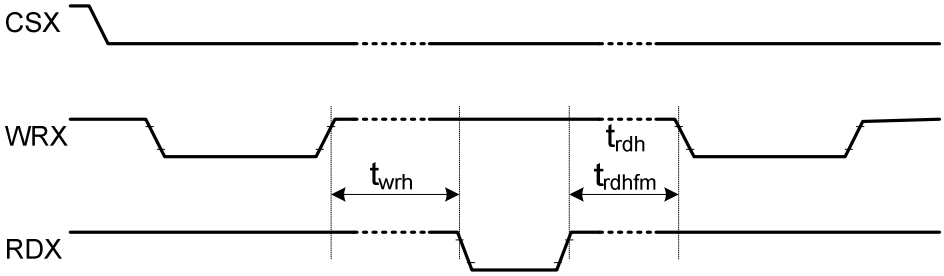
1. Ta = -30 to 70℃, VDDI = 1.65V to 3.3V, VDDA = 2.6V to 4.8V, VSSAM = GND= 0V
2. Logic high and low levels are specified as 30% and 70% of VDDI for input signals.
3. Input signal rising and falling time:



4. The CSX timing:

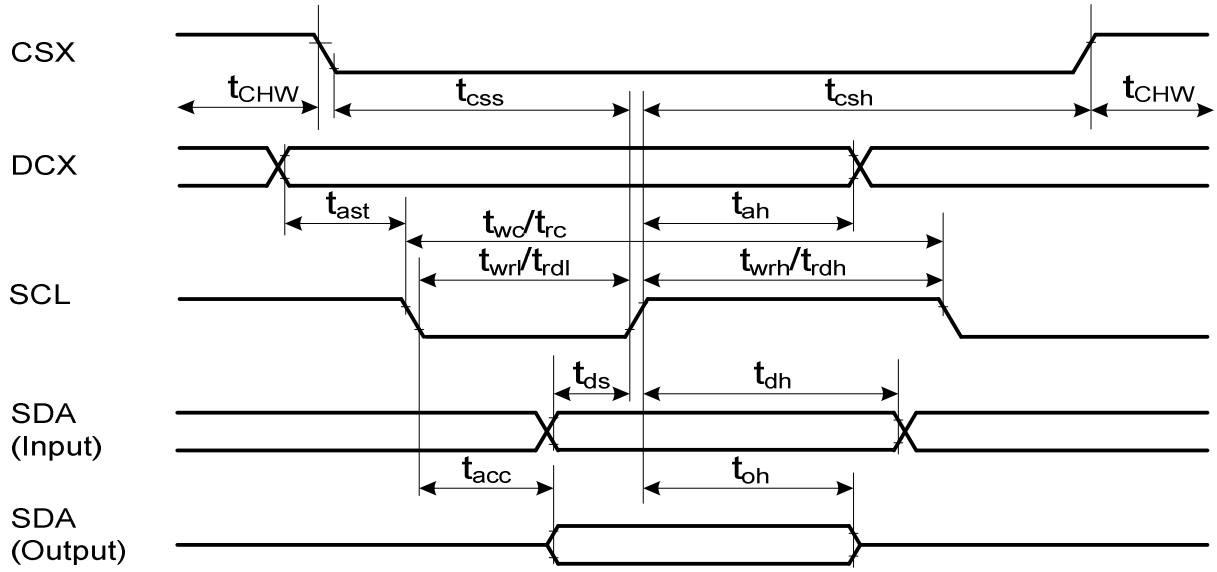


5. The Write-to-Read or the Read-to-Write timing:



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19.3.7. Display Serial Interface Timing Characteristics (3-line SPI system)



Signal	Symbol	Parameter	min	max	Unit	Description
CSX	tcss	Chip select time (Write)	15	-	ns	
	tcsh	Chip select hold time (Read)	15	-	ns	
	tCHW	CS "H" pulse width	40	-	ns	
SCL	twc	Serial clock cycle (Write)	30	-	ns	
	twrh	SCL “H” pulse width (Write)	10	-	ns	
	twrl	SCL “L” pulse width (Write)	10	-	ns	
	trc	Serial clock cycle (Read)	150	-	ns	
	trdh	SCL “H” pulse width (Read)	60	-	ns	
	trdl	SCL “L” pulse width (Read)	60	-	ns	
DCX	tas	DCX setup time	10	-	ns	
	tah	DCX hold time (Write/Read)	10	-	ns	
SDA (Input)	tds	Data setup time (Write)	10	-	ns	
	tdh	Data hold time (Write)	10	-	ns	
SDA (Output)	tacc	Access time (Read)	10	50	ns	CL = 30pF (maximum)
	toh	Output disable time (Read)	15	50	ns	CL = 8pF (minimum)

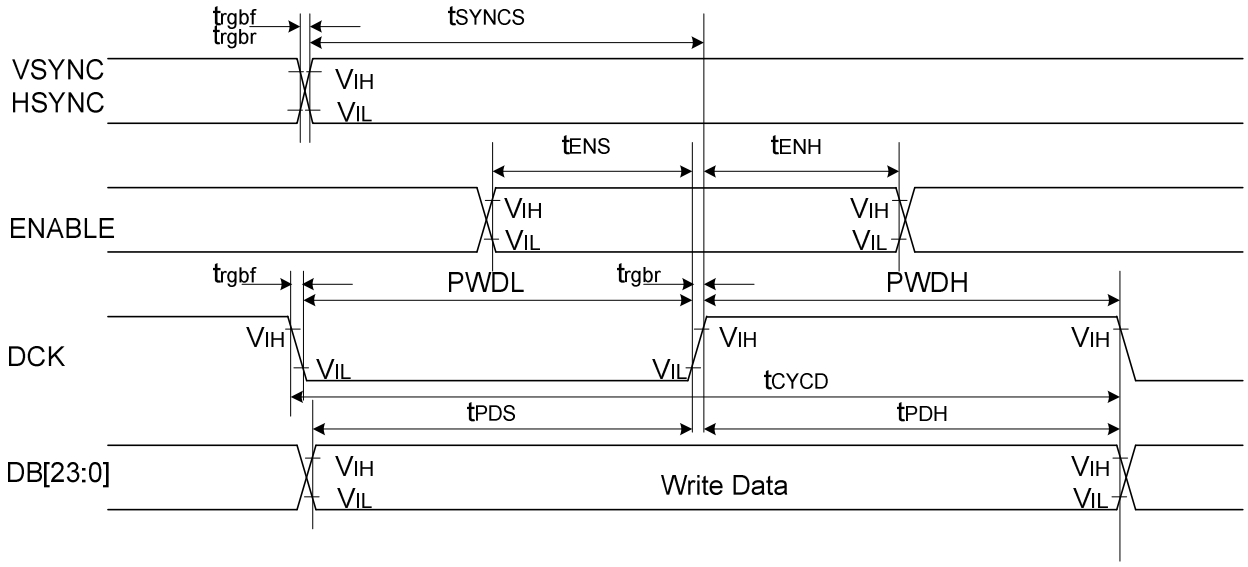
Notes:

1. Ta = -30 to 70℃, VDDI = 1.65V to 3.3V, VDDA = 2.6V to 4.8V, VSSAM = GND = 0V, T = 10+/-0.5ns.

2. Does not include signal rising and falling times.

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19.3.8. Parallel 24/18/16-bit RGB Interface Timing Characteristics



Signal	Symbol	Parameter	min	max	Unit	Description
VSYNC/ HSYNC	t_{SYNCS}	VSYNC/HSYNC setup time	5	-	ns	24/18/16-bit bus RGB interface mode
	t_{SYNCH}	VSYNC/HSYNC hold time	5	-	ns	
ENABLE	t_{ENS}	ENABLE setup time	5	-	ns	
	t_{ENH}	ENABLE hold time	5	-	ns	
DB [23:0]	t_{POS}	Data setup time	5	-	ns	
	t_{PDH}	Data hold time	5	-	ns	
DCK	PWDH	DCK high-level period	13	-	ns	
	PWDL	DCK low-level period	13	-	ns	
	t_{CYCD}	DCK cycle time	28	-	ns	
	t_{rgbr} , t_{rgbf}	DCK,HSYNC,VSYNC rise/fall time	-	15	ns	

Note: Ta = -30 to 70℃, VDDI = 1.65V to 3.3V, VDDA = 2.6V to 4.8V, VSSAM = GND = 0V

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19.3.9. DSI Timing Characteristics

19.3.10. High Speed Mode – Clock Channel Timing

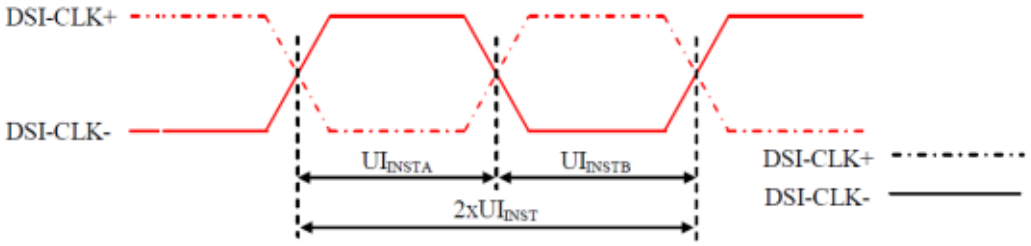


Figure 191: DSI Clock Channel Timing

Table 51: DSI Clock Channel Timing

Signal	Symbol	Parameter	Min	Max	Unit
DSI-CLK+/-	$2xUI_{INST}$	Double UI instantaneous	4	25	ns
DSI-CLK+/-	UI_{INSTA},UI_{INSTB}	UI instantaneous Half	2	12.5	ns

Note: UI = UI_{INSTA} = UI_{INSTB}

19.3.11. High Speed Mode – Data Clock Channel Timing

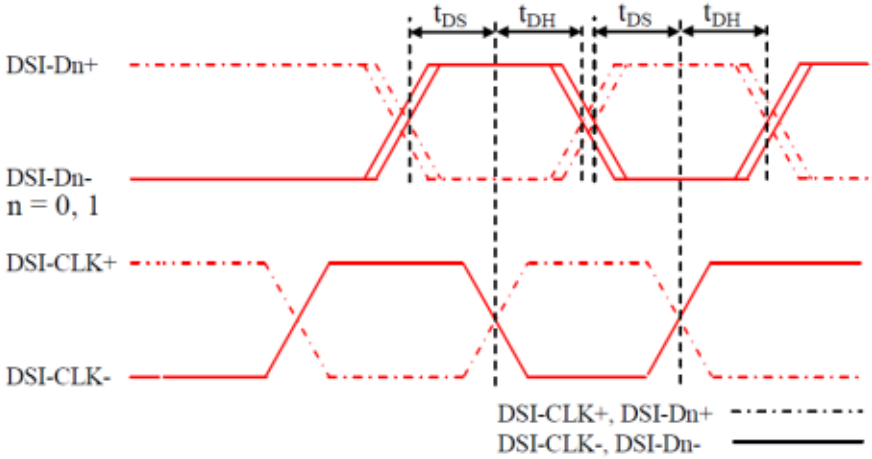


Figure 192: DSI Data to Clock Channel Timings

Table 52: DSI Data to Clock Channel Timings

Signal	Symbol	Parameter	Min	Max
DSI-Dn+/- , n=0 or 1	t_{DS}	Data to Clock Setup time	0.15xUI	-
	t_{DH}	Clock to Data Hold Time	0.15xUI	-

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19.3.12. High Speed Mode – Rising and Falling Timings

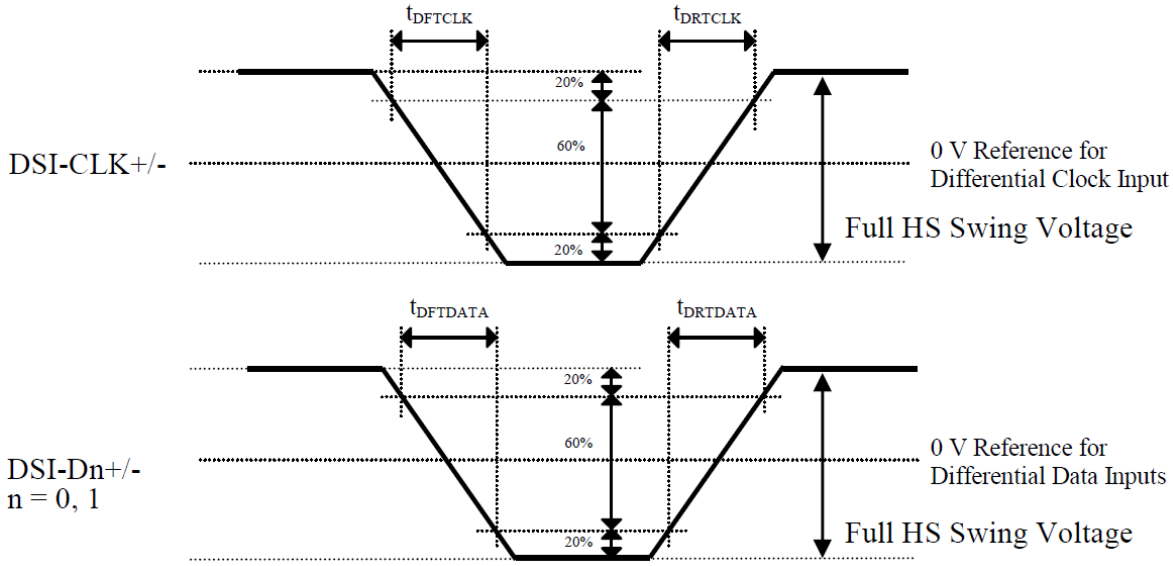


Figure 193: Rising and Falling Timings on Clock and Data Channels

Table 53: Rising and Falling Timings on Clock and Data Channels

Parameter	Symbol	Condition	Specification			
			Min	Typ	Max	Unit
Differential Rising Time for Clock	t_{DRTCLK}	DSI-CLK+/-	-	-	150 (Note 1)	ps
Differential Rising Time for Data	$t_{DRTDATA}$	DSI-Dn+/- n=0 and 1	-	-	150 (Note 1)	ps
Differential Falling Time for Clock	t_{DFTCLK}	DSI-CLK+/-	-	-	150 (Note 1)	ps
Differential Falling Time for Data	$t_{DFTDATA}$	DSI-Dn+/- n=0 and 1	-	-	150 (Note 1)	ps

Note: The display module has to meet timing requirements, what are defined for the transmitter (MPU) on MIPI D-Phy standard.

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19.3.13. Low Speed Mode – Bus Turn Around

Lower Power Mode and its State Periods on the Bus Turnaround (BTA) from the MPU to the Display Module (ILI9806G) sequence are illustrated below for reference purposes.

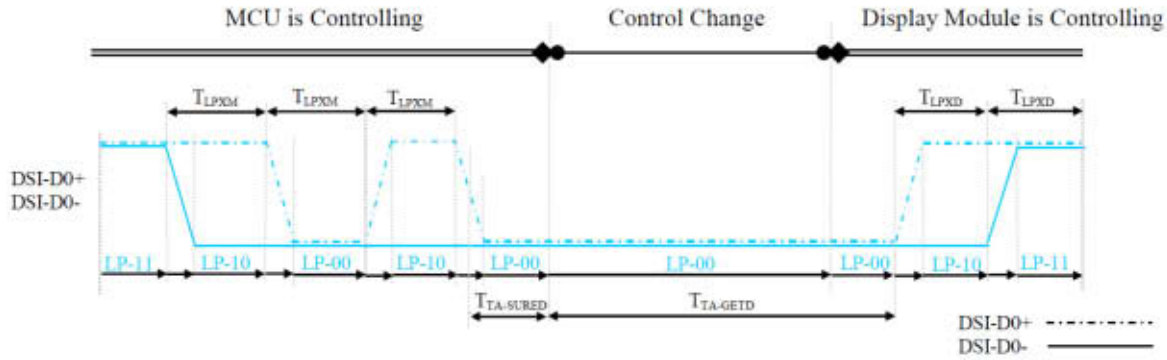


Figure 194: BTA from the MPU to the Display Module

Lower Power Mode and its State Periods on the Bus Turnaround (BTA) from the Display Module (ILI9806G) to the MPU sequence are illustrated below for reference purposes.

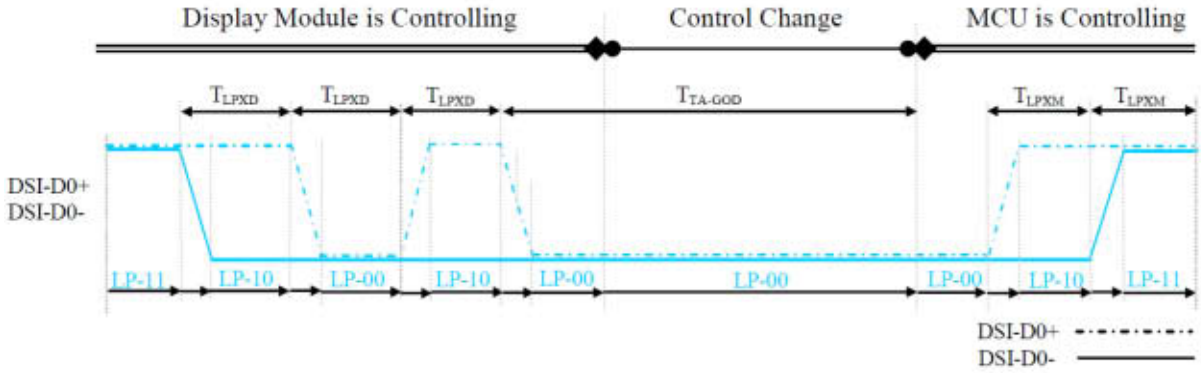


Figure 195: BTA from the Display Module to the MPU

Table 54: Low Power State Period Timings – A

Signal	Symbol	Description	Min	Max	Unit
DSI-D0+/-	T_{LPXM}	Length of LP-00, LP-01, LP-10 or LP-11 periods MPU ➔ Display Module (ILI9806G)	50	75	ns
DSI-D0+/-	T_{LPXD}	Length of LP-00, LP-01, LP-10 or LP-11 periods Display Module (ILI9806G) ➔ MPU	50	75	ns
DSI-D0+/-	$T_{TA-SURED}$	Time-out before the Display Module (ILI9806G) starts driving	T_{LPXD}	$2xT_{LPXD}$	ns

Table 55: Low Power State Period Timings – B

Signal	Symbol	Description	Time	Unit
DSI-D0+/-	$T_{TA-GETD}$	Time to drive LP-00 by Display Module (ILI9806G)	$5xT_{LPXD}$	ns
DSI-D0+/-	T_{TA-GOD}	Time to drive LP-00 after turnaround request - MPU	$4xT_{LPXD}$	ns

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19.3.14. Data Lanes from Low Power Mode to High Speed Mode

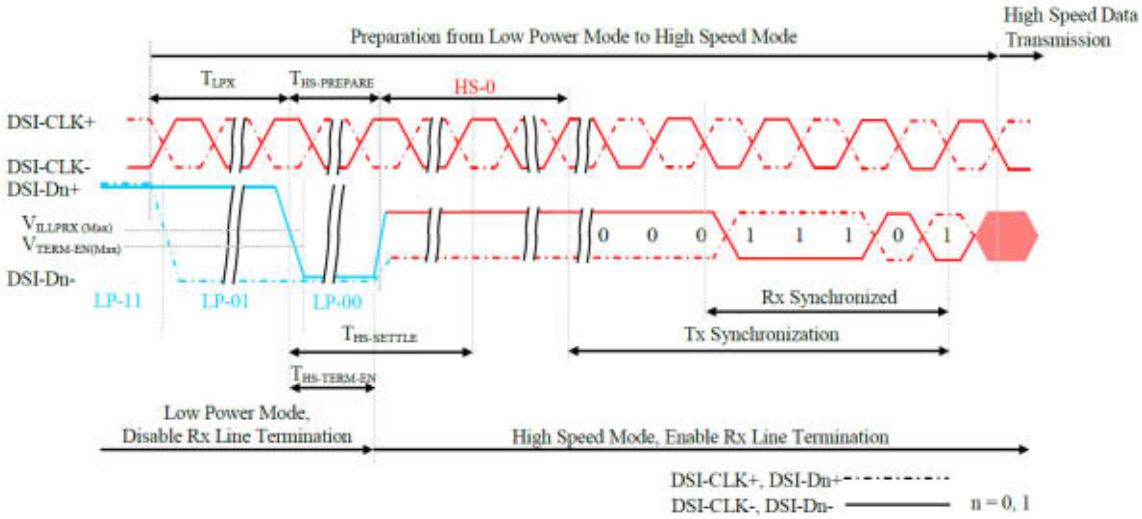


Figure 196: Data Lanes - Low Power Mode to High Speed Mode Timings

Table 56: Data Lanes - Low Power Mode to High Speed Mode Timings

Signal	Symbol	Description	Min	Max	Unit
DSI-Dn+/-, n=0 or 1	T_{LPX}	Length of any Low Power State Period	50	-	ns
DSI-Dn+/-, n=0 or 1	$T_{HS-PREPARE}$	Time to drive LP-00 to prepare for HS Transmission	40+4xUI	85+6xUI	ns
DSI-Dn+/-, n=0 or 1	$T_{HS-TERM-EN}$	Time to enable Data Lane Receiver line termination measured from when Dn crosses VILMAX	-	35+4xUI	ns

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19.3.15. Data Lanes from High Speed Mode to Low Power Mode

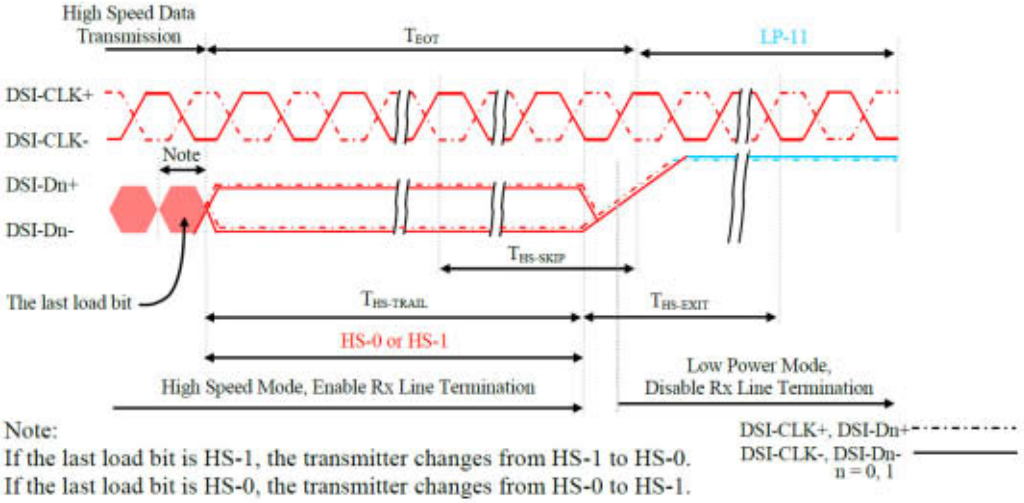


Figure 197: Data Lanes - High Speed Mode to Low Power Mode Timings

Table 57: Data Lanes - High Speed Mode to Low Power Mode Timings

Signal	Symbol	Description	Min	Max	Unit
DSI-Dn+/-, n=0 or 1	$T_{HS-SKIP}$	Time-Out at Display Module (ILI9806G) to ignore transition period of EoT	40	55+4xUI	ns
DSI-Dn+/-, n=0 or 1	$T_{HS-EXIT}$	Time to driver LP-11 after HS burst	100	-	ns

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19.3.16. DSI Clock Burst – High Speed Mode to/from Low Power Mode

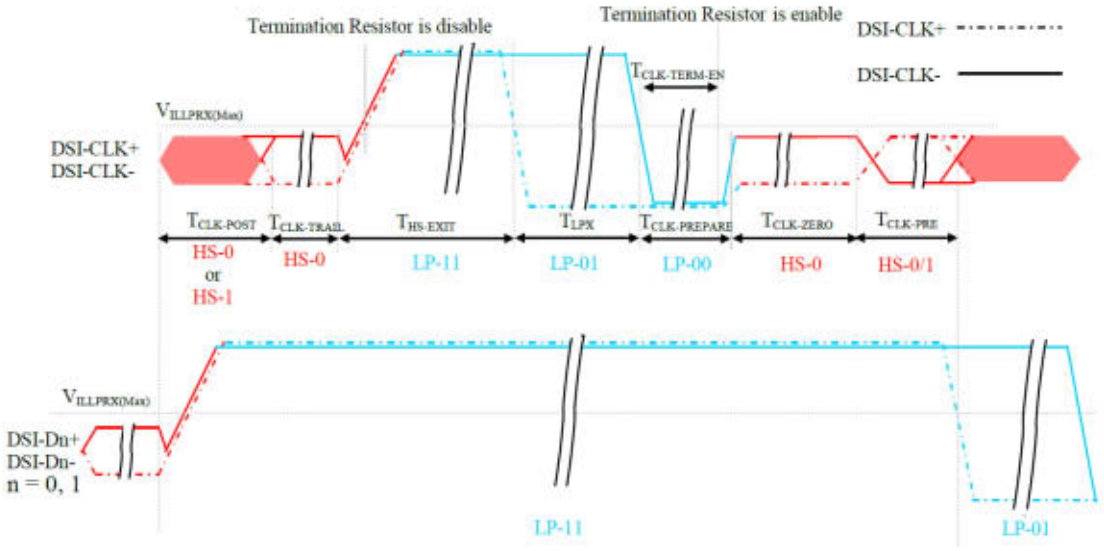


Figure 198: Clock Lanes - High Speed Mode to/from Low Power Mode Timings

Table 58: Clock Lanes - High Speed Mode to/from Low Power Mode Timings

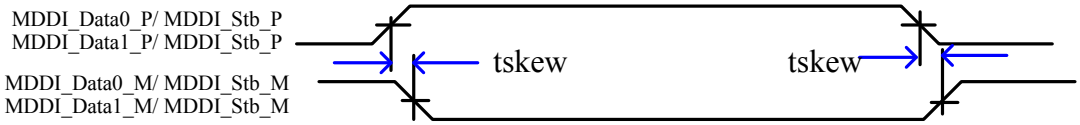
Signal	Symbol	Description	Min	Max	Unit
DSI-CLK+/-	$T_{CLK-POST}$	Time that the MPU shall continue sending HS clock after the last associated Data Lanes has transitioned to LP mode	60+52xUI	-	ns
DSI-CLK+/-	$T_{CLK-TRAIL}$	Time to drive HS differential state after last payload clock bit of a HS transmission burst	60	-	ns
DSI-CLK+/-	$T_{HS-EXIT}$	Time to drive LP-11 after HS burst	100	-	ns
DSI-CLK+/-	$T_{CLK-PREPARE}$	Time to drive LP-00 to prepare for HS transmission	38	95	ns
DSI-CLK+/-	$T_{CLK-TERM-EN}$	Time-out at Clock Lane to enable HS termination	-	38	ns
DSI-CLK+/-	$T_{CLK-PREPARE}$	Minimum lead HS-0 drive period before starting Clock	300	-	ns
DSI-CLK+/-	$T_{CLK-PRE}$	Time that the HS clock shall be driven prior to any associated Data Lane beginning the transition from LP to HS mode	8xUI	-	ns

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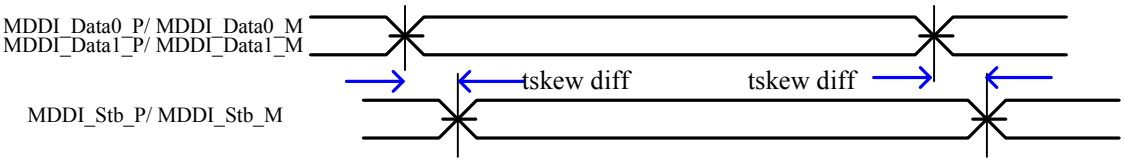
19.3.17. MDDI Timing Characteristics

Item	Symbol	Rating			Unit
		Min	Typ	Max	
Data transfer rate	1/t Bit	-	-	400	Mbps
Differential Transfer Input Skew	| ± tskew|	-	-	50	ps
Data_ Strobe Input Skew	| ± tskew diff|	-	-	0.45*tBIT	

Skew between positive and negative



Skew between Data and Strobe



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20. Application Circuit

20.1. Reference Circuit

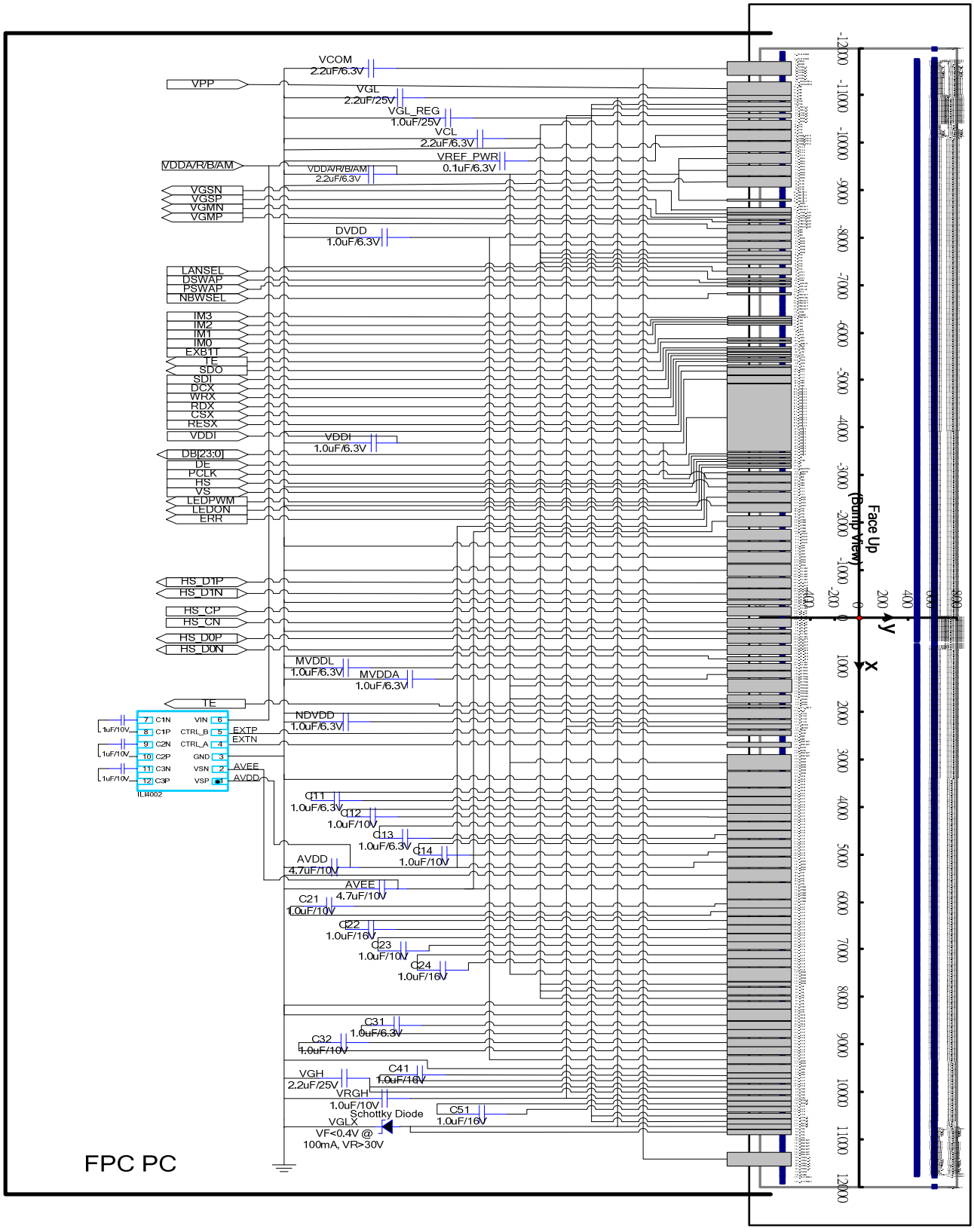


Figure 199: Reference Circuit

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20.2. External Component

Table 59 shows specifications of external elements connected to the power supply circuit of the ILI9806G.

Table 59: External Component Table

Pad Name	Connection	Typical Value
VDDI	Stable Capacitor to GND	1.0 µF (Max 6.3V)
VDDA, VDDR, VDDB, VDDAM	Stable Capacitor to GND	2.2 µF (Max 6.3V)
DVDD	Stable Capacitor to GND	1.0 µF (Max 6.3V)
NDVDD	Stable Capacitor to GND	1.0 µF (Max 6.3V)
MVDDA	Stable Capacitor to GND	1.0 µF (Max 6.3V)
MVDDL	Stable Capacitor to GND	1.0 µF (Max 6.3V)
C11P, C11N	Flying Capacitor	1.0 µF (Max 6.3V)
C12P, C12N	Flying Capacitor	1.0 µF (Max 10V)
C13P, C13N	Flying Capacitor	1.0 µF (Max 6.3V)
C14P, C14N	Flying Capacitor	1.0 µF (Max 10V)
C21P, C21N	Flying Capacitor	1.0 µF (Max 10V)
C22P, C22N	Flying Capacitor	1.0 µF (Max 16V)
C23P, C23N	Flying Capacitor	1.0 µF (Max 10V)
C24P, C24N	Flying Capacitor	1.0 µF (Max 16V)
C31P, C31N	Flying Capacitor	1.0 µF (Max 6.3V)
C32P, C32N	Flying Capacitor	1.0 µF (Max 10V)
C41P, C41N	Flying Capacitor	1.0 µF (Max 16V)
C51P, C51N	Flying Capacitor	1.0 µF (Max 16V)
AVDD	Stable Capacitor to GND	4.7 µF (Max 10V)
AVEE	Stable Capacitor to GND	4.7 µF (Max 10V)
VCL	Stable Capacitor to GND	2.2 µF (Max 6.3V)
VGH	Stable Capacitor to GND	2.2 µF (Max 25V)
VGL	Stable Capacitor to GND	2.2 µF (Max 25V)
	Schottky Diode to GND	VF=0.52V / IF=1A , at TA=25℃ Rated Current= 1A, VR=40V (optional)
LVGL	Stable Capacitor to GND	1.0 µF (Max 25V)
VRGH	Stable Capacitor to GND	1.0 µF (Max 10V)
VCOM	Stable Capacitor to GND	2.2 µF (Max 6.3V)
VREF_PWR	Stable Capacitor to GND	0.1 µF (Max 6.3V)

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21. Revision History

Version No.	Date	Page	Description
V100	2014/04/17	—	New Creative

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